(e) Lubricate the threads of the bleed plug [52] and restrictor [54] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(f) Install the bleed plug [52] in the lower retract port.

(g) Tighten the bleed plug [52] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(h) Install the restrictor [54] in the upper retract port.

NOTE: The restrictor can be installed in either direction.

(i) Tighten the restrictor [54] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

SUBTASK 78-31-03-020-019-G00

(9) Connect and tighten the hydraulic lines:

(a) Remove the protective covers from the hydraulic fittings.

(b) Connect the elbow [50] to the tee fitting [56] and the restrictor [54].

NOTE: Hand tighten the nuts on the elbow at this time.

(c) Tighten the bolts on the clamp blocks that hold the upper and lower retract lines to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(d) Tighten the nuts on the elbow [50] at the tee fitting [56] and the restrictor [54] as follows:

1) Tighten the nuts to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the nuts.

3) Tighten the nuts again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(e) Temporarily install the upper and lower sync shaft and tubing:

NOTE: To manually retract the thrust reverser, the sync shafts must be temporarily installed.

1) Do this task: Sync Shaft Inspection, TASK 78-31-04-200-801-G00.

2) If there is contamination, remove the contamination from the sync shaft with a cotton wiper, G00034.

3) Move the coupling nut and tube adaptor along the tubing.

4) Insert the sync shaft into the tubing.

a) Make sure that the sync shaft engages the internal spline in the lower and upper actuator port.

5) Flex the sync shaft and insert the opposite end into the internal spline in the middle actuator port.

6) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

NOTE: This is a temporary installation and it is not necessary to double-torque the coupling nut.

SUBTASK 78-31-03-420-011-G00

(10) Install the sleeve lock proximity sensor [73] (TASK 78-34-03-400-801-G00).

SUBTASK 78-31-03-700-001-G00

(11) Do this task: Sleeve Lock Proximity Sensor Adjustment and Test, TASK 78-34-03-800-801-G00.

SUBTASK 78-31-03-980-004-G00

(12) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

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SUBTASK 78-31-03-420-012-G00

(13) Rig the upper and lower sync shafts as follows:

(a) Do this task: Sync Shaft Removal, TASK 78-31-04-000-801-G00.

(b) Do this task: Sync Shaft Installation, TASK 78-31-04-400-801-G00.

1) Make sure that the thrust reverser is completely retracted when rigging the sync shafts.

NOTE: This task uses the thrust reverser maintenance lock equipment, SPL-9002. There are two -5 wire bundle cable accessories in the B78009 tool assembly.

SUBTASK 78-31-03-440-003-G00

(14) If not done, do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-03-710-003-G00

(15) Do the operational test of the thrust reverser (TASK 78-31-00-700-801-G00).

SUBTASK 78-31-03-210-004-G00

(16) Examine the thrust reverser area for hydraulic fluid leaks (TASK 29-00-00-790-801).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-03-410-007-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-03-040-020-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

TASK 78-31-03-000-804-G00

8. Lower Hydraulic Actuator Removal

(Figure 403)

A. General

(1) This task gives the instructions to remove the lower hydraulic actuator.

(2) There are three hydraulic actuators on each torque box of the left and right thrust reversers on an engine.

(3) The upper and middle hydraulic actuators are locking actuators, the lower hydraulic actuators is a non-locking actuator.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

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(Continued)

Reference	Title
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)
STD-7895	Wrench - Allen, 3/16 inch
STD-14286	Allen Wrench - Hex, 1/4 Inch

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

G. Prepare for the Removal

SUBTASK 78-31-03-040-021-G00

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

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SUBTASK 78-31-03-860-003-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-03-040-022-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-03-040-023-G00

(4) Remove the hydraulic power from the applicable engine as follows (TASK 29-11-00-860-805):

(a) For Engine 1, remove the system A hydraulic power.

(b) For Engine 2, remove the system B hydraulic power.

SUBTASK 78-31-03-040-024-G00

(5) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-31-03-040-025-G00

(6) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-31-03-010-018-G00

(7) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-03-010-025-G00

(8) Remove the applicable access door [15].

Open these access panels:

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

SUBTASK 78-31-03-980-005-G00

(9) Expose the gimbal pins and prepare for the use of the actuator removal and installation tool:

NOTE: The thrust reverser sleeve must be extended to remove the gimbal pins from the aft side of the torque box and to use the actuator removal and installation tool.



CAUTION

DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN THE PERMITTED LIMIT. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. DAMAGE CAN OCCUR.

(a) For the inboard thrust reverser sleeve, manually extend the thrust reverser sleeve no more than 15 in. (381 mm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(b) For the outboard thrust reverser sleeve, manually extend the thrust reverser sleeve to the fully extended position (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing and if the sleeve is fully extended, the tool will be easier to use.

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SUBTASK 78-31-03-020-020-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(10) Drain the hydraulic fluid from the hydraulic lines (Figure 403 (Sheet 1), View B):

NOTE: To decrease hydraulic fluid spray when the coupling nuts are loosened, wrap cotton wiper, G00034, around the wrench, nut and hydraulic line.

(a) Wrap a cotton wiper, G00034, around the electrical connector on the sync lock receptacle on the lower actuator.

NOTE: The cloth will catch the hydraulic fluid and prevent damage to the electrical connector.

(b) Loosen the two bolts on each of the two clamp blocks that hold the lower hydraulic retract line [132].

NOTE: To remove the sleeve fitting on the hydraulic retract line from the actuator port, the hydraulic retract line must be carefully flexed. If the clampblocks are loose, the hydraulic retract line can be flexed easier.

(c) Disconnect the lower hydraulic retract line [132] at the lower actuator [103].

1) Carefully remove the sleeve fitting on the lower hydraulic retract line [132] from the actuator port.

2) let the hydraulic fluid drain in a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

(d) Disconnect the coupling nut on the sync shaft tubing at the lower actuator [103].

1) Let the hydraulic fluid drain into the 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

2) Move the coupling nut and tube adapter back along the tubing to get access to the sync shaft end.

3) Flex the sync shaft to remove it from the internal spline port in the lower actuator [103].

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CAUTION

DO NOT LET DIRT OR OTHER CONTAMINATION GET ON THE SYNC SHAFT. DAMAGE TO THE SYNC SHAFT CAN OCCUR.

4) Remove the sync shaft.

NOTE: Make sure that dirt or contamination does not get on the sync shaft.

(e) Remove the union [110] and packing [111] from the upper retract port of the lower actuator (Figure 403 (Sheet 1), View B).

NOTE: Keep the union [110] for the installation.

(f) Discard the packing [111].

(g) Install protective covers on the hydraulic lines and actuator ports.

SUBTASK 78-31-03-020-021-G00

(11) Remove the sync lock [105] as follows:

(a) Disconnect the electrical connector from the sync lock receptacle as follows (Figure 403 (Sheet 1), View B):

1) For the left thrust reverser, disconnect the electrical connector D1008.

2) For the right thrust reverser, disconnect the electrical connector D1016.

(b) Remove the nut [106] and washer [107] from the adapter plug [108].

NOTE: You do not have to remove the two bolts that hold the bracket to the sync lock. After the coupling nut is loose, the bracket will move off the stud.

(c) Loosen the coupling nut on the sync lock [105].

(d) Remove the sync lock [105] and bracket.

(e) Remove the adapter plug [108] and packing [109].

(f) Discard the packing [109].

(g) Install protective covers on the sync lock [105] and lower actuator [103].

H. Lower Actuator Removal

SUBTASK 78-31-03-020-022-G00

(1) Remove the two gimbal pins [120] that attach the lower actuator [103] to the gimbal ring on the aft side of the torque box as follows (Figure 403 (Sheet 2), View E):

(a) If you use the clamp assembly tool [42] (actuator tool, SPL-2439) to remove the gimbal pins [120], do these steps (Figure 403 (Sheet 4), View E):

NOTE: Use a allen wrench, STD-7895 to turn the clamp block screws [40] and use a 1/4 inch allen wrench, STD-14286 to turn the adjustment screw [41].

1) Loosen the clamp block screws [40] sufficiently so that the clamp block will fit over the lower actuator [103].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

3) Tilt the aft end of the clamp assembly tool down and over the aft cascade support ring.

4) Lower the forward end of the clamp assembly tool until the clamp block engages the lower actuator [103].

5) Tighten the adjustment screw [41] to move the clamp block aft.

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a) Move the clamp block until the aft edge touches where the diameter of the lower actuator [103] increases.

6) Tighten the clamp block screws [40].

7) Tighten the adjustment screw [41] to compress the fire seal [123] or fire seal [125] against the torque box.

(b) Remove a bolt [121] and washer [122] from each of the gimbal pins [120].

1) Remove the gimbal pins [120].

(c) If you used the clamp assembly tool [42] (actuator tool, SPL-2439), loosen the adjustment screw [41] and clamp block screw [40] to remove the tool.

SUBTASK 78-31-03-010-026-G00

(2) Retract the inboard and outboard thrust reverser sleeve to the stowed position (TASK 78-31-00-980-804-G00).

SUBTASK 78-31-03-020-029-G00

(3) Disconnect the rod end of the lower actuator [103] as follows (Figure 403 (Sheet 2), View H):

(a) Remove the nut [118], bolt [112], washer [113], washer [117], and the bushing [114] that attach the rod end of the actuator to the attach fitting on the sleeve.

(b) Disconnect the rod end of the lower actuator [103] from the actuator attach fitting.

SUBTASK 78-31-03-010-011-G00

(4) Do these steps to get the necessary clearance to remove the lower actuator [103]:

WARNING MAKE SURE THAT YOU MONITOR THE POSITION OF THE INBOARD THRUST REVERSER WHILE IT OPENS. MAKE SURE THAT THE INBOARD THRUST REVERSER DOES NOT TOUCH THE WING LEADING EDGE. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(a) For the inboard thrust reverser, open the thrust reverser.

(b) Make sure that the leading edge flaps are completely retracted.

NOTE: Without hydraulics to hold the flaps in the retract position, the weight of the flaps can cause them to extend a small amount.

1) If it is necessary, do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

(c) For the removal of the lower actuator [103] on the inboard thrust reverser, do this task:TASK 78-31-00-010-801-G00.

1) Install the actuator safety lock assembly, SPL-14551, on the inboard thrust reverser only.

a) Make sure to use safety lock PN C78035–3. This is the full open inboard tool.

SUBTASK 78-31-03-010-016-G00

(5) For the removal of the lower actuator [103] on the outboard thrust reverser, do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

(a) Make sure to install the actuator safety lock assembly, SPL-14551.

1) Make sure to use safety lock PN C78035–2. This is the full open outboard tool.

NOTE: The outboard thrust reverser will not touch the leading edge of the wing.

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SUBTASK 78-31-03-010-012-G00

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(6) Carefully remove the lower actuator [103].

SUBTASK 78-31-03-020-023-G00

(7) Remove the fire shield [124] and the fire seal [123] from the lower actuator [103] as follows (Figure 403 (Sheet 3), View F):

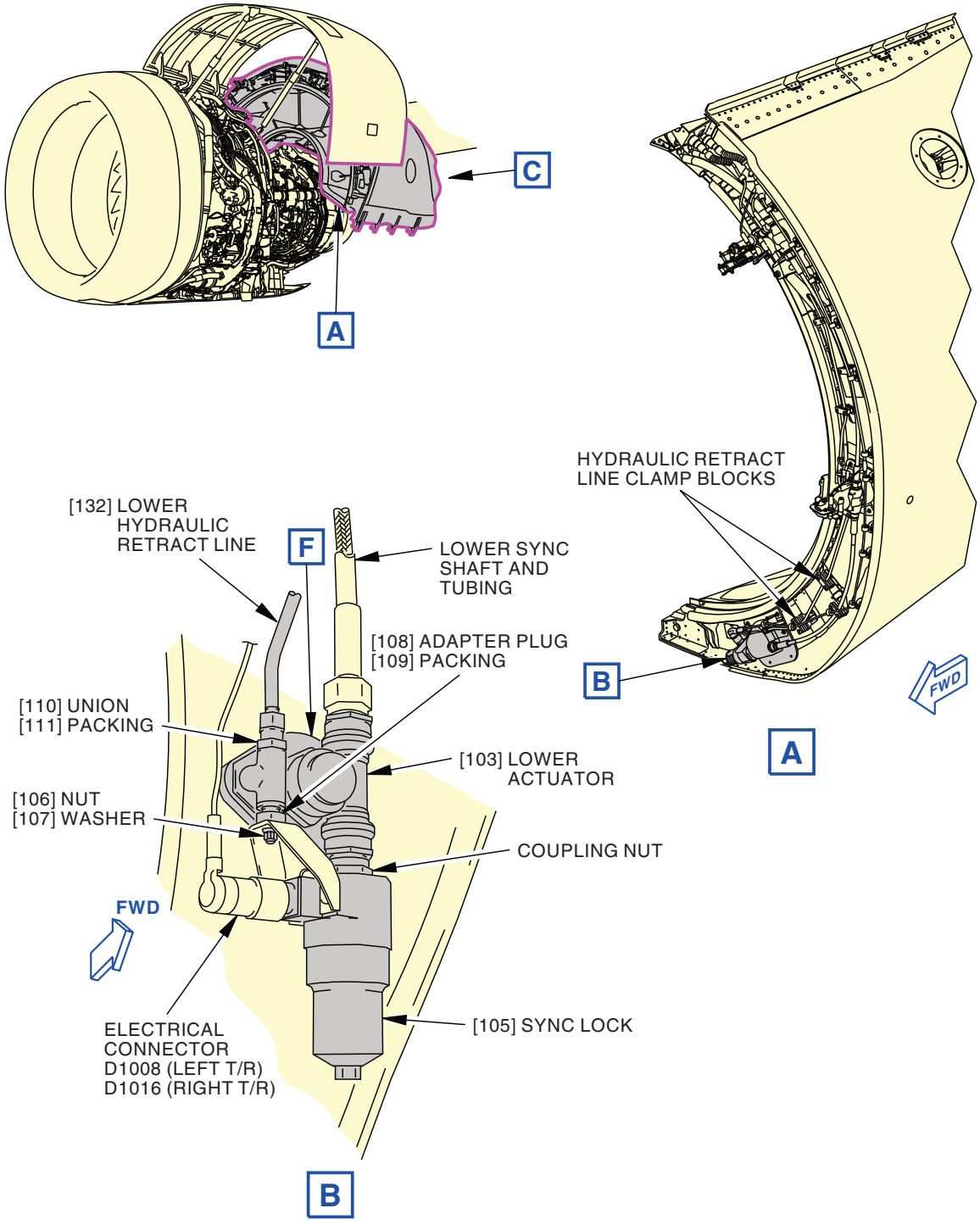
(a) Examine the fire seal [123] for damage.

(b) If you find no damage, keep the fire seal for the subsequent installation.

(c) If you find damage, replace the fire seal [123].

——— END OF TASK ———

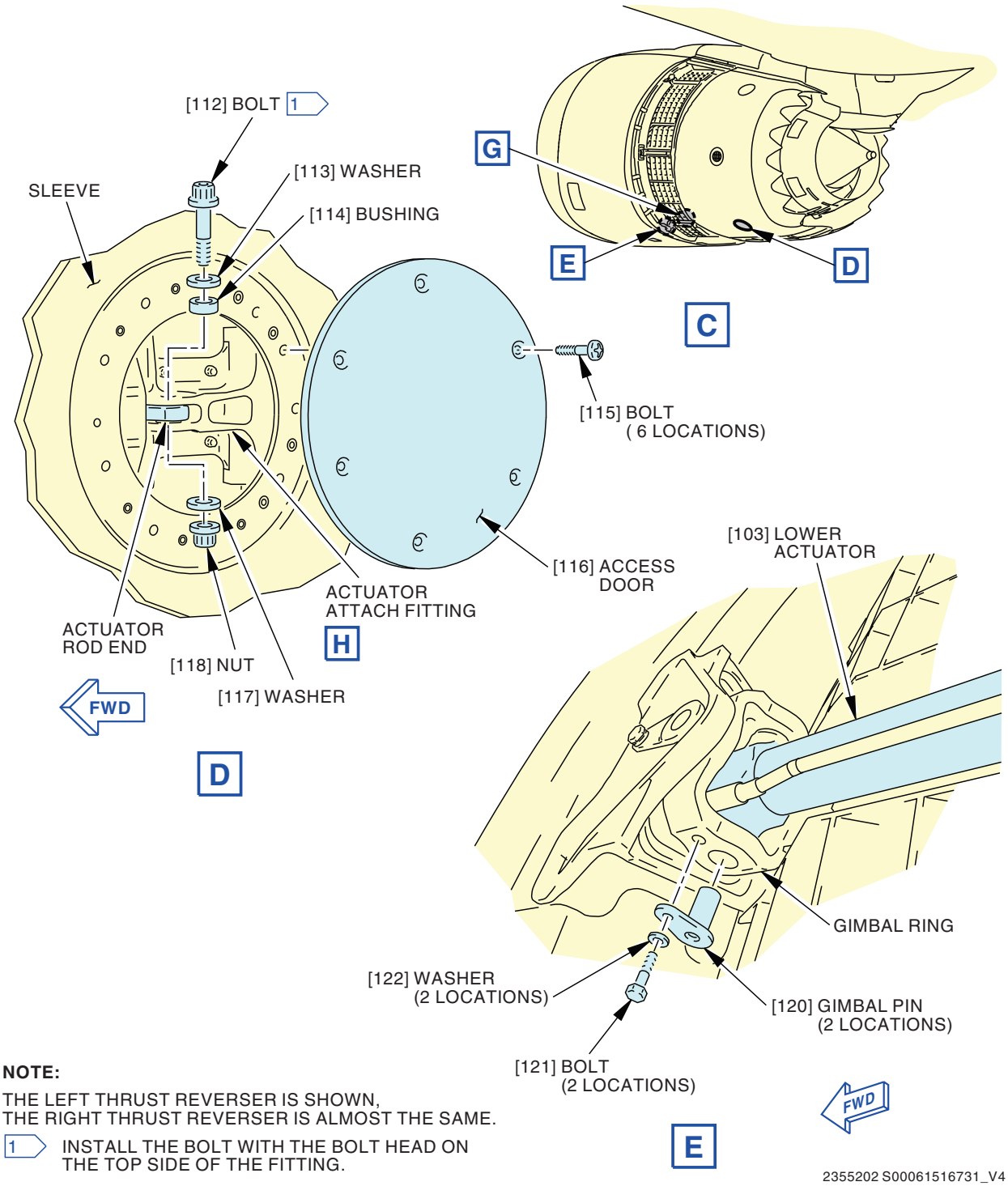
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355201 S00061516730_V2

Lower Actuator Installation
Figure 403/78-31-03-990-803-G00 (Sheet 1 of 5)

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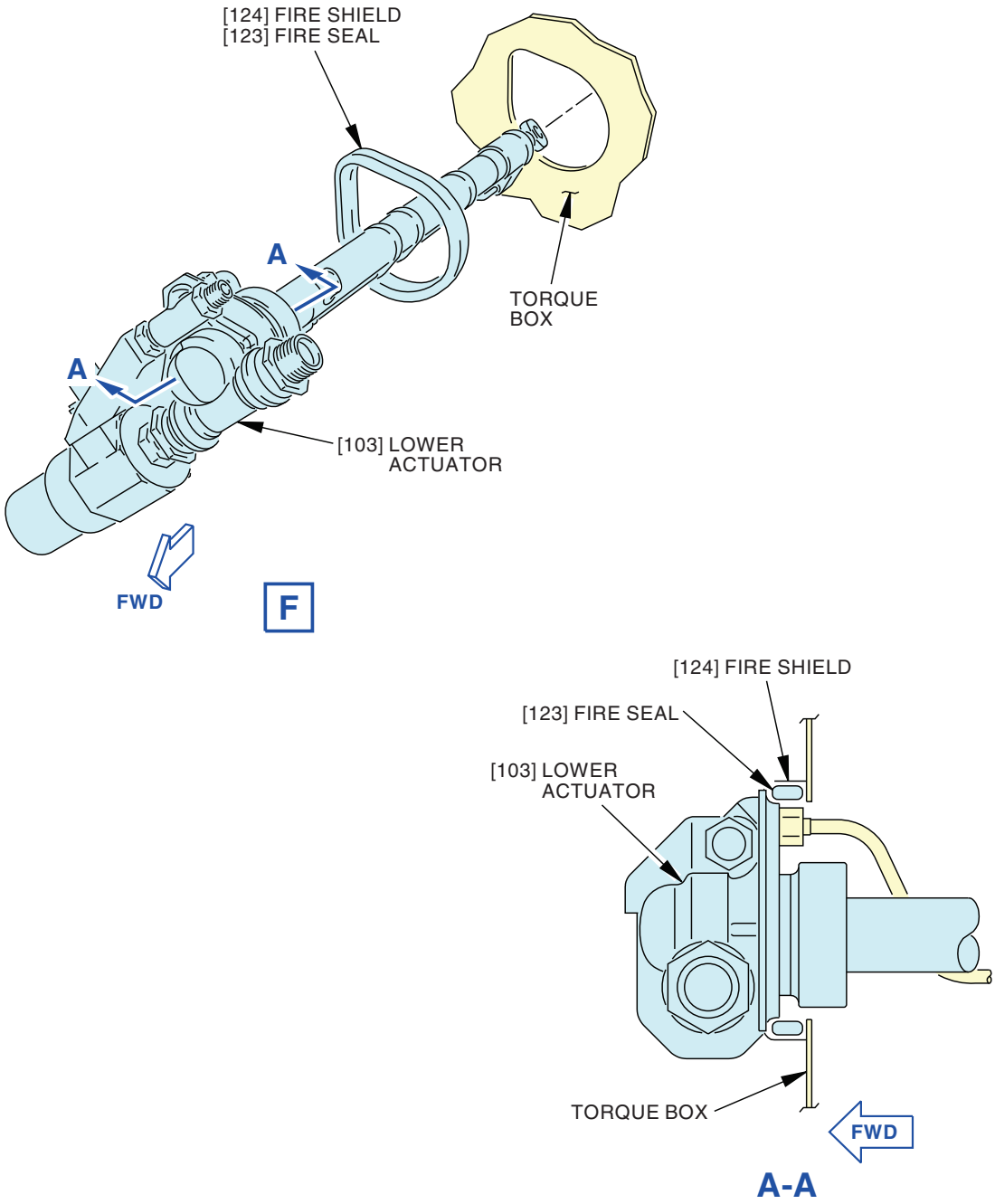
NOTE:

THE LEFT THRUST REVERSER IS SHOWN,
THE RIGHT THRUST REVERSER IS ALMOST THE SAME.

1 INSTALL THE BOLT WITH THE BOLT HEAD ON THE TOP SIDE OF THE FITTING.

Lower Actuator Installation
Figure 403/78-31-03-990-803-G00 (Sheet 2 of 5)

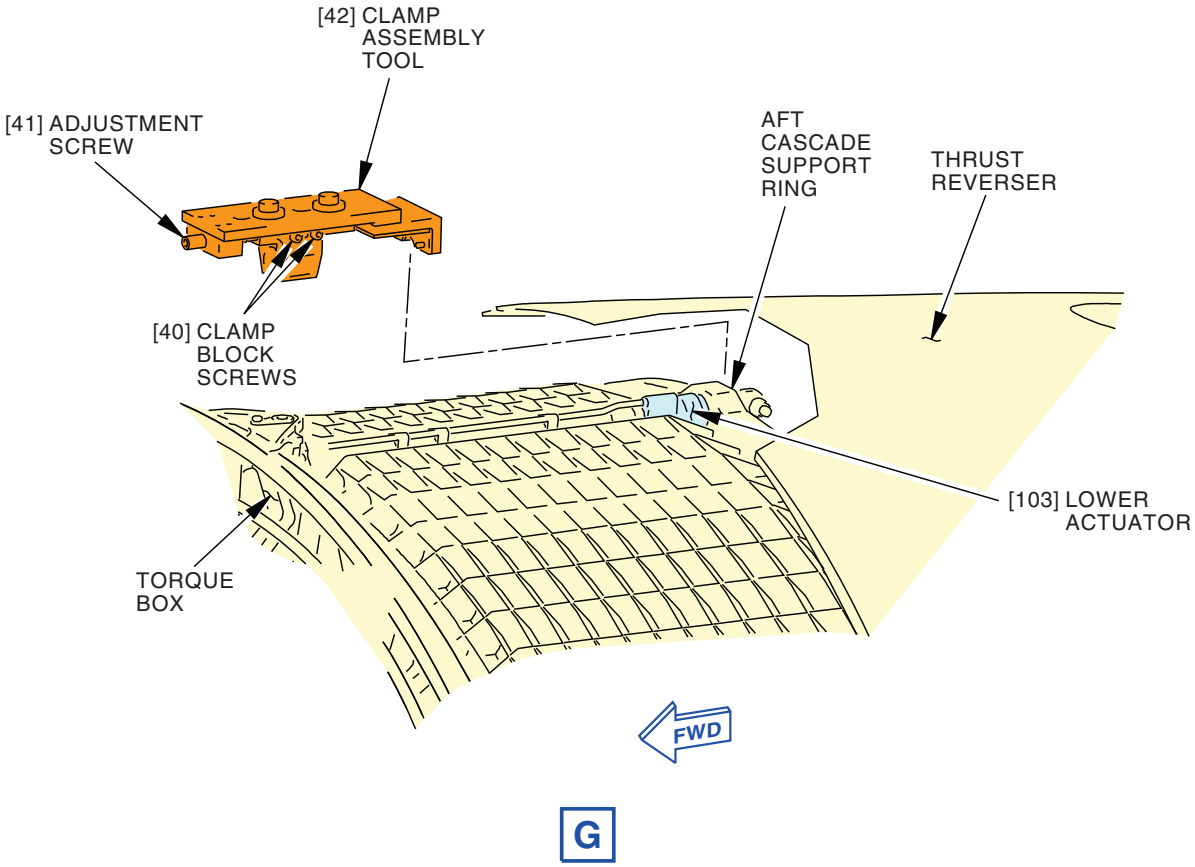
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355203 S00061516732_V2

Lower Actuator Installation
Figure 403/78-31-03-990-803-G00 (Sheet 3 of 5)

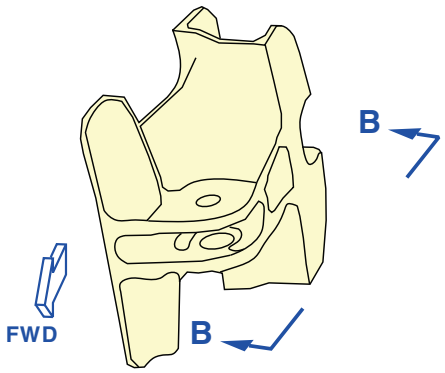
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355204 S00061516733_V2

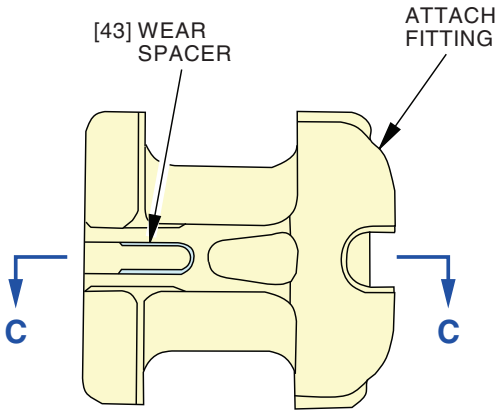
Lower Actuator Installation
Figure 403/78-31-03-990-803-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

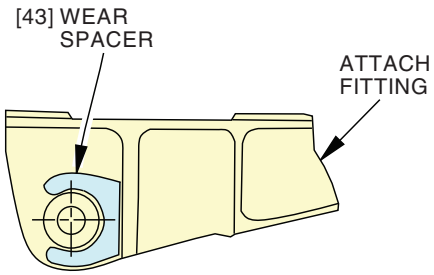


ACTUATOR ATTACH FITTING

H



B-B



C-C

2355205 S00061516734_V2

Lower Actuator Installation
Figure 403/78-31-03-990-803-G00 (Sheet 5 of 5)

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TASK 78-31-03-400-804-G00

9. Lower Hydraulic Actuator Installation

(Figure 403)

A. General

(1) This task gives the instructions to install the lower actuator.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-04-000-801-G00	Sync Shaft Removal (P/B 401)
78-31-04-200-801-G00	Sync Shaft Inspection (P/B 601)
78-31-04-400-801-G00	Sync Shaft Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2439	Tool - Thrust Reverser Actuator, Installation/Removal Part #: C78025-30 Supplier: 81205
SPL-9002	Lock Equipment - Thrust Reverser Maintenance Part #: B78009-26 Supplier: 81205
SPL-14551	Lock - Thrust Reverser Opening Actuator, Fully Extended, LEAP-1B Part #: C78035-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
109	Packing	78-31-03-01-160	PTF ALL
111	Packing	78-31-03-01-150	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

H. Lower Hydraulic Actuator Installation

SUBTASK 78-31-03-420-013-G00

(1) Install the fire seal [123] and fire shield [124] on the lower actuator [103] as follows:

(a) Put the fire seal [123] in the fire shield [124].

(b) Put the fire seal [123] and fire shield [124] over the rod end of the lower actuator [103] as follows:

1) Make sure that the fire seal [123] will contact the forward side of the torque box.

SUBTASK 78-31-03-420-014-G00

CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(2) Install the lower actuator [103] as follows:

NOTE: Recommended to not turn the actuator rod end. For actuators without a Linear Variable Differential Transformer (LVDT), accidental rotation of the rod end will not cause detrimental effects.

(a) Pull the rod end out of the lower actuator [103] by hand to extend as follows:

NOTE: Do not let the rod end turn when you extend it manually. Make sure that you hold the rod end when you extend the actuator manually. Do not use hydraulic power to extend the rod end.

1) For the inboard and outboard thrust reverser, extend the rod end approximately the same length as the one removed.

NOTE: It is also acceptable to extend the rod 1 in. (3 cm) - 2 in. (5 cm) longer to facilitate installation of the rod end bolt.

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CAUTION

DO NOT LET THE PUSH-ROD HIT THE AFT CASCADE-SUPPORT-RING OR THE TORQUE BOX. DAMAGE TO THE PUSH-ROD CAN OCCUR.

(b) Carefully insert the rod end through the opening in the torque box and aft cascade support ring.

SUBTASK 78-31-03-420-029-G00

(3) Align the rod end of the lower actuator [103] with the clevis on the attach fitting as follows:

(a) Make sure that the actuator attach fitting has a wear spacer [43].

1) If it is necessary, attach a wear spacer [43] into place with sealant, A00803.

NOTE: Seal around the edge of the fitting and inner sleeve.

(b) Install the bushing [114], washer [113], washer [117], bolt [112], and nut [118] that attach the rod end.

(c) Tighten the nut [118] to 370 in-lb (41.8 N·m) - 690 in-lb (78.0 N·m).

SUBTASK 78-31-03-860-004-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

(a) Remove the actuator safety lock assembly, SPL-14551, from the inboard thrust reverser only.

(b) Do not do the thrust reverser or leading edge activation.

(c) Do not close the fan cowl panels at this time.

SUBTASK 78-31-03-010-028-G00

(5) For the inboard thrust reverser, manually extend the sleeve no more than 15 in. (38 cm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(a) For the outboard thrust reverser sleeve, manually fully extend the thrust reverser sleeve (TASK 78-31-00-980-803-G00).

NOTE: The outboard thrust reverser sleeve will not touch the leading edge of the wing. The tool will be easier to use with the sleeve fully extended.

SUBTASK 78-31-03-420-015-G00

(6) Install the two gimbal pins [120] that attach the lower actuator [103] to the gimbal ring on the aft side of the torque box as follows:

(a) Apply grease, D00633, on the shanks of the two gimbal pins [120].

(b) Align the attachment holes in the lower actuator [103] and the gimbal ring.

(c) If you use the clamp assembly tool [42] (actuator tool, SPL-2439) to compress the fire seal [123] or fire seal [125] against the torque box, do these steps:

1) Loosen the clamp block screws [40] sufficiently, so that the clamp block will fit over the lower actuator [103].

2) Make sure that the adjustment screw [41] is retracted.

NOTE: The adjustment screw is on the forward end of the tool.

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3) Tilt the aft end of the clamp assembly tool [42] down and over the aft cascade support ring.

4) Lower the forward end of the clamp assembly tool [42] until the clamp block engages the lower actuator [103].

5) Tighten the adjustment screw [41] to move the clamp block aft.

NOTE: Move the clamp block until the aft edge touches where the diameter of the lower actuator increases.

6) Tighten the clamp block screws [40].

7) Tighten the adjustment screw [41] to compress the fire seal [123] or fire seal [125] against the torque box.

8) Turn the housing one direction or the other to align the attachment holes.

(d) Manually compress the fire seal [123] or fire seal [125] against the torque box as follows:

1) Push aft on the forward end of the lower actuator [103] to compress the fire seal [123] or fire seal [125] against the torque box.

2) Turn the housing one direction or the other to align the attachment holes.

(e) Install the two gimbal pins [120].

(f) Install a washer [122] and a bolt [121] in each of the gimbal pins [120].

(g) Tighten the bolts [121] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(h) If you used the clamp assembly tool [42], loosen the adjustment screw [41] and clamp block screws [40] to remove the tool.

SUBTASK 78-31-03-410-008-G00

(7) Install the applicable access doors [116]:

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

(a) Make sure that the installation surfaces are clean.

(b) Use sealant, A50110 to create a prepack or injection seal between the access door [116] and the installation surface on the thrust reverser.

(c) Put the access door [116] in the correct position to align the bolt holes.

(d) Install the six bolts [115] to attach the access door [116].

(e) Tighten the bolts [115] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-03-020-024-G00

(8) Do these steps if the replacement lower actuator [103] does not have the union [110] installed:

(a) Remove the protective covers from the lower actuator ports.

(b) Lubricate a new packing [111] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(c) Install the new packing [111] on the union [110].

(d) Lubricate the threads of the union [110] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(e) Install the union [110] in the upper retract port.

(f) Tighten the union [110] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

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SUBTASK 78-31-03-420-017-G00

(9) Install the sync lock [105] and the adapter plug [108] as follows:

(a) Lubricate a new packing [109] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the new packing [109] on the adapter plug [108].

(c) Lubricate the threads of the adapter plug [108] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(d) Install the adapter plug [108] in the lower retract port of the lower actuator.

(e) Tighten the adapter plug [108] to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(f) Put the sync lock [105] in the correct position.

NOTE: Make sure that the fastener hole in the bracket aligns with the stud on the adapter plug.

(g) Hand tighten the coupling nut on the sync lock [105].

(h) Install the washer [107] and nut [106] that connect the bracket to the adapter plug stud.

(i) Tighten the nut [106] to 65 in-lb (7.3 N·m) - 100 in-lb (11.3 N·m).

CAUTION

USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(j) Tighten the coupling nut on the sync lock [105] as follows:

1) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

SUBTASK 78-31-03-020-025-G00

(10) Connect the hydraulic lines as follows:

(a) Remove the protective covers from the hydraulic lines.

(b) Connect the lower hydraulic retract line [132] to the upper retract port of the lower actuator [103].

1) Hand tighten the coupling nut at this time.

(c) Tighten the two bolts on each of the two clamp blocks that hold the lower hydraulic retract line [132] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(d) Tighten the coupling nut on the lower hydraulic retract line [132] as follows:

1) Tighten the coupling nut to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 256 in-lb (28.9 N·m) - 283 in-lb (32.0 N·m).

(e) Temporarily install the lower sync shaft and tubing:

NOTE: To manually retract the thrust reverser, the sync shafts must be temporarily installed.

1) Do this task: Sync Shaft Inspection, TASK 78-31-04-200-801-G00.

2) If there is contamination, remove the contamination from the sync shaft with a cotton wiper, G00034.

3) Move the coupling nut and tube adaptors along the tubing.

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4) Insert the sync shaft into the tubing.

NOTE: Make sure that the sync shaft engages the internal spline in the middle actuator port.

5) Flex the sync shaft and insert the opposite end into the internal spline in the lower actuator port.

6) Tighten the coupling nut to 855 in-lb (96.6 N·m) - 945 in-lb (106.8 N·m).

NOTE: This is a temporary installation and it is not necessary to double-torque the coupling nuts.

SUBTASK 78-31-03-420-018-G00

(11) Connect the electrical connectors to the sync lock receptacle as follows:

(a) For the left thrust reverser, connect the electrical connector, D1008.

(b) For the right thrust reverser, connect the electrical connector, D1016.

SUBTASK 78-31-03-980-008-G00

(12) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

SUBTASK 78-31-03-420-019-G00

(13) Rig the lower sync shaft between the lower actuator [103] and the middle actuator [51] as follows:

(a) Do this task: Sync Shaft Removal, TASK 78-31-04-000-801-G00.

(b) Do this task: Sync Shaft Installation, TASK 78-31-04-400-801-G00.

1) Make sure that the thrust reverser is completely retracted when rigging the sync shafts.

NOTE: This task uses the thrust reverser maintenance lock equipment, SPL-9002. There are two -5 wire bundle cable assemblies in the B78009 tool assembly.

SUBTASK 78-31-03-440-004-G00

(14) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-03-710-004-G00

(15) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

(a) Operate the thrust reverser through the deploy and stow cycles until the sleeves move smoothly.

(b) Examine the thrust reverser area for hydraulic fluid leaks (TASK 29-00-00-790-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-03-410-009-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-03-040-030-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

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THRUST REVERSER HYDRAULIC ACTUATORS - INSPECTION/CHECK

1. General

A. This procedure has one task.

(1) Thrust reverser hydraulic actuator rod end inspection.

TASK 78-31-03-610-801-G00

2. Thrust Reverser Hydraulic Actuator Rod End Inspection

(Figure 601)

A. General

(1) This task gives the instructions for the visual inspection of the hydraulic actuator rod ends.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-02-400-801-G00	Translating Sleeve Installation (P/B 401)
78-31-03-000-801-G00	Thrust Reverser Hydraulic Actuator Removal (Selection) (P/B 401)
78-31-03-400-801-G00	Thrust Reverser Hydraulic Actuator Installation (Selection) (P/B 401)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

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(Continued)

Number	Name/Location
426ER	Right Thrust Reverser Actuator (Middle), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-03-040-031-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do the deactivation procedure for the thrust reverser for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-31-03-010-013-G00

(2) Do these steps to remove the access door for the applicable actuator:

(a) To remove the upper actuator access doors [2], do these steps:

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2

1) Remove the lower Krueger flap deflector (TASK 78-31-09-010-801-G00).

2) Remove the six bolts [1].

(b) To remove the upper actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(c) To remove the middle actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(d) To remove the lower actuator access doors [2], remove the six bolts [1].

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

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SUBTASK 78-31-03-020-027-G00

CAUTION: DO NOT LET THE ROD ENDS OF THE HYDRAULIC ACTUATOR TURN WHEN YOU REMOVE THE BOLTS. IF THE ROD END TURNS, IT CAN CAUSE DAMAGE TO THE HYDRAULIC ACTUATOR AND THRUST REVERSER STRUCTURE.

(3) Disconnect the actuator rod end from the actuator attach fitting on the translating sleeve (TASK 78-31-03-000-801-G00).

NOTE: It is necessary to disconnect the actuator rod end for the inspection.

G. Thrust Reverser Hydraulic Actuator Rod End Inspection

SUBTASK 78-31-03-212-001-G00

(1) Examine the applicable rod end for these conditions:

(a) Cracks

1) Not permitted.

(b) Nicks

1) Not permitted.

(c) Dents

1) Not permitted.

(d) Scratches

1) Not permitted.

(e) Bushing damage

1) Not permitted.

SUBTASK 78-31-03-869-001-G00

(2) If you find a condition that is not permitted, replace the hydraulic actuator (TASK 78-31-03-000-801-G00 and TASK 78-31-03-400-801-G00).

SUBTASK 78-31-03-210-001-G00

(3) Examine the applicable rod end bolt and nut for a hardware that is missing, cracked, or bent.

(a) If a hardware is missing, cracked, or bent, replace the bolt or nut.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-03-420-021-G00

CAUTION: DO NOT TURN THE ACTUATOR ROD END. THE ACTUATOR LENGTH IS SET. IF YOU TURN THE ACTUATOR ROD END, DAMAGE TO THE ACTUATOR OR THE STRUCTURE CAN OCCUR.

(1) Connect the actuator rod end to the actuator attach fitting on the sleeve (TASK 78-31-02-400-801-G00).

SUBTASK 78-31-03-410-010-G00

(2) Do these steps to install the access door for the applicable actuator:

(a) For the upper actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415DL	Left Thrust Reverser Actuator (Upper), Engine 1

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(Continued)

Number	Name/Location
416DR	Right Thrust Reverser Actuator (Upper), Engine 1
425DL	Left Thrust Reverser Actuator (Upper), Engine 2
426DR	Right Thrust Reverser Actuator (Upper), Engine 2

(b) For the middle actuator, put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415EL	Left Thrust Reverser Actuator (Middle), Engine 1
416ER	Right Thrust Reverser Actuator (Middle), Engine 1
425EL	Left Thrust Reverser Actuator (Middle), Engine 2
426ER	Right Thrust Reverser Actuator (Middle), Engine 2

(c) For the lower actuator access door [2], do the steps that follow:

1) Make sure that the installation surfaces are clean.
2) Use sealant, A50110 to create a prepack or injection seal between the access door [2] and the installation surface on the thrust reverser.
3) Put the access doors [2] in the correct position to align the bolt holes.

Number	Name/Location
415FL	Left Thrust Reverser Actuator (Lower), Engine 1
416FR	Right Thrust Reverser Actuator (Lower), Engine 1
425FL	Left Thrust Reverser Actuator (Lower), Engine 2
426FR	Right Thrust Reverser Actuator (Lower), Engine 2

(d) Install the six bolts [1] to attach the applicable access door [2].

1) Tighten the bolts [1] to 30.0 in-lb (3.4 N·m) - 50.0 in-lb (5.6 N·m).

SUBTASK 78-31-03-410-013-G00

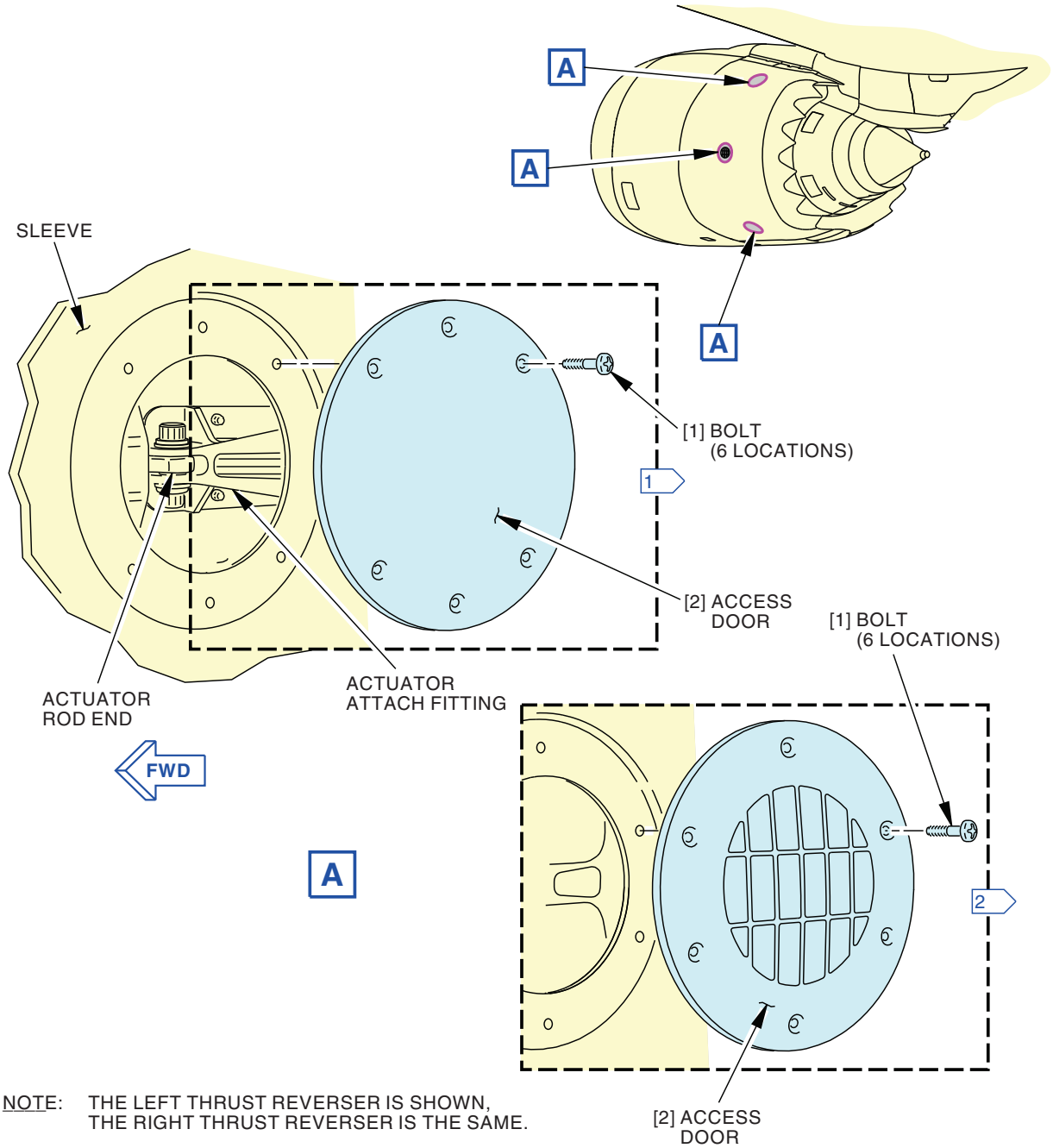
(3) If removed, install the lower Krueger flap deflector (TASK 78-31-09-420-801-G00).

SUBTASK 78-31-03-440-001-G00

(4) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

——— END OF TASK ———

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NOTE: THE LEFT THRUST REVERSER IS SHOWN, THE RIGHT THRUST REVERSER IS THE SAME.

1 UPPER AND LOWER ACCESS DOOR

2 MIDDLE ACCESS DOOR

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Thrust Reverser Hydraulic Actuator Rod End Inspection
Figure 601/78-31-03-990-804-G00

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THRUST REVERSER HYDRAULIC ACTUATORS - REPAIRS

1. General

A. This procedure has one task.

(1) Manual Unlock Handle Assembly Replacement.

TASK 78-31-03-300-801-G00

2. Manual Unlock Handle Assembly Replacement

(Figure 801, Figure 802, and Figure 803)

A. General

(1) This task gives the instructions to replace the manual unlock handle assembly on the upper and center locking hydraulic actuator.

(2) The locking actuator must be in the stowed and locked position to do this task.

B. References

Reference	Title
71-00-00-710-806-G00	Test No. 17 - Electronic Engine Control (EEC) System Test (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)
78-31-03-000-802-G00	Upper Locking Hydraulic Actuator Removal (P/B 401)
78-31-03-000-803-G00	Middle Locking Hydraulic Actuator Removal (P/B 401)
78-31-03-400-802-G00	Upper Locking Hydraulic Actuator Installation (P/B 401)
78-31-03-400-803-G00	Middle Locking Hydraulic Actuator Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4156	Locator Assembly - Proximity Sensor (Included in Overhaul Set) Part #: B78015-39 Supplier: 81205
SPL-15435	Gap Measurement Gauge - TRAS Locking Actuator Part #: C78030-18 Supplier: 81205

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

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E. Prepare for the Replacement

SUBTASK 78-31-03-860-005-G00

(1) For on-wing replacement of the manual unlock handle assembly [1], do these steps:

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(c) Make sure that the thrust reverser sleeve is in the stowed and locked position.

SUBTASK 78-31-03-000-001-G00

(2) For off-wing replacement of the manual unlock handle assembly [1], do these steps:

(a) Remove the applicable locking hydraulic actuator (TASK 78-31-03-000-802-G00 or TASK 78-31-03-000-803-G00).

(b) Retract and lock the actuator.

F. Manual Lockout Handle Assembly Replacement

SUBTASK 78-31-03-020-026-G00

(1) Do these steps to remove the defective :

(a) Mark the position of the handle relative to the actuator housing so that the new handle assembly is installed in the same position.

NOTE: In the locked position, the handle lever is approximately vertical to the axis of the actuator.

(b) Remove the two screws [9] that attach the detent housing [2] to the actuator housing [13].

(c) Remove the detent housing [2], manual unlock pin [10], spring [11] and spring guide [12].

(d) Remove the nut [6] from the manual unlock handle assembly [1].

(e) Remove the shim [5], unlock lever [7], torsion spring [8], and the manual unlock handle assembly [1].

NOTE: It is not necessary to remove the bushing on each side of the housing.

(f) Remove the two rivets [4] and the target [3] from the manual unlock handle assembly [1].

NOTE: Remove the target for installation on the new manual unlock handle assembly.

SUBTASK 78-31-03-420-020-G00

(2) Do these steps to install the new manual unlock handle assembly [1]:

(a) If it is necessary, install the target [3] with the two rivets [4] on the new manual unlock handle assembly [1].

(b) Install the torsion spring [8] on the new manual unlock handle assembly [1].

NOTE: Locate one leg of the torsion spring [8] in the hole.

(c) Install the unlock lever [7] in the locked position in the actuator housing [13].

(d) Align the manual unlock handle assembly [1] with the mark (locked position) and insert the manual unlock handle assembly [1] through the bushing until the splines just engage.

1) Pull the free leg of the torsion spring [8] around to align with the hole in the actuator housing [13].

2) Push the manual unlock handle assembly [1] through until it bottoms on the bushing [21].

NOTE: Make sure that the installation of torsion spring is correct.

(e) Install the shim [5] with the counterbore towards the actuator housing [13].

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1) Measure the gap G between the shim [5] and the bushing [20].
   a) If the gap G is more than 0.004 in. (0.102 mm), make a new shim [5].
   b) If there is no gap and the shim face is in contact with the bushing [20], trim the shim [5] as follows:
      <1> Measure F (the distance from the shoulder end of the lockout handle shaft) and subtract 0.002 in. (0.051 mm) to get the shim thickness (S3) you will need.
      <2> Trim the shim face that contacts the bushing [20] to obtain the S3 thickness (the gap tolerance is 0.001 in. (0.025 mm) to 0.004 in. (0.102 mm).

      NOTE: A new shim is oversized. The shim face will be in contact with the bushing. You will need to trim it to obtain the 0.002 in. (0.051 mm) gap.

(f) Install and tighten the nut [6] to 20 in-lb (2.26 N·m) - 30 in-lb (3.39 N·m).

(g) Pull the manual unlock handle assembly [1] to the unlocked position.

(h) Make sure that the unlock lever [7] and manual unlock handle assembly [1] operate correctly with no binding.

(i) Put the manual unlock handle assembly [1] back to the locked position.

(j) Make sure that the actuator is in the stowed and locked position.

(k) Remove the six bolts [14] from the applicable access panel cover [15].

(l) Remove the upper or center access panel cover [15] from the TR sleeve [19] to get access to the actuator rod end [18].

(m) Make sure that the spacer is within 0.007 in. (0.18 mm) - 0.033 in. (0.84 mm) of the rod stop [16].

(n) Do these steps to install the detent housing [2] as follows:
   1) Install the spring guide [12], spring [11], manual unlock pin [10], and detent housing [2] in the actuator housing [13].
   2) Install the two screws [9] that attach the detent housing [2] to the actuator housing [13].
   3) Tighten the two screws [9] to 27 in-lb (3.05 N·m) - 30 in-lb (3.39 N·m).
   4) Install lockwire to the screws [9].

(o) Pull the manual unlock handle assembly [1] to the unlocked position, then push and release the manual unlock pin [10].
   1) Make sure that manual unlock pin [10] does not bind and springs back smoothly.

(p) Put the manual unlock handle assembly [1] back to the locked position.

(q) Make sure that the actuator is in the stowed and locked position as follows:
   1) Look through the access area and see if the rod stop [16] is in contact with the gland nut [17].

(r) Do these steps to install the target [3] on the manual unlock handle assembly [1]:

NOTE: It is recommended to use proximity sensor locator, SPL-4156, or gap gauge, SPL-15435, to put the target in position on the manual unlock handle assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply and keep a slight pressure to the manual unlock handle assembly [1] in the unlock position.

NOTE: Keep light pressure against the internal lock sleeve but not move the sleeve.

2) Make sure that the target is 0.28 in. (7.11 mm) - 0.30 in. (7.62 mm) from the center of the bracket sensor hole.

3) Install the target [3] on the handle with the two rivets [4].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-03-410-011-G00

(1) For on-wing replacement of the manual unlock handle assembly [1], do these steps:

(a) Install the access panel cover [15] on the TR sleeve [19] with six bolts [14].

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-03-400-001-G00

(2) For off-wing replacement of the manual unlock handle assembly [1], install the locking hydraulic actuator (TASK 78-31-03-400-802-G00 or TASK 78-31-03-400-803-G00).

SUBTASK 78-31-03-710-006-G00

(3) For on-wing replacement of the manual unlock handle assembly [1], do these checks:

(a) Do this task:Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

NOTE: Operate the thrust reverser through the deploy and stow cycles until the sleeves move smoothly.

(b) Do this task: Test No. 17 - Electronic Engine Control (EEC) System Test, TASK 71-00-00-710-806-G00.

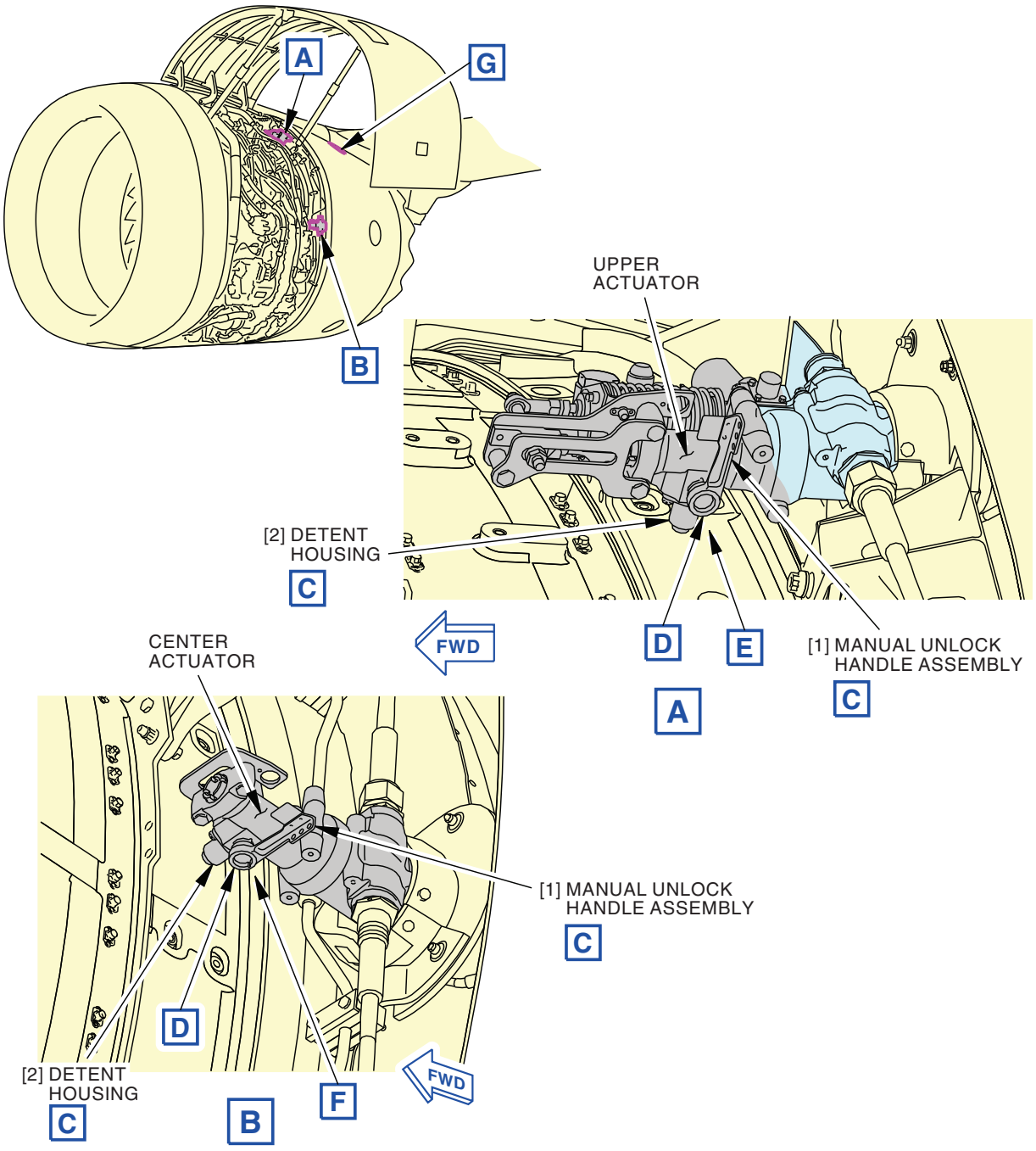
NOTE: This check will make sure that the electrical connections for the LVDT's are correct.

1) Make sure that no LVDT maintenance messages show.

a) If a maintenance message shows, do the applicable procedure in the IFIM for that maintenance message.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



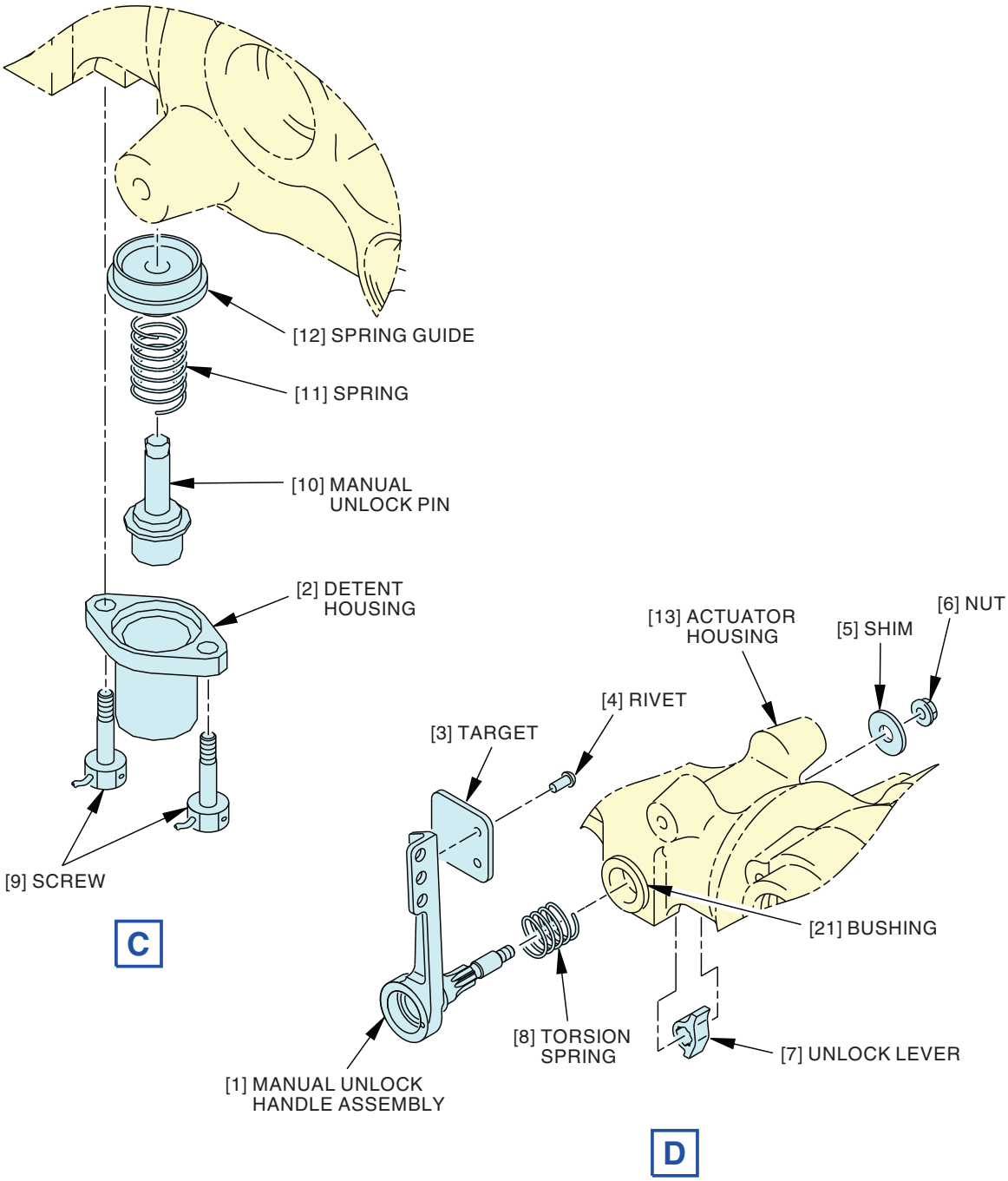
NOTE:

LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE.

2355207 S00061516741_V2

Manual Unlock Handle Assembly Repair
Figure 801/78-31-03-990-805-G00 (Sheet 1 of 4)

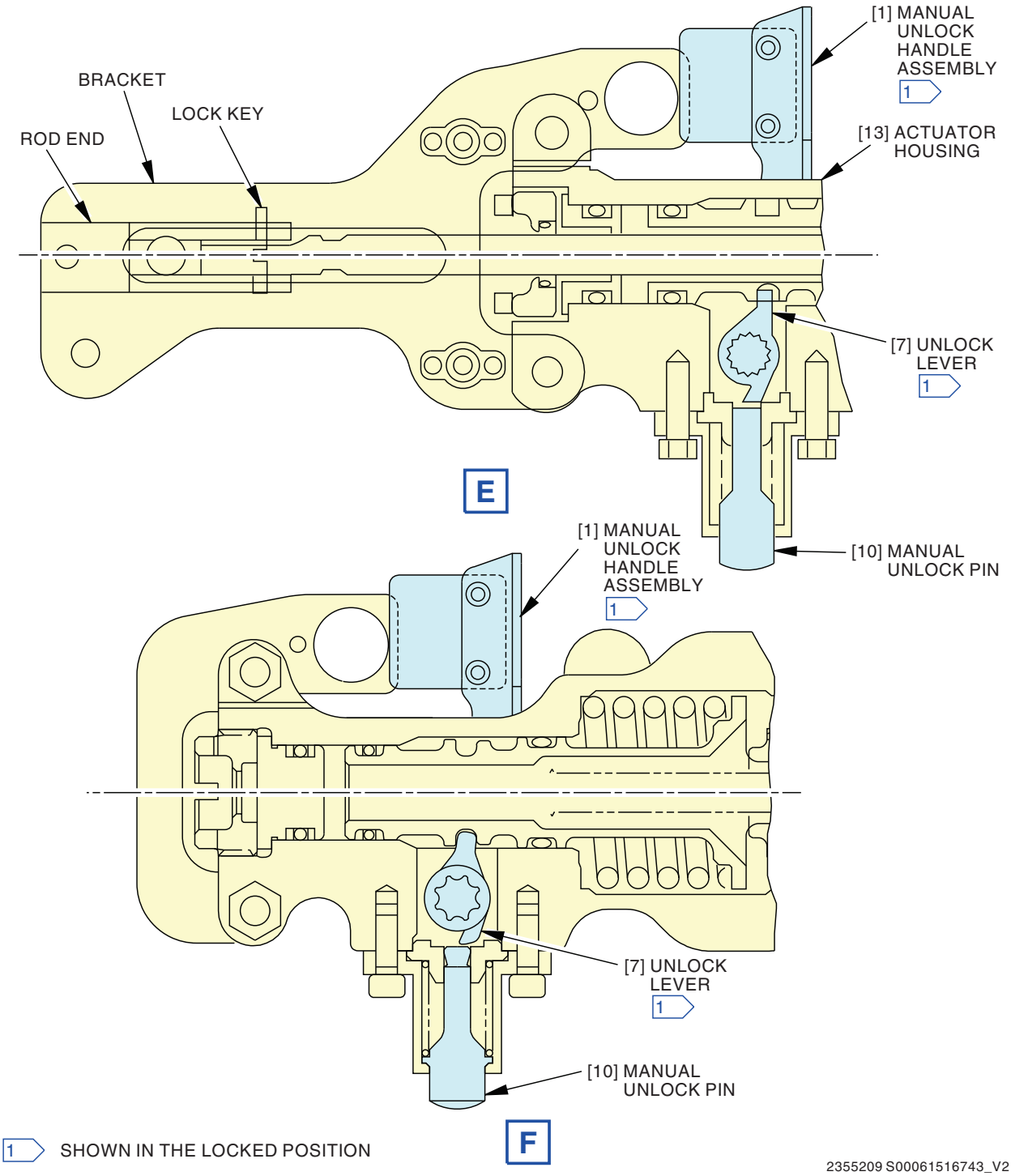
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355208 S00061516742_V2

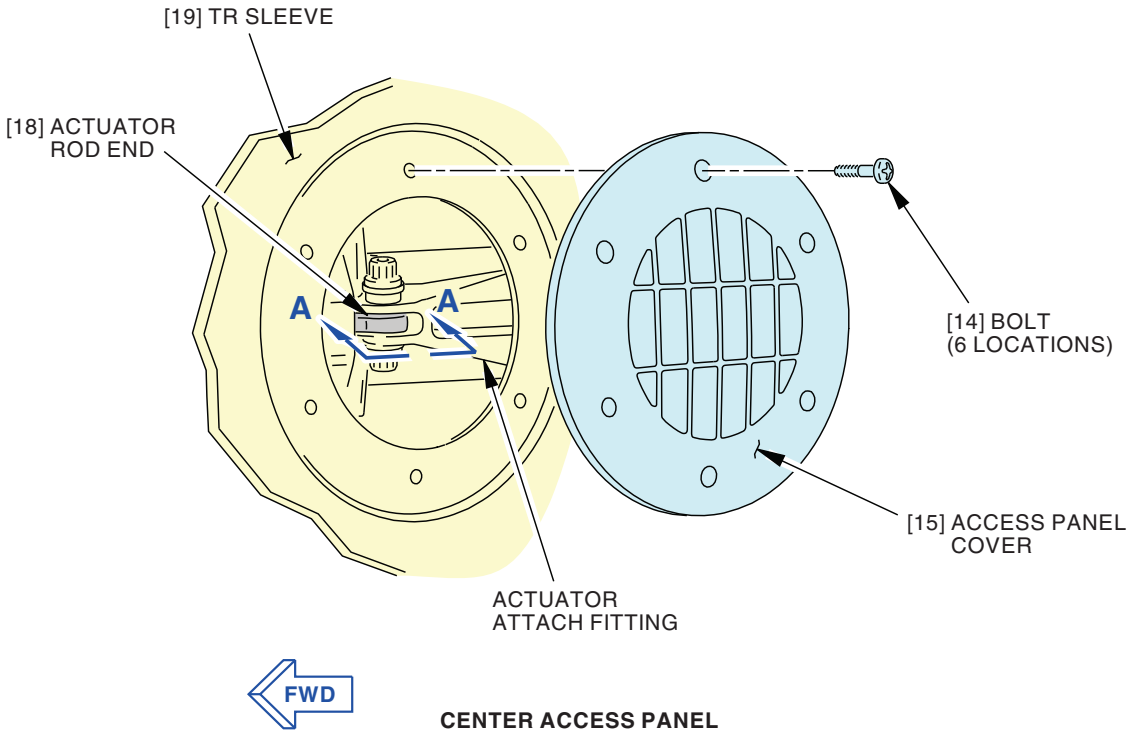
Manual Unlock Handle Assembly Repair
Figure 801/78-31-03-990-805-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



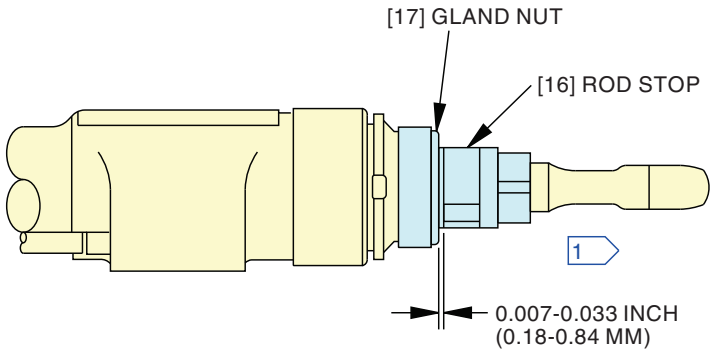
Manual Unlock Handle Assembly Repair
Figure 801/78-31-03-990-805-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CENTER ACCESS PANEL

G



CENTER LOCKING ACTUATOR

A-A

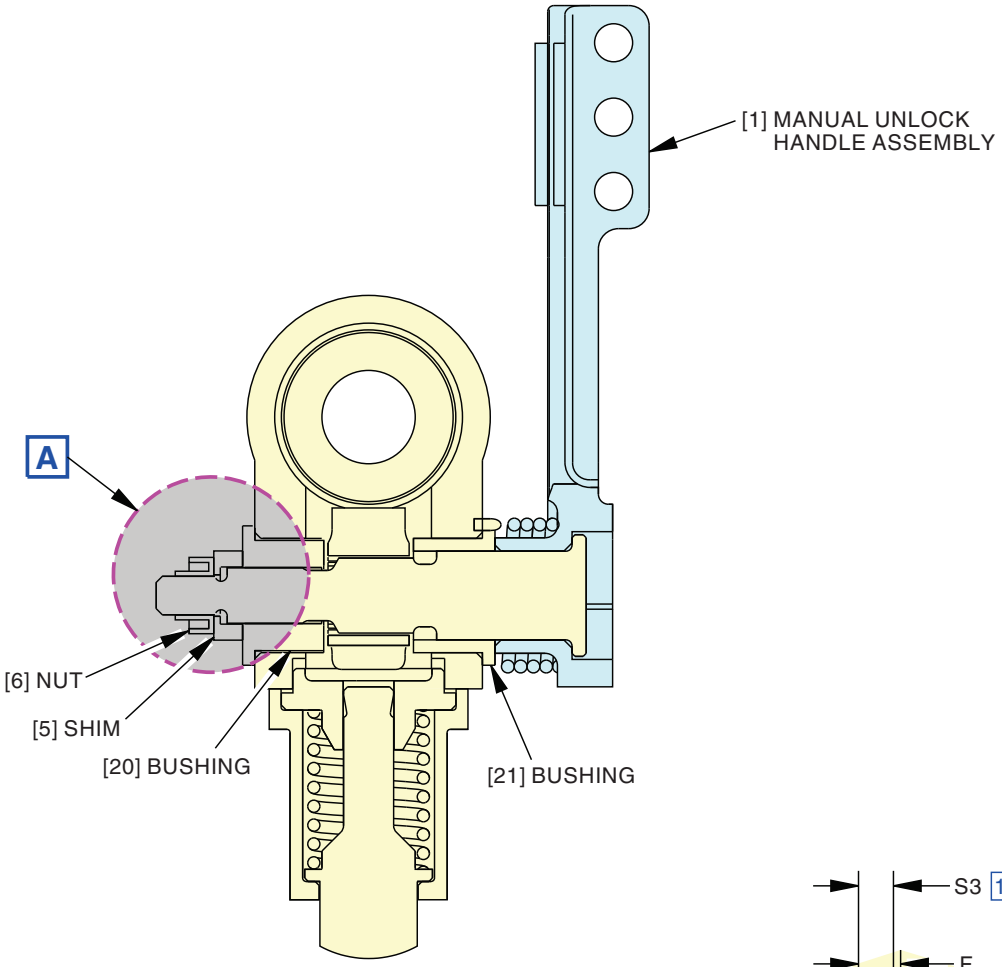
NOTE:

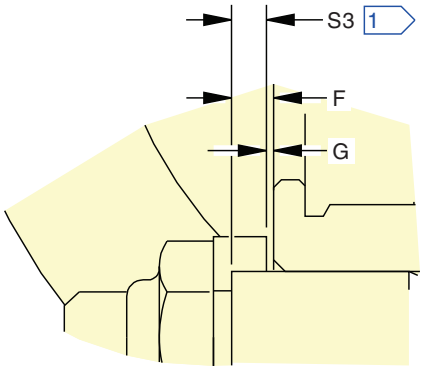
1 THE SAME GAP DIMENSION APPLIES TO THE UPPER LOCKING ACTUATOR

2355210 S00061516744_V2

Manual Unlock Handle Assembly Repair
Figure 801/78-31-03-990-805-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(EXAMPLE)

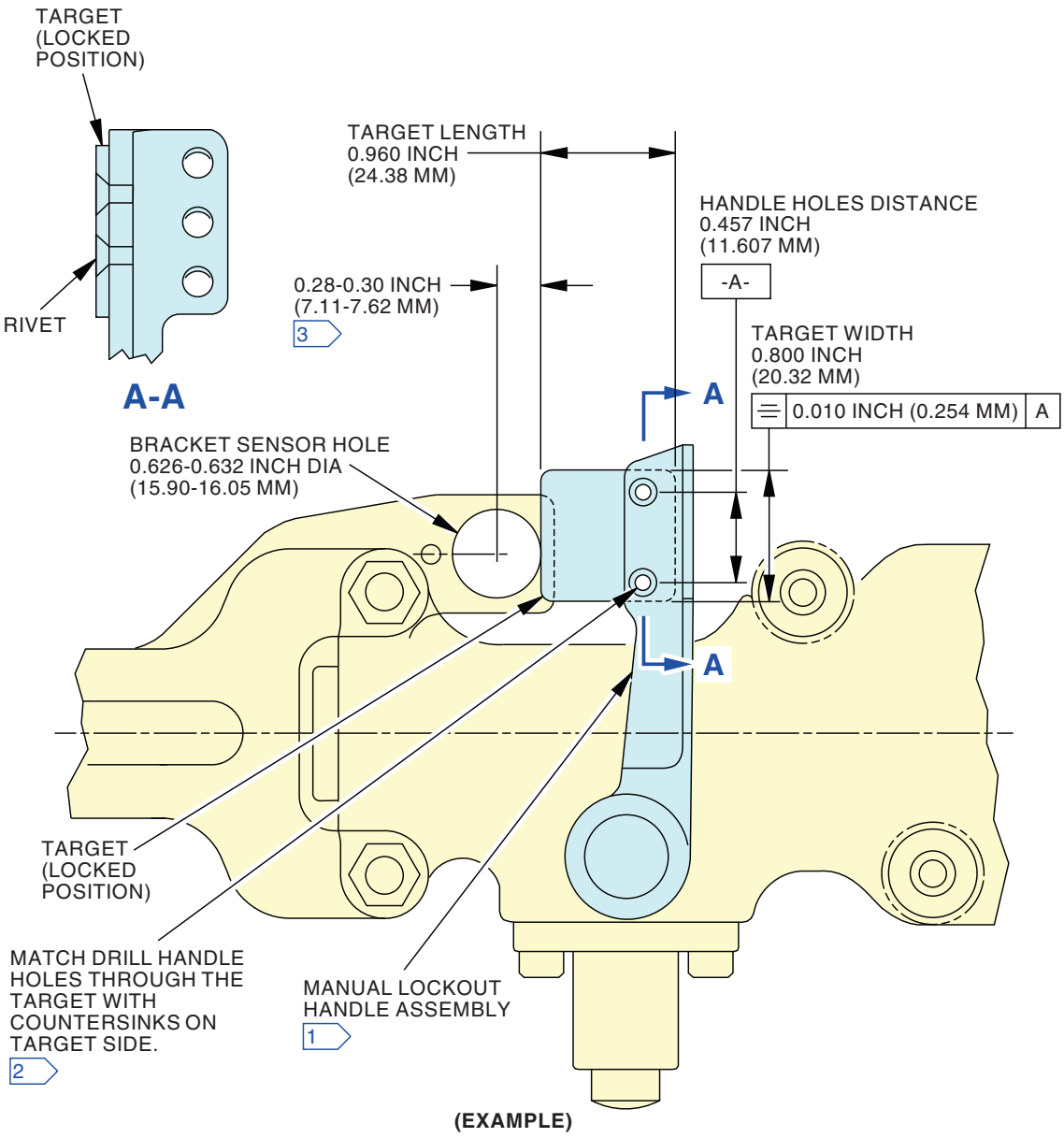
A

1 DIMENSION S3 IS THE NECESSARY SHIM THICKNESS. ADJUST COUNTERBORE DEPTH OF SHIM TO DIMENSION S3.

2355211 S00061516745_V2

Shim Installation
Figure 802/78-31-03-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

1. APPROXIMATELY PERPENDICULAR WITH ACTUATOR CENTERLINE WHEN PISTON IS IN FULLY LOCKED POSITION.
2. MARK TARGET HOLES AGAINST THE HANDLE. DRILL TARGET HOLES, INSTALL TARGET TO HANDLE ON BENCH.
3. MEASUREMENT SHOULD BE MADE WITH HANDLE IN RELAXED POSITION. DO NOT MOVE HANDLE FROM REST POSITION BEFORE MAKING MEASUREMENT.

2355212 S00061516746_V1

Target Installation
Figure 803/78-31-03-990-807-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SYNC SHAFT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Sync Shaft Removal

(2) Sync Shaft Installation.

TASK 78-31-04-000-801-G00

2. Sync Shaft Removal

(Figure 401, Figure 402)

A. General

(1) This task is the removal of the sync shafts from the left or right thrust reverser on an engine.

(2) The sync shafts are located between the upper and middle actuators and between the middle and lower actuators on each thrust reverser torque box.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Prepare for the Removal

SUBTASK 78-31-04-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-04-860-001-G00

(2) For engine 1, open this circuit breaker and install a safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-04-860-002-G00

(3) For engine 2, open this circuit breaker and install a safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-04-860-003-G00

(4) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-31-04-860-004-G00

(5) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-31-04-010-001-G00

(6) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

SUBTASK 78-31-04-020-001-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE UNION, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBE AND UNIONS CAN OCCUR.

(7) Do these steps to drain the hydraulic fluid from the hydraulic lines:

NOTE: To decrease the hydraulic fluid spray when the coupling nuts are loosened, wrap a cotton wiper, G00034, around the wrench, the coupling nut and the hydraulic line.

(a) Wrap a cotton wiper, G00034, around the electrical connector on the sync lock receptacle on the lower actuator.

NOTE: The cotton wiper, G00034, will catch the hydraulic fluid and prevent contamination of the electrical connector.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Disconnect the sync shaft tubing (deploy tube) at the lower actuator.

(c) Disconnect the hydraulic retract line (stow tube) at the lower actuator.

(d) Let the hydraulic fluid drain into a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110.

(e) Loosen the coupling nuts on the deploy and stow line flexhoses at the upper actuator.

NOTE: This will let air into the hydraulic lines to let the hydraulic fluid drain.

SUBTASK 78-31-04-420-001-G00

(8) Torque the coupling nut on the retract line flexhose at the upper actuator.

(a) Torque the coupling nut to 270 ±13 in-lb (31 ±2 N·m).

(b) Loosen the coupling nut.

(c) Torque the coupling nut again to 270 ±13 in-lb (31 ±2 N·m).

SUBTASK 78-31-04-420-002-G00

(9) Torque the coupling nut on the deploy line flexhose at the upper actuator.

(a) Torque the coupling nut to 900 ±45 in-lb (102 ±5 N·m).

(b) Loosen the coupling nut.

(c) Torque the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

SUBTASK 78-31-04-420-003-G00

(10) Re-connect the stow line at the lower actuator.

(a) Torque the coupling nut to 270 ±13 in-lb (31 ±2 N·m).

(b) Loosen the coupling nut.

(c) Torque the coupling nut again to 270 ±13 in-lb (31 ±2 N·m).

SUBTASK 78-31-04-420-004-G00

(11) If the lower sync shaft [4] will not be removed, re-connect the sync shaft tubing at the lower actuator.

(a) Torque the coupling nut to 900 ±45 in-lb (102 ±5 N·m).

(b) Loosen the coupling nut.

(c) Torque the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

SUBTASK 78-31-04-010-002-G00

(12) If the upper sync shaft [4] and tubing will be removed, do this step to remove the upper clamp block assembly.

(a) Remove the two bolts [9], two washers [10], clamp block retainer [8], clamp block [11], clamp block [12], two sleeve spacers [13], clamp block retainer [7] and spacer [6].

NOTE: To eliminate tubing preload during installation, it is important to keep the clamp block retainer [8], clamp block retainer [7], clamp block [11], clamp block [12], and spacer [6] together and to note their position for the subsequent installation.

SUBTASK 78-31-04-010-003-G00

(13) If the lower sync shaft [4] and tubing will be removed, do this step to remove the lower clamp block assembly.

NOTE: To eliminate tubing preload during installation, it is important to keep the clamp block retainer [8], clamp block retainer [7], clamp block [11], clamp block [12], and spacer [6] together and to note their position for the subsequent installation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the bolt [9], bolt [15], two washers [10], wiring clamp [14], and clamp block retainer [8].

(b) Remove the clamp block [11], clamp block [12], two sleeve spacers [13], clamp block retainer [7] and spacer [6].

G. Sync Shaft Removal

SUBTASK 78-31-04-020-002-G00

CAUTION

DO NOT LET DIRT OR OTHER CONTAMINATION GET ON THE SYNC SHAFT. DAMAGE TO THE SYNC SHAFT CAN OCCUR.

(1) Do these steps to remove the sync shaft [4]:

(a) Disconnect the coupling nuts at the two ends of the upper tube [2] or lower tube [3].

(b) Move the coupling nuts and tube adapters back along the tube [2] or tube [3] to get access to the ends of the sync shaft [4].

(c) Flex the sync shaft [4] to remove it from the splined receiver in the actuator port [1].

(d) Remove the sync shaft [4] and the tube [2] or tube [3] together.

SUBTASK 78-31-04-420-005-G00

WARNING

MAKE SURE THAT THE TWO SYNC SHAFT TUBES ARE INSTALLED BETWEEN THE ACTUATORS. IF THE TUBES ARE NOT INSTALLED WHEN YOU SUPPLY RETRACT PRESSURE, PRESSURIZED HYDRAULIC FLUID CAN HIT PERSONNEL. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these steps to temporarily install the upper tube [2] or lower tube [3]:

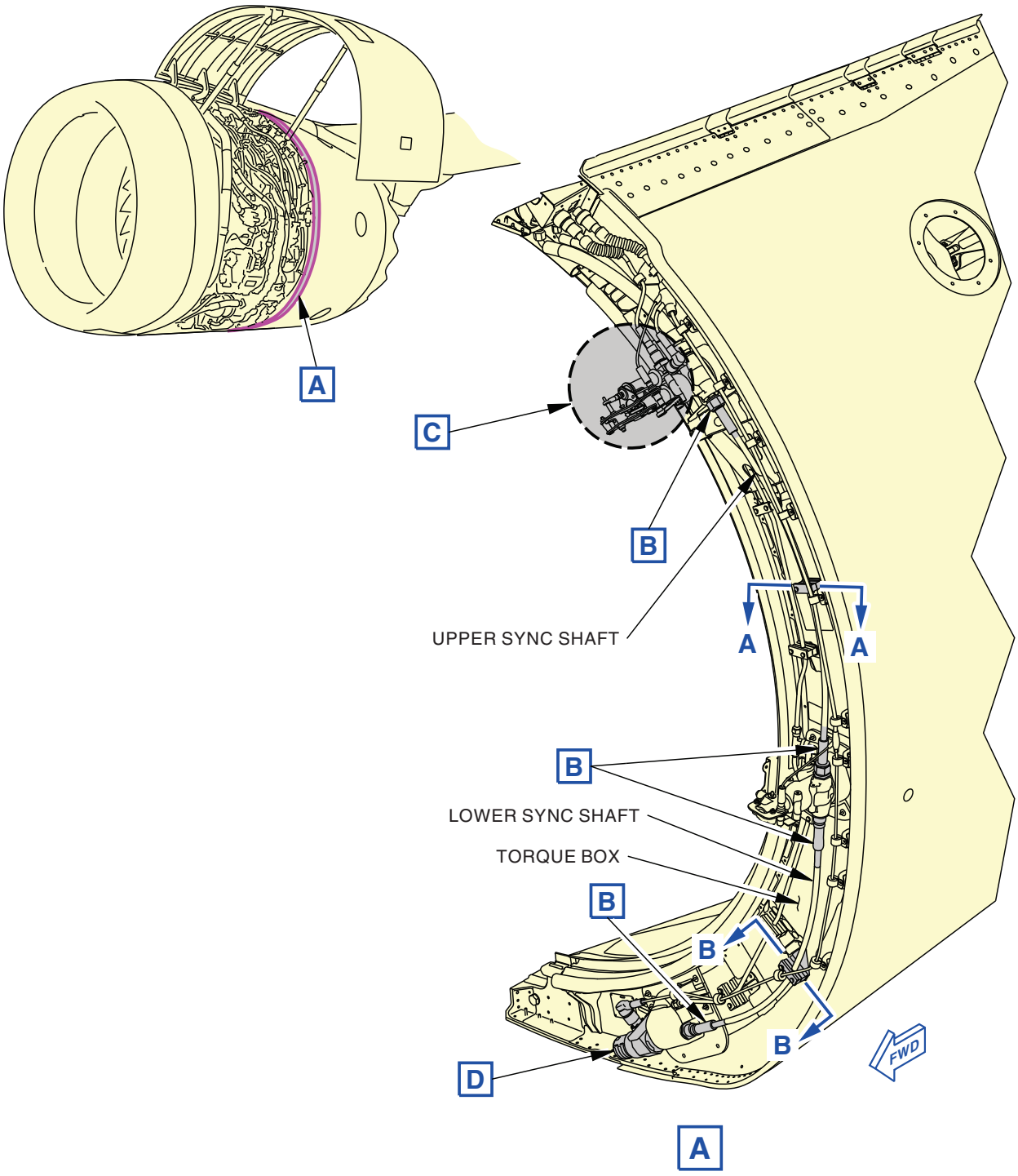
(a) Put the upper tube [2] or lower tube [3], without the sync shaft [4], in its position between the actuators.

(b) Torque the coupling nuts to 900 ±45 in-lb (102 ±5 N·m).

NOTE: This is a temporary installation and it is not necessary to torque, loosen, then torque again (double-torque) the coupling nuts.

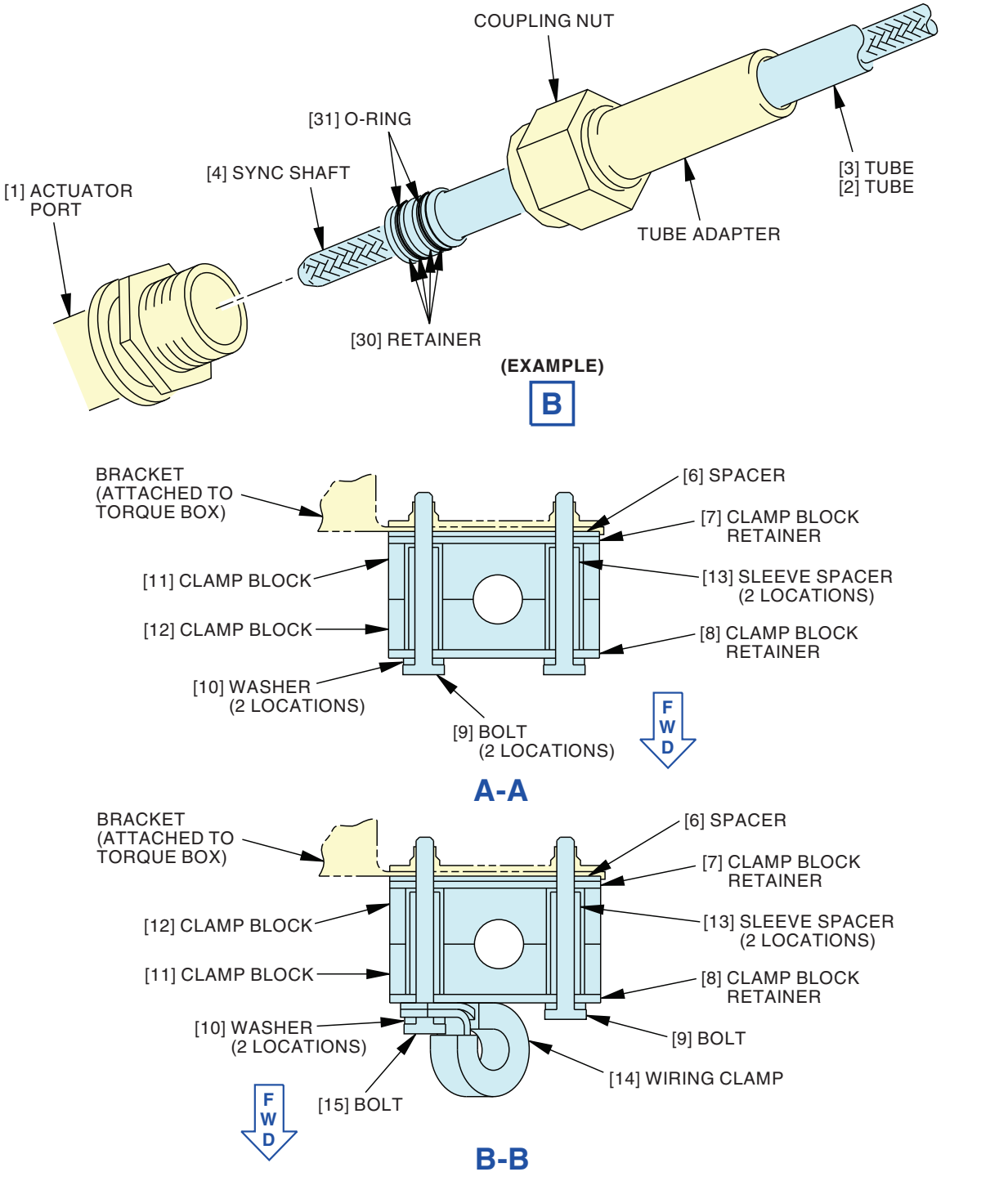
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



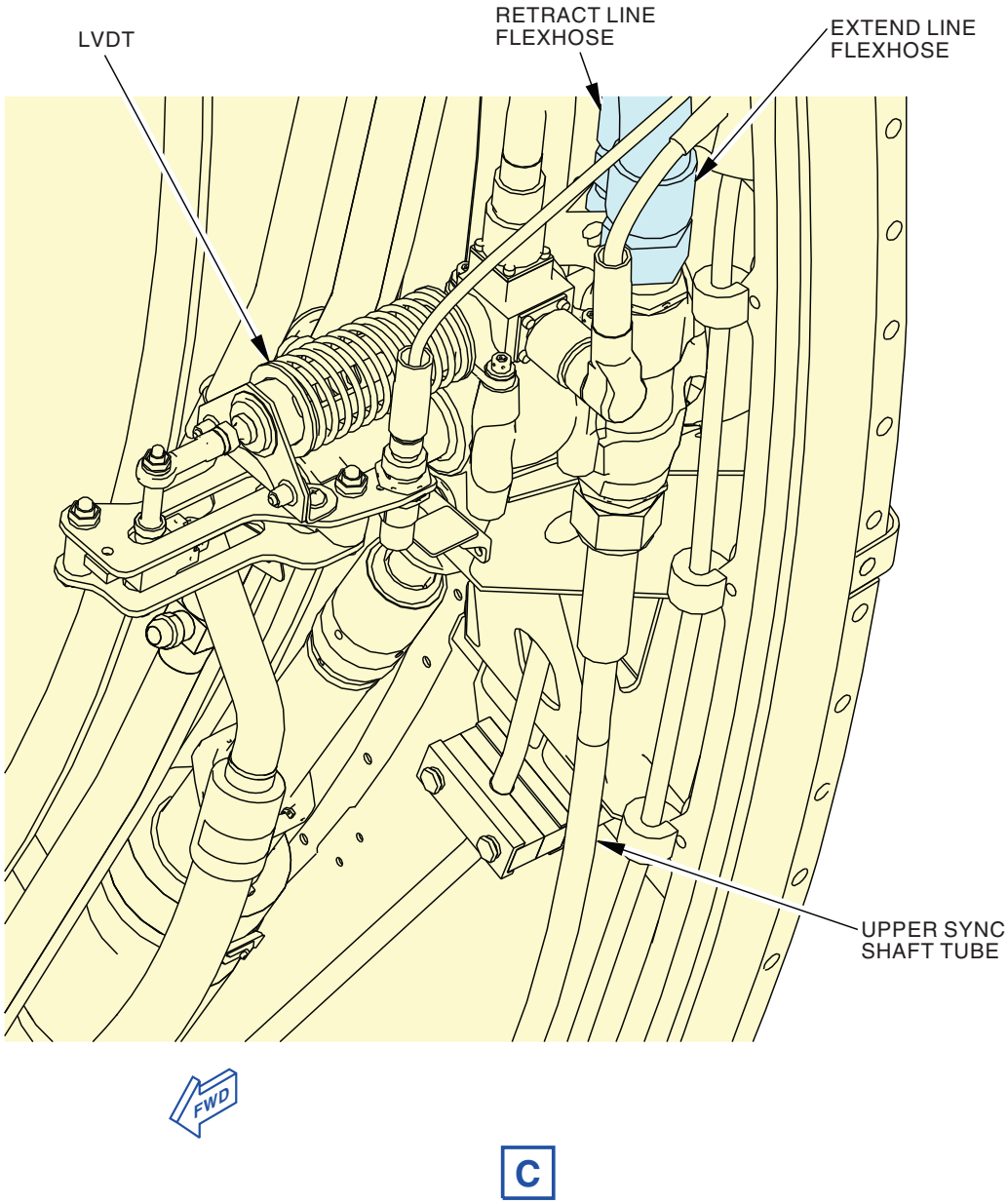
Sync Shaft Installation
Figure 401/78-31-04-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



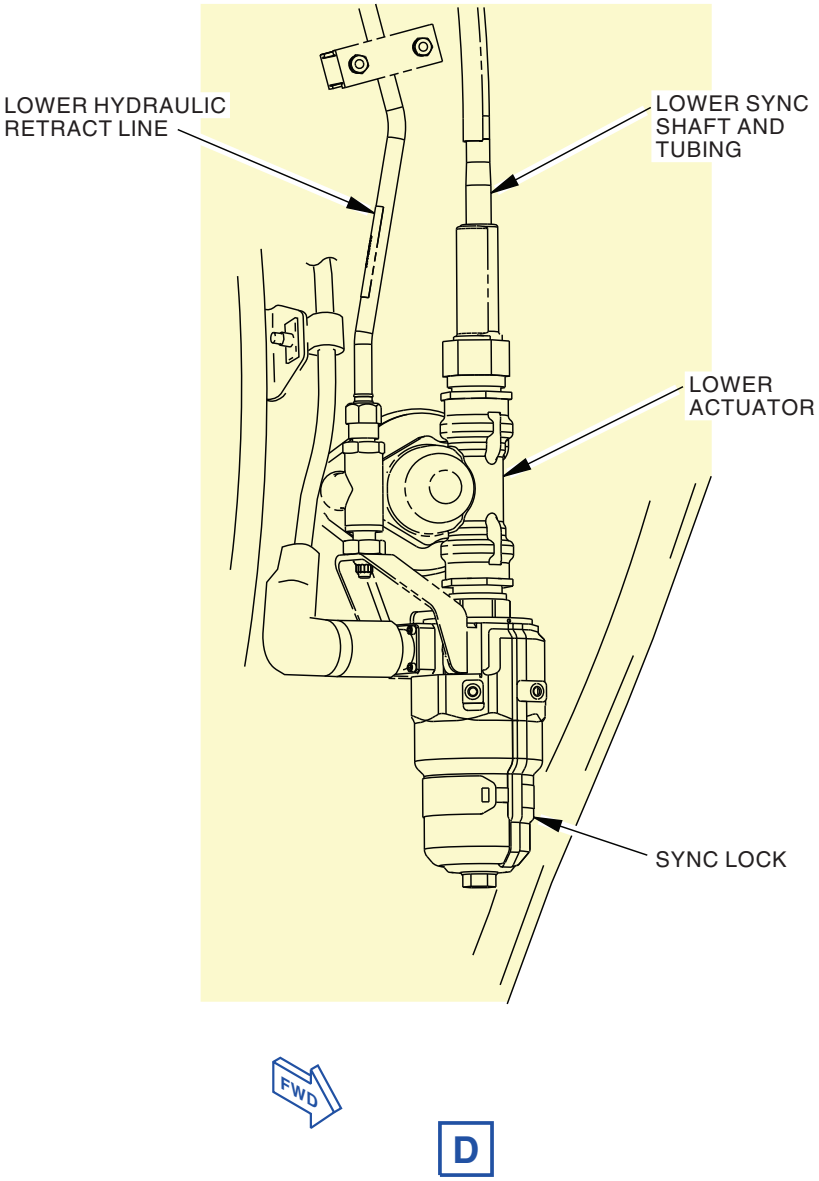
Sync Shaft Installation
Figure 401/78-31-04-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Sync Shaft Installation
Figure 401/78-31-04-990-801-G00 (Sheet 3 of 4)

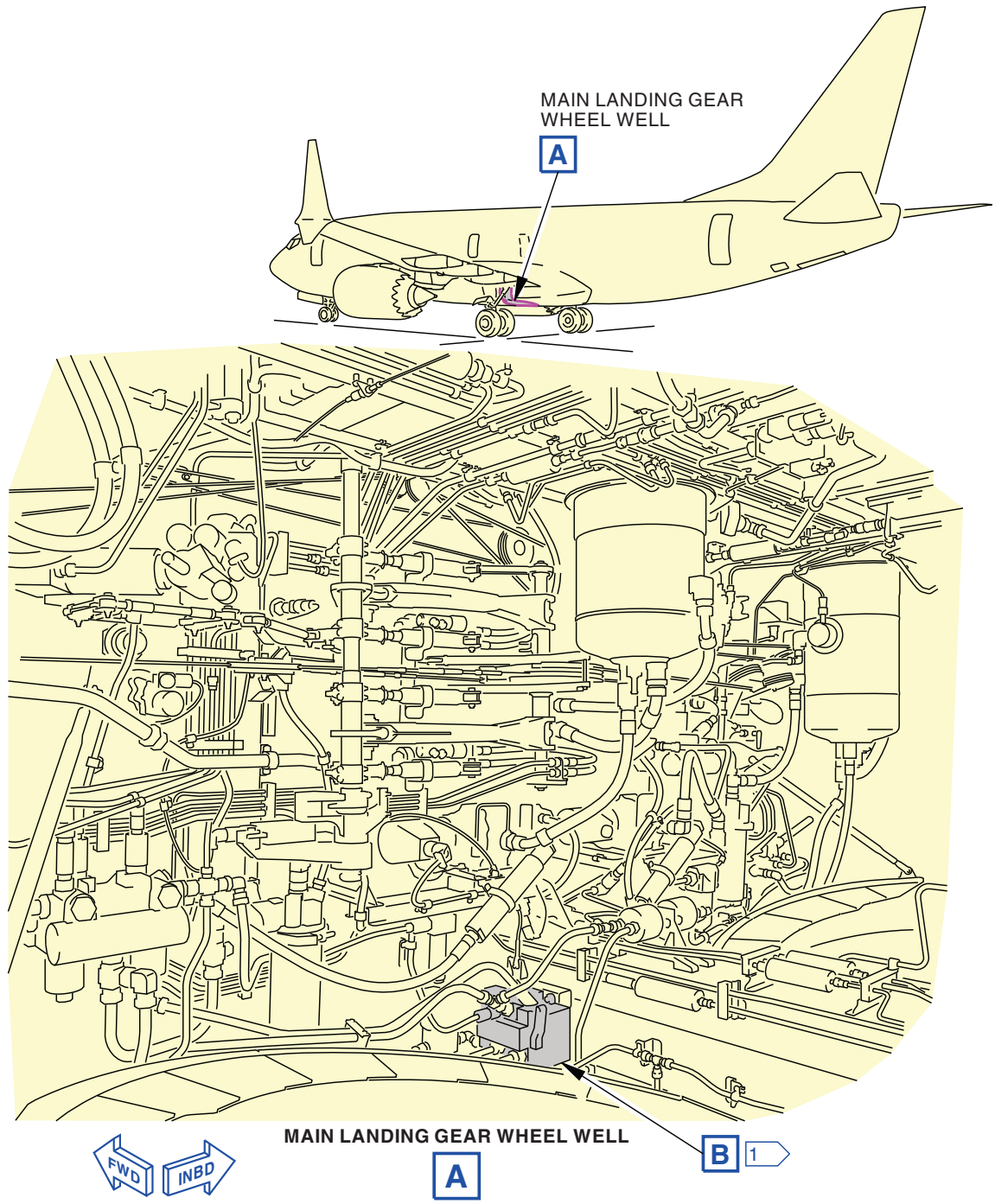
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2553074 S0000608247_V1

Sync Shaft Installation
Figure 401/78-31-04-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

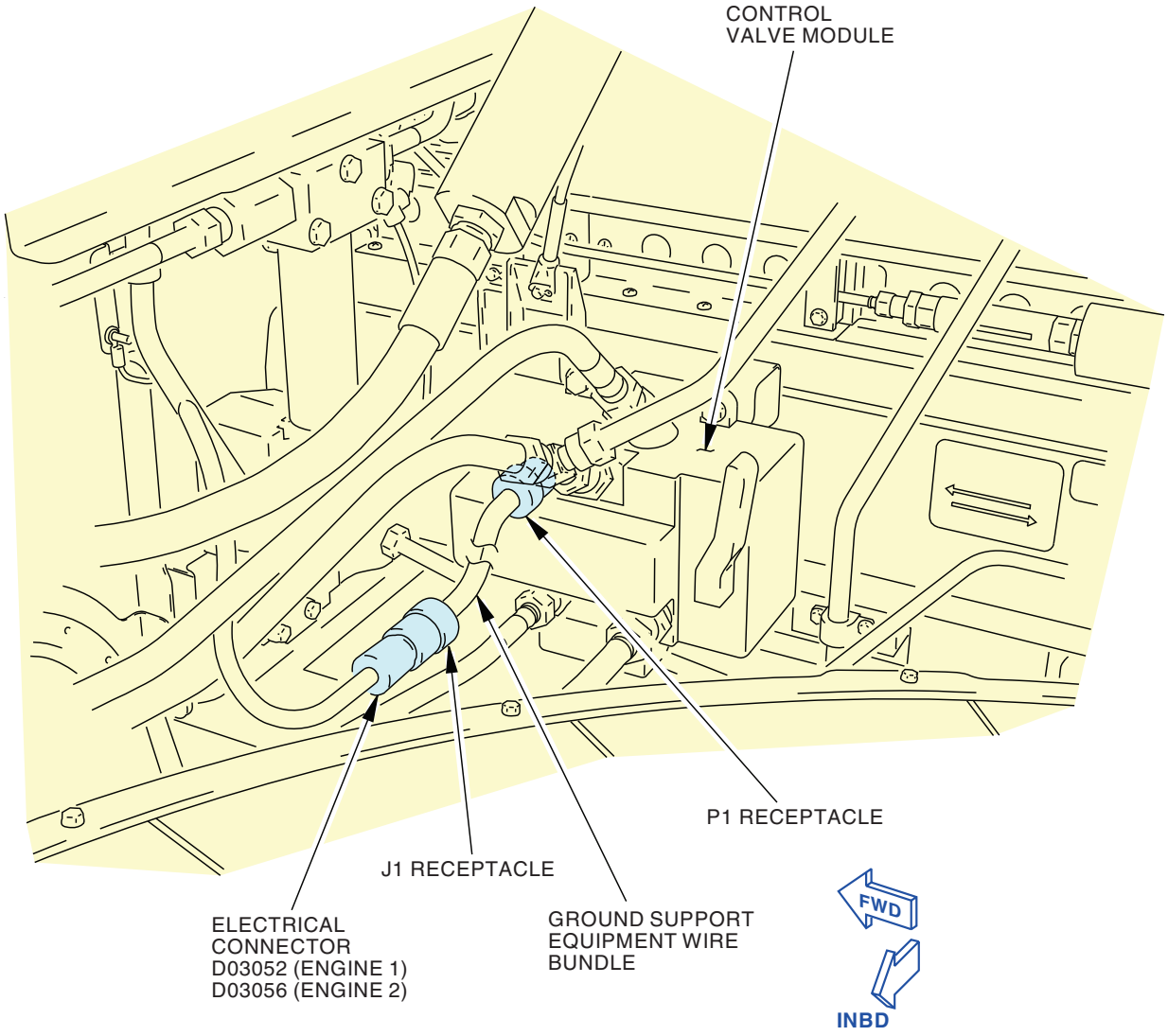


1 ENGINE 1 CONTROL VALVE MODULE IS SHOWN,
ENGINE 2 CONTROL VALVE MODULE IS ON THE RIGHT SIDE OF THE KEEL BEAM.

2355215 S00061516752_V1

Ground Support Equipment Wire Bundle Installation
Figure 402/78-31-04-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355216 S00061516753_V1

Ground Support Equipment Wire Bundle Installation
Figure 402/78-31-04-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-04-400-801-G00

3. Sync Shaft Installation

(Figure 401, Figure 402)

A. General

(1) After you install the sync shaft and tubing, you must do the thrust reverser normal operational test.

(2) To correctly rig the sync shaft, hydraulic pressure must be applied to the retract side of the thrust reverser actuators when the sync shaft is installed.

(3) The upper and lower sync shafts are interchangeable and can be reversed end for end and left thrust reverser half to right thrust reverser half.

B. References

Reference	Title
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-9002	Lock Equipment - Thrust Reverser Maintenance Part #: B78009-26 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00316	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I, ASTM D-3735 Type I
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
30	Retainer	78-31-04-01-095	PTF ALL
		78-31-04-02-095	PTF ALL
31	O-ring	78-31-04-01-100	PTF ALL
		78-31-04-02-100	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Prepare for the Installation

SUBTASK 78-31-04-480-001-G00

WARNING MAKE SURE THAT YOU DEACTIVATE THE DEPLOY CONTROL CIRCUIT. THIS WILL MAKE SURE THAT THE HYDRAULIC PRESSURE IS ON THE RETRACT LINE ONLY. IF YOU DO NOT, THE TR SLEEVES CAN DEPLOY. INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Connect the B78009-5 cable assembly from the thrust reverser maintenance lock equipment, SPL-9002, to the control valve module as follows:

NOTE: The B78009-5 cable assembly is part of the thrust reverser maintenance lock equipment. There are two -5 wire bundle cable assemblies in the B78009 tool. The cable assembly disables the directional control valve stow and deploy solenoid allowing only the arm solenoid to be energized.

(a) Disconnect the electrical connector from the applicable control valve module receptacle.

1) For Engine 1, disconnect the electrical connector, D03052.

2) For Engine 2, disconnect the electrical connector, D03056.

(b) Connect the P1 connector of the -5 cable assembly to the control valve module receptacle.

(c) Connect the electrical connector to the J1 receptacle of the -5 cable assembly.

SUBTASK 78-31-04-860-005-G00

(2) Do these steps to supply hydraulic power to the retract side of the actuators:

(a) Make sure that the tubing is temporarily installed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(c) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

1) For engine 1, pressurize hydraulic system A.

2) For engine 2, pressurize hydraulic system B.

(e) Start the thrust reverser cycle special function (TASK 71-00-00-700-816-G00).

NOTE: Thrust reverser cycle special function will time out after 5 minutes of no thrust lever movement. If this occurs, restart thrust reverser cycle special function.

(f) Move the applicable reverse thrust lever up and aft to the extend (deploy) position.

NOTE: This will unlock the sync lock.

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE THRUST REVERSER. IF YOU PARTIALLY DEPLOY THE THRUST REVERSER, IT WILL RETRACT WHEN YOU REMOVE THE LOCKPIN. INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Remove the lockpin from the manual isolation valve handle on the control valve module.

NOTE: Because the cable assembly is installed that disables the deploy actuation circuit, the thrust reverser will retract (stow) as soon as the lockpin is removed from the manual isolation valve handle. This will keep hydraulic pressure on the stow side of the actuators during the flex shaft installation.

H. Sync Shaft Installation

SUBTASK 78-31-04-420-006-G00

(1) Do these steps to install the sync shaft [4] and tube [2] or tube [3]:

(a) Remove the upper tube [2] or lower tube [3] that was temporarily installed.

1) Replace the two O-rings [31] and the four retainers [30] on the upper tube [2] or lower tube [3] as follows:

a) Slide the coupling nut and tube adapter away from the end of the tube.

b) Remove the two O-rings [31] and four retainers [30].

<1> Discard the two O-rings [31] and four retainers [30].

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WARNING ALIPHATIC NAPHTHA IS THE RECOMMENDED SOLVENT AND IS ALSO POISONOUS. IF YOU USE NAPHTHA IN A CONFINED SPACE, MECHANICAL VENTILATION IS MANDATORY. RESPIRATORY DEVICES AND SKIN PROTECTION ARE NECESSARY. IF YOU DO NOT FOLLOW THESE INSTRUCTIONS, INJURY TO PERSONS CAN OCCUR.

c) Clean the seal surfaces and grooves with a clean cotton wiper, G00034, moistened with solvent, B00316.

<1> Wipe all the solvent off the surfaces with a clean, dry cotton wiper, G00034.

NOTE: Do not let the solvent, B00316, dry on the surfaces.

d) Lubricate the new O-rings [31] and new retainers [30] with MCS 352B fluid, D00054, or hydraulic fluid, D00153.

e) Install the two new O-rings [31] and four new retainers [30].

NOTE: Each O-ring [31] is installed with two retainers [30]. There is a retainer [30] on each side of the O-ring [31].

f) Slide the coupling nut and tube adapter toward the end of the tube [2] or tube [3].

(b) If there is contamination on the sync shaft [4], remove the contamination with a cotton wiper, G00034.

(c) Insert the sync shaft [4] into the tube [2] or tube [3].

(d) Move the coupling nuts and tube adapters back along the tube [2] or tube [3] to get access to the sync shaft [4] ends.

(e) Insert one end of the sync shaft [4] into the internal spline in the actuator port [1].

(f) Flex the sync shaft [4] and insert the opposite end into the internal spline in the other actuator port [1].

(g) Tighten the coupling nuts enough to engage one or two threads at this time.

(h) Center the tube [2] or tube [3] in the coupling nuts and adapters.

(i) Do these steps to tighten the coupling nuts:

1) Tighten the coupling nut to 900 ±45 in-lb (102 ±5 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

SUBTASK 78-31-04-410-001-G00

(2) To install the clamp blocks, do these steps:

NOTE: To eliminate tubing preload, it is important to install the clamp block retainer, clamp block and spacers back in their original positions.

(a) For the upper clamp block assembly, install the spacer [6], clamp block retainer [7], clamp block [12], clamp block [11], two sleeve spacers [13], clamp block retainer [8], two washers [10], and two bolts [9].

(b) For the lower clamp block assembly, install the spacer [6], clamp block retainer [7], clamp block [12], clamp block [11], two sleeve spacers [13], clamp block retainer [8], wiring clamp [14], two washers [10], bolt [9] and bolt [15].

(c) Tighten the bolts [9] and bolt [15] to 32.5 ±2.5 in-lb (3.7 ±0.3 N·m).

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SUBTASK 78-31-04-040-002-G00

(3) Do these steps again:

(a) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(b) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(d) For Engine 1, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(e) For Engine 2, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

SUBTASK 78-31-04-860-010-G00

(4) Make sure that the thrust reverser is fully closed (TASK 78-31-00-010-802-G00 and TASK 78-31-00-980-802-G00).

SUBTASK 78-31-04-860-006-G00

(5) Move the applicable reverse thrust lever forward and down to the retract (stow) position.

NOTE: If there is a small gap, the thrust reverser is not fully closed and the manual unlock lever cannot be moved.

SUBTASK 78-31-04-080-001-G00

(6) Do these steps to disconnect the thrust reverser maintenance lock equipment, SPL-9002, B78009-5 cable assembly, from the control valve module:

(a) Disconnect the P1 connector of the -5 cable assembly from the control valve module receptacle.

(b) Disconnect the electrical connector from the J1 receptacle of the -5 cable assembly.

(c) Connect the electrical connector to the control valve module.

1) For Engine 1, connect the electrical connector, D03052.

2) For Engine 2, connect the electrical connector, D03056.

SUBTASK 78-31-04-980-001-G00

(7) Manually extend and retract the thrust reverser through one cycle.

(a) Do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

(b) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

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I. Sync Shaft Installation Test

SUBTASK 78-31-04-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-04-710-001-G00

(2) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

(a) Remove the DO NOT OPERATE tag, STD-858, from the reverse thrust lever.

(b) Operate the thrust reverser through the extend (deploy) and retract (stow) cycles until all of the air is removed from the thrust reverser hydraulic system.

NOTE: This is shown by a smooth movement of the sleeves.

1) Do a check of the hydraulic tubes and connections for hydraulic fluid leaks.

a) If you find leaks, do a check of the hydraulic system for external leakage (TASK 29-00-00-790-801).

NOTE: Refer to the tube connections, static seals, and other dynamic seals sections of the procedure for the leakage limits.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-04-410-002-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SYNC SHAFT - INSPECTION/CHECK

1. General

A. This procedure contains a visual inspection of the Sync Shafts.

TASK 78-31-04-200-801-G00

2. Sync Shaft Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the thrust reverser sync shaft.

B. References

Reference	Title
78-31-04-000-801-G00	Sync Shaft Removal (P/B 401)
78-31-04-400-801-G00	Sync Shaft Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-3912	Lens - Magnifying, 5x, Hand Held

D. Prepare for the Inspection

SUBTASK 78-31-04-020-003-G00

(1) Remove the sync shaft that you will examine (TASK 78-31-04-000-801-G00).

E. Sync Shaft Inspection

SUBTASK 78-31-04-210-002-G00

(1) Use a hand held 5x magnifying lens, STD-3912 to do a visual inspection of the sync shaft wires:

(a) Look for a broken wire.

1) If you find a broken wire, replace the sync shaft.

(b) Look for wire strand separations.

1) Permitted, if the width of the strand separation is equal to or less than the diameter of the wire strand.

2) If the width of the strand separation is greater than the diameter of the wire strand, replace the sync shaft.

SUBTASK 78-31-04-210-003-G00

(2) Look for twists in the square ends of the sync shaft.

(a) If you find a twist in the square end, replace the sync shaft.

SUBTASK 78-31-04-210-004-G00

(3) Look for kinks in the sync shaft.

(a) Permitted, if you find a kink that is less than or equal to 0.020 in. (0.508 mm).

(b) If you find a kink that is greater than 0.020 in. (0.508 mm), replace the sync shaft.

SUBTASK 78-31-04-210-005-G00

(4) Look for abraded or worn areas on the surface of the sync shaft.

(a) If you find abraded or worn areas, then calculate the amount of face wear W as follows:

1) Measure the diameter D2 of the sync shaft at the maximum depth of the worn area.

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2) Measure the diameter D1 of the sync shaft in an adjacent area that have no damage.

3) Subtract diameter D2 from diameter D1 and then divide by 2.

NOTE: (D1 - D2) /2 = W.

(b) Permitted, If the face wear W is less than or equal to 0.013 in. (0.330 mm).

(c) If the face wear W is greater than 0.013 in. (0.330 mm), replace the sync shaft.

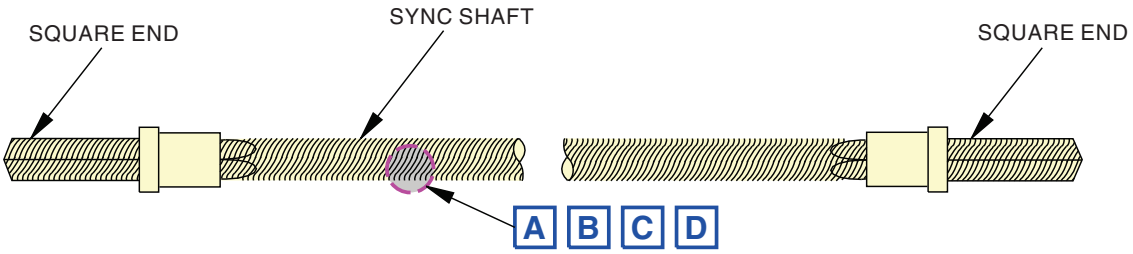
F. Put the Airplane Back to Its Usual Condition

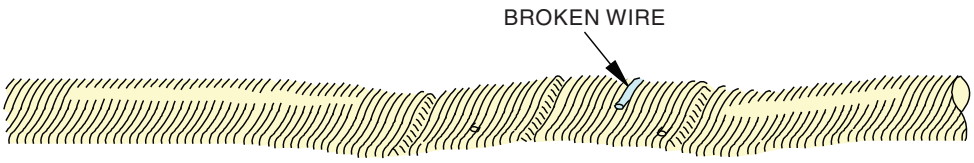
SUBTASK 78-31-04-420-007-G00

(1) Install the serviceable and replacement sync shafts (TASK 78-31-04-400-801-G00).

——— END OF TASK ———

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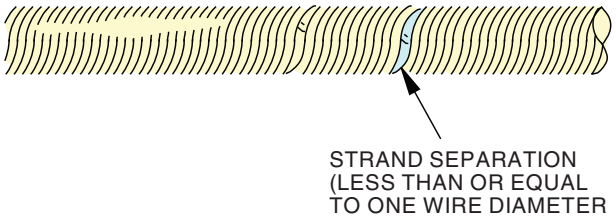
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A

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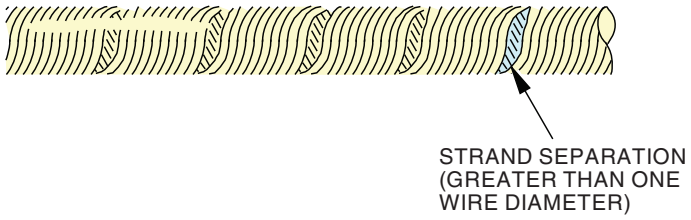
Sync Shaft Inspection
Figure 601/78-31-04-990-803-G00 (Sheet 1 of 4)

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(PERMITTED)

B



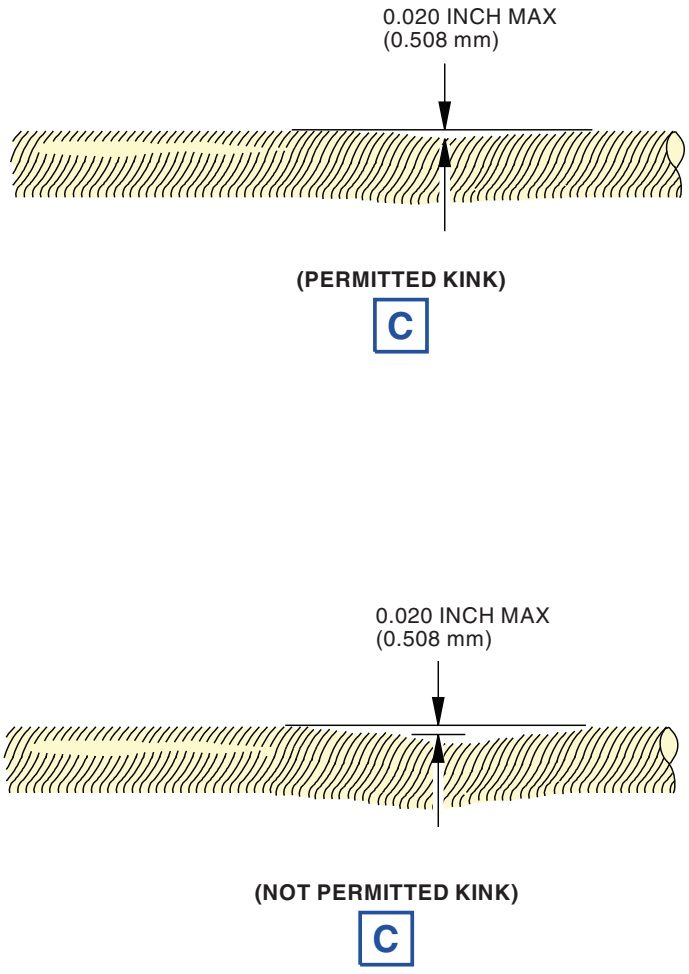
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B

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Sync Shaft Inspection
Figure 601/78-31-04-990-803-G00 (Sheet 2 of 4)

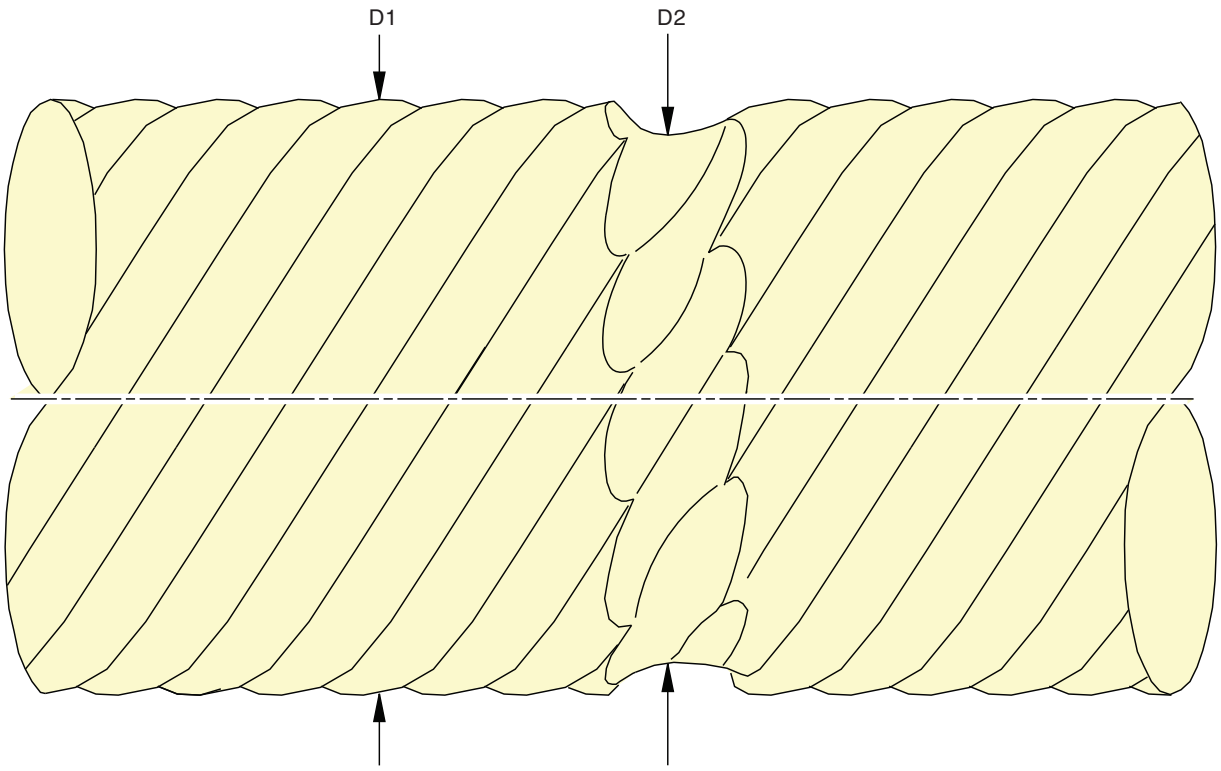
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Sync Shaft Inspection
Figure 601/78-31-04-990-803-G00 (Sheet 3 of 4)

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(FACE WEAR)

D

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Sync Shaft Inspection
Figure 601/78-31-04-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CASCADES - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the cascades.

(2) The installation of the cascades.

TASK 78-31-05-000-801-G00

2. Cascade Removal

(Figure 401, Figure 402)

A. General

(1) This task is for the removal of the cascades from the left or right thrust reverser on an engine.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Removal

SUBTASK 78-31-05-840-001-G00

(1) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00, to expose the thrust reverser cascades.

SUBTASK 78-31-05-040-001-G00

WARNING DO THE THRUST REVERSER DEACTIVATION PROCEDURE FOR GROUND MAINTENANCE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00, to deactivate the thrust reverser for ground maintenance.

E. Cascade Removal

SUBTASK 78-31-05-000-001-G00

(1) Do these steps to remove the applicable cascade:

(a) Remove the bolts [30] and washers [31] that attach the cascade to the aft cascade support ring.

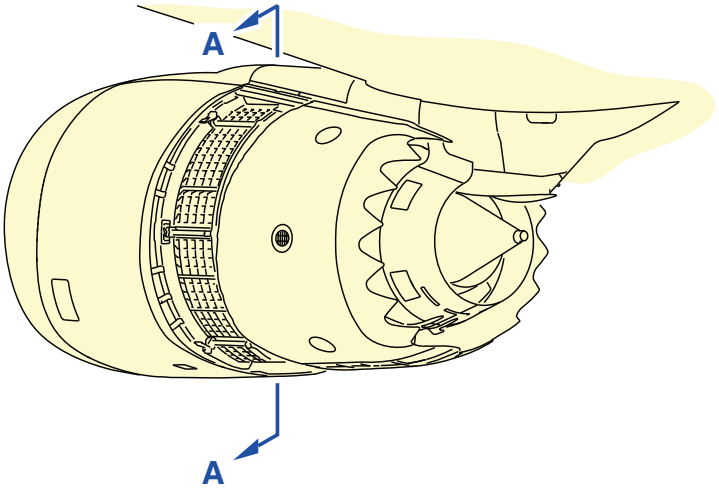
(b) Remove the bolts [30] and washers [31] that attach the cascade to the torque box.

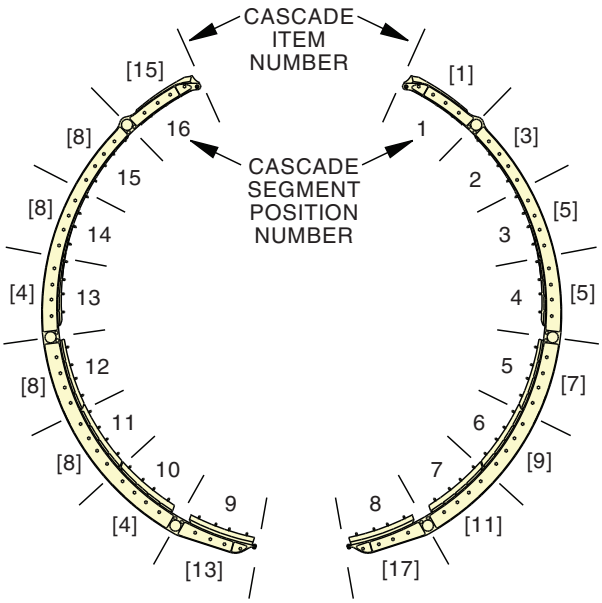
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(c) Remove the cascade [1], cascade [2], cascade [3], cascade [4], cascade [5], cascade [6], cascade [7], cascade [8], cascade [9], cascade [10], cascade [11], cascade [12], cascade [13], cascade [14], cascade [15], cascade [16], or cascade [17].

——— END OF TASK ———

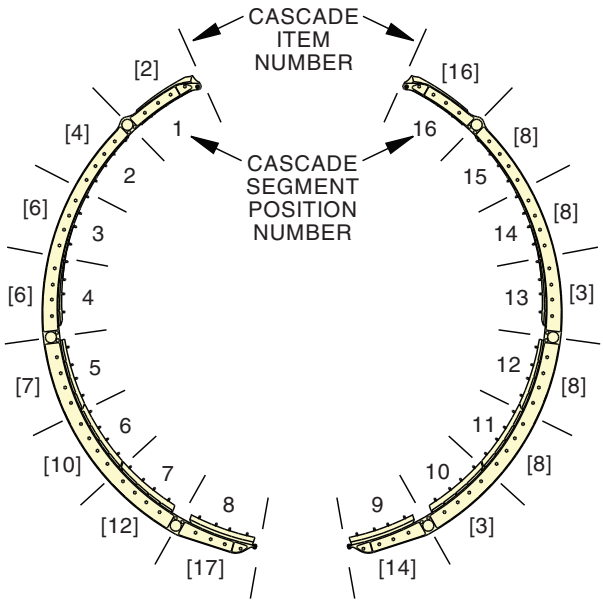
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**ENGINE POSITION 1,
THRUST REVERSER CASCADES
(AFT LOOKING FORWARD)**

A-A



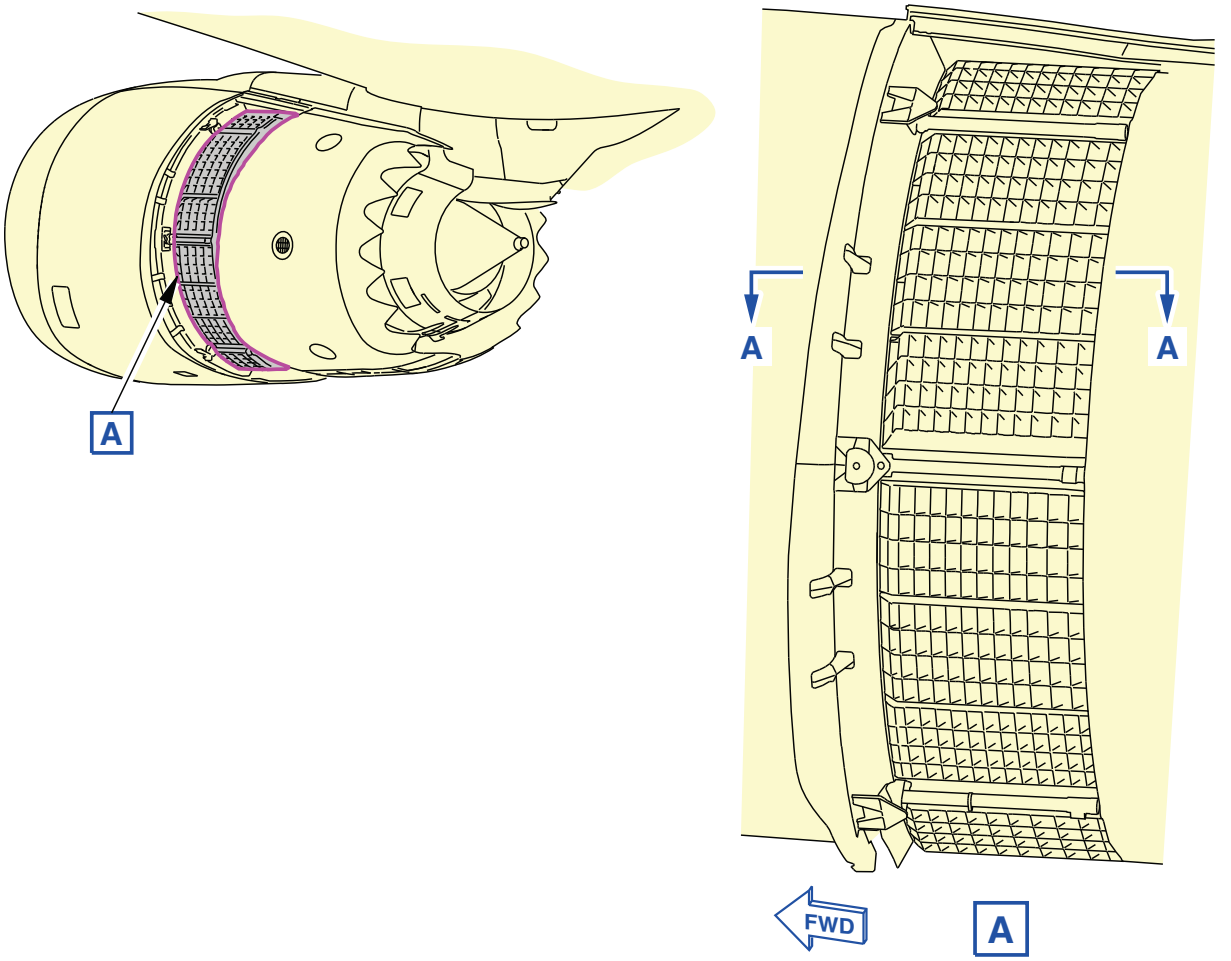
**ENGINE POSITION 2,
THRUST REVERSER CASCADES
(AFT LOOKING FORWARD)**

A-A

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**Cascade Positions
Figure 401/78-31-05-990-802-G00**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cascade Installation
Figure 402/78-31-05-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-05-400-801-G00

3. Cascade Installation

(Figure 401, Figure 402))

A. References

Reference	Title
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)

B. Consumable Materials

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Cascade Installation

SUBTASK 78-31-05-420-001-G00

CAUTION

MAKE SURE THE CORRECT CASCADES ARE INSTALLED AT THEIR APPLICABLE LOCATIONS. INCORRECT CASCADES INSTALLATION CAN EFFECT THE ENGINE PERFORMANCE.

(1) Use the table that follows to identify the correct cascade part number for the cascade position on the engine.

Cascade Position and Part Number Cross Reference Table

Cascade Item Number	Cascade Segment Position Number	Cascade Part Number
1	Engine 1, #1	LP5001
2	Engine 2, #1	LP5002
3	Engine 1, #2 Engine 2, #10 and #13	LP5003
4	Engine 1, #10 and #13 Engine 2, #2	LP5004
5	Engine 1, #3 and #4	LP5005 (Combination Cascade)
6	Engine 2, #3 and #4	LP5006 (Combination Cascade)
7	Engine 1, #5 Engine 2, #5	LP5007
8	Engine 1, #11, #12, #14 and #15 Engine 2, #11, #12, #14 and #15	LP5008
9	Engine 1, #6	LP5009
10	Engine 2, #6	LP5010

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Cascade Position and Part Number Cross Reference Table (Continued)

Cascade Item Number	Cascade Segment Position Number	Cascade Part Number
11	Engine 1, #7	LP5011
12	Engine 2, #7	LP5012
13	Engine 1, #9	LP5013
14	Engine 2, #9	LP5014
15	Engine 1, #16	LP5015
16	Engine 2, #16	LP5016
17	Engine 1, #8 Engine 2, #8	LP5017 (Blocked)

SUBTASK 78-31-05-000-002-G00

(2) Do these steps to install the cascade that was identified in the table:

(a) Apply primer, C00259, to the fastener holes in the cascade, the cascade support ring and the torque box.

1) Let the primer dry for 30 minutes.

(b) Lubricate the bolts [30] with grease, D00633.

(c) Install in its position cascade [1], cascade [2], cascade [3], cascade [4], cascade [5], cascade [6], cascade [7], cascade [8], cascade [9], cascade [10], cascade [11], cascade [12], cascade [13], cascade [14], cascade [15], cascade [16], or cascade [17].

(d) Align the bolt holes in the cascade with the fastener holes in the cascade support ring and the torque box.

(e) Install the bolts [30] and washers [31] in each fastener location to hold the cascade in its position.

(f) Tighten the bolts [30] at the torque box and the aft cascade support ring to 62.5 ±12.5 in-lb (7.1 ±1.5 N·m).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-05-840-002-G00

(1) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00, to stow the thrust reverser.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CASCADE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inpection of the thrust reverser cascade segments.

TASK 78-31-05-200-801-G00

2. Thrust Reverser Cascade Segments Inspection

(Figure 601)

A. General

(1) The cascade segments are made of a graphite/epoxy material. The cascade segments are mounted between the forward torque box and the aft cascade support ring. When the thrust reverser is stowed, the cascade segments are located between the outer and inner skin of the thrust reverser sleeve.

(2) During reverse thrust, the strongbacks transfer the thrust load from the cascade vanes to the forward torque box through the attachment flanges.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)
78-31-05-000-801-G00	Cascade Removal (P/B 401)
78-31-05-400-801-G00	Cascade Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Inspection

SUBTASK 78-31-05-010-001-G00

(1) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00, to expose the thrust reverser cascades.

SUBTASK 78-31-05-010-002-G00

WARNING

YOU MUST DO THE THRUST REVERSER DEACTIVATION PROCEDURE FOR GROUND MAINTENANCE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(2) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00, to deactivate the thrust reverser for ground maintenance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Thrust Reverser Cascade Segments Inspection

SUBTASK 78-31-05-210-001-G00

(1) Examine the cascade segments for damage, tears, cracks or delaminated areas in the cascade vanes, strongbacks and attachment flanges.

(a) If damage, tears, cracks or delaminated areas in the cascade vanes, strongbacks or attachment flanges is found, replace the cascade segment.

These are the tasks:

Cascade Removal, TASK 78-31-05-000-801-G00,

Cascade Installation, TASK 78-31-05-400-801-G00.

(b) Contact Boeing to report the kind of damage and delamination found for repair disposition.

NOTE: If it is possible, document the damage with digital photographs that can be sent by email, or provide a sketch of the damage that can be faxed.

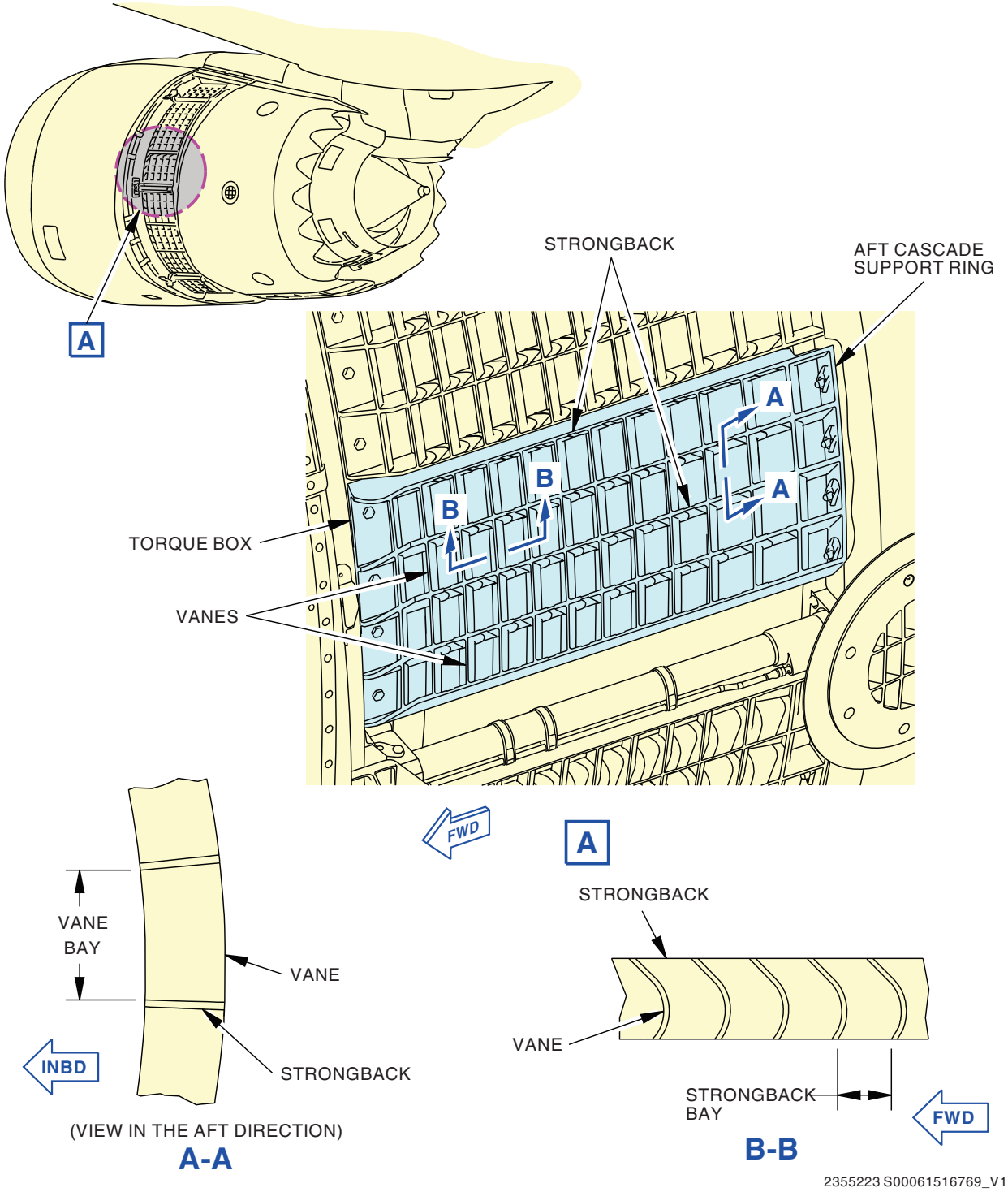
F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-05-440-002-G00

(1) Do this task to stow the thrust reverser: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Cascade Segment Inspection
Figure 601/78-31-05-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BLOCKER DOORS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the blocker door

(2) An installation of the blocker door.

TASK 78-31-06-000-801-G00

2. Blocker Door Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the blocker door.

B. References

Reference	Title
78-31-07-000-801-G00	Blocker Door Drag Link Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Blocker Door Removal

SUBTASK 78-31-06-010-002-G00

(1) Disconnect the drag link at the blocker door (TASK 78-31-07-000-801-G00).

SUBTASK 78-31-06-020-001-G00

CAUTION

MAKE SURE THAT YOU PUT PROTECTION ON THE FAN DUCT WALLS AND BLOCKER DOORS. DAMAGE CAN OCCUR IF A TOOL OR DRAG LINK HITS THE WALL OR DOORS.

(2) Do these steps to remove the blocker door:

(a) Make sure that there is protective mat, STD-585, on the fan duct walls and blocker doors.

(b) Disconnect the applicable hinges:

NOTE: Each blocker door has two hinges with different fastening set.

1) At the hinge [1], remove the nut [8], washers [6], bushings [7], anti-rotation bracket [5], and bolt [4] (View B, Figure 401).

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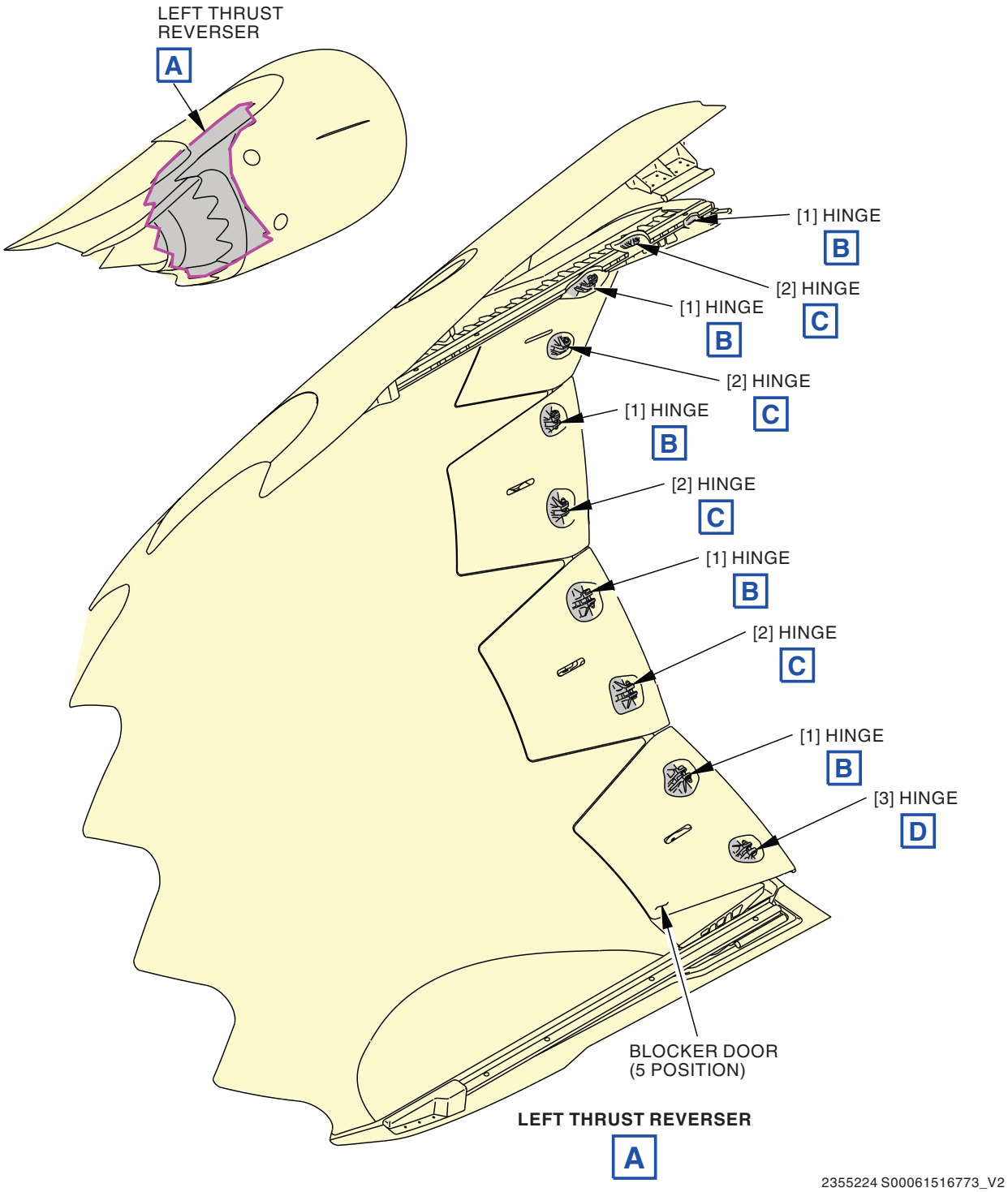
2) At the hinge [2] and hinge [3], remove the nut [8], washers [6], bushing [7], washer [9], and bolt [4] (View C and D, Figure 401).

NOTE: The hinge [2] has the bushing installed at the bolt head. The hinge [3] has the bushing installed at the threaded end of the bolt.

(c) Remove the blocker door.

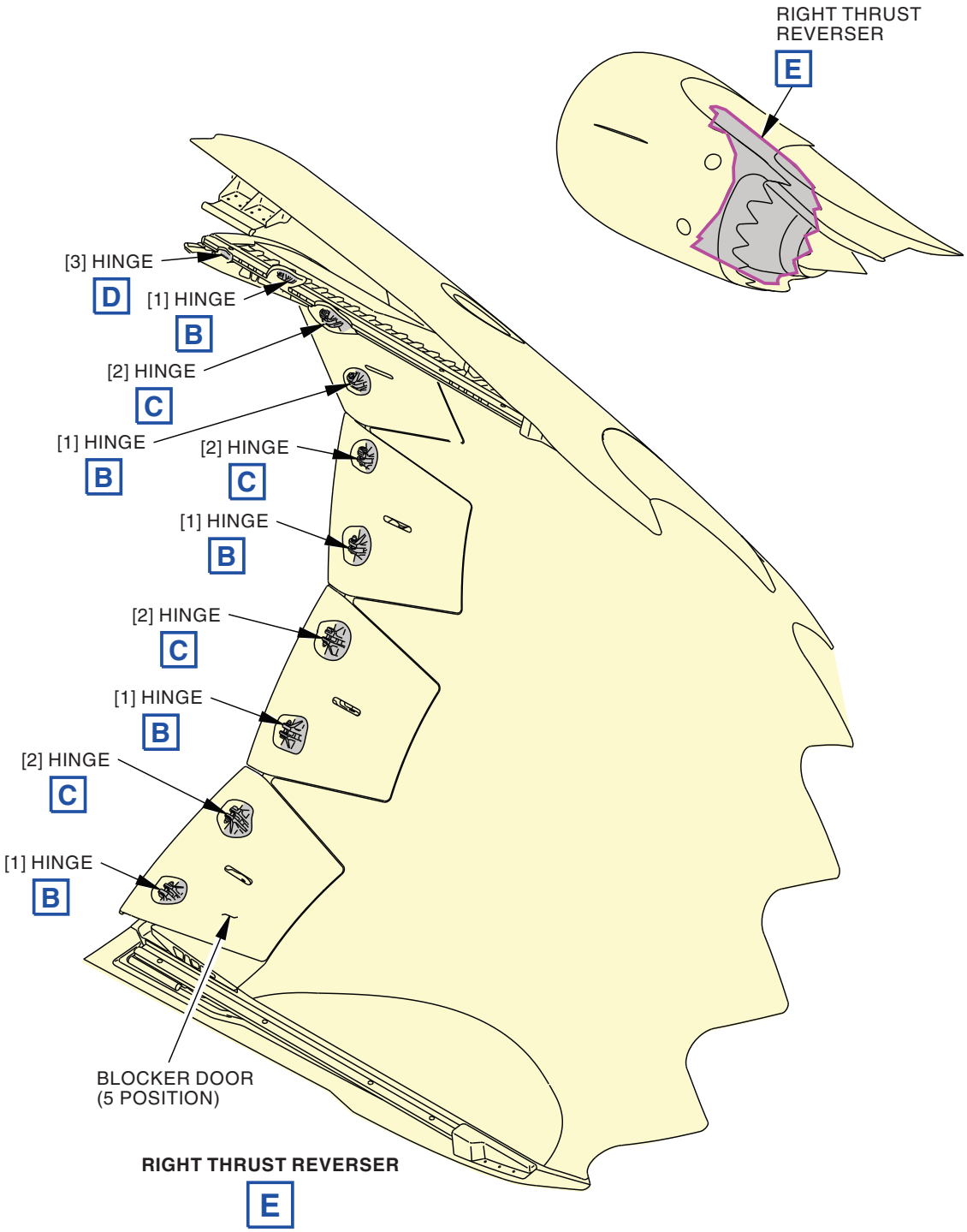
——— END OF TASK ———

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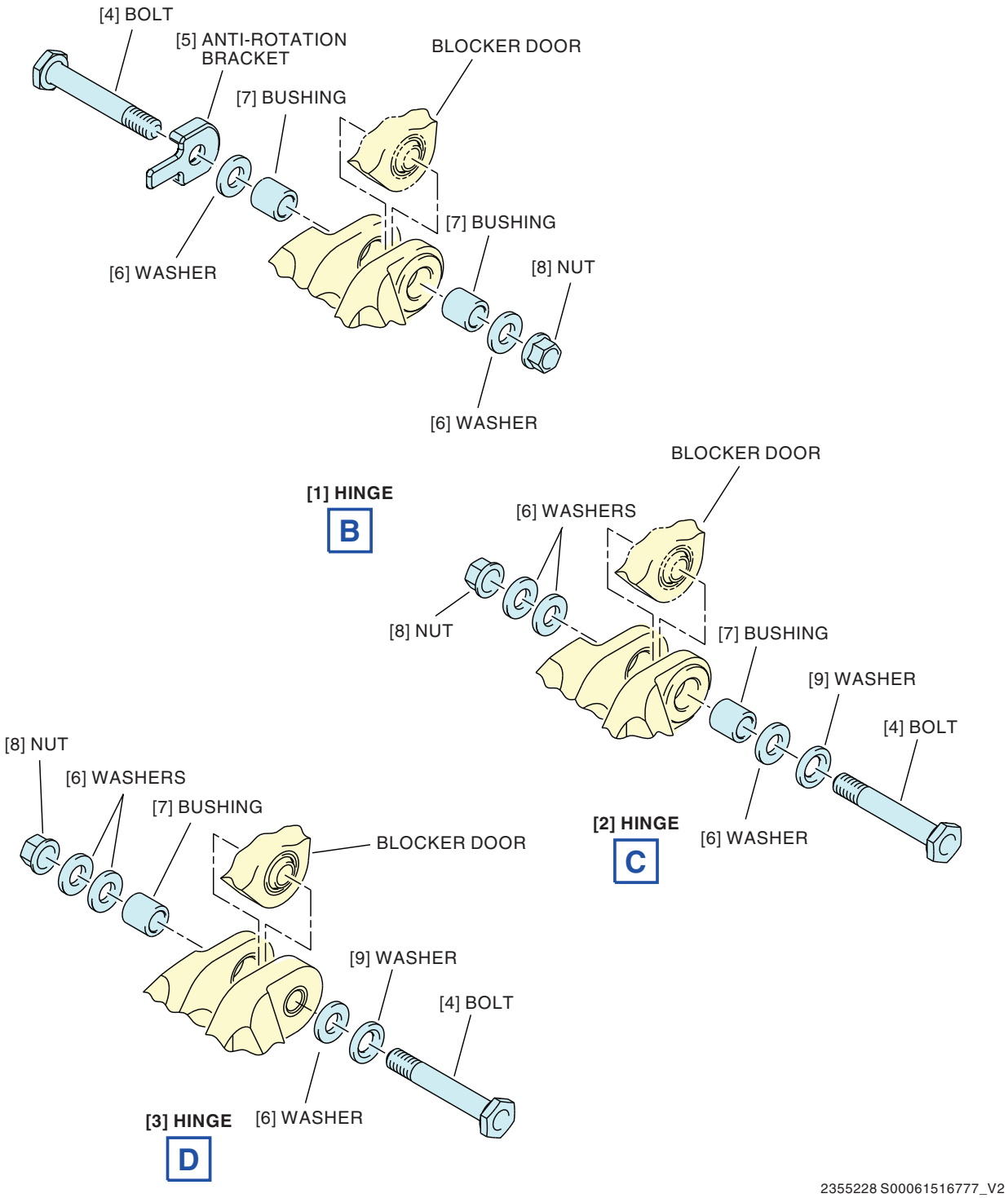
Blocker Door Installation
Figure 401/78-31-06-990-801-G00 (Sheet 1 of 3)

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Blocker Door Installation
Figure 401/78-31-06-990-801-G00 (Sheet 2 of 3)

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Blocker Door Installation
Figure 401/78-31-06-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-06-400-801-G00

3. Blocker Door Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the blocker door.

B. References

Reference	Title
78-31-07-400-801-G00	Blocker Door Drag Link Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Blocker Door Installation

SUBTASK 78-31-06-420-001-G00

CAUTION MAKE SURE THAT YOU PUT PROTECTION ON THE FAN DUCT WALLS AND BLOCKER DOORS. DAMAGE CAN OCCUR IF A TOOL OR DRAG LINK HITS THE WALL OR DOORS.

(1) Do these steps to install the blocker door:

(a) Make sure that there is protective mat, STD-585, on the fan duct walls and blocker doors.

(b) Align the fastener holes in the blocker door hinge and the sleeve hinge fitting.

(c) Connect the applicable hinges as follows:

NOTE: Each blocker door has two hinges with a different fastening set.

1) At the hinge [1], install the bolt [4], anti-rotation bracket [5], washers [6], bushings [7], and nut [8] (View B, Figure 401).

2) At the hinge [2] and hinge [3], install the bolt [4], washer [9], washers [6], bushing [7], and nut [8] (View C and D, Figure 401).

NOTE: The hinge [2] has the bushing installed at the bolt head. The hinge [3] has the bushing installed at the threaded end of the bolt.

3) Make sure that the anti-rotation bracket [5] fit in the slot in the acoustic panel assembly.

4) Tighten the nut [8] to 65 in-lb (7.3 N·m) - 70 in-lb (7.9 N·m).

SUBTASK 78-31-06-700-001-G00

(2) Do this check of the blocker door:

NOTE: If there is side movement of the blocker door or if more than three threads show at the end of the bolt at the clamped hinge, this is an indication that the hinge is not clamped.

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(a) Make sure that there is no side movement of the blocker door at the hinges.

(b) Make sure that three threads or less show at the end of the bolt [4].

1) If there is side movement of the blocker door at the hinges or more than three threads show at the end of the bolt [4], add more washers [6] at the nut [8].

2) Tighten the nut [8] to 65 in-lb (7.3 N·m) - 70 in-lb (7.9 N·m).

SUBTASK 78-31-06-420-002-G00

(3) Connect the drag link at the blocker door (Blocker Door Drag Link Installation, TASK 78-31-07-400-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BLOCKER DOORS - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A visual check of the blocker doors.

(2) A visual check of the blocker door hinge attachments and bearings.

TASK 78-31-06-200-801-G00

2. Blocker Door Visual Inspection

(Figure 601, Figure 602, Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions on how to do a visual check of the blocker doors.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)
78-31-01-960-802-G00	Blocker Door Support Assembly Replacement (P/B 801)
78-31-06 P/B 401	BLOCKER DOORS - REMOVAL/INSTALLATION
78-31-24 P/B 401	BLOCKER DOOR AERO SEAL - REMOVAL/INSTALLATION

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 78-31-06-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-06-010-004-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Blocker Door Visual Inspection

SUBTASK 78-31-06-210-001-G00

(1) Look for missing blocker doors.

SUBTASK 78-31-06-400-001-G00

(2) If blocker doors are missing, replace the blocker doors (PAGEBLOCK 78-31-06/401).

SUBTASK 78-31-06-210-002-G00

(3) Examine the blocker doors for the damage that follows:

(a) holes

(b) cracks

(c) nicks

(d) gouges

(e) scratches

(f) dents

(g) edge damage.

SUBTASK 78-31-06-910-001-G00

(4) If you find damage, refer to the SRM SUBJECT 54-30-01 for the permitted limits and repair procedures.

SUBTASK 78-31-06-860-001-G00

(5) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

SUBTASK 78-31-06-210-004-G00

(6) Examine the blocker door support assemblies.

(a) If one support assembly is missing below a blocker door, the Continue-In-Service limit is a maximum of 14 days.

NOTE: The limit of 14 days is an action to prevent damage. The limit makes it necessary to replace the first missing support because it is possible that the second support could also detach. Two blocker door supports prevent the vibration of the closed blocker door. One blocker door support is permitted because the door is supported. If the second support detaches, the blocker door will vibrate which will cause wear and damage to the blocker door and the thrust reverser sleeve.

(b) If two support assemblies are missing below a blocker door, replace the support assemblies (TASK 78-31-01-960-802-G00).

NOTE: Continued operation, for a maximum of 14 days, is permitted if one support assembly is re-installed. If a new part is not available, one support assembly from an adjacent door, that has two support assemblies, can be removed and installed.

SUBTASK 78-31-06-210-005-G00

(7) Examine the blocker door aero seal for missing material, cuts, gouges, and holes through the material (Figure 603).

(a) If the blocker door aero seal is damaged, replace the seal (PAGEBLOCK 78-31-24/401) or contact Boeing.

SUBTASK 78-31-06-860-002-G00

(8) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-06-410-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-06-860-004-G00

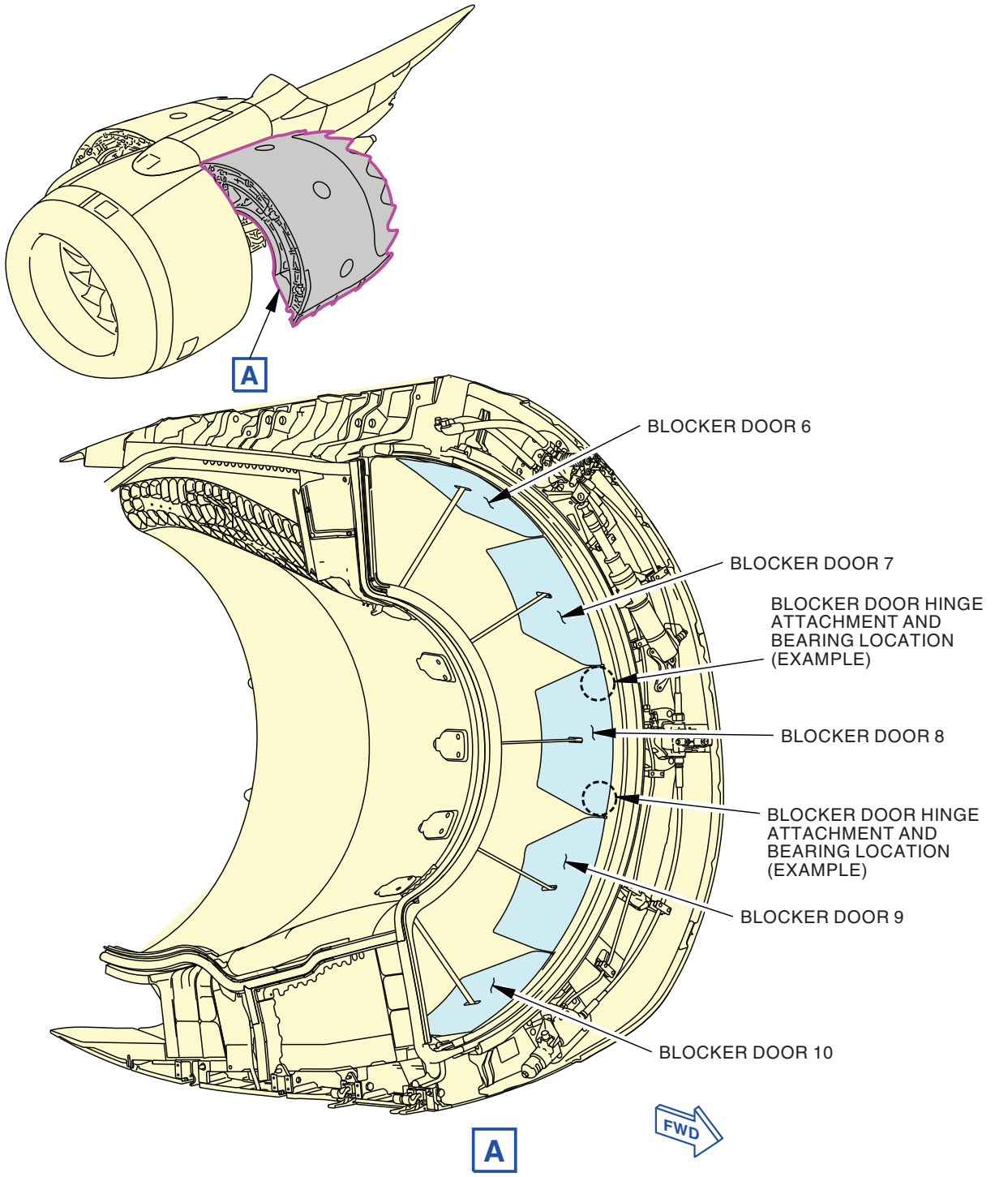
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

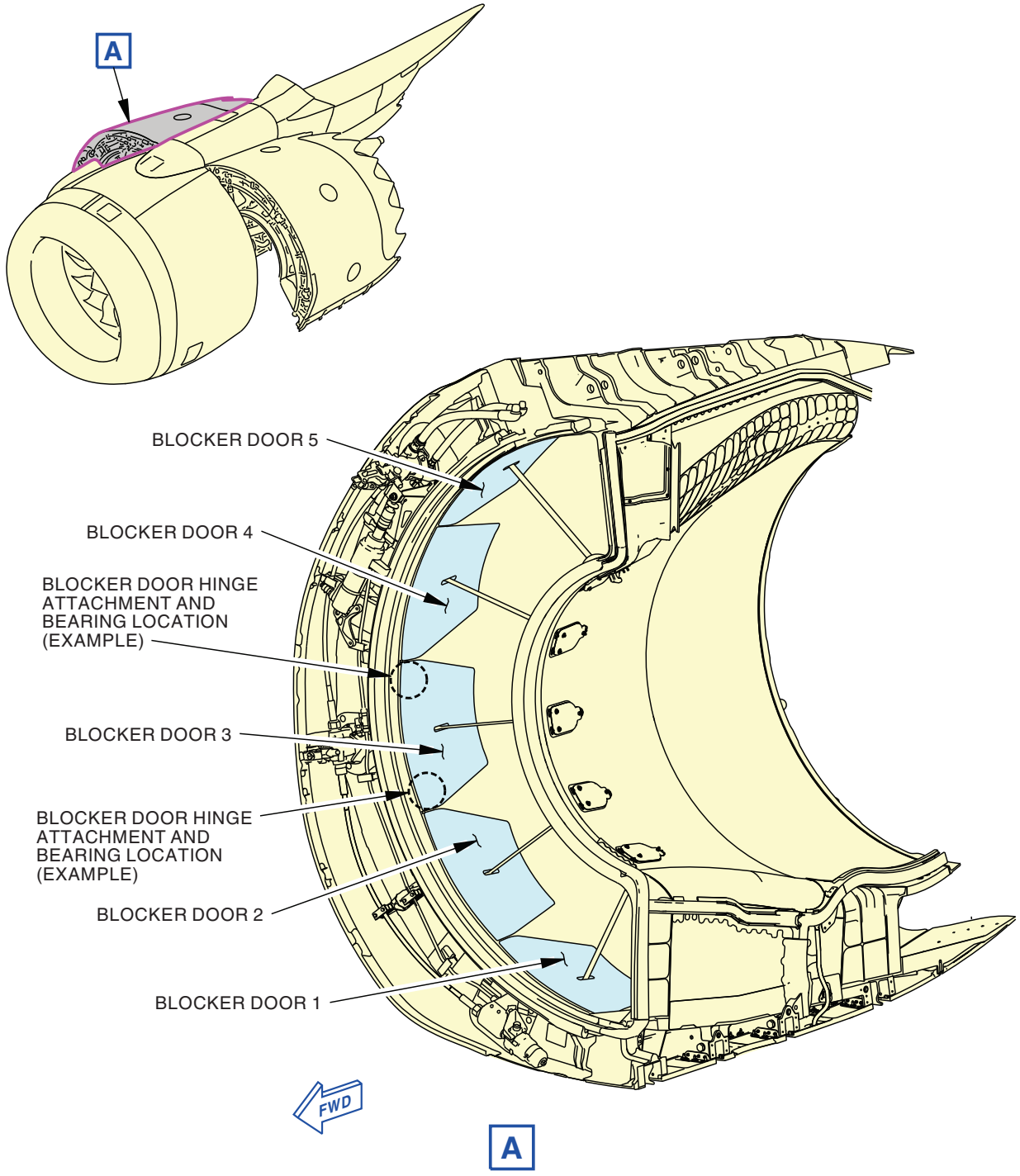
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Thrust Reverser Blocker Door Inspection
Figure 601/78-31-06-990-802-G00

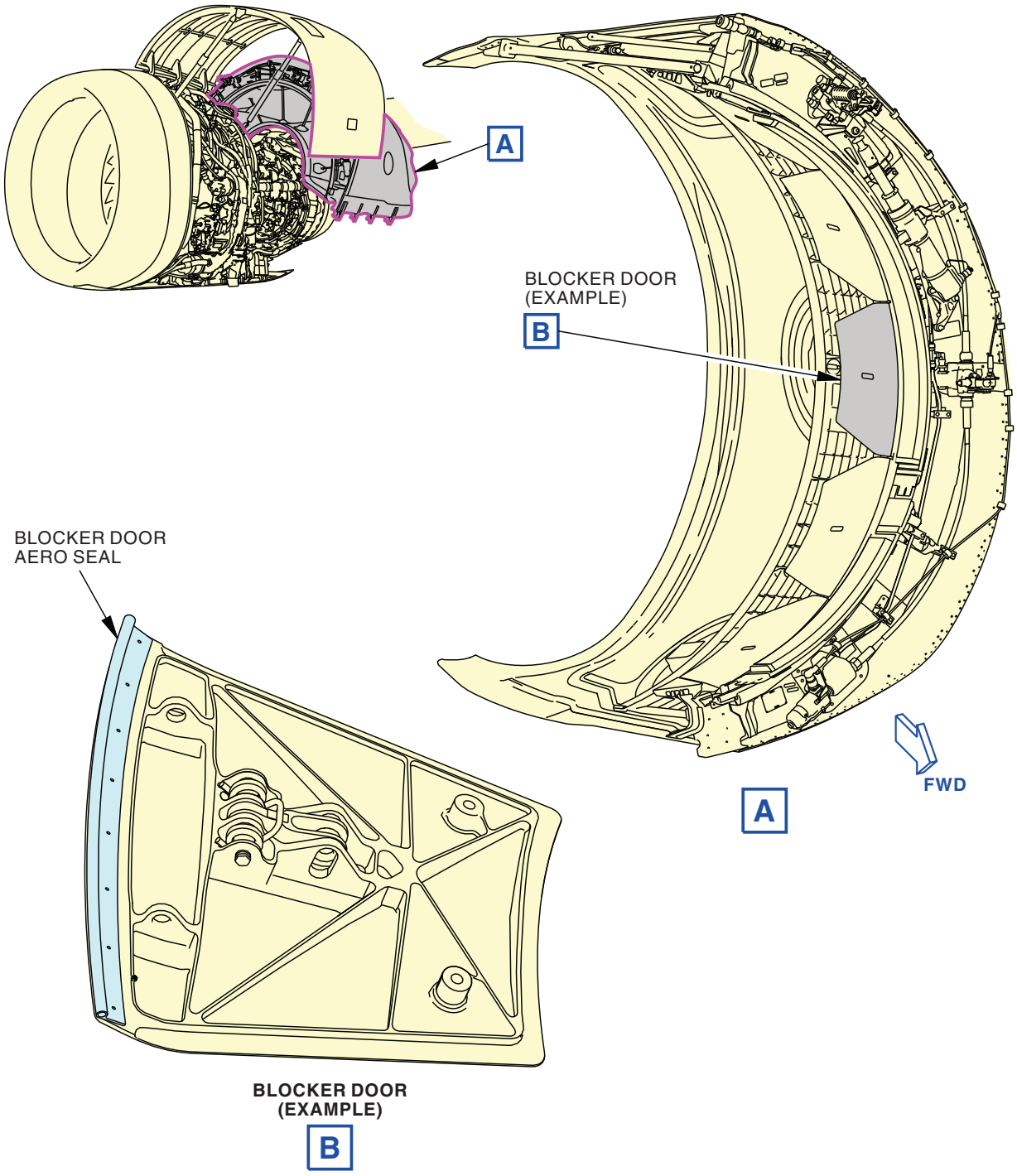
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355230 S00061516782_V2

Right Thrust Reverser Blocker Door Inspection
Figure 602/78-31-06-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2971999 S0000750745_V1

Blocker Door Aero Seal Inspection
Figure 603/78-31-06-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-06-211-801-G00

3. Blocker Door Hinge Attachments and Bearings Inspection

(Figure 601, Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do a visual check of the blocker door hinge attachments and bearings.

B. References

Reference	Title
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Inspection

SUBTASK 78-31-06-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-06-860-006-G00

(2) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

F. Blocker Door Hinge Attachments and Bearings Inspection

SUBTASK 78-31-06-211-001-G00

(1) Do a detailed inspection of the thrust reverser blocker door hinge attachments and bearings for fretting and cracking.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-06-860-007-G00

(1) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

SUBTASK 78-31-06-860-008-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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BLOCKER DOOR DRAG LINKS - MAINTENANCE PRACTICES

1. General

A. This procedure has one task:

(1) An inspection of the drag link spherical bearing.

TASK 78-31-07-900-801-G00

2. Drag Link Spherical Bearing Inspection

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions on how to remove and examine the spherical bearing in the drag link.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-00-980-805-G00	Thrust Reverser ,Operation - Extend (Power Procedure) (P/B 201)
78-31-00-980-806-G00	Thrust Reverser Operation - Retract (Power Procedure) (P/B 201)
78-31-07-000-801-G00	Blocker Door Drag Link Removal (P/B 401)
78-31-07-400-801-G00	Blocker Door Drag Link Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers

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D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02329	Tape - Aluminum Foil, Pressure Sensitive - Vibration Damping Tape 434	

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 78-31-07-040-001-G00

(1) Deactivate the leading edge flaps and slats in the retract position (TASK 27-81-00-040-801).

SUBTASK 78-31-07-010-005-G00

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSER: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THIS SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Open the applicable left or right thrust reverser (TASK 78-31-00-010-801-G00).

(a) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Install the actuator safety 15 degree lock assembly, SPL-14993, on the inboard thrust reverser's extended piston rod to lock the thrust reverser at a 15-degree open position.

NOTE: The part number of the actuator safety lock is C78035-14. The part number (C78035-14) is for the set of two actuator safety locks.

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SUBTASK 78-31-07-010-009-G00

CAUTION

DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN THE PERMITTED LIMIT. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. DAMAGE CAN OCCUR.

(3) Manually extend the outboard thrust reverser sleeve and inboard thrust reverser sleeve no more than 15 in. (381.0 mm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(a) Monitor the position of the thrust reverser sleeve as it extends to make sure that it does not touch the leading edge of the wing.

NOTE: While the inboard thrust reverser is open to the 15-degree position, the maximum sleeve extension is 15 in. (381 mm) to avoid contact with the leading edge.

NOTE: The sleeve must be partially extended to release the load on the drag link. This also permits you to get to the hardware that attaches the drag link to the blocker door.

SUBTASK 78-31-07-840-001-G00

CAUTION

MAKE SURE THAT YOU PUT PROTECTION ON THE FAN DUCT WALLS AND BLOCKER DOORS. DAMAGE CAN OCCUR IF A TOOL OR DRAG LINK HITS THE WALL OR DOORS.

(4) Put protective mat, STD-585, on the fan duct walls and blocker doors.

SUBTASK 78-31-07-010-007-G00

(5) Get access to the front of the thrust reverser duct.

NOTE: To facilitate the access from the front of the thrust reverser duct, disconnect one blocker door at a time.

SUBTASK 78-31-07-020-002-G00

(6) From the front of the thrust reverser duct, disconnect the drag link [1] at the inner wall (TASK 78-31-07-000-801-G00).

SUBTASK 78-31-07-420-003-G00

(7) Install a cotton wiper, G00034, with Vibration Damping Tape 434 tape, G02329, around the end of the drag link [1] to prevent damage to the inner wall.

SUBTASK 78-31-07-210-004-G00

(8) Make sure that the blocker door can rotate freely with the attached drag link [1].

SUBTASK 78-31-07-480-001-G00

(9) Rotate and set the blocker door approximately perpendicular to its stowed position.

(a) Use Vibration Damping Tape 434 tape, G02329, to hold the blocker door in this position.

SUBTASK 78-31-07-010-006-G00

(10) Examine the drag link [1] for these conditions:

NOTE: Record the drag links location and condition.

(a) Loose nuts [6] on the drag link bolts [4].

(b) Drag link bolts [4] that have a shank length that is too long.

(c) If you find one or more of these conditions, replace the drag link [1].

SUBTASK 78-31-07-020-003-G00

(11) Disconnect the drag link [1] at the blocker door (TASK 78-31-07-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Drag Link Spherical Bearing Inspection

SUBTASK 78-31-07-010-004-G00

(1) Open the applicable blocker door.

SUBTASK 78-31-07-010-010-G00

(2) Remove the cotton wiper, G00034, with the Vibration Damping Tape 434 tape, G02329, from the end of the drag link [1].

SUBTASK 78-31-07-210-001-G00

(3) Examine the drag link [1] for these conditions:

NOTE: Record the drag links location and condition.

(a) The bushing [5] and bushing [14] in the pivot links of the blocker door are worn.

(b) The bushing [11] in the drag link [1] anchor fittings are worn.

(c) The race of the spherical bearings are loose in the drag link [1].

(d) The pivot links or anchor fittings have cracks or other damage.

(e) If you find one or more of these conditions, replace the drag link [1].

SUBTASK 78-31-07-210-003-G00

(4) Examine the ball and the spherical bearing race for pits, scratches, tears in the liner or missing liner.

NOTE: The spherical bearing consists of a ball and a race. The ball is not removable on the liner spherical bearing. The liner is attached to the inside of the race.

(a) Turn the ball 90°.

(b) Clean the ball with a cotton wiper, G00034.

(c) If the ball has pits or scratches, replace the drag link [1].

(d) If you find tears in the liner or missing liner on the race, replace the drag link [1].

NOTE: The liner material from the race may transfer to the ball which will be light brown in color. The transfer of liner material on to the ball is permitted. The liner bearing will not corrode. Do not mistake the liner material, dirt or grease as corrosion on the bearing.

NOTE: The re-application of the solid film lubricant is not recommended.

SUBTASK 78-31-07-420-001-G00

(5) Install the serviceable or new drag link [1] at the blocker door and inner wall (TASK 78-31-07-400-801-G00).

SUBTASK 78-31-07-010-011-G00

(6) Remove the protective mat, STD-585, from the fan duct walls and blocker doors.

SUBTASK 78-31-07-010-012-G00

(7) Manually retract the thrust reverser sleeves (TASK 78-31-00-980-804-G00).

SUBTASK 78-31-07-410-006-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Close and latch the thrust reversers (TASK 78-31-00-010-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Remove the actuator safety 15 degree lock assembly, SPL-14993, from the inboard thrust reversers only.

(c) Do not do the thrust reverser or leading edge activation.

(d) Do not close the fan cowl panels at this time.

SUBTASK 78-31-07-980-002-G00

(9) Manually translate the sleeves through an extend and retract cycle as follows:

(a) Do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

(b) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

(c) Make sure that the blocker doors and drag links move smoothly.

I. Drag Link Spherical Bearing Installation Test

SUBTASK 78-31-07-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-07-710-001-G00

(2) Operate the thrust reverser a minimum of three cycles to make sure that the drag link and blocker door move correctly.

(a) Do this task: Thrust Reverser ,Operation - Extend (Power Procedure), TASK 78-31-00-980-805-G00.

(b) Do this task: Thrust Reverser Operation - Retract (Power Procedure), TASK 78-31-00-980-806-G00.

(c) Make sure that the blocker doors and drag links operate correctly.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-07-000-002-G00

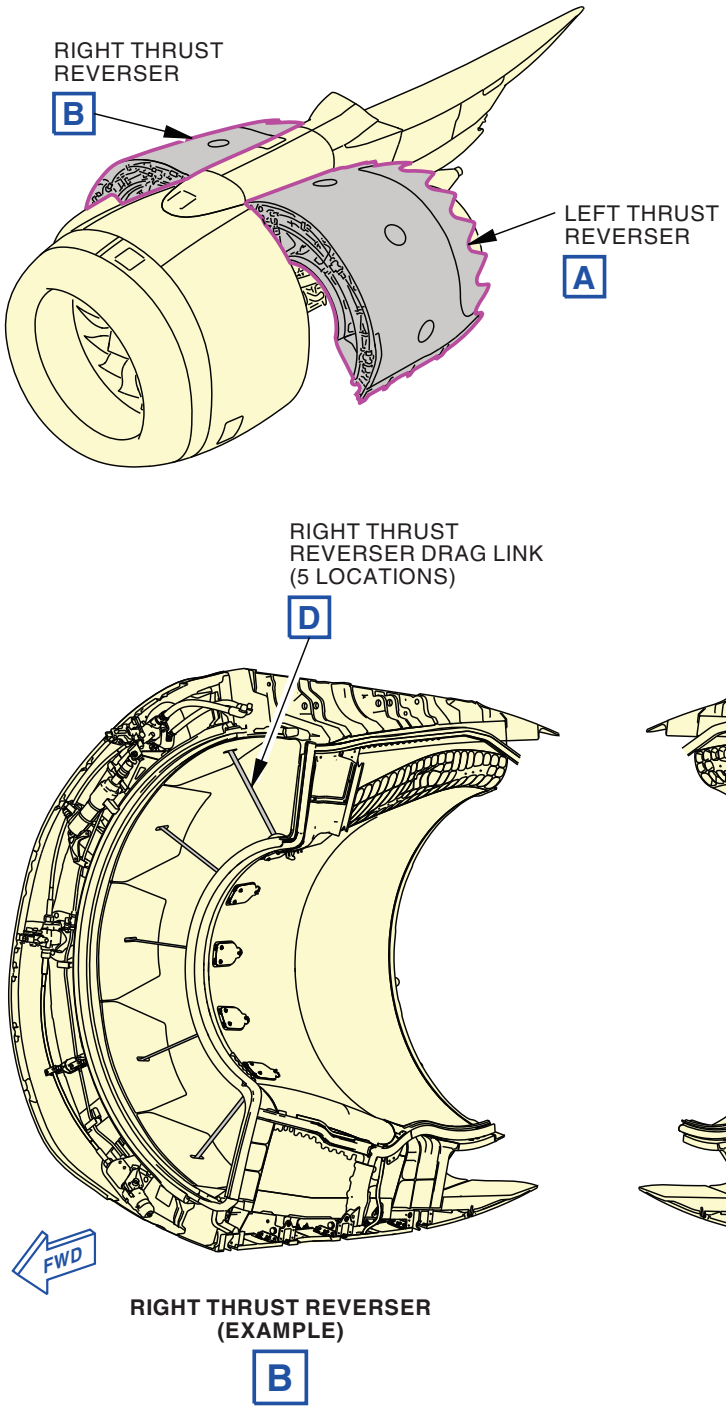
(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-07-000-003-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

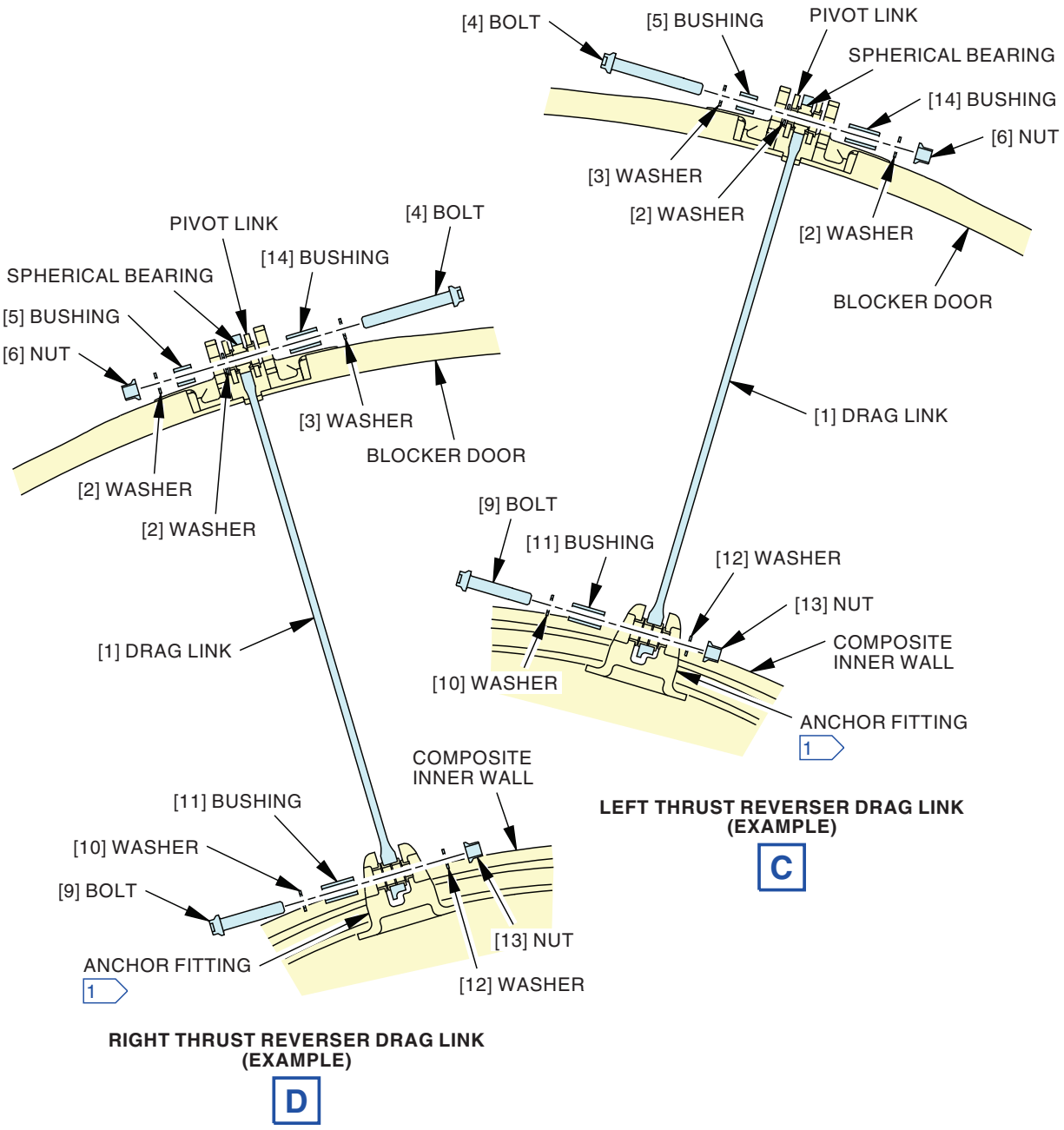
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Drag Link Spherical Bearing Installation
Figure 201/78-31-07-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

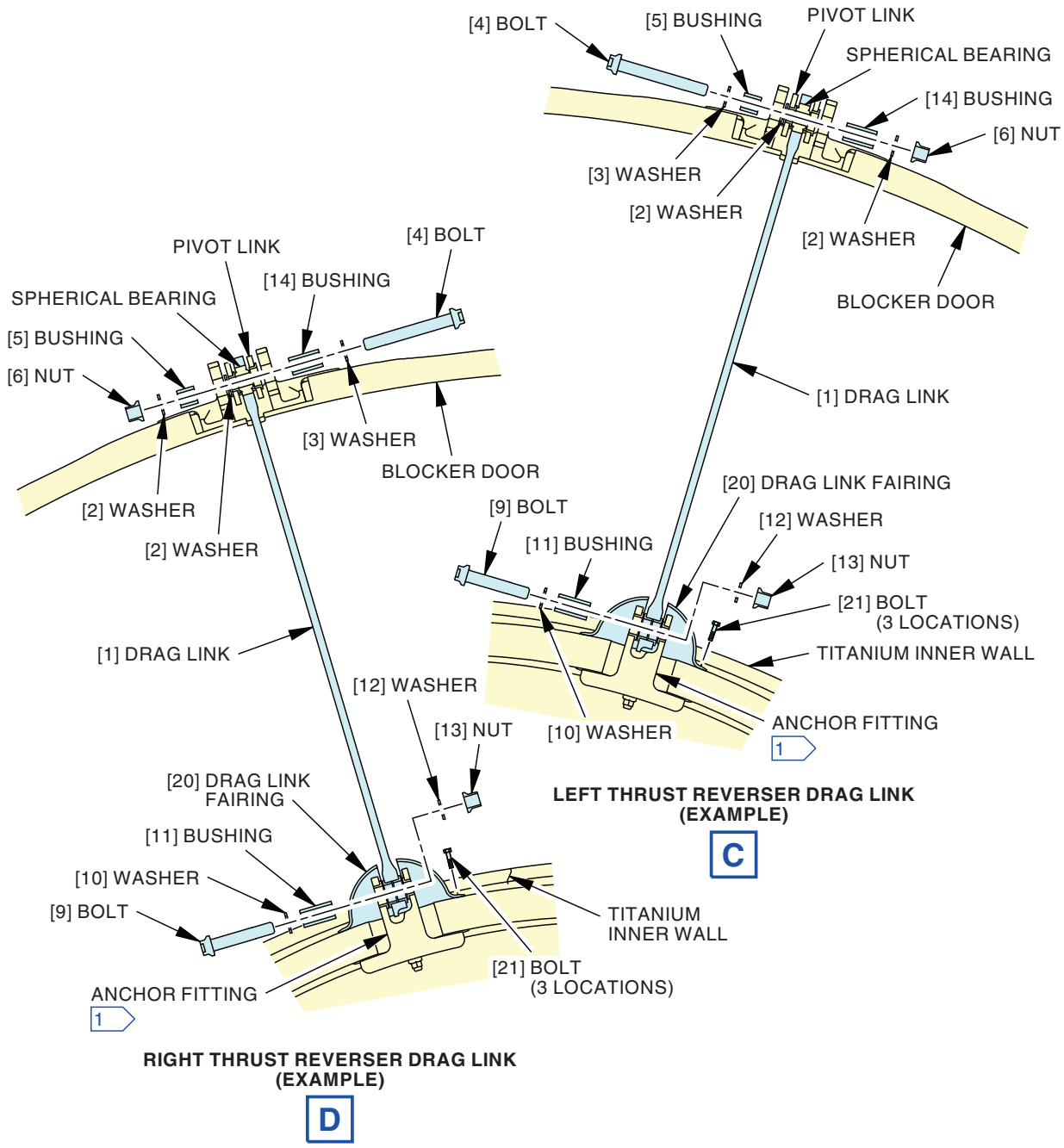


1 WRAP CLOTH AROUND THE ANCHOR FITTING TO PROTECT THE INNER WALL

3042238 S0000809606_V1

Drag Link Spherical Bearing Installation
Figure 201/78-31-07-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 WRAP CLOTH AROUND THE ANCHOR FITTING TO PROTECT THE INNER WALL

3042306 S0000809617_V1

Drag Link Spherical Bearing Installation
Figure 201/78-31-07-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BLOCKER DOOR DRAG LINKS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the blocker door drag links.

(2) The installation of the blocker door drag links.

TASK 78-31-07-000-801-G00

2. Blocker Door Drag Link Removal

(Figure 401)

A. General

(1) This task is for the removal of the blocker door drag links from the left or right thrust reverser on an engine. There are five blocker door drag links on each thrust reverser and the removal task is the same for all of the blocker door drag links.

(2) The blocker door drag links are between the blocker doors and the inner wall of the thrust reverser fan duct.

(3) For this task the blocker door drag links will be referred to as the drag links.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02329	Tape - Aluminum Foil, Pressure Sensitive - Vibration Damping Tape 434	

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right

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(Continued)

Zone	Area
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Prepare for the Removal

SUBTASK 78-31-07-010-002-G00

(1) Deactivate the leading edge flaps and slats in the retract position (TASK 27-81-00-040-801).

SUBTASK 78-31-07-980-003-G00

WARNING

DO THESE SPECIFIED TASKS IN THE CORRECT SEQUENCE BEFORE YOU OPEN THE THRUST REVERSER: RETRACT THE LEADING EDGE, DO THE DEACTIVATION PROCEDURES FOR THE LEADING EDGE AND THE THRUST REVERSER (FOR GROUND MAINTENANCE), AND OPEN THE FAN COWL PANELS. IF YOU DO NOT OBEY THIS SEQUENCE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

(a) Install the actuator safety 15 degree lock assembly, SPL-14993, on the inboard thrust reverser's extended piston rod to lock the thrust reverser at a 15-degree open position.

NOTE: The part number of the actuator safety lock is C78035-14. The part number (C78035-14) is for the set of two actuator safety locks.

SUBTASK 78-31-07-010-008-G00

CAUTION

DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN 15 INCHES. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(3) Manually extend the outboard thrust reverser sleeve and inboard thrust reverser sleeve no more than 15 in. (381 mm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

(a) Monitor the position of the thrust reverser sleeve as it is extends to make sure that it does not touch the leading edge of the wing.

NOTE: While the inboard thrust reverser is open to the 15-degree position, the maximum sleeve extension is 15 in. (381 mm) to avoid contact with the leading edge.

NOTE: The sleeve must be partially extended to release the load on the drag link. This also permits you to get to the hardware that attaches the drag link to the blocker door.

G. Blocker Door Drag Link Removal

SUBTASK 78-31-07-020-001-G00

CAUTION

MAKE SURE THAT YOU PUT PROTECTION ON THE FAN DUCT WALLS AND BLOCKER DOORS. DAMAGE CAN OCCUR IF A TOOL OR DRAG LINK HITS THE WALL OR DOORS.

(1) Do these steps to remove the drag link [1], (Figure 401):

(a) Put protective mat, STD-585 on the fan duct walls and blocker doors.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Put cotton wiper, G00034, around the anchor fitting and drag link at the inner wall.

NOTE: When the drag link is disconnected from the blocker door, it can move forward or aft and fall against the inner wall. This cause damage to the inner wall composite panel.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

(c) Disconnect the drag link [1] at the inner wall, as follows:

1) Remove the nut [13], washer [12], bushing [11], washer [10], and bolt [9] from the anchor fitting.

2) Remove the drag link [1].

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

(d) Disconnect the drag link [1] at the inner wall, as follows:

1) Remove the three bolts [21] and the drag link fairing [20] from the inner wall.

2) Remove the nut [26], washer [25], bushing [24], washer [23], and bolt [22] from the anchor fitting.

3) Remove the drag link [1].

PTF ALL

(e) Make sure that the blocker door can rotate freely with the attached drag link [1].

(f) Rotate and set the blocker door against the outer wall of the fan duct and approximately perpendicular to its stowed position.

1) Use Vibration Damping Tape 434 tape, G02329, to hold the blocker door in this position.

(g) Disconnect the drag link [1] at the blocker door, as follows:

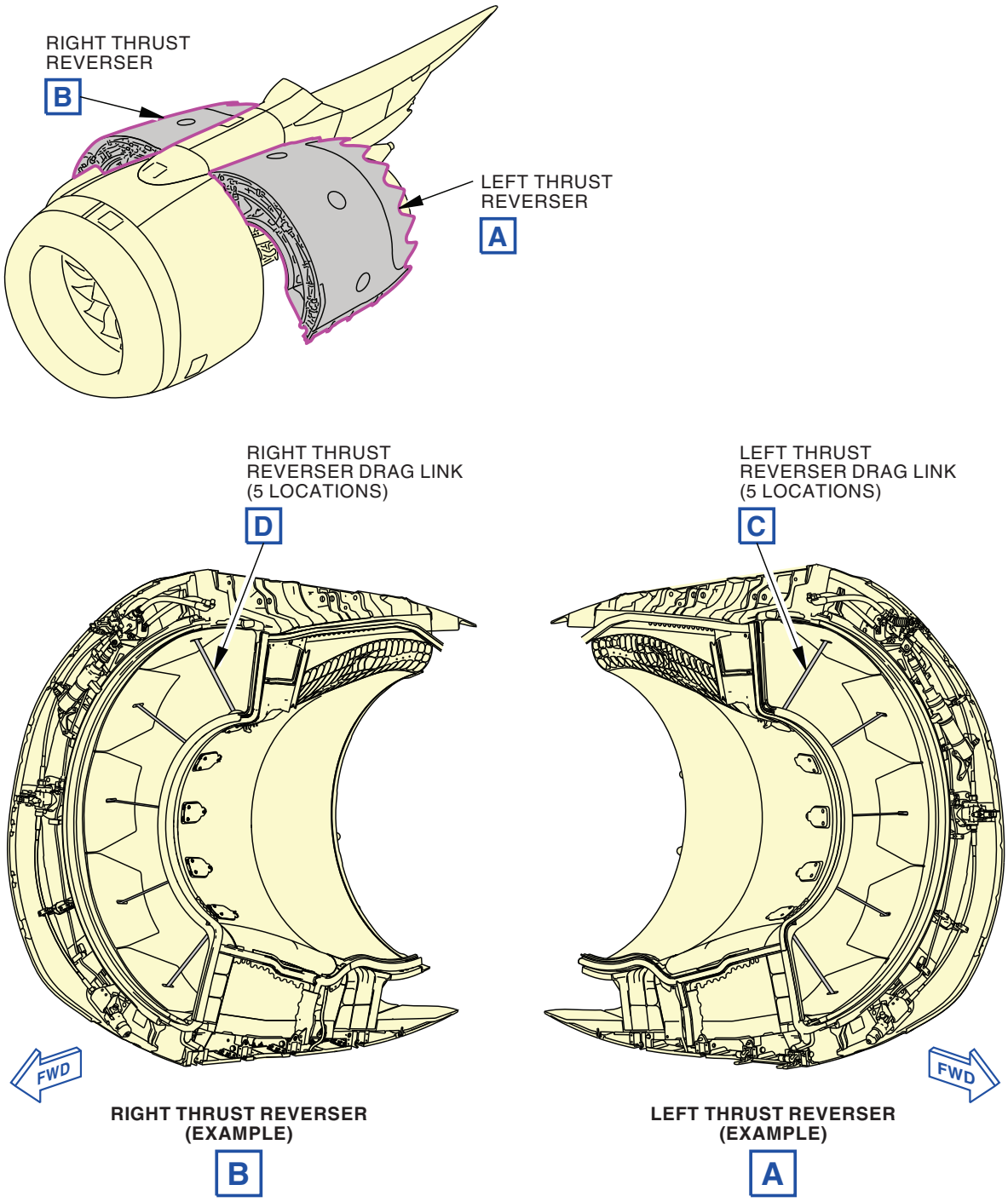
1) For the left thrust reverser, remove the nut [6], washer [2], bushing [14], washer [2], bushing [5], washer [3], and bolt [4] from the pivot link.

2) For the right thrust reverser, remove the nut [6], washer [2], bushing [5], washer [2], bushing [14], washer [3], and bolt [4] from the pivot link.

3) Put a tie strap through the spherical bearing to hold the ball in its position.

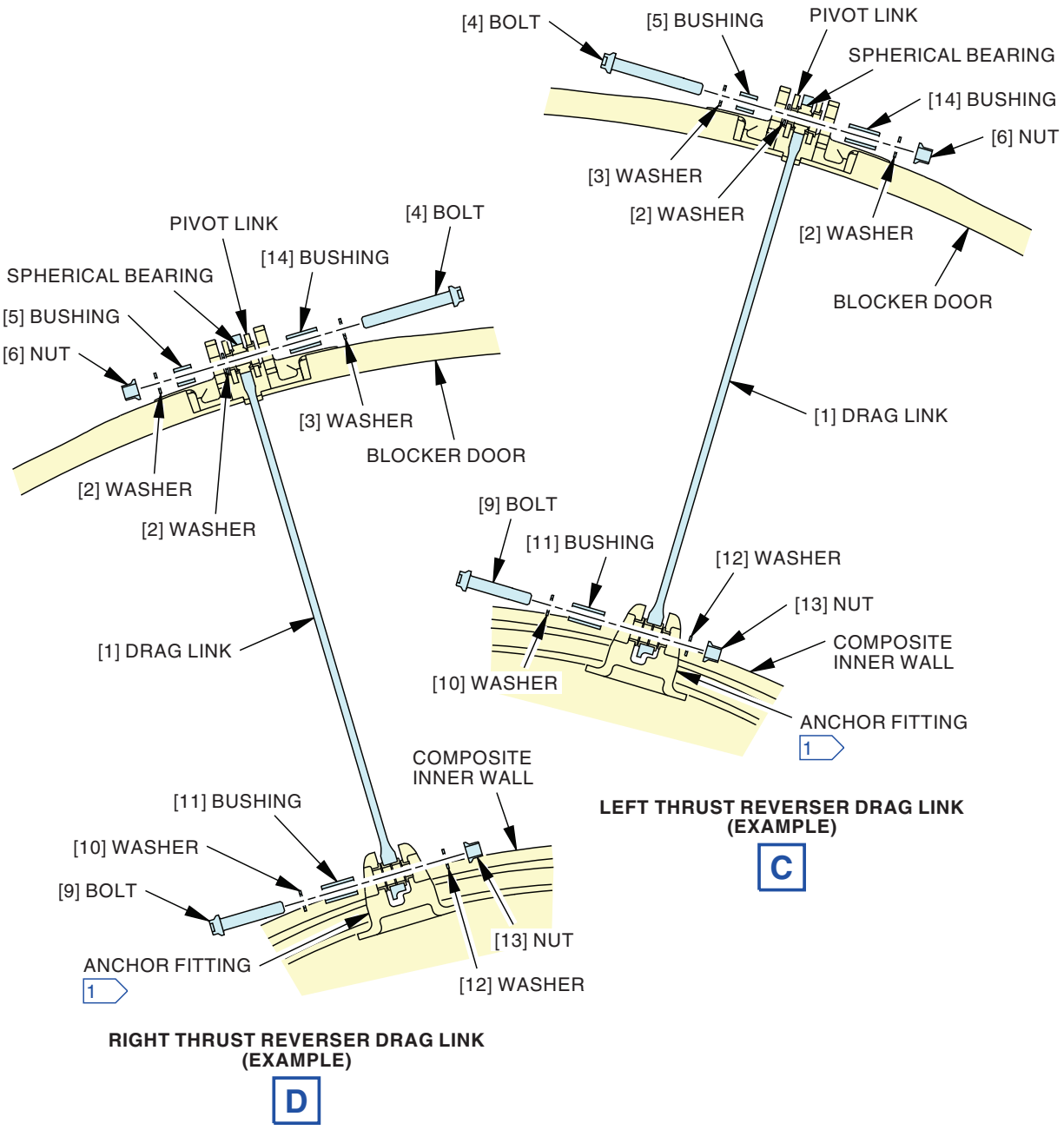
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drag Link Installation
Figure 401/78-31-07-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

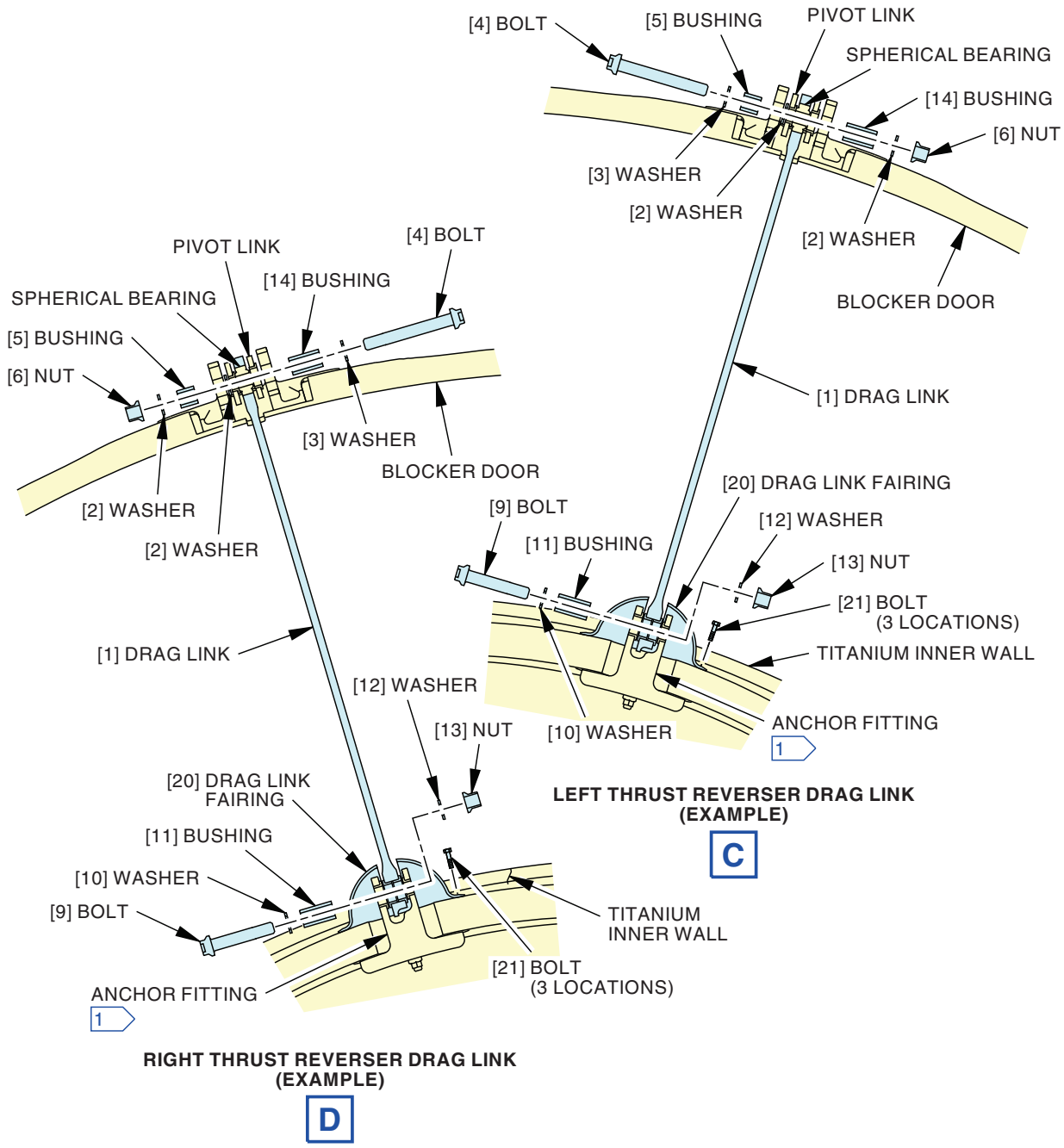


1 WRAP CLOTH AROUND THE ANCHOR FITTING TO PROTECT THE INNER WALL

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Drag Link Installation
Figure 401/78-31-07-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drag Link Installation
Figure 401/78-31-07-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-07-400-801-G00

3. Blocker Door Drag Link Installation

(Figure 401)

A. General

(1) This task is for the installation of the blocker door drag links on the left or right thrust reverser on an engine. The installation is the same for all of the drag links.

(2) For this task the blocker door drag links will be referred to as drag links.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-00-980-805-G00	Thrust Reverser ,Operation - Extend (Power Procedure) (P/B 201)
78-31-00-980-806-G00	Thrust Reverser Operation - Retract (Power Procedure) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02329	Tape - Aluminum Foil, Pressure Sensitive - Vibration Damping Tape 434	

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Blocker Door Drag Link Installation

SUBTASK 78-31-07-420-002-G00

(1) Examine the bottom of the replacement drag link to find the forward edge of the drag link (Figure 401).

NOTE: There is a raised key feature on the side of the drag link that attaches to the drag link anchor fitting. The key only goes across half of the bottom of the drag link. The key moves in a slot in the anchor fitting and should prevent the drag link from being installed in the incorrect direction. The key can be missing from the drag link if key was damaged.

(a) If the key is missing, look at the contours of the leading edge and trailing edge of the drag link and the cut away in the contours near the ends of the drag link (Figure 401).

NOTE: There are flanged bushings installed in the end of the drag link that attaches to the anchor fitting. There is a spherical bearing installed in the end of the drag link that attaches to the pivot fitting in the blocker door.

SUBTASK 78-31-07-000-001-G00

CAUTION

MAKE SURE THAT YOU PUT PROTECTION ON THE FAN DUCT WALLS AND BLOCKER DOORS. DAMAGE CAN OCCUR IF A TOOL OR DRAG LINK HITS THE WALL OR DOORS.

(2) Install the drag link [1], as follows:

(a) Make sure that there is protective mat, STD-585 on the fan duct walls and the blocker doors.

(b) Connect the drag link [1] at the blocker door, as follows:

1) Remove the Vibration Damping Tape 434 tape, G02329, from the blocker door.

2) Remove the tie strap, if you installed it through the spherical bearing to hold the ball in its position.

3) Align the drag link and pivot link bolt holes.

4) For the left thrust reverser, install the bolt [4], washer [3], bushing [5], washer [2], bushing [14], washer [2], and nut [6].

NOTE: The bolt [4] must be installed with the bolt head on the top position.

a) Tighten the nut [6] to 388 in-lb (44 N·m) - 412.0 in-lb (46.5 N·m).

5) For the right thrust reverser, install the bolt [4], washer [3], bushing [14], washer [2], bushing [5], washer [2], and nut [6].

NOTE: The bolt [4] must be installed with the bolt head in the top position.

a) Tighten the nut [6] to 388 in-lb (44 N·m) - 412.0 in-lb (46.5 N·m).

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

(c) Connect the drag link [1] at the inner wall, as follows:

1) Align the drag link [1] and anchor fitting bolt holes.

2) Install the bolt [9], washer [10], bushing [11], washer [12], and nut [13] to the anchor fitting.

NOTE: The bushing [11] only goes in on one side of the anchor fitting.

a) Make sure that the bolt [9] head is on the same side as the bushing [11].

b) Tighten the nut [13] to 388 in-lb (44 N·m) - 412.0 in-lb (46.5 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

(d) Connect the drag link [1] at the inner wall, as follows:

1) Align the drag link [1] and anchor fitting bolt holes.

2) Install the bolt [9], washer [10], bushing [11], washer [12], and nut [13] to the anchor fitting.

NOTE: The bushing [11] only goes in on one side of the anchor fitting.

a) Make sure that the bolt [9] head is on the same side as the bushing [11].

b) Tighten the nut [13] to 388 in-lb (44 N·m) - 412.0 in-lb (46.5 N·m).

3) Install the drag link fairing [20] and the three bolts [21] to the inner wall.

PTF ALL

(e) Remove the cotton wiper, G00034, from the anchor fitting at the inner wall and the protective mat, STD-585 from the fan duct walls and blocker doors.

(f) Manually retract the thrust reverser sleeves (TASK 78-31-00-980-804-G00).

SUBTASK 78-31-07-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Close and latch the thrust reversers (TASK 78-31-00-010-802-G00).

(a) Remove the actuator safety 15 degree lock assembly, SPL-14993, from the inboard thrust reversers only.

(b) Do not do the thrust reverser or leading edge activation.

(c) Do not close the fan cowl panels at this time.

SUBTASK 78-31-07-980-004-G00

(4) Manually translate the sleeve through an extend and retract cycle.

(a) Do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

(b) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

(c) Make sure that the blocker doors and drag links move smoothly.

G. Blocker Door Drag Link Installation Test

SUBTASK 78-31-07-440-003-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-07-710-002-G00

(2) Operate the thrust reverser a minimum of three cycles to make sure that the drag link and blocker door move correctly.

(a) Do this task: Thrust Reverser ,Operation - Extend (Power Procedure), TASK 78-31-00-980-805-G00.

(b) Do this task: Thrust Reverser Operation - Retract (Power Procedure), TASK 78-31-00-980-806-G00.

(c) Make sure that the blocker doors and drag links operate correctly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-07-410-004-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-07-440-004-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER OPENING ACTUATOR - MAINTENANCE PRACTICES

1. General

A. This procedure has one task:

(1) To fill and bleed the thrust reverser opening actuator.

TASK 78-31-08-870-801-G00

2. Fill and Bleed Procedure

(Figure 201)

A. General

(1) This task gives the instructions how to fill and bleed the thrust reverser opening actuator with oil.

(2) This task is necessary when the opening actuator is replaced or when the opening actuator retracts quickly more than 0.5 in. (1.3 cm). The quick retraction indicates that there is air or a vacuum in the opening actuator. It can also indicate damage to the actuator.

B. References

Reference	Title
78-31-08-000-801-G00	Thrust Reverser Opening Actuator Removal (P/B 401)
78-31-08-400-801-G00	Thrust Reverser Opening Actuator Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2417	Pump - Hand, Cowl Opening Part #: A78019-64 Supplier: 81205 Part #: C78005-68 Supplier: 81205 Part #: FT-HDC-101200D Supplier: 46A73

D. Consumable Materials

Reference	Description	Specification
D00068	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699 Class STD (Standard)
D00071	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-7808 Grade 3

E. Fill and Bleed Procedure

SUBTASK 78-31-08-020-001-G00

(1) Do this task: Thrust Reverser Opening Actuator Removal, TASK 78-31-08-000-801-G00.

SUBTASK 78-31-08-020-003-G00

(2) Remove the dust cover [2] from the inlet fitting [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-31-08-610-001-G00

CAUTION

MAKE SURE THAT THE HAND PUMP OR MANUAL PUMP CONTAINS CLEAN OIL WITHOUT CONTAMINATION. IF YOU DO NOT OBEY, OIL CONTAMINATION CAN CAUSE DAMAGE TO EQUIPMENT.

(3) Prepare the cowl opening hand pump, SPL-2417:

NOTE: If available, it is recommended to use C78005-60. This will help to prevent damage to the thrust reverser equipment.

(a) Make sure that the hand pump is full of oil, D00071, or oil, D00068.

(b) Close the return valve on the hand pump.

(c) Connect the hand pump hose to the inlet fitting [1] on the opening actuator.

SUBTASK 78-31-08-870-001-G00

(4) Do these steps to fill and bleed the opening actuator:

(a) Hold the opening actuator vertically so that the inlet fitting [1] is at the top.

(b) Operate the hand pump to fully extend the opening actuator until there is large increase in resistance and operate the hand pump handle one more time through its full travel.

(c) Open the return valve on the pump.

(d) Move the locking collar by pushing towards the rod end bearing to unlock the opening actuator.

(e) Keep the opening actuator vertically with inlet fitting [1] at the top and use hand to push the piston rod in to the fully retracted position.

(f) Immediately close the return valve on the hand pump.

(g) Disconnect the hand pump hose from the inlet fitting [1] on the opening actuator.

SUBTASK 78-31-08-420-003-G00

(5) Install the dust cover [2] on the inlet fitting [1].

SUBTASK 78-31-08-720-001-G00

(6) Do these steps to make sure that there is no air or vacuum in the opening actuator:

(a) Slowly pull the actuator piston rod with your hand until the actuator lock collar engages.

(b) After 60 seconds, move the actuator lock collar to disengage the lock and do these steps:

1) Push the actuator piston rod back to the fully retracted position.

2) Make sure that the actuator piston rod moves smoothly to the fully retracted position.

NOTE: Usually the actuator piston rod will initially move quickly a small amount (less than 0.5 in. (1.3 cm)).

3) If the actuator piston rod quickly retracts more than 0.5 in. (1.3 cm), repeat the fill and bleed procedure.

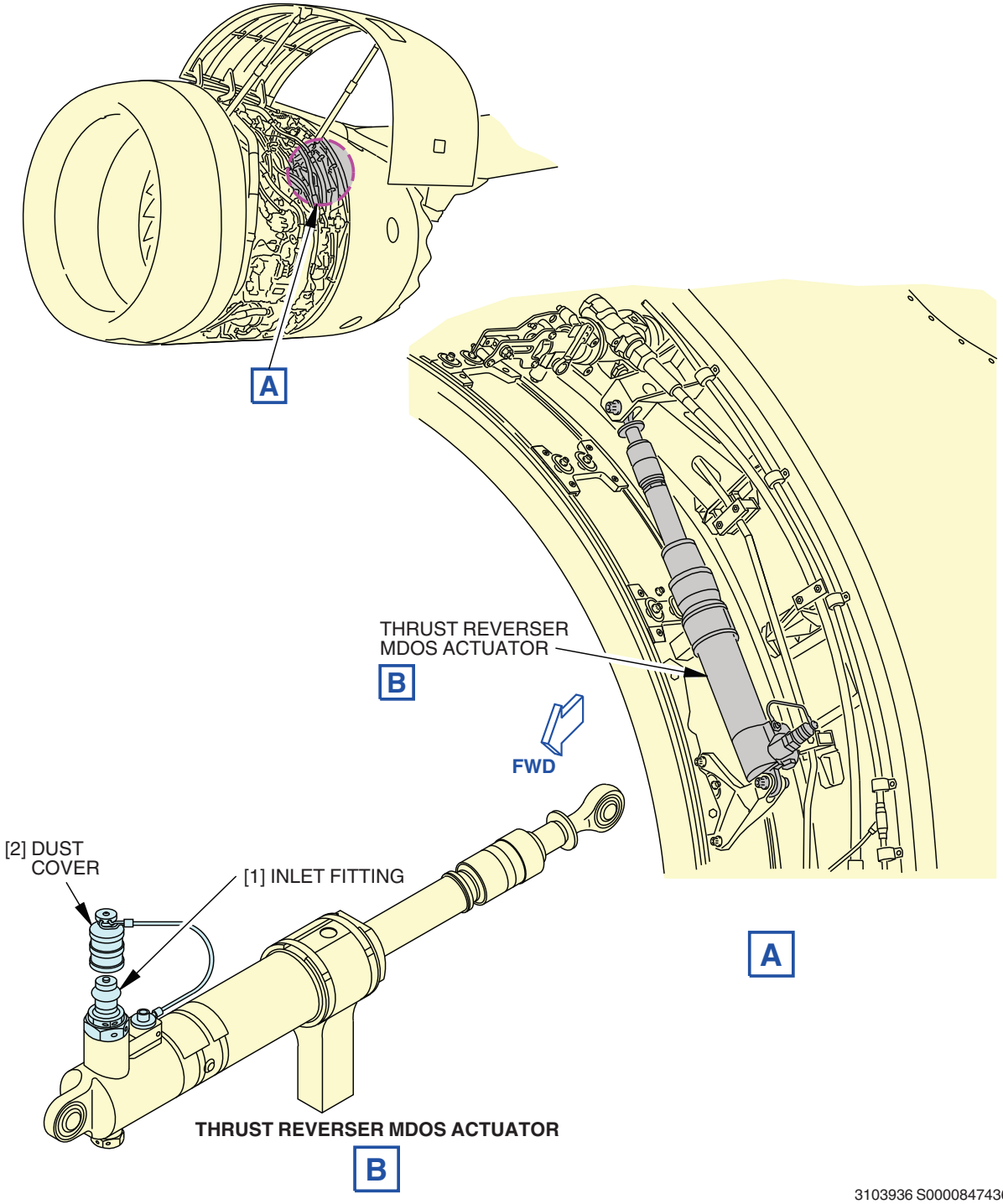
4) If there is still too much movement, replace the opening actuator.

SUBTASK 78-31-08-420-001-G00

(7) Do this task: Thrust Reverser Opening Actuator Installation, TASK 78-31-08-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fill and Bleed Thrust Reverser Opening Actuator
Figure 201/78-31-08-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER OPENING ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the thrust reverser opening actuator

(2) An installation of the thrust reverser opening actuator.

TASK 78-31-08-000-801-G00

2. Thrust Reverser Opening Actuator Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the thrust reverser opening actuator from the left or right thrust reverser on an engine.

(2) In this task, the thrust reverser opening actuator will be referred to as the Manual Door Opening System (MDOS) actuator.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Removal

SUBTASK 78-31-08-040-001-G00

CAUTION

RETRACT THE LEADING EDGE SLATS. DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE SLATS BEFORE YOU OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THESE INSTRUCTIONS, DAMAGE TO THE LEADING EDGE SLATS AND THRUST REVERSERS CAN OCCUR.

(1) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 78-31-08-040-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

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SUBTASK 78-31-08-040-003-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-08-010-001-G00

(4) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

E. Opening Actuator Removal

SUBTASK 78-31-08-020-002-G00

(1) Remove the thrust reverser MDOS actuator [1], do these steps:

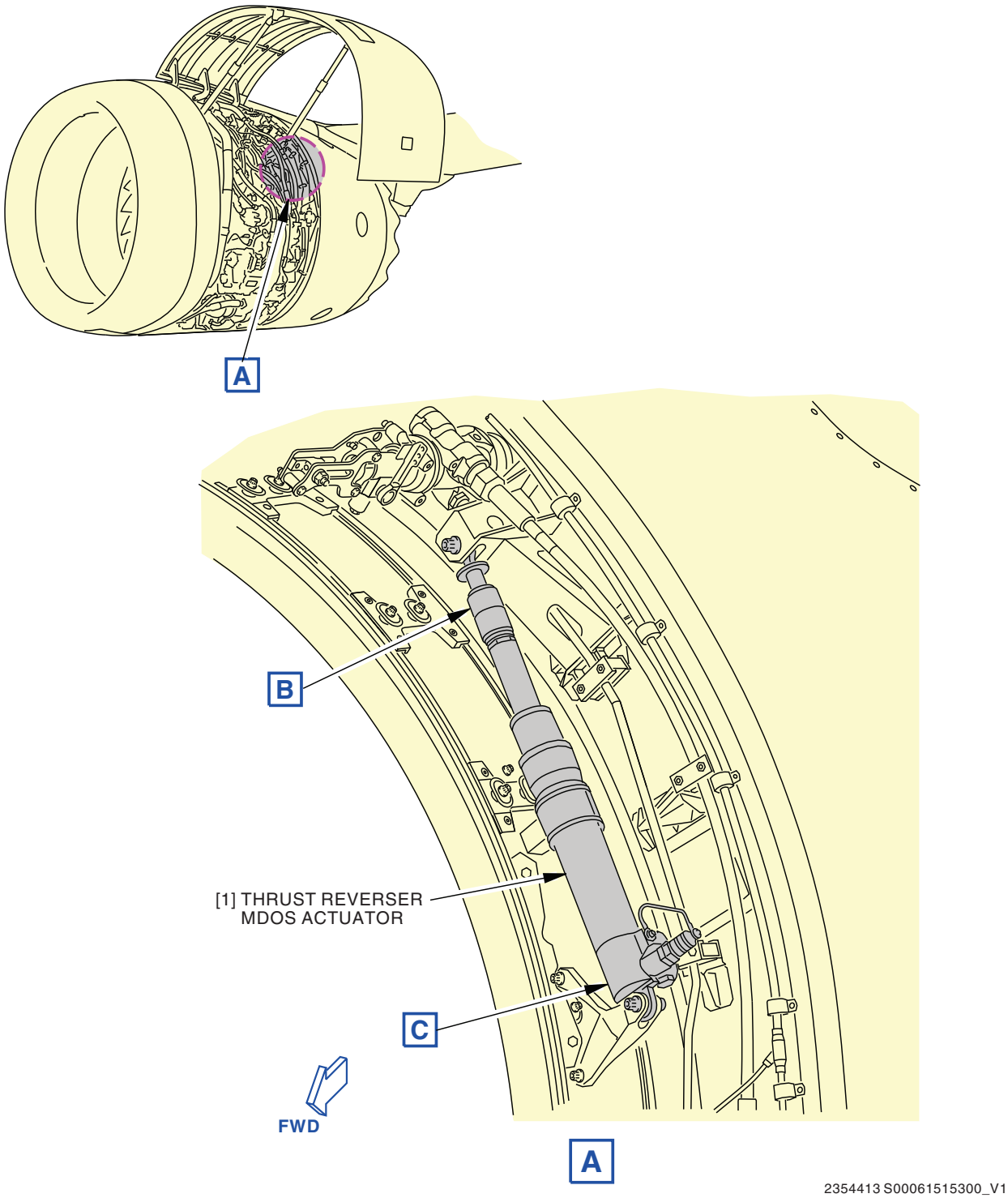
(a) Remove the bolt [7], washer [8], bushing [9], aligment washer [12], washer [10], and nut [11], from the fitting on the fan case.

(b) Remove the bolt [2], washer [3], bushing [4], barrel nut [5], and barrel retainer [6] from the fitting on the thrust reverser torque box.

(c) Remove the thrust reverser MDOS actuator [1].

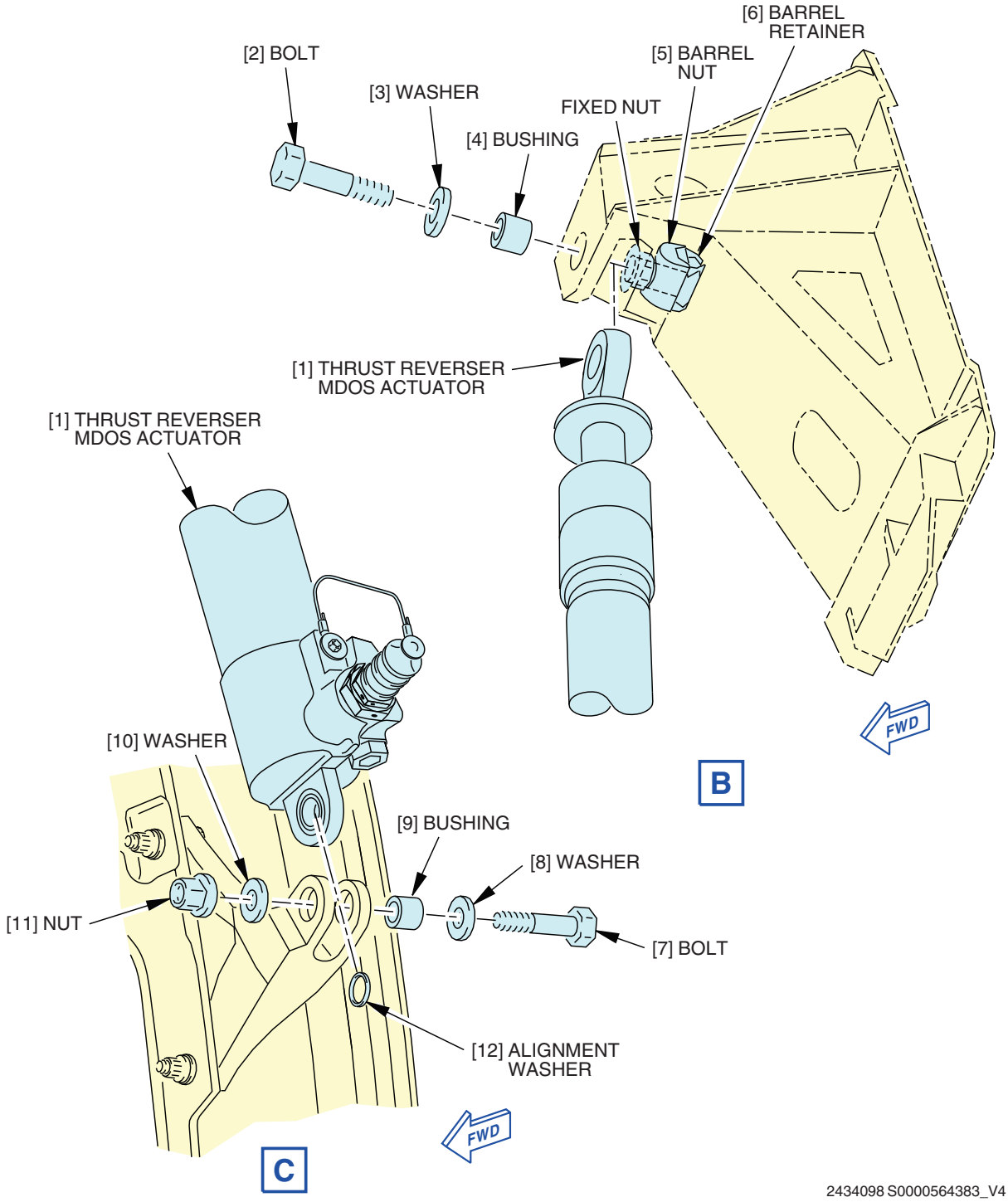
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser MDOS Actuator Installation
Figure 401/78-31-08-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser MDOS Actuator Installation
Figure 401/78-31-08-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-08-400-801-G00

3. Thrust Reverser Opening Actuator Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the thrust reverser opening actuator to the left or right thrust reverser on an engine.

(2) The opening actuators for the inboard and outboard thrust reversers are different part numbers. Make sure that you install the correct actuator for the position.

B. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-08-870-801-G00	Fill and Bleed Procedure (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2417	Pump - Hand, Cowl Opening Part #: A78019-64 Supplier: 81205 Part #: C78005-68 Supplier: 81205 Part #: FT-HDC-101200D Supplier: 46A73

D. Consumable Materials

Reference	Description	Specification
D00068	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-23699 Class STD (Standard)
D00071	Oil - Aircraft Turbine Engine, Synthetic Base	MIL-PRF-7808 Grade 3
D00633	Grease - Aircraft General Purpose	BMS3-33

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Opening Actuator Installation

SUBTASK 78-31-08-870-002-G00

WARNING

DO THIS TASK TO MAKE SURE THAT THERE IS MINIMUM AIR OR A VACUUM IN THE OPENING ACTUATOR. AIR OR A VACUUM IN THE OPENING ACTUATOR COULD CAUSE THE THRUST REVERSER TO CLOSE SUDDENLY WHEN YOU OPEN OR CLOSE THE THRUST REVERSER. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT COULD OCCUR.

(1) If you will replace the thrust reverser MDOS actuator [1], fill and bleed the replacement thrust reverser MDOS actuator [1] before you install it (TASK 78-31-08-870-801-G00).

SUBTASK 78-31-08-420-002-G00

(2) Install the thrust reverser MDOS actuator [1] as follows:

(a) Apply grease, D00633, to the shank of the bolt [2] and bolt [7].

NOTE: Do not get grease on the threads of the bolts.

(b) Align the actuator rod end with the attach fitting on the thrust reverser torque box.

1) Install the bolt [2], washer [3], bushing [4], barrel nut [5], and barrel retainer [6].

a) Do a check of the barrel nut [5] for the minimum breakaway torque (TASK 20-50-11-910-801).

<1> If it is necessary, replace the barrel nut [5] and barrel retainer [6].

b) Tighten the bolt [2] to 340.0 in-lb (38.4 N·m) - 361.0 in-lb (40.8 N·m).

(c) Align the thrust reverser MDOS actuator [1] with the attach fitting on the fan case.

1) Install the bolt [7], washer [8], bushing [9], aligment washer [12], washer [10], and nut [11].

a) Make sure that you install the aligment washer [12] with the teflon surface against the actuator spherical bearing.

NOTE: The alignment washer has a rubber and teflon layer, the thinner layer is teflon with a smoother surface.

b) Tighten the nut [11] to 290.0 in-lb (32.8 N·m) - 310.0 in-lb (35.0 N·m).

G. Opening Actuator Installation Test

SUBTASK 78-31-08-710-001-G00

WARNING

DO NOT OPEN OR CLOSE THE THRUST REVERSER IN HIGH WINDS OR SUDDEN-STRONG-WINDS CONDITIONS. MAKE SURE THAT THE CONSTANT WINDS OR SUDDEN WINDS ARE LESS THAN 40 KNOTS. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING

DO NOT STAY BETWEEN THE THRUST REVERSER AND THE ENGINE WHEN YOU DO THE INSTALLATION TEST FOR THE OPENING ACTUATOR. INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT COULD OCCUR IF THE THRUST REVERSER SUDDENLY CLOSES.

(1) Test the thrust reverser MDOS actuator [1], do these steps:

NOTE: You will open the thrust reverser to the half open position and then lower it. You will open it again to the full open position and the opening actuator locks, and then lower the thrust reverser.

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CAUTION

MAKE SURE THAT THE HAND PUMP OR MANUAL PUMP CONTAINS CLEAN OIL WITHOUT CONTAMINATION. IF YOU DO NOT OBEY, OIL CONTAMINATION CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Make sure that the cowl opening hand pump, SPL-2417, is full of oil, D00071, or oil, D00068.

NOTE: If available, it is recommended to use C78005-60. This will help to prevent damage to the thrust reverser equipment.

(b) Remove the dust cap from the inlet fitting on the thrust reverser MDOS actuator [1].

(c) Connect the hand pump to the inlet fitting.

PTF ALL POST SB 737-78-1095

1) Route the hand pump hose at the MDOS-GSE hose support.

NOTE: The MDOS-GSE hose support is near the top of the oil tank and AGB.

PTF ALL

(d) Cycle the thrust reverser to the half open position as follows:

1) With the return valve on the hand pump closed, operate the hand pump to extend the thrust reverser MDOS actuator [1] and lift the thrust reverser to the half open position.

2) Open the hand pump return valve and let the thrust reverser lower.

(e) Cycle the thrust reverser to the fully open position as follows:

1) With the return valve on the hand pump closed, operate the hand pump to extend the thrust reverser MDOS actuator [1] and lift the thrust reverser to the fully open position.

NOTE: Listen for the click sound of the lock collar.

2) These are the indications that the thrust reverser is in the fully open position and thrust reverser MDOS actuator [1] is locked:

a) Make sure that the word LOCKED shows on the bottom of the extended piston.

b) Make sure that you can see the red band on the actuator rod.

3) Open the return valve on the hand pump to let the weight of the thrust reverser rest on the locked thrust reverser MDOS actuator [1].

WARNING

MAKE SURE THAT THE RETURN VALVE ON THE HAND PUMP IS CLOSED WHEN THE THRUST REVERSER IS HELD BY THE LOCKED OPENING ACTUATOR. IF THE RETURN VALVE IS LEFT OPEN, THE ENGINE OIL CAN CONTINUE TO DRAIN OUT OF THE ACTUATOR INTO THE HAND PUMP. DECREASED ENGINE OIL CAN PERMIT THE THRUST REVERSER TO CLOSE QUICKLY DURING SUBSEQUENT OPERATIONS TO CLOSE THE THRUST REVERSER. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

4) After the thrust reverser rests on the thrust reverser MDOS actuator [1], close the return valve on the hand pump.

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WARNING

ALWAYS EXTEND THE OPENING ACTUATOR TO LIFT THE WEIGHT OF THE THRUST REVERSER OFF THE OPENING ACTUATOR LOCK. DO NOT OPEN THE RETURN VALVE ON THE HAND PUMP UNTIL YOU DISENGAGE THE ACTUATOR LOCK COLLAR. THIS WILL PREVENT INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

5) With the return valve closed on the hand pump, operate the hand pump to release the load on the thrust reverser MDOS actuator [1].
6) Push up on the actuator lock collar to disengage the lock.
7) Open the hand pump return valve and let the thrust reverser lower.
8) Make sure that the thrust reverser closes fully.
9) Disconnect the hand pump from the thrust reverser MDOS actuator [1].
10) Install dust caps on the inlet fitting on the thrust reverser MDOS actuator [1] and hand pump hose.
11) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-08-040-004-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-08-040-005-G00

(2) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-08-040-006-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

KRUEGER FLAP DEFLECTOR AND PLUG - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Krueger flap deflector removal

(2) Krueger flap deflector installation.

TASK 78-31-09-010-801-G00

2. Krueger Flap Deflector Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Krueger flap deflector from the translating sleeve on the inboard thrust reverser on an engine.

(2) Make sure that the Krueger flap deflectors are installed only on the inboard translating sleeve.

(3) It is possible to install the Krueger flap deflector on the left or right translating sleeve.

(4) On the translating sleeve without the Krueger flap deflectors, plugs are installed in the mounting holes.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Prepare for the Removal

SUBTASK 78-31-09-010-001-G00

WARNING OBEY ALL OF THE WARNINGS AND CAUTIONS IN THE REFERENCED PROCEDURE. IF YOU DO NOT OBEY THE WARNINGS AND CAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

SUBTASK 78-31-09-040-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

E. Krueger Flap Deflector Removal

SUBTASK 78-31-09-020-001-G00

(1) Do these steps to remove the applicable Krueger flap deflector [1]:

NOTE: Be sure not to drop hardware, bolts, washers or nuts into the thrust reverser structure or through the cascade vanes.

You get access to these bolts from the inside wall of the outer cowl.

(a) Remove the 12 bolts [2] that attach the upper Krueger flap deflector [1] to the self-locking nuts on the outer cowl.

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(b) Remove the eight bolts [3] that attach the lower Krueger flap deflector [1] to the self-locking nuts on the outer cowl.

(c) Remove the Krueger flap deflector [1].

SUBTASK 78-31-09-020-004-G00

(2) If the Krueger flap deflector [1] is not to be replaced on this translating sleeve, do the steps that follow:

(a) Make sure that the surfaces of the translating sleeve and the Krueger flap deflector [1] holes are clean.

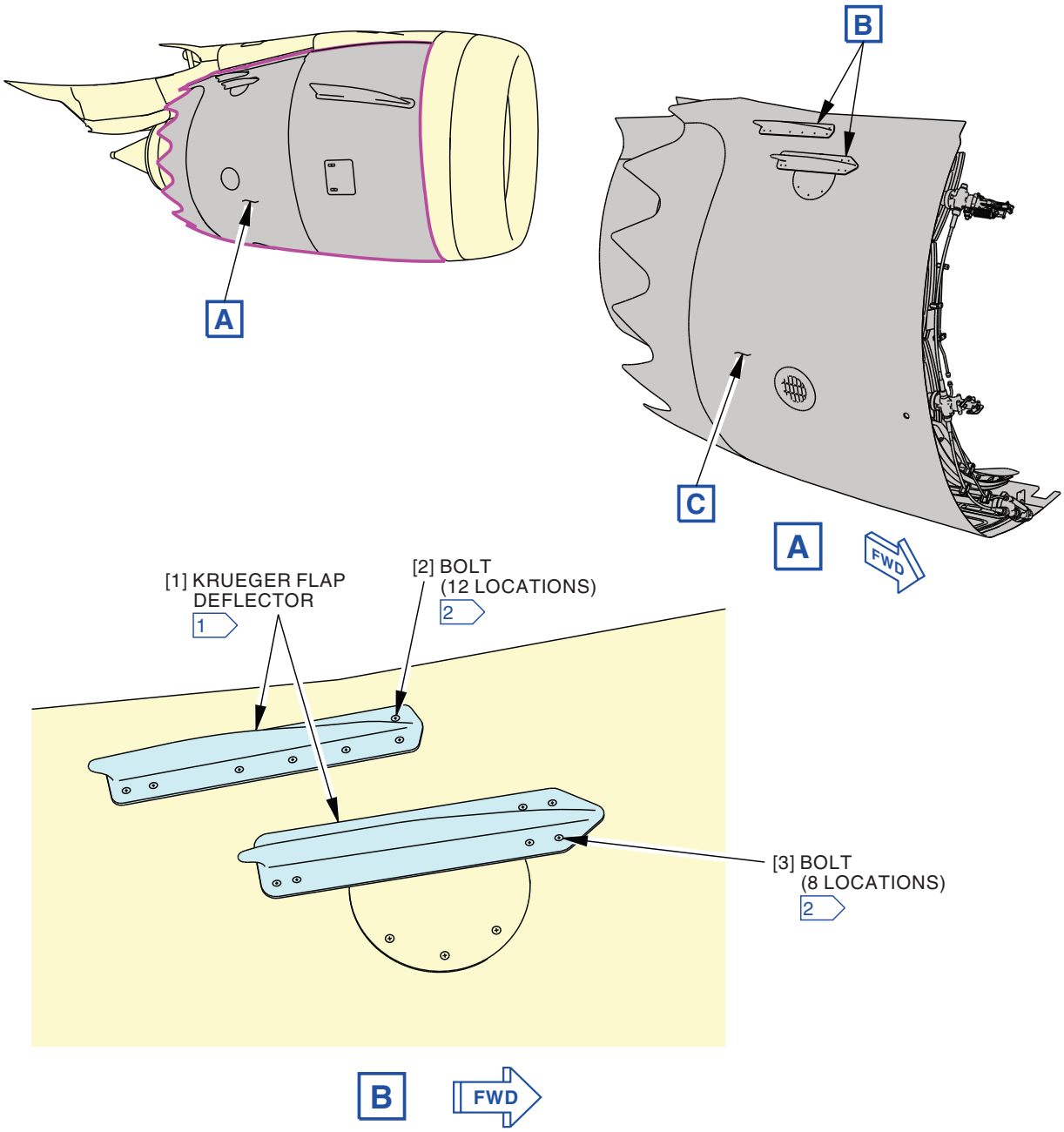
(b) Install the plugs [4] and plugs [5] into the open holes on the skin.

(c) Tighten the plug [4] and plug [5] to 60 ±5 in-lb (6.8 ±0.6 N·m).

(d) Apply the corrosion inhibiting material, G50136, to the plug fasteners.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



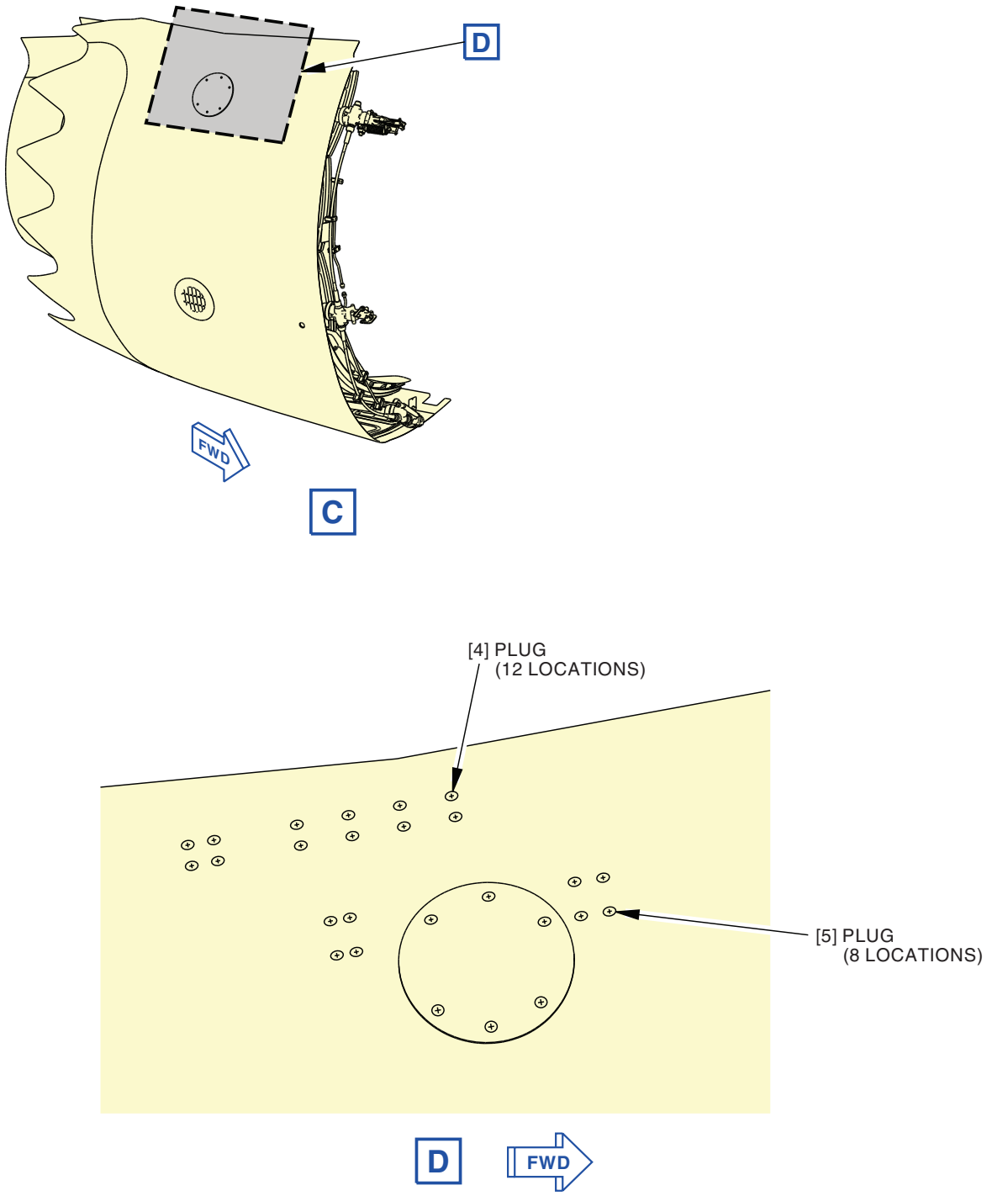
NOTE:

1. THE KRUEGER FLAP DEFLECTORS ARE INSTALLED ON THE INBOARD SIDE OF THE THRUST REVERSER. THE KRUEGER FLAP DEFLECTORS ARE INTERCHANGEABLE ON THE LEFT OR RIGHT TRANSLATING SLEEVE.
2. DESIGNATED BOND FASTENER RESISTANCE CHECK.

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Krueger Flap Deflector Installation
Figure 401/78-31-09-990-801-G00 (Sheet 1 of 2)

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Krueger Flap Deflector Installation
Figure 401/78-31-09-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-09-420-801-G00

3. Krueger Flap Deflector Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Krueger flap deflector to the translating sleeve on the inboard thrust reverser on an engine.

(2) Make sure that the Krueger flap deflectors are installed only on the inboard translating sleeve.

(3) It is possible to install the Krueger flap deflector on the left or right translating sleeve.

(4) On the translating sleeve without the Krueger flap deflectors, plugs are installed in the mounting holes.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
51-22-00-160-801	Prepare For Sealing (P/B 201)
51-22-00-390-802	Removable Faying (Mated) Surface Seal Application (P/B 201)
78-31-00-980-802-G00	Thrust Reverser Operation - Retract (Selection) (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

D. Consumable Materials

Reference	Description	Specification
A00436	Sealant - Fuel Tank	BMS5-45 (Supersedes BMS5-26)
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

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E. Prepare for the Installation

SUBTASK 78-31-09-010-002-G00

(1) If the translating sleeve had the plug fasteners installed, remove the plug fasteners as follows:

NOTE: Be sure not to drop hardware, bolts, washers or nuts into the thrust reverser structure or through the cascade vanes.

(a) Remove the plugs [4] and plugs [5].

SUBTASK 78-31-09-390-001-G00

(2) Do this task: Prepare For Sealing, TASK 51-22-00-160-801.

(a) Make sure that the mating surfaces of the outer cowl and the Krueger flap deflector [1] are clean.

F. Krueger Flap Deflector Installation

SUBTASK 78-31-09-420-003-G00

(1) Do these steps to install the applicable Krueger flap deflector [1]:

NOTE: Be sure not to drop hardware, bolts, washers or nuts into the thrust reverser structure or through the cascade vanes.

(a) Put the Krueger flap deflector [1] in the correct position on the outer cowl.

(b) Put Scotch Flatback Masking Tape 250, G00270, on the outer cowl around the periphery of the Krueger flap deflector [1].

(c) Remove the Krueger flap deflector [1].

(d) Apply faying surface seal with corrosion inhibiting material, G50136, to the outer cowl surface in the area where the Krueger flap deflector [1] will be installed (TASK 51-22-00-390-802).

(e) Remove the Scotch Flatback Masking Tape 250, G00270.

(f) Apply sealant, A00436, to the surfaces of the Krueger flap deflector [1] that will be against the outer cowl.

(g) Put the Krueger flap deflector [1] in the correct position on the outer cowl.

1) Apply pressure to the Krueger flap deflector [1] until the sealant pushes out around the deflector.

a) Make sure that the sealant pushes out around the entire periphery of the Krueger flap deflector [1].

(h) Apply the corrosion inhibiting material, G50136, to the shank and the threads of the bolts.

(i) Install the 12 bolts [2] that attach the upper Krueger flap deflector [1] to the self-locking nuts on the outer cowl.

1) Tighten the bolts [2] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

(j) Install the eight bolts [3] that attach the lower Krueger flap deflector [1] to the self-locking nuts on the outer cowl.

1) Tighten the bolts [3] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

(k) Measure the electrical bonding resistance between the fastener head and Krueger flap deflector [1] (SWPM 20-20-00).

1) Use the non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.005 ohm (5.0 milliohm) or less.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-09-440-001-G00

(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 78-31-09-410-001-G00

WARNING

OBEY ALL OF THE WARNINGS AND CAUTIONS IN THE REFERENCED PROCEDURE. IF YOU DO NOT OBEY THE WARNINGS AND CAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Thrust Reverser Operation - Retract (Selection), TASK 78-31-00-980-802-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

KRUEGER FLAP DEFLECTOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Krueger flap deflector inspection.

TASK 78-31-09-200-801-G00

2. Krueger Flap Deflector Inspection

A. General

(1) This task provides the instructions on how to do a visual inspection of the Krueger flap deflector that is on the inboard thrust reverser sleeve at the top of the sleeve.

(2) If the damage is more than the specified limits, the Krueger flap deflector must be replaced.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-09-010-801-G00	Krueger Flap Deflector Removal (P/B 401)
78-31-09-420-801-G00	Krueger Flap Deflector Installation (P/B 401)

C. Prepare for the Inspection

SUBTASK 78-31-09-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

D. Krueger Flap Deflector Inspection

SUBTASK 78-31-09-210-001-G00

(1) Do a check of the Krueger flap deflector for obvious damage:

(a) Scratches, gouges and worn areas are permitted if the depth of the damage is not more than 0.050 in. (1.27 mm).

(b) If the depth of the damage is more than 0.050 in. (1.27 mm), replace the Krueger flap deflector.

These are the tasks:

- Krueger Flap Deflector Removal, TASK 78-31-09-010-801-G00
- Krueger Flap Deflector Installation, TASK 78-31-09-420-801-G00.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-09-410-002-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUBSTRIP - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Rubstrip removal

(2) Rubstrip installation.

TASK 78-31-10-000-801-G00

2. Rubstrip Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the rubstrips from the torque box on the left or right thrust reverser on an engine.

(2) The forward rubstrips interface with the fan cowl panel and the aft rubstrips interface with the thrust reverser translating sleeve.

(3) For this procedure, the rubstrips and shims will be referred to as follows:

Table 401/78-31-10-993-801-G00

RUBSTRIP NUMBER	INTERFACES WITH	REFERRED TO AS
1 1A	Fan Cowl Panel	Forward-Upper Rubstrip Forward-Upper Shim
2 2A	Fan Cowl Panel	Forward-Mid Rubstrip Forward-Mid Shim
3 3A	Fan Cowl Panel	Forward-Lower Rubstrip Forward-Lower Shim
4 4A	Fan Cowl Panel	Forward-Bottom Rubstrip Forward-Bottom Shim
5 5A	Fan Cowl Panel	Forward-Top Rubstrip Forward-Top Shim
6 6A	Fan Cowl Panel	Forward-Bottom Beam Rubstrip Forward-Bottom Beam Shim
7 7A	Translating Sleeve	Aft-Upper Rubstrip Aft-Upper Shim
8 8A	Translating Sleeve	Aft-Upper-Mid Rubstrip Aft-Upper-Mid Shim
9 9A	Translating Sleeve	Aft-Mid Rubstrip Aft-Mid Shim
10 10A	Translating Sleeve	Aft-Lower-Mid Rubstrip Aft-Lower-Mid Shim
11 11A	Translating Sleeve	Aft-Upper-Lower Rubstrip Aft-Upper-Lower Shim

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Table 401/78-31-10-993-801-G00 (Continued)

RUBSTRIP NUMBER	INTERFACES WITH	REFERRED TO AS
12 12A	Translating Sleeve	Aft-Lower Rubstrip Aft-Lower Shim
13 13A	Translating Sleeve	Aft-Bottom Rubstrip Aft-Bottom Shim
14 14A	Translating Sleeve	Aft Guide Block Aft Guide Shim

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)

C. Prepare the Removal

SUBTASK 78-31-10-040-001-G00

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-10-010-001-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

SUBTASK 78-31-10-840-001-G00

(3) Manually extend the thrust reverser sleeve approximately 4 in. (10 cm) to expose the aft rubstrips, do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

D. Forward Rubstrip Removal

SUBTASK 78-31-10-020-022-G00

(1) For all of the forward rubstrips, make sure that you keep the shims for the subsequent installation.

NOTE: If the original shim is used with the replacement rubstrip, there will be less adjustment to get the necessary rubstrip height dimensions.

SUBTASK 78-31-10-020-023-G00

(2) To remove the forward-top rubstrip [5], do these steps (View C):

(a) Remove all of the sealant that bonds the forward-top rubstrip [5] to the hinge beam.

(b) Remove the three rivets [15].

NOTE: Use the standard overhaul practices to remove the rivets.

(c) Remove the forward-top rubstrip [5] and forward-top shim [5A].

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1) Keep the forward-top shim [5A] for the subsequent installation.

SUBTASK 78-31-10-020-024-G00

(3) To remove the forward-upper rubstrip [1], do these steps (View C):

(a) Remove the four bolts [16], washers [17], and nuts [18] from the four upper locations.

(b) Remove the remaining 14 bolts [19], washers [17], and nuts [18].

(c) Remove the forward-upper rubstrip [1] and forward-upper shim [1A].

1) Keep the forward-upper shim [1A] for the subsequent installation.

SUBTASK 78-31-10-020-025-G00

(4) To remove the forward-mid rubstrip [2], do these steps (View D):

(a) Remove the 21 bolts [19], washers [17], and nuts [18].

(b) Remove the two bolts [20], washers [17], and nuts [18].

(c) Remove the forward-mid rubstrip [2] and forward-mid shim [2A].

1) Keep the forward-mid shim [2A] for the subsequent installation.

SUBTASK 78-31-10-020-026-G00

(5) To remove the forward-lower rubstrip [3], do these steps (View E):

(a) Remove the 18 bolts [19], washers [17], and nuts [18].

(b) Remove the two bolts [20], washers [17], and nuts [18].

(c) Remove the forward-lower rubstrip [3] and forward-lower shim [3A].

1) Keep the forward-lower shim [3A] for the subsequent installation.

SUBTASK 78-31-10-020-027-G00

(6) To remove the forward-bottom rubstrip [4], do these steps (View F):

(a) Remove the 11 bolts [19], washers [17], and nuts [18].

(b) Remove the forward-bottom rubstrip [4] and forward-bottom shim [4A].

1) Keep the forward-bottom shim [4A] for the subsequent installation.

SUBTASK 78-31-10-020-028-G00

(7) To remove the forward-bottom beam rubstrip [6], do these steps (View F):

(a) Remove all of the sealant that bonds the forward-bottom beam rubstrip [6] and the aft-bottom rubstrip [13] to the latch beam.

(b) Remove the eight bolts [16], washers [17], and nuts [18] from the eight upper locations.

(c) Remove the remaining four bolts [19], washers [17], and nuts [18].

(d) Remove the forward-bottom beam rubstrip [6] and forward-bottom beam shim [6A].

1) Keep the forward-bottom beam shim [6A] for the subsequent installation.

E. Aft Rubstrip Removal

SUBTASK 78-31-10-020-007-G00

(1) For all of the aft rubstrips, make sure that you keep the shims for the subsequent installation.

NOTE: If the original shim is used with the replacement rubstrip, there will be less adjustment to get the necessary rubstrip height dimensions.

SUBTASK 78-31-10-020-008-G00

(2) To remove the aft-upper rubstrip [7], do these steps (View C):

(a) Remove the nine bolts [21], washers [22], and nuts [23].

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(b) Remove the aft-upper rubstrip [7] and aft-upper shim [7A].

1) Keep the aft-upper shim [7A] for the subsequent installation.

SUBTASK 78-31-10-020-009-G00

(3) To remove the aft-upper-mid rubstrip [8], do these steps (View C):

(a) Remove the 11 bolts [21], washers [22], and nuts [23].

(b) Remove the aft-upper-mid rubstrip [8] and aft-upper-mid shim [8A].

1) Keep the aft-upper-mid shim [8A] for the subsequent installation.

SUBTASK 78-31-10-020-010-G00

(4) To remove the aft-mid rubstrip [9], do these steps (View D):

(a) Remove the 10 bolts [23], washers [24], and nuts [25].

(b) Remove the aft-mid rubstrip [9] and aft-mid shim [9A].

1) Keep the aft-mid shim [9A] for the subsequent installation.

SUBTASK 78-31-10-020-011-G00

(5) To remove the aft-lower-mid rubstrip [10], do these steps (View D):

(a) Remove the 10 bolts [23], washers [24], and nuts [25].

(b) Remove the four bolts [24], washers [22], and nuts [23].

(c) Remove the aft-lower-mid rubstrip [10] and aft-lower-mid shim [10A].

1) Keep the aft-lower-mid shim [10A] for the subsequent installation.

SUBTASK 78-31-10-020-012-G00

(6) To remove the aft-upper-lower rubstrip [11], do these steps (View E):

(a) Remove the seven bolts [21], washers [22], and nuts [23].

(b) Remove the aft-upper-lower rubstrip [11] and aft-upper-lower shim [11A].

1) Keep the aft-upper-lower shim [11A] for the subsequent installation.

SUBTASK 78-31-10-020-021-G00

(7) To remove the aft-lower rubstrip [12], do these steps (View F):

(a) Remove the four bolts [21], washers [22], and nuts [23].

(b) Remove the aft-lower rubstrip [12] and aft-lower shim [12A].

1) Keep the aft-lower shim [12A] for the subsequent installation.

SUBTASK 78-31-10-020-013-G00

(8) To remove the aft-bottom rubstrip [13], do these steps (View F):

(a) Remove the 11 bolts [21], washers [22], and nuts [23].

(b) Remove the aft-bottom rubstrip [13] and aft-bottom shim [13A].

1) Keep the aft-bottom shim [13A] for the subsequent installation.

SUBTASK 78-31-10-020-014-G00

(9) To remove the aft guide block [14], do these steps (View C, D, E, and F):

NOTE: There are five aft guide blocks on each of the left and right thrust reversers on an engine.

(a) Remove the two bolts [21], washers [22], and nuts [23].

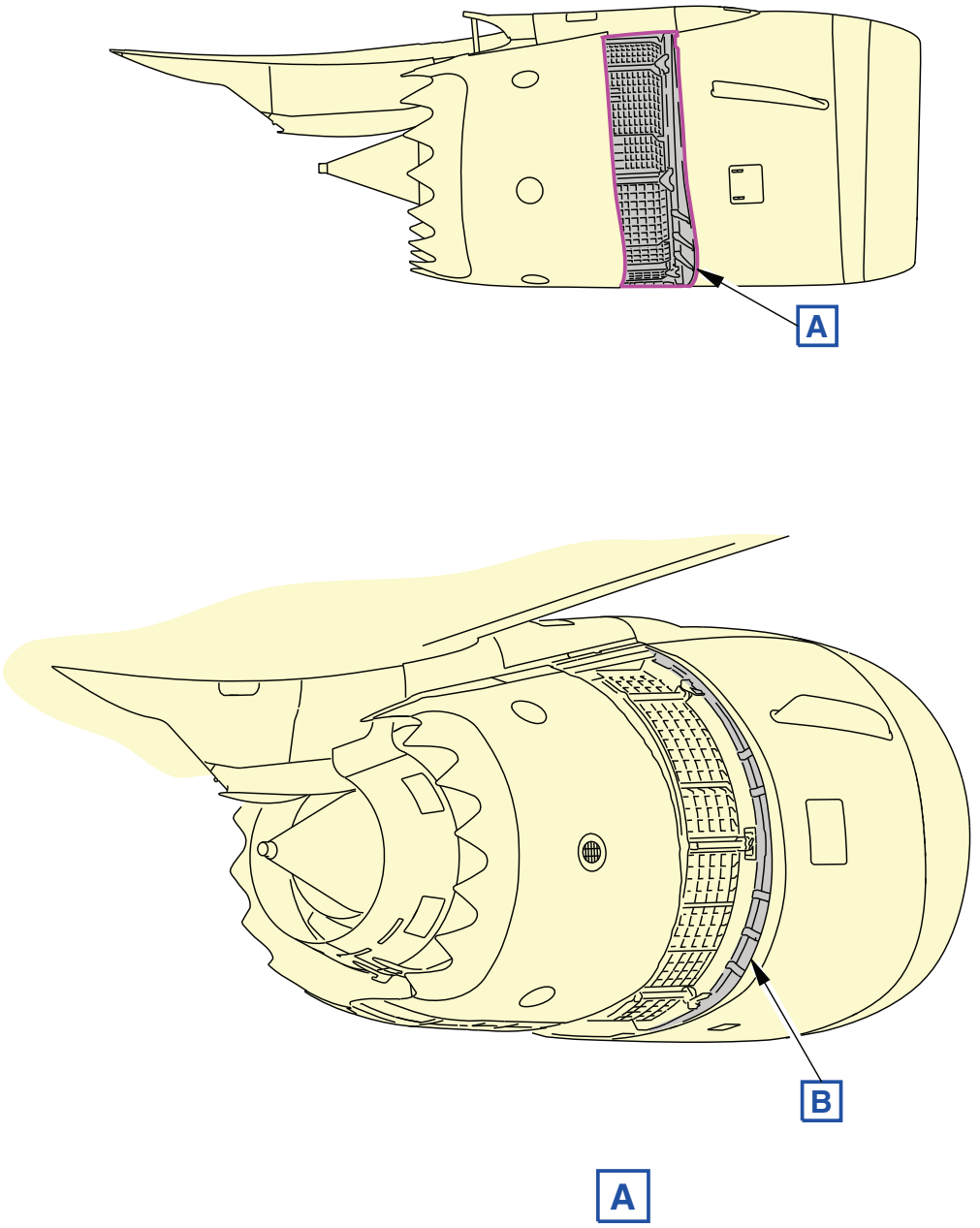
(b) Remove the aft guide block [14] and aft guide shim [14A].

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1) Keep the aft guide shim [14A] for the subsequent installation.

——— END OF TASK ———

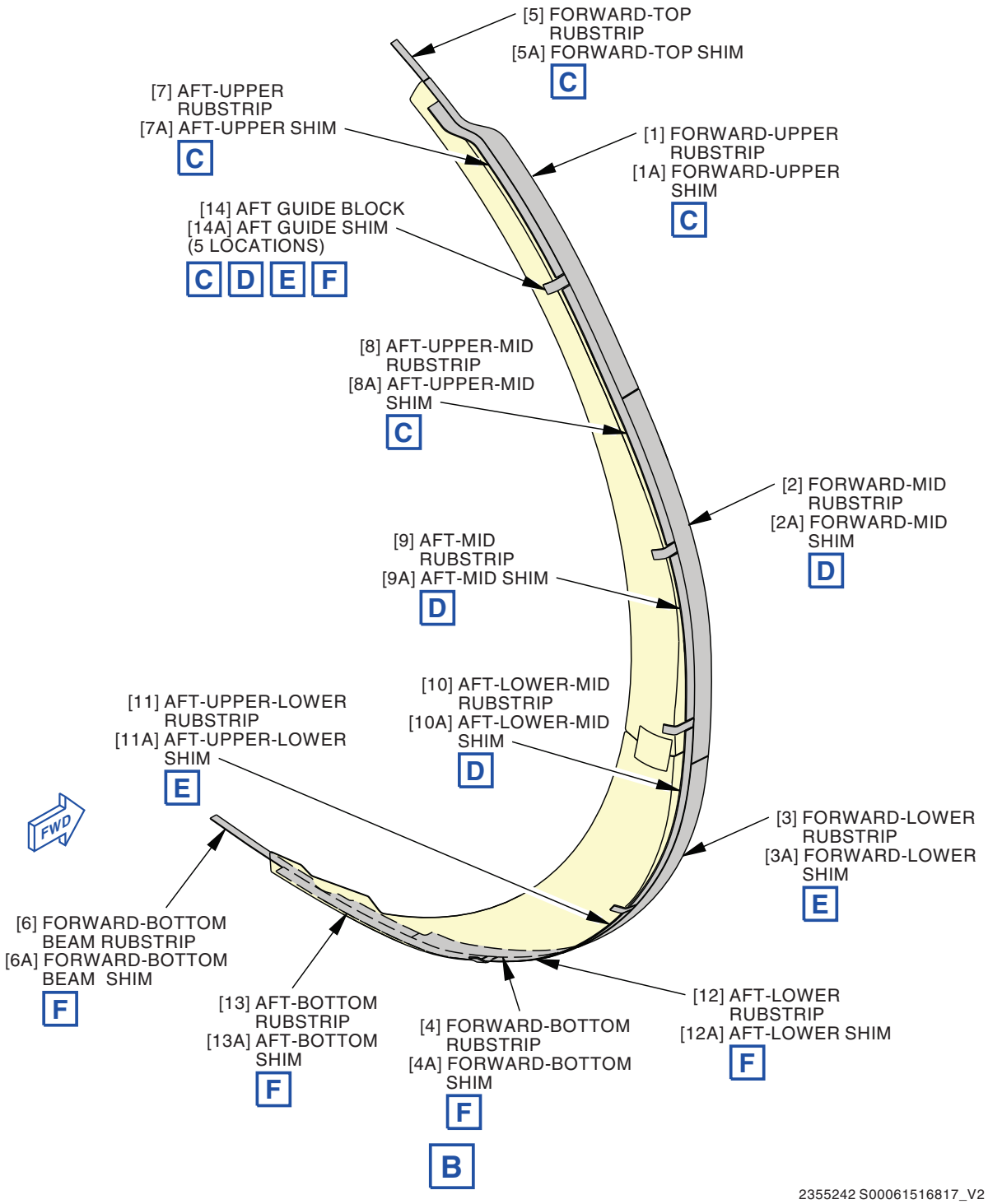
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2355241 S00061516816_V1

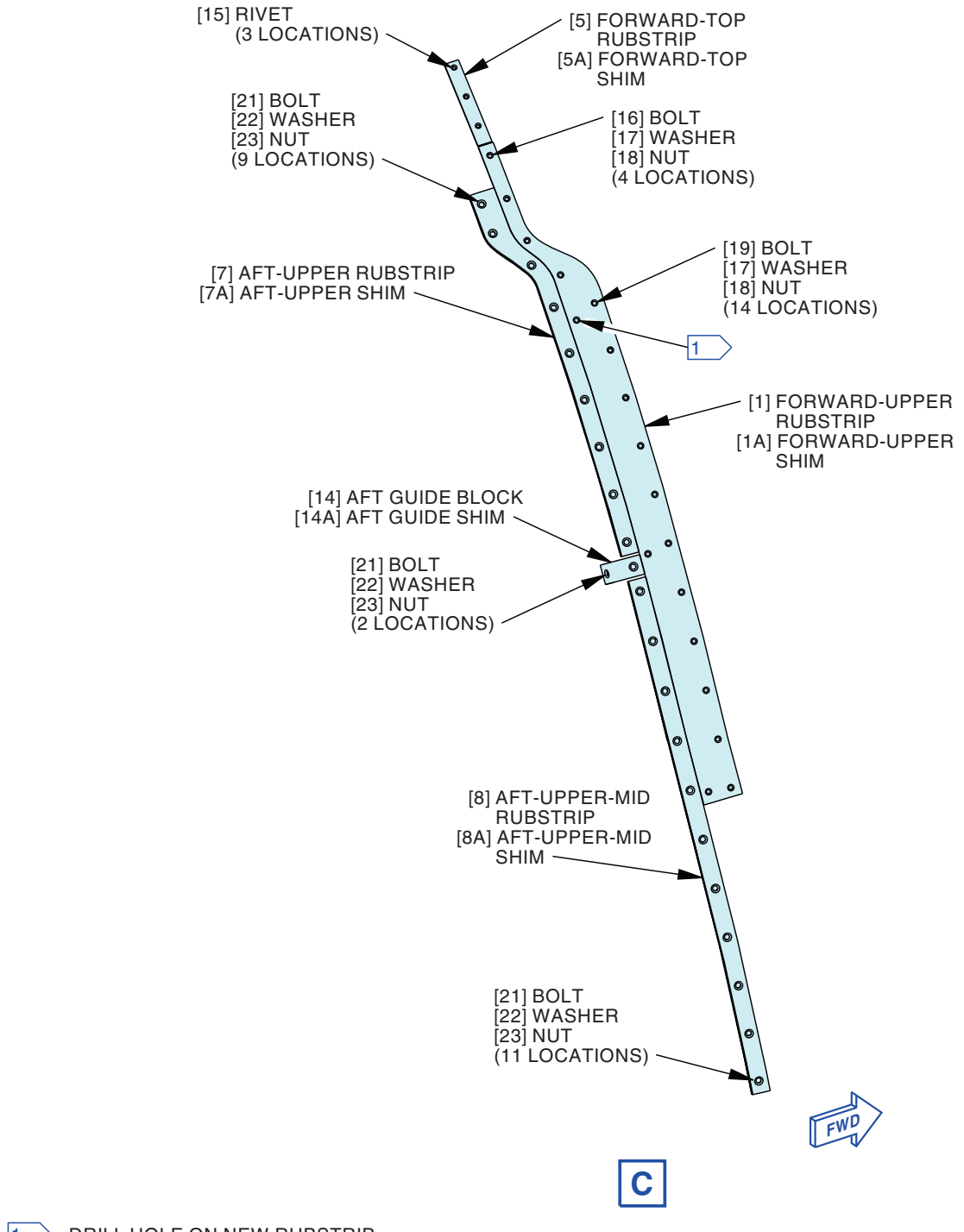
Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 1 of 6)

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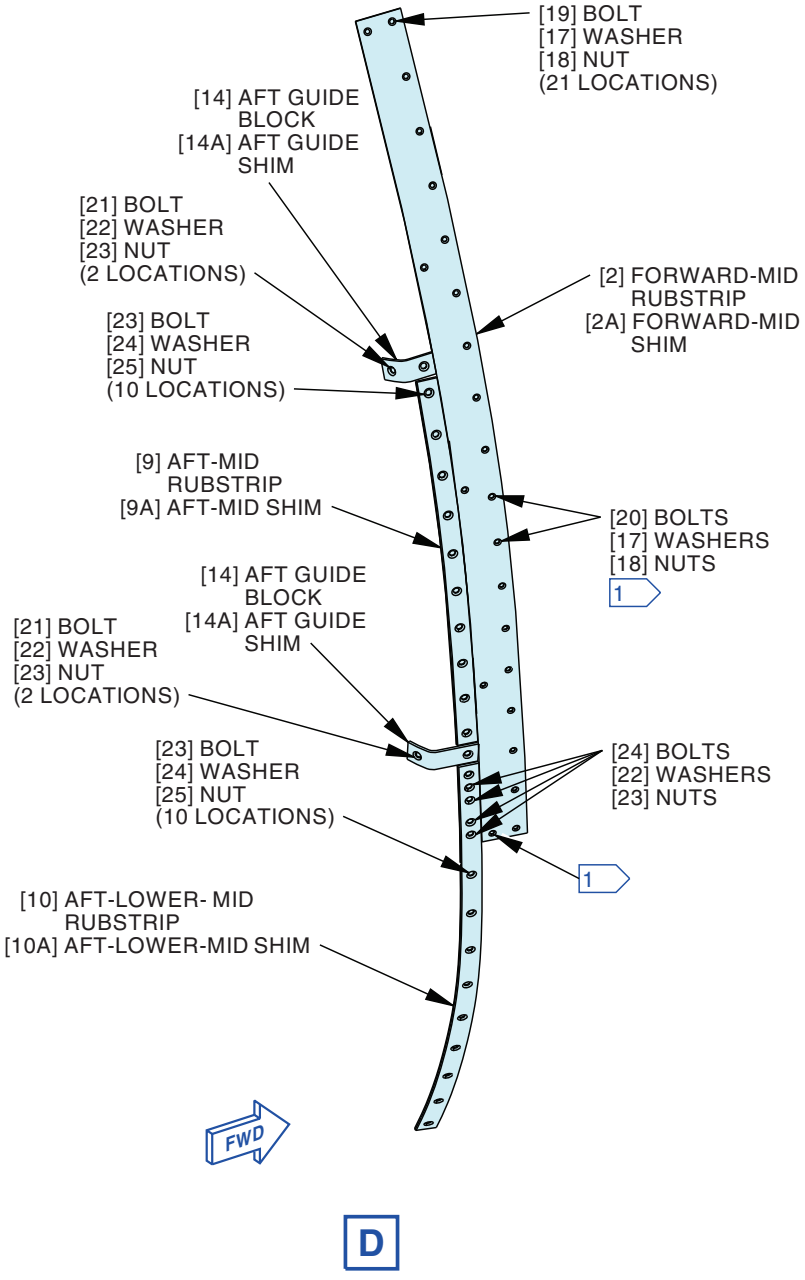
Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 2 of 6)

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Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 3 of 6)

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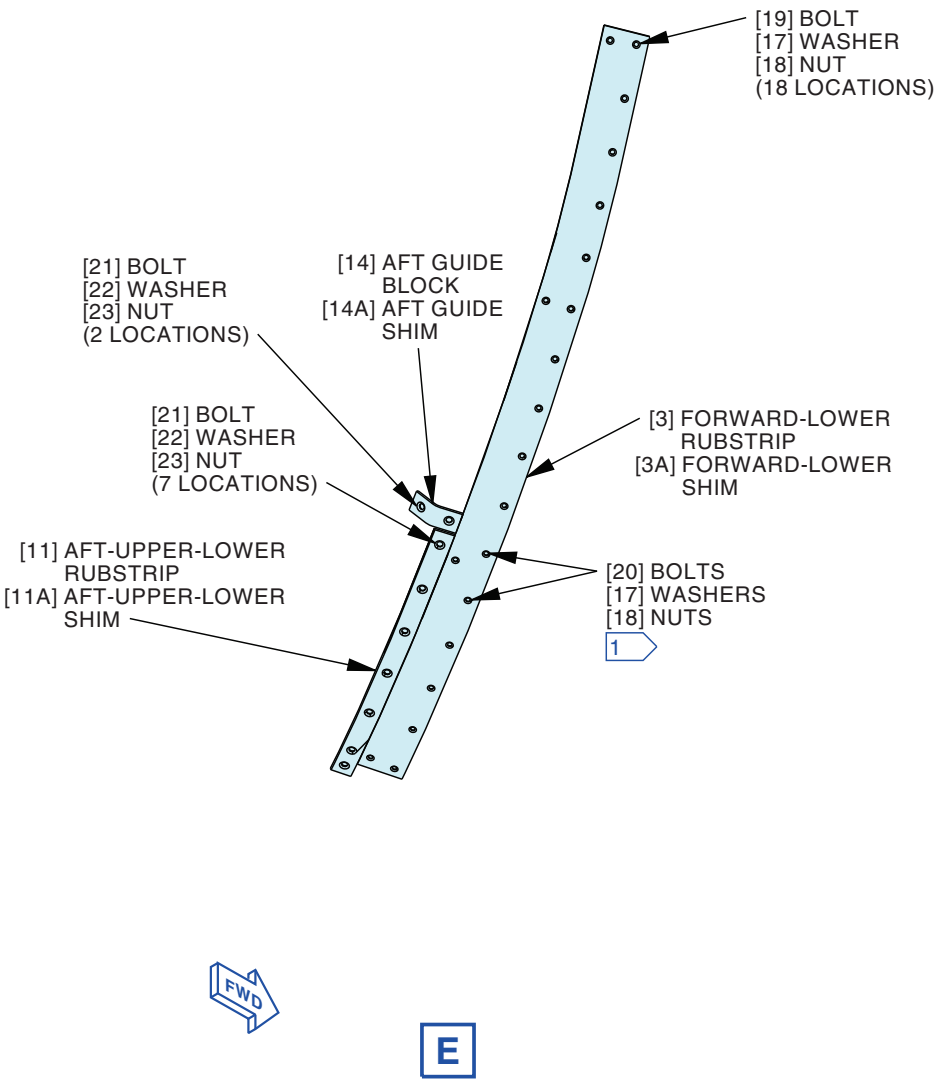


1 DRILL HOLE ON NEW RUBSTRIP

2355244 S00061516819_V3

Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

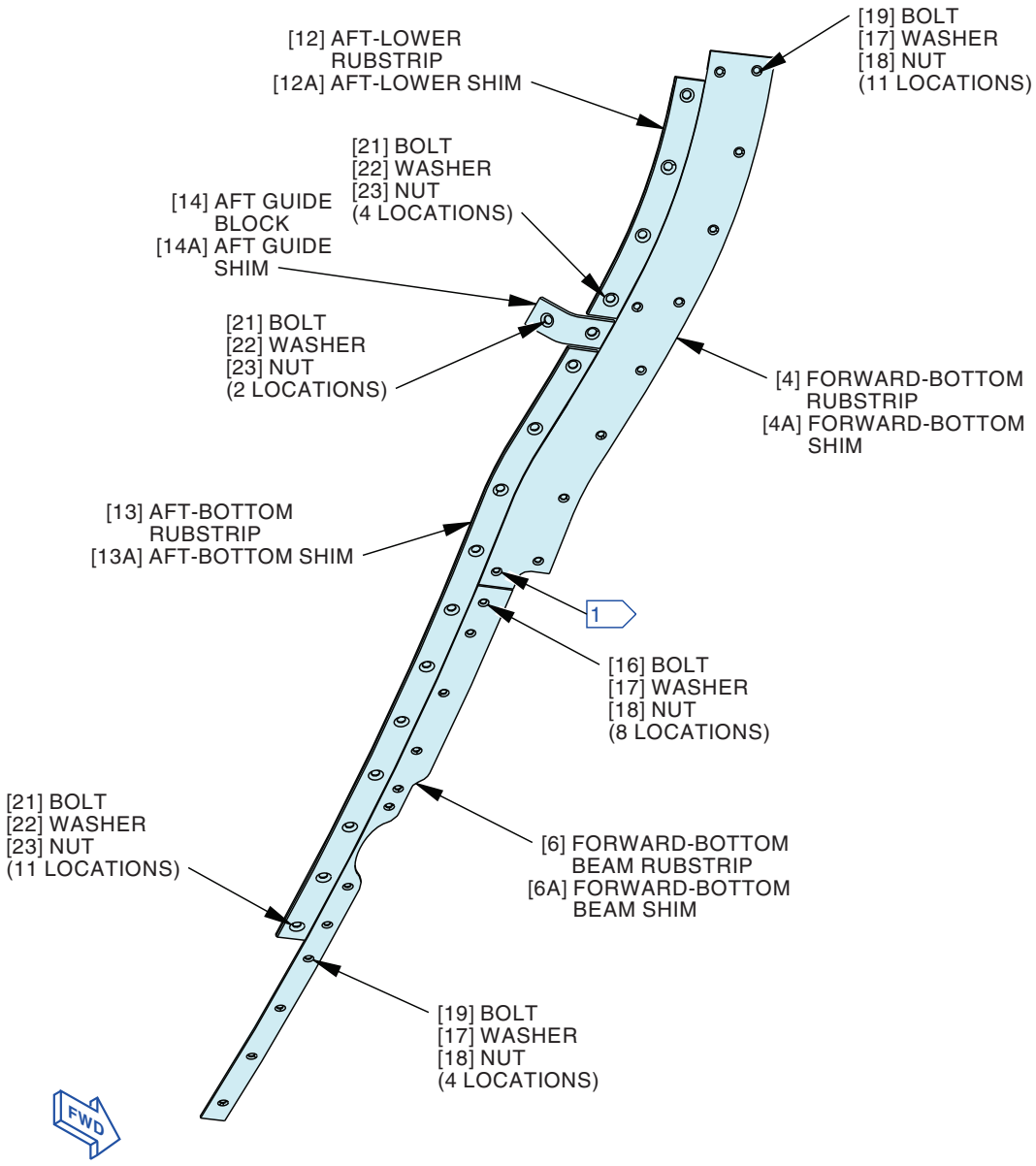


1 DRILL HOLE ON NEW RUBSTRIP

2355245 S00061516820_V3

Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 5 of 6)

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Rubstrip Installation
Figure 401/78-31-10-990-801-G00 (Sheet 6 of 6)

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TASK 78-31-10-400-801-G00

3. Rubstrip Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the rubstrips on the torque box for the left or right thrust reverser.

(2) If all of the rubstrips are removed, install the aft rubstrips and shims before you install the forward rubstrips and shims.

B. References

Reference	Title
71-11-05-821-801-G00	Fan Cowl Latch Adjustment (P/B 501)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63

D. Prepare for the Installation

SUBTASK 78-31-10-420-002-G00

(1) If necessary, temporarily install the replacement aft rubstrip with the applicable shim.

NOTE: If the original shim is used with the replacement rubstrip, there will be less adjustment to get the necessary rubstrip height dimensions.

SUBTASK 78-31-10-420-003-G00

(2) For all of the aft rubstrips, there can be 0.00 in. (0.00 mm) - 0.060 in. (1.52 mm) measurement between the rubstrips.

E. Aft Rubstrip Installation

SUBTASK 78-31-10-420-004-G00

(1) To install the aft-upper rubstrip [7] and aft-upper shim [7A], do these steps:

(a) Align the aft-upper rubstrip [7] and aft-upper shim [7A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-upper rubstrip [7] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the nine bolts [21], washers [22], and nuts [23].

(d) Make sure that the measurement between the aft-upper rubstrip [7] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

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SUBTASK 78-31-10-420-005-G00

(2) To install the aft-upper-mid rubstrip [8] and aft-upper-mid shim [8A], do these steps:

(a) Align the aft-upper-mid rubstrip [8] and aft-upper-mid shim [8A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-upper-mid rubstrip [8] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the 11 bolts [21], washers [22], and nuts [23].

(d) Make sure that the measurement between the aft-upper-mid rubstrip [8] and the translating sleeve is still in the limit.

1) If the measurement is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 78-31-10-420-006-G00

(3) To install the aft-mid rubstrip [9] and aft-mid shim [9A], do these steps:

(a) Align the aft-mid rubstrip [9] and aft-mid shim [9A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-mid rubstrip [9] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the 10 bolts [23], washers [24], and nuts [25].

(d) Make sure that the measurement between the aft-mid rubstrip [9] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 78-31-10-420-007-G00

(4) To install the aft-lower-mid rubstrip [10] and aft-lower-mid shim [10A], do these steps:

(a) Align the aft-lower-mid rubstrip [10] and aft-lower-mid shim [10A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-lower-mid rubstrip [10] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the 10 bolts [23], washers [24], and nuts [25].

(d) Install the four bolts [24], washers [22], and nuts [23].

(e) Make sure that the measurement between the aft-lower-mid rubstrip [10] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

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SUBTASK 78-31-10-420-008-G00

(5) To install the aft-upper-lower rubstrip [11] and aft-upper-lower rubstrip [11], do these steps:

(a) Align the aft-upper-lower rubstrip [11] and aft-upper-lower shim [11A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-upper-lower rubstrip [11] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the seven bolts [21], washers [22], and nuts [23].

(d) Make sure that the measurement between the aft-upper-lower rubstrip [11] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 78-31-10-420-012-G00

(6) To install the aft-lower rubstrip [12] and aft-lower shim [12A], do these steps:

(a) Align the aft-lower rubstrip [12] and aft-lower shim [12A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-lower rubstrip [12] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the four bolts [21], washers [22], and nuts [23].

(d) Make sure that the measurement between the aft-lower rubstrip [12] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 78-31-10-420-009-G00

(7) To install the aft-bottom rubstrip [13] and aft-bottom shim [13A], do these steps:

(a) Align the aft-bottom aft-bottom rubstrip [13] and aft-bottom shim [13A] with the holes on the thrust reverser structure.

NOTE: The shim is installed under the rubstrip.

(b) Make sure that the measurement between the aft-bottom rubstrip [13] and the translating sleeve is in the limit.

1) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

(c) Install the 11 bolts [21], washers [22], and nuts [23].

(d) Make sure that the measurement between the aft-bottom rubstrip [13] and the translating sleeve is still in the limit.

1) If the dimension is correct, tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

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SUBTASK 78-31-10-420-010-G00

(8) To install the aft guide block [14], do these steps:

NOTE: There are five aft guide blocks on the torque box on the left and right thrust reverser on an engine.

(a) Align the aft guide block [14] and aft guide shim [14A] with the holes on the thrust reverser structure.

1) If the measurement between the aft guide block [14] and the adjacent rubstrips is not 0.00 in. (0.00 mm) - 0.060 in. (1.52 mm), cut the aft guide block [14] to get the correct dimension.

(b) Measure the measurement between the height of the aft guide block [14] and the height of the adjacent aft rubstrips.

(c) Make sure that the height of the aft guide block [14] is 0.000 in. (0.000 mm) - 0.020 in. (0.508 mm) below the adjacent aft rubstrips.

1) If the dimension is not correct, add shim or remove the lamination skins from the shim to get the correct dimension.

NOTE: A shim must be installed under the aft guide block.

(d) Install the two bolts [21], washers [22], and nuts [23].

1) Tighten the nuts [23] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

(e) Slowly retract the translating sleeve manually to the completely retracted position.

1) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

2) Make sure that the translating sleeve moves smoothly over the aft guide blocks.

3) Stop the retraction of the translating sleeve if there is interference with the aft guide blocks or rubstrips.

(f) If there is interference, do the Translating Sleeve to Aft Rubstrip Dimension Check again.

F. Translating Sleeve to Aft Rubstrip Dimension Check

SUBTASK 78-31-10-220-001-G00

(1) Do these steps to measure the measurement between the aft rubstrips and the translating sleeve:

(a) Slowly retract the translating sleeve manually to the completely retracted position.

1) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

2) Make sure that the translating sleeve moves smoothly over the rubstrips.

3) Stop the retraction of the translating sleeve if there is interference with the rubstrips.

4) If the rubstrip interferes with the movement of the translating sleeve, do these steps:

a) Measure and mark the area on the rubstrip where it is too high.

b) Manually extend the translating sleeve to get access to the rubstrip.

<1> Do this task: Thrust Reverser Operation - Extend (Manual Procedure), TASK 78-31-00-980-803-G00.

c) Remove the aft rubstrip and shim.

d) Remove the laminations from the shim until the rubstrip is at the correct height.

e) Make sure that you do not remove too many of the lamination skins.

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<1> The measurement between the translating sleeve and the rubstrip must be no more that 0.0300 in. (0.7620 mm).

f) Re-install the rubstrip and shim.

g) Manually retract the translating sleeve until it is in the completely retracted position.

h) Do the above steps again if there is still interference.

(b) Use a feeler gage to measure the measurement between the translating sleeve and each of the rubstrips.

NOTE: Take measurements at several locations on each rubstrip.

1) Record the measurements found at each location for each of the rubstrips.

2) The measurement between the aft rubstrip and the translating sleeve must be 0.0000 in. (0.0000 mm) - 0.0300 in. (0.7620 mm).

3) If the measurement is not in the limits, add a new shim or remove laminations from the original shim to get the necessary measurement.

G. Forward Rubstrip Installation

SUBTASK 78-31-10-420-011-G00

(1) For all of the rubstrips, there can be 0.00 in. (0.00 mm) - 0.060 in. (1.52 mm) measurement between the rubstrips.

SUBTASK 78-31-10-420-013-G00

(2) To install the forward-top rubstrip [5] and forward-top shim [5A] do these steps:

(a) Use the forward-top shim [5A] if it is necessary, 0.032 in. (0.813 mm) maximum to keep forward-top rubstrip [5] between 0.000 in. (0.000 mm) - 0.020 in. (0.508 mm) below the adjacent rubstrip.

(b) Align the forward-top rubstrip [5] and forward-top shim [5A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the three rivets [15].

NOTE: The uppermost rivet is the largest of the three rivets [15].

NOTE: Use the standard overhaul practices to install the rivets.

(d) Apply a 0.0525 in. (1.3335 mm) fillet seal around the gap between the forward-top rubstrip [5] and the edge of the hinge beam with sealant, A00160.

1) Apply the fillet seal for a minimum of 2.0 in. (50.8 mm) from the top of the forward-top rubstrip [5]

2) Remove excess sealant and smooth the transitions.

SUBTASK 78-31-10-420-014-G00

(3) To install the forward-upper rubstrip [1] and forward-upper shim [1A], do these steps:

(a) Lubricate the threads of the bolts [16] and bolts [19] with the sealant, A00160.

(b) Align the forward-upper rubstrip [1] and forward-upper shim [1A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the four bolts [16], washers [17], and nuts [18] to the four upper locations.

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

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(d) Install the remaining 14 bolts [19], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

SUBTASK 78-31-10-420-015-G00

(4) To install the forward-mid rubstrip [2] and forward-mid shim [2A], do these steps:

(a) Lubricate the threads of the bolts [19] and bolts [20] with the sealant, A00160.

(b) Align the forward-mid rubstrip [2] and forward-mid shim [2A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the 21 bolts [19], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

(d) Install the two bolts [20], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

SUBTASK 78-31-10-420-016-G00

(5) To install the forward-lower rubstrip [3] and forward-lower shim [3A], do these steps:

(a) Lubricate the threads of the bolts [19] and bolts [20] with the sealant, A00160.

(b) Align the forward-lower rubstrip [3] and forward-lower shim [3A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the 18 bolts [19], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

(d) Install the two bolts [20], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

SUBTASK 78-31-10-420-017-G00

(6) To install the forward-bottom rubstrip [4] and forward-bottom shim [4A], do these steps:

(a) Lubricate the threads of the bolts [19] with the sealant, A00160.

(b) Align the forward-bottom rubstrip [4] and forward-bottom shim [4A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the 11 bolts [19], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

SUBTASK 78-31-10-420-018-G00

(7) To install the forward-bottom beam rubstrip [6] and forward-bottom beam shim [6A], do these steps:

(a) Lubricate the threads of the bolts [16] and bolts [19] with the sealant, A00160.

(b) Align the forward-bottom beam rubstrip [6] and forward-bottom beam shim [6A] with the holes on the fan cowl structure.

NOTE: The shim is installed under the rubstrip.

(c) Install the eight bolts [16], washers [17], and nuts [18] to the eight upper locations.

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

(d) Install the remaining four bolts [19], washers [17], and nuts [18].

1) Tighten the nuts [18] to 10 in-lb (1.1 N·m) - 18 in-lb (2.0 N·m).

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(e) Fillet seal the gap below the forward-bottom beam rubstrip [6] and the end of the aft-bottom rubstrip [13] using sealant, A00160.

1) Remove excess sealant and smooth the transitions.

SUBTASK 78-31-10-820-001-G00

(8) Do this task: Fan Cowl Latch Adjustment, TASK 71-11-05-821-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-10-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

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TENSION LATCHES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Tension Latch Removal

(2) Tension Latch Installation.

TASK 78-31-11-000-801-G00

2. Tension Latch Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the tension latches from the thrust reverser.

(a) There are a total of nine tension latches.

(2) For this task a tension latch will be referred to as a latch.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6634	Knife - Plastic

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-31-11-010-002-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Latch Removal

SUBTASK 78-31-11-020-001-G00

(1) Do these steps to remove the applicable latch [1] (No. 1-6):

(a) If necessary, use a plastic knife, STD-6634 to break the bond of the sealant that is around the washers.

NOTE: After you break the bond of the sealant from the attach fitting, the sealant will come off with the washers.

(b) Remove the nut [8].

(c) Remove the washer [6] and washers [7] as necessary.

NOTE: Keep the washers for the installation.

(d) Remove the bolt.

1) To remove the latch No. 1 and latch No. 2 (which are adjacent), remove bolt [5].

NOTE: The latch No. 1 and latch No. 2 are held in their positions with one bolt. The bolt connects the two latches to the attach fitting.

2) To remove the latch No. 3, remove bolt [4].

3) To remove the latch No. 4, remove bolt [3].

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4) To remove the latch No. 5 and latch No. 6 (which are adjacent), remove the bolt [2].

NOTE: The latch No. 5 and latch No. 6 are held in their positions with one bolt. The bolt connects the two latches to the attach fitting.

(e) Remove the latch.

(f) If necessary, remove the sealant that remains from the fitting assembly with solvent, B00062.

SUBTASK 78-31-11-020-002-G00

(2) Do these steps to remove the applicable latch [1] (No. 7-9):

(a) Remove the latch No. 7 as follows:

1) Remove the cotter pin [24] and nut [15].
2) Remove the washer [16] and washer [17].
3) Remove the bolt [10] and washer [12].
4) Remove the latch No. 7.

(b) Remove the latch No. 8 as follows:

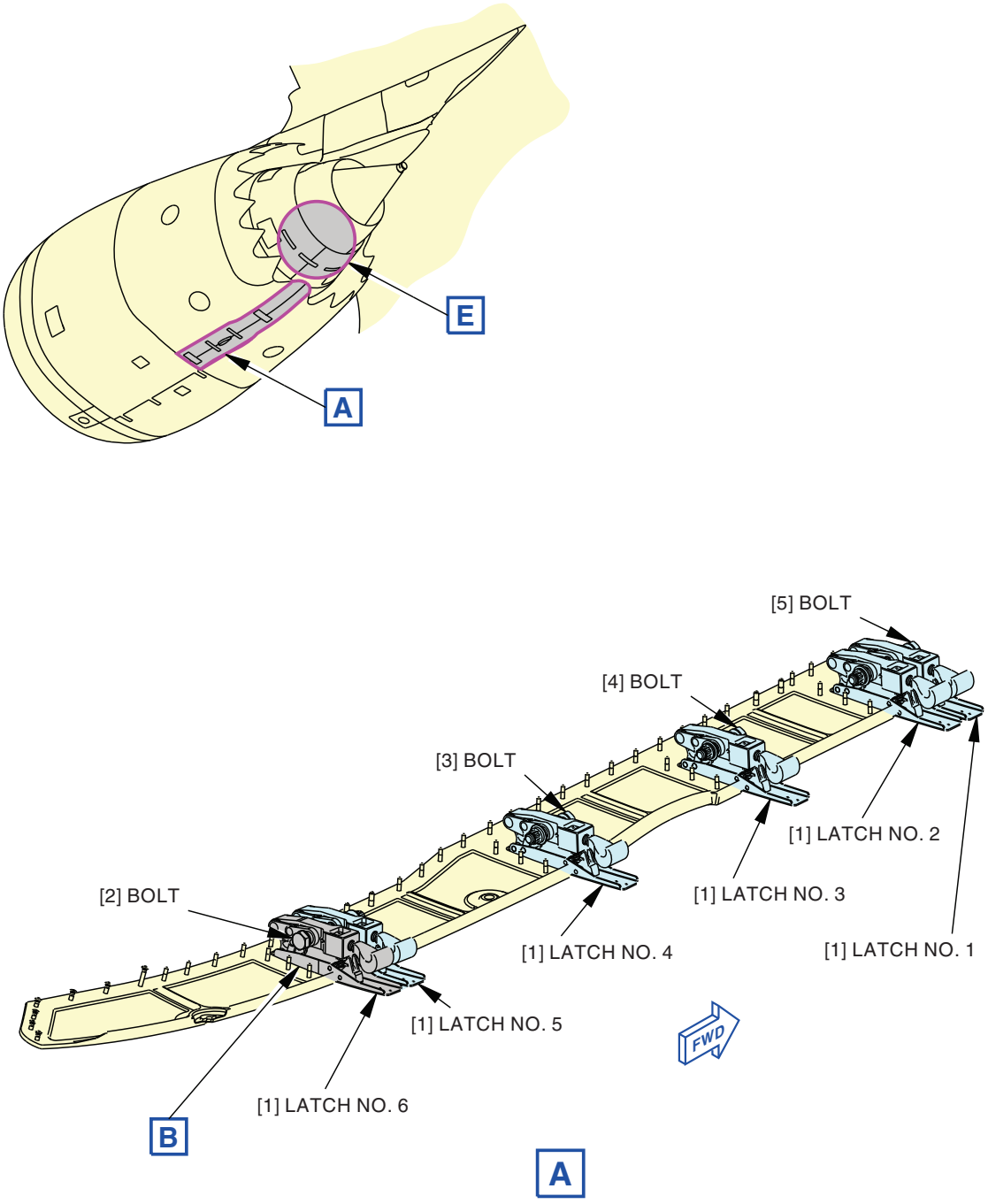
1) Remove the cotter pin [24] and nut [18].
2) Remove the washer [19] and two washers [20].
3) Remove the bolt [9] and washer [13].
4) Remove the latch No. 8.

(c) Remove the latch No. 9 as follows:

1) Remove the cotter pin [24] and nut [21].
2) Remove the washer [22] and two washers [23].
3) Remove the bolt [11] and washer [14].
4) Remove the latch No. 9.

——— END OF TASK ———

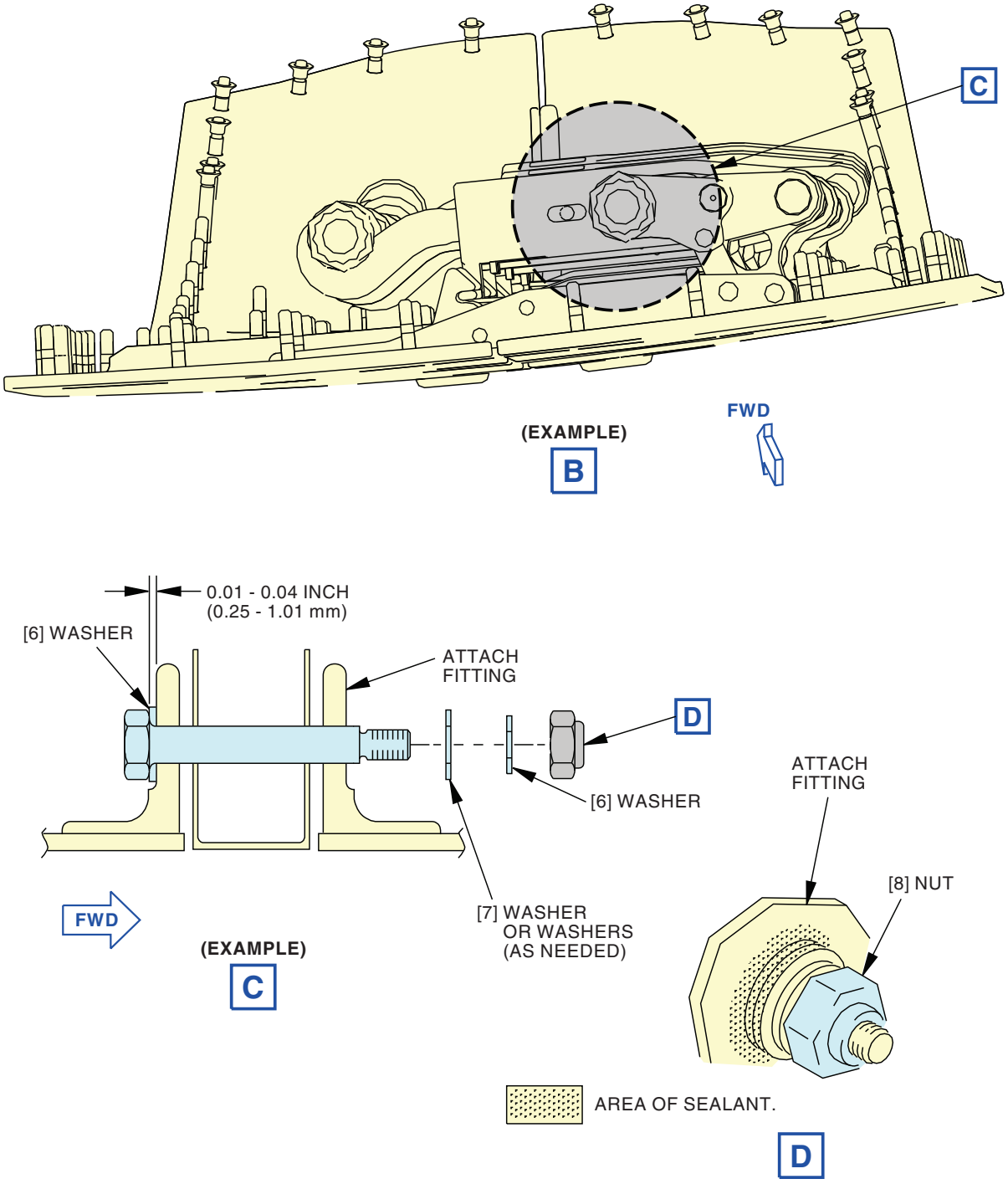
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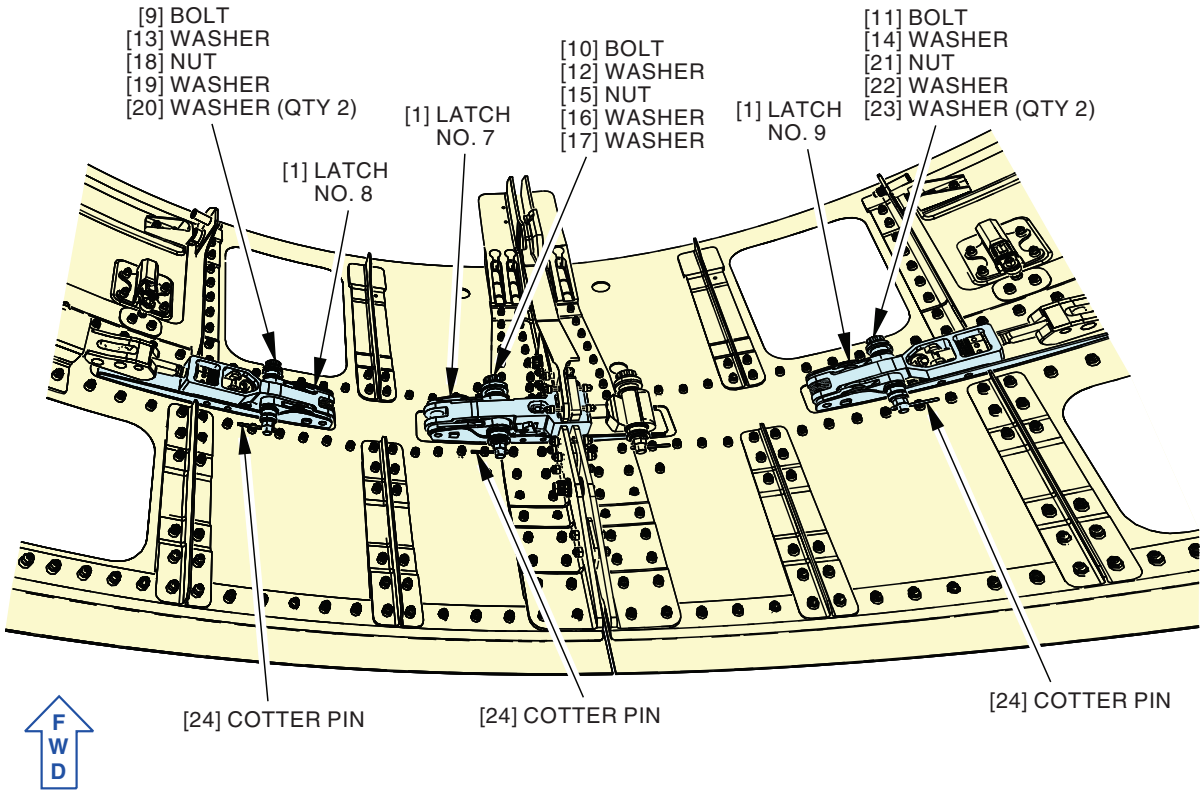
Thrust Reverser Tension Latch Installation
Figure 401/78-31-11-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Tension Latch Installation
Figure 401/78-31-11-990-801-G00 (Sheet 2 of 3)

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(AFT COWL STRUCTURES REMOVED FOR CLARITY)

E

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Thrust Reverser Tension Latch Installation
Figure 401/78-31-11-990-801-G00 (Sheet 3 of 3)

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TASK 78-31-11-400-801-G00

3. Tension Latch Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the tension latches on the thrust reverser.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-11-820-801-G00	Latches Adjustment (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1107	Gauge - Feeler, 0.0 - 0.5 Inch, Readable to 1/1000th

D. Consumable Materials

Reference	Description	Specification
A00436	Sealant - Fuel Tank	BMS5-45 (Supersedes BMS5-26)
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)

E. Latch Installation

SUBTASK 78-31-11-010-001-G00

(1) Make sure the thrust reverser is open (TASK 78-31-00-010-801-G00).

SUBTASK 78-31-11-420-001-G00

(2) Do these steps to install the applicable latch [1] (No. 1-6):

(a) If necessary, remove the sealant that remains with solvent, B00062.

(b) Install the latch [1] in the attach fitting.

NOTE: The latches are installed with unique different length bolts. Make sure that you use the correct bolt with the applicable latch or latches.

(c) Install the applicable bolt with the washer [6] under the bolt head.

1) For latch No. 1 and latch No.2 (adjacent), use bolt [5]:

NOTE: Latch No. 1 and latch No. 2 are held in their positions with one bolt. The bolt connects the two latches to the attach fitting.

2) For latch No. 3, use bolt [4].

3) For latch No. 4, use bolt [3].

4) For latch No. 5 and latch No. 6 (adjacent), use bolt [2].

NOTE: Latch No. 5 and latch No. 6 are held in their positions with one bolt. The bolt connects the two latches to the attach fitting.

(d) Install the washer or washers [7] that were removed in the removal task.

(e) Install the washer [6] and the nut [8].

1) Hand tighten the nut only.

(f) Use a 0.0 - 0.5 Inch feeler gauge, STD-1107 to measure the distance between the bolt head and the attach fitting (View C).

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1) When you measure the distance, move the bolt until the nut holds the washers against the attach fitting.

NOTE: Alternative: you can move the head of bolt and washer against the fitting to measure the distance.

2) The distance must be 0.01 in. (0.25 mm) - 0.04 in. (1.02 mm) (correct axial movement).

3) If the dimension is not correct, add or remove the applicable washers [7] as necessary.

NOTE: Only latches No.1 and 2 are permitted to have three washers. Latch No. 3-6 can have a maximum of two washers.

a) The thickness of the available washers are: 0.016 in. (0.406 mm), 0.032 in. (0.813 mm), and 0.063 in. (1.600 mm).

(g) Tighten the nut to 400 in-lb (45 N·m) - 430 in-lb (49 N·m) or the bolt head to 500 in-lb (56 N·m) - 530 in-lb (60 N·m).

(h) Apply a fillet of sealant, A00436 around bolt shank between the washers and the attach fitting (View D).

SUBTASK 78-31-11-420-002-G00

(3) Do these steps to install the applicable latch [1] (No. 7-9):

(a) Install the latch No. 7 as follows:

1) Hold the latch No. 7 in the attach fitting.

2) Install the washer [12] and bolt [10].

3) Install the washer [16], washer [17], and nut [15].

4) Tighten the nut [15] to 194 in-lb (22 N·m) - 206 in-lb (23 N·m).

5) Install the cotter pin [24].

(b) Install the latch No. 8 as follows:

1) Hold the latch No. 8 in the attach fitting.

2) Install the washer [13] and bolt [9].

3) Install the washer [19], two washers [20], and nut [18].

4) Tighten the nut [18] to 121 in-lb (14 N·m) - 129 in-lb (15 N·m).

5) Install the cotter pin [24].

(c) Install the latch No. 9 as follows:

1) Hold the latch No. 9 in the attach fitting.

2) Install the bolt [11] and washer [14].

3) Install the washer [22], two washers [23], and nut [21].

4) Tighten the nut [21] to 121 in-lb (14 N·m) - 129 in-lb (15 N·m).

5) Install the cotter pin [24].

SUBTASK 78-31-11-410-001-G00

WARNING OBEY THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE PROCEDURE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Close the Thrust Reverser (TASK 78-31-00-010-802-G00).

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SUBTASK 78-31-11-820-001-G00

(5) Adjust the latches (TASK 78-31-11-820-801-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TENSION LATCHES - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) The adjustment of the latches.

TASK 78-31-11-820-801-G00

2. Latches Adjustment

(Figure 501)

A. General

(1) This task is for the adjustment of the thrust reverser tension latches.

(2) The thrust reverser tension latches are at the bottom centerline of the thrust reverser.

(3) For this task the thrust reverser tension latch will be called the latch.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1184	Scale - Spring, 0-100 Lbs, Tension
STD-1330	Wrench - Hexdrive, Allen Wrench

D. Prepare for the Adjustment

SUBTASK 78-31-11-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

E. Latches Adjustment

SUBTASK 78-31-11-820-007-G00

(1) Make sure that all the latches are closed.

SUBTASK 78-31-11-820-002-G00

(2) Do these steps to adjust the latches No.1 through No.7:

(a) Open the latch hook to disengage the hook from the latch pin.

CAUTION DO NOT USE MORE THAN 100 POUNDS FORCE ON THE LATCH HANDLE. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE LATCH, LATCH KEEPER, OR THE FAN COWL PANEL.

(b) Apply 40 lbf (178 N) - 60 lbf (267 N) of force with a spring scale (0-100 Lbs), STD-1184, no more than 1.0 in. (25.4 mm) from the end of the latch handle to close it.

(c) If the force is not in the limits, do these steps to adjust the latch:

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1) Disengage the latch.
2) Put an hexdrive allen wrench, STD-1330, in the latch load adjustment wheel in the latch housing.
   a) Turn the hexdrive allen wrench, STD-1330, clockwise to increase the latch tension.
   b) Turn the hexdrive allen wrench, STD-1330, counterclockwise to decrease the latch tension.
3) Do the above steps again until the latch tension is in the limits.
4) Make sure that the latch remains closed as the adjustment for the other latches is done.

SUBTASK 78-31-11-820-005-G00

(3) Do these steps to adjust the latches No.8 through No.9 as follows:

(a) Make sure that the latches No.1 through No.7 closed after the latch adjustment is completed.

(b) Open the latch hook to disengage the hook from the latch pin.

CAUTION DO NOT USE MORE THAN 100 POUNDS FORCE ON THE LATCH HANDLE. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE LATCH, LATCH KEEPER, OR THE FAN COWL PANEL.

(c) Apply 40 lbf (178 N) - 60 lbf (267 N) of force with a spring scale (0-100 Lbs), STD-1184, no more than 1.0 in. (25.4 mm) from the end of the latch handle to close it.

(d) If the force is not in the limits, do these steps to adjust the latch:
   1) Disengage the latch.
   2) Put a hexdrive allen wrench, STD-1330, in the latch load adjustment wheel in the latch housing.
      a) Turn the hexdrive allen wrench, STD-1330, clockwise to increase the latch tension.
      b) Turn the hexdrive allen wrench, STD-1330, counterclockwise to decrease the latch tension.

(e) Do the above steps again until the latch tension is in the limits.

(f) Make sure that the latch remains closed as the adjustment for the other latches is done.

SUBTASK 78-31-11-820-003-G00

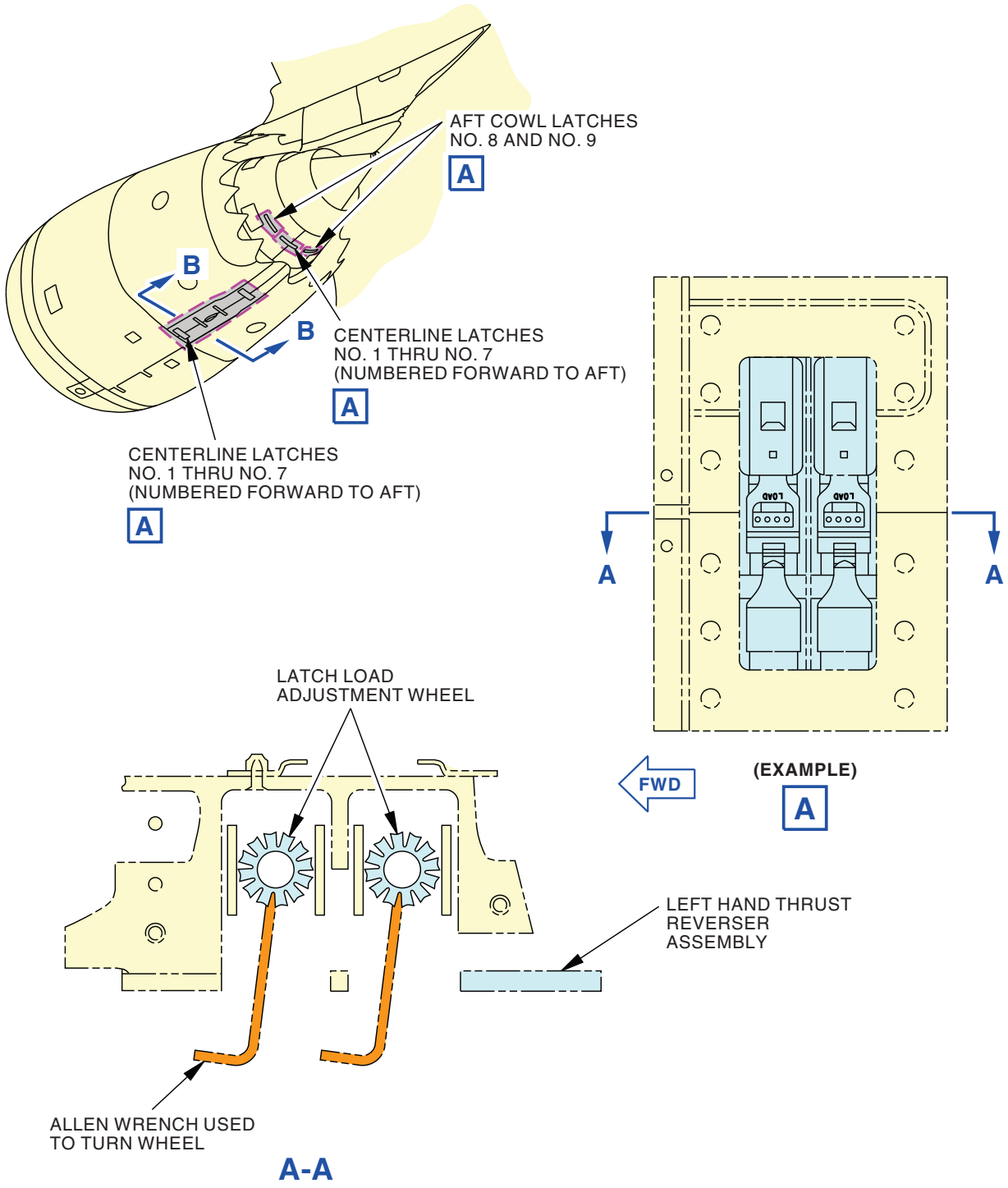
(4) Do a check of all of the latches again to make sure that the tension for each latch is still in the limits.

SUBTASK 78-31-11-440-001-G00

(5) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

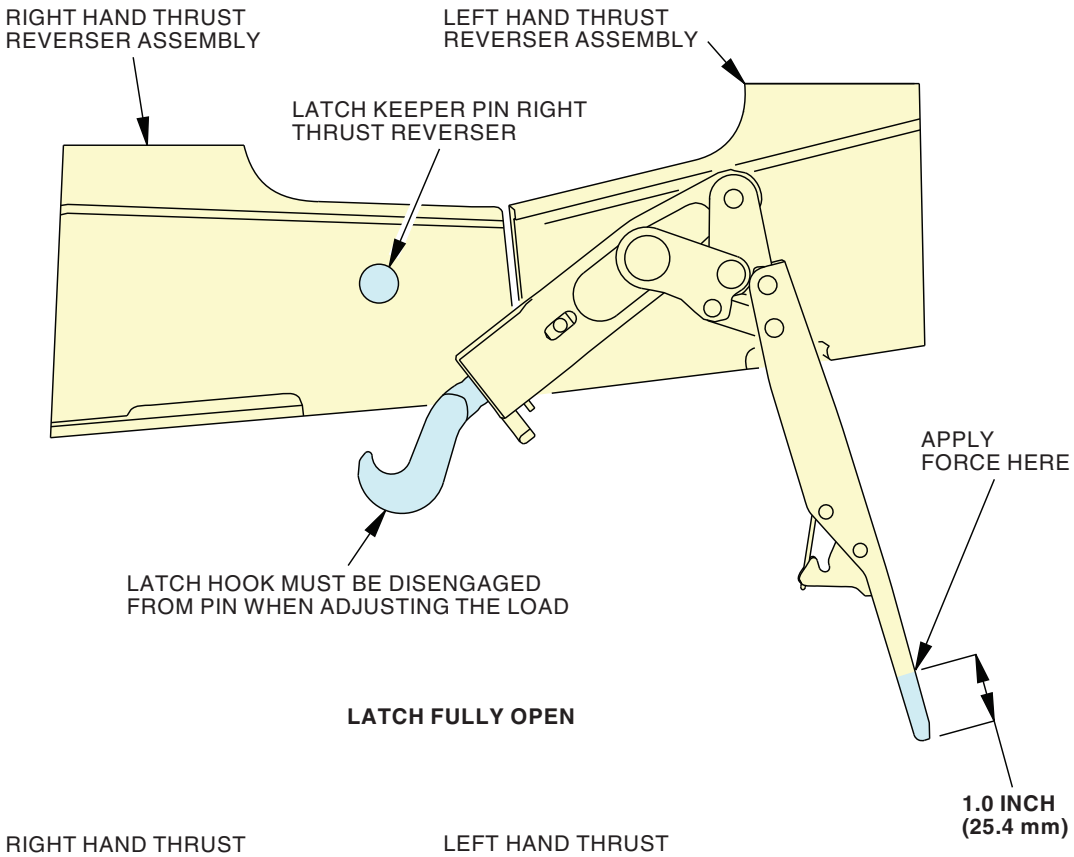
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Thrust Reverser Tension Latch Adjustment
Figure 501/78-31-11-990-802-G00 (Sheet 1 of 2)

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B-B

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Thrust Reverser Tension Latch Adjustment
Figure 501/78-31-11-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FIRE SEAL - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the thrust reverser fire seal.

(2) The installation of the thrust reverser fire seal.

TASK 78-31-12-000-801-G00

2. Fire Seal Removal

(Figure 401, Figure 402, Figure 403, Figure 404)

A. General

(1) This task is for the removal of the fire seal from the left or right thrust reverser.

(2) For this task the forward firewall blockseal will be referred to as the blockseal.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-765	Scraper - Plastic

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-31-12-010-001-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL

G. Left Side Thrust Reverser Fire Seal Removal

SUBTASK 78-31-12-020-018-G00

(1) Do these steps to remove the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] from the left side thrust reverser (View B, Figure 401):

(a) Carefully remove the sealant from the area where the forward firewall blockseal [5] is inserted into the forward firewall fire resistance blockseal [6].

(b) Remove all of the sealant that bonds the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] to the fireseal retainer [7] (View A-A, Figure 401).

(c) Slide the forward firewall blockseal [5] out of the fireseal retainer [7].

(d) Slide the forward firewall fire resistance blockseal [6] out of the fireseal retainer [7].

SUBTASK 78-31-12-020-001-G00

(2) Do these steps to remove the upper fire seal [1] from the left side thrust reverser (Figure 401):

(a) Remove all of the sealant that bonds the upper fire seal [1] to the fireseal retainer [7] (View A-A, Figure 401).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the upper fire seal [1] to remove approximately 3.0 in. (76.2 mm) of the upper fire seal [1] from the fireseal retainer [7].

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the upper fire seal [1] from the fireseal retainer [7] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the upper fire seal [1] to pull the rest upper fire seal [1] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the upper fire seal [1] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the upper fire seal [1] with a mirror.

(f) Remove the upper fire seal [1] from the fireseal retainer [7].

SUBTASK 78-31-12-020-012-G00

(3) Do these steps to remove the barrel fire seal [2] from the left side thrust reverser (Figure 401):

(a) Remove all of the sealant that bonds the barrel fire seal [2] to the fireseal retainer [7] (View A-A, Figure 401).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the barrel fire seal [2] to remove approximately 3.0 in. (76.2 mm) of the barrel fire seal [2] from the fireseal retainer [7].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the barrel fire seal [2] from the fireseal retainer [7] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the barrel fire seal [2] to pull the rest barrel fire seal [2] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the barrel fire seal [2] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the barrel fire seal [2] with a mirror.

(f) Remove the barrel fire seal [2] from the fireseal retainer [7].

SUBTASK 78-31-12-020-013-G00

(4) Do these steps to remove the lower fire seal [3] from the left side thrust reverser (Figure 401):

(a) Remove all of the sealant that bonds the lower fire seal [3] to the fireseal retainer [7] (View A-A, Figure 401).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the lower fire seal [3] to remove approximately 3.0 in. (76.2 mm) of the lower fire seal [3] from the fireseal retainer [7].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the lower fire seal [3] from the fireseal retainer [7] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the lower fire seal [3] to pull the rest lower fire seal [3] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the lower fire seal [3] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the lower fire seal [3] with a mirror.

(f) Remove the lower fire seal [3] from the fireseal retainer [7].

SUBTASK 78-31-12-020-019-G00

(5) Do these steps to remove the fire seal [9] from the left side thrust reverser (View B-B and View C, Figure 401):

(a) Remove all of the sealant that bonds the fire seal [9] to the fireseal retainer [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

(b) Slide the fire seal [9] out of the fireseal retainer [7].

SUBTASK 78-31-12-020-014-G00

(6) Do these steps to remove the forward latch beam fire seal [8] from the left side thrust reverser (View D, Figure 401):

(a) Remove the two bolts [10], two washers [11], and two collars [12].

(b) Use the plastic scraper, STD-765, to remove the forward latch beam fire seal [8] from the structure.

H. Right Side Thrust Reverser Fire Seal Removal

SUBTASK 78-31-12-020-020-G00

(1) Do these steps to remove the forward firewall fire resistance blockseal [16] and forward firewall blockseal [17] from the right side thrust reverser (View B, Figure 402):

(a) Carefully remove the sealant from the area where the forward firewall blockseal [17] is inserted into the forward firewall fire resistance blockseal [16].

(b) Remove all of the sealant that bonds the forward firewall blockseal [17] and forward firewall fire resistance blockseal [16] to the fireseal retainer [18] (View A-A, Figure 402).

(c) Slide the forward firewall blockseal [17] out of the fireseal retainer [18].

(d) Slide the forward firewall fire resistance blockseal [16] out of the fireseal retainer [18].

SUBTASK 78-31-12-020-021-G00

(2) Do these steps to remove the upper fire seal [13] from the right side thrust reverser (Figure 402):

(a) Remove all of the sealant that bonds the upper fire seal [13] to the fireseal retainer [18] (View A-A, Figure 402).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the upper fire seal [13] to remove approximately 3.0 in. (76.2 mm) of the upper fire seal [13] from the fireseal retainer [18].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the upper fire seal [13] from the fireseal retainer [18] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the upper fire seal [13] to pull the rest upper fire seal [13] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the upper fire seal [13] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the upper fire seal [13] with a mirror.

(f) Remove the upper fire seal [13] from the fireseal retainer [18].

SUBTASK 78-31-12-020-022-G00

(3) Do these steps to remove the barrel fire seal [14] from the right side thrust reverser (Figure 402):

(a) Remove all of the sealant that bonds the barrel fire seal [14] to the fireseal retainer [7] (View A-A, Figure 402).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the barrel fire seal [14] to remove approximately 3.0 in. (76.2 mm) of the barrel fire seal [14] from the fireseal retainer [18].

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the barrel fire seal [14] from the fireseal retainer [18] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the barrel fire seal [14] to pull the rest barrel fire seal [14] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the barrel fire seal [14] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the barrel fire seal [14] with a mirror.

(f) Remove the barrel fire seal [14] from the fireseal retainer [18].

SUBTASK 78-31-12-020-023-G00

(4) Do these steps to remove the lower fire seal [15] from the right side thrust reverser (Figure 402):

(a) Remove all of the sealant that bonds the lower fire seal [15] to the fireseal retainer [18] (View A-A, Figure 402).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the lower fire seal [15] to remove approximately 3.0 in. (76.2 mm) of the lower fire seal [3] from the fireseal retainer [18].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the lower fire seal [15] from the fireseal retainer [18] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the lower fire seal [15] to pull the rest lower fire seal [15] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the lower fire seal [15] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the lower fire seal [15] with a mirror.

(f) Remove the lower fire seal [15] from the fireseal retainer [18].

SUBTASK 78-31-12-020-024-G00

(5) Do these steps to remove the fire seal [19] from the right side thrust reverser (View B-B, Figure 402):

(a) Remove all of the sealant that bonds the fire seal [19] to the fireseal retainer [18].

(b) Slide the fire seal [19] out of the fireseal retainer [18].

SUBTASK 78-31-12-020-025-G00

(6) Do these steps to remove the forward latch beam fire seal [20] from the right side thrust reverser (View C, Figure 402):

(a) Remove the two bolts [21], two washers [22], and two collars [23].

(b) Use the plastic scraper, STD-765, to remove the forward latch beam fire seal [20] from the structure.

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PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL

I. Left Thrust Reverser Fire Seal Removal

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL COMPOSITE INNER WALL

SUBTASK 78-31-12-020-030-G00

(1) Do these steps to remove the pressure seal [42] from the left side thrust reverser (View A, Figure 403):

(a) Carefully remove the sealant from the area around the pressure seal [42].

(b) Remove all of the sealant that bonds the pressure seal [42] to the inner wall.

(c) Use the plastic scraper, STD-765, to remove the pressure seal [42] from the inner wall.

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL

SUBTASK 78-31-12-020-002-G00

(2) Do these steps to remove the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] from the left side thrust reverser (View B, Figure 403):

(a) Carefully remove the sealant from the area where the forward fireseal blockseal [25] is inserted into the forward firewall fire resistance blockseal [26].

(b) Remove all of the sealant that bonds the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] to the fireseal retainer [27] (View A-A, Figure 403).

(c) Slide the forward fireseal blockseal [25] out of the fireseal retainer [27].

(d) Slide the forward firewall fire resistance blockseal [26] out of the fireseal retainer [27].

SUBTASK 78-31-12-020-003-G00

(3) Do these steps to remove the main fire seal [24] from the left side thrust reverser (Figure 403):

(a) Remove all of the sealant that bonds the main fire seal [24] to the fireseal retainer [27] (View A-A, Figure 403).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the main fire seal [24] to remove approximately 3.0 in. (76.2 mm) of the main fire seal [24] from the fireseal retainer [27].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the main fire seal [24] from the fireseal retainer [27] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the main fire seal [24] to pull the rest main fire seal [24] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the main fire seal [24] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the main fire seal [24] with a mirror.

(f) Remove the main fire seal [24] from the fireseal retainer [27].

SUBTASK 78-31-12-020-015-G00

(4) Do these steps to remove the fire seal [28] from the left side thrust reverser (View B-B, Figure 403):

(a) Remove all of the sealant that bonds the fire seal [28] to the fireseal retainer [27].

(b) Slide the fire seal [28] out of the fireseal retainer [27].

SUBTASK 78-31-12-020-008-G00

(5) Do these steps to remove the forward latch beam fire seal [20] from the left side thrust reverser (View C, Figure 403):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

(a) Remove the two bolts [21], two washers [22], and two collars [23].

(b) Use the plastic scraper, STD-765, to remove the forward latch beam fire seal [20] from the structure.

J. Right Thrust Reverser Fire Seal Removal

SUBTASK 78-31-12-020-026-G00

(1) Do these steps to remove the forward firewall fire resistance blockseal [34] and forward firewall blockseal [35] from the right side thrust reverser (View B, Figure 404):

(a) Carefully remove the sealant from the area where the forward firewall blockseal [35] is inserted into the forward firewall fire resistance blockseal [34].

(b) Remove all of the sealant that bonds the forward firewall blockseal [35] and forward firewall fire resistance blockseal [34] to the fireseal retainer [36] (View A-A, Figure 404).

(c) Slide the forward firewall blockseal [35] out of the fireseal retainer [36].

(d) Slide the forward firewall fire resistance blockseal [34] out of the fireseal retainer [36].

SUBTASK 78-31-12-020-027-G00

(2) Do these steps to remove the main fire seal [33] from the right side thrust reverser (Figure 404).

(a) Remove all of the sealant that bonds the main fire seal [33] to the fireseal retainer [36] (View A-A, Figure 404).

(b) Insert a plastic scraper, STD-765, into the unsealed aft edge of the main fire seal [33] to remove approximately 3.0 in. (76.2 mm) of the main fire seal [33] from the fireseal retainer [27].

(c) Use the plastic scraper, STD-765, to remove the upper unsealed edge of the main fire seal [33] from the fireseal retainer [36] in the direction from the aft edge to the forward edge.

(d) Use the removed aft 3.0 in. (76.2 mm) of the main fire seal [33] to pull the rest main fire seal [33] outwards and up at a 45 degree angle from the insulation blanket.

(e) Continue to remove the main fire seal [33] at the same time a second person will break any sealant beads with a sharpened plastic scraper, STD-765.

1) If it is necessary, monitor the removal of the main fire seal [33] with a mirror.

(f) Remove the main fire seal [33] from the fireseal retainer [36].

SUBTASK 78-31-12-020-028-G00

(3) Do these steps to remove the fire seal [37] from the right side thrust reverser (View B-B, Figure 404):

(a) Remove all of the sealant that bonds the fire seal [37] to the fireseal retainer [36].

(b) Slide the fire seal [37] out of the fireseal retainer [36].

SUBTASK 78-31-12-020-029-G00

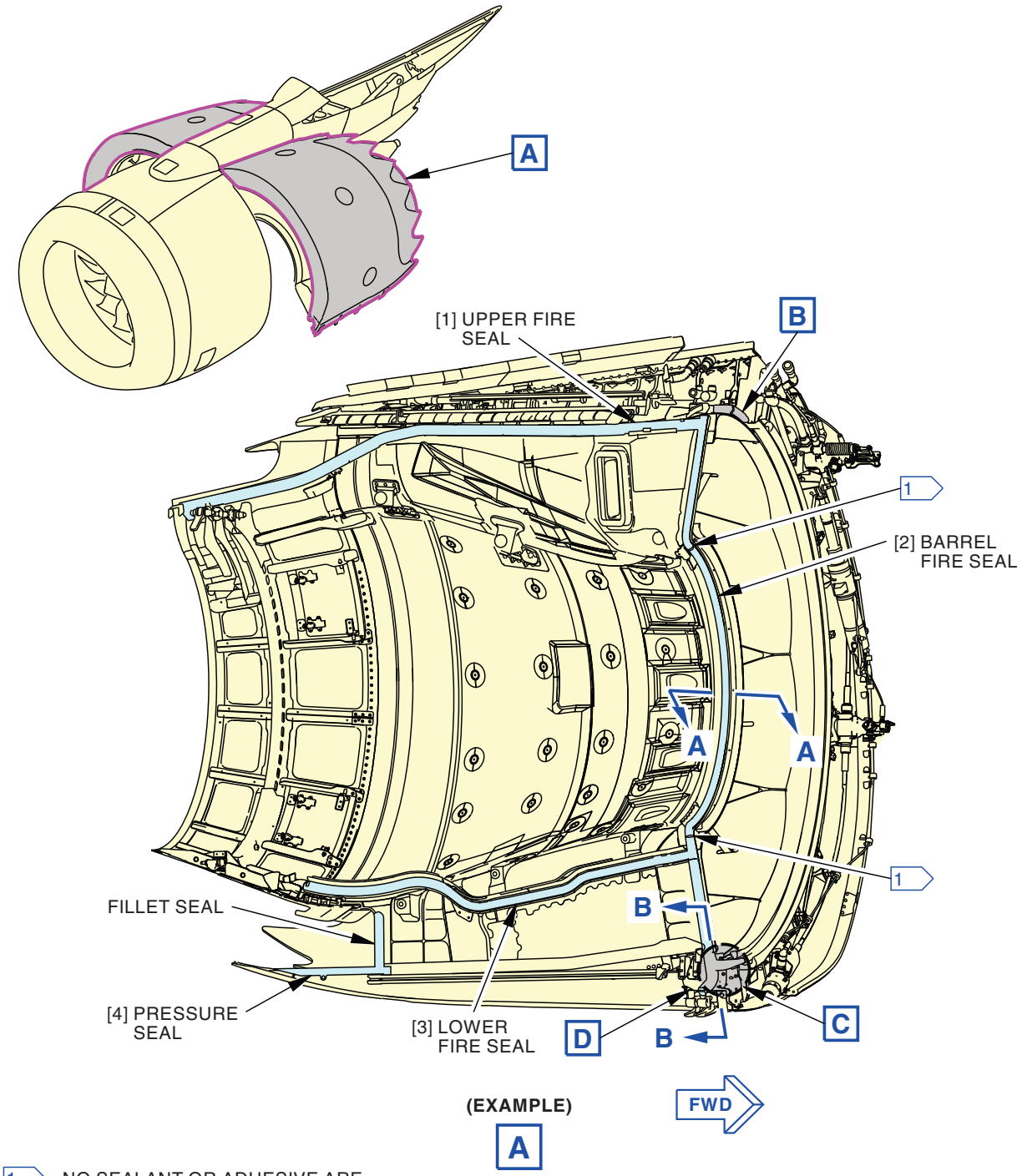
(4) Do these steps to remove the forward latch beam fire seal [38] from the right side thrust reverser (View C, Figure 404):

(a) Remove the two bolts [39], two washers [40], and two collars [41].

(b) Use the plastic scraper, STD-765, to remove the forward latch beam fire seal [38] from the structure.

END OF TASK

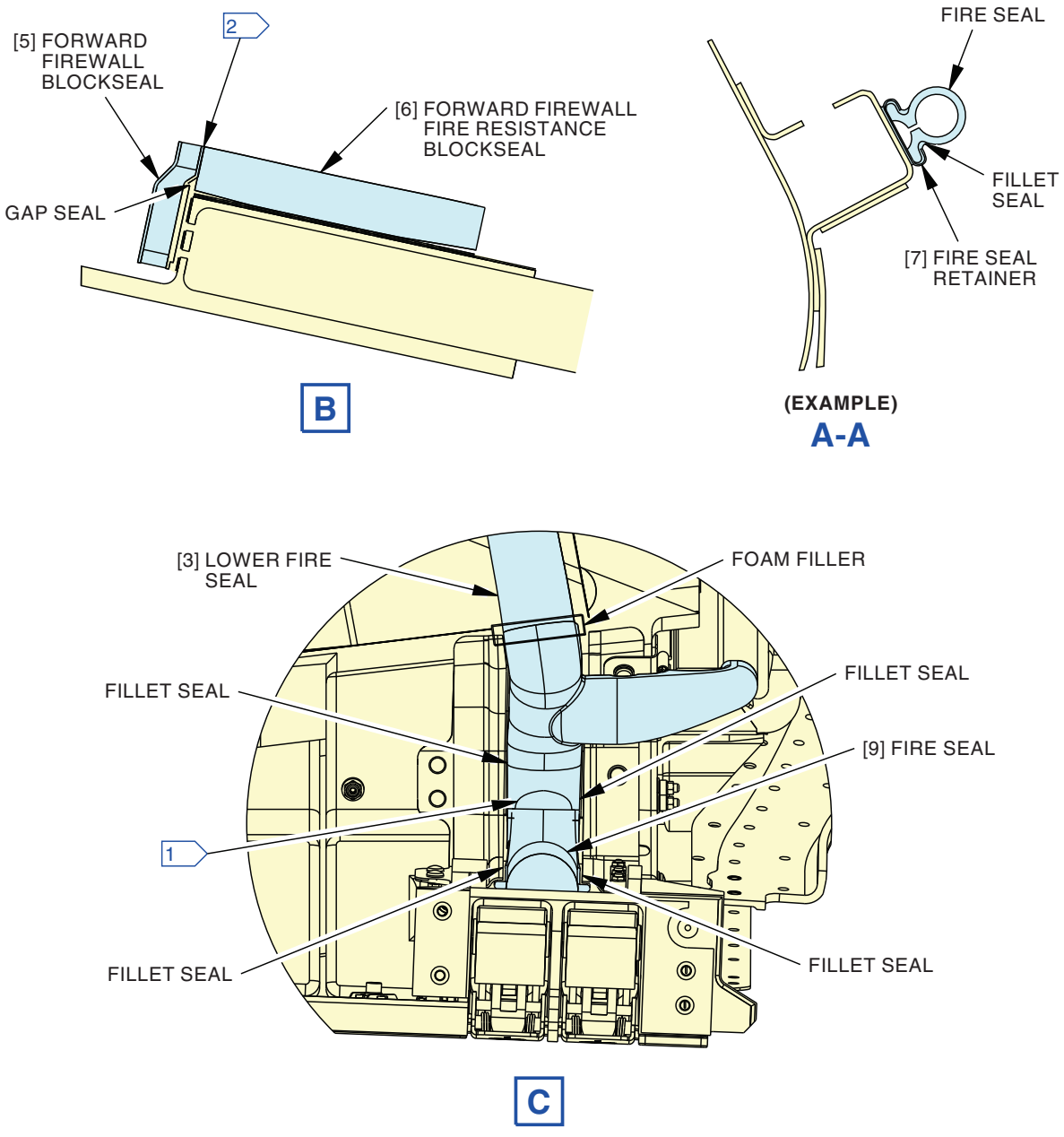
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2978007 S0000756445_V1

Left Side Thrust Reverser Fire Seal Installation
Figure 401/78-31-12-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

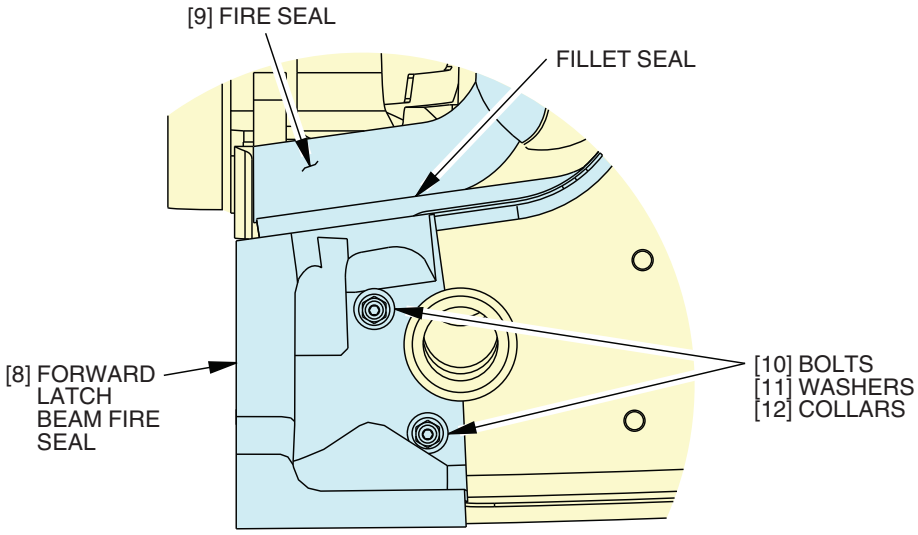


2 THE SURFACES MUST BE FLUSH BEFORE YOU APPLY ADHESIVE.

2355254 S00061516839_V3

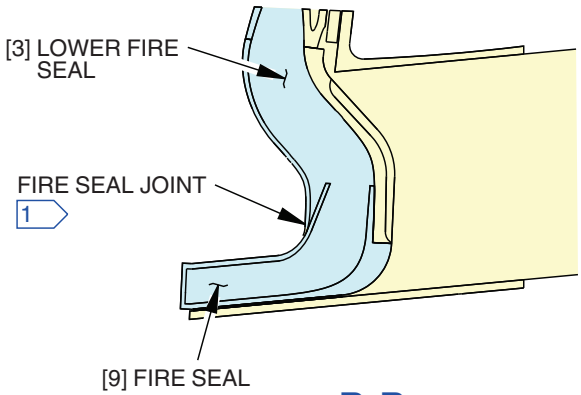
Left Side Thrust Reverser Fire Seal Installation
Figure 401/78-31-12-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

D

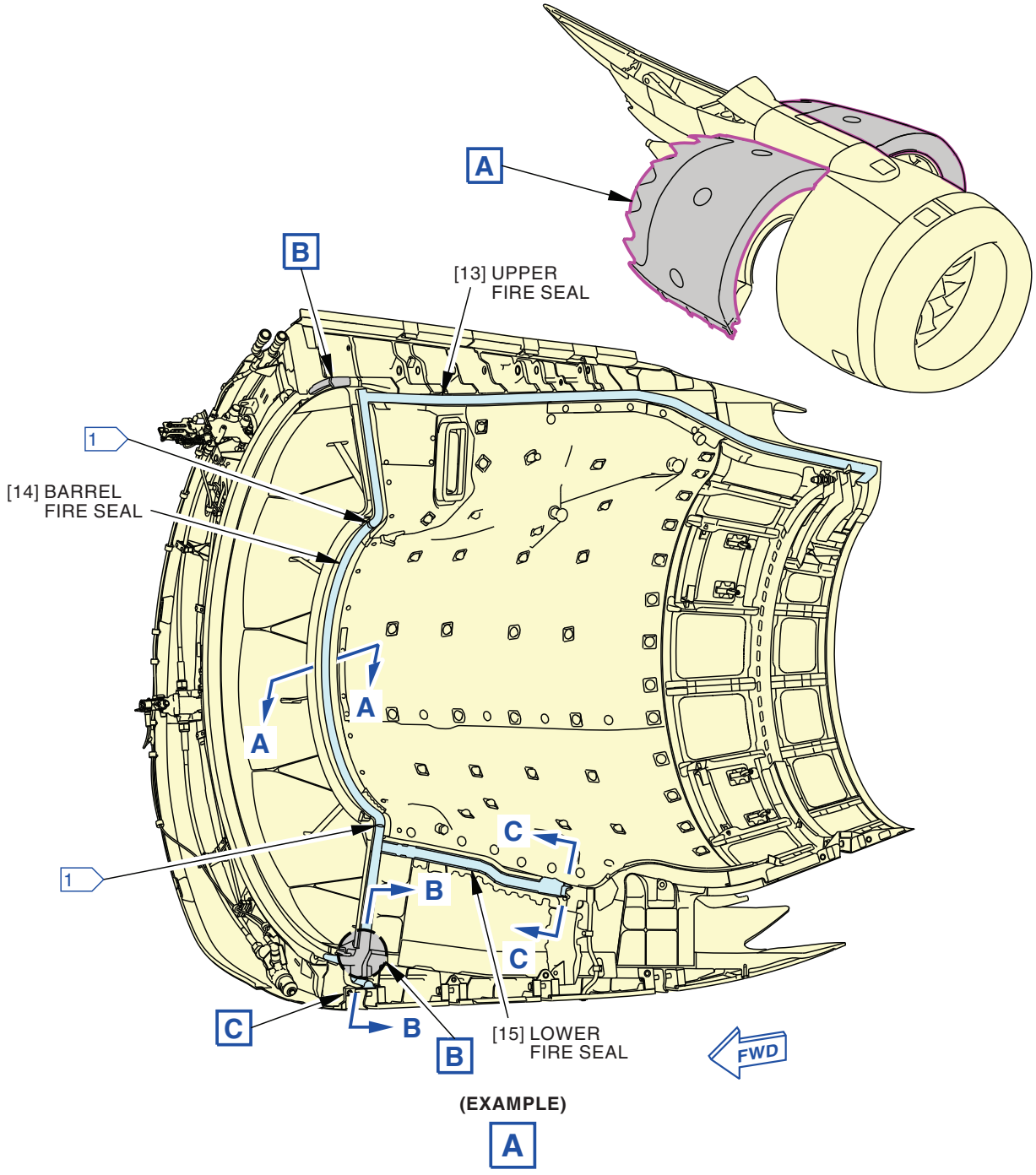


B-B

2968935 S0000748079_V2

Left Side Thrust Reverser Fire Seal Installation
Figure 401/78-31-12-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

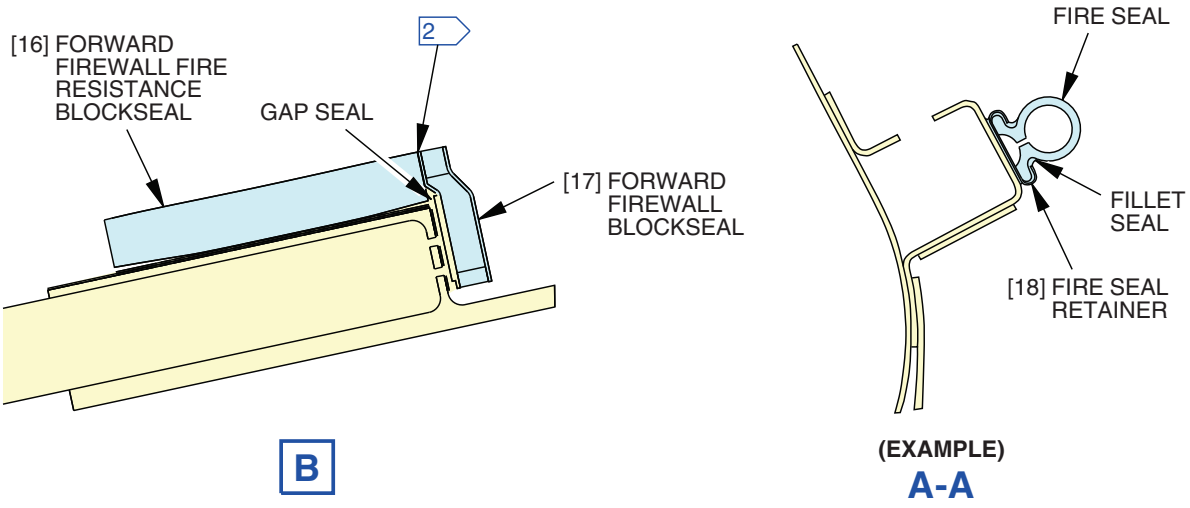


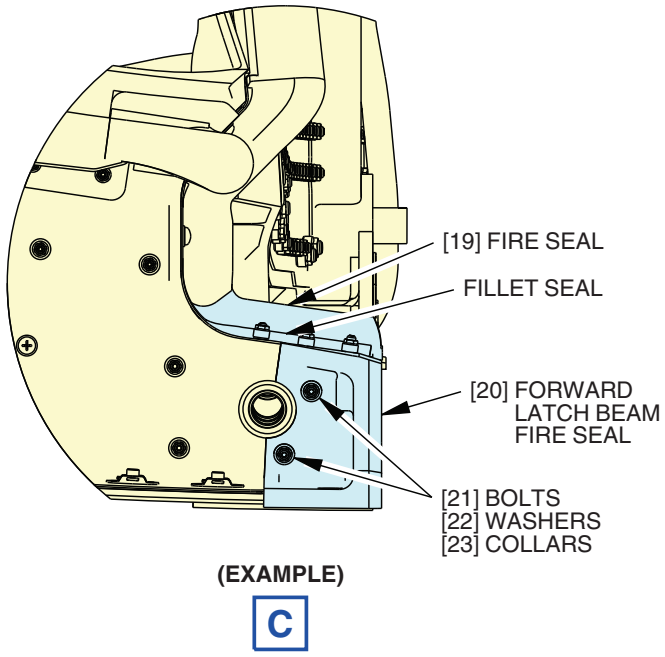
1 NO SEALANT OR ADHESIVE ARE ACCEPTABLE TO JOINT BETWEEN SEAL.

2355256 S00061516841_V5

Right Side Thrust Reverser Fire Seal Installation
Figure 402/78-31-12-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



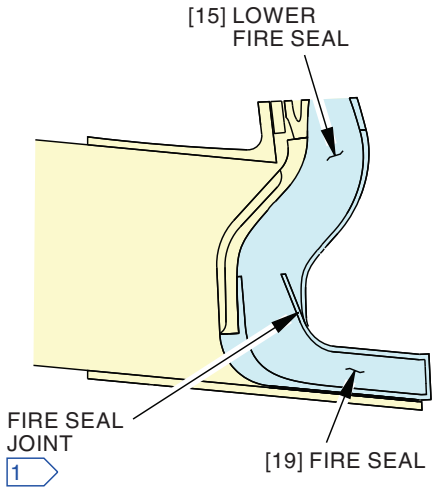


2 THE SURFACES MUST BE FLUSH BEFORE YOU APPLY ADHESIVE.

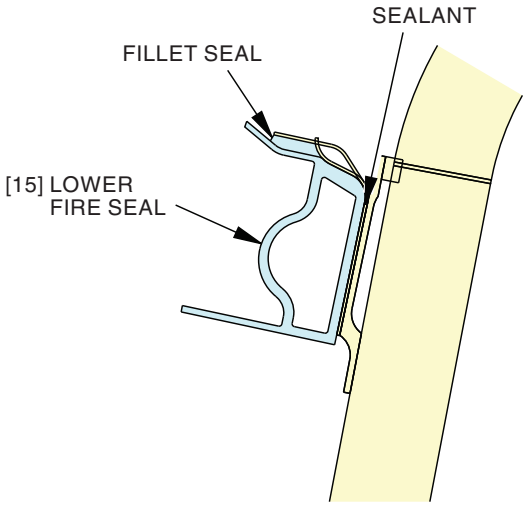
2355257 S00061516842_V5

Right Side Thrust Reverser Fire Seal Installation
Figure 402/78-31-12-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B

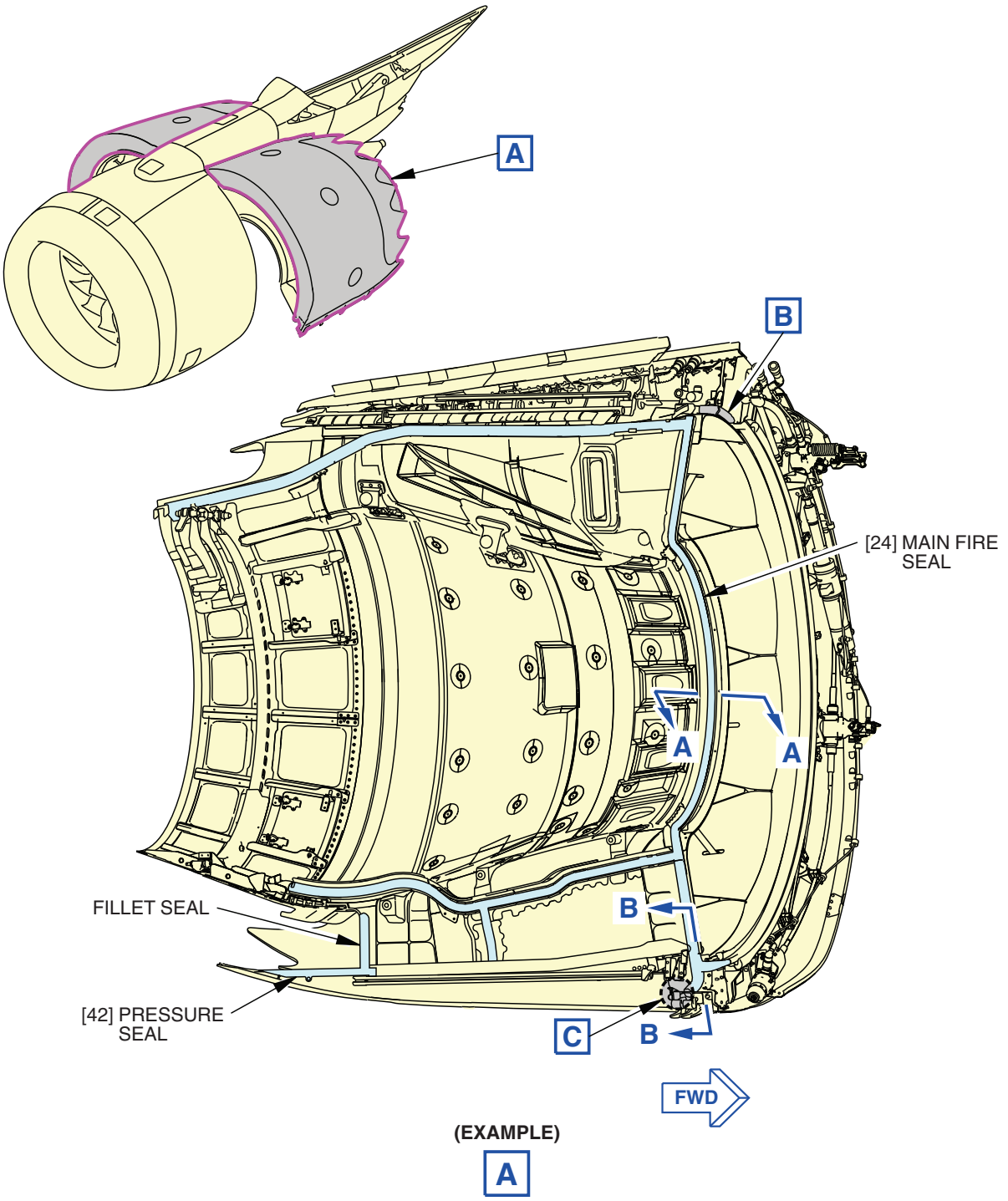


C-C

2968948 S0000748080_V1

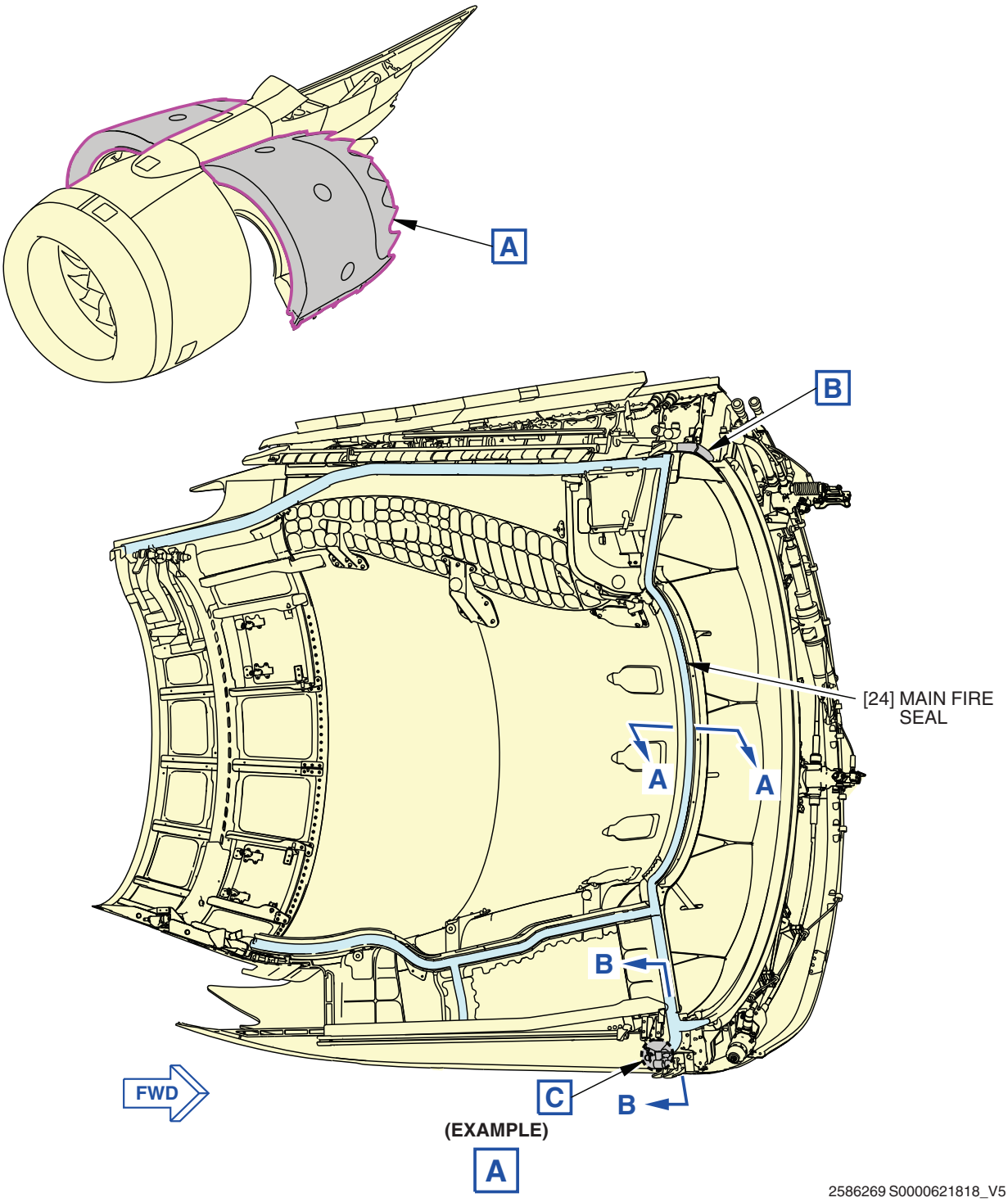
Right Side Thrust Reverser Fire Seal Installation
Figure 402/78-31-12-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



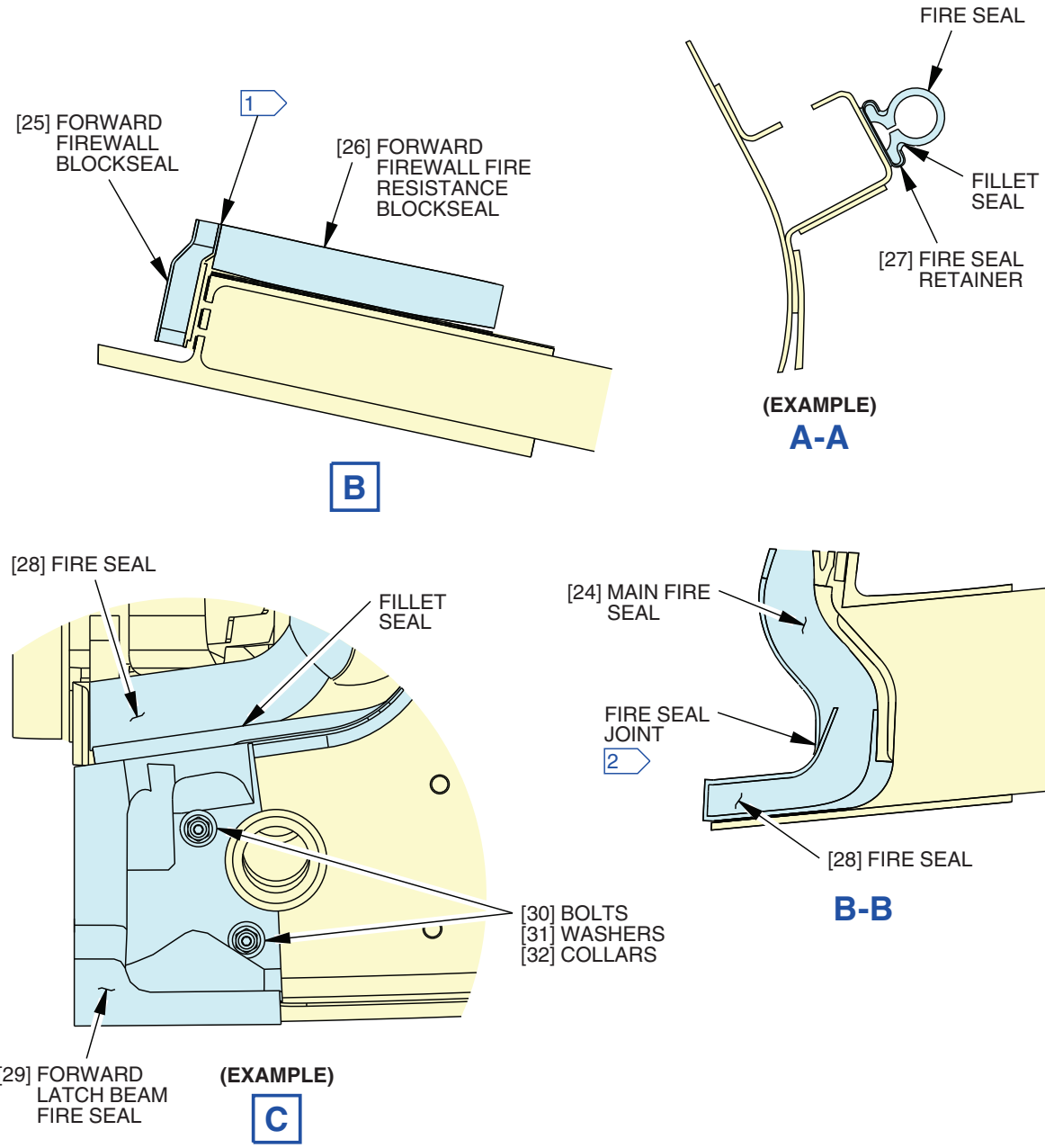
Left Side Thrust Reverser Fire Seal Installation
Figure 403/78-31-12-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Side Thrust Reverser Fire Seal Installation
Figure 403/78-31-12-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



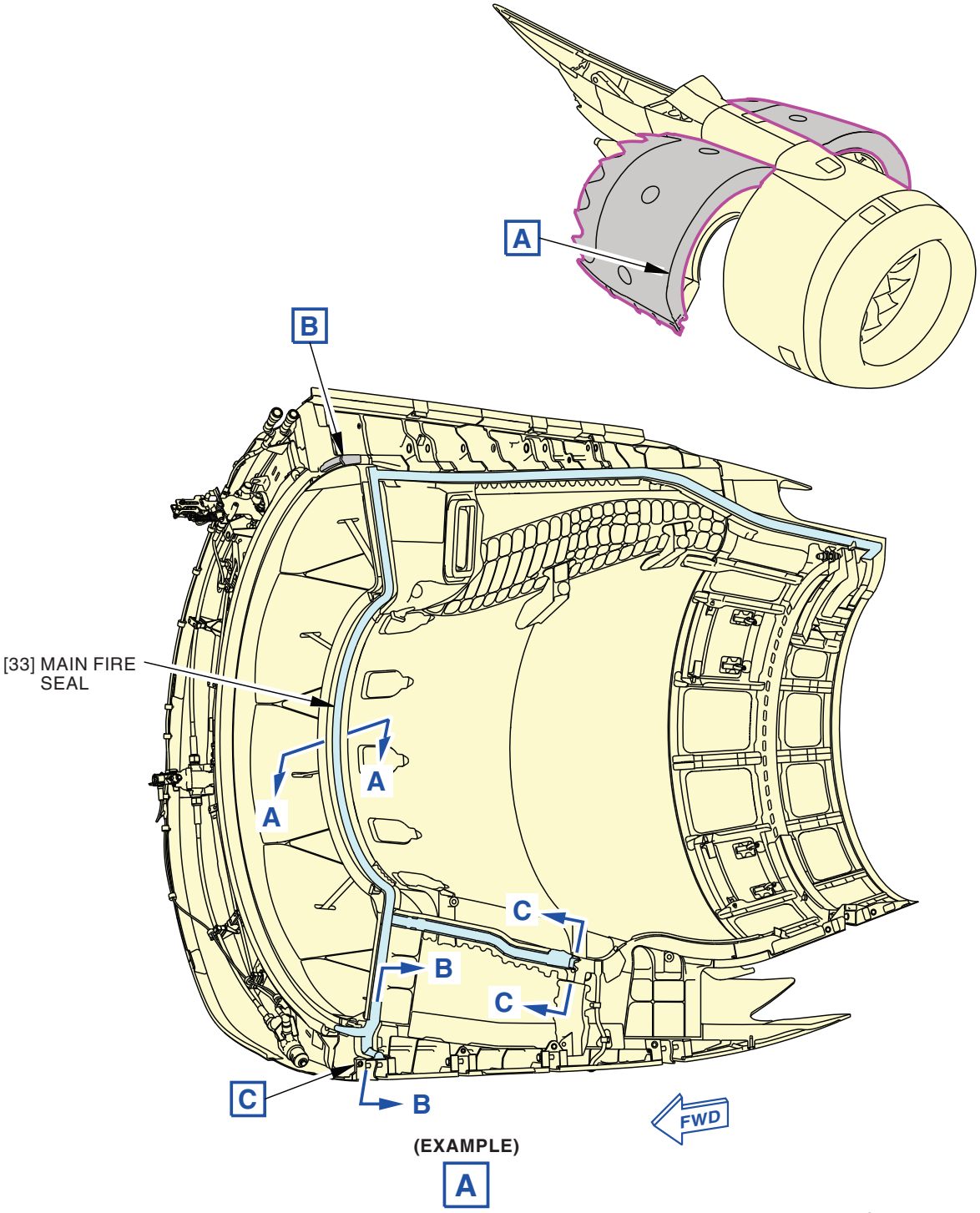
1 THE SURFACES MUST BE FLUSH BEFORE YOU APPLY ADHESIVE.

2 NO SEALANT OR ADHESIVE ARE ACCEPTABLE TO JOINT BETWEEN SEAL.

2948126 S0000725055_V3

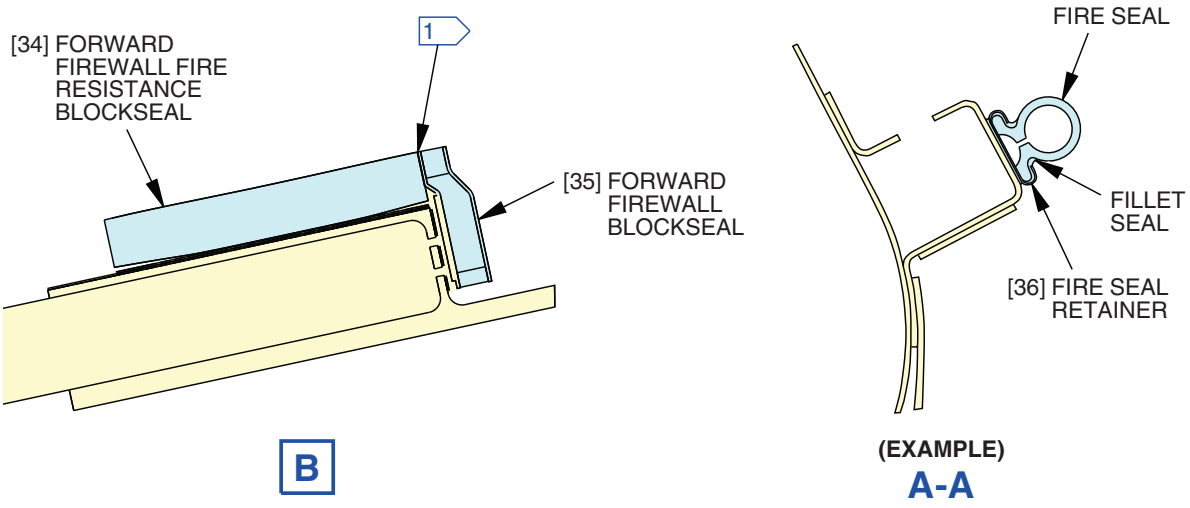
Left Side Thrust Reverser Fire Seal Installation
Figure 403/78-31-12-990-806-G00 (Sheet 3 of 3)

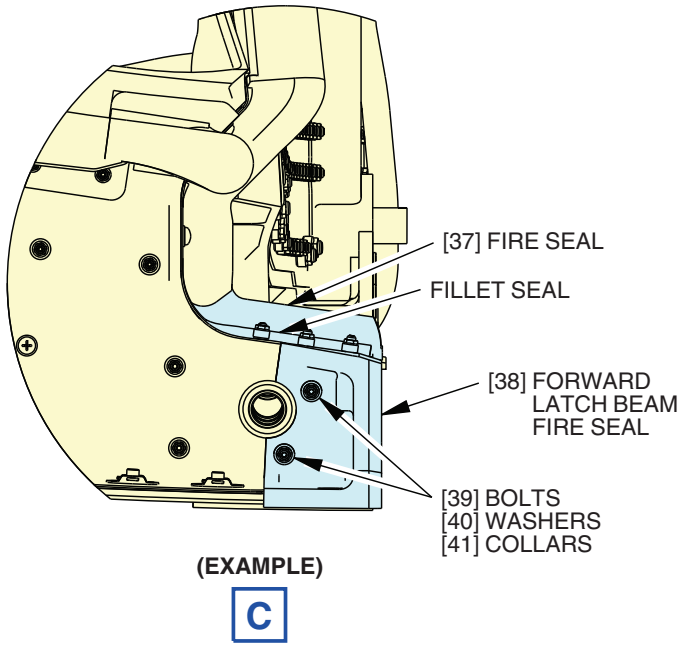
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Side Thrust Reverser Fire Seal Installation
Figure 404/78-31-12-990-807-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



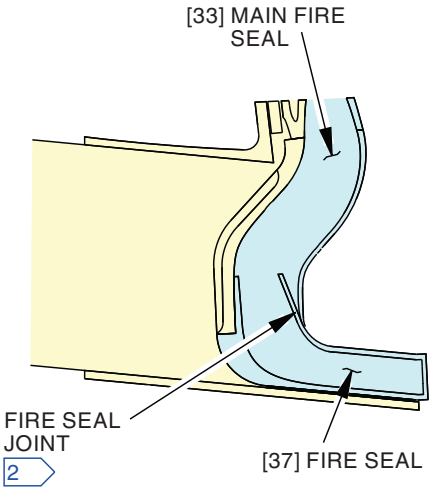


1 THE SURFACES MUST BE FLUSH BEFORE YOU APPLY ADHESIVE.

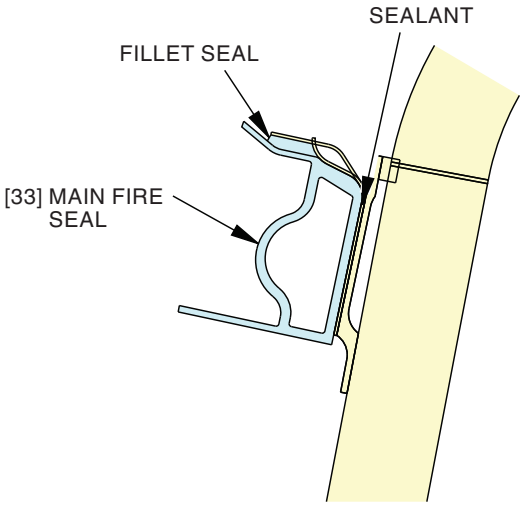
2948138 S0000725056_V3

Right Side Thrust Reverser Fire Seal Installation
Figure 404/78-31-12-990-807-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B



C-C

2 NO SEALANT OR ADHESIVE ARE ACCEPTABLE TO JOINT BETWEEN SEAL.

2969047 S0000748082_V1

Right Side Thrust Reverser Fire Seal Installation
Figure 404/78-31-12-990-807-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-12-400-801-G00

3. Fire Seal Installation

(Figure 401, Figure 402, Figure 403, Figure 404)

A. General

(1) This task provides instruction for the installation of the fire seals for the left or right thrust reverser.

(2) It is important that all sealants are applied correctly. The incorrect application of sealants can decrease the fire protection of the fire seals.

(3) For this task the lower fire seal/aero blockseal will be referred to as the blockseal.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-12-800-801-G00	Fire Seal Compression Check (P/B 601)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4127	Durometer - Shore A Part #: H-1000-A Supplier: 1GGG5
STD-765	Scraper - Plastic

D. Consumable Materials

Reference	Description	Specification
A00081	Adhesive - Silicone Rubber - RTV 106	BAC5010 Type 74
A50096	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type II
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Upper fire seal	78-31-12-05-165	PTF ALL
2	Barrel fire seal	78-31-12-05-160	PTF ALL
3	Lower fire seal	78-31-12-05-170	PTF ALL
4	Pressure seal	78-31-12-05-142	PTF ALL
13	Upper fire seal	78-31-12-06-095	PTF ALL
14	Barrel fire seal	78-31-12-06-090	PTF ALL
15	Lower fire seal	78-31-12-06-100	PTF ALL
24	Main fire seal	Not Specified	

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
33	Main fire seal	Not Specified	
42	Pressure seal	78-31-12-05-135	PTF ALL
		78-31-12-05-142	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Installation

SUBTASK 78-31-12-010-002-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

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1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Left Side Thrust Reverser Fire Seal Installation

SUBTASK 78-31-12-420-013-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Make sure that the new pressure seal [4] is installed on the left side thrust reverser (View A, Figure 401).

(a) If the new pressure seal [4] is installed, skip these steps.

(b) If the pressure seal [42] is installed, do these steps (Figure 403):

1) Remove the old pressure seal [42] refer to Paragraph I. (1) (TASK 78-31-12-000-801-G00).

2) Clean all the sealant residue from the inner wall with solvent, B00062.

3) Install the pressure seal [4] on the inner wall (Figure 401).

4) Apply RTV 106 adhesive, A00081, to bond the mounting surface of the pressure seal [4] to inner wall.

a) Completely seal all gaps under bond joint to ensure a continuous bond.

5) Apply RTV 106 adhesive, A00081, to form fillet seal around contour of pressure seal [4].

NOTE: Do not mix materials for the bonding and fillet sealing.

PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL

SUBTASK 78-31-12-420-001-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do these steps to install the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] on the left side thrust reverser (View B, Figure 401):

(a) Clean all the sealant residue from the fireseal retainer [7] with solvent, B00062.

(b) Slide the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] into the fireseal retainer [7] sections.

(c) Position the forward firewall blockseal [5] with end surface flush with the surface of forward firewall fire resistance blockseal [6].

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

(d) Make sure that the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] are flush.

(e) Apply sealant, A50096, along the forward firewall blockseal [5] and forward firewall fire resistance blockseal [6] to seal gaps.

NOTE: The indication of the fillet seal is not required.

SUBTASK 78-31-12-420-002-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Do these steps to install the upper fire seal [1] on the left side thrust reverser (Figure 401):

NOTE: The joint between the upper fire seal and barrel fire seal slips. Do not apply sealant in this area.

(a) Clean all the sealant residue from the fireseal retainer [7] with solvent, B00062.

(b) Compress the upper fire seal [1] foot into the fireseal retainer [7] along the length of the fireseal retainer [7].

1) Use a plastic scraper, STD-765, with a twisting motion to push the upper fire seal [1] foot into the fireseal retainer [7].

(c) Do these steps to seal the edges of the upper fire seal [1]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the upper fire seal [1] and fireseal retainer [7].

3) Completely fill all gaps between the fireseal retainer [7] sections to ensure a continuous seal.

SUBTASK 78-31-12-400-001-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Do these steps to install the barrel fire seal [2] on the left side thrust reverser (Figure 401):

NOTE: The joints between the barrel fire seal, upper fire seal, and lower fire seal slips. Do not apply sealant in these areas.

(a) Clean all the sealant residue from the fireseal retainer [7] with solvent, B00062.

(b) Compress the barrel fire seal [2] foot into the fireseal retainer [7] along the length of the fireseal retainer [7].

1) Use a plastic scraper, STD-765, with a twisting motion to push the barrel fire seal [2] foot into the fireseal retainer [7].

(c) Do these steps to seal the edges of the barrel fire seal [2]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

1) Clean the areas to be sealed with solvent, B00062.
2) Apply RTV 106 adhesive, A00081, to form fillet seal between the barrel fire seal [2] and fireseal retainer [7].
3) Completely fill all gaps between fireseal retainer [7] sections to ensure a continuous seal.

SUBTASK 78-31-12-400-002-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Do these steps to install the lower fire seal [3] on the left side thrust reverser (Figure 401):

NOTE: The joint between the barrel fire seal and lower fire seal slips. Do not apply sealant in this area.

(a) Clean all the sealant residue from the fireseal retainer [7] with solvent, B00062.

(b) Compress the lower fire seal [3] foot into the fireseal retainer [7] along the length of the fireseal retainer [7].

1) Use a plastic scraper, STD-765, with a twisting motion to push the lower fire seal [3] foot into the fireseal retainer [7].

(c) Do these steps to seal the edges of the lower fire seal [3]:

1) Clean the areas to be sealed with solvent, B00062.
2) Apply RTV 106 adhesive, A00081, to form fillet seal between the lower fire seal [3] and fireseal retainer [7].
3) Completely fill all gaps between the fireseal retainer [7] sections to ensure a continuous seal.

(d) Apply RTV 106 adhesive, A00081, to mounting surface of the foam filler.

1) Completely seal all gaps under bond joint to ensure a continuous bond.

SUBTASK 78-31-12-420-014-G00

(6) Do these steps to install the fire seal [9] on the left side thrust reverser (View B-B and View C, Figure 401):

(a) Clean all the sealant residue from the fireseal retainer [7] with solvent, B00062.

(b) Compress the fire seal [9] foot into the fireseal retainer [7] along the length of the fireseal retainer [7].

1) Use a plastic scraper, STD-765, with a twisting motion to push the fire seal [9] foot into the fireseal retainer [7].

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-400-003-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Do these steps to install the forward latch beam fire seal [8] on the left side thrust reverser (View D, Figure 401):

(a) Apply RTV 106 adhesive, A00081, to the mating surface of the forward latch beam fire seal [8].

(b) Install the two bolts [10], washers [11], and collars [12].

1) Hand tighten the two collars [12] until two washers [11] is snug against the surface of seal.

2) Tighten the two collars [12] one half revolution.

3) Make sure that there is minimum amount of the squeeze-out.

(c) Make sure that each fastener is not loose.

(d) Make sure that the two bolts [10] show thread protrusion through the two collars [12].

SUBTASK 78-31-12-390-001-G00

(8) Let all the sealant cure as follows:

(a) Let RTV 106 adhesive, A00081, cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

WARNING

DO NOT USE DECREASED CURE TIME FOR A SERVICE BULLETIN WITH AN AIRWORTHINESS DIRECTIVE AGAINST IT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

1) To decrease the cure time, do this:

NOTE: The cure time may be decreased if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

NOTE: The cure time is dependent on both heat and humidity conditions. Cure time decreases with higher heat, only if the local humidity remains high. In low humidity conditions, the seal may take more time to cure.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

c) After 4 hours, do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

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<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

d) If after 4 hours the seal is not cured, check again every 4 hours until the durometer verifies that the sealant is cured. Do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

(b) Let sealant, A50096, cure at 72°F (22°C) to 82°F (28°C) for 14 hours.

SUBTASK 78-31-12-750-001-G00

(9) Do this task: Fire Seal Compression Check, TASK 78-31-12-800-801-G00.

J. Right Side Thrust Reverser Fire Seal Installation

SUBTASK 78-31-12-420-015-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do these steps to install the forward firewall fire resistance blockseal [16] and forward firewall blockseal [17] on the right side thrust reverser (View B, Figure 402):

(a) Clean all the sealant residue from the fireseal retainer [18] with solvent, B00062.

(b) Slide the forward firewall fire resistance blockseal [16] and forward firewall blockseal [17] into the fireseal retainer [18] sections.

(c) Position the forward firewall blockseal [17] with end surface flush with the surface of forward firewall fire resistance blockseal [16].

(d) Make sure that the forward firewall fire resistance blockseal [16] and forward firewall blockseal [17] are flush.

(e) Apply sealant, A50096, along the forward firewall fire resistance blockseal [16] and forward firewall blockseal [17] to seal gaps.

NOTE: The indication of the fillet seal is not required.

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-420-016-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do these steps to install the upper fire seal [13] on the right side thrust reverser (Figure 402):

NOTE: The joint between the upper fire seal and barrel fire seal slips. Do not apply sealant in this area.

(a) Clean all the sealant residue from the fireseal retainer [18] with solvent, B00062.

(b) Compress the upper fire seal [13] foot into the fireseal retainer [18] along the length of the fireseal retainer [18].

1) Use a plastic scraper, STD-765, with a twisting motion to push the upper fire seal [13] foot into the fireseal retainer [18].

(c) Do these steps to seal the edges of the upper fire seal [13]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the upper fire seal [13] and fireseal retainer [18].

3) Completely fill all gaps between the fireseal retainer [18] sections to ensure a continuous seal.

SUBTASK 78-31-12-420-020-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Do these steps to install the barrel fire seal [14] on the right side thrust reverser (Figure 402):

NOTE: The joints between the barrel fire seal, upper fire seal, and lower fire seal slips. Do not apply sealant in these areas.

(a) Clean all the sealant residue from the fireseal retainer [18] with solvent, B00062.

(b) Compress the barrel fire seal [14] foot into the fireseal retainer [18] along the length of the fireseal retainer [18].

1) Use a plastic scraper, STD-765, with a twisting motion to push the barrel fire seal [14] foot into the fireseal retainer [18].

(c) Do these steps to seal the edges of the barrel fire seal [14]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the barrel fire seal [14] and fireseal retainer [18].

3) Completely fill all gaps between the fireseal retainer [18] sections to ensure a continuous seal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-420-021-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Do these steps to install the lower fire seal [15] on the right side thrust reverser (Figure 402):

NOTE: The joint between the barrel fire seal and lower fire seal slips. Do not apply sealant in this area.

(a) Clean all the sealant residue from the fireseal retainer [18] with solvent, B00062.

(b) Compress the lower fire seal [15] foot into the fireseal retainer [18] along the length of the fireseal retainer [18].

1) Use a plastic scraper, STD-765, with a twisting motion to push the lower fire seal [15] foot into the fireseal retainer [18].

(c) Do these steps to seal the edges of the lower fire seal [15]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the lower fire seal [15] and fireseal retainer [18].

3) Completely fill all gaps between the fireseal retainer [18] sections to ensure a continuous seal.

(d) Apply RTV 106 adhesive, A00081, to mounting surface of the foam filler.

1) Completely seal all gaps under bond joint to ensure a continuous bond.

(e) Apply RTV 106 adhesive, A00081, or sealant, A50096, between the lower fire seal [15] and fire seal support (View C-C, Figure 402).

SUBTASK 78-31-12-420-022-G00

(5) Do these steps to install the fire seal [19] on the right side thrust reverser (View B-B, Figure 402):

(a) Clean all the sealant residue from the fireseal retainer [18] with solvent, B00062.

(b) Compress the fire seal [19] foot into the fireseal retainer [18] along the length of the fireseal retainer [18].

1) Use a plastic scraper, STD-765, with a twisting motion to push the fire seal [19] foot into the fireseal retainer [18].

(c) Apply RTV 106 adhesive, A00081, to mounting surface of fire seal [19] and fireseal retainer [18].

(d) Make sure that there are non-continuous gaps greater than 0.03 in. (0.76 mm) between the fire seal flange and Thrust Reverser structure.

(e) If continuous gap or non-continuous gap greater than 0.03 in. (0.76 mm) exist, trim 0.100 in. (2.54 mm) maximum of fire seal [19] at the contact of lower fire seal [15].

1) Apply lightly coat RTV 106 adhesive, A00081, on the trimmed edge.

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-420-023-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Do these steps to install the forward latch beam fire seal [20] on the right side thrust reverser (View C, Figure 402):

(a) Apply RTV 106 adhesive, A00081, to the mating surface of the forward latch beam fire seal [20].

(b) Install the two bolts [21], washers [22], and collars [23].

1) Hand tighten the two collars [23] until two washers [22] is snug against the surface of seal.

2) Tighten the two collars [23] one half revolution.

3) Make sure that there is minimum amount of the squeeze-out.

(c) Make sure that each fastener is not loose.

(d) Make sure that the two bolts [21] show thread protrusion through the two collars [23].

SUBTASK 78-31-12-420-017-G00

(7) Let all the sealant cure as follows:

(a) Let RTV 106 adhesive, A00081, cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

WARNING

DO NOT USE DECREASED CURE TIME FOR A SERVICE BULLETIN WITH AN AIRWORTHINESS DIRECTIVE AGAINST IT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

1) To decrease the cure time to do this:

NOTE: The cure time may be decreased to 14 hours if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

NOTE: The cure time is dependent on both heat and humidity conditions. Cure time decreases with higher heat, only if the local humidity remains high. In low humidity conditions, the seal may take more time to cure.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

c) After 4 hours, do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

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PTF ALL; AIRPLANES WITH 3 PIECE FIRE SEAL (Continued)

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

d) If after 4 hours the seal is not cured, check again every 4 hours until the durometer verifies that the sealant is cured. Do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

(b) Let sealant, A50096, cure at 72°F (22°C) to 82°F (28°C) for 14 hours.

SUBTASK 78-31-12-750-008-G00

(8) Do this task: Fire Seal Compression Check, TASK 78-31-12-800-801-G00.

PTF ALL

K. Left Side Thrust Reverser Fire Seal Installation

SUBTASK 78-31-12-420-024-G00

(1) Check the type of the main fire seal [24] is installed.

(a) If the main fire seal [24] is replaced with new upper fire seal [1], new barrel fire seal [2], and new lower fire seal [3], refer to Paragraph I.

(b) If the main fire seal [24] is replaced with new main fire seal [24], continue the procedure.

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL

SUBTASK 78-31-12-420-003-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do these steps to install the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] on the left side thrust reverser (View B, Figure 403):

(a) Clean all the sealant residue from the fireseal retainer [27] with solvent, B00062.

(b) Slide the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] into the fireseal retainer [27] sections.

(c) Position the forward fireseal blockseal [25] with end surface flush with the surface of forward firewall fire resistance blockseal [26].

(d) Make sure that the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] are flush.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

(e) Apply sealant, A50096, along the forward fireseal blockseal [25] and forward firewall fire resistance blockseal [26] to seal gaps.

NOTE: The indication of the fillet seal is not required.

SUBTASK 78-31-12-420-011-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Do these steps to install the main fire seal [24] on the left side thrust reverser (Figure 403):

(a) Clean all the sealant residue from the fireseal retainer [27] with solvent, B00062.

(b) Compress the main fire seal [24] foot into the fireseal retainer [27] along the length of the fireseal retainer [27].

1) Use a plastic scraper, STD-765, with a twisting motion to push the main fire seal [24] foot into the fireseal retainer [27].

(c) Do these steps to seal the edges of the main fire seal [24]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the main fire seal [24] and fireseal retainer [27].

3) Completely fill all gaps between the fireseal retainer [27] sections to ensure a continuous seal.

SUBTASK 78-31-12-420-004-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Do these steps to install the fire seal [28] on the left side thrust reverser (View B-B, Figure 403):

NOTE: The joints between the main fire seal fire seal slips. Do not apply sealant in these areas.

(a) Clean all the sealant residue from the fireseal retainer [27] with solvent, B00062.

(b) Compress the fire seal [28] foot into the fireseal retainer [27] along the length of the fireseal retainer [27].

1) Use a plastic scraper, STD-765, with a twisting motion to push the fire seal [28] foot into the fireseal retainer [27].

(c) Apply RTV 106 adhesive, A00081, to form fillet seal between the fire seal [28] and fireseal retainer [27].

(d) Completely fill all gaps between the fireseal retainer [27] sections to ensure a continuous seal.

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PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-420-018-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Do these steps to install the forward latch beam fire seal [29] on the left side thrust reverser (View C, Figure 403):

(a) Apply RTV 106 adhesive, A00081, to the mating surface of the forward latch beam fire seal [29].

(b) Install the two bolts [30], washers [31], and collars [32].

1) Hand tighten the two collars [32] until two washers [31] is snug against the surface of seal.

2) Tighten the two collars [32] one half revolution.

3) Make sure that there is minimum amount of the squeeze-out.

(c) Make sure that each fastener is not loose.

(d) Make sure that the two washers [31] show thread protrusion through the two collars [32].

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL COMPOSITE INNER WALL

SUBTASK 78-31-12-420-026-G00

(6) Do these steps to install the pressure seal [42] on the left side thrust reverser (View A, Figure 403):

(a) Clean all the sealant residue from the inner wall with solvent, B00062.

(b) Install the pressure seal [42] on the inner wall.

(c) Apply RTV 106 adhesive, A00081, to bond the mounting surface of the pressure seal [42] to inner wall.

1) Completely seal all gaps under bond joint to ensure a continuous bond.

(d) Apply RTV 106 adhesive, A00081, to form fillet seal around contour of pressure seal [42].

NOTE: Do not mix materials for the bonding and fillet sealing.

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL

SUBTASK 78-31-12-390-004-G00

(7) Let all the sealant cure as follows:

(a) Let RTV 106 adhesive, A00081, cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

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PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

WARNING DO NOT USE DECREASED CURE TIME FOR A SERVICE BULLETIN WITH AN AIRWORTHINESS DIRECTIVE AGAINST IT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

1) To decrease the cure time to do this:

NOTE: The cure time may be decreased to 14 hours if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

NOTE: The cure time is dependent on both heat and humidity conditions. Cure time decreases with higher heat, only if the local humidity remains high. In low humidity conditions, the seal may take more time to cure.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

c) After 4 hours, do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

d) If after 4 hours the seal is not cured, check again every 4 hours until the durometer verifies that the sealant is cured. Do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

(b) Let sealant, A50096 cure at 72°F (22°C) to 82°F (28°C) for 14 hours.

SUBTASK 78-31-12-750-006-G00

(8) Do this task: Fire Seal Compression Check, TASK 78-31-12-800-801-G00.

PTF ALL

L. Right Thrust Reverser Fire Seal Installation

SUBTASK 78-31-12-420-025-G00

(1) Check the type of the main fire seal [33] is installed.

(a) If the main fire seal [33] is replaced with new upper fire seal [13], new barrel fire seal [14], and new lower fire seal [15], refer to Paragraph J.

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(b) If the new main fire seal [33] is replaced with main fire seal [33], continue the procedure.

PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL

SUBTASK 78-31-12-420-005-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do these steps to install the forward firewall fire resistance blockseal [34] and forward firewall blockseal [35] on the right side thrust reverser (View B, Figure 404):

(a) Clean all the sealant residue from the fireseal retainer [27] with solvent, B00062.

(b) Slide the forward firewall blockseal [35] and forward firewall fire resistance blockseal [34] into the fireseal retainer [36] sections.

(c) Position the forward firewall blockseal [35] with end surface flush with the surface of forward firewall fire resistance blockseal [34].

(d) Make sure that the forward firewall blockseal [35] and forward firewall fire resistance blockseal [34] are flush.

(e) Apply sealant, A50096, along the forward firewall fire resistance blockseal [34] and forward firewall blockseal [35] to seal gaps.

NOTE: The indication of the fillet seal is not required.

SUBTASK 78-31-12-420-012-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Do these steps to install the main fire seal [33] on the right side thrust reverser (Figure 404):

(a) Clean all the sealant residue from the fireseal retainer [36] with solvent, B00062.

(b) Compress the main fire seal [33] foot into the fireseal retainer [36] along the length of the fireseal retainer [36].

1) Use a plastic scraper, STD-765, with a twisting motion to push the main fire seal [33] foot into the fireseal retainer [36].

(c) Do these steps to seal the edges of the main fire seal [33]:

1) Clean the areas to be sealed with solvent, B00062.

2) Apply RTV 106 adhesive, A00081, to form fillet seal between the main fire seal [33] and fireseal retainer [36].

3) Completely fill all gaps between the fireseal retainer [36] sections to ensure a continuous seal.

(d) Apply RTV 106 adhesive, A00081, or sealant, A50096, between the main fire seal [33] and fire seal support (View C-C, Figure 404).

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PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-420-006-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(4) Do these steps to install the fire seal [37] on the right side thrust reverser (View B-B, Figure 404):

NOTE: The joints between the main fire seal fire seal slips. Do not apply sealant in these areas.

(a) Clean all the sealant residue from the fireseal retainer [36] with solvent, B00062.

(b) Compress the fire seal [37] foot into the fireseal retainer [27] along the length of the fireseal retainer [36].

1) Use a plastic scraper, STD-765, with a twisting motion to push the fire seal [37] foot into the fireseal retainer [27].

(c) Apply RTV 106 adhesive, A00081, to the mating surface of the fire seal [37] and fireseal retainer [36].

(d) Make sure that there are non-continuous gaps greater than 0.03 in. (0.76 mm) between the fire seal flange and thrust reverser structure.

(e) If continuous gap or non-continuous gap greater than 0.03 in. (0.76 mm) exist, trim 0.100 in. (2.54 mm) maximum of fire seal [37] at the contact of main fire seal [33].

1) Apply lightly coat RTV 106 adhesive, A00081, on the trimmed edge.

SUBTASK 78-31-12-420-019-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Do these steps to install the forward latch beam fire seal [38] on the right side thrust reverser (View C, Figure 404):

(a) Apply RTV 106 adhesive, A00081, to the mating surface of the forward latch beam fire seal [38].

(b) Install the two bolts [39], washers [40], and collars [41].

1) Hand tighten the two collars [41] until two washers [40] is snug against the surface of seal.

2) Tighten the two collars [41] one half revolution.

3) Make sure that there is minimum amount of the squeeze-out.

(c) Make sure that each fastener is not loose.

(d) Make sure that the two washers [40] show thread protrusion through the two collars [41].

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PTF ALL; AIRPLANES WITH 1 PIECE FIRE SEAL (Continued)

SUBTASK 78-31-12-390-003-G00

(6) Let all the sealant cure to the time limits as follows:

(a) Let RTV 106 adhesive, A00081, cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

WARNING DO NOT USE DECREASED CURE TIME FOR A SERVICE BULLETIN WITH AN AIRWORTHINESS DIRECTIVE AGAINST IT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

1) To decrease the cure time to do this:

NOTE: The cure time may be decreased to 14 hours if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

NOTE: The cure time is dependent on both heat and humidity conditions. Cure time decreases with higher heat, only if the local humidity remains high. In low humidity conditions, the seal may take more time to cure.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

c) After 4 hours, do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

d) If after 4 hours the seal is not cured, check again every 4 hours until the durometer verifies that the sealant is cured. Do these steps:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127, indication is more than 30, the seal is cured.

<3> Remove the test strip.

NOTE: The durometer indication of more than 30 is required to make sure the seal is cured.

(b) Let sealant, A50096, cure at 72°F (22°C) to 82°F (28°C) for 14 hours.

SUBTASK 78-31-12-750-005-G00

(7) Do this task: Fire Seal Compression Check, TASK 78-31-12-800-801-G00.

PTF ALL

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M. Put the Airplane Back to its Usual Condition

SUBTASK 78-31-12-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

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FIRE SEAL - INSPECTION/CHECK

1. General

A. This procedure has three tasks:

(1) A visual inspection of the fire seal.

(2) A detailed inspection of the fire seal.

(3) A compression check of the fire seal.

TASK 78-31-12-200-801-G00

2. Fire Seal Inspection (Visual Check)

(Figure 601)

A. General

(1) The fire seal is on the inner surface of each thrust reverser on an engine.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Preparation

SUBTASK 78-31-12-010-003-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Procedure

SUBTASK 78-31-12-210-001-G00

(1) Look for missing fire seal and for obvious damage.

(a) If you find damaged or missing fire seal, do this task: Fire Seal Inspection (Detailed), TASK 78-31-12-200-802-G00.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-12-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

TASK 78-31-12-200-802-G00

3. Fire Seal Inspection (Detailed)

(Figure 601)

A. General

(1) The fire seal is on the inner surface of each thrust reverser on an engine.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-12 P/B 401	FIRE SEAL - REMOVAL/INSTALLATION
78-31-12-300-801-G00	Fire Seal Repair (P/B 801)
78-31-12-400-801-G00	Fire Seal Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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E. Preparation

SUBTASK 78-31-12-010-004-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Procedure

SUBTASK 78-31-12-220-001-G00

(1) Visually examine the fire seal for the following types of damage:

(a) Cuts, frayed material, missing or loose fire seal, and missing sealant.

SUBTASK 78-31-12-960-002-G00

(2) If you find damage to the AFT area of the top section of the fire seal, replace the fire seal (Figure 602, TASK 78-31-12-300-801-G00).

SUBTASK 78-31-12-960-001-G00

(3) If you find damage or missing fire seal, replace the fire seal (PAGEBLOCK 78-31-12/401).

SUBTASK 78-31-12-211-001-G00

(4) If there is damage to the insulation at the fire seal at the upper bifurcation, do this task: Fire Seal Inspection (Visual Check), TASK 78-31-12-200-801-G00.

SUBTASK 78-31-12-390-002-G00

(5) If sealant is missing, replace the sealant (Fire Seal Installation, TASK 78-31-12-400-801-G00).

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SUBTASK 78-31-12-220-002-G00

(6) Visually examine the fire seal for loose, missing, or damaged fire seal retainers.

(a) If there is loose, missing, or damaged fire seal retainers, tighten or replace the fire seal retainers.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-12-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

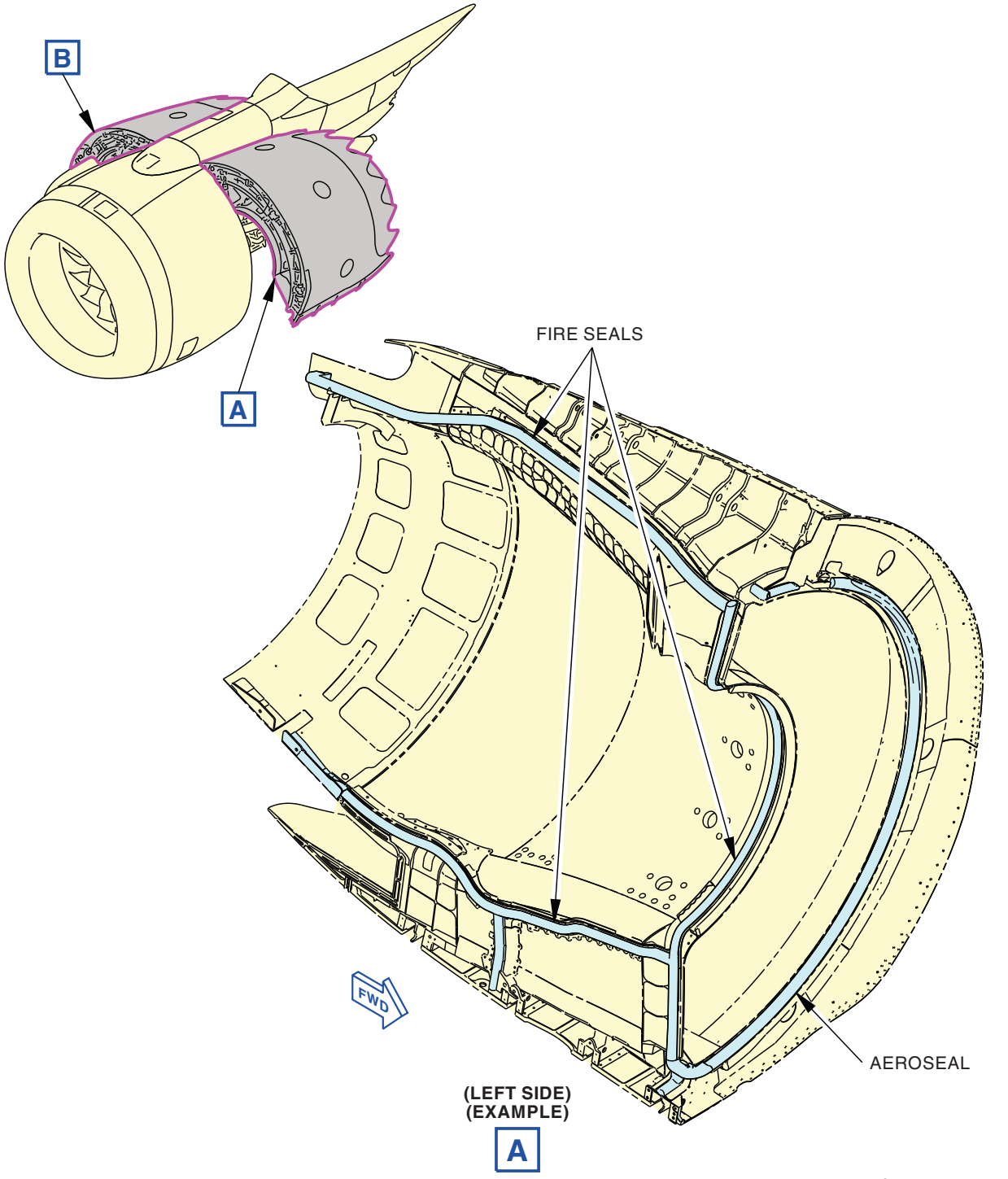
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

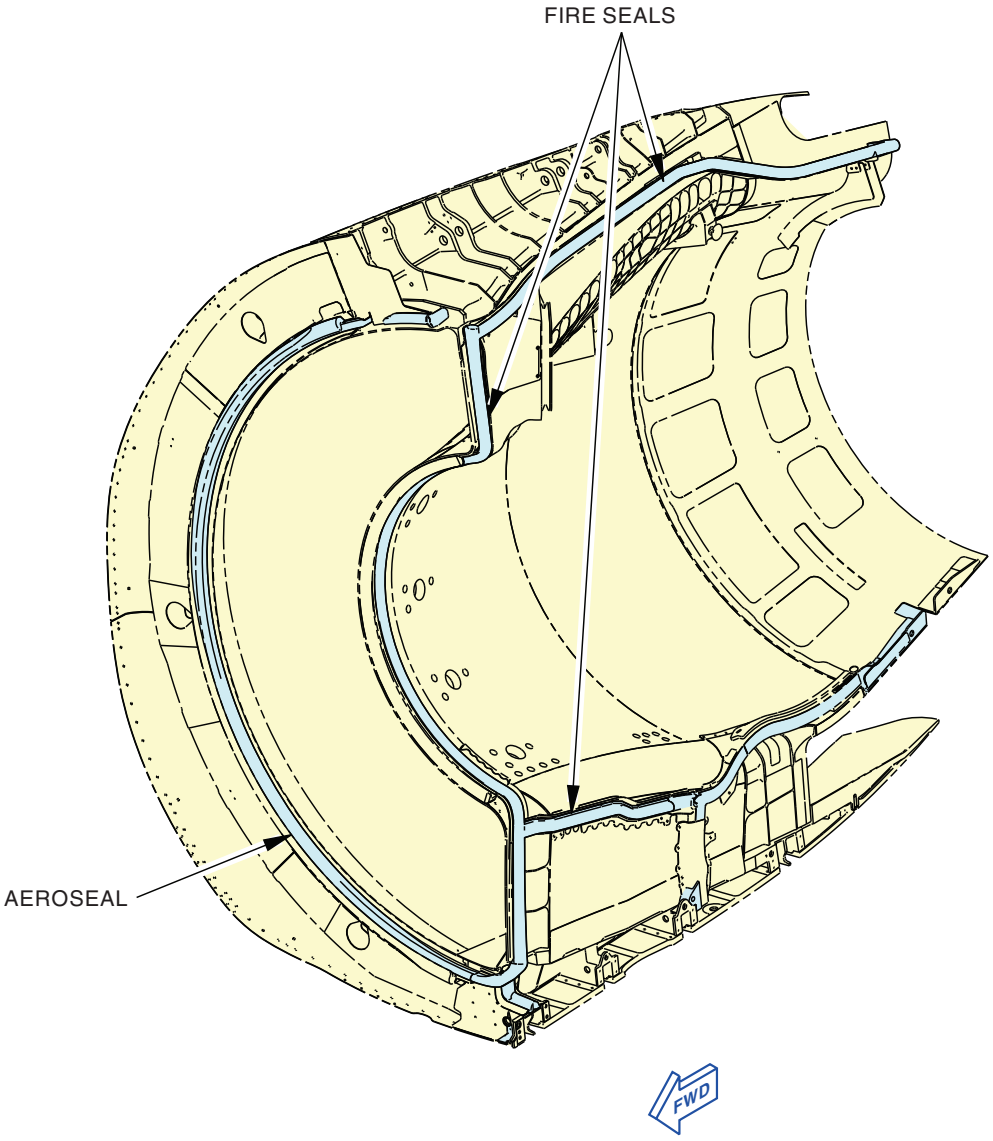
END OF TASK

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Thrust Reverser Fire Seal Inspection
Figure 601/78-31-12-990-804-G00 (Sheet 1 of 2)

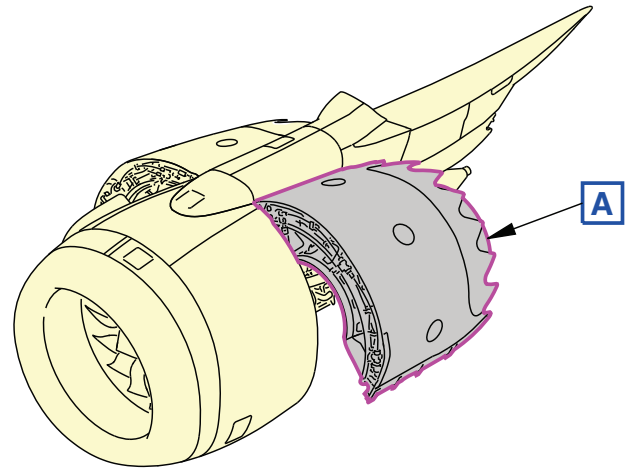
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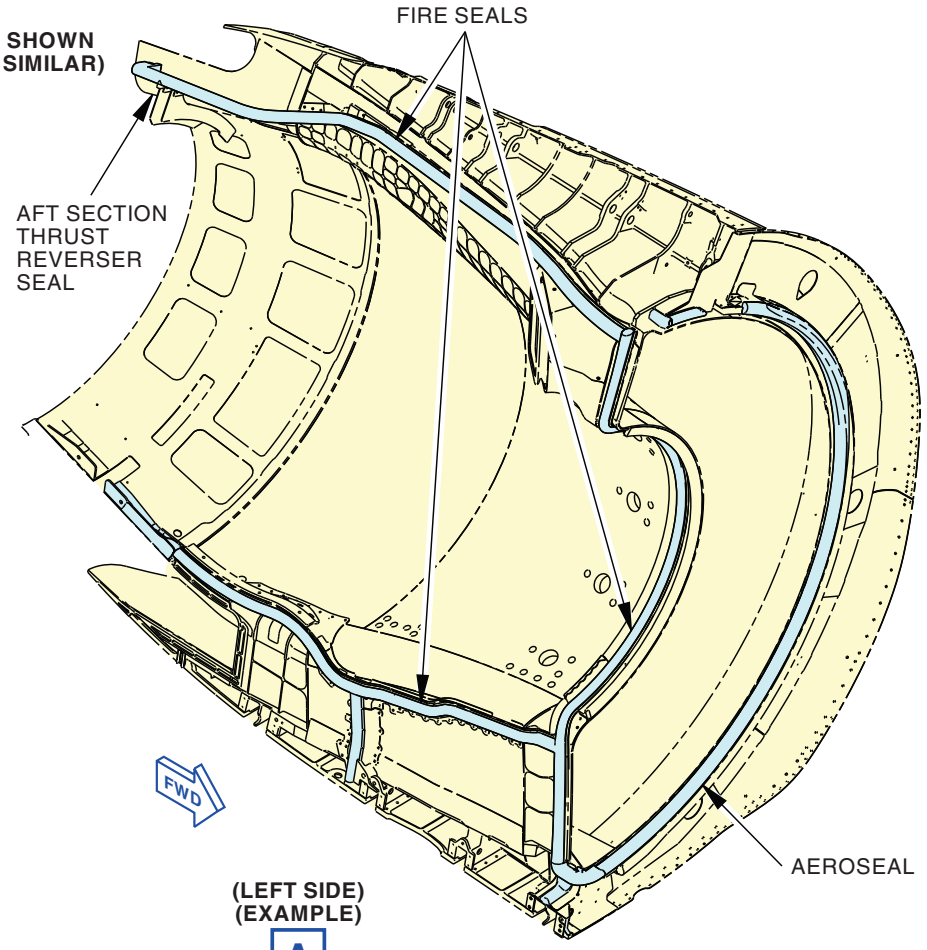
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Thrust Reverser Fire Seal Inspection
Figure 601/78-31-12-990-804-G00 (Sheet 2 of 2)

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(LEFT THRUST REVERSER SHOWN
RIGHT THRUST REVERSER SIMILAR)



(LEFT SIDE)
(EXAMPLE)

A

2938137 S0000713509_V1

Detailed Inspection of Engine Thrust Reverser
Figure 602/78-31-12-990-808-G00

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TASK 78-31-12-800-801-G00

4. Fire Seal Compression Check

(Figure 603)

A. General

(1) This task provides instruction to check the compression limits for the fire seal on the inner surface of each thrust reverser on an engine.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4127	Durometer - Shore A Part #: H-1000-A Supplier: 1GGG5

D. Consumable Materials

Reference	Description	Specification
A00081	Adhesive - Silicone Rubber - RTV 106	BAC5010 Type 74
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G02380	Developer - Inspection - Met-L-Chek D-70	AMS 2644 Form D Type 1, Form E Type 2
G02415	Agent - Parting, Paste Wax (Johnson's Paste Wax or equivalent)	
G50146	Developer - Non-acqueous - Dubl-Check D-100	SAE AMS2644

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

F. Prepare for the Fire Seal Compression Check

SUBTASK 78-31-12-010-005-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-12-750-002-G00

(2) Make sure that the fire seal sealant has cured to the time limits that follow:

(a) The RTV 106 adhesive, A00081 must cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

1) To decrease the cure time to do this:

NOTE: The cure time may be decreased to 4 hours if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

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b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

c) After 4 hours, do the following:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

NOTE: When the type A shore A durometer, COM-4127 indication is more than 30, the seal is cured.

<2> Remove the test strip.

(b) The sealant, A00160 must cure at 72°F (22°C) to 82°F (28°C) for these types and minimum times:

1) (Type I) 48 hours.

2) (Type II, Class -1/2) 4 hours.

SUBTASK 78-31-12-750-003-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Do these steps to prepare for the fire seal compression check:

(a) Use solvent, B00062 to clean the fire seals on the left and right thrust reverser and the areas on the strut, engine, and lower edge of the right thrust reverser where the fire seals will compress when closed.

(b) For Zones I, II, and III, apply Met-L-Chek D-70 developer, G02380 or Dubl-Check D-100 developer, G50146 approximately 2.0 in. (50.8 mm) wide on the fire seal.

(c) For Zone IV (right hand side lower bifurcation fire seal), apply Met-L-Chek D-70 developer, G02380 or Dubl-Check D-100 developer, G50146 along the length of the fire seal (not on the engine side fire seal).

(d) Make sure that the developer is dry.

NOTE: The developer becomes a white powder with a dull finish when it is dry.

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS WHEN YOU CLOSE THE THRUST REVERSERS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(e) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00).

1) Close and latch the thrust reverser; but do not do the thrust reverser or leading edge activation at this time.

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(f) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00

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G. Fire Seal Compression Check

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

SUBTASK 78-31-12-750-004-G00

(1) Measure the width of the developer that is on the fire seal. The minimum acceptable limits are:

(a) Zone I - 0.35 in. (8.89 mm)

(b) Zone II - 0.25 in. (6.35 mm)

(c) Zone III - 0.20 in. (5.08 mm)

(d) Zone IV - A continuous seal footprint is permitted (no limits).

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

SUBTASK 78-31-12-750-007-G00

(2) Measure the width of the developer that is on the fire seals.

(a) Make sure that the minimum width of the developer is 0.35 in. (8.89 mm) along the full length of the fire seals.

(b) If the above limit is not met, use the steps below to measure the developer in each section:

1) Divide the fire seal into four sections as follows (Figure 603):

a) The fire seal on the top edge of a thrust reverser where it compresses against the strut.

b) The fire seal on the forward edge of a thrust reverser where it compresses against the engine.

c) The fire seal on the bottom edge of the left thrust reverser where it compresses against the lower edge of the right thrust reverser.

d) The fire seal on the right hand side lower bifurcation.

2) For each section, areas where the developer is 0.20 in. (5.08 mm) to 0.35 in. (8.89 mm) wide is permitted with these conditions:

a) The length of each area is not more than 1.50 in. (38.1 mm).

b) The total length of all areas is not more that 3.00 in. (76.2 mm).

3) For each section, areas where the developer is less that 0.20 in. (5.08 mm) wide is permitted with these conditions:

a) The length of each area is not more than 0.25 in. (6.35 mm).

b) The total length of all areas is not more than 1.00 in. (25.4 mm).

PTF ALL

SUBTASK 78-31-12-390-005-G00

(3) If a section of the fire seal does not have an acceptable width of the developer (footprint), do the following steps:

(a) Apply paste wax parting agent, G02415, on the seal depressor surface.

(b) Extend paste wax parting agent, G02415, minimum of 3 in. (7.62 cm) beyond the noted discrepant area.

(c) Let the parting agent dry.

(d) Apply a continuous and uniform layer of the sealant, A00160, on the fire seal.

1) Make sure that layer is not more than 0.25 in. (0.64 cm) thick.

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2) Make sure that sealant extends to 1 in. (2.54 cm) in each direction past the discrepant area.

(e) Close and latch the thrust reverser (TASK 78-31-00-010-802-G00).

(f) Let the sealant fully dry.

(g) Open the thrust reverser (TASK 78-31-00-010-801-G00).

(h) Remove the parting agent.

(i) Make sure that width of the developer is acceptable.

1) Do the foot print verification setup to make sure that there are no remaining gaps at the discrepant location (TASK 78-31-01-820-801-G00).

SUBTASK 78-31-12-560-001-G00

(4) If the fire seal developer is not in the limits, do this task: Thrust Reverser Adjustment, TASK 78-31-01-820-801-G00.

(a) If the fire seal developer is in the limits, continue with this procedure.

NOTE: It is not necessary to clean off the developer.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-12-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

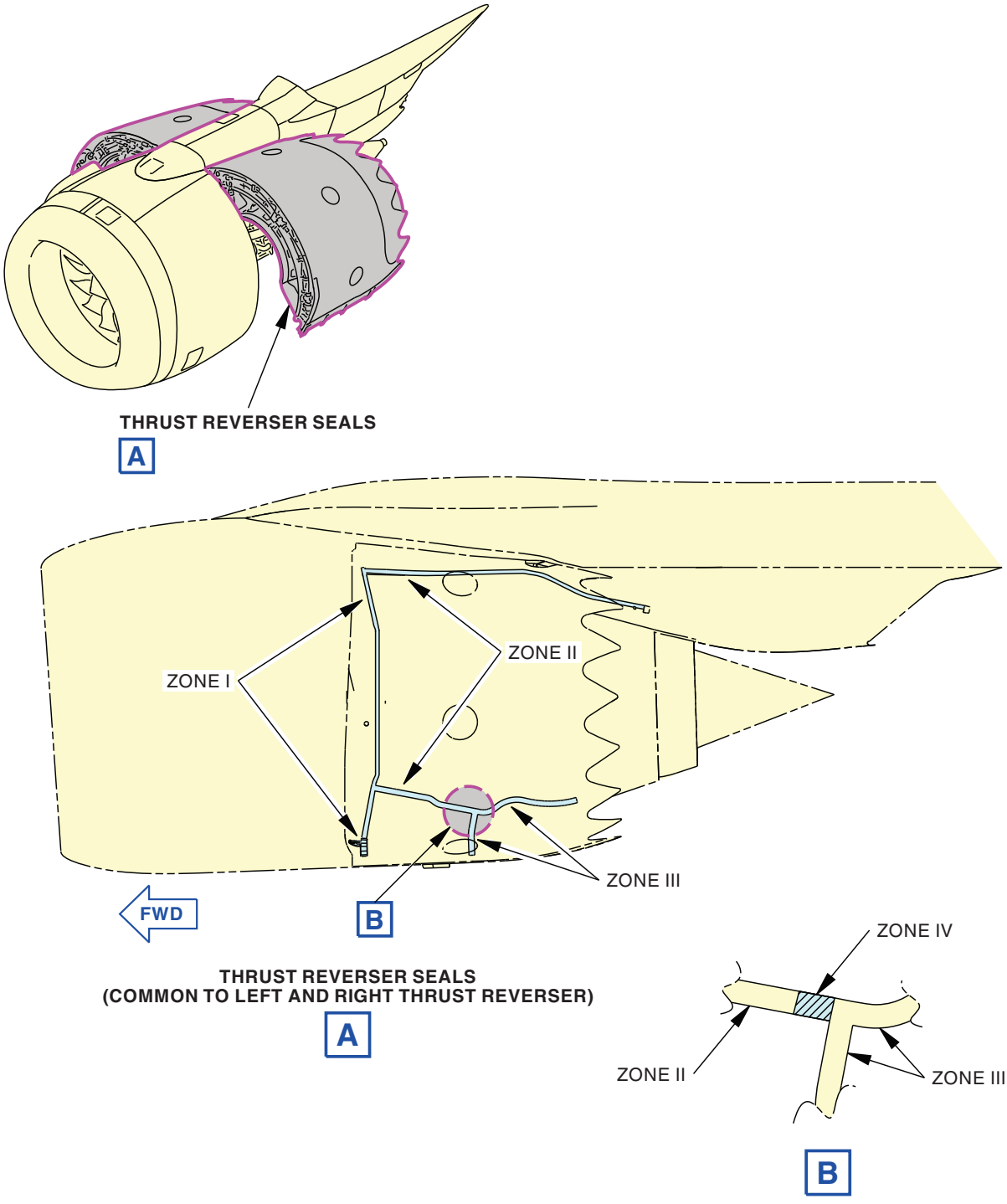
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

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Fire Seal Compression Check
Figure 603/78-31-12-990-805-G00

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FIRE SEAL - REPAIR

1. General

A. This procedure has one task:

(1) The repair of the fire seal on the thrust reverser.

TASK 78-31-12-300-801-G00

2. Fire Seal Repair

A. General

(1) This task gives the instructions for a repair of damage found on the aft area of the top section of the fire seal.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-12-800-801-G00	Fire Seal Compression Check (P/B 601)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4127	Durometer - Shore A Part #: H-1000-A Supplier: 1GGG5
SPL-19243	Aid - Cutting Part #: C78044-1 Supplier: 81205
SPL-19244	Mandrel - Repair, Fire seal Part #: C78045-1 Supplier: 81205
STD-194	Container - 1 Quart (1 Liter), Calibrated
STD-551	Knife - Razor
STD-552	Knife - Utility
STD-625	Pen
STD-765	Scraper - Plastic

D. Consumable Materials

Reference	Description	Specification
A00081	Adhesive - Silicone Rubber - RTV 106	BAC5010 Type 74
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G50527	Water	

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left

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(Continued)

Zone	Area
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Damage Limits for the Repair

SUBTASK 78-31-12-200-001-G00

(1) Do a detailed inspection of the thrust reverser fire seal (Figure 801).

NOTE: If the damage to the fire seal is within the damage limits, you can do a repair on the fire seal.

(a) Examples of damage that requires repair:

1) Torn fabric
2) Missing fabric
3) Holes worn through seal

(b) Examples of acceptable damage that do not require repair:

1) Surface whitening or surface wear
2) Labyrinth blade contact if not worn through the seal.

H. Prepare for the Repair

WARNING DO ALL OF THE SPECIFIED TASKS IN THE CORRECT SEQUENCE TO OPEN THE THRUST REVERSER(S). IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

SUBTASK 78-31-12-010-006-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

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1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Thrust Reverser Fire Seal Repair

NOTE: The repair procedures for the LH Thrust Reverser Fire Seal and the RH Thrust Reverser Fire Seal are the same. Make sure to use the appropriate repair seal, cutting aid and mandrel for the applicable Thrust Reverser Fire Seal.

SUBTASK 78-31-12-020-006-G00

CAUTION

BE CAREFUL WHEN YOU REMOVE THE FIRE SEAL FROM THE RETAINER OR REMOVE SEALANT. DAMAGE TO THE FIRE SEALS CAN OCCUR.

(1) Do these steps to remove the damaged section of the thrust reverser seal (Figure 802).

NOTE: See the table below for a list of the parts necessary to complete the Thrust Reverser Fire Seal repair.

KIT 002A0077		
PART NUMBER	NAME	QUANTITY
002A0077–1	TOP KIT - REPAIR SEAL LH	
315A6001–1	FIRE SEAL REPAIR SPLICE LH	1
315A6001-3	SEAL PLUG LH	1
002A0077–2	TOP KIT - REPAIR SEAL RH	
315A6001-2	FIRE SEAL REPAIR SPLICE RH	1
315A6001-4	SEAL PLUG RH	1

(a) Use a plastic scraper, STD-765 to remove no more than 20 in. (508 mm) of the existing fire seal from the fire seal retainer.

(b) Remove the RTV 106 adhesive, A00081 between the existing fire seal and the seal retainer.

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1) Make sure no damage occurs to the fire seal retainer.

(c) Partially insert base of the repair seal into lower side of retainer.

1) Make sure seal block is flush with the support and the seal nub is aligned with the back up tab.

(d) Using marker pen, STD-625, or similar, draw a line on the seal retainer at the end of the repair seal.

(e) Partially reinstall the existing fire seal in the lower side of the retainer to mark cut line.

(f) Using a marker pen, STD-625, or similar, draw a line on the existing fire seal.

1) Draw the line around the crown of the seal.

(g) Do these steps to cut the existing fire seal:

1) Cut fire seal halfway between marked cut line and seal nub to allow the seal cutting aid, SPL-19243 to be installed.

2) Align the cutting aid, SPL-19243 to the cut line.

3) Make sure the cut line on the fire seal is perpendicular.

4) Using a utility knife, STD-552 or razor knife, STD-551, insert the blade tip into the cutting aid, SPL-19243 slot and pierce the crown of the seal.

NOTE: The crown must be pierced initially in order to obtain a clean cut.

(h) Clean any residual RTV 106 adhesive, A00081 from the existing fire seal and fire seal retainer with solvent, B00062.

SUBTASK 78-31-12-020-007-G00

WARNING DO ALL OF THE SPECIFIED TASKS IN THE CORRECT SEQUENCE TO OPEN THE THRUST REVERSER(S). IF YOU DO NOT OBEY THIS INSTRUCTION, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these steps to remove a damaged repair seal (Figure 803).

(a) Use a plastic scraper, STD-765 to remove no more than 20 in. (508 mm) of the fire seal from the fire seal retainer.

(b) Remove the RTV 106 adhesive, A00081 between the fire seal and the seal retainer.

1) Make sure no damage occurs to the fire seal retainer.

(c) Fold the repair seal over the original fire seal at the split so that the crowns are touching.

(d) Insert a blunt tool into the space created at the bend in the seal splice.

(e) Pry the seal splice out of the original fire seal.

(f) Gently pull the fire seal down along the original fire seal until the seal flap separates from the original fire seal.

SUBTASK 78-31-12-420-007-G00

(3) Do these steps to verify that the repair seal splice joint will fit properly (Figure 804):

(a) Dip the first 4 inches of the mandrel, SPL-19244 into a 1 Quart (1 l) calibrated container, STD-194 of water, G50527 and insert it into the rear of the repair seal until the tip of the mandrel, SPL-19244 is flush with the end of the splice.

(b) Pinch the sides of the existing seal to create a circular opening to facilitate the insertion of the repair seal.

NOTE: Slight back and forth twisting of the repair seal and pinching of the existing seal will ease the insertion of the splice joint.

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(c) Make sure the block end of the repair seal is touching the end of the seal retainer and measure the gap between the repair seal and the existing seal.

NOTE: Maximum gap 0.10 inch.

SUBTASK 78-31-12-420-008-G00

(4) Do these steps to install the repair seal:

(a) Pull the repair seal out of the existing seal and fold the flap of the repair seal back to gain access to the OD (outside diameter) of the splice tube.

1) Apply a thin coat of RTV 106 adhesive, A00081 to the base (0.25-0.50 inches) of the OD of the splice tube and to the end face of the repair seal and existing seal.

NOTE: A thin coat of RTV 106 sealant is 1 to 5 mils thick.

2) Minimal adhesive should be used to fully cover the surface.

NOTE: Excessive sealant application to repair splice may lead to difficulty removing the seal end for a future repair.

(b) Pinch the sides of the existing seal to create a circular opening and insert the splice tube into the trimmed original seal such that the repair seal and original seal are butt joined.

NOTE: Maximum gap between repair seal and original seal is 0.10 inch.

1) Make sure repair seal is installed over cross tie strap.

2) Smooth RTV 106 adhesive, A00081 in the gap between the existing fire seal and repair seal and remove any excess RTV 106 adhesive, A00081.

3) Insert the base of modified seal into the upper retainer and work the other side of the base into the lower side of the retainer using a plastic scrapper as an aid.

(c) Secure repair seal flap.

1) Apply a thick coat of RTV 106 adhesive, A00081 to the underside of the flap. Use enough RTV 106 adhesive, A00081 to achieve squeeze out around the periphery of the flap.

2) Fold the seal flap down and bond it to the original fire seal.

a) Make sure there is squeeze out.

b) If the corners of the flap do not remain in contact with the OD of the fire seal, it may be necessary to apply pressure to flap during heat cure.

c) Fair RTV 106 adhesive, A00081 approximately 1 inch from edge of flap.

d) Make sure transition and surface of RTV 106 adhesive, A00081 is smooth.

NOTE: If RTV 106 adhesive, A00081 surface is not smooth, the repair may fail the compression check and require reapplication of RTV 106 adhesive, A00081.

3) Apply heat to the RTV 106 adhesive, A00081, between 120 degrees Fahrenheit (48.9 degrees Centigrade) and a maximum of 150 degrees Fahrenheit (65.6 degrees Centigrade) for 10 minutes to set and accelerate cure RTV 106 adhesive, A00081.

NOTE: It may be necessary to apply pressure to flap during heat cure.

(d) The mandrel, SPL-19244 can be removed immediately after heat cure is complete.

SUBTASK 78-31-12-700-001-G00

(5) Do a compression check on the thrust reverser fire seal in the repair area, do this task Fire Seal Compression Check, TASK 78-31-12-800-801-G00.

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(a) Make sure the RTV 106 adhesive, A00081 has a tack free surface before the compression check.

(b) Make sure there is continuous seal contact within the repair area.

(c) Areas of no seal contact (no developer transfer) up to 0.125 inches along the length are acceptable.

(d) If compression test does not pass, clean the developer off of the seal prior to the RTV 106 adhesive, A00081 re-application, re-fair RTV 106 adhesive, A00081 and repeat the inspection.

SUBTASK 78-31-12-420-009-G00

(6) Install the seal plug.

(a) Apply a thin coat of sealant, A00160 or RTV 106 adhesive, A00081 to the outside of the seal plug fully covering the bonding surface.

(b) Insert the seal plug into the open end of the fire seal repair splice until it is flush with the end of the fire seal repair splice.

1) Coat the remaining seal plug surface with sealant, A00160 or RTV 106 adhesive, A00081 such that the end of the seal plug is no longer visible.

SUBTASK 78-31-12-420-010-G00

(7) Apply sealant, A00160 or RTV 106 adhesive, A00081 to the following areas:

(a) Repair seal

(b) Areas where previous sealant, A00160 or RTV 106 adhesive, A00081 were broken for seal removal

(c) Bead of sealant, A00160 or RTV 106 adhesive, A00081 to the edge of the seal and the seal retainer

(d) Between the block end and upper skin strap

(e) Between the block end and the fire seal support stiffener

(f) Bead of sealant, A00160 or RTV 106 adhesive, A00081 around the base of the block end to upper skin strap and block end to seal support.

SUBTASK 78-31-12-860-001-G00

(8) Let all the sealant cure to the time limits that follow:

(a) Let the RTV 106 adhesive, A00081 cure a minimum of 24 hours at 65°F (18°C) to 100°F (38°C).

WARNING DO NOT USE DECREASED CURE TIME FOR A SERVICE BULLETIN WITH AN AIRWORTHINESS DIRECTIVE AGAINST IT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

1) To decrease the cure time to do this:

NOTE: The cure time may be decreased to 4 hours if the seal area can be locally heated to maximum of 140°F (60°C). Use low temperature heat lamps and a thermocouple to measure the temperature.

a) Make a test strip of sealant 0.25 in. (0.64 cm) minimum thickness and put adjacent to the repair.

b) Heat the seal area to a maximum of 140°F (60°C).

NOTE: Temperatures more than 140°F (60°C) will cause bubbles to be in the sealant.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) After 4 hours, do this:

<1> Measure the hardness of the test strip with a type A shore A durometer, COM-4127.

<2> When the type A shore A durometer, COM-4127 indication is more than 30, the seal is cured.

<3> Remove the test strip.

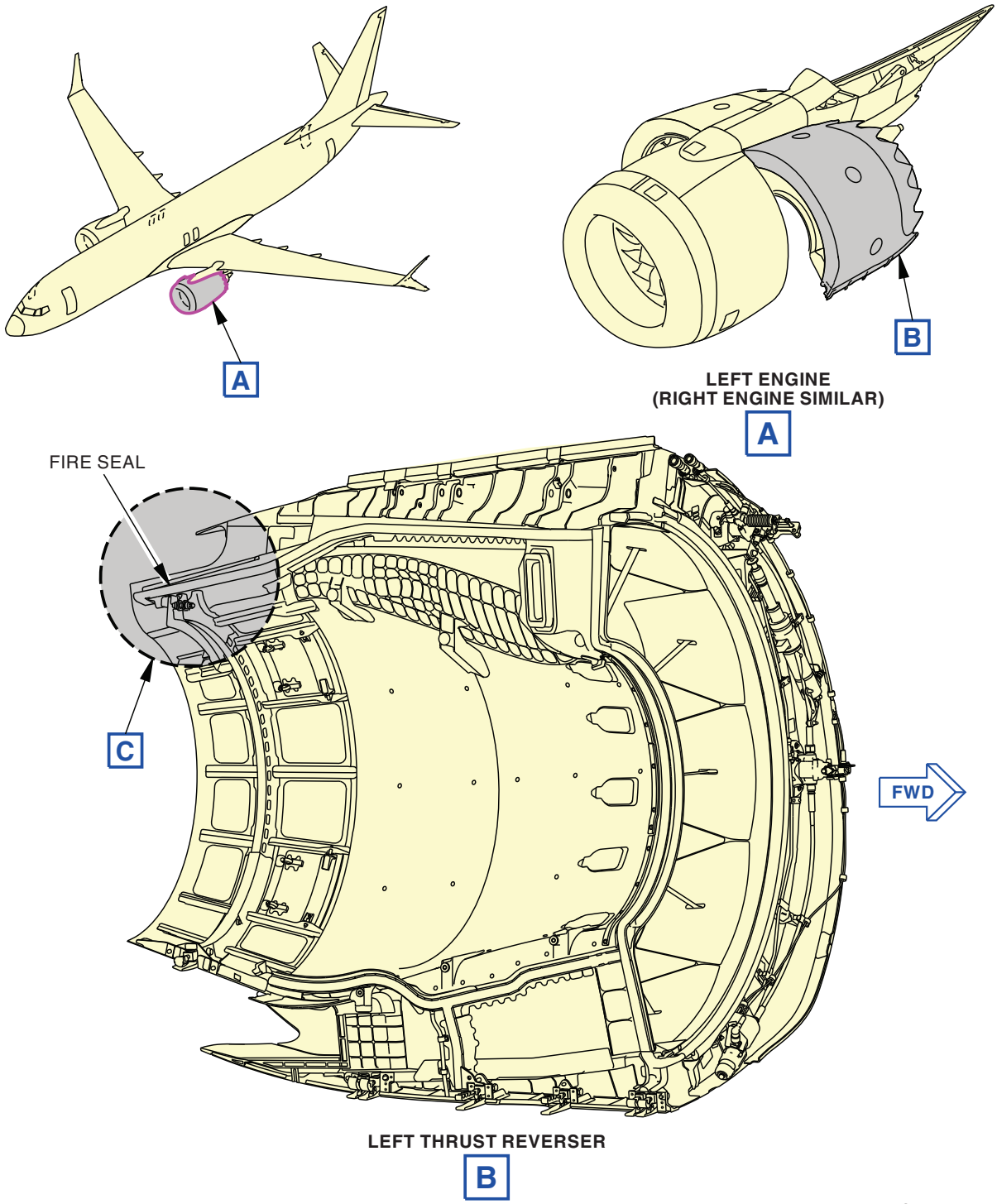
(b) Let the sealant, A00160 cure at 72°F (22°C) to 82°F (28°C) for these Types and minimum times:

1) (Type I) 48 hours.

2) (Type II, Class -1/2) 4 hours.

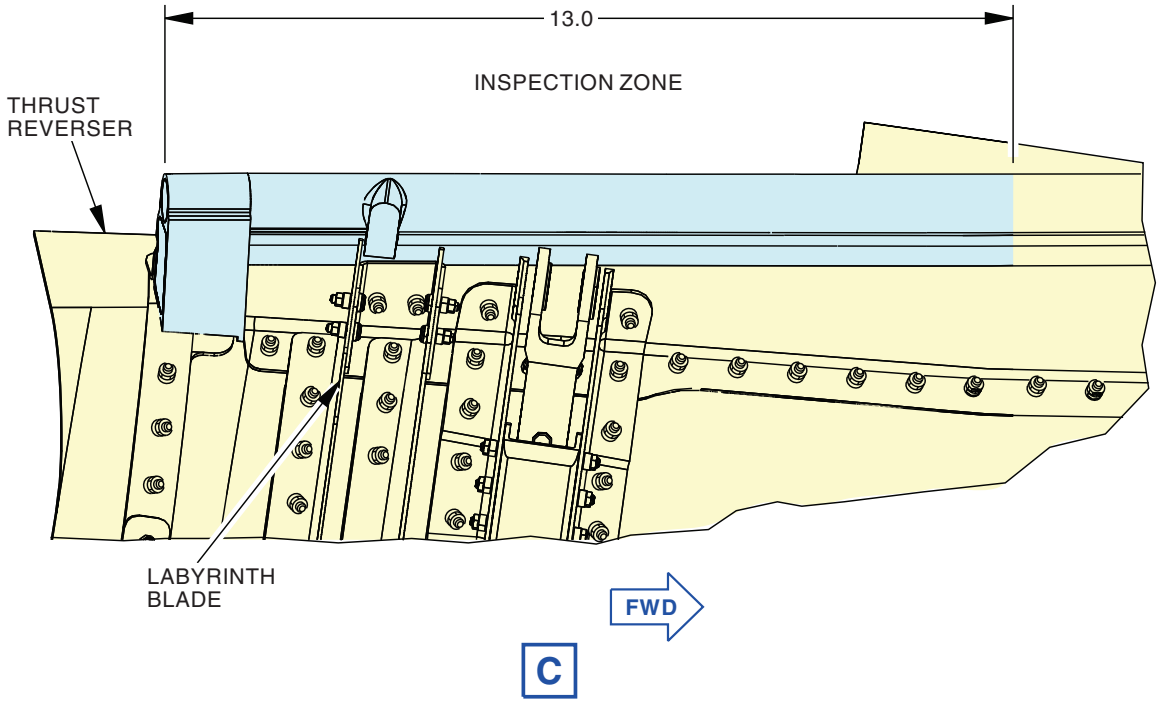
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Detailed Inspection (LH Shown, RH Similar)
Figure 801/78-31-12-990-809-G00 (Sheet 1 of 2)

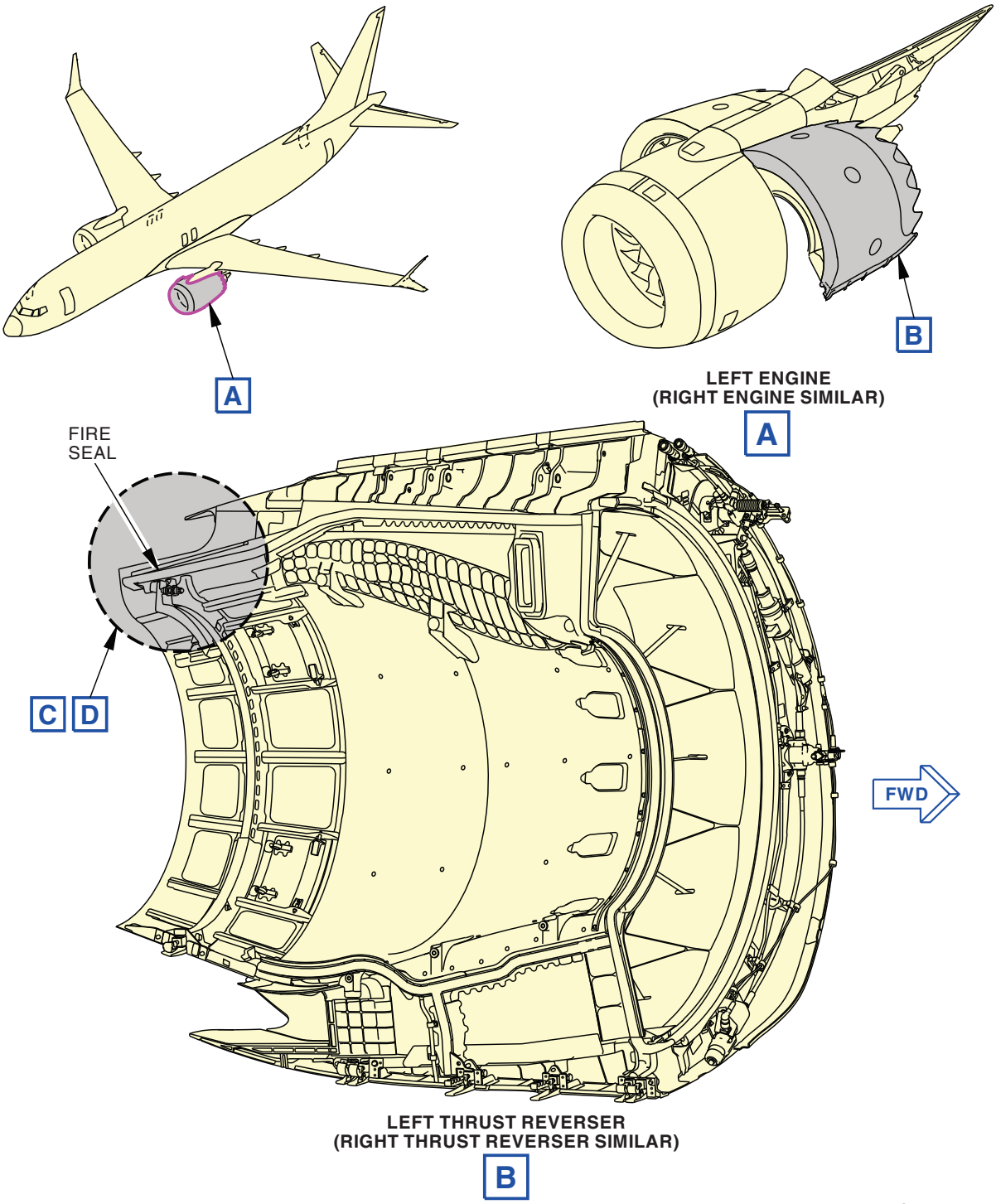
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Detailed Inspection (LH Shown, RH Similar)
Figure 801/78-31-12-990-809-G00 (Sheet 2 of 2)

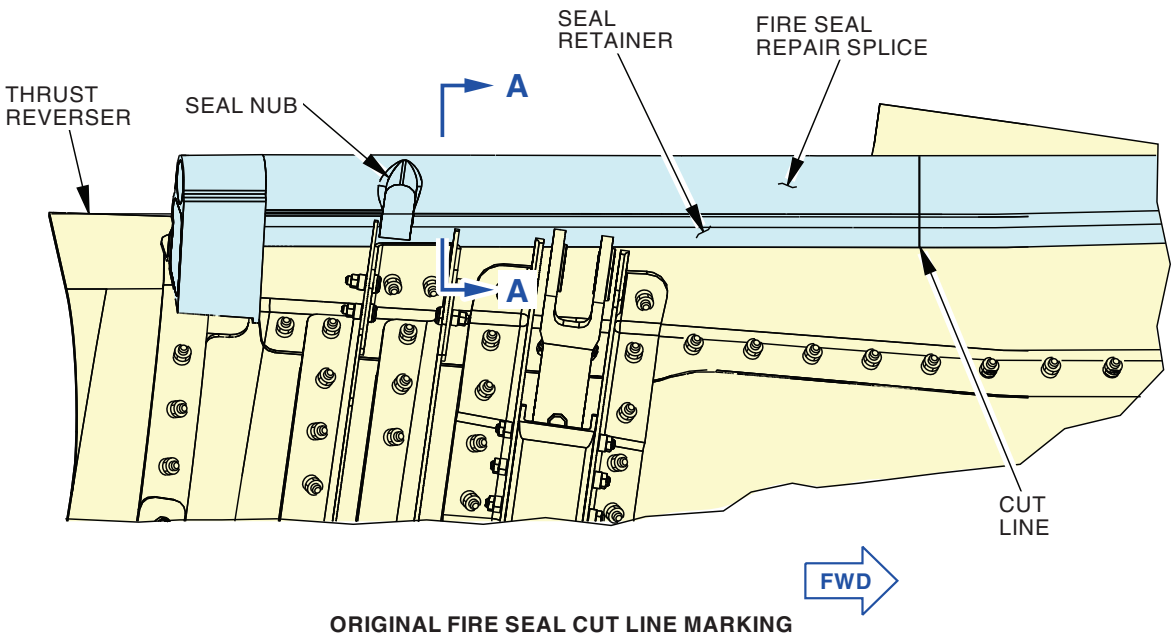
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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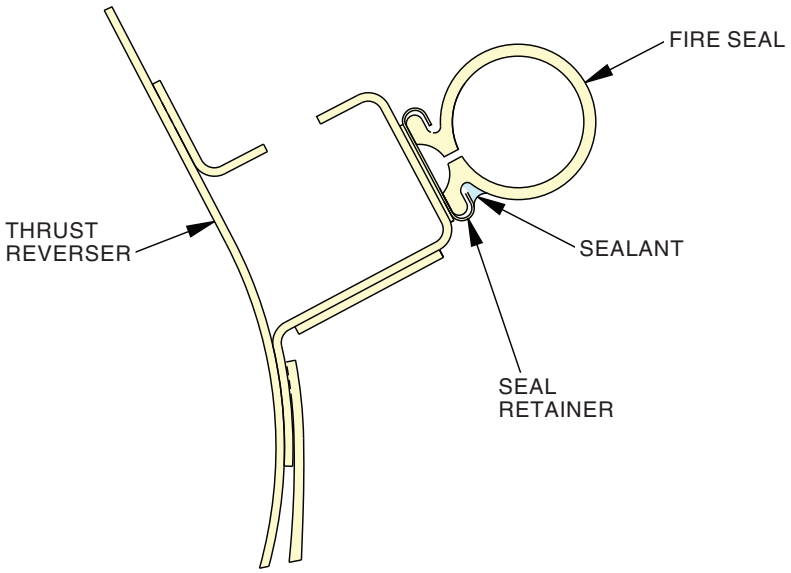
Thrust Reverser Damaged Fire Seal Removal
Figure 802/78-31-12-990-810-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ORIGINAL FIRE SEAL CUT LINE MARKING

C

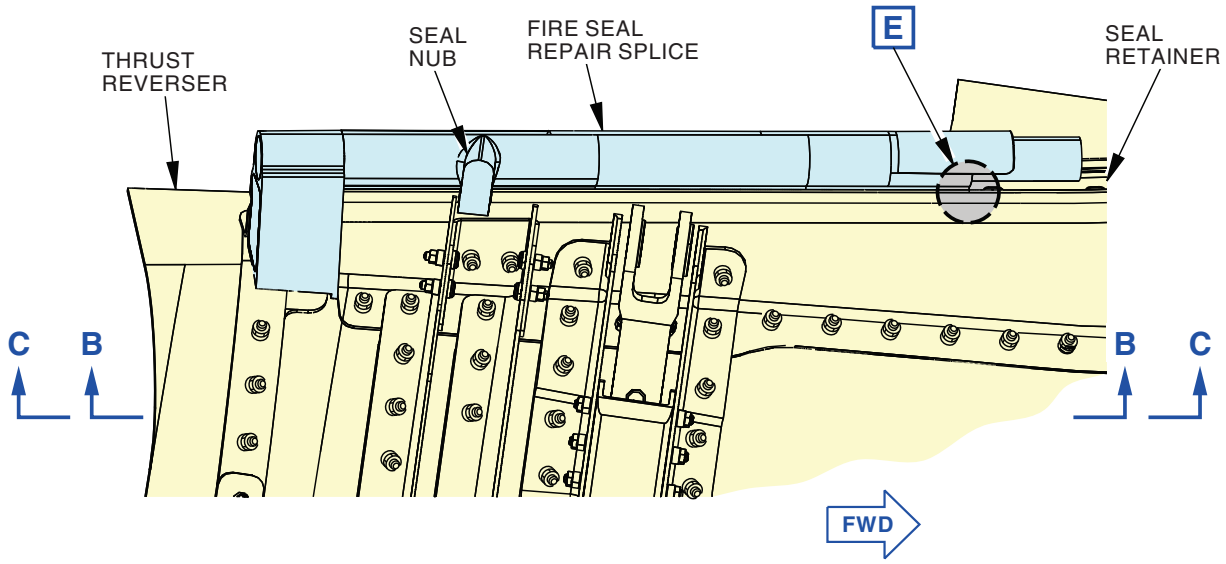


A-A

2938049 S0000713496_V1

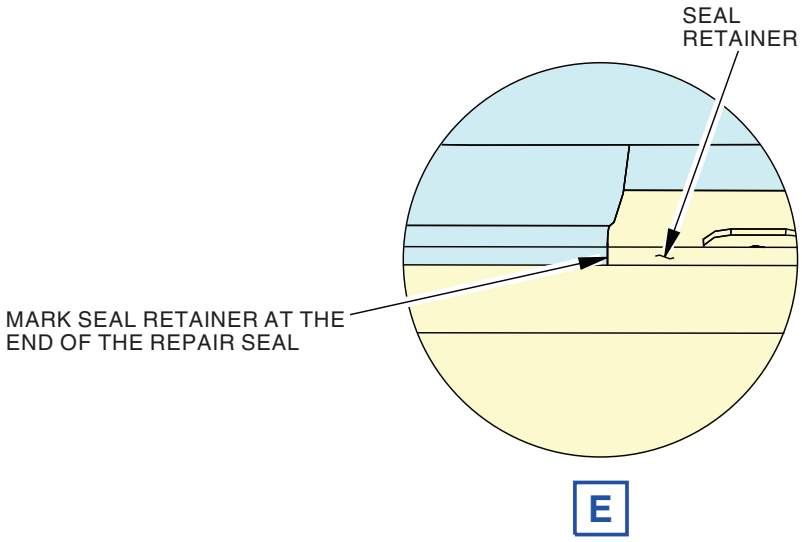
Thrust Reverser Damaged Fire Seal Removal
Figure 802/78-31-12-990-810-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TEMPORARY FIRE SEAL REPAIR SPLICE INSTALLATION FOR CUT LINE MARKING

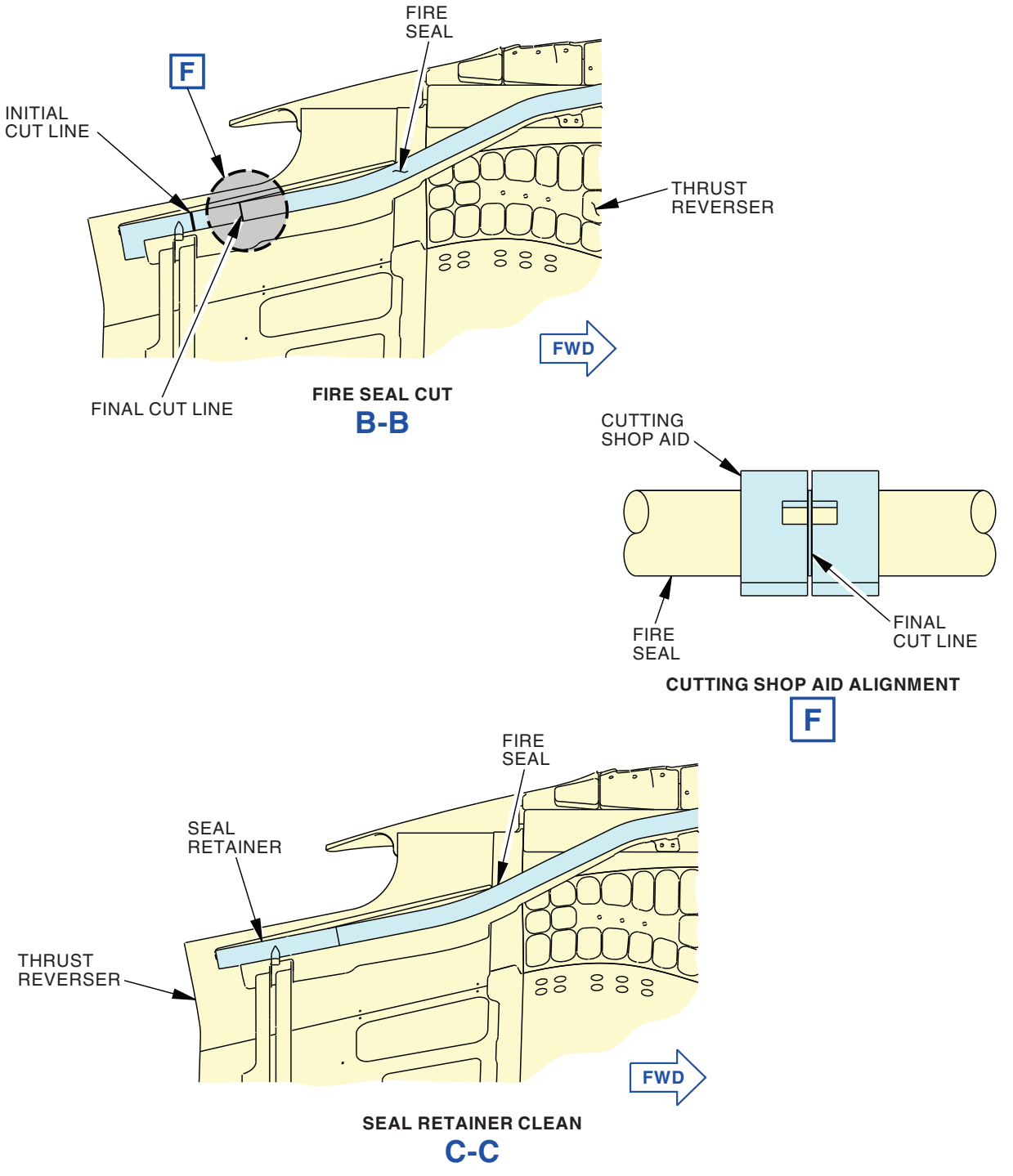
D



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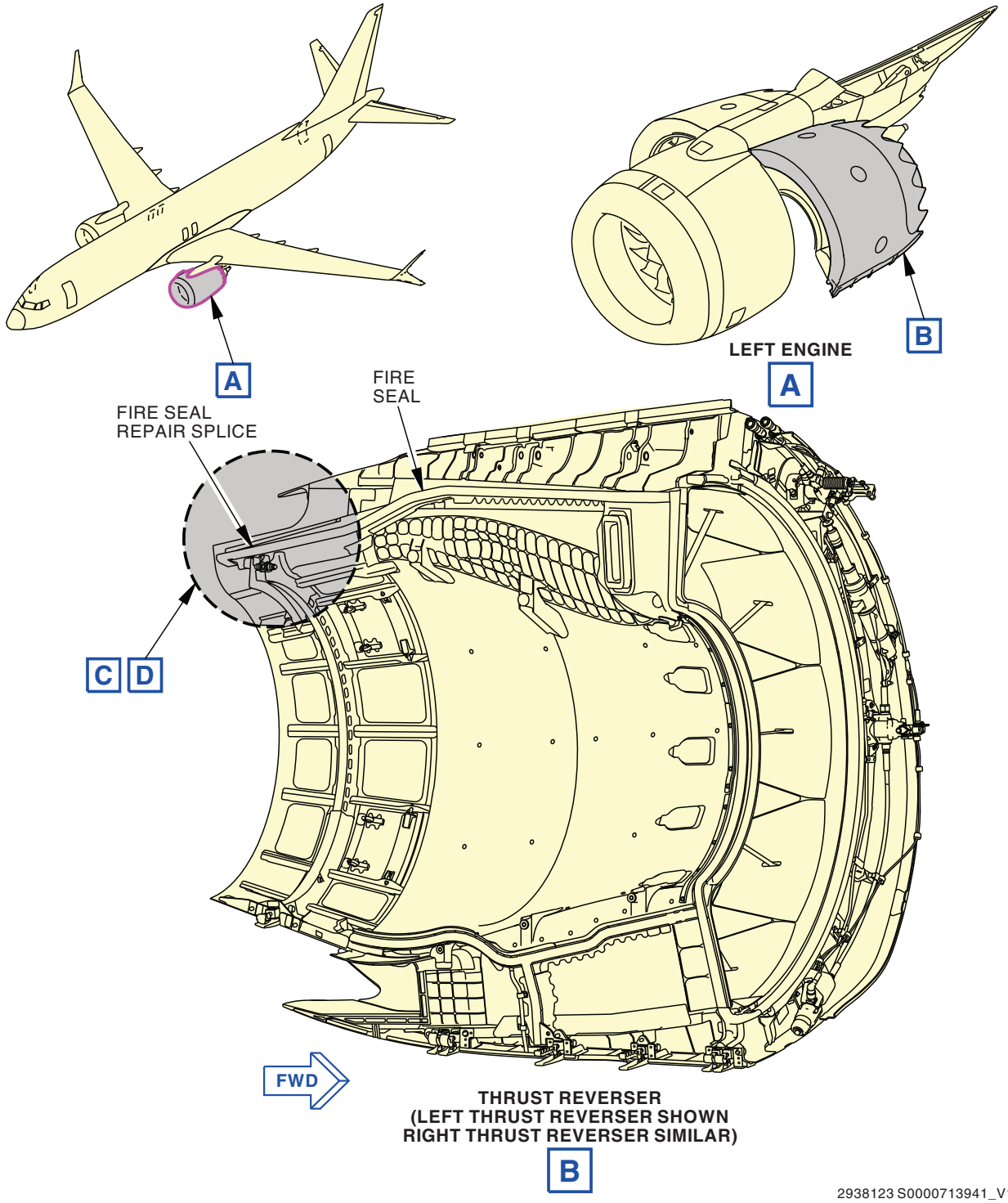
Thrust Reverser Damaged Fire Seal Removal
Figure 802/78-31-12-990-810-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



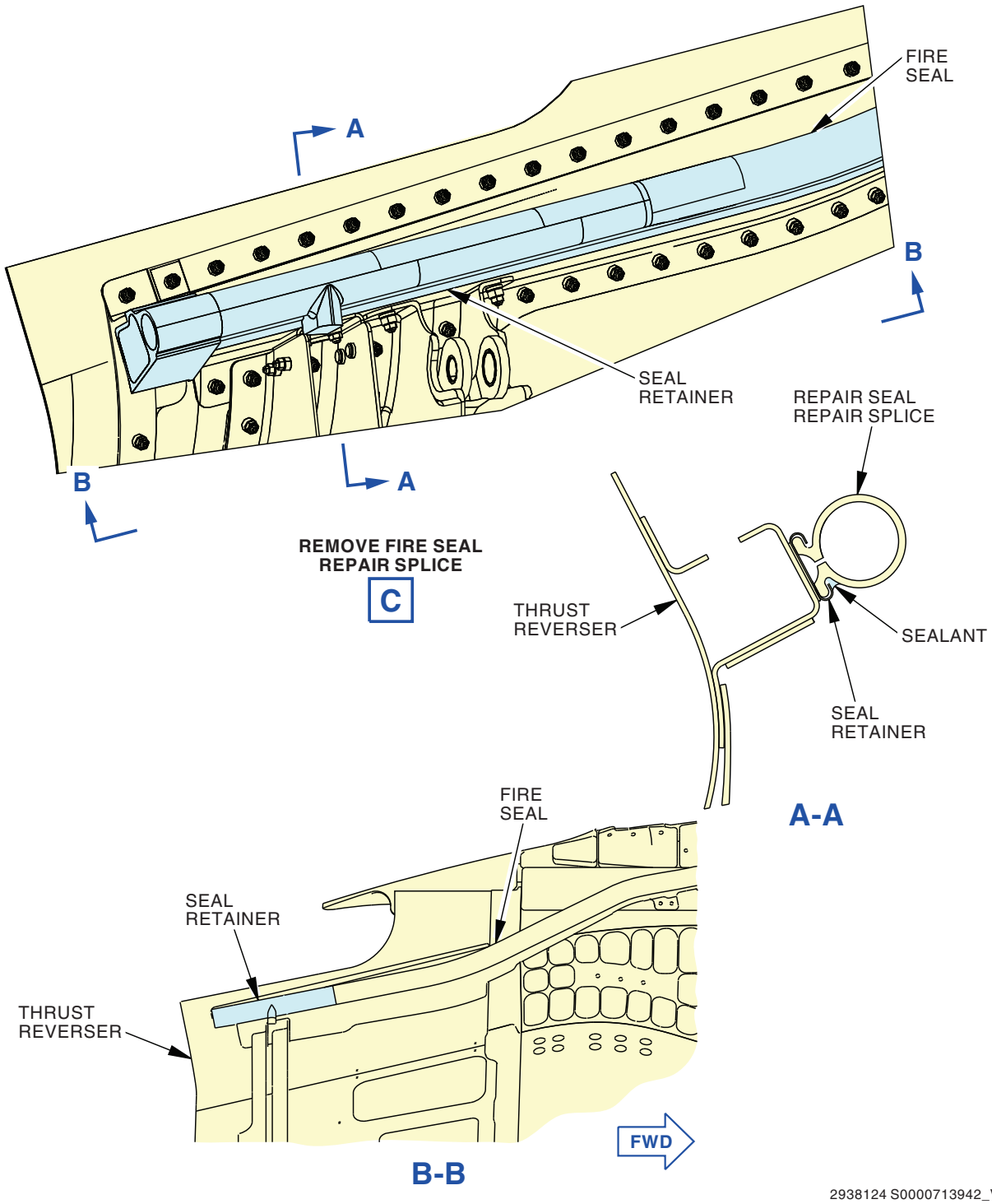
Thrust Reverser Damaged Fire Seal Removal
Figure 802/78-31-12-990-810-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



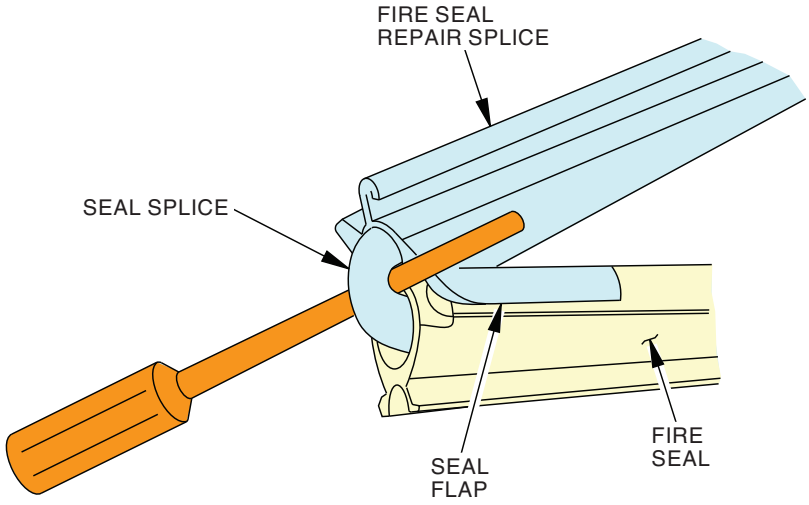
Thrust Reverser Damaged Repair Splice Removal
Figure 803/78-31-12-990-812-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



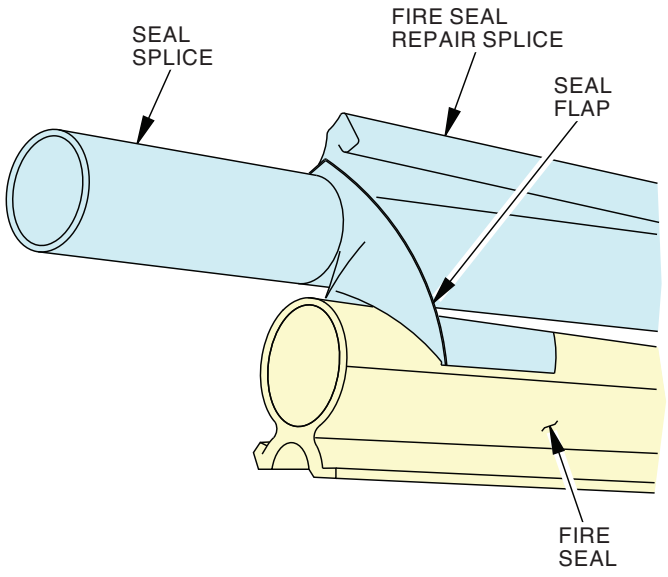
Thrust Reverser Damaged Repair Splice Removal
Figure 803/78-31-12-990-812-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**SEAL SPLICE REMOVAL
(BEFORE)**

D



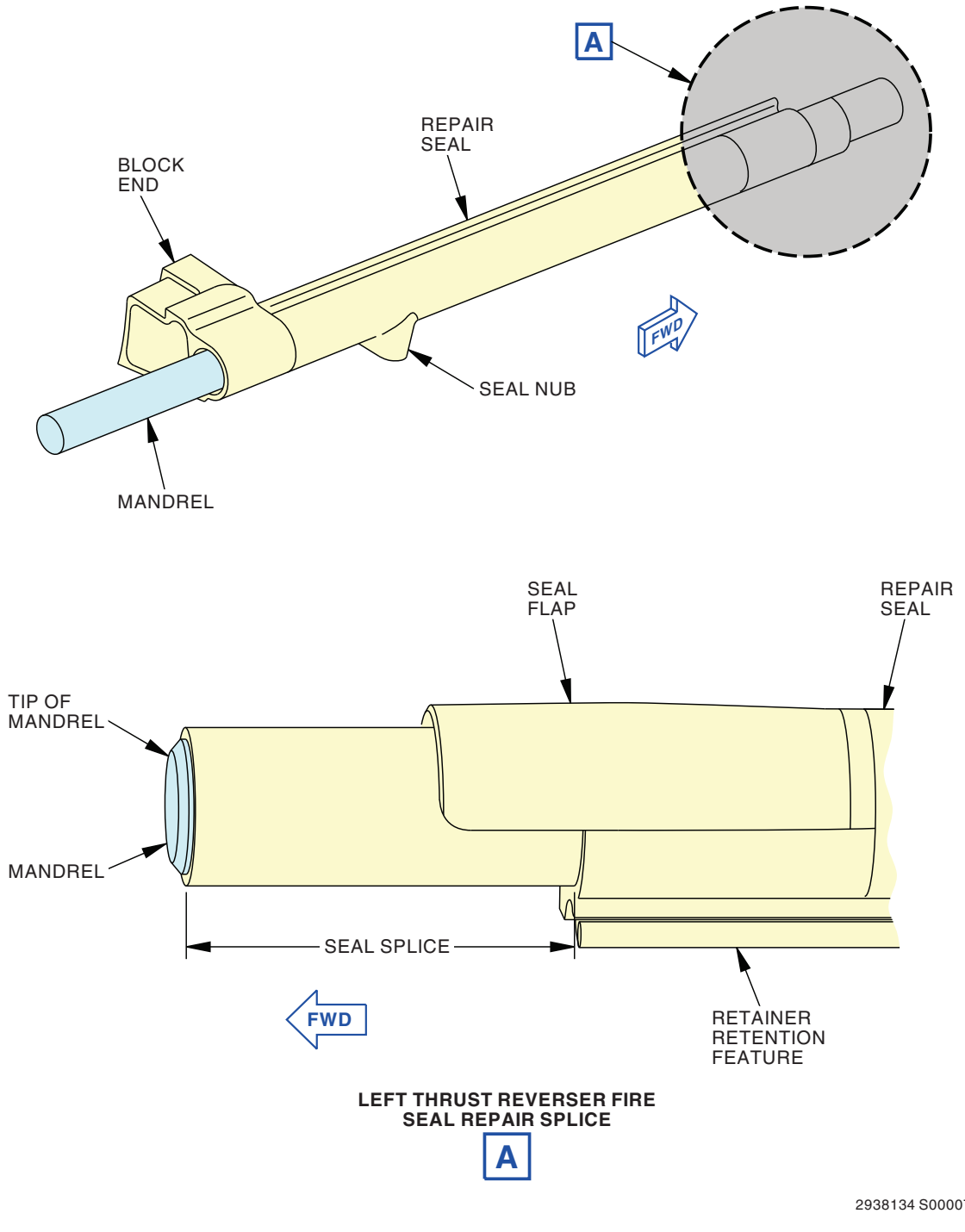
**SEAL SPLICE REMOVAL
(AFTER)**

D

2938125 S0000713943_V1

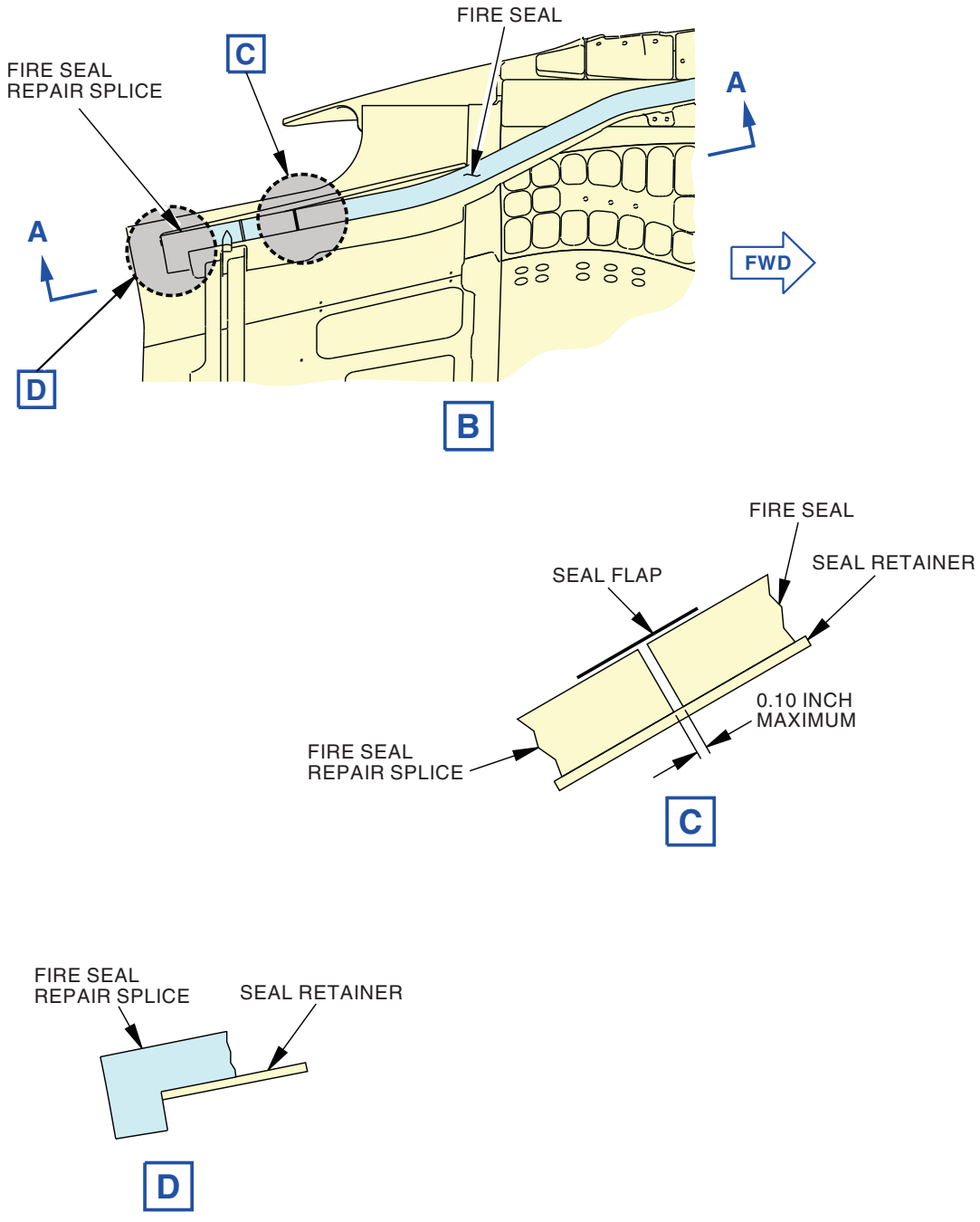
**Thrust Reverser Damaged Repair Splice Removal
Figure 803/78-31-12-990-812-G00 (Sheet 3 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Seal Repair Splice
Figure 804/78-31-12-990-813-G00 (Sheet 1 of 3)

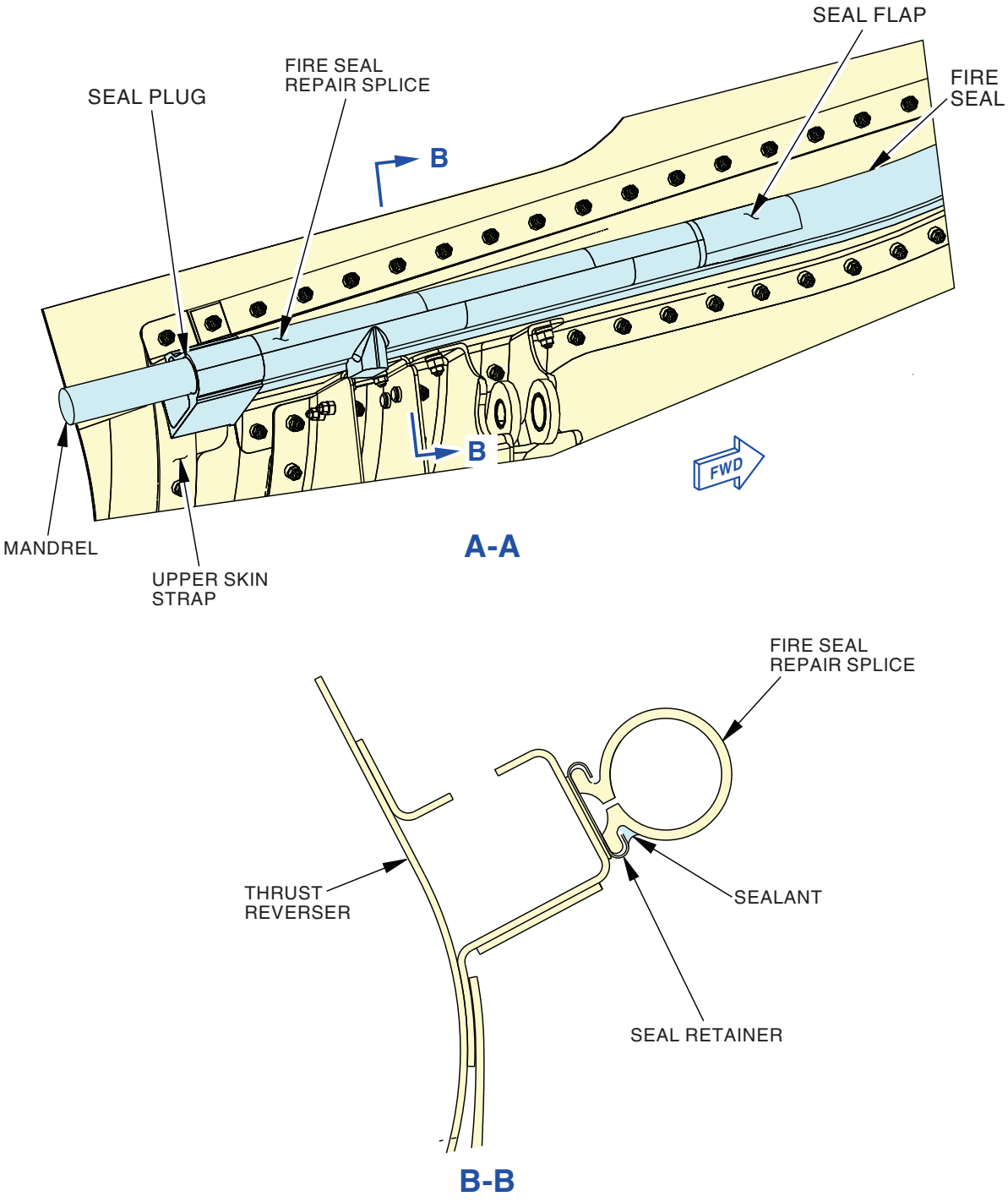
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Seal Repair Splice
Figure 804/78-31-12-990-813-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Seal Repair Splice
Figure 804/78-31-12-990-813-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INSULATION BLANKET - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the insulation blankets.

(2) The installation of the insulation blankets.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

B. The thrust reversers have insulation blankets at the following locations:

(1) GSE thermal barrier

(2) Precooler inlet duct at the forward upper bifurcation

(3) Fire seal support along the lower bifurcation

(4) Upper Inner wall

(5) Lower Inner wall.

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

C. The thrust reversers have insulation blankets at the following locations:

(1) Precooler thermal barrier on the precooler inlet duct

(2) GSE precooler thermal barrier is on the precooler thermal barrier

(a) The GSE precooler thermal barrier is referred to as the small precooler thermal barrier in these tasks.

(3) Fireseal blanket installed on the lower fireseal support.

(a) The installation of the left fireseal blanket different to the right fireseal blanket.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

TASK 78-31-13-000-801-G00

2. Insulation Blankets - Removal

(Figure 401)

A. General

(1) This task is for the removal of the insulation blankets from the left and right thrust reversers.

(2) The procedures are the same for the left and right thrust reversers.

CAUTION

REMOVE AND INSTALL THE INSULATION BLANKETS IN THE CORRECT ORDER GIVEN IN THESE PROCEDURES. IF THE ORDER OF REMOVAL AND INSTALLATION IS NOT FOLLOWED, DAMAGE TO EQUIPMENT CAN OCCUR.

(3) The insulation blankets are as follows:

(a) GSE Blanket

(b) Precooler Blanket

(c) Fire Seal Support Blanket

(d) Upper TPS Panel Blanket

(e) Lower TPS Panel Blanket.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00634	Solvent - Stabilized Limonene Cleaner	BMS11-10 Type 1, 2, or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-31-13-010-001-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:



WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Thrust Reverser GSE Insulation Blanket Removal

SUBTASK 78-31-13-020-003-G00

(1) Remove the GSE blanket [1] as follows:

(a) Remove the two screws [6], two washers [7], two screws [8], two washers [7], nut [11], and washer [7].

CAUTION HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(b) Remove the GSE blanket [1].

(c) Keep all removed hardware for installation.

SUBTASK 78-31-13-100-001-G00

(2) Clean the area and remove all contamination as follows:

(a) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

(b) Wipe the structure with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

(c) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

(d) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

H. Thrust Reverser Precooler Insulation Blanket Removal

SUBTASK 78-31-13-020-001-G00

(1) Remove the precooler blanket [2] as follows:

(a) Remove the two screws [8] and two washers [7].

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(b) Remove the nut [10] and washer [7].

(c) Remove the receiver cup [29], washer [30] and up to five shims [31].

(d) Remove the screw [6] and washer [7] at two locations.

(e) Remove the screw [15] and washer [12].

(f) Remove the two screws [38] and two washers [7].

(g) Remove the precooler drain tube fitting [36].

(h) Remove the three screws [13] and three washers [12].

(i) Remove the nut [10] and washer [14].

(j) Remove the three caps [22], three nuts [10], and three washers [12].

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(k) Remove the precooler blanket [2].

(l) Keep all removed hardware for installation.

SUBTASK 78-31-13-020-014-G00

(2) Clean the area and remove all contamination as follows:

(a) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

(b) Wipe the structure with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

(c) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

(d) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

I. Thrust Reverser Fire Seal Support Insulation Blanket Removal

SUBTASK 78-31-13-160-004-G00

(1) Remove the fireseal support blanket [3] as follows:

(a) Remove the screw [15] and washer [7] from the fireseal support blanket [3].

(b) Remove the three screws [13] and the three washers [12] from the fireseal support blanket [3].

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(c) Remove the fireseal support blanket [3].

(d) Keep all removed hardware for installation.

SUBTASK 78-31-13-160-011-G00

(2) Clean the area and remove all contamination as follows:

(a) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(b) Wipe the structure with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

(c) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

(d) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

J. Thrust Reverser Upper TPS Panel Blanket Removal

SUBTASK 78-31-13-020-004-G00

CAUTION

DO NOT INTERMIX LEFT AND RIGHT FLAME SHIELDS. MAKE SURE THAT YOU REMOVE AND INSTALL LEFT AND RIGHT FLAME SHIELDS ON THE CORRECT HALF OF THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove the two flame shields [17] and two flame shields [20] as follows:

(a) Remove five screws [6], five washers [7], screw [15] and washer [7] from the two flame shields [17].

(b) Remove the two flame shields [17].

(c) Remove the five nuts [10] and five washers [12] from the two flame shields [20].

(d) Remove the two flame shields [20].

SUBTASK 78-31-13-020-018-G00

CAUTION

DO NOT APPLY A SIDE LOAD WHEN YOU REMOVE THE SPACER FITTING. APPLY THE TORQUE TO THE FITTING. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Remove the spacer fitting [23], washer [24] and up to three washers [25] at two locations.

SUBTASK 78-31-13-020-006-G00

(3) Remove the upper TPS panel blanket [5] as follows:

(a) Remove the 26 caps [22], 26 nuts [10], and 26 washers [12] from the upper TPS panel blanket [5].

(b) Remove the four nuts [10] and four washers [14].

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(c) Remove the upper TPS panel blanket [5].

SUBTASK 78-31-13-020-019-G00

(4) Keep the all removed hardware for installation.

SUBTASK 78-31-13-160-005-G00

(5) Clean the area and remove all contamination as follows:

(a) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

(b) Wipe the structure with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(c) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

(d) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

K. Thrust Reverser Lower TPS Panel Blanket Removal

SUBTASK 78-31-13-020-008-G00

CAUTION

DO NOT APPLY A SIDE-LOAD TO THE MAST FITTING LUGS. BE CAREFUL TO REMOVE OR INSTALL THE BOLTS THAT ATTACH THE CONTROL RODS OF THE ELEVATOR TO THE MAST FITTING TO MAKE SURE NOT TO APPLY A SIDE-LOAD. IF YOU APPLY A SIDE-LOAD, DAMAGE WILL OCCUR AT THE BOTTOM OF THE MAST FITTING LUGS.

(1) Remove the spacer fitting [27], washer [26] and up to three washers [28] at two locations.

SUBTASK 78-31-13-020-009-G00

CAUTION

DO NOT INTERMIX LEFT AND RIGHT FLAME SHIELDS. MAKE SURE THAT YOU REMOVE AND INSTALL LEFT AND RIGHT FLAME SHIELDS ON THE CORRECT HALF OF THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Remove the lower TPS panel blanket [4] as follows:

(a) Remove four screws [6] and four washers [7] from the flame shield [18].

(b) Remove the flame shield [18].

(c) Remove the six nuts [10] and six washers [14].

(d) Remove the screw [6] and washer [14].

(e) Remove the screw [19] and washer [14].

(f) Remove the 12 caps [22], 12 nuts [10], and 12 washers [12].

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(g) Remove the lower TPS panel blanket [4].

SUBTASK 78-31-13-020-020-G00

(3) Keep the all removed hardware for installation.

SUBTASK 78-31-13-160-007-G00

(4) Clean the area and remove all contamination as follows:

(a) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

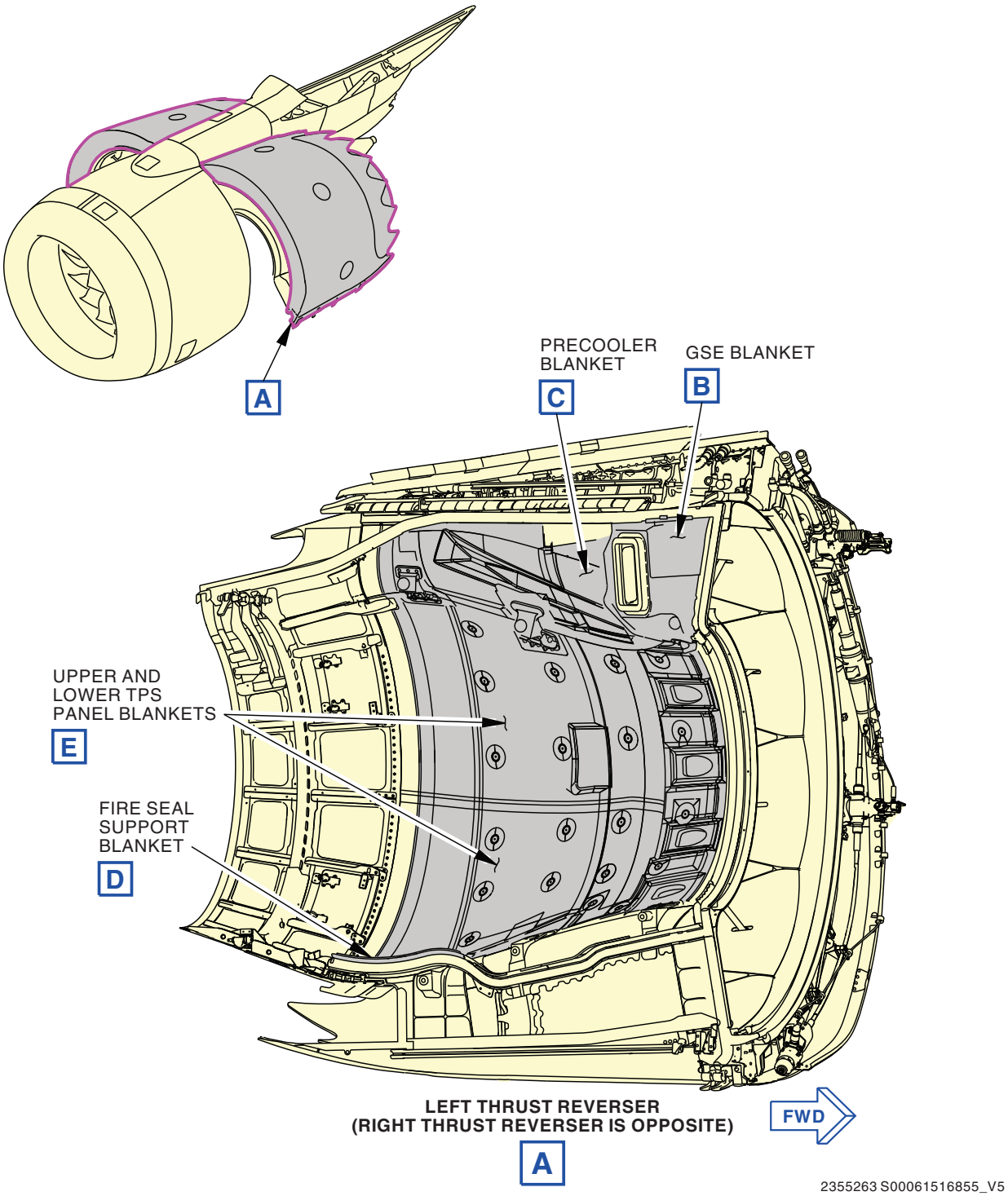
(b) Wipe the structure with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

(c) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

(d) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

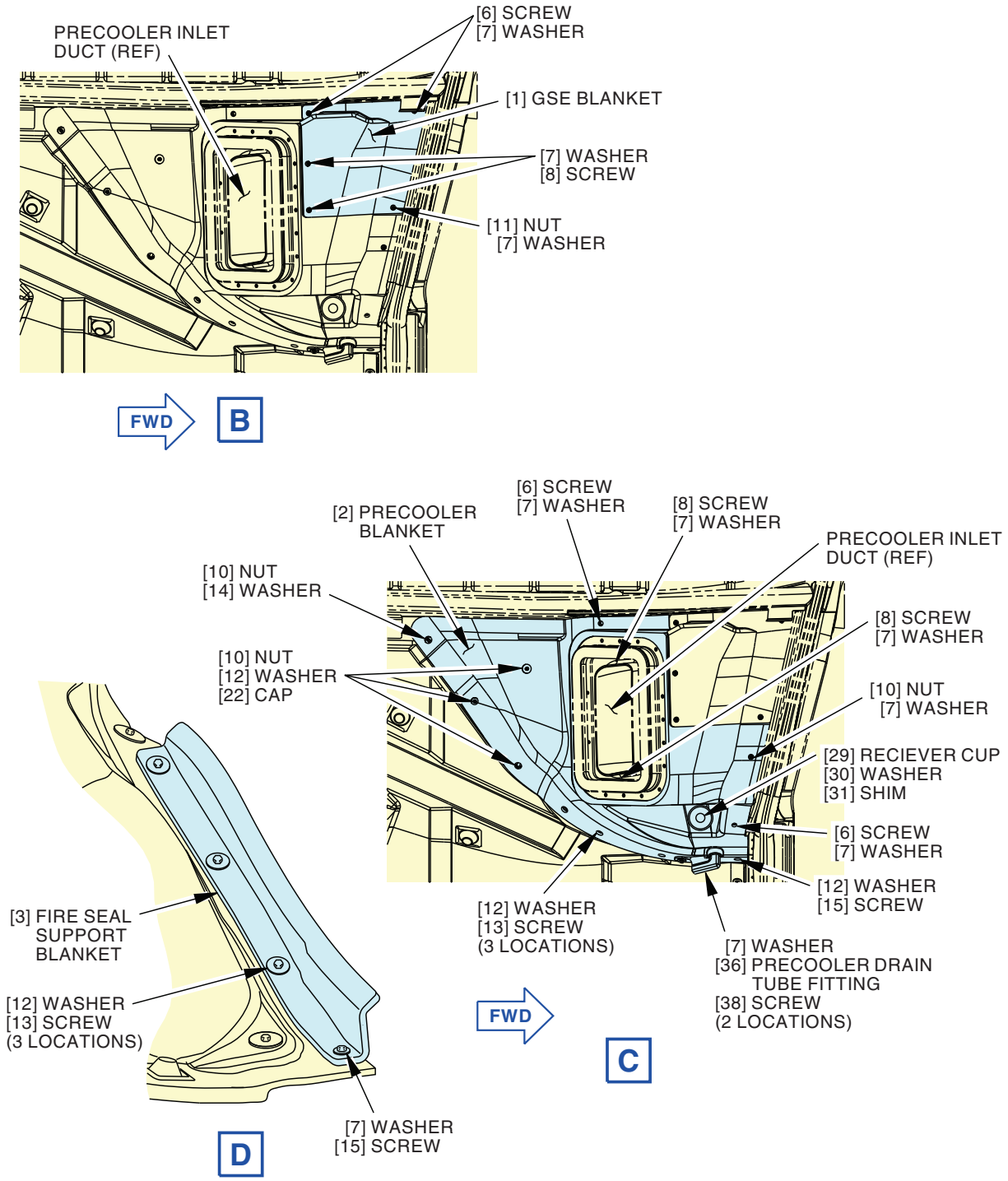
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Insulation Blanket Installation
Figure 401/78-31-13-990-801-G00 (Sheet 1 of 4)

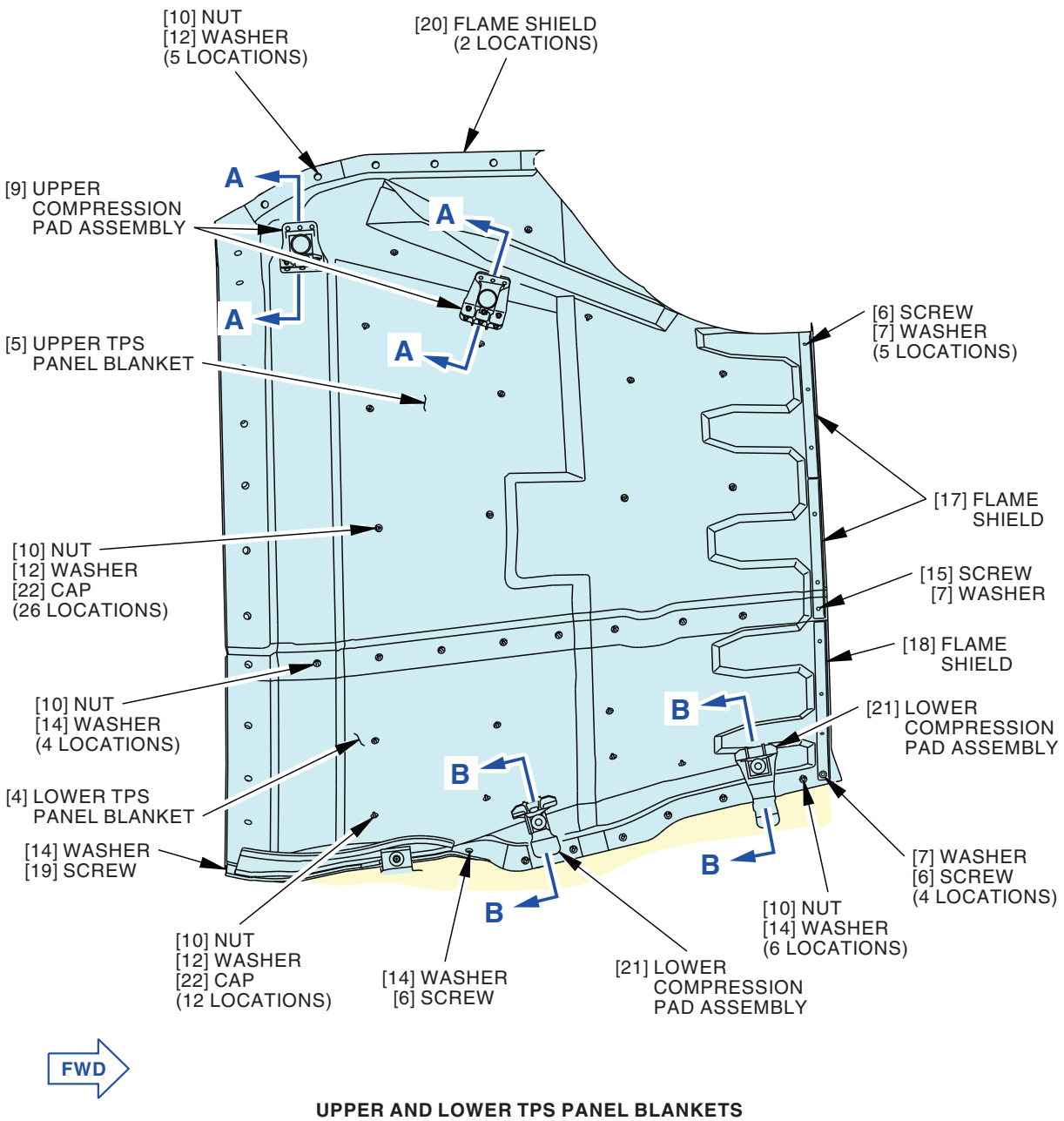
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Thrust Reverser Insulation Blanket Installation
Figure 401/78-31-13-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



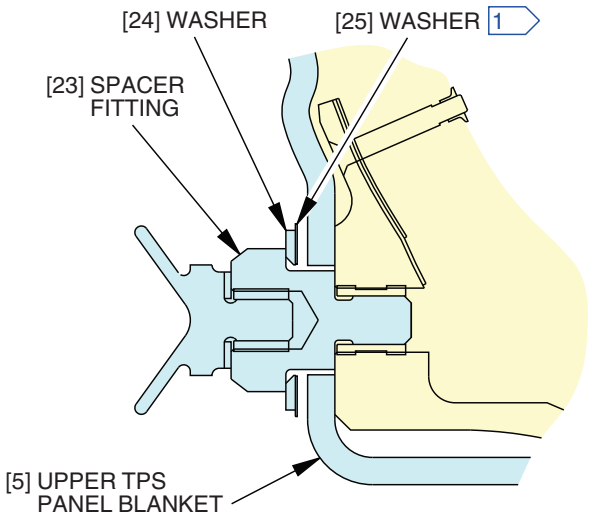
UPPER AND LOWER TPS PANEL BLANKETS

E

2541077 S0000603865_V4

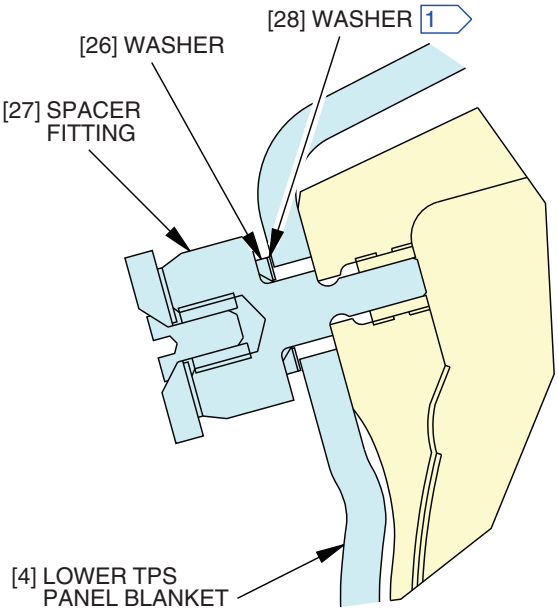
Thrust Reverser Insulation Blanket Installation
Figure 401/78-31-13-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UPPER COMPRESSION PAD ASSEMBLY

A-A



LOWER COMPRESSION PAD ASSEMBLY

B-B

1 UP TO THREE WASHERS MAY BE INSTALLED

2541133 S0000604141_V5

Thrust Reverser Insulation Blanket Installation
Figure 401/78-31-13-990-801-G00 (Sheet 4 of 4)

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TASK 78-31-13-400-801-G00

3. Insulation Blankets - Installation

(Figure 401)

A. General

CAUTION

FASTENERS CAN BE DIFFERENT IN LENGTH. IF YOU USE THE INCORRECT FASTENERS, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

REMOVE AND INSTALL THE INSULATION BLANKETS IN THE CORRECT ORDER GIVEN IN THESE PROCEDURES. IF THE ORDER OF REMOVAL AND INSTALLATION IS NOT FOLLOWED, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) This task is for the installation of the insulation blankets on the left and right thrust reverser with a composite inner wall on an engine.

(2) The procedures are the same for the left and right thrust reversers.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00634	Solvent - Stabilized Limonene Cleaner	BMS11-10 Type 1, 2, or 3
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(Continued)

Zone	Area
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Installation

SUBTASK 78-31-13-410-004-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Thrust Reverser Lower TPS Panel Blanket Installation

SUBTASK 78-31-13-420-003-G00

(1) Make sure that the area of the lower TPS panel blanket [4] is clean from contamination.

(a) If it is necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-013-G00

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(2) Position the lower TPS panel blanket [4] on the applicable thrust reverser.

SUBTASK 78-31-13-420-014-G00

CAUTION

WHEN YOU INSTALL THE SPACER FITTING, DO NOT CROSS THREAD OR APPLY A SIDE LOAD. APPLY TORQUE TO THE FITTING. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Install the washer [26] and spacer fitting [27] at two locations.

(a) If it is necessary, install a maximum of three washers [28] to accommodate the differences in the Thermal Protection System (TPS) thickness.

1) Make sure that there is no gap between the lower TPS panel blanket [4] and washer [26].

(b) If it is necessary, apply Never Seez, D50294, to the threads of the spacer fitting [27].

(c) Torque the spacer fitting [27] to 40 in-lb (4.5 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 78-31-13-420-015-G00

(4) Install the lower TPS panel blanket [4] as follows:

(a) Install 12 caps [22] between 12 washers [12] and lower TPS panel blanket [4].

1) Orient hinged edge facing the forward direction.

(b) Install the 12 nuts [10].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the 12 nuts [10] prior to installation.

2) Tighten the 12 nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(c) Once final nut torque is met, fold cover over the 12 caps [22] and firmly fold flaps to secure cover.

(d) Install the screw [19] and washer [14].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of screw [19] prior to installation.

2) Tighten the screw [19] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(e) Install the screw [6] and washer [14].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of screw [6] prior to installation.

2) Tighten the screw [6] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(f) Install the six nuts [10], and six washers [14].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the six nuts [10] prior to installation.

2) Tighten the six nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

SUBTASK 78-31-13-420-008-G00

(5) Make sure that the area of the flame shield [18] is clean of any contamination.

(a) If it is necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-019-G00

CAUTION

DO NOT INTERMIX LEFT AND RIGHT FLAME SHIELDS. MAKE SURE THAT YOU REMOVE AND INSTALL LEFT AND RIGHT FLAME SHIELDS ON THE CORRECT HALF OF THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(6) Install the flame shield [18] as follows:

(a) Position the flame shield [18] on the applicable thrust reverser.

(b) Install the four screws [6] and four washers [7] through the flame shield [18].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the four screws [6] prior to installation..

2) Tighten the four screws [6] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

H. Thrust Reverser Upper TPS Panel Blanket Installation

SUBTASK 78-31-13-160-008-G00

(1) Make sure that the area of the upper TPS panel blanket [5] is clean from contamination.

(a) If it is necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-005-G00

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(2) Position the upper TPS panel blanket [5] on the applicable thrust reverser.

SUBTASK 78-31-13-420-021-G00

CAUTION

WHEN YOU INSTALL THE SPACER FITTING, DO NOT CROSS THREAD OR APPLY A SIDE LOAD. APPLY TORQUE TO THE FITTING. IF YOU DO NOT OBEY INSTRUCTIONS, DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Install the washer [24] and spacer fitting [23] at two locations.

(a) If it is necessary, install a maximum of three washers [25] to accommodate the differences in the TPS thickness.

1) Make sure that there is no gap between the upper TPS panel blanket [5] and washer [24].

(b) Apply Never Seez, D50294, to the threads of the spacer fitting [23].

(c) Torque the spacer fitting [23] to 150 in-lb (16.9 N·m) - 165 in-lb (18.6 N·m).

SUBTASK 78-31-13-420-022-G00

(4) Install the upper TPS panel blanket [5] as follows:

(a) Install the four nuts [10] and four washers [14].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the four nuts [10].

2) Tighten the four nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(b) Install the 26 caps [22] between 26 washers [12] and upper TPS panel blanket [5].

1) Orient hinged edge facing the forward direction.

(c) Install the 26 nuts [10].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the 26 nuts [10] prior to installation.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

2) Tighten the 26 nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(d) Once final nut torque is met, fold cover over the 26 caps [22] and firmly fold flaps to secure cover.

SUBTASK 78-31-13-160-002-G00

(5) Make sure that the areas of the two flame shields [17] and two flame shields [20] are clean from all contamination.

(a) If it is necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-160-013-G00

CAUTION

DO NOT INTERMIX LEFT AND RIGHT FLAME SHIELDS. MAKE SURE THAT YOU REMOVE AND INSTALL LEFT AND RIGHT FLAME SHIELDS ON THE CORRECT HALF OF THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(6) Install the two flame shields [20] as follows:

(a) Position the two flame shields [20] on the applicable thrust reverser.

(b) Install the two flame shields [20], five nuts [10], and five washers [12].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the five nuts [10].

2) Tighten the five nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

SUBTASK 78-31-13-160-015-G00

CAUTION

DO NOT INTERMIX LEFT AND RIGHT FLAME SHIELDS. MAKE SURE THAT YOU REMOVE AND INSTALL LEFT AND RIGHT FLAME SHIELDS ON THE CORRECT HALF OF THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(7) Install the two flame shields [17] as follows:

(a) Position the two flame shields [17] on the applicable thrust reverser.

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CAUTION

FASTENERS CAN BE DIFFERENT IN LENGTH. IF YOU USE THE INCORRECT FASTENERS, DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Install the two flame shields [17], five screws [6], five washers [12], screw [15], and washer [7].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the five screws [6] and screw [15].

2) Tighten the five screws [6] and screw [15] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

I. Thrust Reverser Fireseal Support Insulation Blanket Installation

SUBTASK 78-31-13-160-003-G00

(1) Make sure that the area of the fireseal support blanket [3] is clean from all contamination.

(a) If it is necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-002-G00

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(2) Install the fireseal support blanket [3] as follows:

(a) Position the fireseal support blanket [3] on the applicable thrust reverser.

(b) Install the three screws [13] and three washers [12] on the fireseal support blanket [3].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the three screws [13] prior to installation.

2) Tighten the three screws [13] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(c) Install the screw [15] and washer [7] on the fireseal support blanket [3].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of thescrew [15] prior to installation.

2) Tighten the screw [15] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

J. Thrust Reverser Precooler Insulation Blanket Installation

SUBTASK 78-31-13-160-001-G00

(1) Make sure that the area of the precooler blanket [2] is free from all contamination.

(a) If it is necessary, remove all contamination as follows:

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1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.
2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.
3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.
4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-001-G00

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(2) Install the precooler blanket [2] as follows:
   (a) Position the precooler blanket [2] on the applicable thrust reverser.
   (b) Install the three caps [22] between the three washers [12] and precooler blanket [2].
      1) Orient hinged edge facing the forward direction.
   (c) Install the three nuts [10].
      1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the three nuts [10] prior to installation.
      2) Tighten the three nuts [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).
   (d) Once final nut torque is met, fold cover over three caps [22] and firmly fold flaps to secure cover.
   (e) Install the nut [10] and washer [14].
      1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the nut [10] prior to installation.
      2) Tighten the nut [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).
   (f) Install the three screws [13] and three washers [12].
      1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the three screws [13] prior to installation.
      2) Tighten the three screws [13] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).
   (g) Position the precooler drain tube fitting [36] on the applicable precooler blanket [2].
   (h) Install the two screws [38] and two washers [7].
      1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the two screws [38] prior to installation.
      2) Tighten the two screws [38] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).
   (i) Install the screw [15] and washer [12].
      1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the screw [15] prior to installation.
      2) Tighten the screw [15] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).
   (j) Install the two screws [6] and two washers [7], at two locations.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the two screws [6] prior to installation.

2) Tighten the two screws [6] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(k) Install up to five shims [31], washer [30], and receiver cup [29] (TASK 78-31-01-820-801-G00).

(l) Install the nut [10] and the washer [7].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the nut [10] prior to installation.

2) Tighten the nut [10] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

(m) Install the screw [8] and washer [7] at two locations.

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the screw [8] prior to installation.

2) Tighten the screw [8] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

K. Thrust Reverser GSE Insulation Blanket Installation

SUBTASK 78-31-13-160-006-G00

(1) Make sure that the area of the GSE blanket [1] is clean from contamination.

(a) If necessary, remove all contamination as follows:

1) Wipe or scrub the structure with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid or oil.

2) Wipe the structure with solvent, B00184, or solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

3) Wipe the surface dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the surface with a new clean cloth and solvent until no contamination is shown on the cloth.

SUBTASK 78-31-13-420-007-G00

CAUTION

HANDLE THE INSULATION BLANKETS WITH CARE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO EQUIPMENT CAN OCCUR

(2) Install the GSE blanket [1] as follows:

(a) Position the GSE blanket [1] on the applicable thrust reverser.

(b) Install the two screws [6], two washers [7], two screws [8], two washers [7], nut [11], and washer [7] to the GSE blanket [1].

1) Apply a continuous and uniform layer of Pure Nickel Special compound, D00006, to the threads of the two screws [6], two screws [8], and nut [11] prior to installation.

2) Tighten the two screws [6], two screws [8], and nut [11] to 20 in-lb (2.3 N·m) - 22 in-lb (2.5 N·m).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

L. Put the Airplane Back to its Usual Condition

SUBTASK 78-31-13-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

END OF TASK

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

TASK 78-31-13-000-802-G00

4. Insulation Blankets - Removal

A. General

(1) This task is for the removal of the insulation blankets from the left and right thrust reversers.

(a) The removal of the fireseal support blanket is different on the left and right thrust reversers.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Removal

SUBTASK 78-31-13-040-001-G00

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-13-010-011-G00

(2) Open the applicable fan cowls:

(TASK 71-11-04-010-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-13-010-012-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Open the applicable thrust reversers:

(TASK 78-31-00-010-801-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Removal of the Precooler Thermal Barrier

SUBTASK 78-31-13-020-024-G00

(1) Remove the small precooler thermal barrier [41]:

(a) Remove the screws [42], washers [43], screws [44], and washers [45].

SUBTASK 78-31-13-020-021-G00

(2) To remove the large precooler thermal barrier [40], do these steps:

(a) Remove the receiver cup [72], washer [73], and shims [74].

NOTE: The shims are optional to pre-rig the compression fittings between the washer and receiver cup (TASK 78-31-01-820-801-G00).

(b) Remove the screw [59] and washer [60].

(c) Remove the nuts [56].

1) If installed, remove the washers [57].

(d) Remove the two nuts [50] and two washers [51].

(e) Remove the nuts [54] and washers [55].

(f) Remove the nut [52] and washer [53].

(g) Remove the screws [46] and washers [47].

(h) Remove the screw [48] and washer [49].

(i) Remove the large precooler thermal barrier [40].

G. Removal of the Fireseal Support Blanket

SUBTASK 78-31-13-020-022-G00

(1) For the right hand side, do these steps to remove the fireseal support blanket [63]:

(a) Remove the screws [64] and washers [65].

SUBTASK 78-31-13-020-023-G00

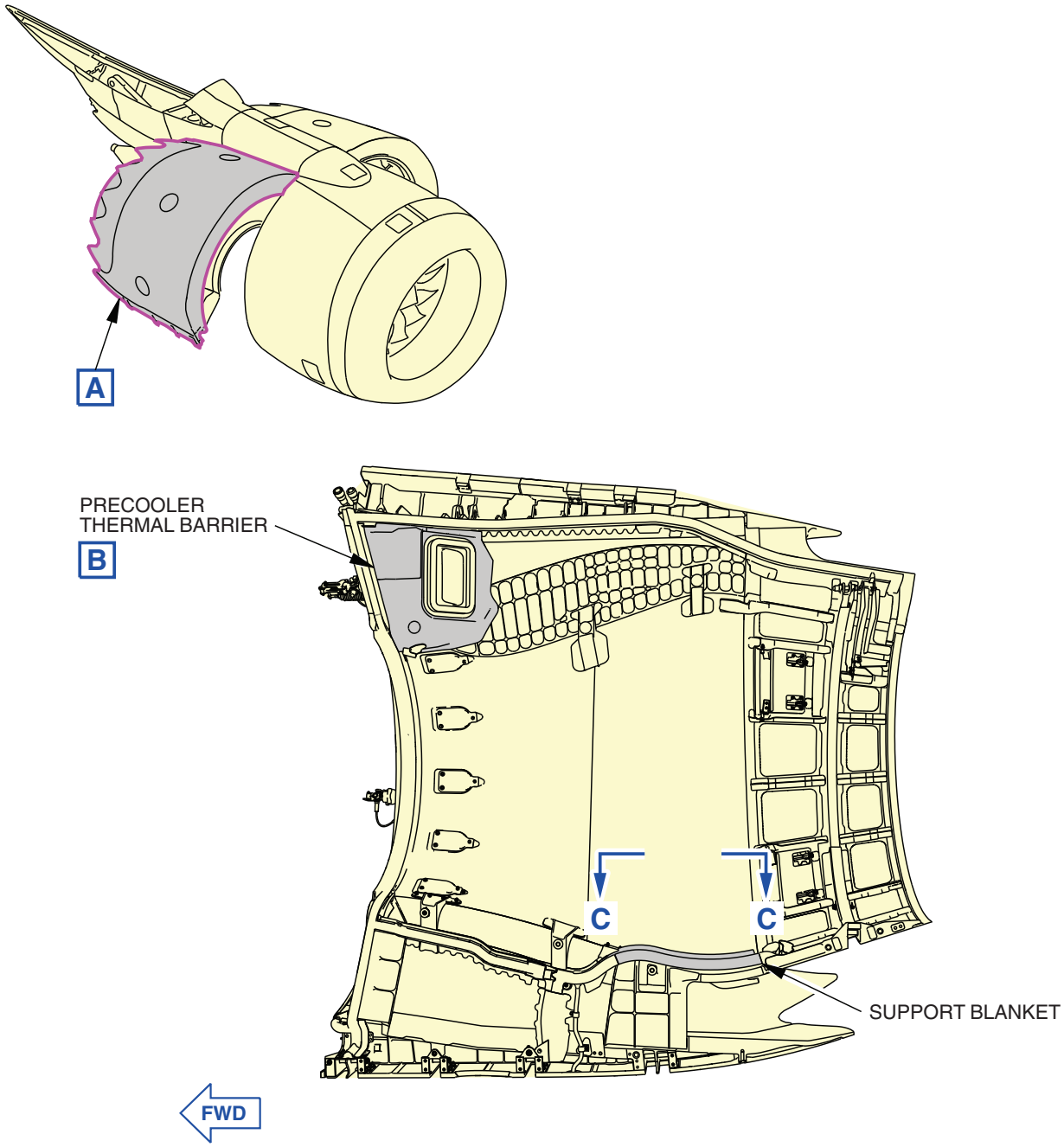
(2) For the left hand side, do these steps to remove the fireseal support blanket [66]:

(a) Remove the screws [67] and washers [68].

(b) Remove the screws [69] and washers [70].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



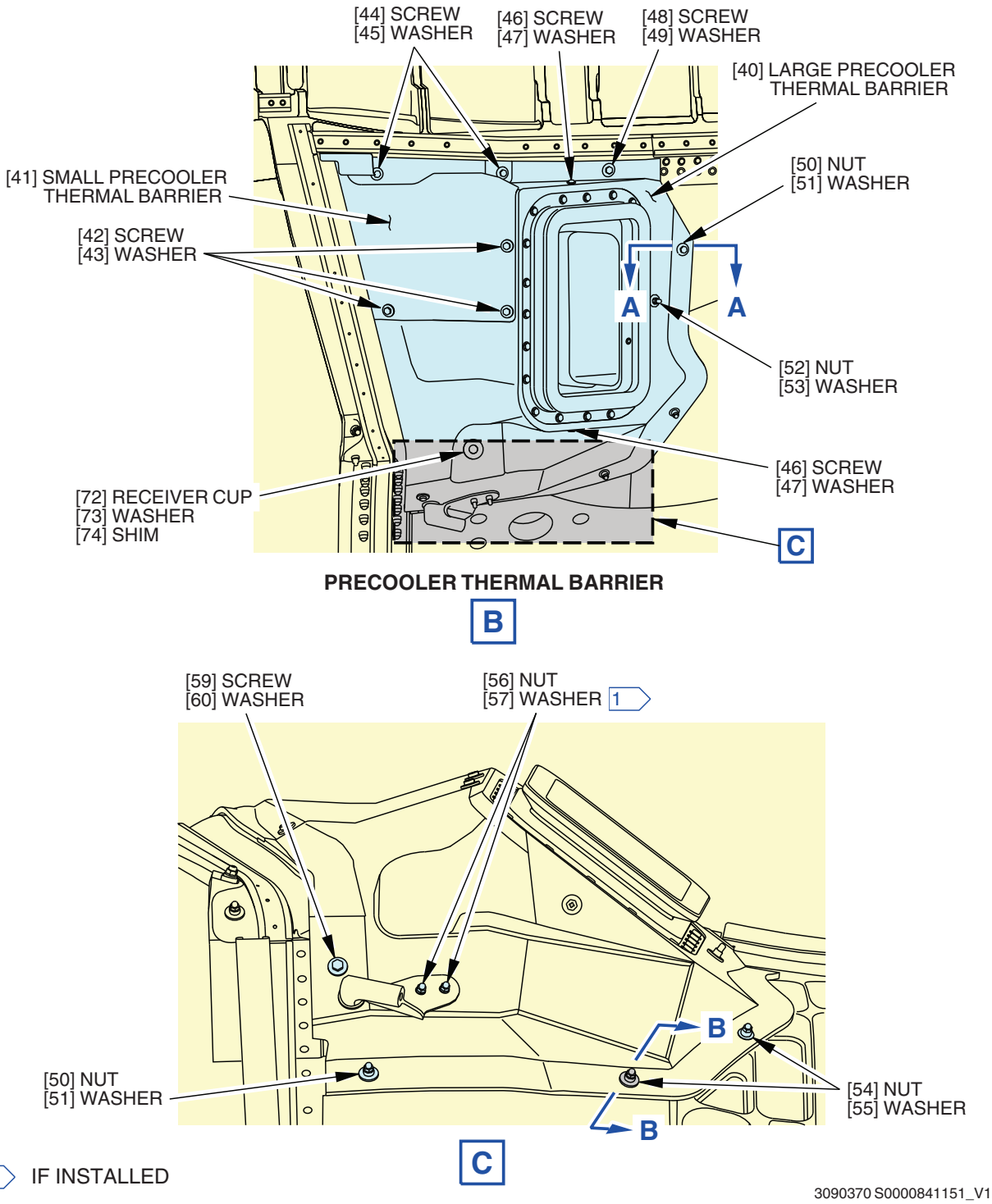
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

A

3090342 S0000841133_V1

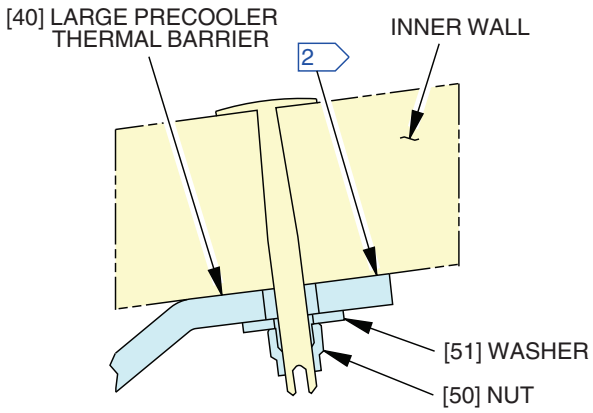
Thrust Reverser Insulation Blanket Installation
Figure 402/78-31-13-990-815-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

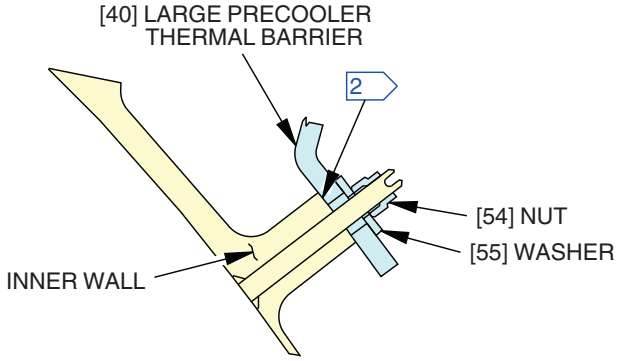


Thrust Reverser Insulation Blanket Installation
Figure 402/78-31-13-990-815-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



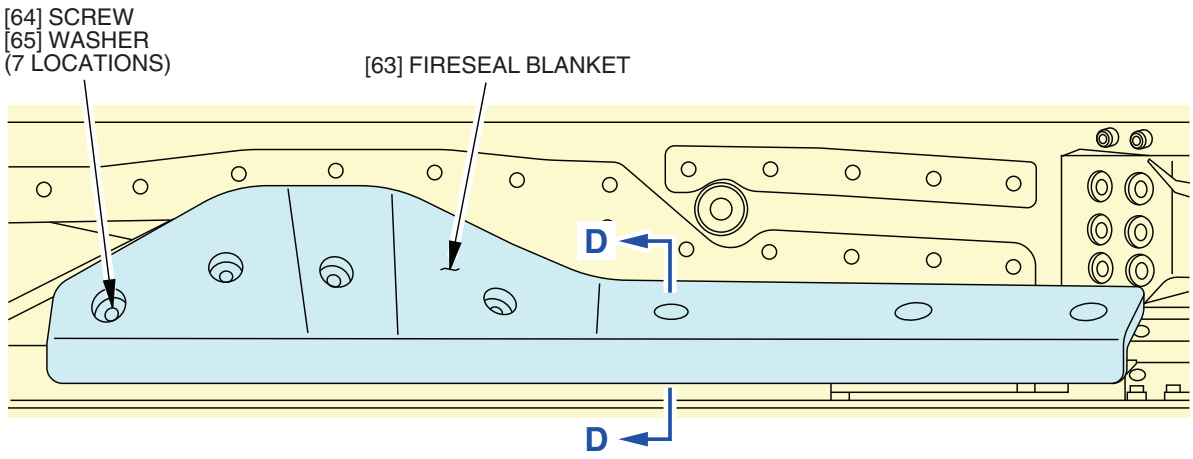
B-B

2 APPLY FAY SURFACE SEAL WITH BMS5-63

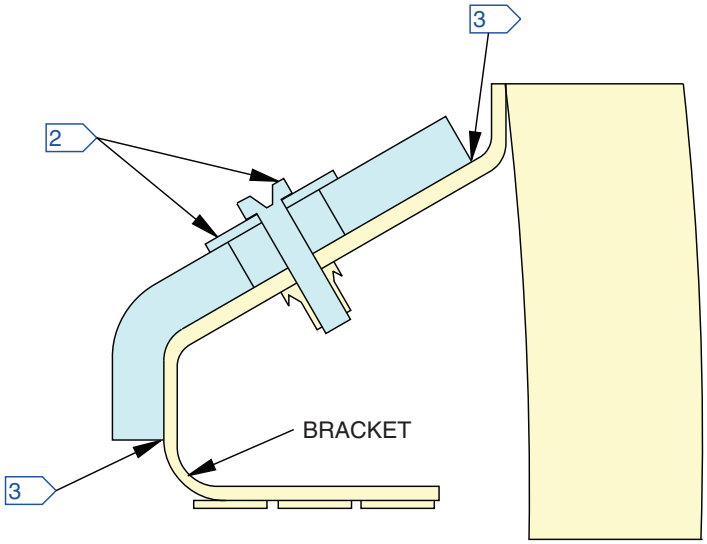
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Thrust Reverser Insulation Blanket Installation
Figure 402/78-31-13-990-815-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIRESEAL BLANKET
(RIGHT HAND SIDE)
C-C



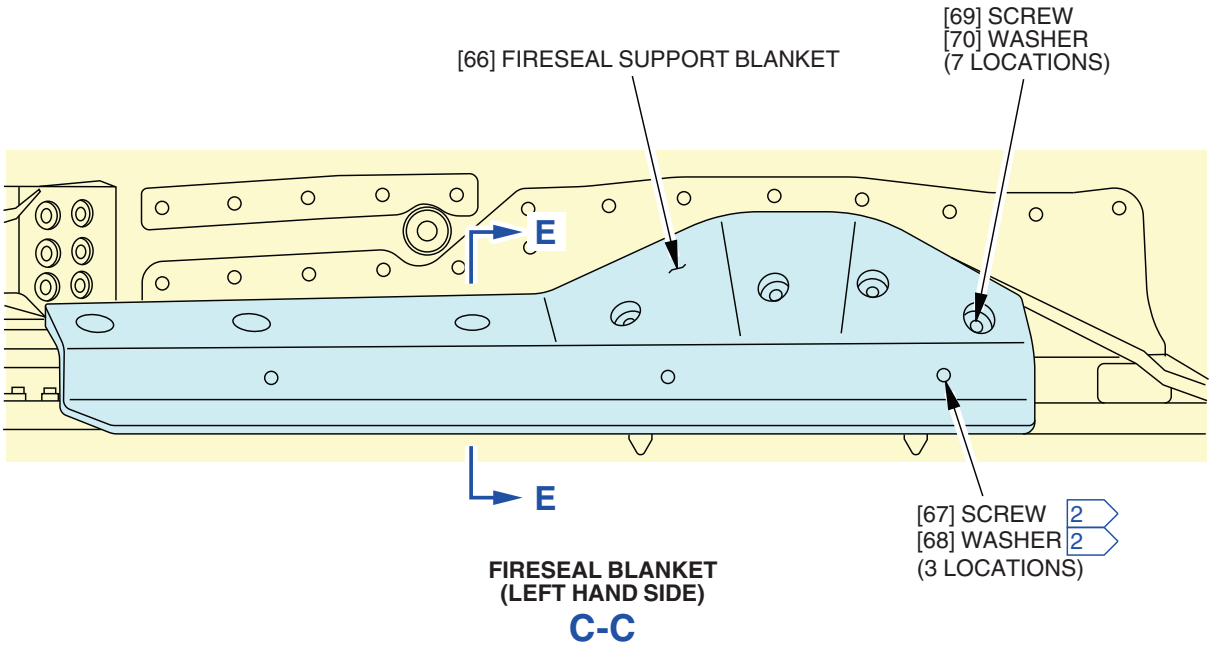
(EXAMPLE)
D-D

2 APPLY FAY SURFACE SEAL WITH BMS5-63
3 APPLY FILLET SEAL WITH BMS5-63

3090862 S0000841155_V1

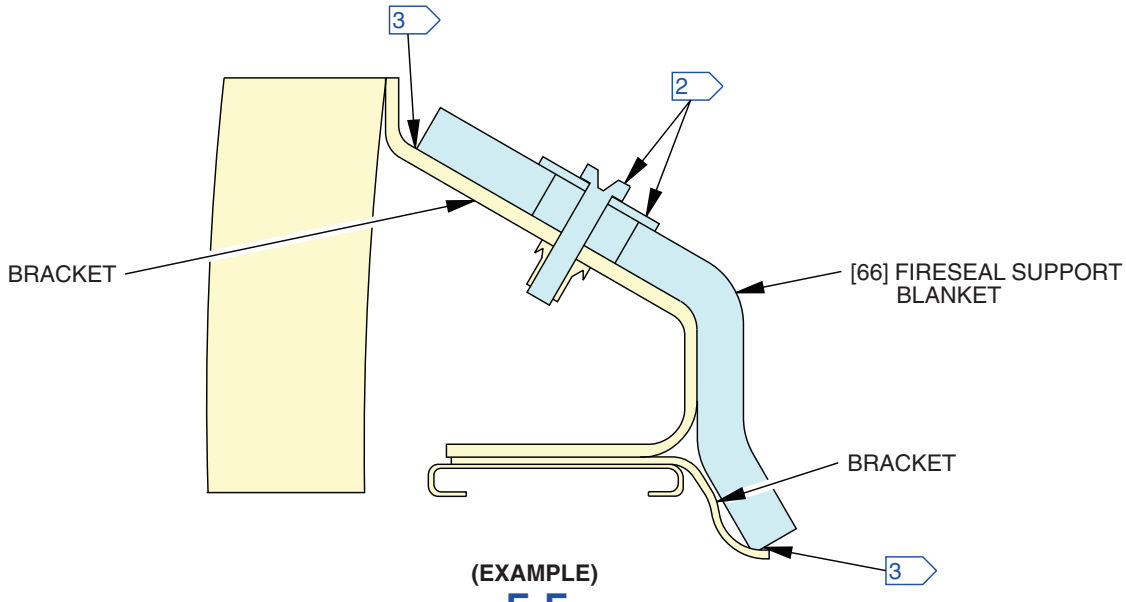
Thrust Reverser Insulation Blanket Installation
Figure 402/78-31-13-990-815-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FIRESEAL BLANKET
(LEFT HAND SIDE)

C-C



(EXAMPLE)

E-E

2 APPLY FAY SURFACE SEAL WITH BMS5-63

3 APPLY FILLET SEAL WITH BMS5-63

3090878 S0000841156_V1

Thrust Reverser Insulation Blanket Installation
Figure 402/78-31-13-990-815-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-13-400-802-G00

5. Insulation Blankets - Installation

A. General

(1) This task includes the steps to install the insulation blankets on the left and right thrust reversers.

(a) The installation of the fireseal support blanket is different on the left and right thrust reverser.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-01-820-801-G00	Thrust Reverser Adjustment (P/B 501)

C. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
B01054	Solvent - Methyl Ethyl Ketone and sec-Butyl Alcohol Blend - (MEK:secButyl Alcohol - 42:58 Percent)	BAC5750, ASTM D740/ASTM D1007
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G50262	Wiper - Cleaning	BMS15-5
G51677	Cloth - 100% Synthetic or Blended Synthetic, Cotton or Cellulose Material	AMS3819 Class 1, 2, or 4, Grade A or B, Form 1

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Installation

SUBTASK 78-31-13-040-002-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-31-13-010-013-G00

(2) Open the applicable fan cowl access panels:

(TASK 71-11-04-010-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-13-010-014-G00

(3) Open the applicable thrust reversers:

(TASK 78-31-00-010-801-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Installation of the Precooler Thermal Barrier

SUBTASK 78-31-13-110-002-G00

(1) Clean the mating surfaces of the large precooler thermal barrier [40] and the inner wall, do these steps:

NOTE: Surfaces that look clean can have contamination on them.

(a) Clean the surfaces with a clean, dry wiper, G50262, or cloth, G51677, to remove loose contamination.

(b) Clean the surfaces with a solvent, B00184, solvent, B00148, solvent, B00666, or solvent, B01054, on a clean wiper, G50262, or cloth, G51677.

(c) While the solvent is on the surface, dry the surface with a clean, dry wiper, G50262, or cloth, G51677.

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(d) Continue to clean the surface until the wiper, G50262, or cloth, G51677, and surfaces are free of contamination.

SUBTASK 78-31-13-390-001-G00

(2) Apply a fay surface seal with sealant, A00160, between the mating surfaces of the large precooler thermal barrier [40] and the inner wall.

SUBTASK 78-31-13-420-036-G00

(3) Do these steps to install the large precooler thermal barrier [40]:

(a) Put the large precooler thermal barrier [40] in its position.

(b) Install the screw [48] and washer [49]

1) Tighten the screw [48] to 35 in-lb (4.0 N·m).

(c) Install the two screws [46] and washers [47].

1) Tighten the screws [46] to 35 in-lb (4.0 N·m).

(d) Install the two nuts [50] and washers [51].

1) Tighten the nuts [50] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

(e) Install the nut [52] and washer [53].

1) Tighten the nut [52] to 29 in-lb (3.3 N·m) - 31 in-lb (3.5 N·m).

(f) Install the two nuts [54] and washers [55].

1) Tighten the nuts [54] to 34 in-lb (3.8 N·m) - 36 in-lb (4.1 N·m).

(g) Install the screw [59] and washer [60].

1) Tighten the screw [59] to 35 in-lb (4.0 N·m).

(h) Install the two nuts [56] and if necessary, the two washers [57].

1) Tighten the nuts [56] to 29 in-lb (3.3 N·m) - 31 in-lb (3.5 N·m).

(i) Install the receiver cup [72]:

1) Apply Pure Nickel Special compound, D00006, to the threads of the receiver cup [72].

2) Install a maximum of five shims [74], the washer [73], and receiver cup [72].

NOTE: The shims are optional to pre-rig the compression fittings between the washer and receiver cup (TASK 78-31-01-820-801-G00).

3) Tighten the receiver cup [72] to 80 in-lb (9.0 N·m) - 95 in-lb (10.7 N·m).

SUBTASK 78-31-13-420-037-G00

(4) Install the small precooler thermal barrier [41]:

(a) Install the two screws [44] and washers [45].

1) Tighten the screws [44] to 35 in-lb (4.0 N·m)

(b) Install the three screws [42] and washers [43].

1) Tighten the screws [42] to 35 in-lb (4.0 N·m).

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

H. Installation of the Fireseal Support Blanket on the Right Side

SUBTASK 78-31-13-110-003-G00

(1) Clean the mating surfaces of the fireseal support blanket [63], bracket, washers [65], and screws [64], do these steps:

NOTE: Surfaces that look clean can have contamination on them.

(a) Clean the surfaces with a clean, dry wiper, G50262, or cloth, G51677, to remove loose contamination.

(b) Clean the surfaces with a solvent, B00184, solvent, B00148, solvent, B00666, or solvent, B01054, on a clean wiper, G50262, or cloth, G51677.

(c) While the solvent is on the surface, dry the surface with a clean, dry wiper, G50262, or cloth, G51677.

(d) Continue to clean the surface until the wiper, G50262, or cloth, G51677, and surfaces are free of contamination.

SUBTASK 78-31-13-420-038-G00

(2) Do these steps to install the fireseal support blanket [63]:

(a) Apply a fay surface seal with sealant, A00160, to the mating surfaces of the six washers [65] and screws [64].

(b) Install the screws [64] and washers [65].

1) Tighten the screws [64] to 35 in-lb (4.0 N·m).

(c) Apply a fillet seal with sealant, A00160, to the mating surfaces of the fireseal support blanket [63] and the bracket.

I. Installation of the Fireseal Support Blanket on the Left Side

SUBTASK 78-31-13-110-004-G00

(1) Clean the mating surfaces of the fireseal support blanket [66], bracket, screws [67], washers [68], screws [69], and washers [70], do these steps:

NOTE: Surfaces that look clean can have contamination on them.

(a) Clean the surfaces with a clean, dry wiper, G50262, or cloth, G51677, to remove loose contamination.

(b) Clean the surfaces with a solvent, B00184, solvent, B00148, solvent, B00666, or solvent, B01054, on a clean wiper, G50262, or cloth, G51677.

(c) While the solvent is on the surface, dry the surface with a clean, dry wiper, G50262, or cloth, G51677.

(d) Continue to clean the surface until the wiper, G50262, or cloth, G51677, and surfaces are free of contamination.

SUBTASK 78-31-13-420-039-G00

(2) Do these steps to install the fireseal support blanket [66]:

(a) Apply a fay surface seal with sealant, A00160, to the mating surfaces of the three screws [67] and washers [68].

(b) Install the screws [67] and washers [68].

1) Tighten the screws [67] to 35 in-lb (4.0 N·m).

(c) Apply a fay surface seal with sealant, A00160, to the mating surfaces of the six screws [69] and washers [70].

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(d) Install the screws [69] and washers [70].

1) Tighten the screws [69] to 35 in-lb (4.0 N·m).

(e) Apply a fillet seal with sealant, A00160, to the mating surfaces of the fireseal support blanket [66] and the brackets.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-13-410-016-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS WHEN YOU CLOSE THE THRUST REVERSERS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Close the applicable thrust reversers:

(TASK 78-31-00-010-802-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-13-410-017-G00

(2) Close the applicable fan cowls:

(TASK 71-11-04-410-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-31-13-440-001-G00

(3) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INSULATION BLANKET - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A detailed inspection of the insulation blankets on the thrust reversers.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

(2) A detailed inspection of the insulation blanket seals on the thrust reversers.

PTF ALL

B. The Insulation Blankets are sometimes referred to as the Thermal Protection System.

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

TASK 78-31-13-200-802-G00

2. Insulation Blankets Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to examine the insulation blankets and panels on the left and right thrust reversers.

(2) The procedures are the same for the left and right thrust reversers.

(3) The insulation blankets are as follows:

(a) GSE Blanket

(b) Precooler Blanket

(c) Fire Seal Support Blanket

(d) Upper Thermal Protection System (TPS) Panel

(e) Lower TPS Panel.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)
78-31-13-300-801-G00	Insulation Blanket Repair (P/B 801)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(Continued)

Zone	Area
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-13-010-005-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

SUBTASK 78-31-13-010-008-G00

(2) Remove the insulation blankets, upper TPS panel, and lower TPS panel (TASK 78-31-13-000-801-G00).

F. Insulation Blankets Inspection

SUBTASK 78-31-13-210-008-G00

(1) Examine both sides of the insulation blanket and panels for damage (Figure 601):

(a) Missing insulation material or fluid contamination

1) Not serviceable.

(b) Cracks and tears

1) Cracks and tears are not permitted.

2) A temporary repair can be done with these conditions:

NOTE: This damage can not be repaired in all locations (TASK 78-31-13-300-801-G00).

a) The tear or crack in the outer metal sheet is not longer than 3.0 in. (76.2 mm).

b) Around each damaged area in all directions, there is a minimum surface distance that is not damaged of not less than 0.5 in. (12.7 mm).

c) The damaged area is not less than 0.5 in. (12.7 mm) from a grommet, sharp bend, attaching parts or edge.

(c) Holes in the outer metal sheet.

1) Holes are not permitted.

2) A temporary repair can be done with these conditions:

NOTE: This damage can not be repaired in all locations (TASK 78-31-13-300-801-G00).

a) The hole in the outer metal sheet is not more than 0.25 in. (6.4 mm) in diameter.

b) Around each damaged area in all directions, there is a minimum surface distance that is not damaged of not less than 0.5 in. (12.7 mm).

c) The damaged area is not less than 0.5 in. (12.7 mm) from a grommet, sharp bend, attaching parts or edge.

SUBTASK 78-31-13-210-007-G00

(2) If the damage is in the limits and conditions, you can do a temporary repair of the insulation blanket and panels on the airplane.

(a) A temporary repair has a Continue-In-Service limit of not more than 500 hours and must be replaced with a permanent repair.

NOTE: For the permitted permanent repair limits and repair procedures, refer to the Component Maintenance Manual (CMM) 78-31-94, Repair 3-1.

(b) Do this task: Insulation Blanket Repair, TASK 78-31-13-300-801-G00.

SUBTASK 78-31-13-210-006-G00

(3) If you find damage to the insulation blanket and panels outside the limits and conditions, unless you are given other instructions, replace the insulation blanket and panels.

(a) Do this task: Insulation Blankets - Removal, TASK 78-31-13-000-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

(b) Do this task: Insulation Blankets - Installation, TASK 78-31-13-400-801-G00.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-13-410-013-G00

(1) Install the insulation blankets, upper TPS panel, and lower TPS panel (TASK 78-31-13-400-801-G00).

SUBTASK 78-31-13-410-009-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

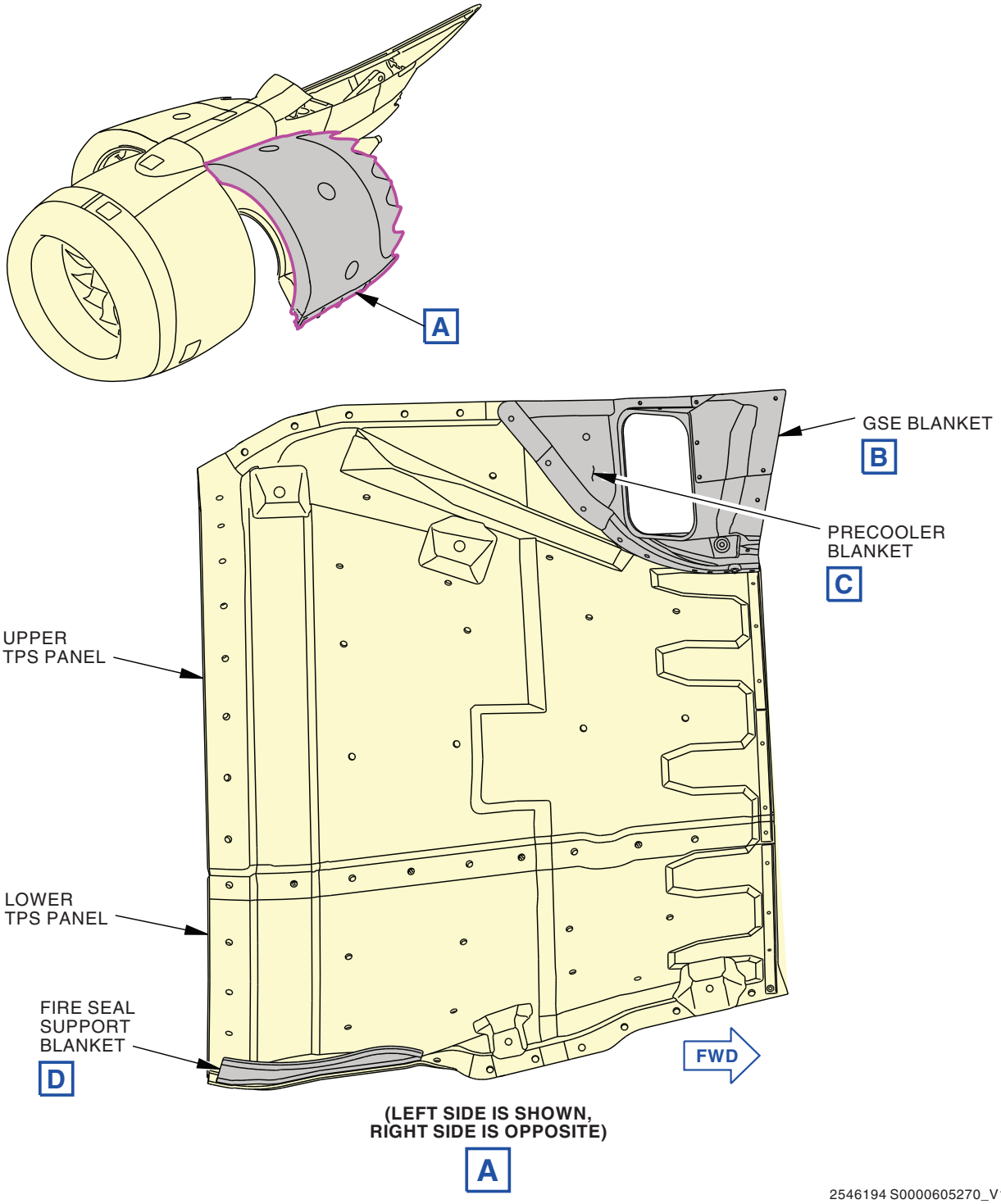
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

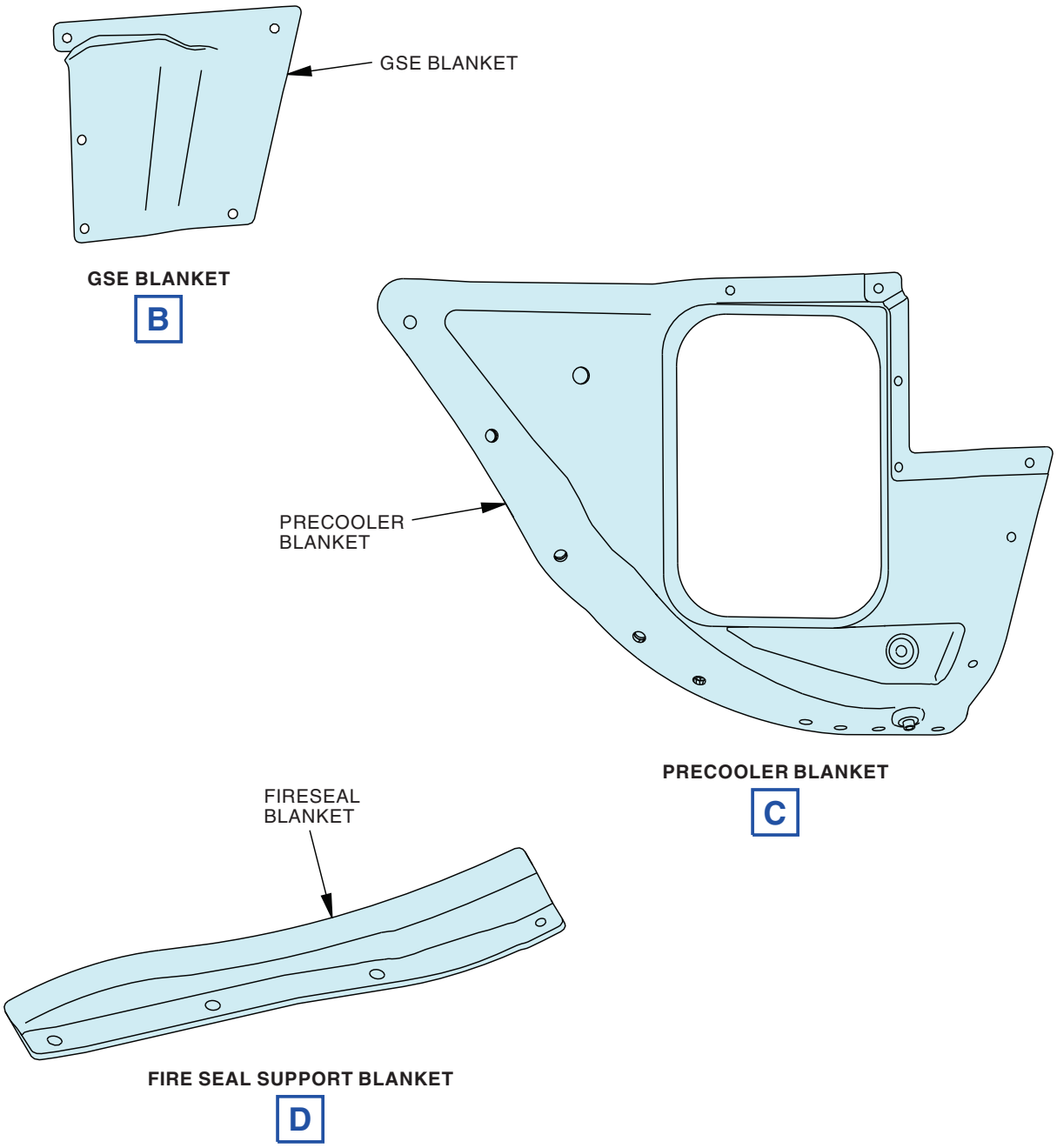
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Insulation Blankets Inspection
Figure 601/78-31-13-990-806-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Insulation Blankets Inspection
Figure 601/78-31-13-990-806-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

TASK 78-31-13-200-803-G00

3. Insulation Blankets Inspection

A. General

(1) This task gives the instructions to examine the insulation blankets and panels on the left and right thrust reversers.

(2) The procedures are the same for the left and right thrust reversers.

(3) The insulation blankets are as follows:

(a) Fireseal Precooler Blanket

(b) Fireseal Precooler Thermal Barrier

(c) Fireseal Support Blanket.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)
78-31-13-300-801-G00	Insulation Blanket Repair (P/B 801)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-13-010-010-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-13-010-015-G00

(2) Remove the insulation blankets, do this task: Insulation Blankets - Removal, TASK 78-31-13-000-801-G00.

F. Insulation Blankets Inspection

SUBTASK 78-31-13-210-009-G00

(1) Examine both sides of the insulation blanket and panels for damage (Figure 602):

(a) Missing insulation material or fluid contamination

1) Not serviceable.

(b) Cracks and tears

1) Cracks and tears are not permitted.

2) A temporary repair can be done with these conditions:

NOTE: This damage can not be repaired in all locations (Insulation Blanket Repair, TASK 78-31-13-300-801-G00).

a) The maximum length of any damage is not more than 5.0 in. (127.0 mm).

b) Around each damaged area in all directions, there is a minimum surface distance that is not damaged of not less than 0.5 in. (12.7 mm).

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

c) The damaged area is not less than 0.5 in. (12.7 mm) from a grommet, sharp bend, attaching parts or edge.

(c) Holes in the outer metal sheet.

1) Holes are not permitted.

2) A temporary repair can be done with these conditions:

NOTE: This damage can not be repaired in all locations (TASK 78-31-13-300-801-G00).

a) The maximum damage size is not more than 3 in^2 (19.4 cm^2).

b) Around each damaged area in all directions, there is a minimum surface distance that is not damaged of not less than 0.5 in. (12.7 mm).

c) The damaged area is not less than 0.5 in. (12.7 mm) from a grommet, sharp bend, attaching parts or edge.

SUBTASK 78-31-13-210-010-G00

(2) If the damage is in the limits and conditions, you can do a temporary repair of the insulation blanket and panels on the airplane.

(a) A temporary repair has a Continue-In-Service limit of not more than 500 hours.

NOTE: A temporary repair must be replaced in less than 500 hours with another temporary repair.

(b) Do this task: Insulation Blanket Repair, TASK 78-31-13-300-801-G00.

SUBTASK 78-31-13-960-002-G00

(3) If you find damage to the insulation blanket and panels outside the limits and conditions, unless you are given other instructions, replace the insulation blanket and panels:

(a) Do this task: Insulation Blankets - Removal, TASK 78-31-13-000-801-G00.

(b) Do this task: Insulation Blankets - Installation, TASK 78-31-13-400-801-G00.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-13-410-018-G00

(1) Install the insulation blankets, do this task: Insulation Blankets - Installation, TASK 78-31-13-400-801-G00.

SUBTASK 78-31-13-410-015-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

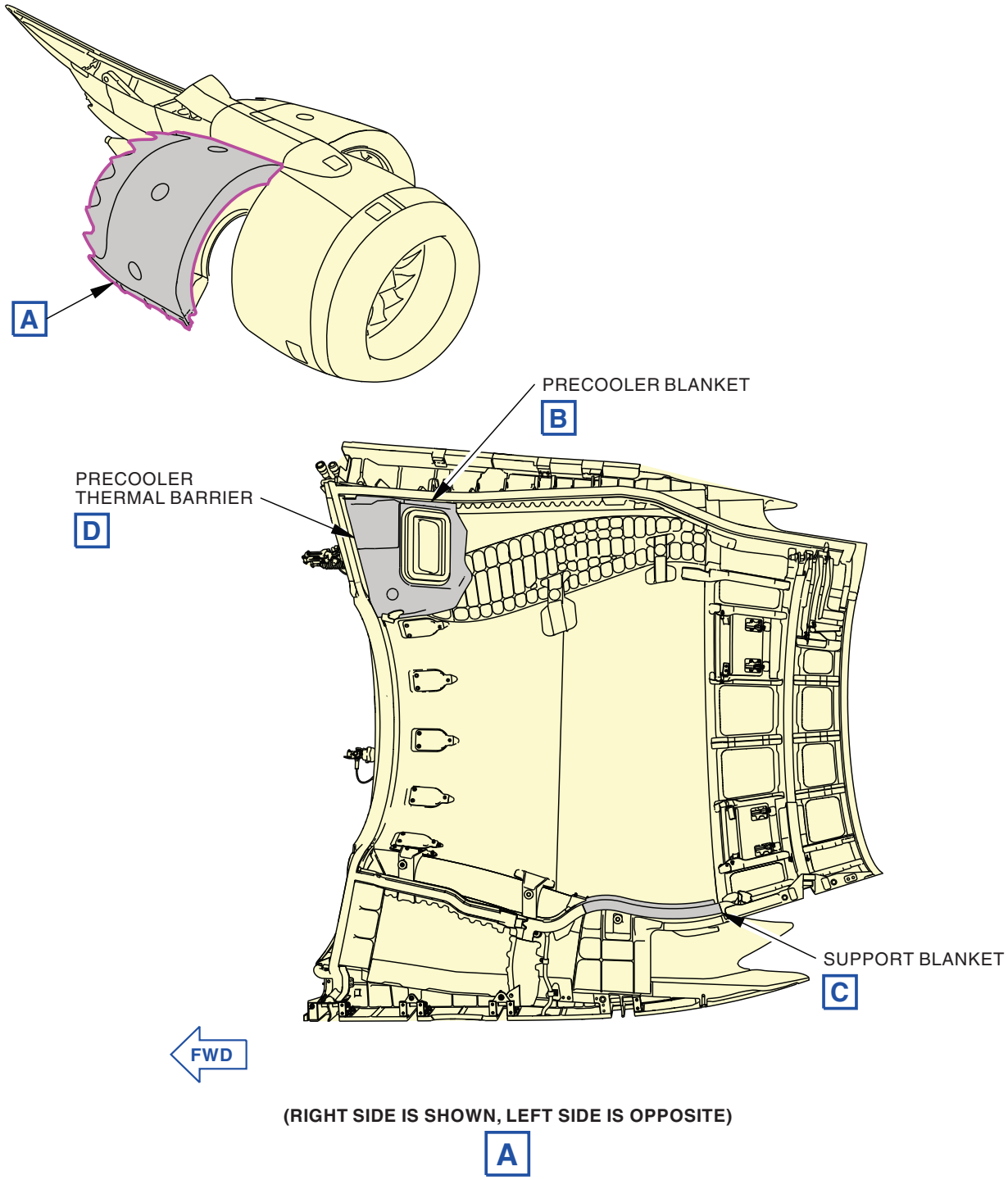
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



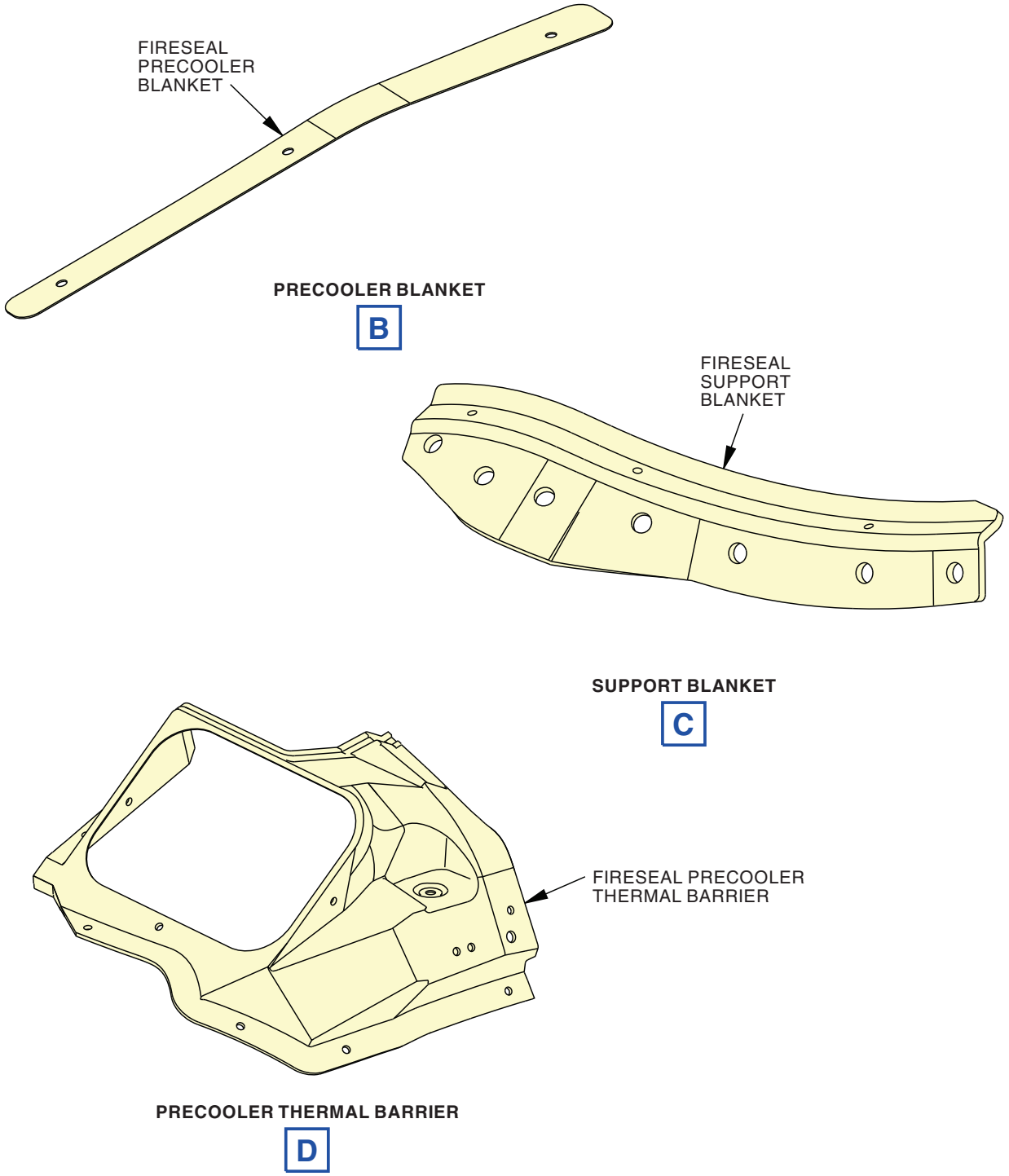
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

A

2965437 S0000744210_V1

Insulation Blankets Inspection
Figure 602/78-31-13-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Insulation Blankets Inspection
Figure 602/78-31-13-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

TASK 78-31-13-211-801-G00

4. Insulation Blanket Seal Inspection

(Figure 603)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for a visual inspection of the seals on the thermal protection system (TPS) panels and insulation blankets on the left and right thrust reversers.

(2) The procedures are the same for the left and right thrust reversers.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-13 P/B 401	INSULATION BLANKET - REMOVAL/INSTALLATION
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-13-010-006-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-13-010-007-G00

(2) Remove the insulation blankets, upper thermal protection system (TPS) panel, and lower TPS panel (TASK 78-31-13-000-801-G00).

F. Thrust Reverser Insulation Blanket Seal Inspection

SUBTASK 78-31-13-211-003-G00

(1) Inspect the insulation blanket seals for damage, missing seals, tears, creases, disbonds, and exposed sealant.

SUBTASK 78-31-13-960-001-G00

(2) If damage is found, replace the insulation blanket (PAGEBLOCK 78-31-13/401).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-13-410-011-G00

(1) Install the insulation blankets, upper thermal protection system (TPS) panel, and lower TPS panel (TASK 78-31-13-400-801-G00).

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

SUBTASK 78-31-13-410-012-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

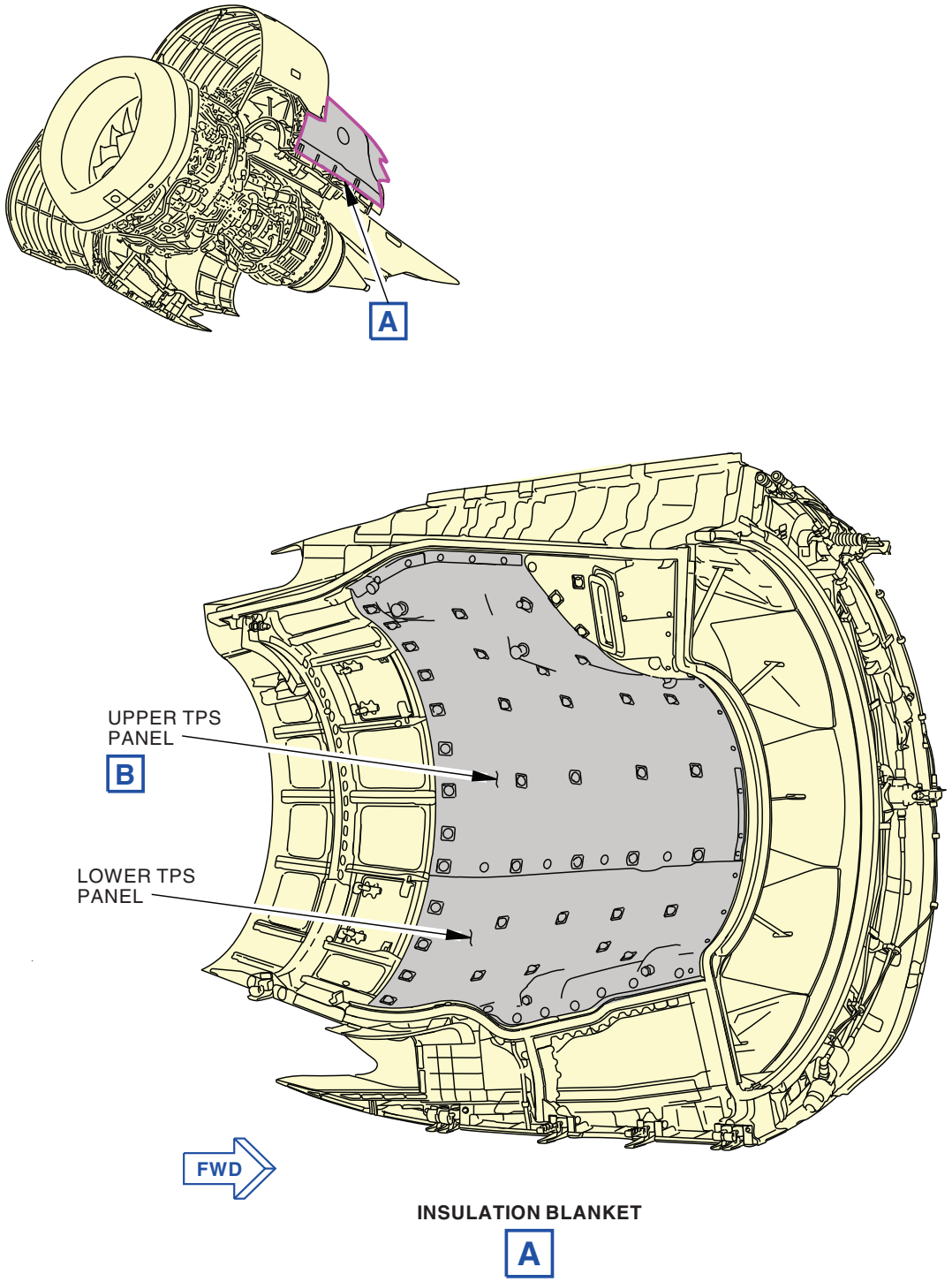
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

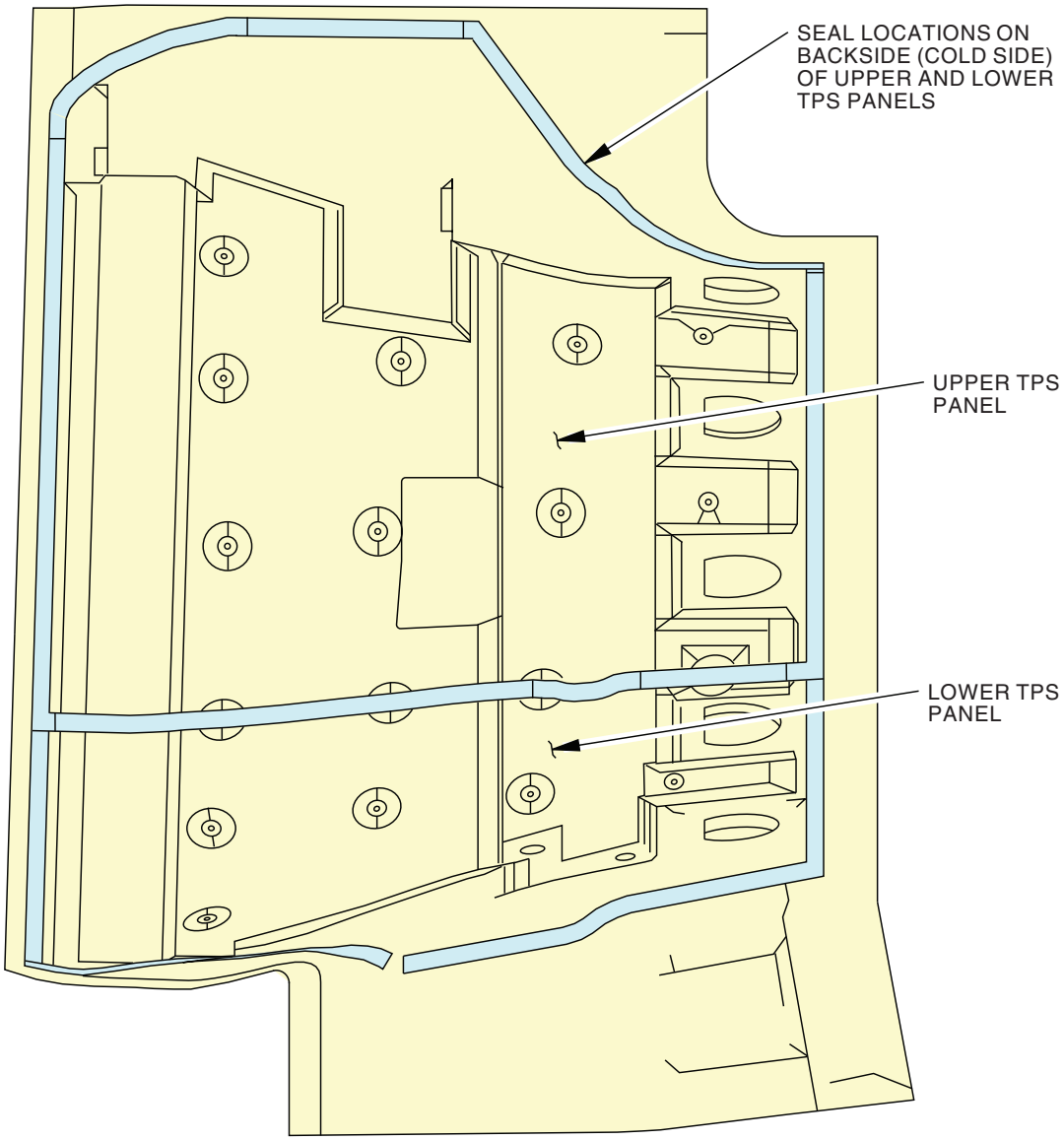
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Insulation Blanket Seals
Figure 603/78-31-13-990-813-G00 (Sheet 1 of 2)

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INSULATION BLANKET SEALS
(VIEW FROM THE OUTBOARD LOOKING IN)

B

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Insulation Blanket Seals
Figure 603/78-31-13-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INSULATION BLANKET - REPAIRS

1. General

A. This procedure has one task:

(1) The temporary repair of the insulation blankets on the thrust reverser.

TASK 78-31-13-300-801-G00

2. Insulation Blanket Repair

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task gives the instructions for a temporary repair of the metal faced (CRES) thermal insulation blankets.

(a) The thermal insulation blanket can remain installed while you do this temporary repair procedure with the thrust reverser in the opened position for the hot side of the blanket.

(2) These blankets are a thermal insulation and fire barrier layer which is necessary to keep the thrust reverser structurally serviceable and can decrease the damage and repair costs from a duct burst or a fire.

(3) The function of the metal face sheets is to prevent fluid contamination of the blanket insulation and to provide a fire barrier.

(4) The function of the polymide sheets on the cold side is to prevent fluid contamination of the blanket insulation.

(5) The thermal performance of the insulation blanket will be reduced if the insulation is exposed to fluid (water, engine oil, hydraulic fluid). The performance is reduced even after the insulation dries after exposure to fluid. It is therefore very important to examine the blanket for fluid penetration in the damaged areas where the metal foil has been compromised.

(6) Clean surfaces are required to get the high strength bonds to hold the adhesive and patch to the blanket. All unwanted materials such as oils, greases, waxes or dirt must be removed.

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(7) The materials in the consumable materials list are used for factory production of the blanket assemblies. Small quantities of these materials from the material suppliers may not be available for a field repair. Substitution of other materials that are not on this consumable material list is not permitted because the substitution of materials could affect the performance of the insulation blanket.

(a) Refer to the supplier for the consumable materials, in the smaller quantities that are necessary, for a field or component repair.

1) Hi-Temp Insulation Inc, 4700 Calle Alto, Camarillo, California 93012, USA.

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

(8) The insulation material is inside a quilted fiberglass fabric bag.

PTF ALL

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

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(Continued)

Reference	Title
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-418	Gloves
STD-809	Spatula
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

D. Consumable Materials

Reference	Description	Specification
A00081	Adhesive - Silicone Rubber - RTV 106	BAC5010 Type 74
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
A00803	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type I
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
C00944	Primer - Firewall - Dapco No. 1-100	BMS5-63 Type I
C00954	Primer - Adhesive Bonding - SS4004P RTV	BAC5010 Type 74
E50001	Solvent - Acetone	ASTM D 329 (Supersedes O-A-51)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00744	Cloth - Emery	
G01306	Gloves - Lint-free	
G50117	Sheet - 301 series, CRES, 0.018 - 0.025 inch Thick	AMS5517, AMS5518, AMS5902, AMS5519
G50769	Insulation - High Temperature Low Thermal Conductivity	BMS9-20
G50797	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type D Style 120

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

G. Damage Limits for the Temporary Repair

SUBTASK 78-31-13-800-001-G00

(1) If the damage to the insulation blanket is within these damage limits, you can do a temporary repair on the airplane.

NOTE: Damage is defined as any penetration through the metal face sheet. Scratches and dents are not considered damage which require repair.

(a) Hot Side:

1) Damage to the inner metal sheet or hot side is less than 2.0 in^2 (1290.3 mm^2).

2) A tear in the inner metal sheet or hot side is shorter than 3.0 in. (76.2 mm).

3) The damaged area is more than 0.5 in. (12.7 mm) from a grommet, sharp bend or edge in the blanket.

4) Around each damaged area in all directions, there is at least 0.5 in. (12.7 mm) of undamaged blanket.

5) The damaged areas must be separated by more than 6.0 in. (152.4 mm).

SUBTASK 78-31-13-350-004-G00

(2) A temporary repair must be replaced within 500 hours with a permanent repair.

NOTE: For the permitted permanent repair limits and repair procedures, refer to the Component Maintenance Manual (CMM) 78-31-94, Repair 3-1.

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

H. Damage Limits for the Temporary Repair

SUBTASK 78-31-13-350-003-G00

(1) If the damage to the insulation blanket is in these limits, you can repair the insulation blanket on the airplane:

NOTE: Use of this repair is permitted in all locations forward of the No. 3 compression pad.

(a) There is no missing insulation material and the quilted fiberglass fabric bag is not damaged.

(b) The insulation material and the quilted fiberglass fabric bag are not wet or stained with hydraulic fluid or oil.

(c) The maximum damage size is not more than 3.0 in^2 (19.4 cm^2).

(d) The maximum length of any damage is not more than 5.0 in. (127.0 mm).

(e) There are no more than 2 temporary repairs within a 12.0 in. (304.8 mm) diameter.

(f) A maximum number of 8 temporary repairs are permitted on each blanket.

(g) Around each damaged area in all directions, there is a at least 0.5 in. (12.7 mm) of damaged blanket.

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(h) The damaged area is not less than 0.5 in. (12.7 mm) from a grommet, attaching parts or edge.

PTF ALL

I. Prepare for the Temporary Repair

SUBTASK 78-31-13-010-003-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL

J. Temporary Repair of the Inner Metal Sheet on the Hot Side of the Thermal Insulation Blanket

SUBTASK 78-31-13-220-001-G00

(1) Make sure that the damaged area of the blanket is within the given damage limits above.

SUBTASK 78-31-13-350-001-G00

(2) Repair the damaged area of the inner metal sheet on the hot side of the blanket, as follows (Figure 801):

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WARNING MAKE SURE THAT YOU USE PROTECTIVE CLOTHING ON THE HAND AND ARM. THIS WILL PREVENT INJURY TO PERSONS.

WARNING USE PROTECTION WHEN YOU REMOVE THE SHARP METAL EDGES FROM THE DAMAGED AREA OF THE OUTER METAL SHEET. SHARP METAL EDGES CAN CAUSE INJURY TO PERSONS.

(a) Remove the sharp edges from the metal sheet and the loose insulation material from the damaged area.

(b) Remove a minimum of four installation fasteners and washers that surround the damage.

1) Keep the installation fasteners and washers for the installation.

(c) Examine the damaged area and find the size of repair patch required.

1) The repair patch must extend a minimum of 1.0 in. (25.4 mm) beyond the fasteners and in all directions around the damaged area by a minimum of 2.0 in. (50.8 mm).

2) Make a pattern for the repair patch with any clean paper.

NOTE: A paper pattern is used to find the size of the patch if the patch must cover a contoured surface.

3) Dry fit the paper pattern over the damaged area to make sure that the repair patch is the correct size and will cover the damaged area.

(d) Use the paper pattern to cut a repair patch from a 0.018 in. (0.457 mm) to 0.025 in. (0.635 mm) thick CRES sheet, G50117 that will overlap the surface repair area.

1) Deburr the edges of the repair patch.

2) Radius the corners of the repair patch a minimum of 0.12 in. (3.05 mm).

(e) Dry fit the repair patch at the repair location.

1) Make a mark of the locations of the fastener holes on the repair patch.

2) Shape the repair patch to match the inside contour of the blanket within 0.030 in. (0.762 mm).

(f) Drill 0.345 in. (8.763 mm) to 0.405 in. (10.287 mm) diameter holes (0.375 in. (9.525 mm) nominal) through the repair patch at the same locations as the fastener locations.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(g) Clean the repair patch.

1) Clean the repair patch with a clean cotton wiper, G00034 moist with solvent, B00062.

a) Continue to clean the surface again with a new, clean cotton wiper, G00034 moist with solvent, B00062 until no contamination is seen on the cloth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

2) Wipe the repair patch with a dry, clean cloth before the solvent evaporates.

(h) Make sure that the cleaned repair patch is kept clean.

NOTE: Apply primer to the cleaned surface within four hours.

1) Use clean lint-free gloves, G01306 when you handle the cleaned repair patch.

(i) Use a clean cotton wiper, G00034 to apply a thin layer of SS4004P RTV primer, C00954 to the cleaned repair patch.

1) Let the primer become dry for a minimum of 30 minutes at 65°F (18°C) to 100°F (38°C).

a) Do not exceed 72 hours to allow the primer to dry.

(j) Make sure that the primed repair patch is kept clean.

1) Use clean, lint-free gloves, STD-418 when you handle the primed repair patch.

(k) Prepare the adhesive.

1) If you use the RTV 106 adhesive, A00081 for the repair, this is a one-part silicone adhesive/sealant which is supplied ready to use.

NOTE: The maximum thickness for one application of RTV 106 adhesive, A00081 is 0.125 in. (3.175 mm), allow the adhesive to completely cure before a second application. If RTV 106 adhesive, A00081 is applied to a thickness of 0.25 in. (6.35 mm) in one application, the outside surface of the RTV 106 adhesive, A00081 forms a thick skin and the inside core of the RTV 106 adhesive, A00081 will not cure correctly.

NOTE: The RTV 106 adhesive, A00081 sealant has a short work life. Exposure to air for more than a few minutes will cause a skin to form which will prevent adhesion on the surfaces.

2) If you use the firewall sealant, A00160 for the repair, use the manufacturers instructions to mix the two-part sealant.

NOTE: After the firewall sealant, A00160 is mixed, there is a limited application time. Firewall sealant, A00160 application time is 4 hours. Do not use the sealant if it has exceeded the application time.

WARNING

DO NOT BREATHE THE DUST FROM THE INSULATION MATERIAL. PUT ON A DUST MASK. INJURIES TO PERSONS CAN OCCUR.

(l) Replace the damaged, contaminated or wet insulation material. (Figure 801)

1) Carefully cut the damaged or wet insulation material from the damaged area.

2) Make sure that you do not cut the polymide sheet on the outside surface of the blanket cold side.

3) Make a new insulation plug, as follows:

a) Cut a piece of quilted microporous insulation, G50769, BMS9-20 Type II Class 16 Grade 3, to the correct size.

b) Make sure that the insulation plug keeps as many stitch lines as possible.

c) Make the insulation plug larger than the hole in the blanket, approximately 0.05 in. (1.27 mm) to 0.15 in. (3.81 mm) more on each side.

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PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

4) Install the replacement insulation material with RTV 106 adhesive, A00081 or firewall sealant, A00160.
   a) Seal the edges and insulation material around the damaged area of the blanket with RTV 106 adhesive, A00081 or firewall sealant, A00160.
   b) Apply RTV 106 adhesive, A00081 or firewall sealant, A00160 to seal the blanket insulation.
   c) Make sure that there are no gaps between the replacement insulation material and the insulation in the blanket.
   d) Fill all gaps with adhesive.

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION

DO NOT GET SOLVENTS ON THE INSULATION MATERIAL. SOLVENTS AND OTHER FLUIDS WILL DECREASE THE PERFORMANCE AND MATERIAL PROPERTIES OF THE INSULATION BLANKET.

(m) Clean the damaged area plus a minimum of 0.5 in. (12.7 mm) in all directions around the damaged area.
   1) Clean the damaged area with a clean cotton wiper, G00034 moist with solvent, B00062.
      a) Continue to clean the surface again with a new, clean cotton wiper, G00034 moist with solvent, B00062 until no contamination is seen on the cloth.
   2) Wipe the repair area with a dry, clean cloth before the solvent evaporates.

(n) Make sure that the cleaned repair surface and the patch are kept clean.

   NOTE: Apply primer to the cleaned surface within four hours.

   1) Use clean lint-free gloves, G01306 when you handle the cleaned repair surface or the patch.

(o) Use a clean cotton wiper, G00034 to apply a thin layer of SS4004P RTV primer, C00954 to the cleaned repair surface and the patch.
   1) Let the primer become dry for a minimum of 30 minutes at 65°F (18°C) to 100°F (38°C).
      a) Do not exceed 72 hours to allow the primer to dry.

(p) Make sure that the primed repair surface and the patch are kept clean.
   1) Use clean, lint-free gloves, STD-418 when you handle the primed repair surface or the patch.

(q) Apply RTV 106 adhesive, A00081 or firewall sealant, A00160 to the primed repair surface of the inner metal sheet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH COMPOSITE INNER WALL (Continued)

1) Use a hardwood or plastic fillet smoothing spatula, STD-810 or spatula, STD-809 to apply a layer of adhesive or sealant.
2) Make sure that the adhesive is a maximum of 0.125 in. (3.175 mm) thick and extends to the edges of the repair patch in all directions.
3) Make sure that the adhesive is smooth.
4) Make sure that the repair patch is installed immediately after the adhesive or sealant is applied.

NOTE: Immediately after the adhesive is applied, the adhesive will begin to cure. Within a few minutes a skin on the surface of the adhesive will form. The patch will not bond correctly if a skin has formed on the adhesive before it is applied.

CAUTION DO NOT APPLY TOO MUCH PRESSURE WHEN YOU APPLY THE PATCH TO THE REPAIR AREA. THE ADHESIVE WILL COME OUT FROM BELOW THE PATCH.

(r) Position and apply the repair patch over the repair area.
1) Install the fasteners and washers.
2) Apply light pressure to the repair patch as the adhesive dries.
3) Remove all extra adhesive from the edges with a clean cotton wiper, G00034 or cloth.

(s) Let the sealant or adhesive cure.
1) Adhesive RTV 106 adhesive, A00081 will cure in a minimum of 24 hours at a minimum temperature of 65°F (18.3°C).

NOTE: The adhesive requires a minimum of 20 percent relative humidity to cure. Because the adhesive requires moisture from the air to cure, do not cover the adhesive. The smell of acetic acid will be present until the adhesive has cured.

2) Firewall sealant, A00803 cure time depends on the type used.

NOTE: Firewall sealant, A00803 cures in 48 hours. Lower temperatures will increase cure time. Higher temperatures will decrease cure time. As an example, at 120°F (49°C), the mixed sealant will cure within 4 hours.

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL

K. Temporary Insulation Blanket Repair

SUBTASK 78-31-13-350-002-G00

(1) Do these steps to repair the damaged area of the outer metal sheet Figure 802:

(a) Make sure that the damaged area is in the limits.

WARNING USE PROTECTION WHEN YOU REMOVE THE SHARP METAL EDGES FROM THE DAMAGED AREA OF THE OUTER METAL SHEET. SHARP METAL EDGES CAN CAUSE INJURY TO PERSONS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(WARNING PRECEDES)

CAUTION

CAREFULLY REMOVE THE SHARP EDGES FROM THE DAMAGED AREA OF THE OUTER METAL SHEET. A TEMPORARY REPAIR AT THIS LOCATION IS NOT PERMITTED IF THE INSULATION MATERIAL OR THE FIBERGLASS FABRIC BAG IS DAMAGED. DAMAGE TO THE INSULATION MATERIAL OR FIBERGLASS FABRIC BAG CAN OCCUR.

(b) If necessary, deburr the edges of the damaged area of the outer metal sheet with a emery cloth, G00744 or equivalent.

(c) Cut a piece of fiberglass fabric, G50797 that will overlap the surface area in all directions around the damaged area by 1.0 in. (25.4 mm) to 1.5 in. (38.1 mm).

1) Trim any edge of the fiberglass patch that overlaps the edge of the blanket or covers a grommet.

2) Cut the corners to make a rounded edge that has a radius of 0.25 in. (6.35 mm).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THESE MATERIALS. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THESE MATERIALS. KEEP THESE MATERIALS AWAY FROM SPARKS, FLAME, AND HEAT. THESE MATERIALS ARE POISONOUS AND FLAMMABLE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION

DO NOT GET SOLVENTS ON THE INSULATION MATERIAL. SOLVENTS AND OTHER FLUIDS WILL DECREASE THE PERFORMANCE AND MATERIAL PROPERTIES OF THE INSULATION BLANKET.

(d) Clean the damaged area and a minimum of 3.0 in. (76.2 mm) in all directions around the damaged area with solvent, B00062, and a cotton wiper, G00034.

1) Make sure that you remove all contaminants (such as hydraulic fluid and oil) around the damaged area. This will make sure that the sealant will adhere properly. If necessary, use a Scotch-Brite or equivalent pad with solvent, E50001. Do not allow the solvent to come in contact with the quilted fiberglass fabric bag.

(e) Clean the surface again with a clean cloth that is moist with solvent and wipe dry with another clean dry cloth while the surface is still moist.

(f) Do the above steps again to clean the fiberglass patch.

(g) Make sure that the cleaned bonding surface and fiberglass patch are kept clean. Use lint-free gloves, G01306, when you touch the cleaned bonding surface or fiberglass patch.

(h) Use a clean piece of cotton wiper, G00034, to apply a thin layer of Dapco No. 1-100 primer, C00944, to the cleaned bonding surface of the outer metal sheet.

1) Let the primer dry for not less than 30 minutes, but for no more than 2 hours at 65°F (18.3°C) to 100°F (37.8°C).

(i) Make sure that the bonding surface and fiberglass patch are kept clean. Use lint-free gloves, G01306, when you touch the bonding surface or fiberglass patch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

(j) Use a hardwood or plastic fillet smoothing spatula, STD-810 to apply a continuous layer of sealant, A00803 to the primed bonding surface of the outer metal sheet.

1) Apply the sealant approximately 0.05 in. (1.3 mm) thick.

2) Make sure that the sealant extends to the edges of the fiberglass patch in all directions.

NOTE: It is acceptable to have the sealant extend slightly beyond the edges of the fiberglass patch.

3) Make sure that the sealant is smooth.

4) Apply the fiberglass patch immediately after the sealant is applied.

CAUTION

DO NOT APPLY TOO MUCH PRESSURE WHEN YOU APPLY THE PATCH TO THE REPAIR AREA. THE ADHESIVE WILL COME OUT FROM BELOW THE PATCH.

(k) Put the fiberglass patch in the correct position over the damaged area.

1) Make sure that you do not use too much pressure when the fiberglass patch is applied.

(l) Use a hardwood or plastic fillet smoothing spatula, STD-810 to make the fiberglass patch smooth over the sealant.

1) Work from side to side to remove air bubbles from under the fiberglass patch.

(m) Use a hardwood or plastic fillet smoothing spatula, STD-810 to apply a continuous layer of sealant, A00803 to the surface of the fiberglass patch.

1) Apply the sealant approximately 0.05 in. (1.3 mm) thick.

2) While you work from one side to the other, gently use a hardwood or plastic fillet smoothing spatula, STD-810 to smooth the sealant over the fiberglass patch.

3) Extend the sealant 0.25 in. (6.35 mm) beyond the edges of the fiberglass patch in all directions.

NOTE: If you use a pre-cured patch for the repair, apply sealant, A00803 over the patch edges, a minimum of 0.25 in. (6.35 mm) on both sides of the patch edge.

(n) Let the sealant dry for a minimum of 48 hours at a minimum of 65°F (18°C).

1) To cure the sealant faster, you can apply heat at a maximum temperature of 125°F (52°C) to the repair for a cure time of 4 hours.

L. Fiberglass Patch (Interim Repair) Inspection Procedure

SUBTASK 78-31-13-750-002-G00

(1) Fiberglass Patch Inspection Time Limits

(a) For Zone 1 (Forward Edge of Blanket to Compression Pad #2)

1) An Initial inspection after fiberglass patch installation must be done at or before 2000 hours.

2) If, after a total of 4000 hours, no deterioration of the fiberglass patch is found, further inspections can be done as part of the heavy maintenance check or equivalent maintenance procedures.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

3) If, after a total of 4000 hours, deterioration of the fiberglass patch is found, repeat inspections must be done at 2000 hour intervals thereafter.

4) If any deterioration of the fiberglass patch is found during any subsequent heavy maintenance check or equivalent inspection, revert back to the 2000 hour inspection interval.

(b) For Zone 2 (Compression Pad #2 to Compression Pad #3)

1) Initial inspection after fiberglass patch installation must be done at or before 1500 hours.

2) Repeat inspections must be done at 1500 hour intervals thereafter.

SUBTASK 78-31-13-750-003-G00

(2) Fiberglass Patch / Sealant Inspection Criteria

(Figure 803)

(a) Cracks in the sealant or the fiberglass patch are not permitted. Replace only the damaged sealant and the damaged fiberglass patch.

(b) If the sealant has hardened but is not cracked, continue with either the 2000 hour or the 1500 hour inspection time limits as specified in each zone, defined in the Fiberglass Patch Inspection Time Limits step above.

(c) For Zone 1 areas, if the sealant remains flexible / pliable / not hardened, inspections can be continued to the next heavy maintenance inspection interval.

(d) Examine the fiberglass patch for the following damage:

1) Peeling of fiberglass patch.

a) Peeling is not permitted.

2) Cracks

a) Cracks are not permitted in the fiberglass patch.

b) If cracks exist in the sealant, inspect the underlying fiberglass patch for indications of cracks. Cracks are not permitted in the fiberglass patch.

c) Micro-surface / superficial cracks in the sealant should be repaired as follows:

<1> Clean the sealant with solvent, B00062, and a cotton wiper, G00034.

<2> Lightly abrade the sealant using Scotch-Brite or equivalent pad.

<3> Reclean the sealant with solvent, B00062, and a cotton wiper, G00034.

<4> Use a brush to apply a light coat of sealant, A00803 to the area to make sure that the cracks are fully sealed.

3) Worn areas

a) Worn areas are permitted if the total depth does not exceed one-half of the Fiberglass Patch thickness. One-half of the fiberglass patch thickness is equivalent to 0.015 in. (0.381 mm).

b) The worn area is smooth (no rough or irregular surfaces).

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PTF ALL; AIRPLANES WITH TITANIUM INNER WALL (Continued)

CAUTION BE CAREFUL WHEN YOU REMOVE THE OLD OR USED SEALANT FROM THE INSULATION BLANKETS. DAMAGE TO THE BLANKETS CAN OCCUR.

4) If you find damage on the patch that is not permitted, carefully remove only the damaged portion of the patch and re-install with a new section of a fiberglass patch.

a) The newly installed repair patch section must overlap the existing patch by a minimum of 0.25 in. (6.35 mm) (Figure 803).

NOTE: The total maximum amount, i.e. thickness of sealant that can be present on a damaged portion must not exceed 0.125 in. (3.175 mm).

PTF ALL

M. Put the Airplane Back to its Usual Condition

SUBTASK 78-31-13-410-010-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

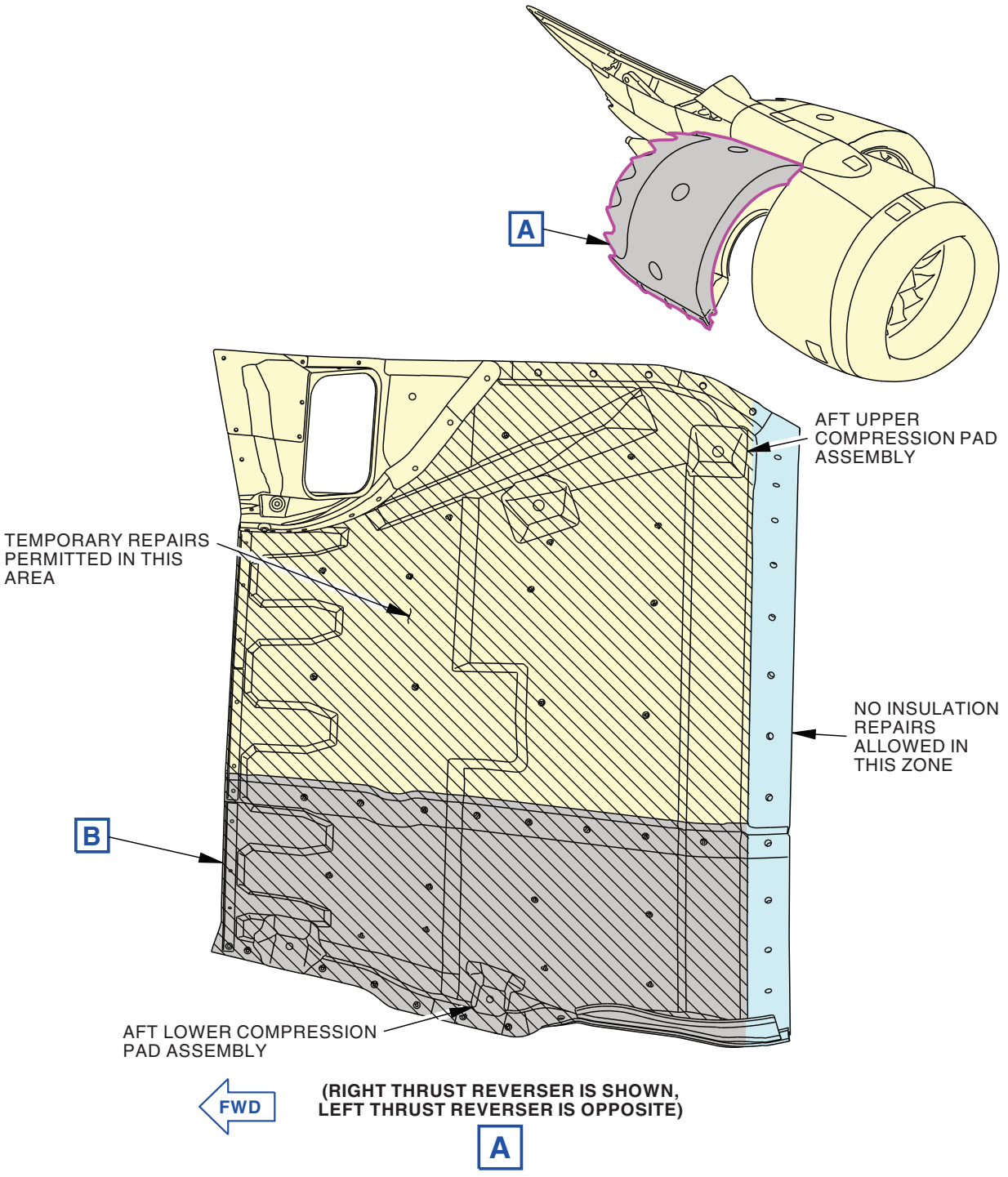
1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

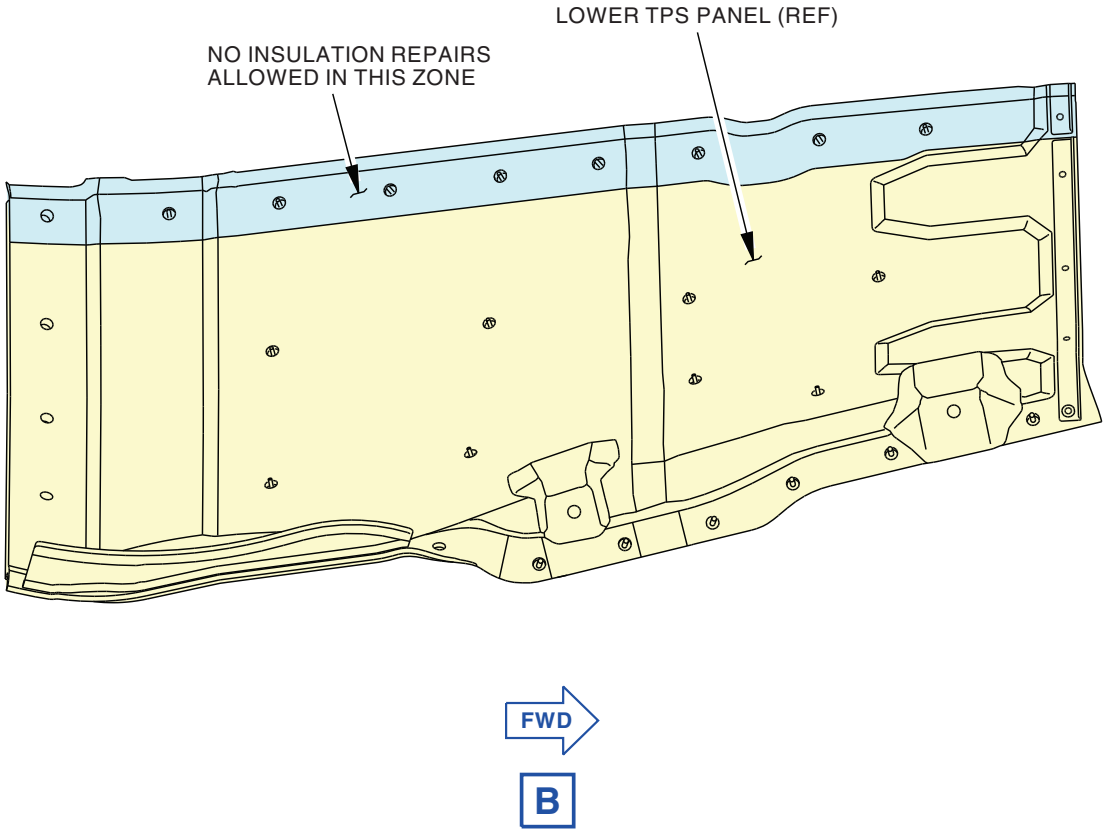
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Temporary Insulation Blanket Repair
Figure 801/78-31-13-990-805-G00 (Sheet 1 of 3)

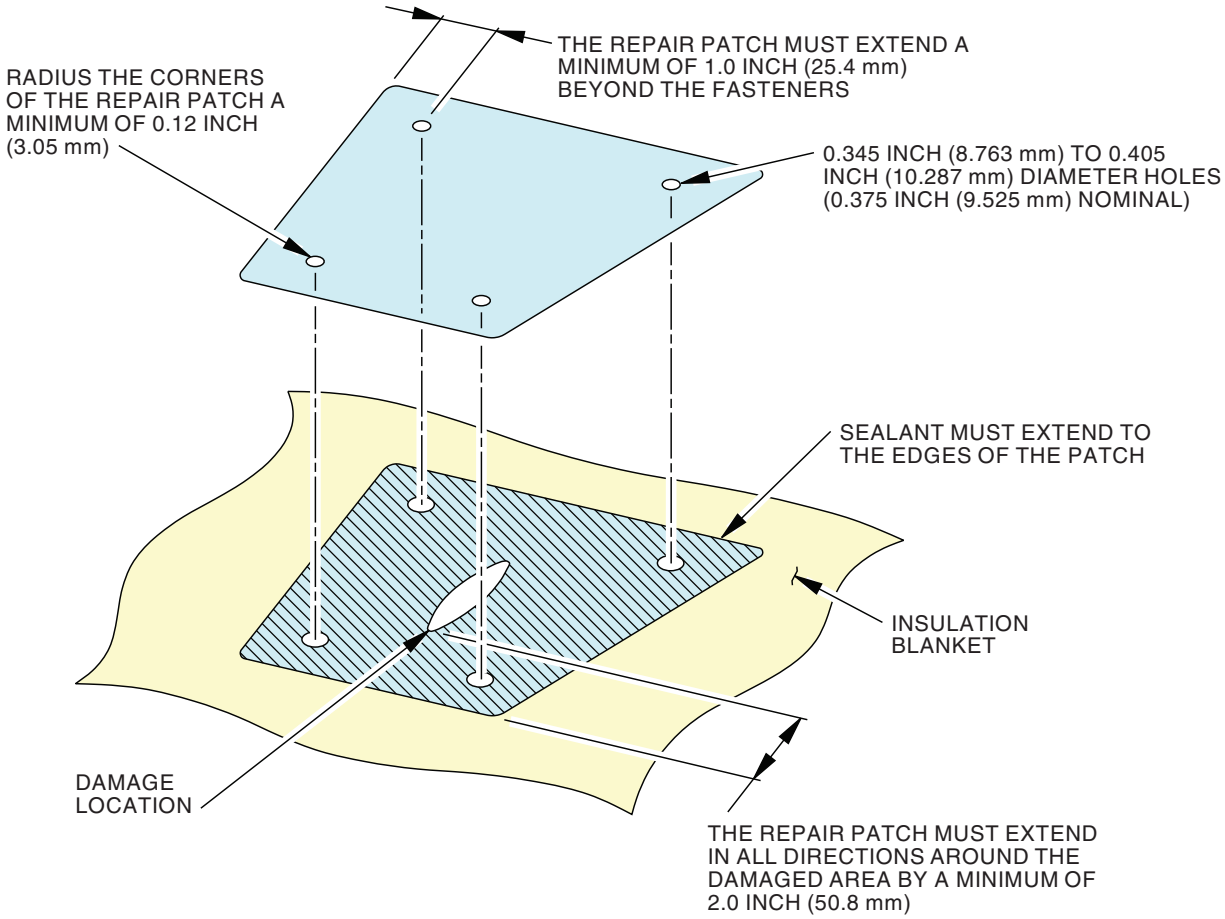
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546141 S0000605486_V1

Temporary Insulation Blanket Repair
Figure 801/78-31-13-990-805-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

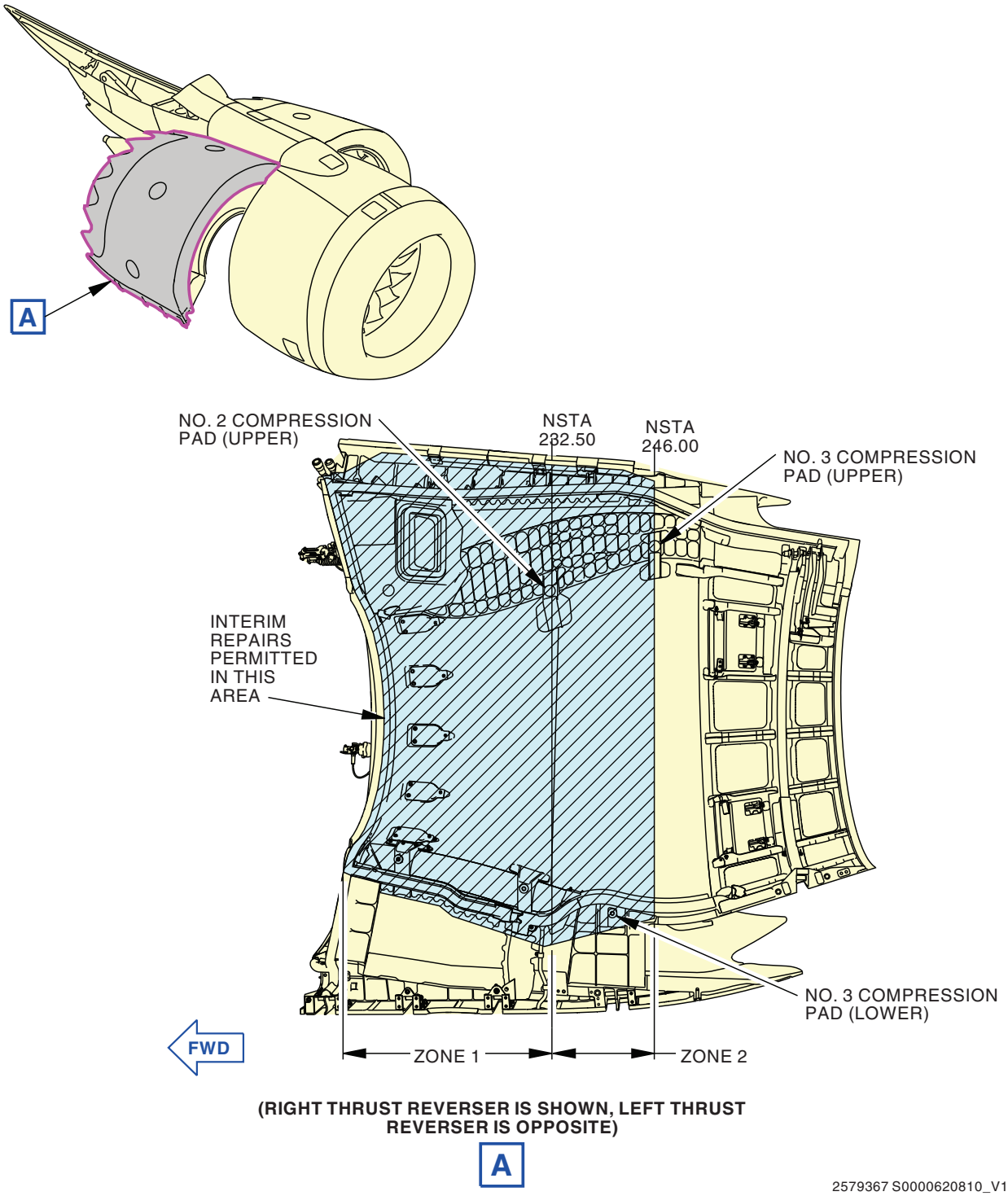


PATCH OVER TEAR OR CRACK

2355277 S00061516874_V2

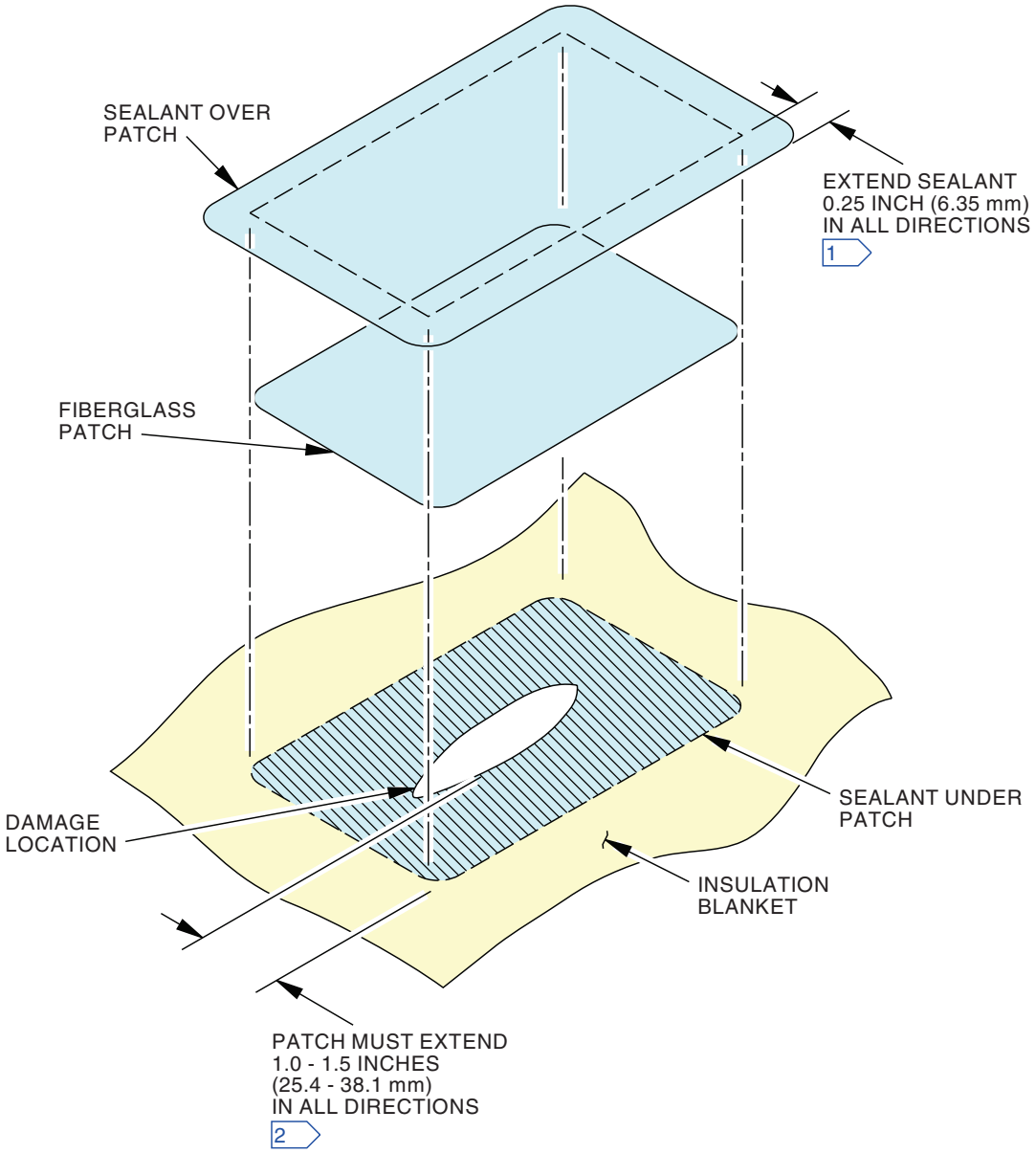
Temporary Insulation Blanket Repair
Figure 801/78-31-13-990-805-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Temporary Insulation Blanket Repair
Figure 802/78-31-13-990-807-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PATCH OVER TEAR OR CRACK

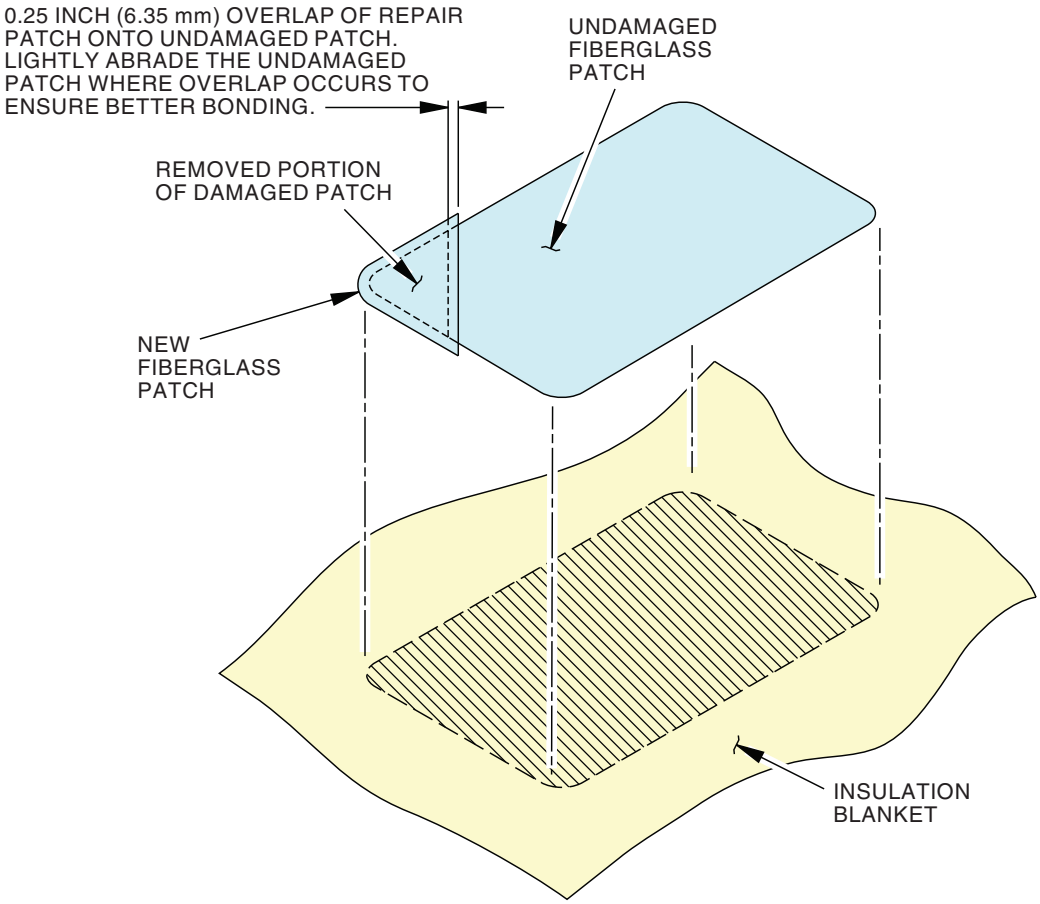
1 A MINIMUM OF 0.25 INCH (6.35 mm) IS ALLOWED WHEN ACCESS IS LIMITED.

2 A MINIMUM OF 0.5 INCH (12.7 mm) IS ALLOWED WHEN ACCESS IS LIMITED.

2579368 S0000620811_V2

Temporary Insulation Blanket Repair
Figure 802/78-31-13-990-807-G00 (Sheet 2 of 2)

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REPAIR OF A DAMAGED PATCH

2579369 S0000620812_V2

Temporary Fiberglass Patch Blanket Repair
Figure 803/78-31-13-990-812-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRECOOLER INLET DUCT GUIDE VANE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the precooler inlet duct guide vane

(2) An installation of the precooler inlet duct guide vane.

TASK 78-31-14-000-801-G00

2. Precooler Inlet Duct Guide Vane Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the precooler inlet duct guide vane.

(2) The precooler inlet duct guide vane is referred to as the guide vane in this task.

(3) There are 2 guide vanes on each engine.

(4) The guide vanes are located at the top of the thrust reverser fan bypass duct on either side of the bifurcation at the front of the precooler inlet.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00634	Solvent - Stabilized Limonene Cleaner	BMS11-10 Type 1, 2, or 3
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Removal

SUBTASK 78-31-14-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-14-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Precooler Inlet Duct Guide Vane Removal

SUBTASK 78-31-14-020-001-G00

(1) Remove the GSE blanket [2] and precooler blanket [3] from the applicable thrust reverser. Do this task: Insulation Blankets - Removal, TASK 78-31-13-000-801-G00.

SUBTASK 78-31-14-020-002-G00

(2) Remove the four nuts [6], two bolts [5], and two bolts [4] from the guide vane [1] on the precooler inlet.

SUBTASK 78-31-14-020-003-G00

(3) Remove the guide vane [1] from the precooler inlet.

(a) Remove all shims from the precooler inlet.

(b) Clean the precooler inlet area, as follows:

1) Wipe or scrub the area with a clean cotton wiper, G00034, to remove all loose soil, grease, hydraulic fluid, or oil.

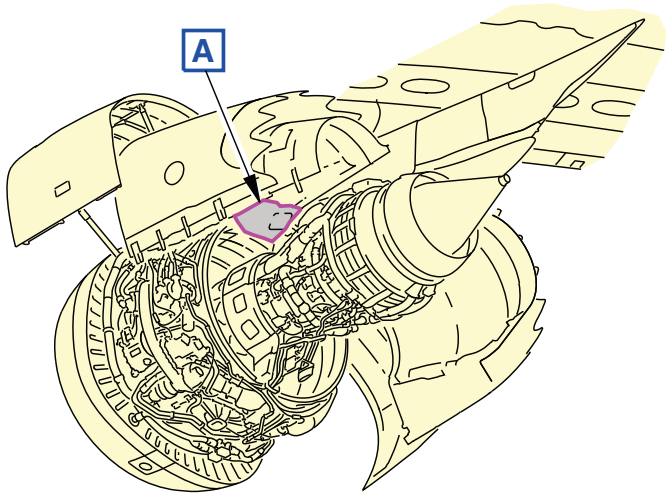
2) Wipe the area with solvent, B00184, solvent, B00148, or Limonene solvent, B00634, on a clean cotton wiper, G00034.

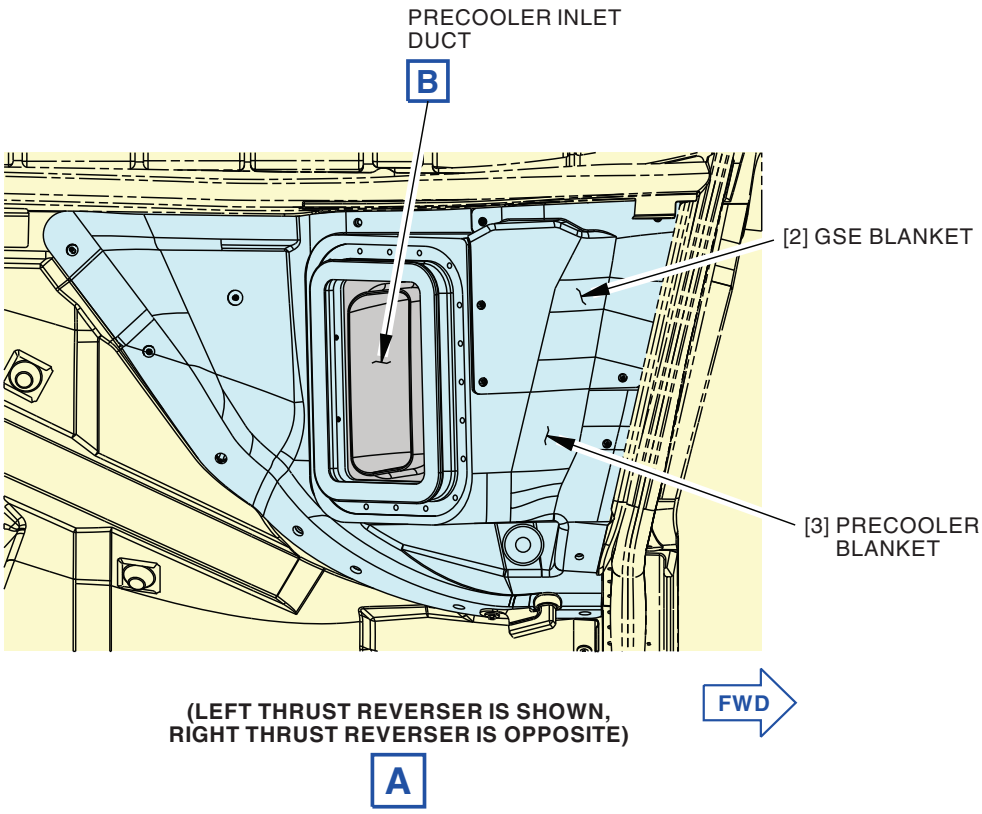
3) Wipe the area dry with a clean cotton wiper, G00034, before the solvent evaporates.

4) Continue to clean the area with a new clean cloth and solvent until no contamination is shown on the cloth.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

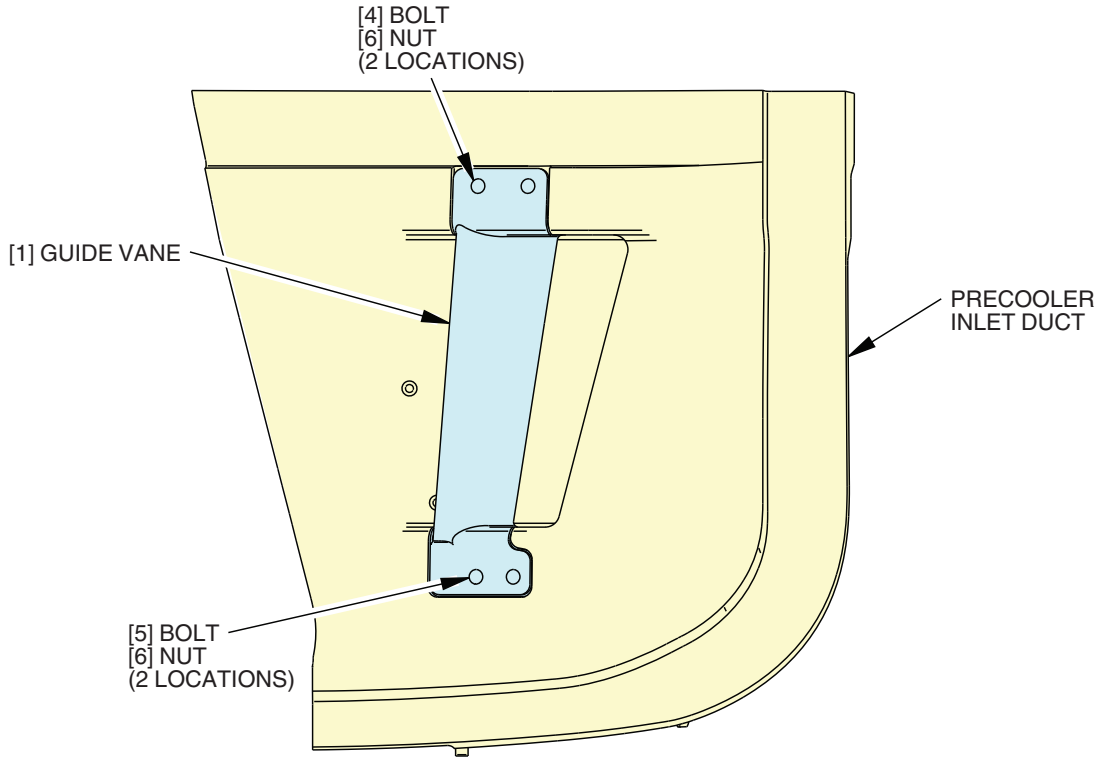




2546554 S0000605716_V2

Precooler Inlet Duct Guide Vane Installation
Figure 401/78-31-14-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD SIDE SHOWN

B

2552990 S0000608484_V3

Precooler Inlet Duct Guide Vane Installation
Figure 401/78-31-14-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-14-400-801-G00

3. Precooler Inlet Duct Guide Vane Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the precooler inlet duct guide vane.

(2) The precooler inlet duct guide vane is referred to as the guide vane in this task.

(3) There are 2 guide vanes on each engine.

(4) The guide vanes are located at the top of the thrust reverser fan bypass duct on either side of the bifurcation at the front of the precooler inlet.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
B50073	Alcohol - Isopropyl	ASTM D 770
G50262	Wiper - Cleaning	BMS15-5

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Guide vane	78-31-14-02-330	PTF ALL
		78-31-14-02-335	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

H. Precooler Inlet Duct Guide Vane Installation

SUBTASK 78-31-14-160-001-G00

(1) If it is necessary, use alcohol, B50073, and a clean wiper, G50262, to remove any contamination from the installation surfaces of the precooler inlet duct and guide vane [1].

SUBTASK 78-31-14-390-002-G00

(2) Apply sealant, A00160, to the installation surfaces of the precooler inlet duct and guide vane [1].

SUBTASK 78-31-14-420-001-G00

(3) Position the guide vane [1] onto the precooler inlet.

(a) Make sure that the installed guide vane [1] will be flush with the precooler inlet duct to ± 0.0050 in. (0.1270 mm).

1) Add shims to the guide vane [1] installation as necessary to meet the flushness requirement.

2) Make sure that the maximum shim thickness is 0.0125 in. (0.3175 mm) or less.

3) If shim is used, apply sealant, A00160, to both sides of the shims and around the edges.

(4) Install the two bolts [4], two bolts [5], and four nuts [6] to attach the guide vane [1] to the precooler inlet.

SUBTASK 78-31-14-390-001-G00

(5) Apply sealant, A00160, to the surfaces of the two bolts [4], two bolts [5], and four nuts [6].

SUBTASK 78-31-14-400-001-G00

(6) Install the GSE blanket [2] and precooler blanket [3]. Do this task: Insulation Blankets - Installation, TASK 78-31-13-400-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-14-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-14-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRECOOLER INLET DUCT SEAL

1. General

A. This procedure has two tasks:

(1) A removal of the precooler inlet duct seal

(2) An installation of the precooler inlet duct seal.

B. The precooler inlet duct seal is referred to as the seal in this procedure.

C. There are 2 seals on each engine.

D. The seals are located on the inside of the left and right thrust reversers on the precooler inlet duct on each engine.

TASK 78-31-15-000-801-G00

2. Precooler Inlet Duct Seal - Removal

(Figure 401)

A. General

(1) This task is the removal of the precooler inlet duct seal.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-13-000-801-G00	Insulation Blankets - Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 78-31-15-040-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-15-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Seal Removal

SUBTASK 78-31-15-020-001-G00

(1) Remove the GSE blanket [2] and precooler blanket [3] from the applicable thrust reverser, do this task: Insulation Blankets - Removal, TASK 78-31-13-000-801-G00.

SUBTASK 78-31-15-020-003-G00

(2) Remove the 20 screws [4] and 20 washers [5] from the precooler inlet.

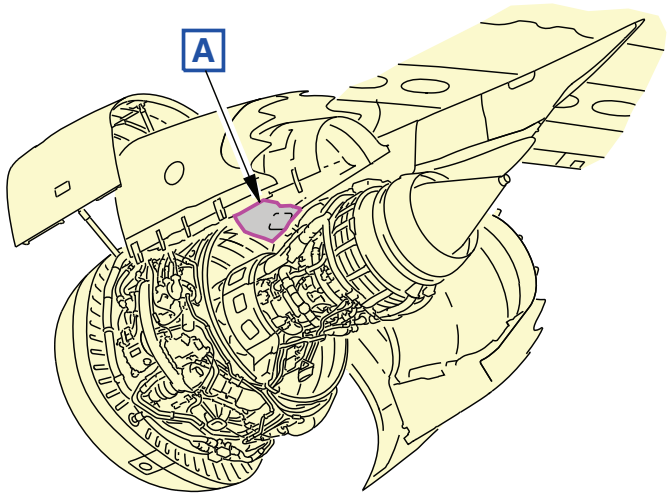
ECCN 9E991 BOEING PROPRIETARY - See title page for details

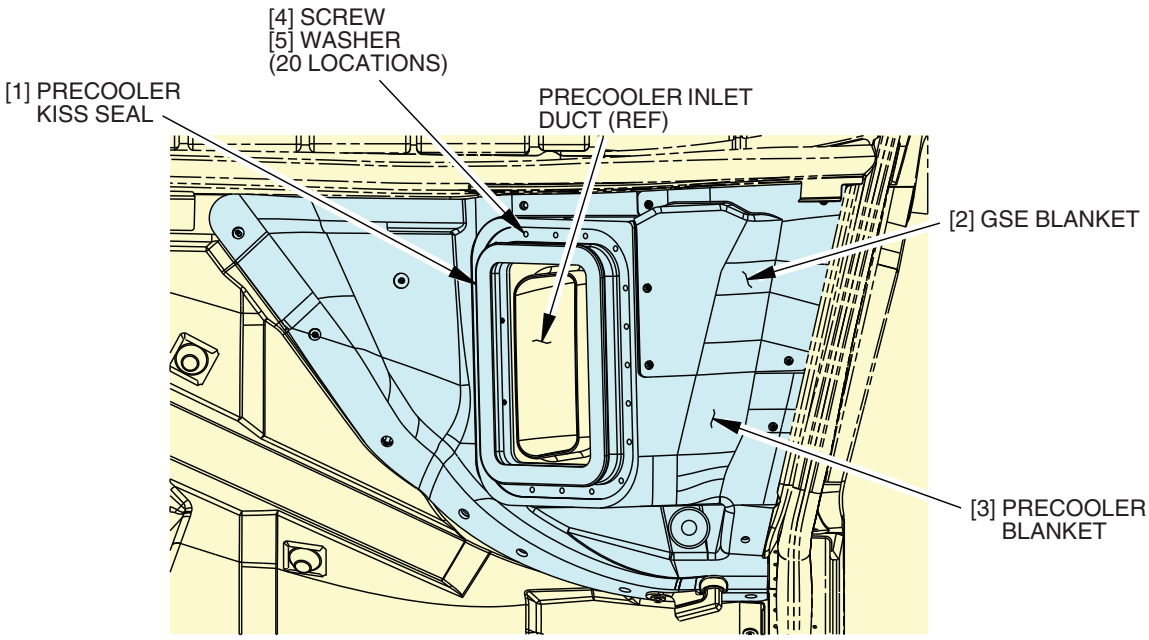
SUBTASK 78-31-15-020-004-G00

(3) Remove the precooler kiss seal [1] from the precooler inlet.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Precooler Inlet Duct Seal Installation
Figure 401/78-31-15-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-15-400-801-G00

3. Precooler Inlet Duct Seal - Installation

(Figure 401)

A. General

(1) This task is the installation of the precooler inlet duct seal.

(2) This task has one or more steps which are a means to satisfy Critical Design Configuration Control Limitation (CDCCL) requirements. A CDCCL note will follow the step to which it applies. Any step or sub-step that precedes or follows a CDCCL identified step is not subject to the CDCCL requirement.

(a) For important information on CDCCL requirements, refer to this task: Airworthiness Limitation Precautions, TASK 36-00-00-910-801.

NOTE: This is applicable to Airworthiness Limitation 36-AWL-04.

B. References

Reference	Title
36-00-00-910-801	Airworthiness Limitation Precautions (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-13-400-801-G00	Insulation Blankets - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Precooler kiss seal	78-31-15-02-050	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

H. Seal Installation

SUBTASK 78-31-15-100-001-G00

36-AWL-04: CDCCL

(1) Make sure that the precooler kiss seal [1] is in good condition and does not have tears or holes that extend beyond one edge of the seal.

NOTE: CDCCL - Refer to the task: Airworthiness Limitation Precautions, TASK 36-00-00-910-801, for important information on Critical Design Configuration Control Limitations (CDCCLs).

NOTE: This is applicable to Airworthiness Limitation 36-AWL-04.

SUBTASK 78-31-15-100-002-G00

(2) Make sure that there is no damage on the area of the precooler kiss seal [1] that is not protected by metal flange.

SUBTASK 78-31-15-100-003-G00

(3) Make sure that the installation surfaces are clean of contamination.

SUBTASK 78-31-15-390-001-G00

(4) Use sealant, A00160, to apply a faying surface seal.

SUBTASK 78-31-15-410-001-G00

(5) Put the precooler kiss seal [1] into position on the precooler inlet duct.

SUBTASK 78-31-15-420-002-G00

(6) Install the 20 washers [5] and 20 screws [4] on the precooler inlet duct.

SUBTASK 78-31-15-410-002-G00

(7) Install the precooler blanket [3] and GSE blanket [2] (TASK 78-31-13-400-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-15-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-15-440-001-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE RELIEF DOOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Pressure relief door removal

(2) Pressure relief door installation.

TASK 78-31-16-000-801-G00

2. Pressure Relief Door Removal

Figure 401, Figure 402

A. General

(1) This task is the removal of the thrust reverser pressure relief door.

(2) There are four pressure relief doors on each engine.

(3) The pressure relief doors are installed on the thrust reverser aft cowl panel assembly, at the 2, 4, 8 and 10 o'clock positions.

(4) The removal procedures is the same for each of the pressure relief doors.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 78-31-16-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-31-16-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Pressure Relief Door Removal

SUBTASK 78-31-16-020-001-G00

(1) Remove the pressure relief door [1] as follows :

(a) Open the two pressure relief door latches [2].

(b) Remove the four nuts [3] that attach the two pressure relief door hinges [4] to the pressure relief door [1].

(c) Remove the two pressure relief door hinges [4], the shims under the hinges (if installed), and the shim washers [6] from the pressure relief door [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

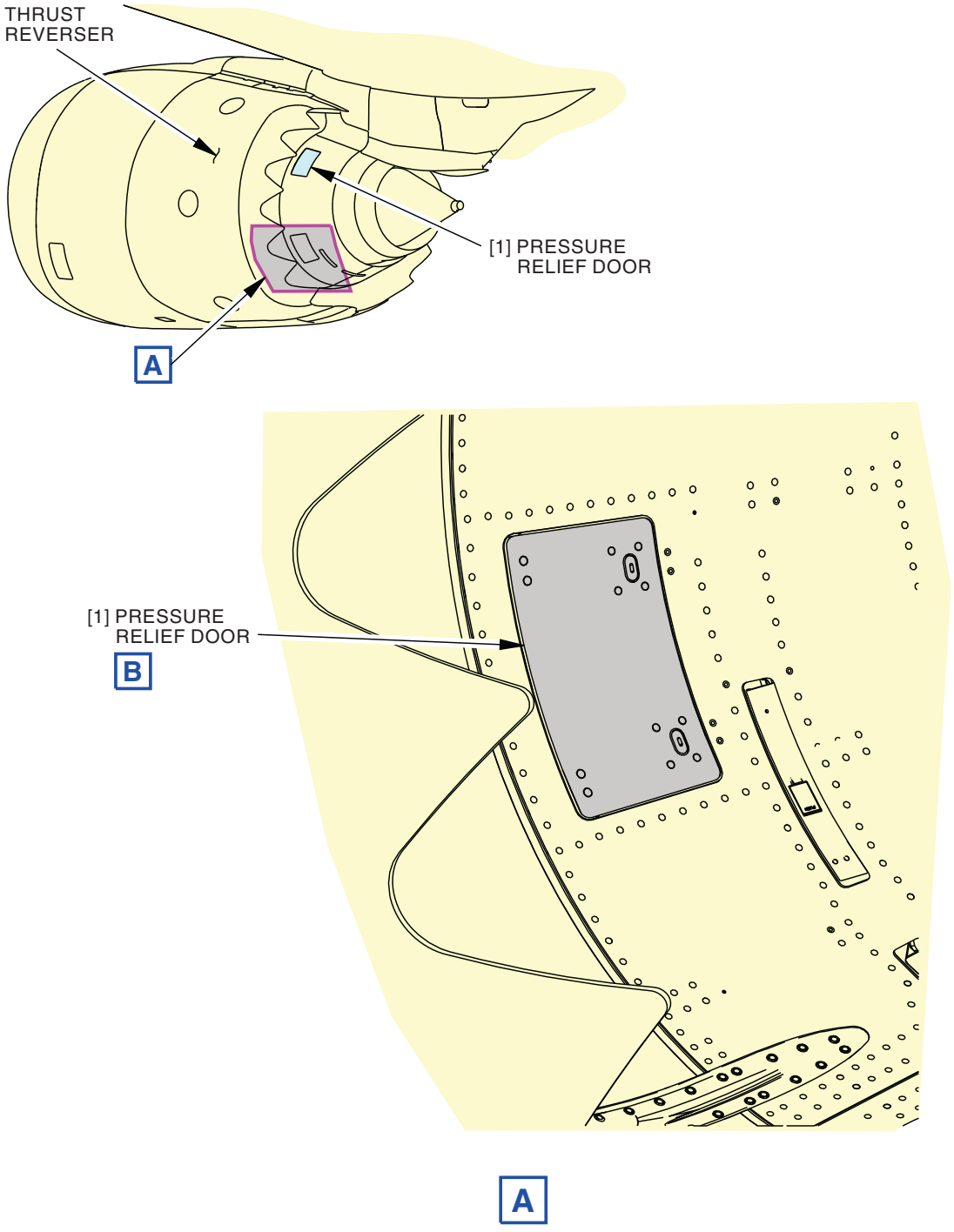
SUBTASK 78-31-16-020-002-G00

(2) If it is necessary, remove the eight nuts [7] that attach the two pressure relief door latches [2] to the pressure relief door [1], and remove the latches from the door.

NOTE: It may be necessary to remove the pressure relief latches so they may be re-installed onto the replacement pressure relief door.

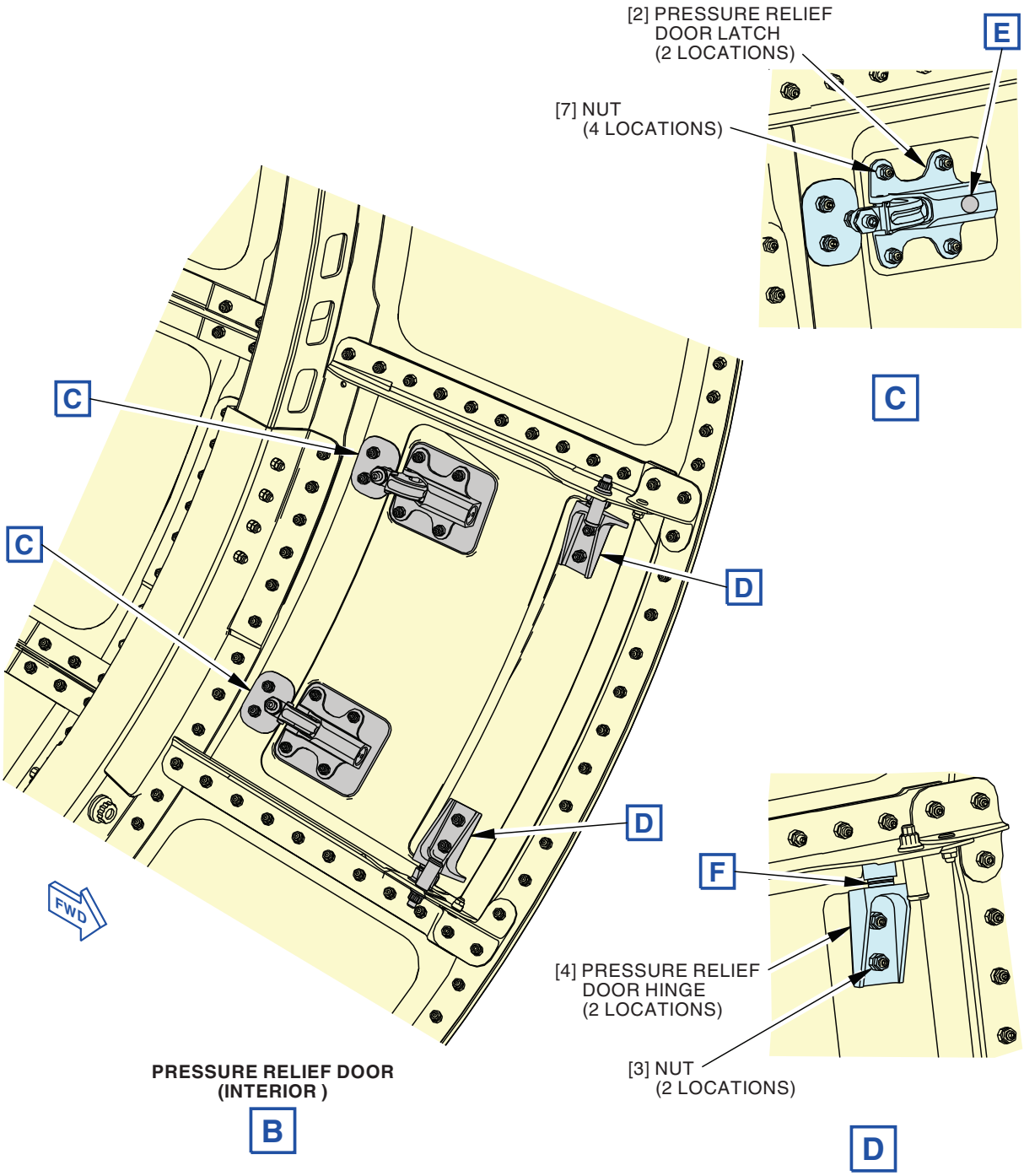
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



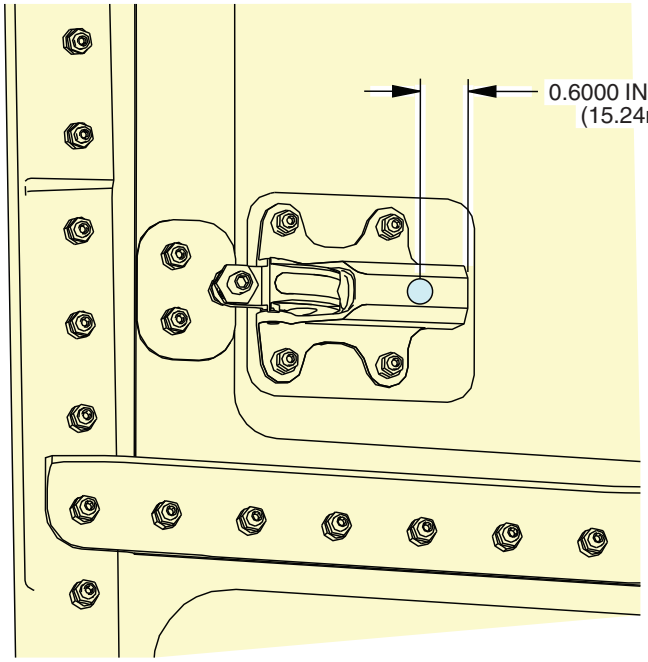
Pressure Relief Door Installation
Figure 401/78-31-16-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

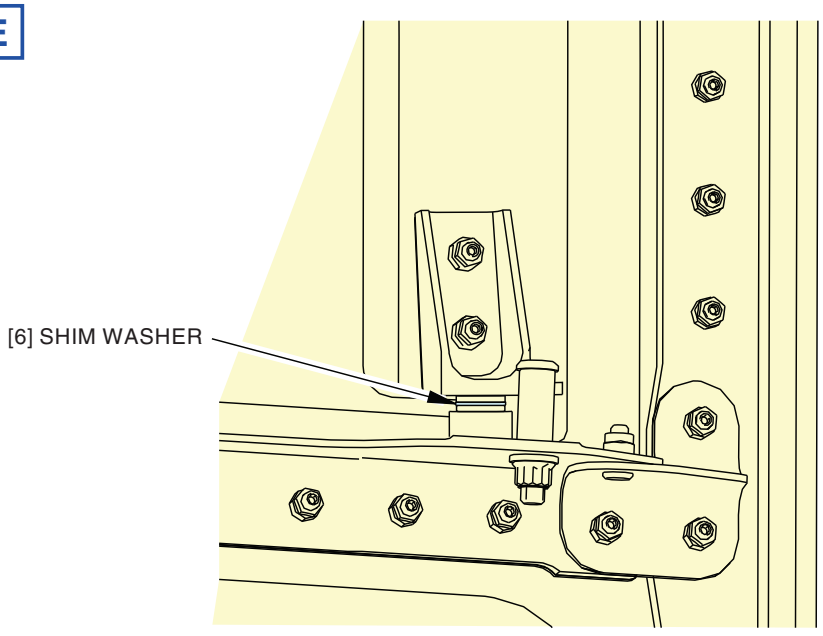


Pressure Relief Door Installation
Figure 401/78-31-16-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



E



F

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Pressure Relief Door Rigging
Figure 402/78-31-16-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-16-400-801-G00

3. Pressure Relief Door Installation

Figure 401 and Figure 402

A. General

(1) This task is the installation of the thrust reverser pressure relief door.

(2) There are four pressure relief doors on each engine.

(3) The pressure relief doors are installed on the thrust reverser aft cowl panel assembly, at the 2, 4, 8 and 10 o'clock positions.

(4) The installation procedure is the same for each of the pressure relief doors.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50177	Lockwire - MS20995N32, Inconel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pressure relief door	78-31-16-01-120	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Pressure Relief Door Installation

SUBTASK 78-31-16-420-001-G00

(1) If it is necessary, install the two pressure relief door latches [2] on the pressure relief door [1] as follows:

(a) Put the two pressure relief door latches [2] in position on the door.

(b) Install the eight nuts [7] and fasteners.

(c) Tighten the eight nuts [7] to 30 ±1 in-lb (3 ±1 N·m).

SUBTASK 78-31-16-420-002-G00

(2) Install the pressure relief door [1] onto the aft cowl assembly as follows:

(a) Make sure that the two pressure relief door latches [2] are in the open position.

(b) Put one pressure relief door hinge [4] and shim washers [6] in position on the aft cowl so that the hinge pin is engaged with the hinge bracket.

1) Adjust the number of shim washers [6] so that the gap between the pressure relief door [1] and the skin is 0.1000 in. (2.5400 mm) with the door in the closed position.

2) Adjust the number of shim washers [6] so that the end play of the door assembly is 0.001 in. (0.0254 mm) to 0.017 in. (0.4318 mm) with the door in the closed position.

(c) Install the two nuts [3] and fasteners, and tighten the nuts hand tight.

(d) Put the other pressure relief door hinge [4] and shim washers [6] in positions on the aft cowl so that the hinge pin is engaged with the hinge bracket.

1) Adjust the number of shim washers [6] so that the gap between the pressure relief door [1] and the skin is 0.1000 in. (2.5400 mm) with the door in the closed position.

2) Adjust the number of shim washers [6] so that the end play of the door assembly is 0.001 in. (0.0254 mm) to 0.017 in. (0.4318 mm) with the door in the closed position.

(e) Install the two nuts [3] and fasteners, and tighten the nuts hand tight.

SUBTASK 78-31-16-820-001-G00

(3) Adjust the pressure relief door as follows:

(a) Make a measurement of the fit and fair gap, and shim as necessary.

1) The maximum gap between the hinges and the door prior to fastener installation is 0.0050 in. (0.1270 mm).

2) The maximum shim thickness is 0.0500 in. (1.2700 mm).

(b) Tighten the four nuts [3] to 35 in-lb (4 N·m).

(c) Make a measurement of the end play on the door hinge, and shim as necessary.

1) Adjust the quantity of shim washers to adjust the gap between the cowl skin and the door assembly.

a) Make sure that there is a 0.1000 in. (2.5400 mm) gap between the pressure relief door [1] and the skins.

2) Make sure that the end play of the door assembly is 0.001 in. (0.025 mm) to 0.017 in. (0.432 mm) with the door in the closed position.

(d) When the adjustments to the pressure relief door are complete, make sure that you tighten the nuts [3] to 35 in-lb (4 N·m) and the nuts [7] to 30 in-lb (3 N·m).

SUBTASK 78-31-16-820-002-G00

(4) Adjust the two pressure relief door latches [2] as follows:

(a) Loosen the jam nut.

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(b) Adjust the bolt on the latch until the striker makes contact with the striker plate.

(c) Tighten the jam nut to 60 ±10 in-lb (7 ±1 N·m).

(d) Install MS20995N32 lockwire, G50177, or MS20995NC32 lockwire, G50347, on the jamnut, the striker, and the bolt.

SUBTASK 78-31-16-710-001-G00

(5) Do a test of the pressure relief door latch function as follows:

(a) Open one of the latches on the pressure relief door.

NOTE: You must perform a test on one latch at a time. The other latch must be in the open position when you test the latch.

(b) Position a spring scale, STD-755, so that you can apply pressure to the latch at the location indicated on Figure 402, View E.

(c) Apply a load to the latch, at a rate of 1 inch per second.

(d) If the latch opens prior to 61 lbf (271 N) or after 89 lbf (396 N) is applied, replace the latch.

(e) Repeat the test on the second latch.

(f) Close the pressure relief door.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-16-840-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-16-840-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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BULLNOSE SEAL - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the thrust reverser bullnose seal.

(2) The installation of the thrust reverser bullnose seal.

TASK 78-31-23-000-801-G00

2. Bullnose Seal Removal

(Figure 401)

A. General

(1) This task is the removal of the bullnose seal from the left or right thrust reverser half.

(2) To get access to the bullnose seal, you must remove the thrust reverser translating sleeve.

B. References

Reference	Title
78-31-02-000-801-G00	Translating Sleeve Removal (P/B 401)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Prepare for the Removal

SUBTASK 78-31-23-020-001-G00

(1) Do this task to remove the translating sleeve from the applicable thrust reverser, Translating Sleeve Removal, TASK 78-31-02-000-801-G00.

E. Bullnose Seal Removal

SUBTASK 78-31-23-020-002-G00

(1) Do the steps that follow to remove the bullnose seal [1] from the applicable thrust reverser translating sleeve:

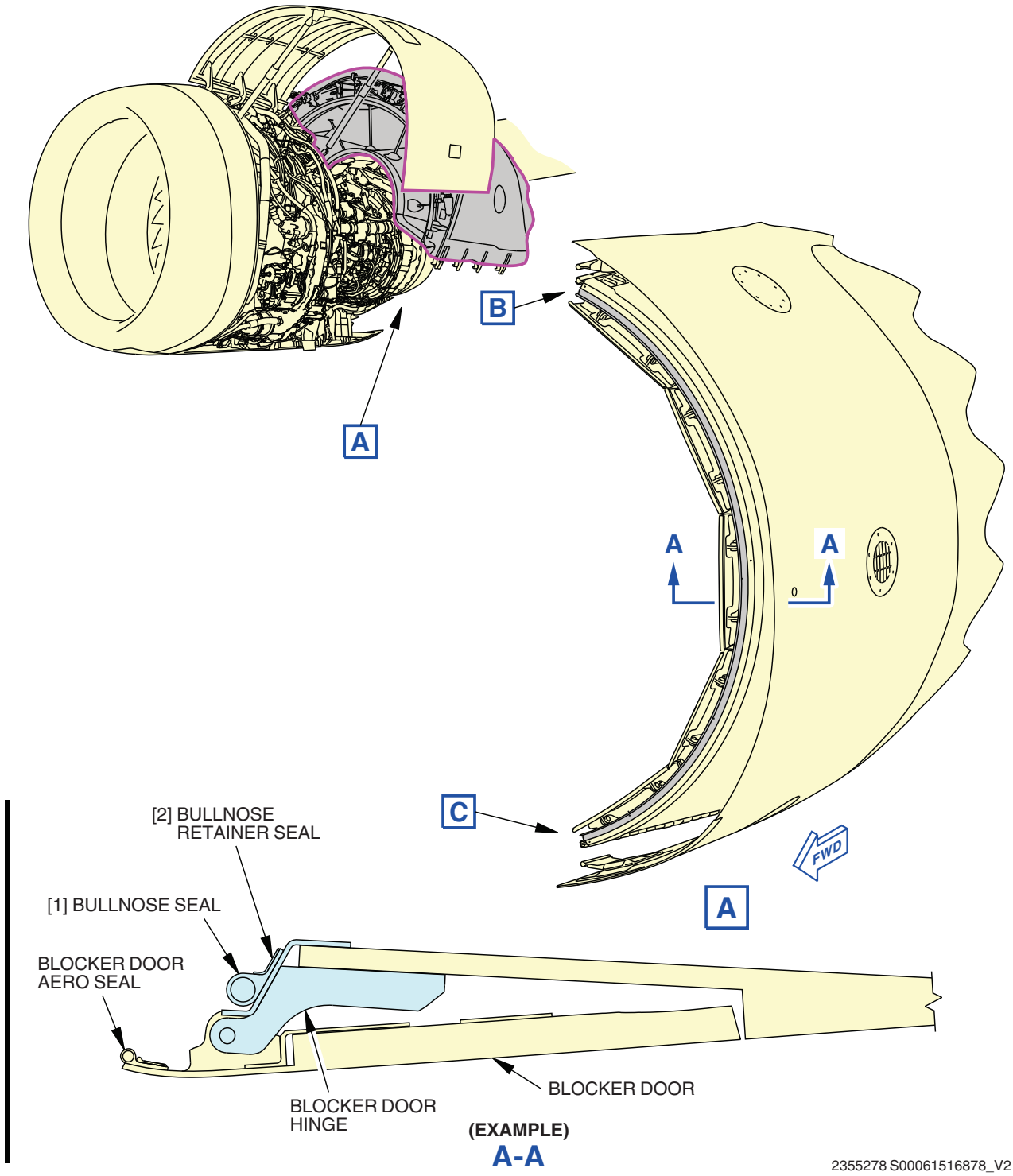
(a) Remove the 31 screws [3] that attach the seal retainer [2] and the bullnose seal [1] to the translating sleeve.

(b) Remove the seal retainer [2] from the translating sleeve.

(c) Remove the bullnose seal [1] from the translating sleeve.

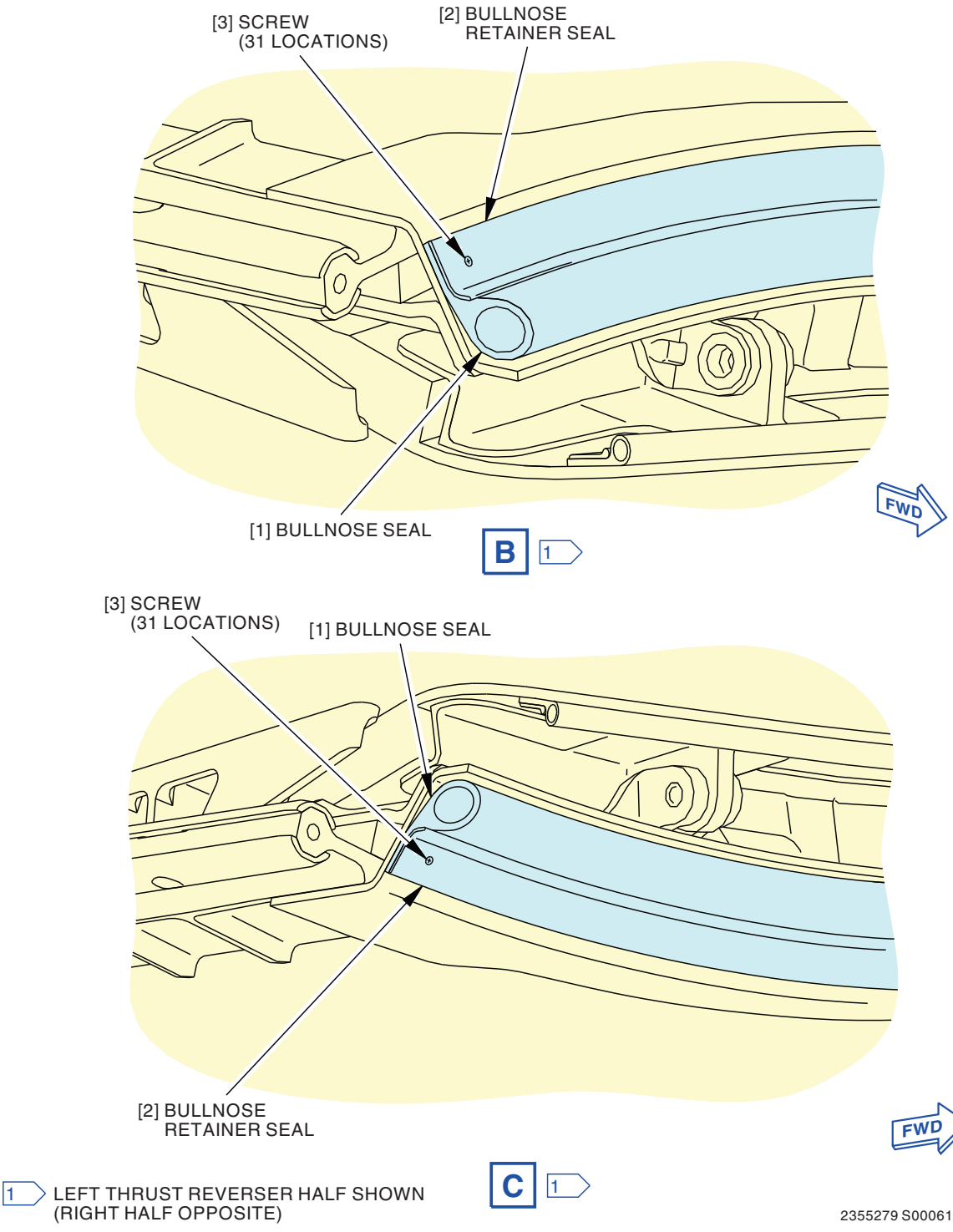
——— END OF TASK ———

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Bullnose Seal Installation
Figure 401/78-31-23-990-801-G00 (Sheet 1 of 2)

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Bullnose Seal Installation
Figure 401/78-31-23-990-801-G00 (Sheet 2 of 2)

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TASK 78-31-23-400-801-G00

3. Bullnose Seal Installation

(Figure 401)

A. General

(1) This task is the installation of the bullnose seal from the left or right thrust reverser on an engine.

(2) To get access to the bullnose seal, you must first remove the thrust reverser translating sleeve.

B. References

Reference	Title
78-31-02-400-801-G00	Translating Sleeve Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Bullnose seal	78-31-23-01-085	PTF ALL

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Bullnose Seal Installation

SUBTASK 78-31-23-020-003-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, IN YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU USE SOLVENTS. KEEP SOLVENTS AWAY FROM SPARKS, FLAME, AND TEMPERATURES ABOVE THE FLASH POINT. SOLVENTS ARE A POISONOUS AND FLAMMABLE MATERIAL WHICH CAN CAUSE INJURY PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do these steps to install the bullnose seal [1] on the applicable translating sleeve:

(a) Clean all contamination from the bullnose seal support ring with a clean cotton wiper, G00034, and solvent, B00062.

(b) Put the bullnose seal [1] into its position on the translating sleeve.

(c) Install the seal retainer [2] over the bullnose seal [1] with the 31 screws [3].

(d) Tighten the screws to 20 ±1 in-lb (2.3 ±0.2 N·m)

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-23-410-001-G00

(1) Install the translating sleeve onto the thrust reverser half (TASK 78-31-02-400-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BULLNOSE SEAL - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) A visual check of the bullnose seal and retainer.

TASK 78-31-23-200-801-G00

2. Bullnose Seal Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions on how to examine the bullnose seal.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-31-23-000-801-G00	Bullnose Seal Removal (P/B 401)
78-31-23-400-801-G00	Bullnose Seal Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14993	Lock - Thrust Reverser Opening Actuator, 15 degree open, LEAP-1B Part #: C78035-14 Supplier: 81205

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

E. Prepare for the Inspection

SUBTASK 78-31-23-040-001-G00

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 78-31-23-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 78-31-23-480-001-G00

(3) Install the actuator safety 15 degree lock assembly, SPL-14993, on the inboard thrust reverser's extended piston rod to lock the thrust reverser at a 15-degree open position.

NOTE: The part number of the actuator safety lock is C78035-14. The part number (C78035-14) is for the set of two actuator safety locks.

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SUBTASK 78-31-23-980-001-G00

CAUTION	DO NOT MANUALLY EXTEND THE INBOARD THRUST REVERSER SLEEVE MORE THAN 15 INCHES. MAKE SURE THAT THE SLEEVE DOES NOT TOUCH THE WING LEADING EDGE WHEN YOU EXTEND THE THRUST REVERSER. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

(4) Do these steps to see the bullnose seal:

NOTE: The sleeve must be partially extended to see the bullnose seal.

(a) Do these steps to manually extend the thrust reverser sleeves:

1) Make sure that the leading edge flaps are completely retracted.

NOTE: Without hydraulics to hold the flaps in the retract position, the weight of the flaps can cause them to extend a small amount.

2) Manually extend the outboard thrust reverser sleeve and inboard thrust reverser sleeve no more than 15 in. (381.0 mm) from the forward edge of the torque box (TASK 78-31-00-980-803-G00).

a) Monitor the position of the thrust reverser sleeve as it extends to make sure that it does not touch the leading edge of the wing.

NOTE: While the inboard thrust reverser is open to the 15-degree position, the maximum sleeve extension is 15 in. (381.0 mm) to avoid contact with the leading edge.

F. Bullnose Seal Inspection

SUBTASK 78-31-23-210-001-G00

(1) Look through the forward end of the thrust reverser to examine the bullnose seal for damage:

(a) Missing material, cuts, gouges and holes that extend through the bullnose seal

1) Not permitted.

2) Replace the bullnose seal (TASK 78-31-23-000-801-G00 and TASK 78-31-23-400-801-G00).

SUBTASK 78-31-23-210-002-G00

(2) Examine the bullnose seal retainer for damage.

(a) Missing metal or distortion

1) Not permitted, replace the seal retainer.

SUBTASK 78-31-23-210-003-G00

(3) Do a check for missing or loose screws and nuts that hold the seal retainer and blocker door hinge in their position.

(a) Missing or loose screws and nuts

1) Not permitted, replace or tighten the nuts that attach the blocker door hinge and the retainer.

a) Tighten the nuts to 25 ±5 in-lb (3 ±1 N·m).

2) Replace or tighten the screws that attach the retainer.

a) Tighten the nuts to 25 ±5 in-lb (3 ±1 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-23-840-001-G00

(1) Do this task: Thrust Reverser Operation - Retract (Manual Procedure), TASK 78-31-00-980-804-G00.

SUBTASK 78-31-23-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS WHEN YOU CLOSE THE THRUST REVERSERS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Close the applicable thrust reverser (TASK 78-31-00-010-802-G00).

(a) Remove the actuator safety 15 degree lock assembly, SPL-14993, from the inboard thrust reverser only.

SUBTASK 78-31-23-410-005-G00

(3) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-31-23-440-002-G00

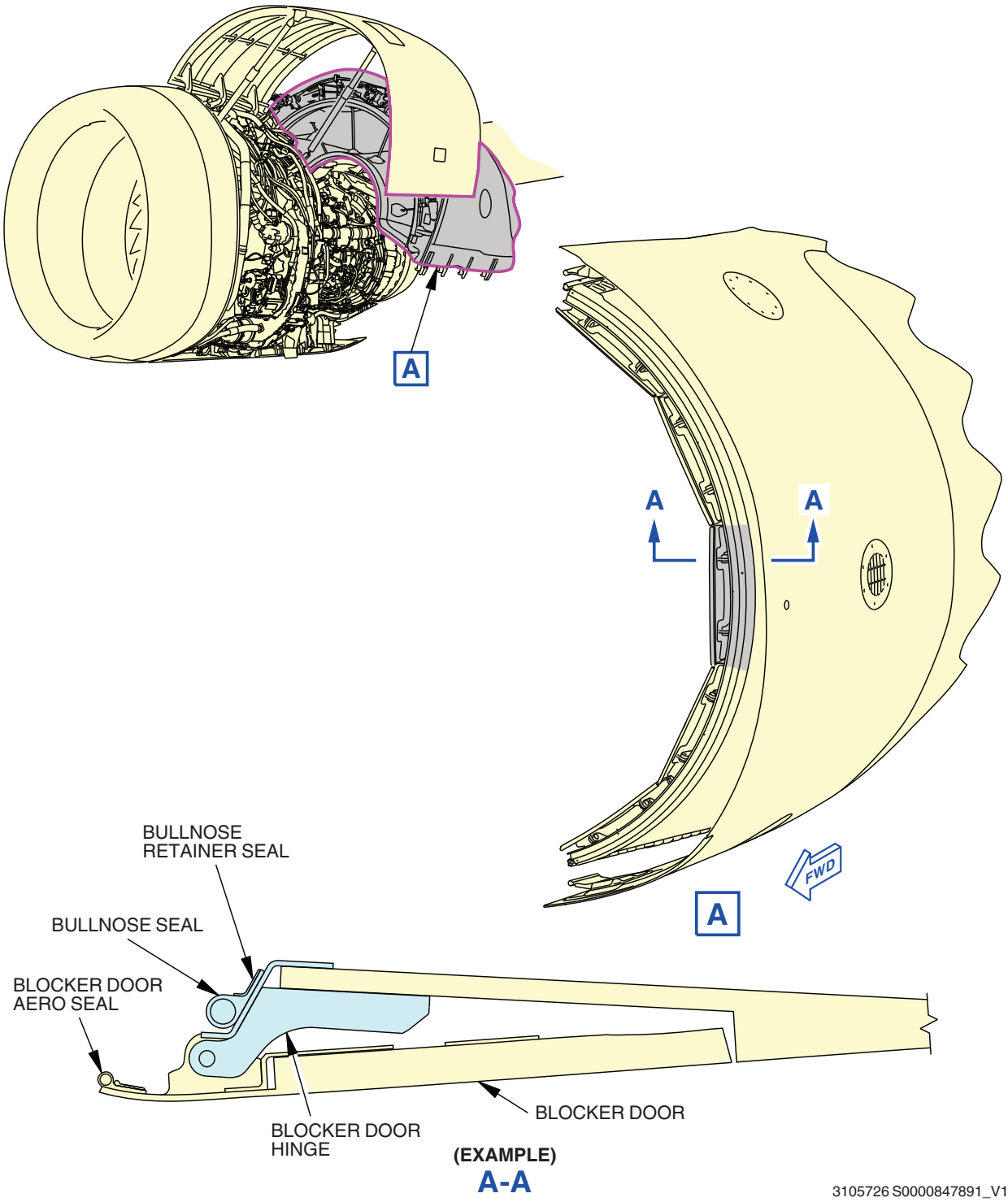
(4) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-31-23-440-003-G00

(5) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Bullnose Seal Inspection
Figure 601/78-31-23-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BLOCKER DOOR AERO SEAL - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the blocker door aero seal.

(2) The installation of the blocker door aero seal.

TASK 78-31-24-000-801-G00

2. Blocker Door Aero Seal Removal

(Figure 401)

A. General

(1) This task removes the blocker door aero seal from the thrust reverser blocker door.

B. References

Reference	Title
78-31-06-000-801-G00	Blocker Door Removal (P/B 401)

C. Prepare for the Removal

SUBTASK 78-31-24-020-001-G00

(1) Remove the applicable blocker door (TASK 78-31-06-000-801-G00).

D. Blocker Door Aero Seal Removal

SUBTASK 78-31-24-020-002-G00

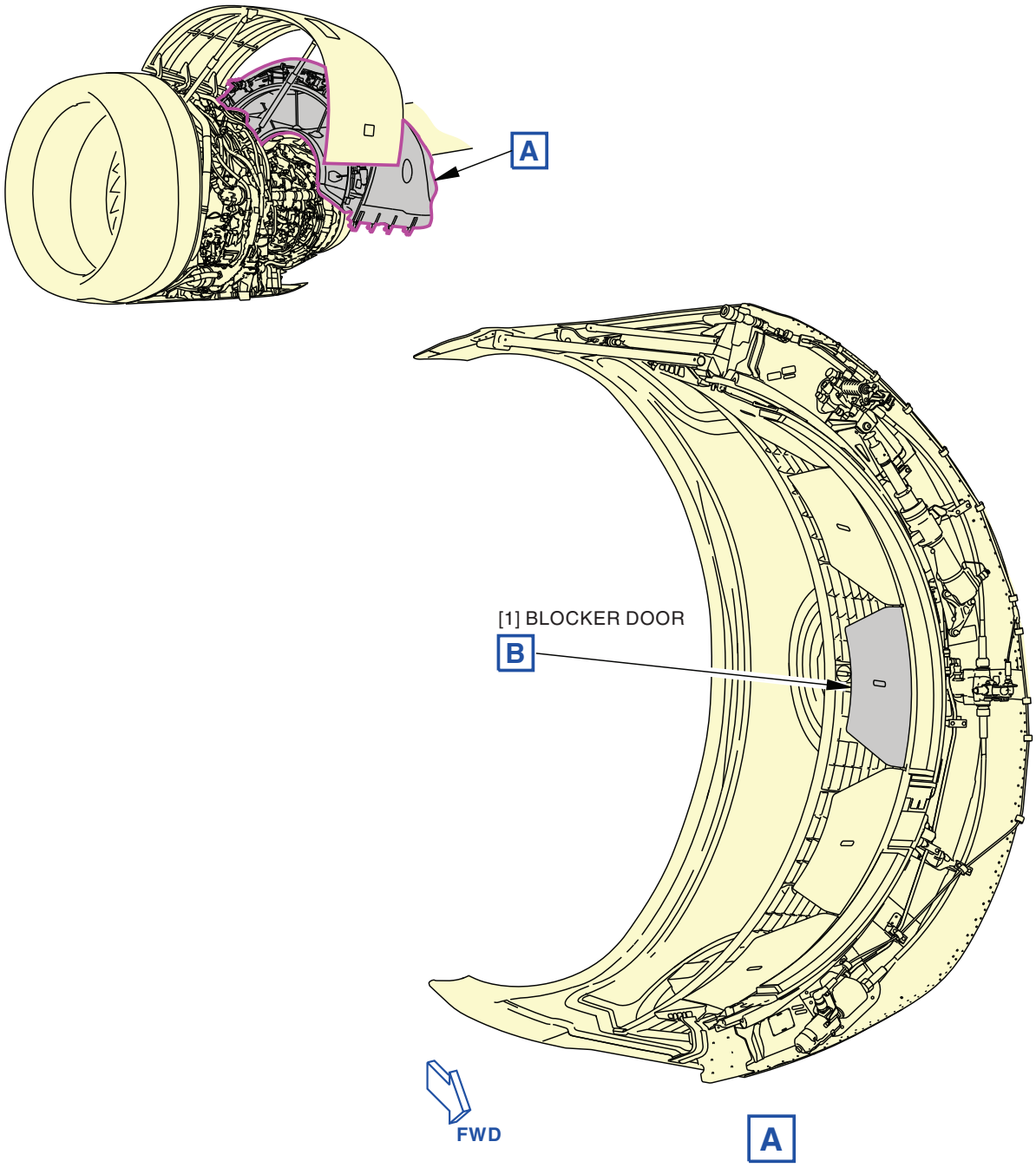
(1) Do the steps that follow to remove the blocker door aero seal [2] from the blocker door [1].

(a) Remove the eight nuts [5] from the bolts [4] that attach the seal retainer [3] and the blocker door aero seal [2] to the blocker door [1].

(b) Remove the seal retainer [3] and the blocker door aero seal [2] from the blocker door [1].

——— END OF TASK ———

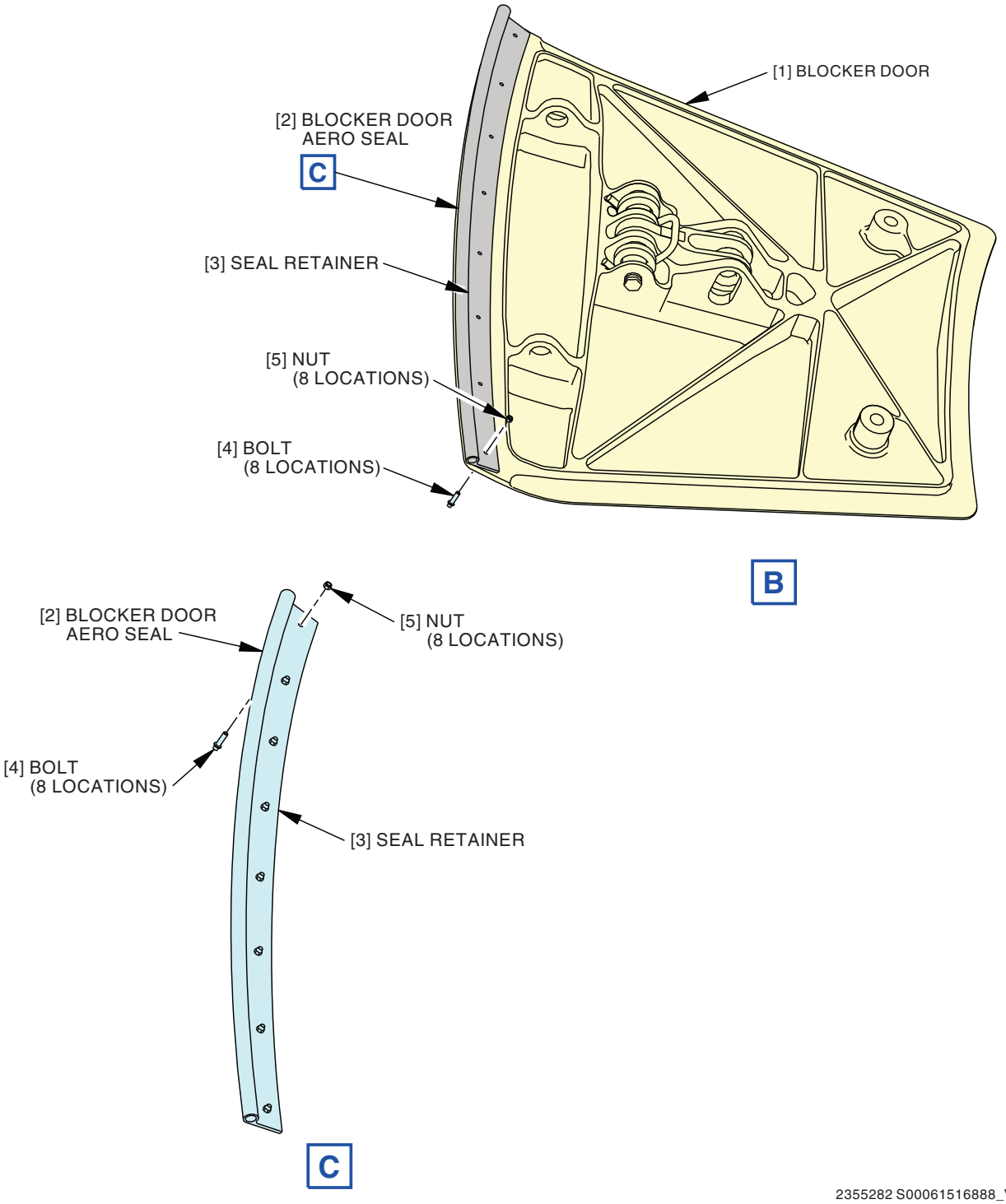
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Blocker Door Aero Seal Installation
Figure 401/78-31-24-990-801-G00 (Sheet 1 of 2)

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Blocker Door Aero Seal Installation
Figure 401/78-31-24-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-31-24-400-801-G00

3. Blocker Door Aero Seal Installation

(Figure 401)

A. General

(1) This task is the installation of the blocker door aero seal onto the thrust reverser blocker door.

B. References

Reference	Title
78-31-06-400-801-G00	Blocker Door Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)

D. Blocker Door Aero Seal Installation

SUBTASK 78-31-24-020-003-G00

(1) Do the steps that follow to install the blocker door aero seal [2] onto the blocker door [1]:

WARNING

DO NOT GET ACETONE, ISOPROPYL ALCOHOL OR CLEANING SOLVENT DESOCLEAN IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE GAS FROM ACETONE, ISOPROPYL ALCOHOL OR CLEANING SOLVENT DESOCLEAN. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE ACETONE, ISOPROPYL ALCOHOL OR CLEANING SOLVENT DESOCLEAN. KEEP ACETONE, ISOPROPYL ALCOHOL AND CLEANING SOLVENT DESOCLEAN AWAY FROM SPARKS, FLAME, AND HEAT. ACETONE, ISOPROPYL ALCOHOL AND CLEANING SOLVENT DESOCLEAN ARE POISONOUS AND FLAMMABLE SOLVENTS WHICH CAN CAUSE INJURY OR DAMAGE.

(a) Clean any contamination from the blocker door aero seal [2], seal retainer [3] and blocker door [1] with solvent, B00062.

(b) Put the seal retainer [3] and blocker door aero seal [2] into their positions on the blocker door [1].

(c) Install the blocker door aero seal [2] and seal retainer [3] onto the blocker door [1] with the eight bolts [4] and nuts [5].

(d) Tighten the eight nuts [5] to 17.5 ±2.5 in-lb (2.0 ±0.3 N·m).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 78-31-24-860-001-G00

(1) Install the blocker door [1] onto the thrust reverser (TASK 78-31-06-400-801-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER CONTROL VALVE MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Control valve module removal

(2) Control valve module installation.

TASK 78-34-01-000-801-G00

2. Control Valve Module Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the control valve module.

(2) There are two control valve modules that control hydraulic power to the thrust reverser hydraulic actuators.

(3) The control valve modules are in the main gear wheel well on the keel beam.

(a) The control valve module for the Engine 1 thrust reverser is on the left side of the keel beam.

(b) The control valve module for the Engine 2 thrust reverser is on the right side of the keel beam.

(4) The thrust reverser control valve module is referred to as the control valve module.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Access Panels

Number	Name/Location
193B	Wheel Well Panel - Forward Inboard

F. Prepare for the Removal

SUBTASK 78-34-01-860-001-G00

(1) For Engine 1, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

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SUBTASK 78-34-01-860-002-G00

(2) For Engine 2, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-34-01-860-003-G00

(3) Make sure that the forward thrust lever is in the idle position.

(a) Install a DO NOT OPERATE tag, STD-858.

SUBTASK 78-34-01-860-004-G00

(4) Make sure that the reverse thrust lever is in the stow position (fully forward and down).

(a) Install a DO NOT OPERATE tag, STD-858.

SUBTASK 78-34-01-860-005-G00

(5) Remove power from the hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-34-01-860-006-G00

(6) Remove power from the standby hydraulic power, do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-34-01-860-007-G00

(7) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-34-01-010-001-G00

(8) If necessary, open this access panel:

Number	Name/Location
193B	Wheel Well Panel - Forward Inboard

G. Control Valve Module Removal

SUBTASK 78-34-01-010-002-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTIONS ARE NOT DIRTY BEFORE YOU CONNECT OR DISCONNECT THEM. SOLID OR LIQUID CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Disconnect the electrical connector [9] from the applicable control valve module [5]:

(a) For Engine 1, disconnect the electrical connector, D3052.

(b) For Engine 2, disconnect the electrical connector, D3056.

(c) Install the protective covers on the electrical connector and the control valve module receptacles.

SUBTASK 78-34-01-010-003-G00

WARNING

DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION USE TWO WRENCHES TO LOOSEN THE FITTINGS. USE THE SECOND WRENCH TO HOLD THE FITTING IN ITS POSITION TO PREVENT DAMAGE TO THE FITTING.

(2) Do these steps to disconnect the hydraulic lines from the control valve module [5].

NOTE: The Thrust Reversers can still operate without the control valve module [5] installed.

CAUTION CATCH THE HYDRAULIC FLUID IN A CONTAINER OR CLOTH WHEN YOU DISCONNECT THE HYDRAULIC LINES. IMMEDIATELY CLEAN ALL SURFACES THAT HYDRAULIC FLUID FALLS ON. HYDRAULIC FLUID CAUSES DAMAGE TO EQUIPMENT.

(a) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 to catch the hydraulic fluid.

(b) For Engine 1 control valve module, disconnect the hydraulic power transfer unit return line (referred to as return hydraulic line) from the tee branch at the return port R.

(c) For Engine 1 and Engine 2 control valve module, disconnect the four hydraulic lines from the control valve module [5].

(d) Install the protective covers on the hydraulic lines.

SUBTASK 78-34-01-020-001-G00

(3) If the new control valve module [5] does not have the union [3] and union [6] or the restrictor check valve [7], do these steps:

NOTE: The new control valve module should have the union installed in the pressure port P.

(a) Remove the restrictor check valve [7], union [3], and union [6] from the control valve module [5] that was removed.

1) Keep the restrictor check valve and the two unions for installation on the new control valve module [5].

2) Remove and discard the packing [8] and the packing [4].

(b) Install the protective covers on the four hydraulic ports on the control valve module [5].

(c) Install the protective covers on the control valve module [5].

SUBTASK 78-34-01-020-002-G00

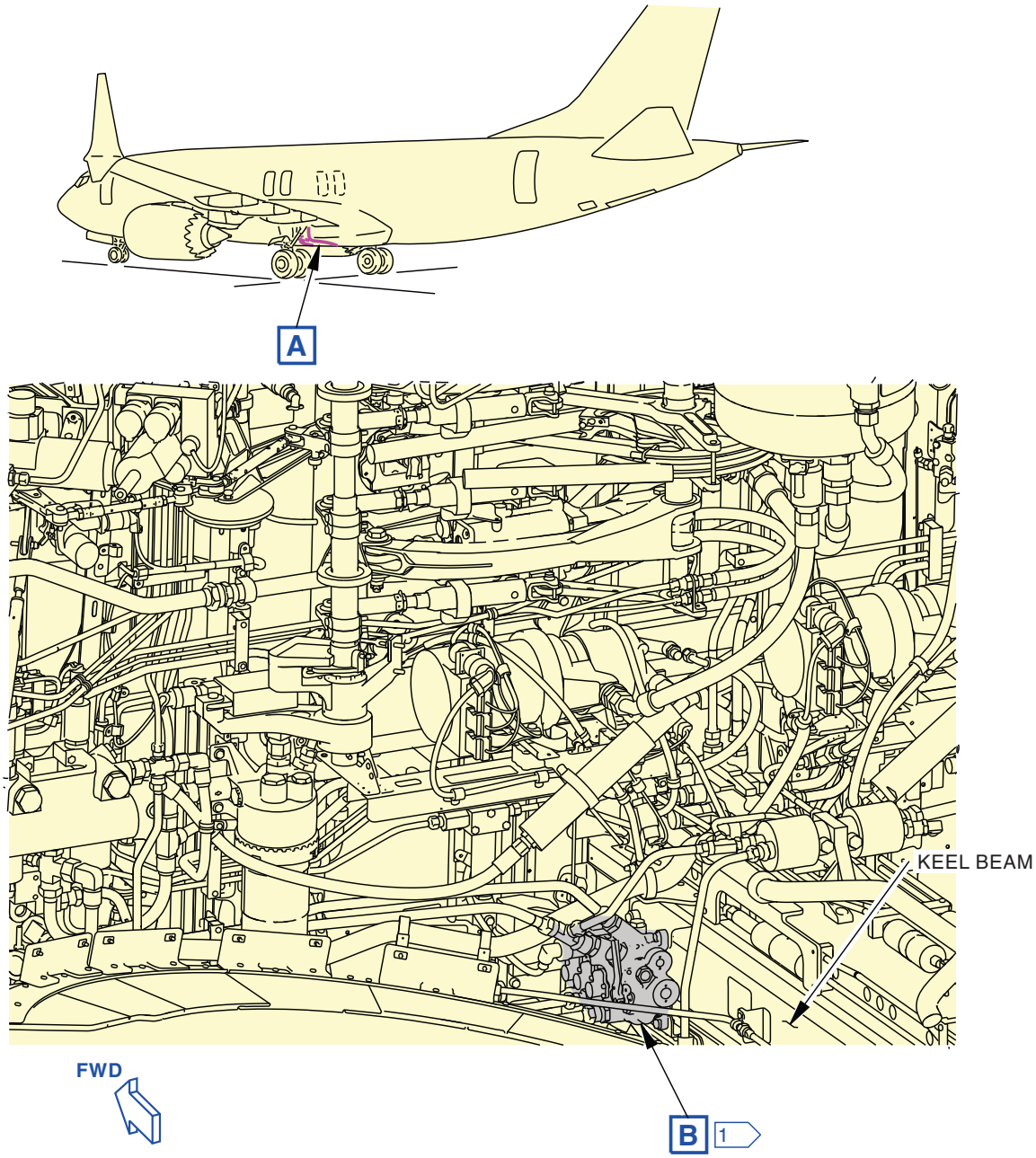
(4) Remove the four bolts [1] and washers [2] that attach the control valve module [5] to the attachment bracket on the keel beam.

SUBTASK 78-34-01-020-003-G00

(5) Remove the control valve module [5].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR WHEEL WELL

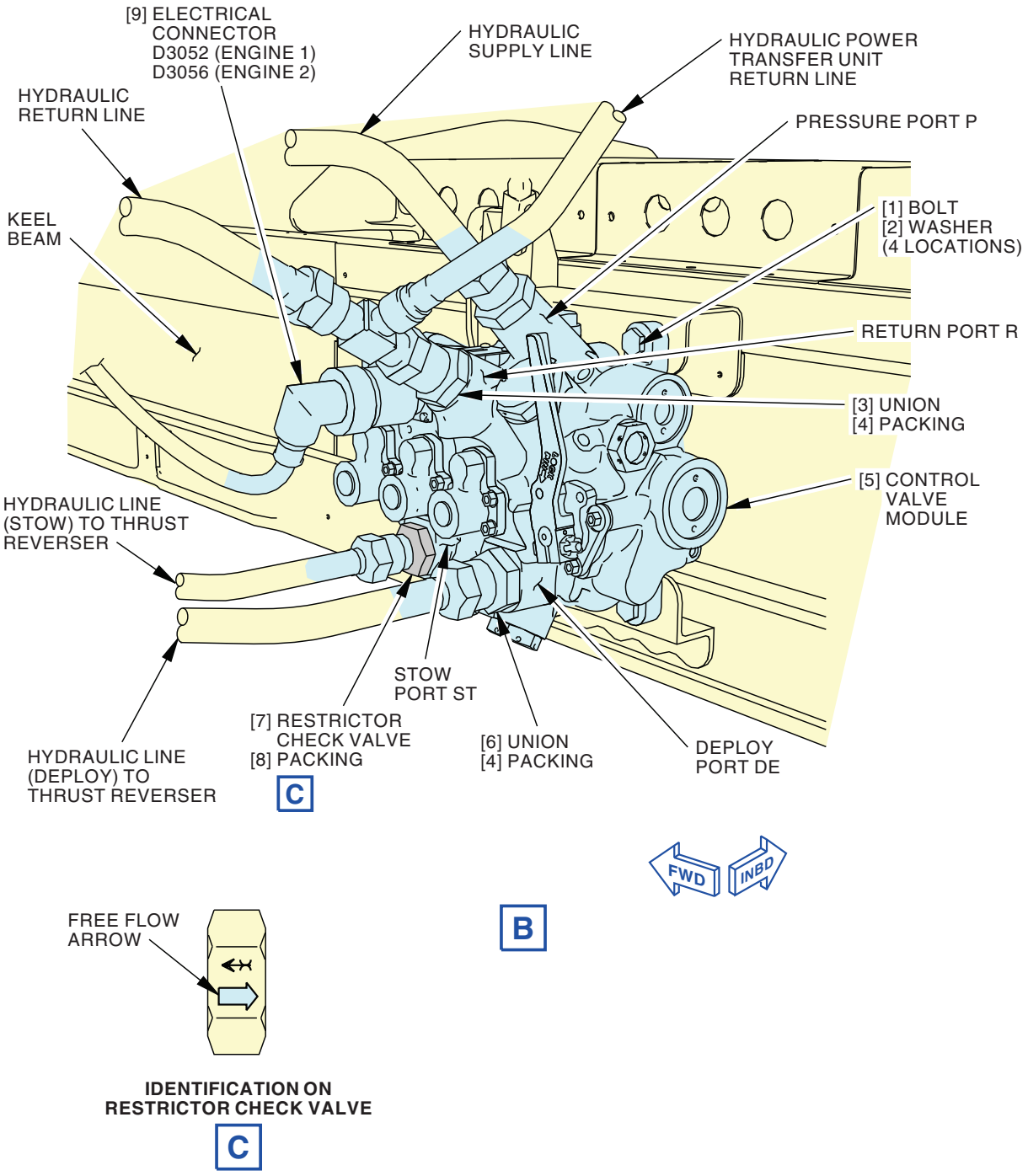
A

1 ENGINE 1 CONTROL VALVE MODULE IS SHOWN, ENGINE 2 CONTROL VALVE IS ON THE RIGHT SIDE OF THE KEEL BEAM.

2355283 S00061516894_V1

Thrust Reverser Control Valve Module Installation
Figure 401/78-34-01-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2355284 S00061516895_V2

Thrust Reverser Control Valve Module Installation
Figure 401/78-34-01-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-01-400-801-G00

3. Control Valve Module Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the control valve module.

(2) The thrust reverser control valve module is referred to as the control valve module.

(3) After you install the control valve module, you must do the Thrust Reverser Normal Operation Test.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Packing	78-34-01-01-035	PTF ALL
5	Control valve module	78-34-01-01-050	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Packing	78-34-01-01-045	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

G. Access Panels

Number	Name/Location
193B	Wheel Well Panel - Forward Inboard

H. Control Valve Module Installation

SUBTASK 78-34-01-420-001-G00

(1) Do these steps to install the applicable control valve module [5]:

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, EYES, OR SKIN. DO NOT BREATHE FUMES FROM SOLVENTS. REFER TO PRODUCT MATERIAL SAFETY DATA SHEETS (MSDS AND LOCAL REQUIREMENTS FOR PROPER HANDLING. SOLVENTS ARE HAZARDOUS MATERIALS AND MAY BE FLAMMABLE OR HARMFUL TO THE ENVIRONMENT.

(a) Clean the sealant residue and the attachment surfaces on the control valve module [5] and the mounting bracket with solvent, B00062.

(b) Put the control valve module [5] in the correct position.

(c) Install the four bolts [1] and washers [2] that attach the control valve module [5] to the attachment bracket on the keel beam.

1) Hand tighten the bolts at this time.

(d) Make sure that there are protective covers on the electrical connector [9] and on the control valve module [5] receptacles.

CAUTION CATCH THE HYDRAULIC FLUID IN A CONTAINER OR CLOTH WHEN YOU DISCONNECT THE HYDRAULIC LINES. IMMEDIATELY CLEAN ALL SURFACES THAT HYDRAULIC FLUID FALLS ON. HYDRAULIC FLUID CAUSES DAMAGE TO EQUIPMENT.

(e) Remove the protective covers from the hydraulic lines on the control valve module [5].

(f) If the new control valve module [5] does not have the union [3] and union [6] or the restrictor check valve [7], do these steps:

1) Use the union [3], union [6], and the restrictor check valve [7] from the control valve module [5] that was removed.

2) Remove the protective covers from the new control valve module [5].

3) Lubricate the packing [8] and two packings [4] with hydraulic fluid, D00153 or MCS 352B fluid, D00054.

4) Install the packing [8] on the restrictor check valve [7].

5) Install a packing [4] on the union [3] and union [6].

6) Lubricate the threads of the restrictor check valve [7], union [3], and union [6] with hydraulic fluid, D00153 or MCS 352B fluid, D00054.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

7) Install the union [6] in the deploy port DE.
   a) Tighten the union to 665 in-lb (75.1 N·m) - 735 in-lb (83.0 N·m).
8) Install the union [3] in the return port R.
   NOTE: This is an aluminum alloy union.
   a) Tighten the union to 342 in-lb (38.6 N·m) - 378 in-lb (42.7 N·m).
9) Install the restrictor check valve [7] in the stow port ST.
   a) Make sure that the large free flow arrow on the restrictor check valve [7] points toward the control valve module [5].
   b) Tighten the restrictor check valve [7] to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

(g) Loosen the four bolts [1].

(h) For Engine 1 and Engine 2 control valve module, loosely connect the four hydraulic lines to the control valve module [5].

(i) For Engine 1 control valve module, loosely connect the return hydraulic line for the hydraulic power transfer unit to the tee at the return port R.

(j) Tighten the bolts [1] to 135 in-lb (15.3 N·m) - 165 in-lb (18.6 N·m).

(k) Do a check of the electrical resistance through the bracket that the control valve module [5] is attached to.
   1) With the intrinsically safe approved bonding meter, COM-1550, do a check of the electrical resistance between the control valve module and the structure where the attachment bracket attaches.
   2) Make sure that the resistance is no more than 0.0025 ohm.

CAUTION USE TWO WRENCHES TO TIGHTEN THE FITTINGS. USE THE SECOND WRENCH TO HOLD THE FITTING IN ITS POSITION TO PREVENT DAMAGE TO THE FITTING.

(l) For Engine 1 and Engine 2 control valve module, tighten the hydraulic lines on the control valve module [5] as follows:
   1) Tighten the coupling nut at the deploy port DE to 665 in-lb (75.1 N·m) - 735 in-lb (83.0 N·m).
   2) Tighten the coupling nut at the stow port ST to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).
   3) Tighten the coupling nut at the return port R to 342 in-lb (38.6 N·m) - 378 in-lb (42.7 N·m).
   4) Tighten the coupling nut at the pressure port P to 665 in-lb (75.1 N·m) - 735 in-lb (83.0 N·m).

(m) For Engine 1 control valve module, tighten the coupling nut at the return port R tee branch to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

SUBTASK 78-34-01-390-001-G00

(2) Fillet seal around the bolt flange on the control valve module [5] and the attach bracket with sealant, A00247 or sealant, A02315.

SUBTASK 78-34-01-860-008-G00

(3) Remove the protective covers from the electrical connector [9] and the control valve module [5] receptacles.

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SUBTASK 78-34-01-410-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTIONS ARE NOT DIRTY BEFORE YOU CONNECT OR DISCONNECT THEM. SOLID OR LIQUID CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Connect the electrical connector [9] to the applicable control valve module [5] receptacles:

(a) For Engine 1, connect the electrical connector, D3052

(b) For Engine 2, connect the electrical connector, D3056.

SUBTASK 78-34-01-610-001-G00

(5) Check the hydraulic fluid level in the appropriate reservoir.

(a) For Engine 1, System A.

(b) For Engine 2, System B.

(c) If necessary, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

I. Control Valve Module Installation Test

SUBTASK 78-34-01-440-001-G00

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) For Engine 1, System A.

(b) For Engine 2, System B.

SUBTASK 78-34-01-860-009-G00

(2) For Engine 1, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT

SUBTASK 78-34-01-860-010-G00

(3) For Engine 2, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-34-01-860-011-G00

(4) Remove the DO-NOT-OPERATE tag from the thrust levers.

SUBTASK 78-34-01-860-012-G00

(5) Remove the DO-NOT-OPERATE tag from the reverse thrust levers.

SUBTASK 78-34-01-710-001-G00

(6) Do this task: TASK 78-31-00-700-801-G00.

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(a) Operate the thrust reverser through the deploy and stow cycle until all of the air is bled from the system and the thrust reverser sleeves move smoothly.

(b) Do a check of the control valve module and hydraulic lines for hydraulic leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-34-01-410-002-G00

(7) If necessary, close this access panel:

Number	Name/Location
193B	Wheel Well Panel - Forward Inboard

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SLEEVE LOCK PROXIMITY SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The thrust reverser sleeve lock proximity sensor removal

(2) The thrust reverser sleeve lock proximity sensor installation.

TASK 78-34-03-000-801-G00

2. Thrust Reverser Sleeve Lock Proximity Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the thrust reverser sleeve lock proximity sensor and wire leads from the left or right thrust reverser on an engine.

(a) The lock sensor is a component of the wire bundle on the thrust reverser.

(2) For this procedure, the thrust reverser sleeve lock proximity sensor will be referred to as the lock sensor.

(3) The upper lock sensor equipment number for the left thrust reverser on an engine is S1200 and the wire bundle number is W1080.

(4) The middle lock sensor equipment number for the left thrust reverser on an engine is S1201 and the wire bundle number is W1080.

(5) The upper lock sensor equipment number for the right thrust reverser on an engine is S1202 and the wire bundle number is W1084.

(6) The middle lock sensor equipment number for the right thrust reverser on an engine is S1203 and the wire bundle number is W1084.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 78-34-03-860-001-G00

(1) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-03-860-002-G00

(2) For Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK

SUBTASK 78-34-03-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-34-03-860-007-G00

(4) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Lock Sensor Removal

SUBTASK 78-34-03-010-002-G00

(1) Disconnect the electrical connectors from the applicable thrust reverser on an engine:

(a) For the left thrust reverser, disconnect these electrical connectors:

1) Disconnect the electrical connector, D30002, from the strut receptacle.

2) Disconnect the electrical connector, D30072, from the LVDT receptacle on the upper actuator.

3) Disconnect the electrical connector, D01008, from the sync lock on the lower actuator.

(b) For the right thrust reverser, disconnect these electrical connectors:

1) Disconnect the electrical connector, D30010, from the strut receptacle.

2) Disconnect the electrical connector, D30078, from the LVDT receptacle on the upper actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Disconnect the electrical connector, D01016, from the sync lock on the lower actuator.

SUBTASK 78-34-03-020-001-G00

(2) Remove the lock sensors [1] from the upper and middle actuators:

(a) Remove the lockwire from the jamnuts [2].

(b) Remove the jamnut [2] from the lock sensor [1].

NOTE: To remove the lock sensor, it is necessary to remove only the jamnut that is adjacent to the sensor target.

(c) Remove the lock sensor [1] and key washer [3] from the locking actuator.

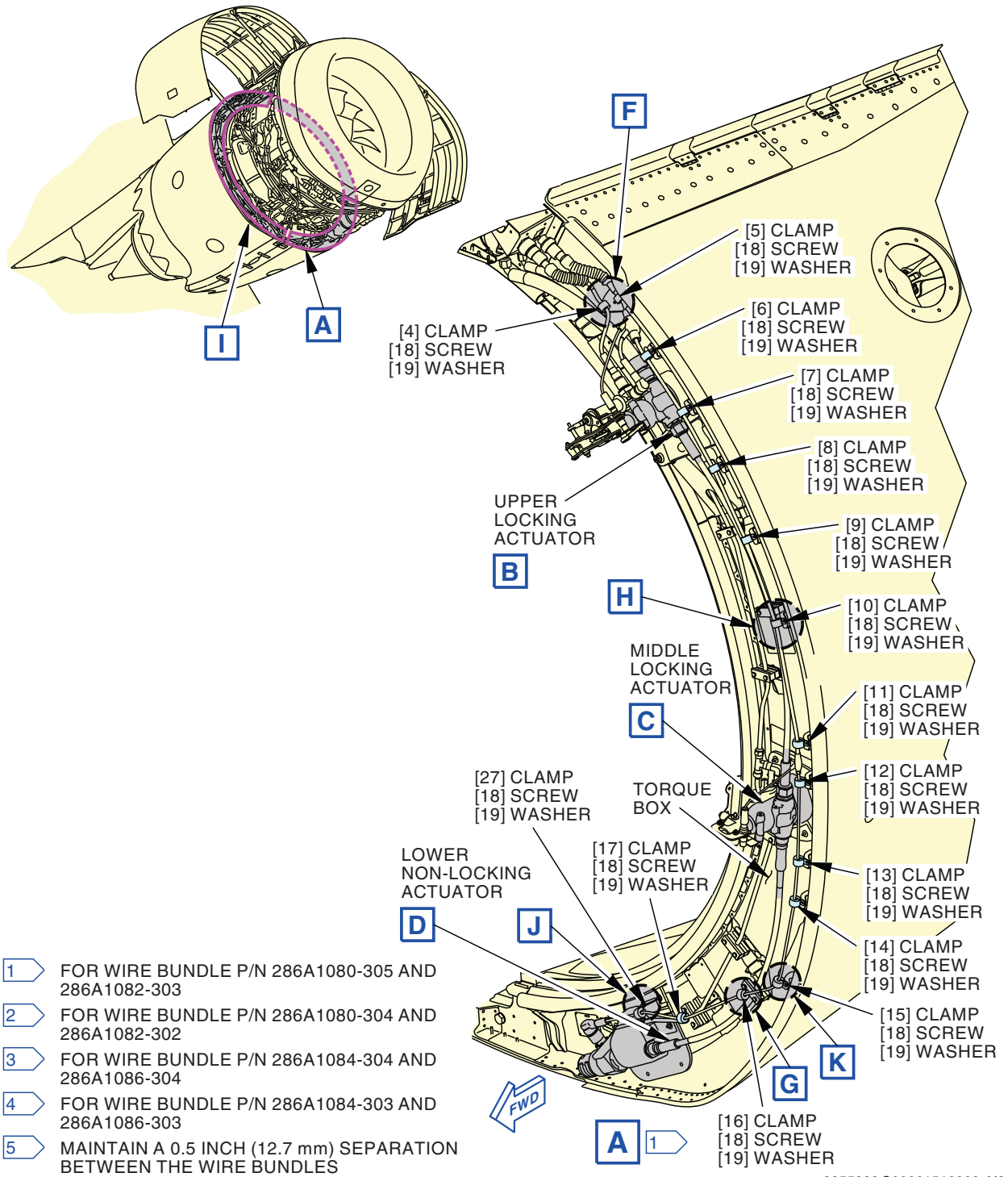
SUBTASK 78-34-03-020-002-G00

(3) Remove the wire bundle from the thrust reverser torque box.

(a) Remove the screws and clamps that attach the wire bundle to the torque box.

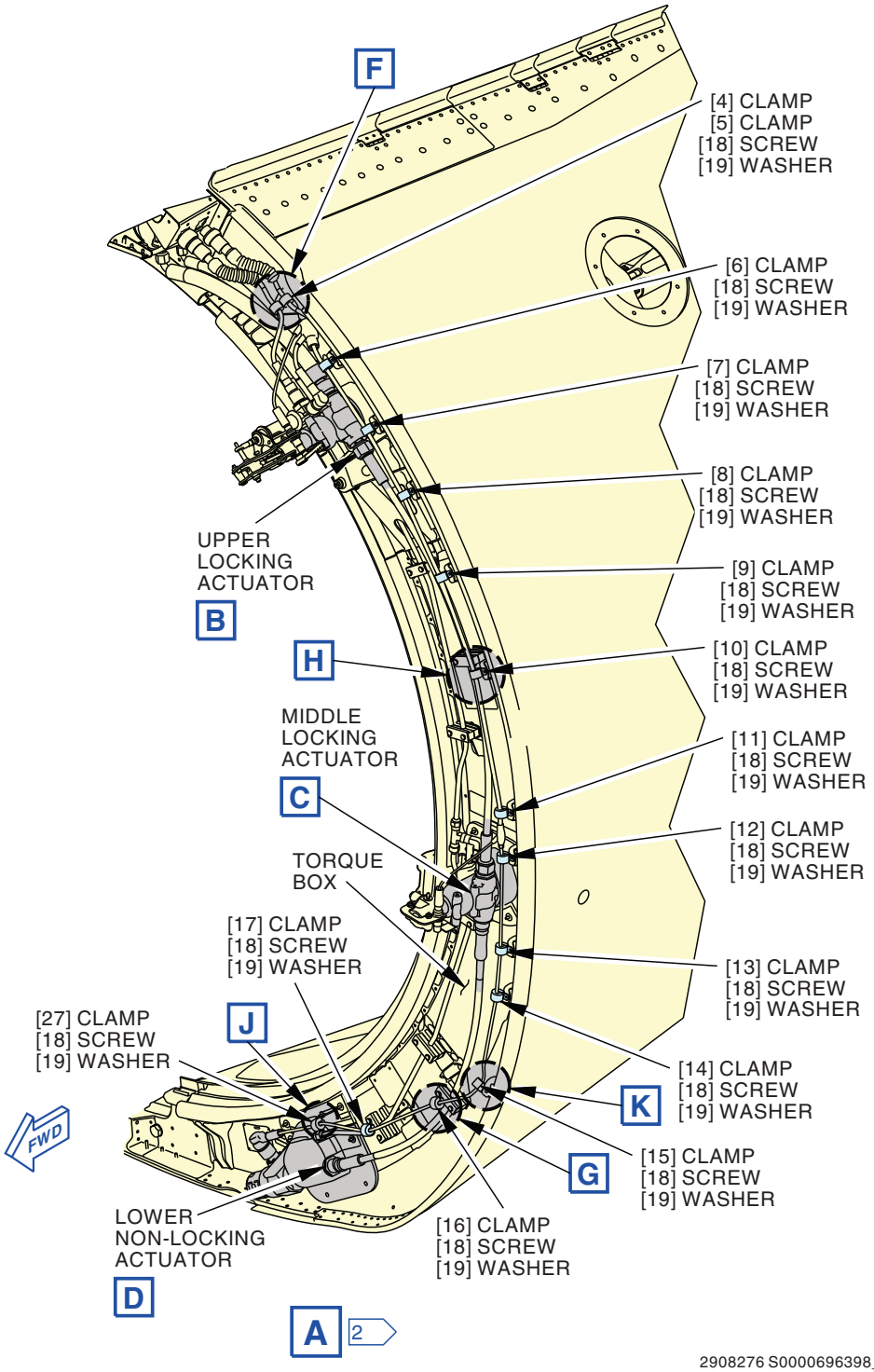
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



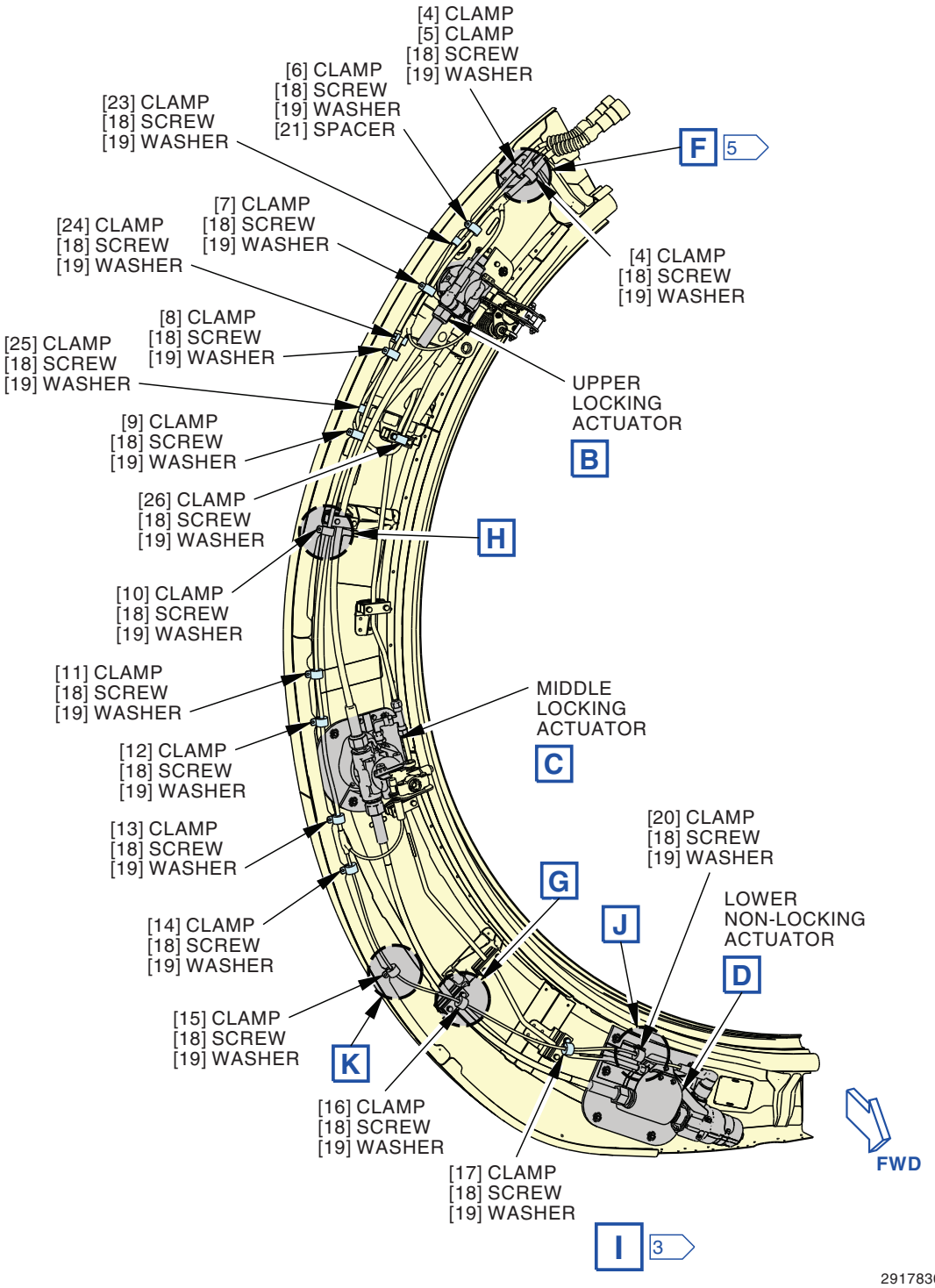
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 1 of 8)

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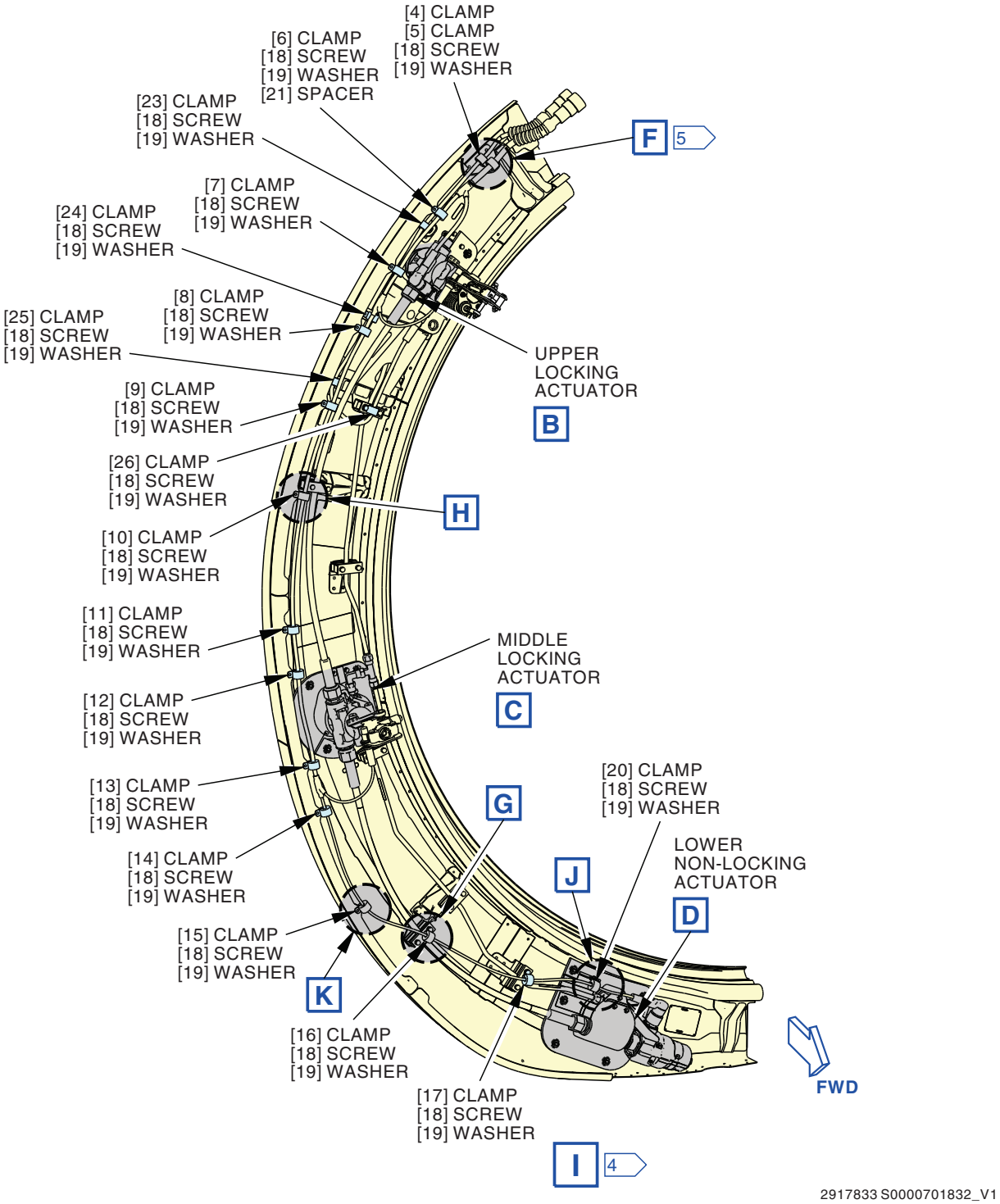
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 2 of 8)

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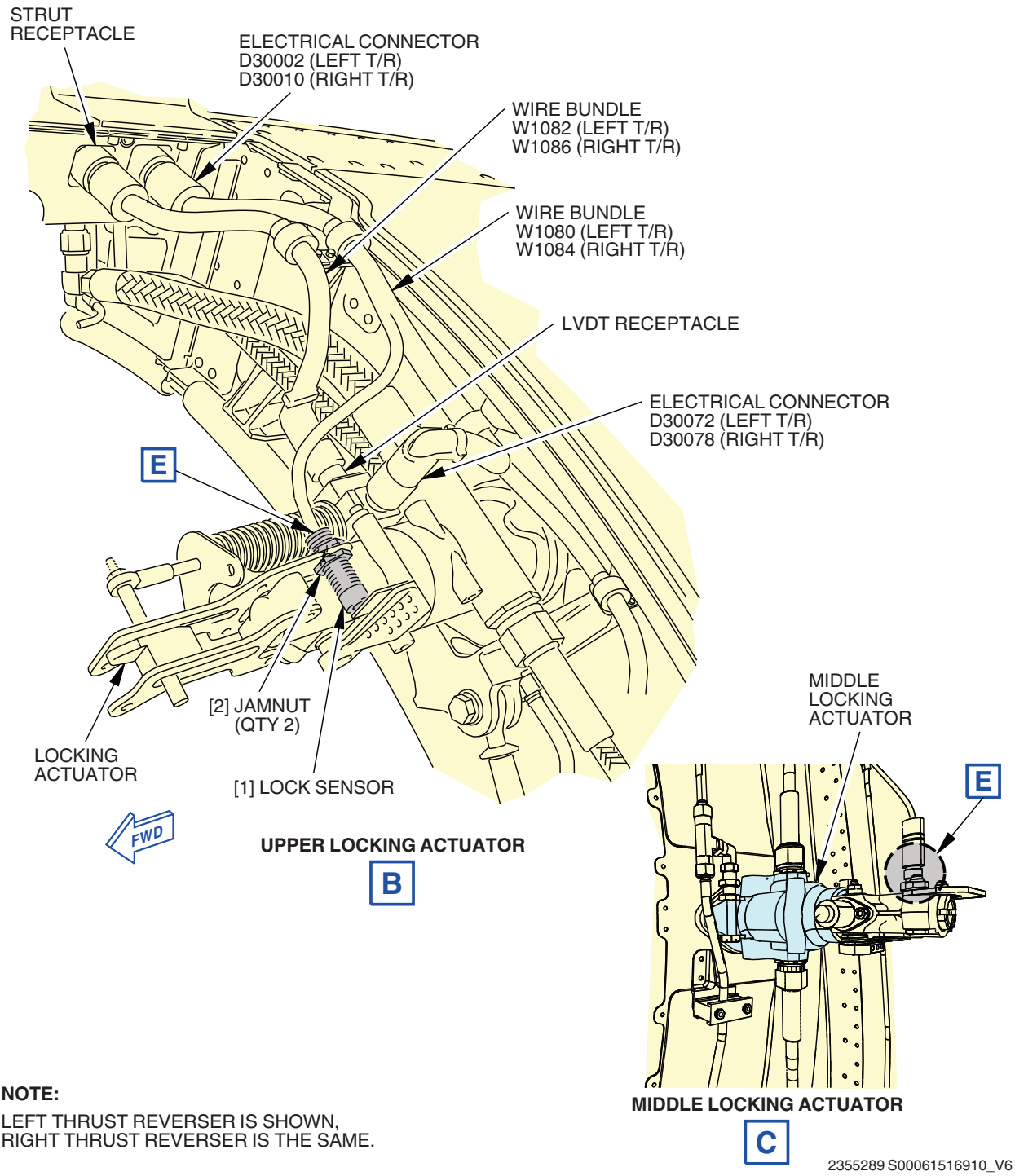
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 3 of 8)

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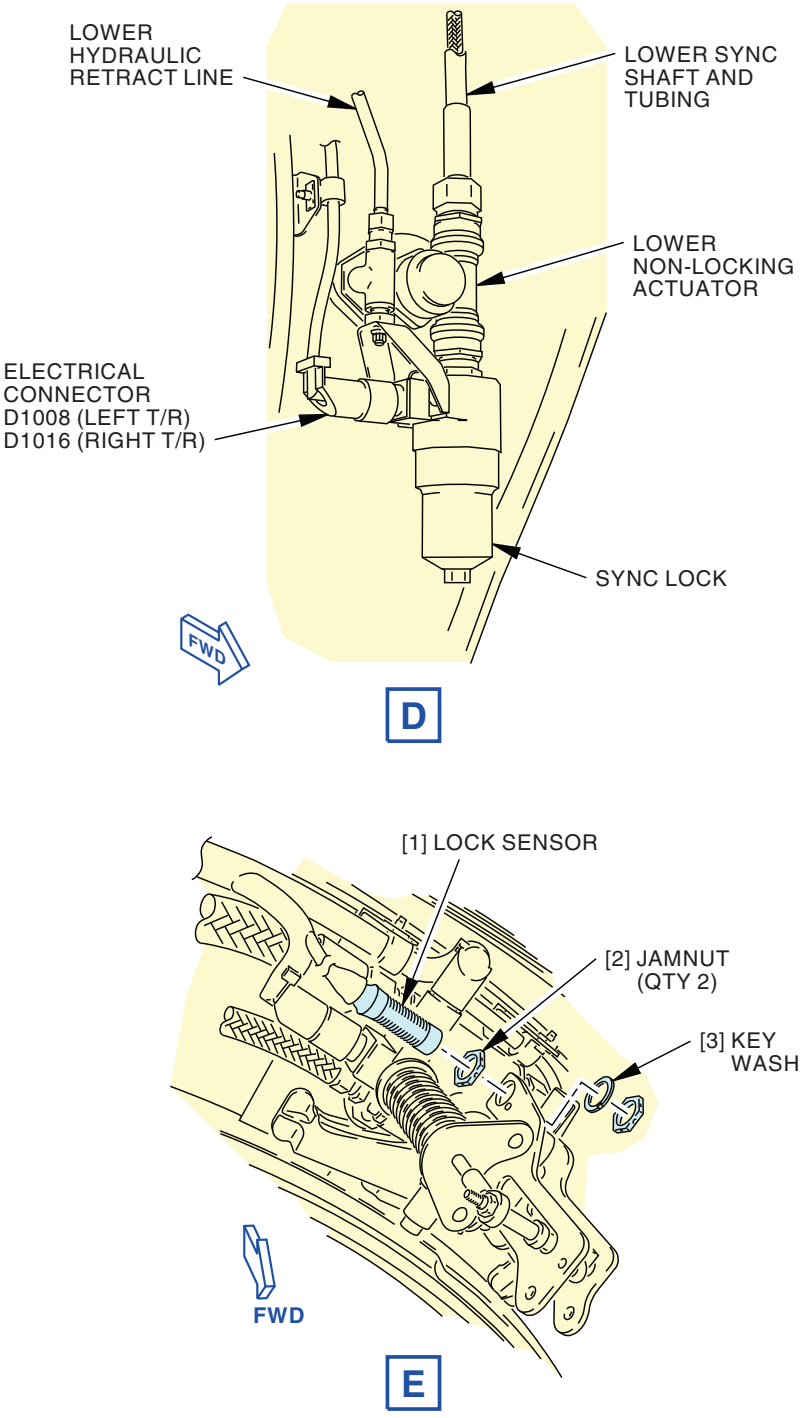
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



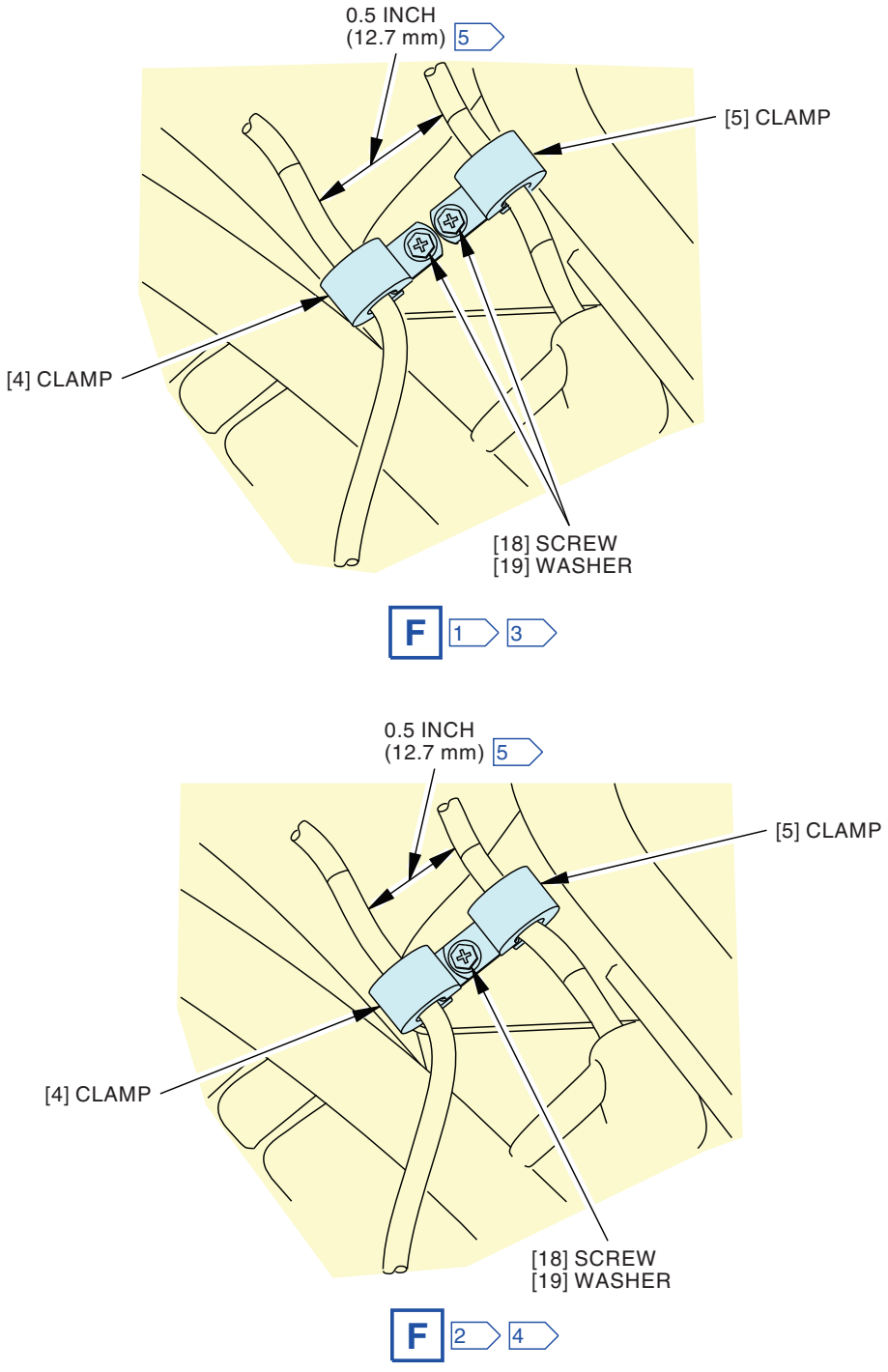
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



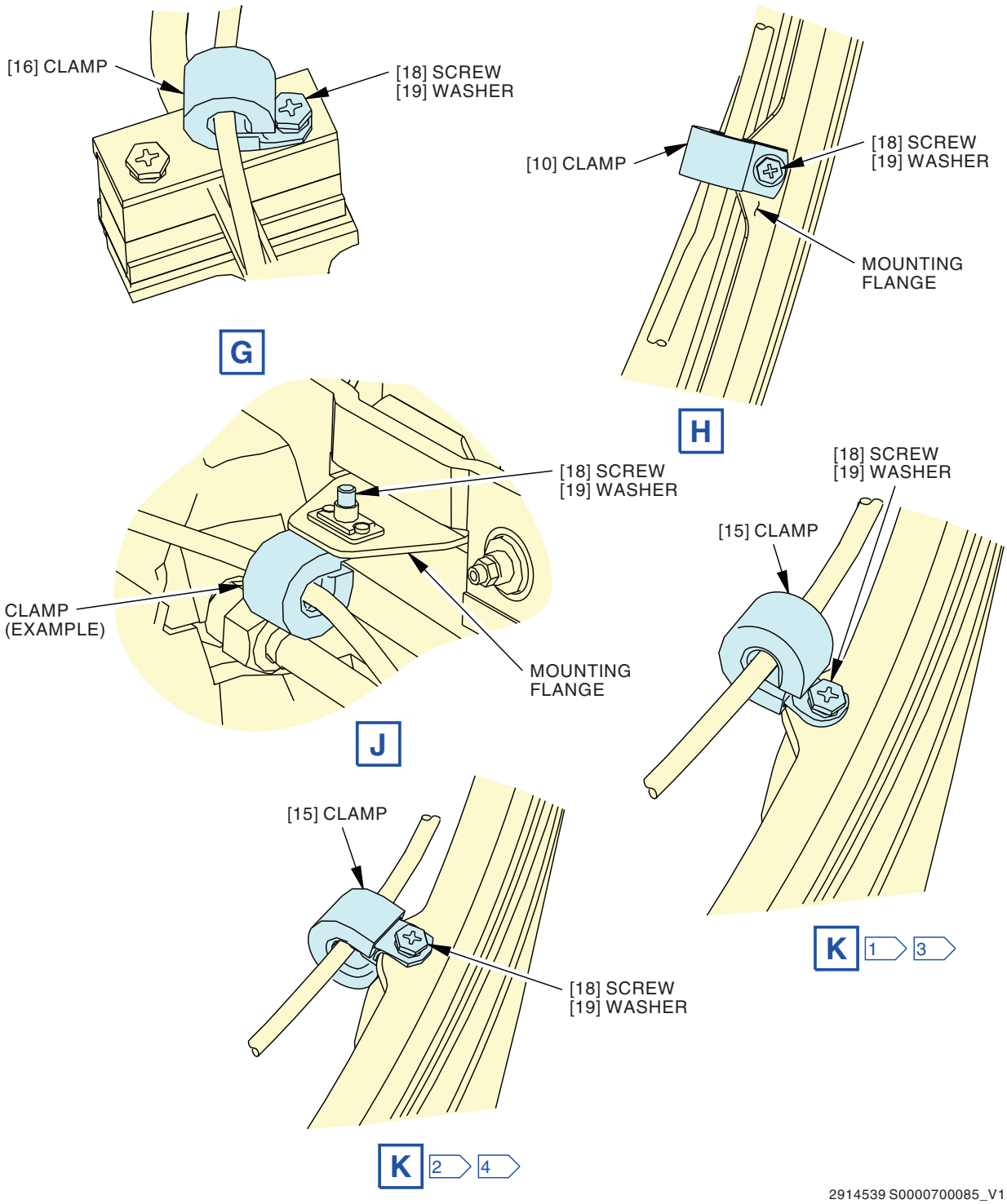
Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 7 of 8)

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Thrust Reverser Sleeve Wire Bundles and Proximity Sensor Installation
Figure 401/78-34-03-990-801-G00 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-03-400-801-G00

3. Thrust Reverser Sleeve Lock Proximity Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the thrust reverser sleeve lock proximity sensors and wire bundle for the left or right thrust reverser on an engine.

B. References

Reference	Title
51-21-72-370-804	BMS10-11 Primer - Application (P/B 701)
51-21-95-300-801	Alodine Treatment Application (P/B 701)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-805-G00	Thrust Reverser Proximity Sensors Operational Test (P/B 501)
78-34-03-800-801-G00	Sleeve Lock Proximity Sensor Adjustment and Test (P/B 501)
SWPM 20-10-11	WIRING ASSEMBLY AND INSTALLATION CONFIGURATION

C. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

D. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

E. Lock Sensor Installation

SUBTASK 78-34-03-860-003-G00

(1) Connect the electrical connectors to the applicable thrust reverser:

(a) For the left thrust reverser, connect these electrical connectors:

1) Connect the electrical connector, D30002, to the strut receptacle.

2) Connect the electrical connector, D30072, to the LVDT receptacle on the upper actuator.

3) Connect the electrical connector, D01008, to the sync lock on the lower actuator.

(b) For the right thrust reverser, connect these electrical connectors:

1) Connect the electrical connector, D30010, to the strut receptacle.

2) Connect the electrical connector, D30078, to the LVDT receptacle on the upper actuator.

3) Connect the electrical connector, D01016, to the sync lock on the lower actuator.

SUBTASK 78-34-03-420-001-G00

(2) Install the lock sensors [1] and wire leads:

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(a) Clean the bracket, key washer [3], and jamnut [2].

(b) Put the upper actuator lock sensor [1] into the bracket.

(c) Install the key washer [3] on the side of the bracket closest to the sensor target.

1) Align the key washer [3] with the index hole in the bracket.

(d) Install the other jamnut [2].

(e) Put the lock sensor [1] against the sensor target.

(f) Hand tighten the jamnut [2] at this time.

NOTE: This is a temporary installation only, the jamnuts will be tightened after the final adjustment of the lock sensor.

(g) Repeat the above steps with the middle actuator lock sensor.

SUBTASK 78-34-03-420-002-G00

(3) Attach the left hand wire bundles to the thrust reverser torque box with the clamps and screws (Figure 401).

(a) Make sure that you maintain a 0.5 in. (12.7 mm) separation between the wire bundles.

(b) Make sure that the wire bundles are installed with appropriate slack (SWPM 20-10-11).

(c) For left hand wire bundles P/N 286A1080-305 and P/N 286A1082-303 install the clamps as follow:

1) Install the clamp [4] for P/N 286A1082-303 with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

2) Install the clamp [5] for P/N 286A1080-305 with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

3) Install the clamp [6] through clamp [17], and clamp [27] for the P/N 286A1080-305 with the screws [18] and washers [19] to attach the wire bundle (View A, Figure 401).

(d) For left hand wire bundles P/N 286A1080-304 and P/N 286A1082-302 install the clamps as follow:

1) Install clamp [4] for P/N 286A1082-302 and clamp [5] for P/N 286A1080-304 in a butterfly configuration with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

2) For the aft hole, do these steps:

NOTE: The aft hole on mounting flange will remain open.

a) Do this task: Alodine Treatment Application, TASK 51-21-95-300-801.

b) Do this task: BMS10-11 Primer - Application, TASK 51-21-72-370-804.

3) For the P/N 286A1080-304, do these steps to install the clamp [15] (View K, Figure 401):

a) Rotate the loop of the clamp [15] on 180 degrees.

NOTE: The clamp loop will be AFT facing.

b) Install the clamp [15] on the mounting flange with the screw [18] and washer [19] to attach the wire bundle.

4) Install the clamp [6] through clamp [14], and clamp [16], clamp [17], clamp [27] for the 286A1080-304 with the screws [18] and washers [19] to attach the wire bundle (View A, Figure 401).

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SUBTASK 78-34-03-420-003-G00

(4) Attach the right hand wire bundles to the thrust reverser torque box with the clamps and screws (Figure 401).

(a) Make sure that you maintain a 0.5 in. (12.7 mm) separation between the wire bundles.

(b) Make sure that the wire bundles are installed with the appropriate slack (SWPM 20-10-11).

(c) For right hand wire bundles P/N 286A1084-304 and P/N 286A1086-304 install the clamps as follow:

1) Install the clamp [4] for P/N 286A1086-304 with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

2) Install the clamp [5] for P/N 286A1084-304 with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

3) Install the clamp [6] for the P/N 286A1084-304 with the screw [18], washer [19], and spacer [21] to attach the wire bundle (View I, Figure 401).

4) Install the clamp [7] through clamp [17], and clamp [20] for the P/N 286A1084-304 with the screws [18] and washers [19] to attach the wire bundle.

5) Install the clamp [23] through clamp [26] for the P/N 286A1086-304 with the screws [18] and washers [19] to attach the wire bundle.

(d) For right hand wire bundles P/N 286A1084-303 and P/N 286A1086-303 install the clamps as follow:

1) Install the clamp [4] for P/N286A1086-303 and clamp [5] for P/N 286A1084-303 in a butterfly configuration with the screw [18] and washer [19] to attach the wire bundle (View F, Figure 401).

2) For the aft hole, do these steps:

NOTE: The aft hole on mounting flange will remain open.

a) Do this task: Alodine Treatment Application, TASK 51-21-95-300-801.

b) Do this task: BMS10-11 Primer - Application, TASK 51-21-72-370-804.

3) For the P/N 286A1084-303, do these steps to install the clamp [15] (View K, Figure 401):

a) Rotate the loop of the clamp [15] on 180 degrees.

NOTE: The clamp loop will be AFT facing.

b) Install the clamp [15] on the mounting flange with the screw [18] and washer [19] to attach the wire bundle.

4) Install the clamp [6] for the P/N 286A1084-303 with the screw [18], washer [19] and spacer [21] to attach the wire bundle (View I, Figure 401).

5) Install the clamp [7] through clamp [14], and clamp [16], clamp [17], clamp [20] for the P/N 286A1084-303 with the screws [18] and washers [19] to attach the wire bundle.

6) Install the clamp [23], clamp [24], clamp [25], and clamp [26] for the P/N 286A1086-303 with the screws [18] and washers [19] to attach the wire bundle.

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SUBTASK 78-34-03-860-004-G00

(5) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-03-860-005-G00

(6) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK

SUBTASK 78-34-03-820-001-G00

(7) Do this task: Sleeve Lock Proximity Sensor Adjustment and Test, TASK 78-34-03-800-801-G00.

SUBTASK 78-34-03-710-001-G00

(8) Do this task: Thrust Reverser Proximity Sensors Operational Test, TASK 78-31-00-700-805-G00.

NOTE: This check will make sure that the electrical connections for the LVDT's are correct.

(a) Make sure that no LVDT maintenance messages show.

1) If a maintenance message shows, do the applicable fault isolation task in the Fault Isolation Manual for that maintenance message.

2) If no maintenance messages show, the electrical connections for the LVDT's are correct.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-03-860-008-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-34-03-860-009-G00

(2) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

——— END OF TASK ———

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THRUST REVERSER SLEEVE LOCK PROXIMITY SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Sleeve Lock Proximity Sensor Adjustment and Test.

TASK 78-34-03-800-801-G00

2. Sleeve Lock Proximity Sensor Adjustment and Test

(Figure 501 and Figure 502)

A. General

(1) This task gives the instructions to adjust the thrust reverser sleeve lock proximity sensor.

(2) The thrust reverser sleeve lock proximity sensor supplies a signal to the Electronic Engine Control (EEC) when the locking actuator is not in the locked position.

(3) To do the sensor adjustment, the manual unlock lever on the locking actuator must be in the unlocked position.

(4) The Thrust Reverser Normal Operation Test must be done after the adjustment(s).

(5) For this task, the thrust reverser sleeve lock proximity sensor will be referred to as the lock sensor.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-03-000-801-G00	Thrust Reverser Hydraulic Actuator Removal (Selection) (P/B 401)
78-31-03-400-801-G00	Thrust Reverser Hydraulic Actuator Installation (Selection) (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

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(Continued)

Reference	Description
SPL-15435	Gap Measurement Gauge - TRAS Locking Actuator Part #: C78030-18 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Adjustment

SUBTASK 78-34-03-040-002-G00

(1) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-34-03-010-004-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-34-03-860-006-G00

(3) Make sure that the thrust reverser sleeve is in the stowed and locked position.

SUBTASK 78-34-03-980-001-G00

(4) Do these steps to manually unlock the locking actuator:

(a) Move the manual unlock lever forward.

(b) Hold the manual unlock lever in the forward position as the detent pin is pushed in.

(c) Release the manual unlock lever.

H. Lock Sensor Adjustment

SUBTASK 78-34-03-820-002-G00

(1) Do the steps that follow to adjust the lock sensor [3]:

(a) Loosen the jamnuts [1].

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1) Make sure that the key washer [2] tab does not come out of the index hole in the attach fitting.

NOTE: The key washer is installed with the tab in the index hole in the bracket. The key washer is installed on the side of the bracket closest to the target.

(b) Adjust the jamnuts [1] to set the lock sensor [3] in the correct position.

1) Use a feeler gage to measure the distance between the end face of the lock sensor [3] and sensor target [4].

2) Make sure that the distance is 0.035 ±0.010 in. (0.889 ±0.254 mm).

(c) Tighten the jamnuts [1] to 45.0 in-lb (5.1 N·m) - 55.0 in-lb (6.2 N·m).

(d) Do a check of the distance again between the end face of the lock sensor [3] and sensor target [4].

1) Make sure that the distance is 0.035 ±0.010 in. (0.889 ±0.254 mm).

(e) Measure the electrical bonding resistance between the bracket and lock sensor [3] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the resistance is 0.003 ohms (3 milliohms) or less.

3) If the electrical bonding resistance is more than 0.003 ohms (3 milliohms), do the following steps:

a) Loosen the jamnuts [1].

b) Do the steps above to loosen the jamnuts [1].

c) Clean the mating surfaces.

d) Tighten the jamnuts [1].

e) Measure the electrical bonding resistance between the bracket and lock sensor [3] again.

(f) Install MS20995C32 lockwire, G01048, on the jamnuts [1] (View B, Figure 501).

(g) If it is necessary, install MS20995C32 lockwire, G01048, on the adapter and adjacent jamnut [1].

(h) Do these steps to lock the locking actuator:

1) Move the manual unlock lever forward.

NOTE: The detent pin will move out of the locked position.

2) Release the manual unlock lever.

I. Lock Sensor Target Check

SUBTASK 78-34-03-820-003-G00

(1) Do these steps to check the sensor target [4] distance from the proximity sensor:

(a) Install the gap gauge, SPL-15435, on the sensor to make sure that the target leading edge is at the correct distance.

1) If the larger cylinder will pass the target leading edge and the target is too far, replace the actuator.

NOTE: The actuator is no longer serviceable.

a) Do this task: Thrust Reverser Hydraulic Actuator Removal (Selection), TASK 78-31-03-000-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Do this task: Thrust Reverser Hydraulic Actuator Installation (Selection), TASK 78-31-03-400-801-G00.

2) If the smaller cylinder will not pass the target leading edge and the target is too close, replace the actuator.

NOTE: The actuator is no longer serviceable.

a) Do this task: Thrust Reverser Hydraulic Actuator Removal (Selection), TASK 78-31-03-000-801-G00.

b) Do this task: Thrust Reverser Hydraulic Actuator Installation (Selection), TASK 78-31-03-400-801-G00.

3) If the larger cylinder will not pass the target leading edge, the target is at the correct distance.

J. Lock Sensor Adjustment Test

SUBTASK 78-34-03-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-34-03-710-002-G00

(2) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

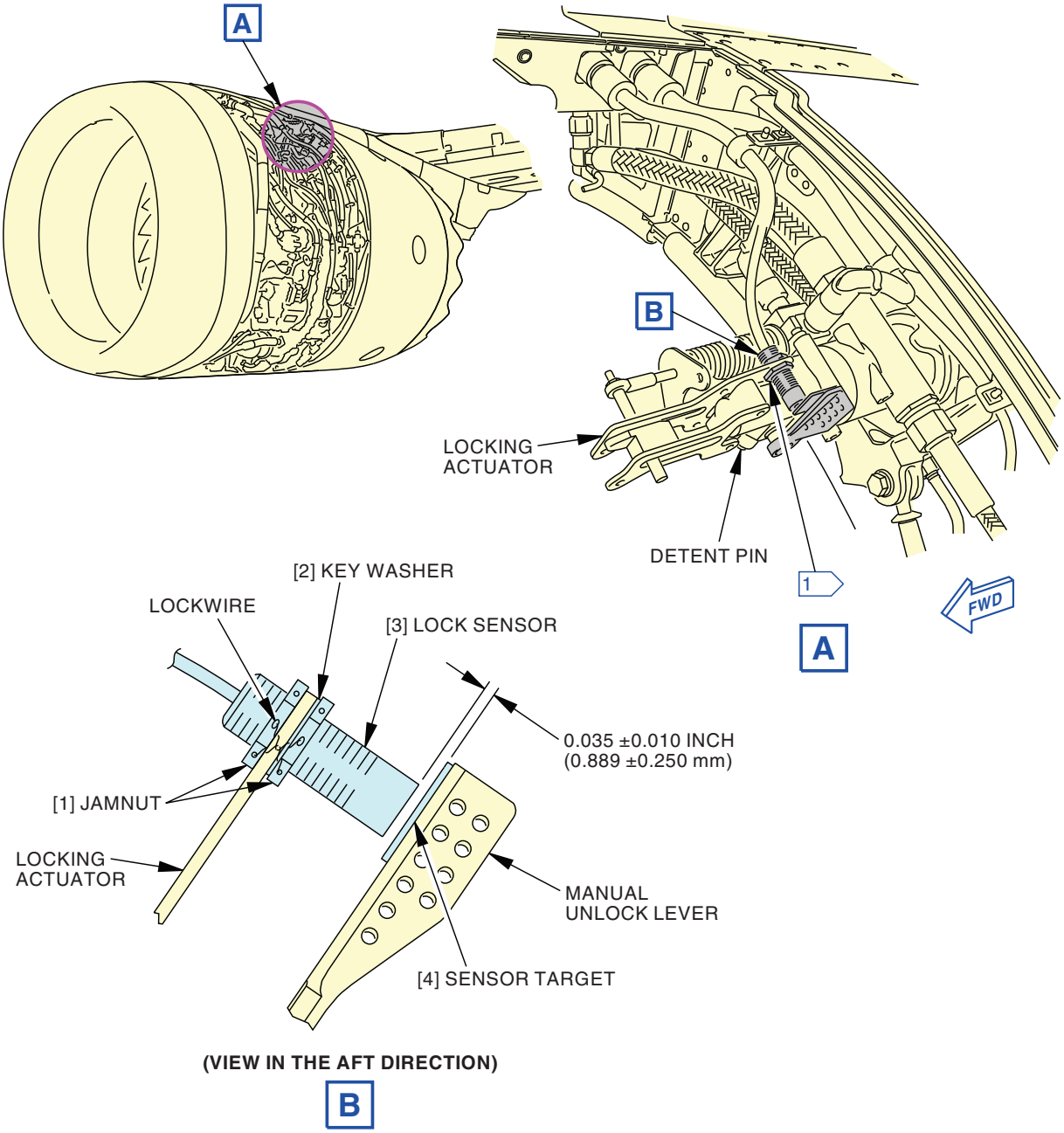
K. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-03-410-001-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

——— END OF TASK ———

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NOTE:

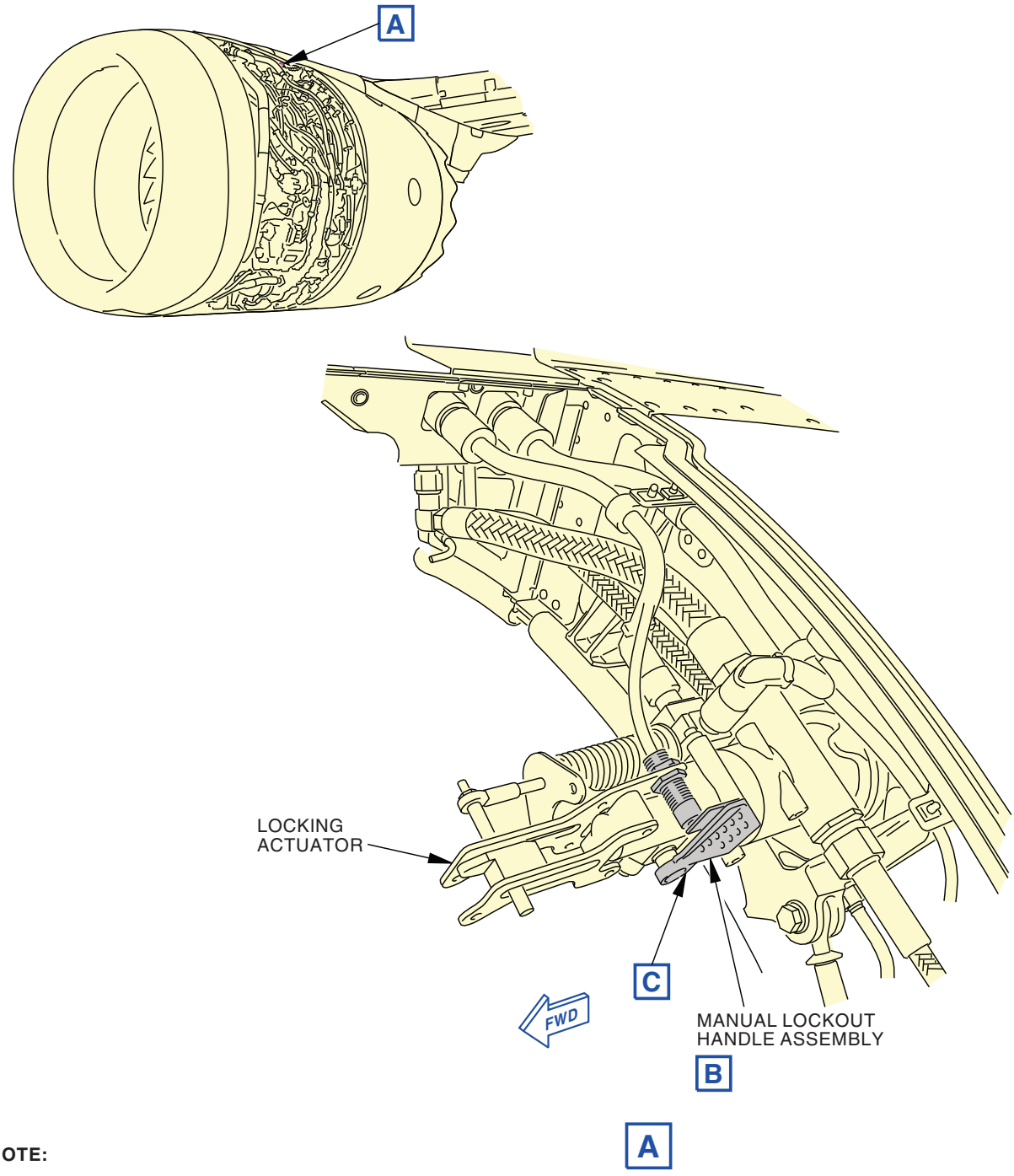
LEFT THRUST REVERSER UPPER LOCKING ACTUATOR IS SHOWN. ALL LOCKING ACTUATORS ARE EQUIVALENT.

1 ELECTRICALLY BOND THE STOW SENSOR TO THIS SIDE OF THE BRACKET. FILLET SEAL AROUND JAMNUT.

2355290 S00061516914_V3

Thrust Reverser Sleeve Lock Proximity Sensor Adjustment
Figure 501/78-34-03-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



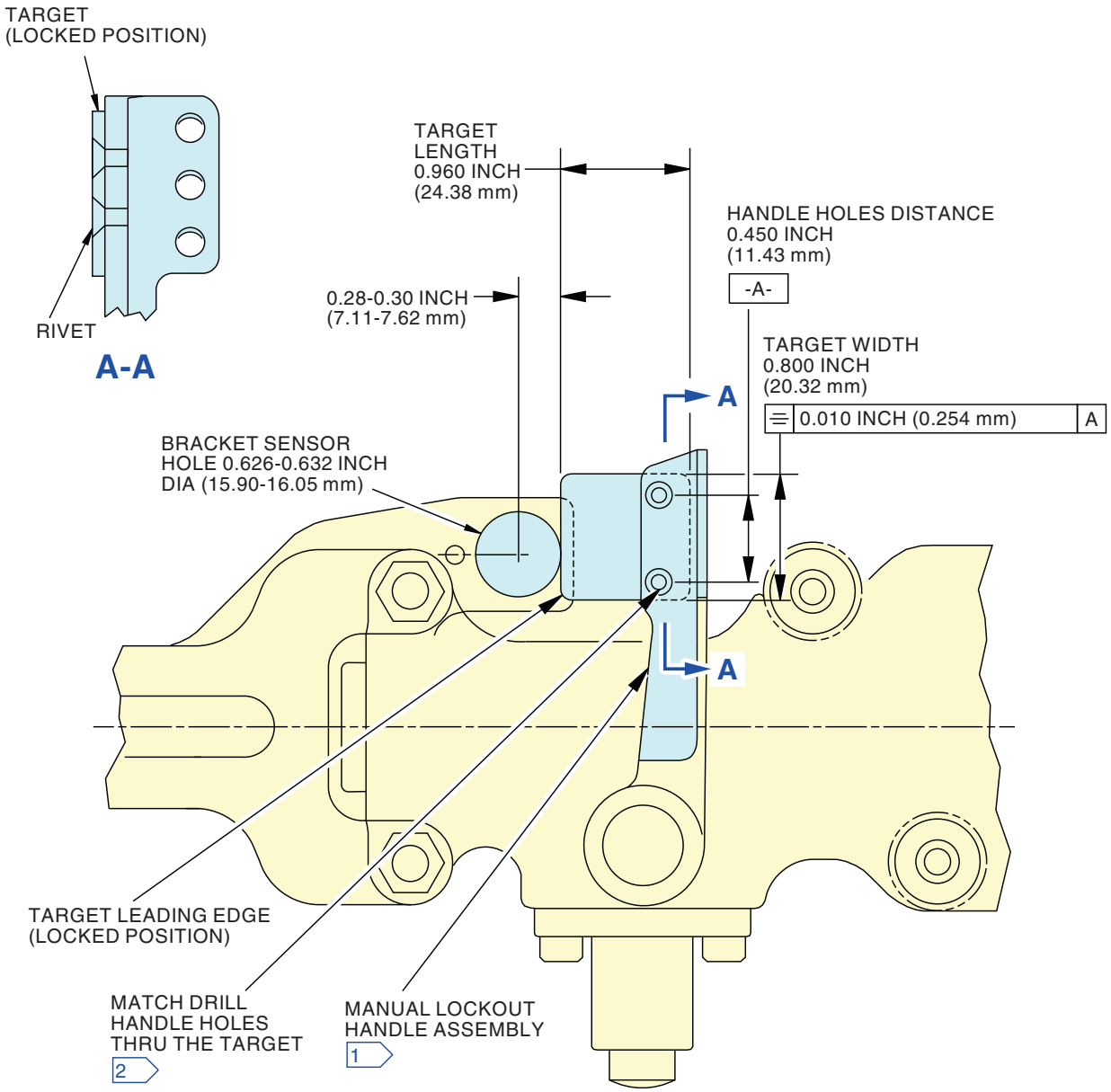
NOTE:

LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE.
(TOOL NOT SHOWN)

2355292 S00061516915_V2

Thrust Reverser Sleeve Lock Proximity Sensor Target Adjustment
Figure 502/78-34-03-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MANUAL LOCKOUT HANDLE ASSEMBLY

B

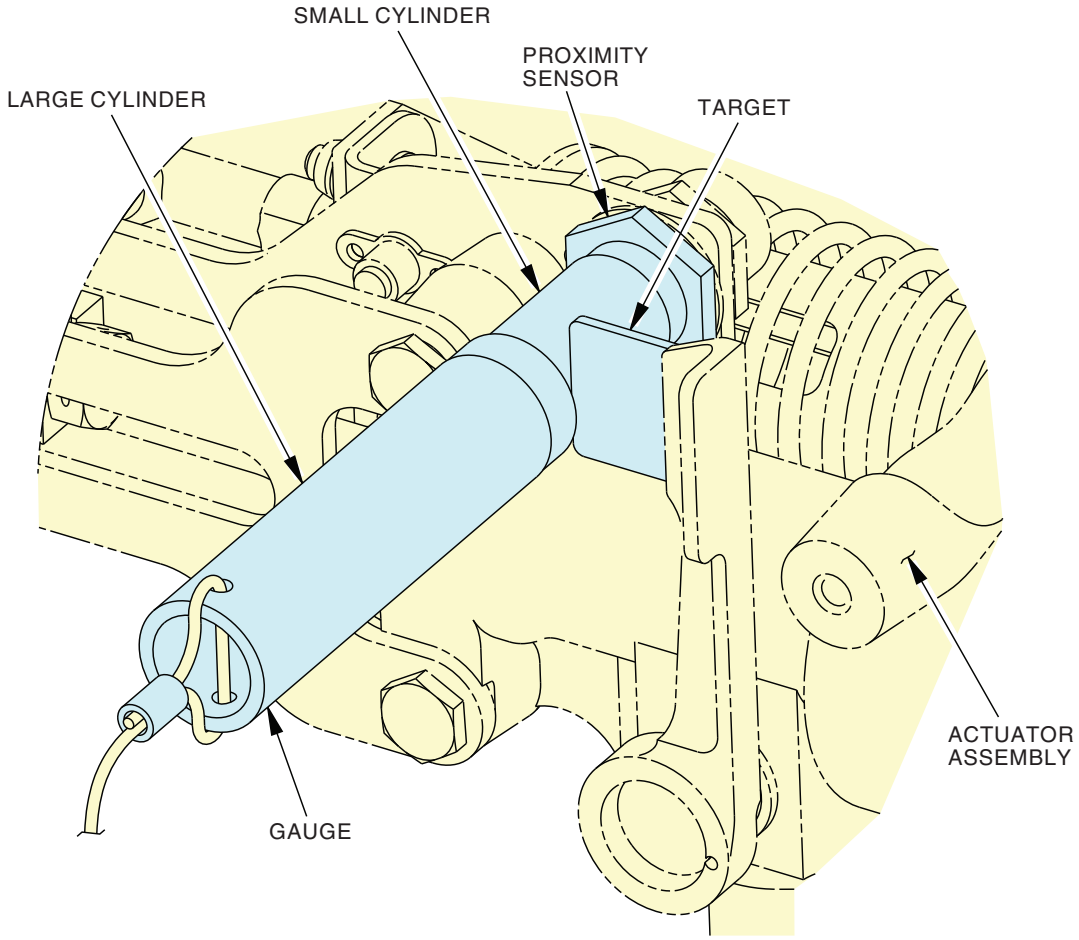
1 APPROXIMATELY PERPENDICULAR WITH ACTUATOR CENTERLINE WHEN PISTON IS IN FULLY LOCKED POSITION.

2 MARK TARGET HOLES AGAINST THE HANDLE. DRILL TARGET HOLES, INSTALL TARGET TO HANDLE ON BENCH.

2355293 S00061516916_V1

Thrust Reverser Sleeve Lock Proximity Sensor Target Adjustment
Figure 502/78-34-03-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

C

2355294 S00061516917_V2

Thrust Reverser Sleeve Lock Proximity Sensor Target Adjustment
Figure 502/78-34-03-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER CONTROL SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Control switch removal

(2) Control switch installation.

TASK 78-34-04-000-801-G00

2. Control Switch Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the thrust reverser control switch.

(2) The thrust reverser control switch is referred to as the control switch.

B. References

Reference	Title
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-805-G00	Thrust Reverser ,Operation - Extend (Power Procedure) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 78-34-04-860-052-G00

(1) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT

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(Continued)

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-34-04-860-053-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-34-04-860-054-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-34-04-860-001-G00

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-34-04-860-003-G00

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKERS FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 78-34-04-860-004-G00

(6) For the applicable engine, make sure that the START LEVER switch is in the CUTOFF position.

(a) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 78-34-04-860-005-G00

(7) Make sure that the applicable thrust lever is in the idle position.

(a) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 78-34-04-860-006-G00

(8) Make sure that the applicable reverse thrust lever is forward and down in the retract (stow) position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-04-860-007-G00

(9) For the applicable engine, move the ENGINE START switch on the forward overhead P5 panel to the CONT position.

NOTE: This supplies power to the EEC which is necessary for the interlock to release.

SUBTASK 78-34-04-860-008-G00

(10) If the thrust reverser will power extend (deploy), do these steps:

NOTE: For the adjustment of the control switch, you must have movement through the full range of the reverse thrust lever. When the interlock releases and the thrust reverser is extended, the reverse thrust lever can move through its entire range.

(a) Do this task: Thrust Reverser ,Operation - Extend (Power Procedure), TASK 78-31-00-980-805-G00.

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-34-04-980-001-G00

(11) If the thrust reverser will not power extend (deploy), do these steps:

NOTE: For the adjustment of the control switch, you must have movement through the full range of the reverse thrust lever. When the interlock releases and the thrust reverser is extended, the reverse thrust lever can move through its entire range.

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Manually extend the thrust reverser (TASK 78-31-00-980-803-G00).

F. Control Switch Removal

SUBTASK 78-34-04-020-001-G00

(1) Do these steps to remove the control switch [1] from the applicable thrust lever:

(a) To get access to the applicable thrust lever cover, move the other thrust lever to the fully forward position.

(b) Remove the four screws [4] that attach the cover to the inboard side of the thrust lever.

(c) Remove the cover from the thrust lever.

(d) Remove the two screws [3] that attach the control switch [1] to the thrust lever.

(e) Remove the two switch plates [2] and the control switch [1].

NOTE: Some configurations of the control switch do not use switch plates.

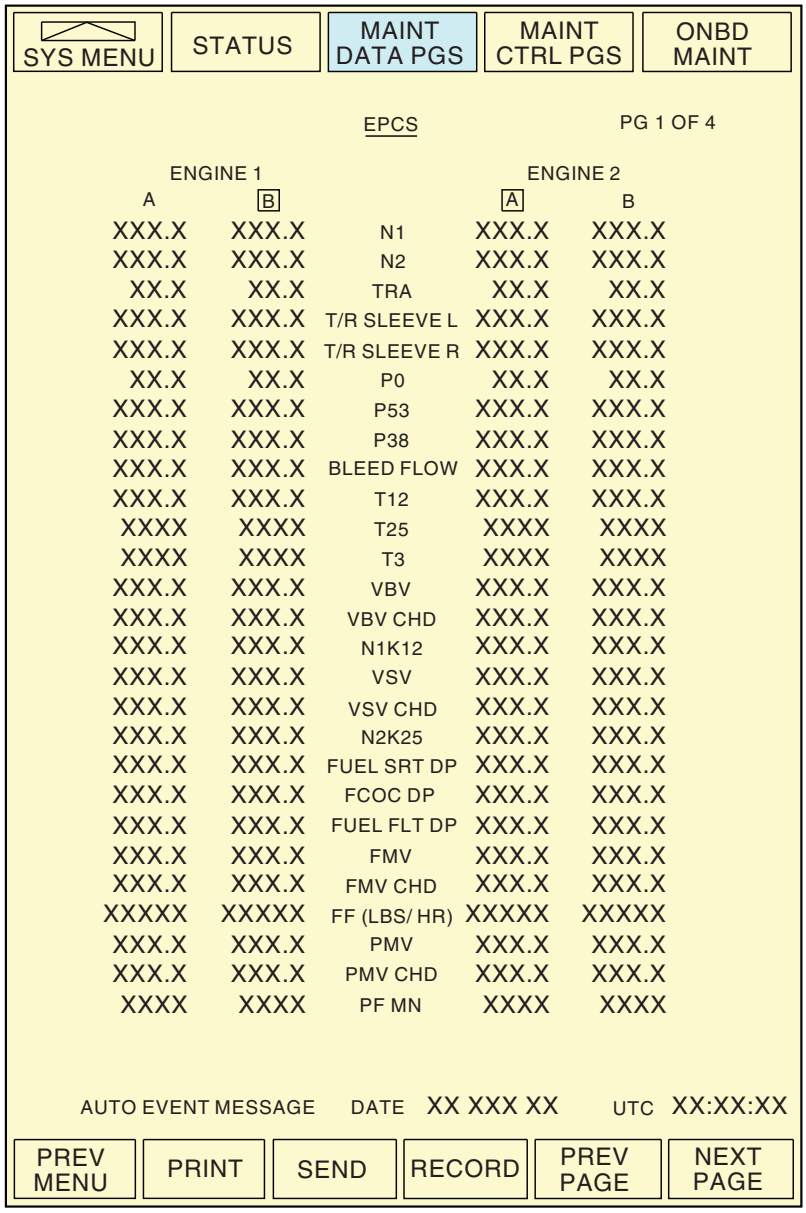
1) Keep the two switch plates [2] for the subsequent installation of the new control switch [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Do the applicable tasks to disconnect the three wire socket contacts from the control switch [1] (SWPM 20).

——— END OF TASK ———

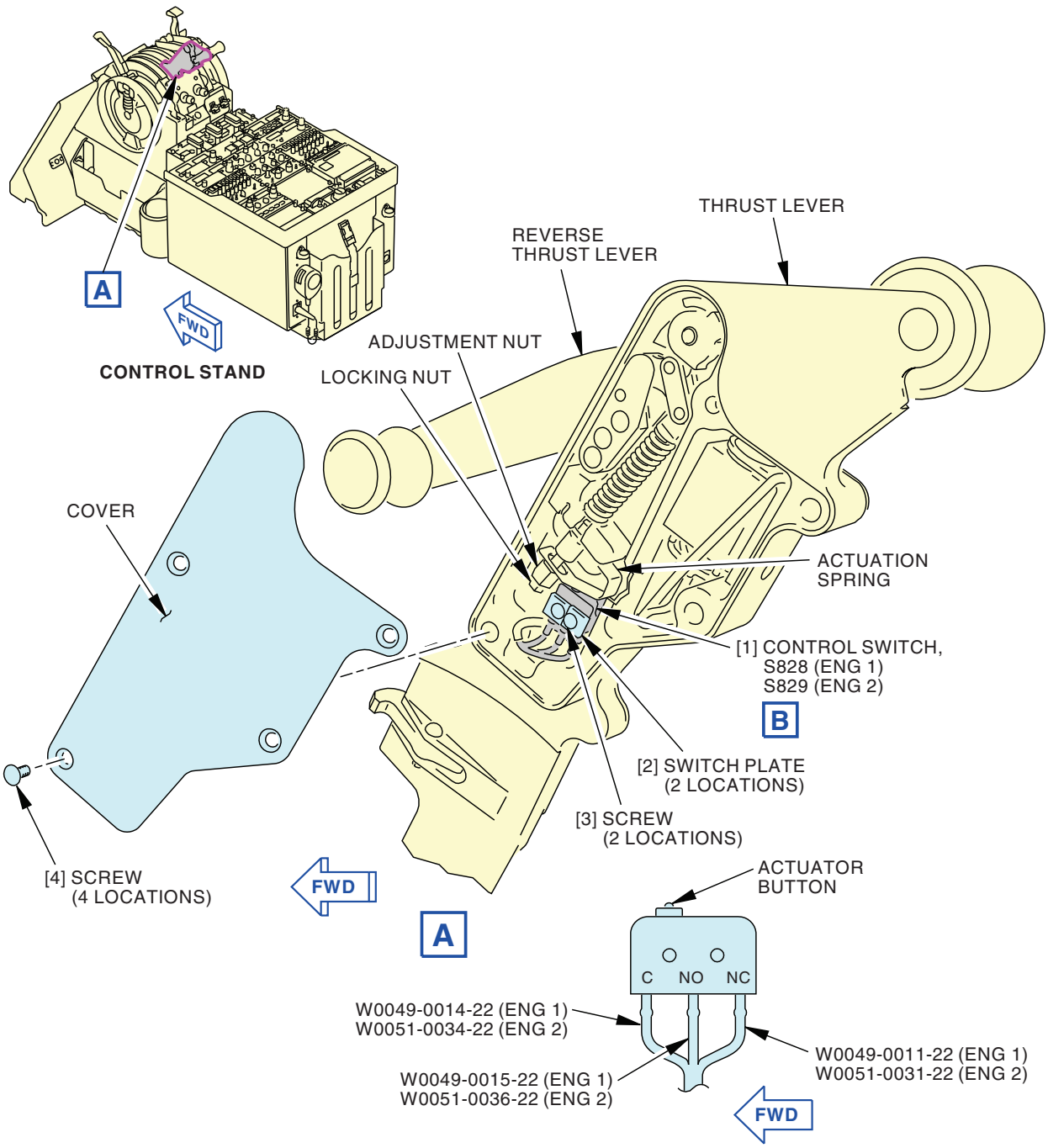
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2354207 S00061514972_V2

Thrust Reverser Control Switch Installation
Figure 401/78-34-04-990-801-G00 (Sheet 1 of 2)

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NOTE:

ENGINE 2 THRUST LEVER IS SHOWN.
ENGINE 1 THRUST LEVER IS OPPOSITE.

2355296 S00061516922_V3

Thrust Reverser Control Switch Installation
Figure 401/78-34-04-990-801-G00 (Sheet 2 of 2)

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TASK 78-34-04-400-801-G00

3. Control Switch Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the thrust reverser control switch.

(2) The thrust reverser control switch is referred to as the control switch.

(3) The TRA (thrust lever resolver angle) values on the Multi-Function Display (MFD) will be used to adjust the control switch.

B. References

Reference	Title
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-806-G00	Thrust Reverser Operation - Retract (Power Procedure) (P/B 201)
78-34-04-700-801-G00	Control Switch Adjustment and Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
STD-442	Gun - Heat, 180 deg. F (82 deg. C) Maximum Output Temperature
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G01148	Sleeve - Insulation, Electrical, Heat Shrinkable - RT-876	
G51210	Tubing - Highly Flame-Retardant, Low-Shrink Temperature, Polyolefin	AMS-DTL-23053/5 Class 1

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Control switch	78-34-04-02-040	PTF ALL
		78-34-04-02A-025	PTF ALL

F. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

G. Control Switch Installation

SUBTASK 78-34-04-420-001-G00

(1) Do these steps to install the control switch [1]:

(a) Put the control switch [1] in its correct position.

1) Make sure that the actuator button is in the correct position.

(b) Install the switch plates [2] on the control switch [1].

NOTE: Some configurations of the control switch do not use switch plates.

(c) Install the screws [3] that attach the switch plates [2] and control switch [1] to the thrust lever.

NOTE: The wire socket contacts are connected after the adjustment of the control switch.

(d) Operate the reverse thrust lever to make sure that the actuating spring does not lose contact or pop off the control switch [1] at the top of the throw.

1) If it is necessary, adjust the locking nut as follows:

a) Loosen the locking nut at the adjustment nut until the actuating spring stays actuated on the control switch [1].

b) Tighten the locking nut to 6 in-lb (0.7 N·m) - 8 in-lb (0.9 N·m).

H. Control Switch Test

SUBTASK 78-34-04-860-009-G00

(1) Make sure that the reverse thrust lever is fully forward and down in the retracted (stowed) position.

SUBTASK 78-34-04-720-001-G00

(2) To do a test of the control switch [1] use the Multi-Function Display (MFD) screen in the flight compartment to show the TRA (thrust lever resolver angle) position values:

NOTE: The relationship between RLA (actual reverse thrust lever angle as measured by a protractor) and TRA is also given in the task.

(a) Select the EPCS page 1 from the Maintenance Data Pages on the Multi-Function Display (MFD).

(b) Use a digital/analog multimeter, COM-1793, to do this continuity check:

1) Make sure that there is an open circuit between terminal C and terminal NO at 36 ±0.25 degrees TRA position value.

NOTE: This TRA position value is the same as 0 degrees RLA.

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(c) As you slowly move the reverse thrust lever toward the extend (deploy) position, do these continuity checks:

NOTE: There is a two second delay between the movement of the reverse thrust lever and the value that is displayed on the MFD.

1) Make sure that as the reverse thrust lever is moved, there is continuity between terminal C and terminal NC between a TRA position value of 36 ±0.25 degrees and 35.50 degrees.

NOTE: This range of TRA position values is the same as 0 to 16 degrees RLA.

2) Stop the movement of the reverse thrust lever when terminal NC to terminal C goes open.

a) Wait for at least 2 seconds.

b) Make sure that the TRA position value is between 35.50 degrees and 35.10 degrees.

NOTE: This range of TRA position values is the same as 16 to 19 degrees RLA.

3) Do a check between terminal C and terminal NO of the control switch:

a) Continue to slowly move the reverse thrust lever until the reverse thrust lever is at the extend (deploy) position.

b) Make sure that the continuity between terminal C and terminal NO is kept during the movement.

4) If the continuity check is not in the limits, do the steps that follow to adjust the control switch.

5) If the continuity check is in the limits, the control switch is serviceable.

I. Control Switch Adjustment

SUBTASK 78-34-04-820-001-G00

(1) Do the adjustment on the control switch [1] (TASK 78-34-04-700-801-G00).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-04-860-011-G00

(1) For the Engine 1 thrust reverser, do these steps to connect the wire socket contacts to the control switch:

(a) Slide a piece of heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, that is approximately 0.60 in. (15.24 mm) over each of the three wire socket contacts.

(b) Connect the W0049-0014-22 socket contact to the C contact.

(c) Connect the W0049-0011-22 socket contact to the NC contact.

(d) Connect the W0049-0015-22 socket contact to the NO contact.

(e) Put the heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, in the correct position over the socket contact and wire.

NOTE: The sleeve must cover the entire socket contact and approximately 0.25 in. (6.35 mm) of the wire.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442, to make the heat shrink sleeve become tight on the socket contact.

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SUBTASK 78-34-04-860-012-G00

(2) For the Engine 2 thrust reverser, do these steps to connect the wire socket contacts to the control switch:

(a) Slide a piece of heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, that is approximately 0.60 in. (15.24 mm) over each of the three wire socket contacts.

(b) Connect the W0051-0034-22 socket contact to the C contact.

(c) Connect the W0051-0031-22 socket contact to the NC contact.

(d) Connect the W0051-0036-22 socket contact to the NO contact.

(e) Put the heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, in the correct position over the socket contact and wire.

NOTE: The sleeve must cover the entire socket contact and approximately 0.25 in. (6.35 mm) of the wire.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442, to make the heat shrink sleeve become tight on the socket contact.

SUBTASK 78-34-04-860-013-G00

(3) Install the cover on the thrust lever.

(a) Install the screws [4] that attach the cover.

SUBTASK 78-34-04-860-014-G00

(4) Make sure that the thrust levers are in the idle position.

SUBTASK 78-34-04-860-015-G00

WARNING

MAKE SURE THAT THERE ARE NO PERSONS OR EQUIPMENT IN THE AREA OF THE THRUST REVERSER. MOVEMENT OF THE THRUST REVERSER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Do these steps to retract (stow) the thrust reverser:

(a) Move the reverse thrust lever up and aft to the extended (deployed) position.

NOTE: The reverse thrust lever must be in the extended (deployed) position to unlock the sync locks. The sync locks will unlock and the thrust reverser will move as soon as the thrust reverser is activated.

(b) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

(c) Do this task: Thrust Reverser Operation - Retract (Power Procedure), TASK 78-31-00-980-806-G00.

SUBTASK 78-34-04-710-001-G00

(6) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

SUBTASK 78-34-04-860-016-G00

(7) If it is not already done, move the ENGINE START switch to the OFF position.

SUBTASK 78-34-04-860-017-G00

(8) If it is not already done, remove the DO NOT OPERATE tag, STD-858, from the thrust lever.

SUBTASK 78-34-04-860-018-G00

(9) If it is not already done, remove the DO NOT OPERATE tag, STD-858, from the START LEVER switch.

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SUBTASK 78-34-04-860-019-G00

(10) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 78-34-04-860-039-G00

(11) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-34-04-860-040-G00

(12) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-34-04-860-041-G00

(13) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-34-04-860-042-G00

(14) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

——— END OF TASK ———

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THRUST REVERSER CONTROL SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Control switch adjustment and test.

TASK 78-34-04-700-801-G00

2. Control Switch Adjustment and Test

(Figure 501)

A. General

(1) This task gives the instructions for the check and adjustment of the thrust reverser control switch.

(2) The thrust reverser control switch is referred to as the control switch.

B. References

Reference	Title
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-980-803-G00	Thrust Reverser Operation - Extend (Manual Procedure) (P/B 201)
78-31-00-980-805-G00	Thrust Reverser ,Operation - Extend (Power Procedure) (P/B 201)
78-31-00-980-806-G00	Thrust Reverser Operation - Retract (Power Procedure) (P/B 201)
78-34-04-000-801-G00	Control Switch Removal (P/B 401)
78-34-04-400-801-G00	Control Switch Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-706	Protractor - Thrust Reverser Levers, Digital Readout Part #: G76002-42 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205
STD-442	Gun - Heat, 180 deg. F (82 deg. C) Maximum Output Temperature
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G01148	Sleeve - Insulation, Electrical, Heat Shrinkable - RT-876	
G51210	Tubing - Highly Flame-Retardant, Low-Shrink Temperature, Polyolefin	AMS-DTL-23053/5 Class 1

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Prepare for the Procedure

SUBTASK 78-34-04-860-049-G00

(1) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

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F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-34-04-860-050-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-34-04-860-051-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-34-04-860-022-G00

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-34-04-860-024-G00

WARNING

MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKERS FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 78-34-04-860-025-G00

(6) Make sure that the start lever for the applicable engine is in the CUTOFF position.

SUBTASK 78-34-04-860-026-G00

(7) Make sure that the applicable thrust lever is in the idle position.

SUBTASK 78-34-04-480-002-G00

(8) Attach a DO NOT OPERATE tag, STD-858 to the start lever and the thrust lever.

SUBTASK 78-34-04-860-027-G00

(9) Make sure that the applicable reverse thrust lever is forward and down in the retract (stow) position.

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SUBTASK 78-34-04-860-029-G00

(10) If the thrust reverser will power extend (deploy), do these steps:

NOTE: For the adjustment of the control switch, you must have movement through the full range of the reverse thrust lever. When the interlock releases and the thrust reverser is extended, the reverse thrust lever can move through its entire range.

(a) Do this task: Thrust Reverser ,Operation - Extend (Power Procedure), TASK 78-31-00-980-805-G00.

(b) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-34-04-980-002-G00

(11) If the thrust reverser will not power extend (deploy), do these steps:

NOTE: For the adjustment of the control switch, you must have movement through the full range of the reverse thrust lever. When the interlock releases and the thrust reverser is extended, the reverse thrust lever can move through its entire range.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Manually extend the thrust reverser (TASK 78-31-00-980-803-G00).

SUBTASK 78-34-04-860-055-G00

(12) Set the applicable EEC MAINT POWER to NORM on the MISC SYSTEM CTRLS page (TASK 73-21-00-800-801-G00)

SUBTASK 78-34-04-720-002-G00

(13) Do these steps to get access to the control switch [1] in the applicable thrust lever:

NOTE: Consider the option of performing the continuity check for engine 1 at connector D10171P and for engine 2 at connector D10173. The connectors are in the aft panel of the control stand. The connectors are accessible through the fwd access hatch (Figure 501 (Sheet 2)).

(a) To get access to the applicable thrust lever cover, move the other thrust lever to the fully forward position.

(b) Remove the screws [4] that attach the cover to the inboard side of the thrust lever.

(c) Remove the cover from the thrust lever.

(d) Disconnect the three wire socket contacts from the control switch [1].

SUBTASK 78-34-04-480-001-G00

(14) Install the thrust reverser lever protractor, SPL-706, on the reverse thrust lever.

(a) Adjust the protractor to show zero degrees with the reverse thrust lever fully forward and down, in the retracted (stowed) position.

G. Control Switch Test

SUBTASK 78-34-04-860-030-G00

(1) Make sure that the reverse thrust lever is fully forward and down in the retracted (stowed) position.

SUBTASK 78-34-04-720-003-G00

(2) Use a digital/analog multimeter, COM-1793, to do these continuity checks:

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(a) Make sure that there is an open circuit between the C and NO terminals and a closed circuit between C and NC terminals of the control switch.

NOTE: If you will do the continuity check at the connector D10171P or connector D10173P, Pin 1 connects to the NC contact, Pin 4 connects to the C contact, Pin 5 connects to the NO contact. All three contacts are on the control switch [1]. Make sure that there is solid connection for the three wires to the control switch [1].

1) If you do not find the open circuit, then do the steps below to adjust the control switch.

2) If you find the open circuit, then continue.

(b) As you slowly move the reverse thrust lever toward the extend (deploy) position, make sure that the circuit between C and NO closes with a Reverse Thrust Lever Angle (RTLA) of 16 to 19 degrees.

1) If the circuit does not close in the limits, then do the steps below to adjust the control switch.

2) If the circuit closes in the limits, then continue.

(c) Make sure that the reverse thrust lever is fully forward and down in the retracted (stowed) position.

(d) Move the reverse thrust lever to 19 degrees RTLA.

1) Make sure that there is an closed circuit between the C and NO terminals and a open circuit between C and NC terminals of the control switch.

(e) Slowly move the reverse thrust lever from 19 degrees to the fully extended (109 ± 2 degrees RTLA) position, make sure that the circuit between C and NO stays closed (continuity) through the full range of movement.

(f) Slowly move the reverse thrust lever from 109 ± 2 degrees to 46 to 50 degrees RTLA, make sure that the circuit between C and NO stays closed (continuity) through the full range of movement.

1) If the continuity is not in the limits, then do the steps below to adjust the control switch.

2) If the continuity is in the limits, then continue.

(g) Move the reverse thrust lever to 19 degrees RTLA.

(h) Slowly move the reverse thrust lever from 19 degrees to the fully extended (109 ± 2 degrees RTLA) position, make sure that the circuit between C and NC stays open the full range of movement.

(i) Slowly move the reverse thrust lever from 109 ± 2 degrees to 46 to 50 degrees RTLA, make sure that the circuit between C and NC stays open (continuity) through the full range of movement.

1) If the continuity is not in the limits, then do the steps below to adjust the control switch.

2) If the continuity is in the limits, then do the step below to put the airplane in its usual condition.

H. Adjust the Control Switch

SUBTASK 78-34-04-820-002-G00

(1) Do these steps to adjust the control switch [1]:

(a) Make sure that the reverse thrust lever is in the stow position (fully forward and down).

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(b) Loosen the lock nut at the adjustment nut.

(c) Do the Test above again and turn the adjustment nut until the continuity checks are in the limits.

1) If you cannot adjust the control switch to operate in the limits, replace the control switch (TASK 78-34-04-000-801-G00 and TASK 78-34-04-400-801-G00).

(d) Tighten the lock nut to 7.0 ±1.0 in-lb (0.79 ±0.11 N·m).

(e) Do the Test again to make sure that the adjustment did not change.

1) If the continuity check is in the limits, the control switch is serviceable.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-04-080-001-G00

(1) Remove the thrust reverser lever protractor, SPL-706, from the reverse thrust lever.

(a) Disconnect the digital/analog multimeter, COM-1793, from the control switch.

SUBTASK 78-34-04-860-031-G00

(2) For the Engine 1 thrust reverser, do these steps to re-connect the wire socket contacts to the control switch:

(a) Slide a piece of heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, that is approximately 0.60 in. (15.24 mm) over each of the three wire socket contacts.

(b) Connect the W0049-0014-22 socket contact to the C contact.

(c) Connect the W0049-0011-22 socket contact to the NC contact.

(d) Connect the W0049-0015-22 socket contact to the NO contact.

(e) Put the heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, in the correct position over the socket contact and wire.

NOTE: The sleeve must cover the entire socket contact and approximately 0.25 in. (6.35 mm) of the wire.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442, to make the heat shrink sleeve become tight on the socket contact.

SUBTASK 78-34-04-860-032-G00

(3) For the Engine 2 thrust reverser, do these steps to re-connect the wire socket contacts to the control switch:

(a) Slide a piece of heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, that is approximately 0.60 in. (15.24 mm) over each of the three wire socket contacts.

(b) Connect the W0051-0034-22 socket contact to the C contact.

(c) Connect the W0051-0031-22 socket contact to the NC contact.

(d) Connect the W0051-0036-22 socket contact to the NO contact.

(e) Put the heat shrinkable RT-876 sleeve, G01148, or Versafit heat shrink tubing, G51210, in the correct position over the socket contact and wire.

NOTE: The sleeve must cover the entire socket contact and approximately 0.25 in. (6.35 mm) of the wire.

1) Use a 180 deg. F (82 deg. C) maximum output temperature heat gun, STD-442, to make the heat shrink sleeve become tight on the socket contact.

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SUBTASK 78-34-04-860-033-G00

(4) Install the cover on the thrust lever.

(a) Install the screws [4] that attach the cover.

SUBTASK 78-34-04-860-034-G00

(5) Make sure that the thrust levers are in the idle position.

SUBTASK 78-34-04-860-045-G00

(6) Remove the DO NOT OPERATE tag, STD-858, from the thrust levers and the start lever.

SUBTASK 78-34-04-860-035-G00

(7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 78-34-04-860-036-G00

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	7	C00210	FLIGHT CONTROL FLAP SHUTOFF VALVES
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-34-04-860-046-G00

(9) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-34-04-860-047-G00

(10) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-34-04-860-048-G00

(11) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-34-04-860-038-G00

WARNING MAKE SURE THAT THERE ARE NO PERSONS OR EQUIPMENT IN THE AREA OF THE THRUST REVERSER. MOVEMENT OF THE THRUST REVERSER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Do these steps to retract (stow) the thrust reverser:

(a) Move the reverse thrust lever up and aft to the extended (deployed) position.

NOTE: The reverse thrust lever must be in the extended (deployed) position to unlock the sync locks. The sync locks will unlock and the thrust reverser will move as soon as the thrust reverser is activated.

(b) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

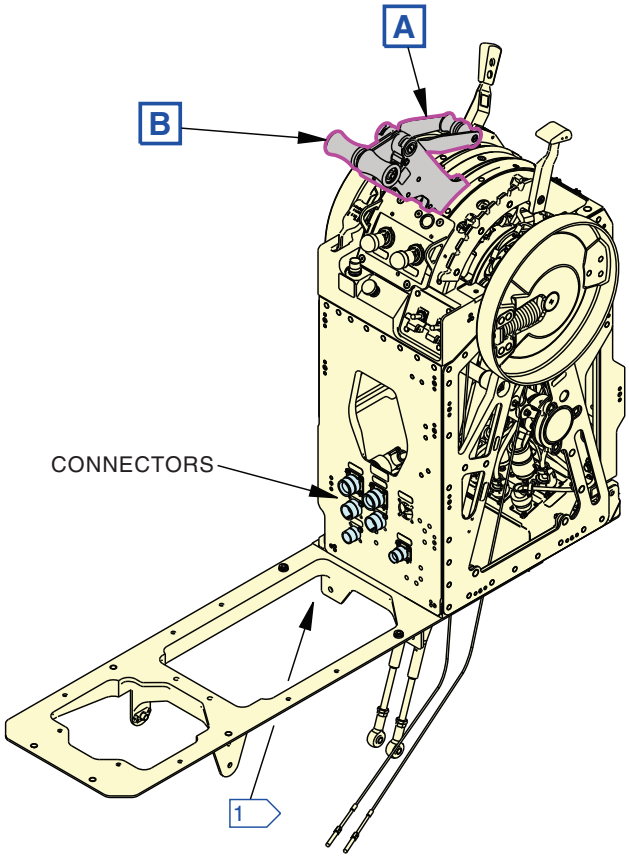
(c) Do this task: Thrust Reverser Operation - Retract (Power Procedure), TASK 78-31-00-980-806-G00.

SUBTASK 78-34-04-710-002-G00

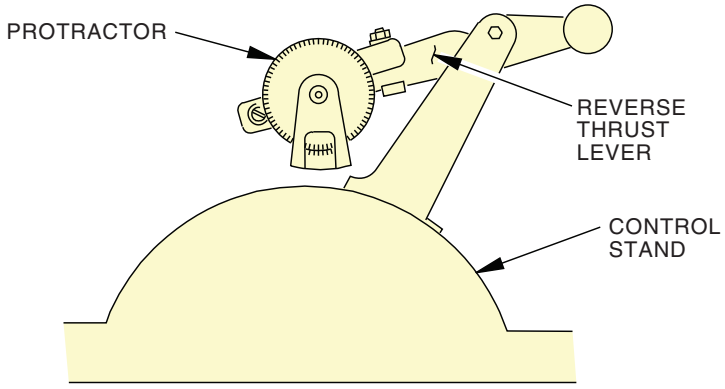
(13) Do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL STAND



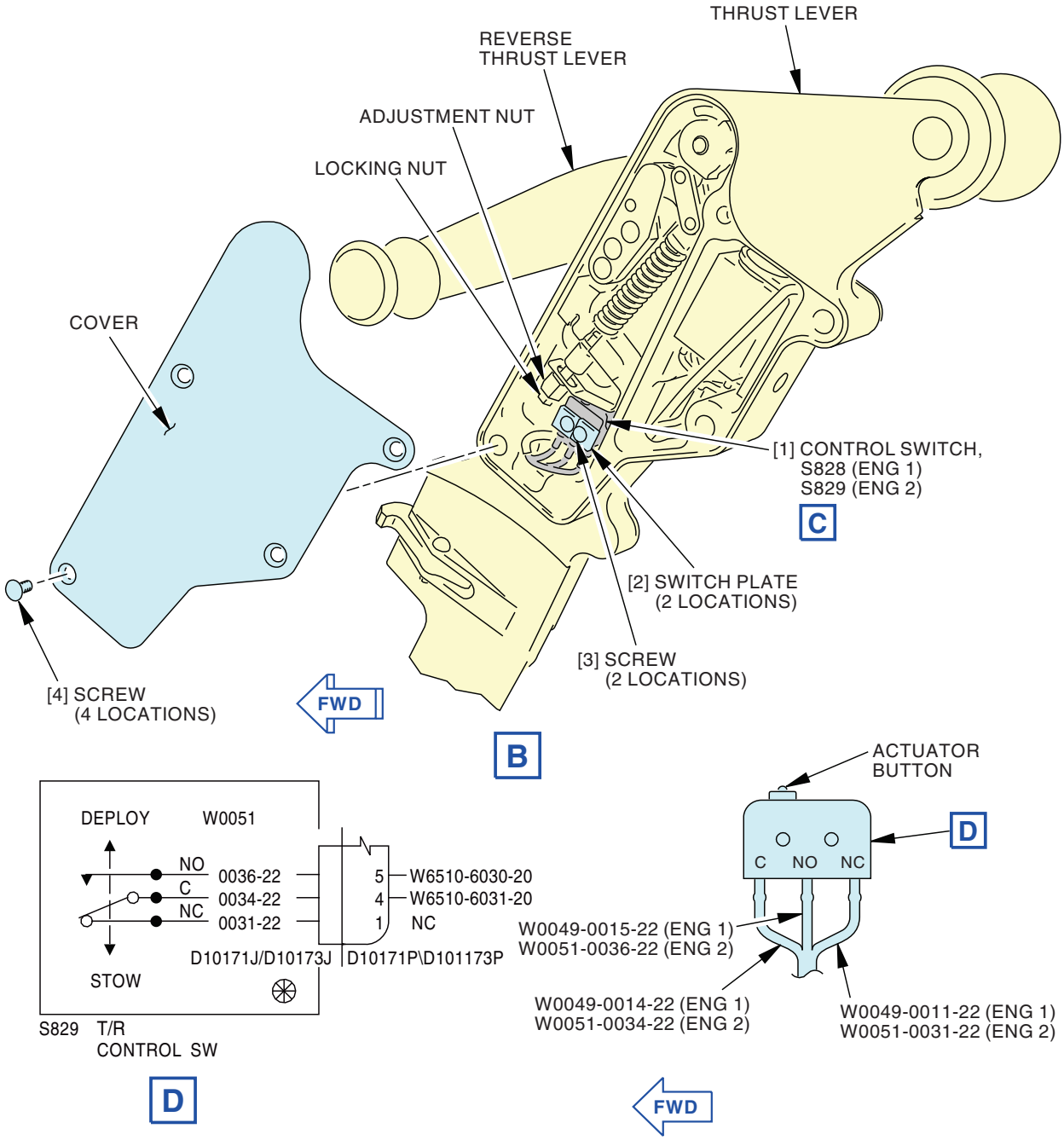
1 ACCESS TO THE CONNECTORS D10171P AND D10173P

A

2355297 S00061516926_V2

Thrust Reverser Control Switch Adjustment
Figure 501/78-34-04-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ENGINE 2 THRUST LEVER IS SHOWN.
ENGINE 1 THRUST LEVER IS OPPOSITE.

2355298 S00061516927_V2

Thrust Reverser Control Switch Adjustment
Figure 501/78-34-04-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER ARM, STOW AND SYNC LOCK SWITCHES - REMOVAL/INSTALLATION

1. General

A. The thrust reverser arm, stow and sync lock switches are on the autothrottle switchpack.

B. The switches for Engine 1 are on the left switchpack and the switches for Engine 2 are on the right switchpack.

C. For Engine 1 and Engine 2, the switch identification numbers are as follows:

(1) The sync lock switch identification number is S4.

(2) The arm switch identification number is S5.

(3) The stow switch identification number is S6.

TASK 78-34-05-000-801-G00

2. Thrust Reverser Arm, Stow and Sync Lock Switches Removal

A. General

(1) For the removal and installation procedures for the thrust reverser arm, stow and sync lock switches, refer to these tasks:

(a) Do this task: Autothrottle Switchpack Assembly Removal, TASK 76-11-07-020-801-G00.

(b) Do this task: Autothrottle Switchpack Assembly Installation, TASK 76-11-07-400-802-G00.

(2) After the installation of the thrust reverser arm, stow or sync lock switches, do this task: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00.

B. References

Reference	Title
76-11-07-020-801-G00	Autothrottle Switchpack Assembly Removal (P/B 401)
76-11-07-400-802-G00	Autothrottle Switchpack Assembly Installation (P/B 401)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VOLUMETRIC HYDRAULIC FUSES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) The standby system volumetric hydraulic fuse removal.

(2) The standby system volumetric hydraulic fuse installation.

B. If a leak occurs, the volumetric hydraulic fuses close and stop hydraulic fluid flow to prevent a complete loss of system hydraulic fluid.

C. Two of the volumetric hydraulic fuses are in the supply lines from the hydraulic standby system and are in the main gear wheel well on the keel beam.

(1) The left fuse is in the supply line to the Engine 1 thrust reverser and the right fuse is in the supply line to the Engine 2 thrust reverser.

(2) For this procedure, the volumetric hydraulic fuses for the standby system will be referred to as the left fuse and the right fuse.

D. There is no on-airplane test for the volumetric hydraulic fuses for the thrust reversers.

(1) The volumetric hydraulic fuses may be tested after removal from the airplane to find if the fuses are functional. Use the functional test procedures contained in the vendor component maintenance instructions.

TASK 78-34-07-000-802-G00

2. Standby System Volumetric Hydraulic Fuse Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task contains these topics:

(a) The removal of the standby system left fuse and the right fuse.

(b) The off-wing functional test of the fuses.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Removal

SUBTASK 78-34-07-864-001-G00

(1) Remove power from the hydraulic system A and B.

(a) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-07-864-002-G00

(2) Remove power from the standby hydraulic system.

(a) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-34-07-864-003-G00

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

F. Standby System Left or Right Fuse Removal

SUBTASK 78-34-07-020-001-G00

WARNING

DO NOT GET THIS MATERIAL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. PUT ON EYE PROTECTION (GOGGLES, OR OTHER APPROVED PROTECTION) AND GLOVES WHEN YOU USE THIS MATERIAL. MAKE SURE THAT THERE IS SUFFICIENT AIRFLOW. THIS MATERIAL CAN CAUSE INJURIES TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(1) Do these steps to remove the left or right hydraulic fuse [5] from the standby pressure system.

(a) Remove the four screws [2], the four washers [3], the four spacers [4], the two channels [1], the clampblock [6] and the clampblock [7] from the ends of the fuse.

(b) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 to catch the hydraulic fluid.

CAUTION

USE TWO WRENCHES TO LOOSEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO LOOSEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(c) Disconnect the coupling nuts at each end of the applicable hydraulic fuse [5].

(d) Remove the hydraulic fuse [5].

(e) Install protective covers on the hydraulic lines and the hydraulic fuse [5].

G. Test the Volumetric Hydraulic Fuses

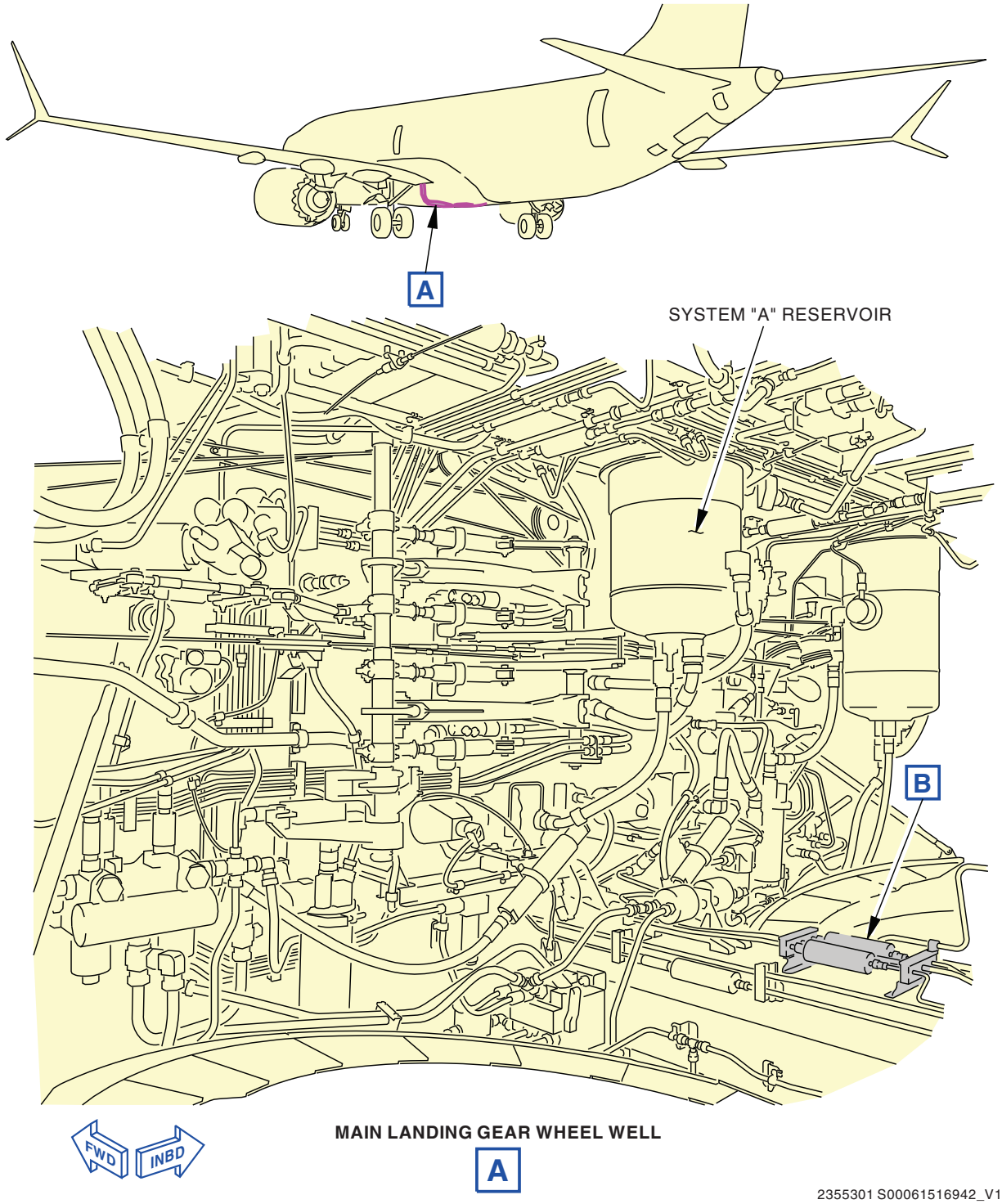
SUBTASK 78-34-07-720-001-G00

(1) Do the functional test of the volumetric hydraulic fuses with the suppliers recommended component maintenance test instructions and test equipment.

NOTE: The suppliers instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the fuses. This is an off-wing bench test.

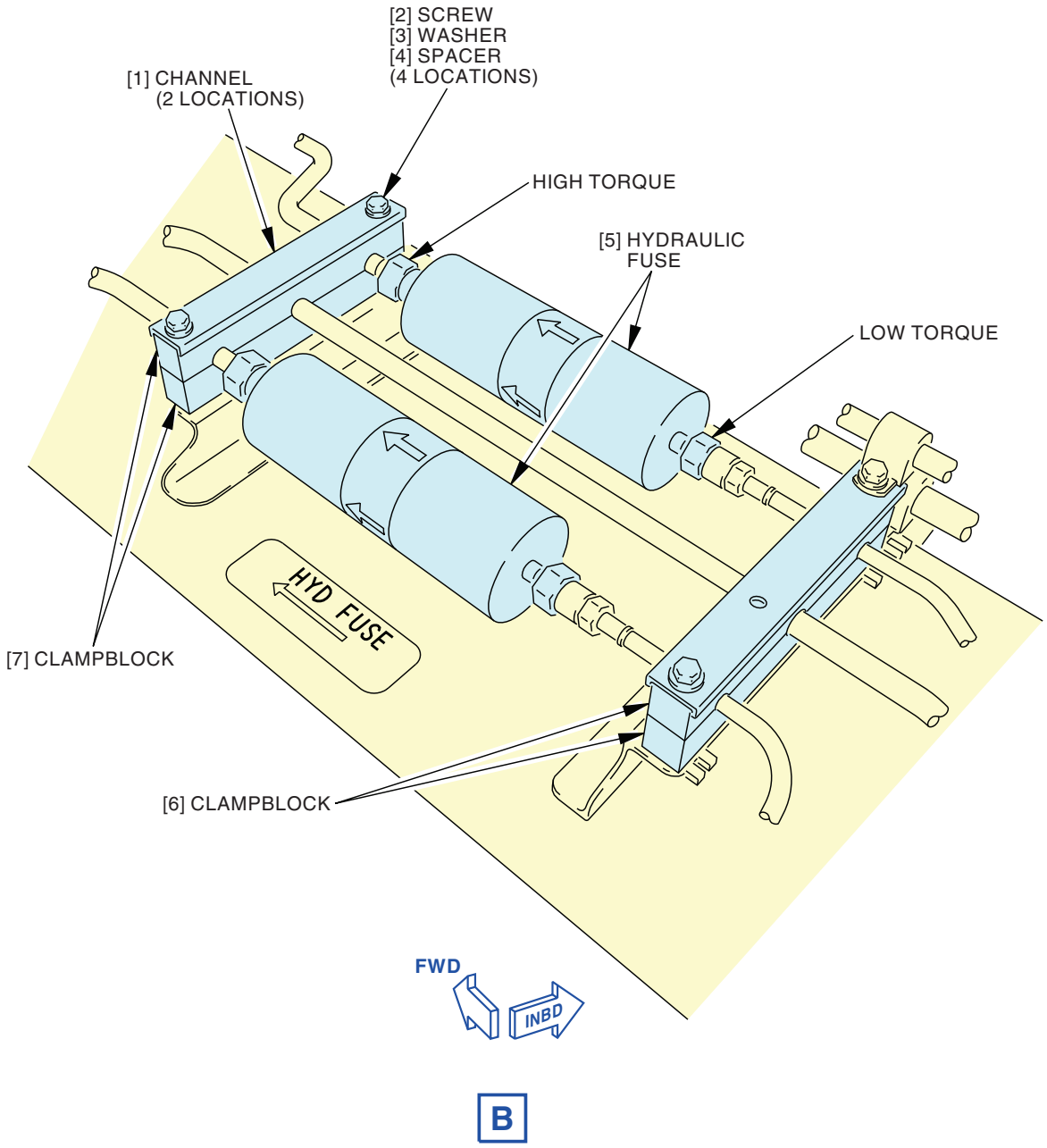
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby System Left and Right Volumetric Hydraulic Fuses Installation
Figure 401/78-34-07-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby System Left and Right Volumetric Hydraulic Fuses Installation
Figure 401/78-34-07-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-07-400-802-G00

3. Standby System Volumetric Hydraulic Fuse Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task contains the installation of the standby system left and right fuse.

B. References

Reference	Title
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
78-31-00-700-803-G00	Thrust Reverser Operational Test (Standby Hydraulic System) (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Hydraulic fuse	78-34-07-01-085	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Standby System Left and Right Fuse Installation

SUBTASK 78-34-07-420-001-G00

WARNING

DO NOT GET THIS MATERIAL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. PUT ON EYE PROTECTION (GOGGLES, OR OTHER APPROVED PROTECTION) AND GLOVES WHEN YOU USE THIS MATERIAL. MAKE SURE THAT THERE IS SUFFICIENT AIRFLOW. KEEP THIS MATERIAL AWAY FROM SPARKS, FLAME, AND HEAT. THIS MATERIAL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Do these steps to install the left or right hydraulic fuse [5]:

(a) Remove the protective covers from the hydraulic lines and the hydraulic fuse [5].

(b) Put the applicable hydraulic fuse [5] in the correct position.

CAUTION

USE TWO WRENCHES TO TIGHTEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(c) Connect the tube coupling nut at the forward end of the fuse.

1) Tighten the coupling nut to 257.0 in-lb (29.0 N·m) to 284.0 in-lb (32.1 N·m).

(d) Connect the tube coupling nut at the aft end of the fuse.

1) Tighten the coupling nut to 133.0 in-lb (15.0 N·m) to 147.0 in-lb (16.6 N·m).

(e) Install the clampblock assembly at the forward end of the fuse.

1) Install the two screws [2], the two washers [3], the two spacers [4], the channel [1] and the clampblock [7].

a) Tighten the screws [2] to 25.0 in-lb (2.8 N·m) to 35.0 in-lb (4.0 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Install the clampblock assembly at the aft end of the fuse.

1) Install the two screws [2], the two washers [3], the two spacers [4], the channel [1] and the clampblock [6].

a) Tighten the screws [2] to 25.0 in-lb (2.8 N·m) to 35.0 in-lb (4.0 N·m).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-07-863-001-G00

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT

(1) Do this task: Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

(a) Do a check of the fuses and hydraulic lines for hydraulic leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-34-07-710-001-G00

(2) Do this task: Thrust Reverser Operational Test (Standby Hydraulic System), TASK 78-31-00-700-803-G00.

(a) Operate the thrust reverser a minimum of three cycles.

(b) Do a check of the fuses and hydraulic lines for hydraulic leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SHUTTLE VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Thrust Reverser Shuttle Valve Removal

(2) Thrust Reverser Shuttle Valve Installation.

TASK 78-34-08-000-801-G00

2. Thrust Reverser Shuttle Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the thrust reverser shuttle valves.

(2) If the primary hydraulic system pressure goes low, the shuttle valve changes the thrust reverser hydraulic operating pressure supply to the standby system.

(3) The shuttle valves are in the main gear wheel well on the forward bulkhead. The shuttle valve for the Engine 1 thrust reverser is on the left side and the Engine 2 thrust reverser shuttle valve is on the right side.

(4) For this procedure, the shuttle valve for the Engine 1 thrust reverser will be referred to as the left shuttle valve and the shuttle valve for the Engine 2 thrust reverser will be referred to as the right shuttle valve.

(5) The left and right shuttle valves are not interchangeable.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Prepare for the Removal

SUBTASK 78-34-08-860-001-G00

(1) Remove power from the hydraulic systems A and B Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-34-08-860-002-G00

(2) Remove power from the standby hydraulic system Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-34-08-860-003-G00

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Left Shuttle Valve Removal

SUBTASK 78-34-08-020-001-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these steps to remove the left shuttle valve [1] from the left side of the forward bulkhead in the main gear wheel well:

(a) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 to catch the hydraulic fluid.

(b) Disconnect the three hydraulic lines from the shuttle valve [1].

1) Disconnect the hydraulic line from the IN B port.

2) Disconnect the hydraulic line from the OUT port.

3) Disconnect the hydraulic line from the IN A port.

(c) Remove the two bolts [2] and two washers [3] that attach the shuttle valve [1] to the forward bulkhead.

(d) Remove the shuttle valve [1].

F. Right Shuttle Valve Removal

SUBTASK 78-34-08-020-002-G00

WARNING DO NOT INSTALL PRESSURE CAPS TO THE HYDRAULIC LINES THAT CONNECT TO CONTROL VALVE MODULE PORTS TO DEACTIVATE THE THRUST REVERSER. THE THRUST REVERSER CAN OPERATE WHEN PRESSURIZED. IF YOU DO NOT OBEY, INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT YOU WEAR PROTECTIVE CLOTHES AND GLOVES WHEN YOU DO WORK ON THE HYDRAULIC SYSTEM. HYDRAULIC FLUID CAN CAUSE A LEAK FROM THE OPEN PORTS OF THE COMPONENT OR FROM THE HYDRAULIC LINES. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these steps to remove the right shuttle valve [4] from the right side of the forward bulkhead in the main gear wheel well:

(a) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 to catch the hydraulic fluid.

(b) Disconnect the three hydraulic lines from the shuttle valve [4].

1) Disconnect the hydraulic line from the IN A port.

2) Disconnect the hydraulic line from the OUT port.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

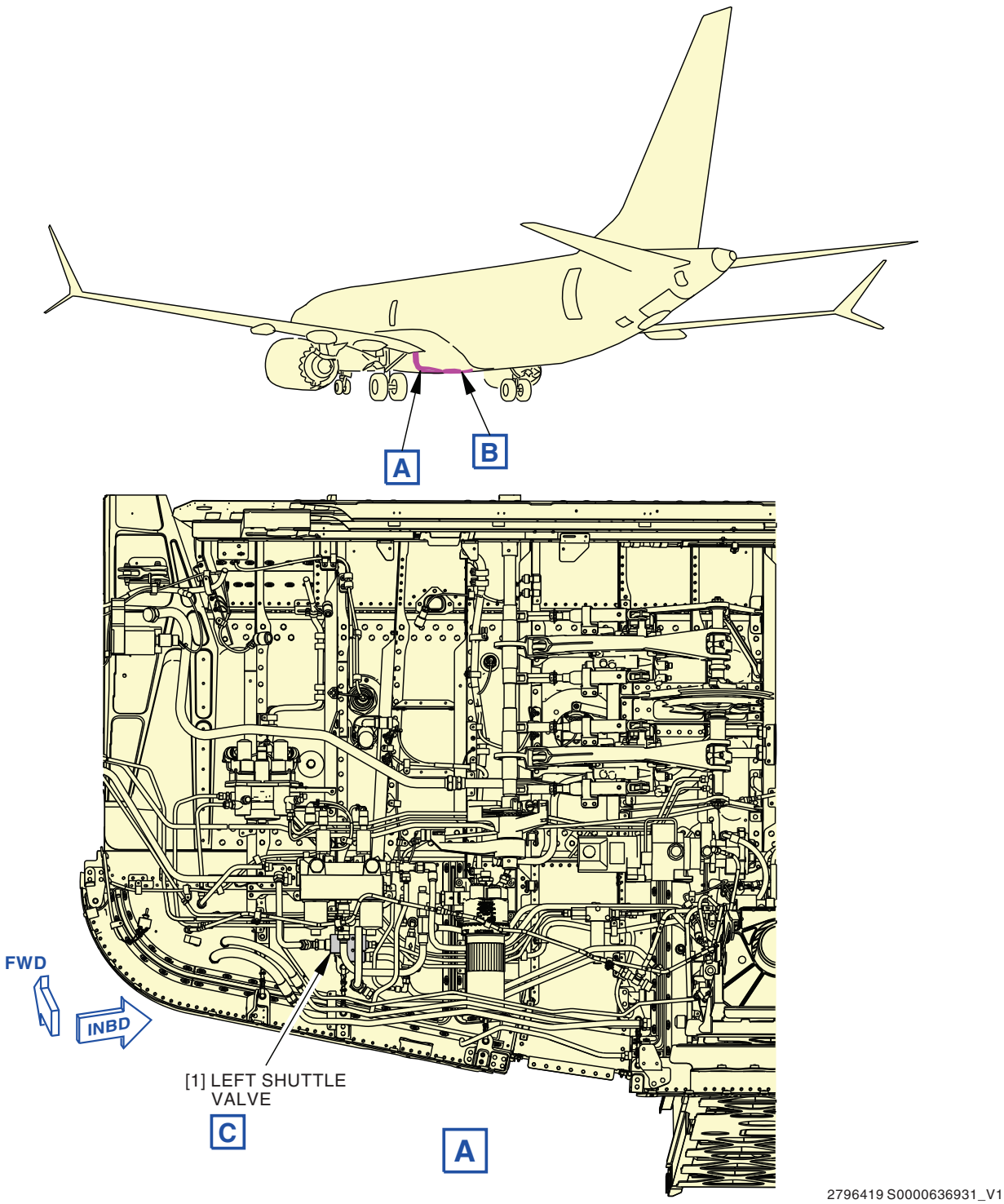
3) Disconnect the hydraulic line from the IN B port.

(c) Remove the two bolts [5] and two washers [6] that attach the shuttle valve [4] to the forward bulkhead.

(d) Remove the shuttle valve [4].

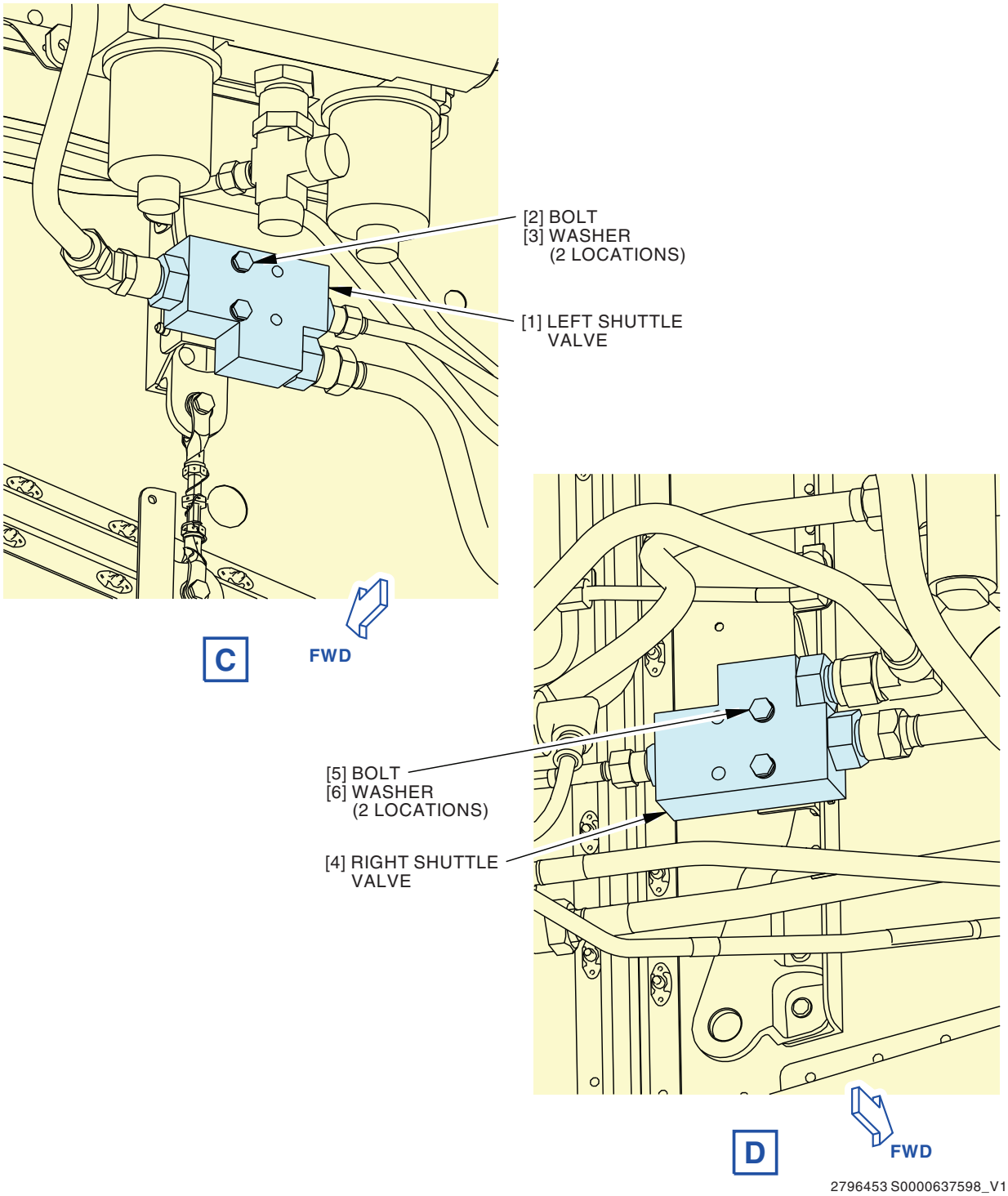
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



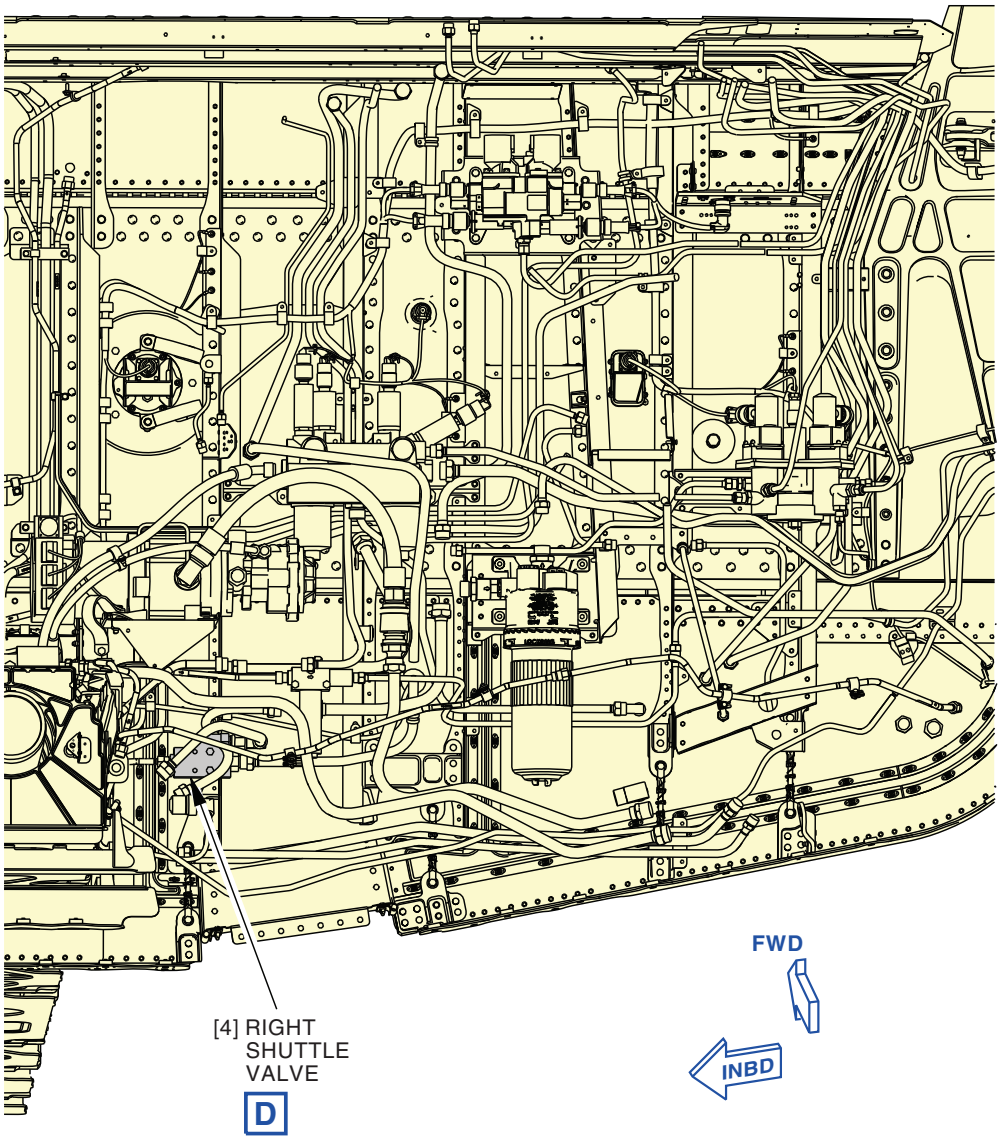
Shuttle Valve Installation
Figure 401/78-34-08-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Shuttle Valve Installation
Figure 401/78-34-08-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Shuttle Valve Installation
Figure 401/78-34-08-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-08-400-801-G00

3. Thrust Reverser Shuttle Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install thrust reverser shuttle valves.

(2) The left and right shuttle valves are not interchangeable.

B. References

Reference	Title
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)
78-31-00-700-803-G00	Thrust Reverser Operational Test (Standby Hydraulic System) (P/B 501)

C. Left Shuttle Valve Installation

SUBTASK 78-34-08-400-001-G00

(1) Do these steps to install the left shuttle valve [1] on the left side of the forward bulkhead in the main gear wheel well:

(a) Put the left shuttle shuttle valve [1] in the correct position.

(b) Install the two washers [3] and two bolts [2] that attach the left shuttle valve [1] to the forward bulkhead.

1) Tighten the bolt [2] to 80 ±8 in-lb (9 ±1 N·m).

SUBTASK 78-34-08-400-002-G00

(2) Do these steps to connect the hydraulic lines to the left shuttle valve [1]:

(a) Remove the protective covers from the hydraulic lines.

(b) Connect the hydraulic line at the OUT port of the left shuttle valve [1].

1) Tighten the coupling nut to 695 ±40 in-lb (79 ±5 N·m).

(c) Connect the hydraulic line at the IN B port of the left shuttle valve [1].

1) Tighten the coupling nut to 270 ±13 in-lb (31 ±1 N·m).

(d) Connect the hydraulic line at the IN A port of the left shuttle valve [1].

1) Tighten the coupling nut to 695 ±40 in-lb (79 ±5 N·m).

D. Right Shuttle Valve Installation

SUBTASK 78-34-08-400-003-G00

(1) Do these steps to install the right shuttle valve [4] on the right side of the forward bulkhead:

(a) Put the right shuttle valve [4] in the correct position.

(b) Install the two washers [6] and two bolts [5] that attach the right shuttle valve [4] to the forward bulkhead.

1) Tighten the bolt [5] to 80 ±8 in-lb (9 ±1 N·m).

SUBTASK 78-34-08-400-004-G00

(2) Do these steps to connect the hydraulic lines to the right shuttle valve [4]:

(a) Remove the protective covers from the hydraulic lines.

(b) Connect the hydraulic line at the OUT port of the right shuttle valve [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the coupling nut to 695 ±40 in-lb (79 ±5 N·m)).

(c) Connect the hydraulic line at the IN B port of the right shuttle valve [4].

1) Tighten the coupling nut to 270 ±13 in-lb (31 ±1 N·m).

(d) Connect the hydraulic line at the IN A port of the right shuttle valve [4].

1) Tighten the coupling nut to 695 ±40 in-lb (79 ±5 N·m).

E. Shuttle Valve Installation Check

SUBTASK 78-34-08-790-001-G00

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply system A hydraulic power for the left shuttle valve or system B hydraulic power for the right shuttle valve Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) Do a check of the hydraulic lines and connections for hydraulic fluid leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-34-08-790-002-G00

(2) Do this task: Thrust Reverser Operational Test (Standby Hydraulic System), TASK 78-31-00-700-803-G00.

(a) Do a check of the hydraulic lines and connections for hydraulic fluid leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-34-08-720-001-G00

(3) Do this task: Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

(a) Operate the thrust reverser through the deploy and stow cycles at least 3 times until the TR sleeves move smoothly.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER CONTROL SWITCH ACTUATION SPRING - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the thrust reverser control switch actuation spring

(2) An installation of the thrust reverser control switch actuation spring.

B. The thrust reverser control switch actuation spring is referred to as the actuation spring.

TASK 78-34-09-000-801-G00

2. Actuation Spring Removal

(Figure 401)

A. References

Reference	Title
78-34-04-000-801-G00	Control Switch Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 78-34-09-020-001-G00

(1) Remove the thrust reverser control switch from the applicable thrust lever.

(a) Do this task: Control Switch Removal, TASK 78-34-04-000-801-G00.

SUBTASK 78-34-09-980-001-G00

(2) Raise the applicable reverse thrust lever to the fully extended (deployed) position.

D. Actuation Spring Removal

SUBTASK 78-34-09-020-002-G00

(1) To remove the actuation spring [1] from the applicable thrust lever, do these steps:

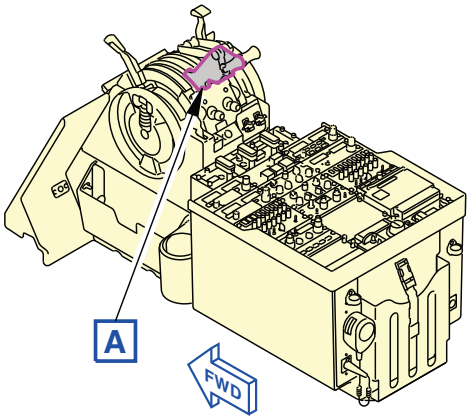
(a) Remove the locking nut [2] from the plunger.

(b) Remove the adjustment nut [3] from the plunger.

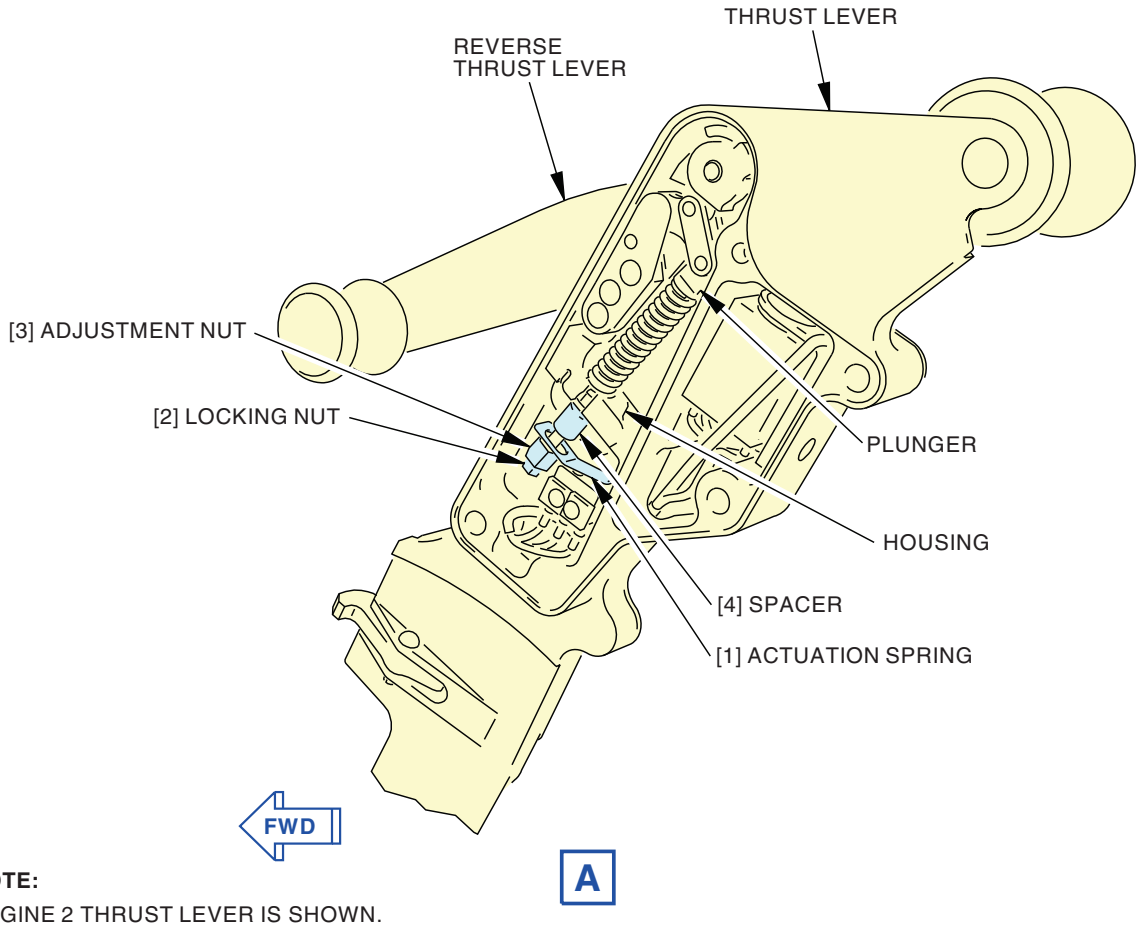
(c) Remove the actuation spring [1] and spacer [4] from the plunger.

——— END OF TASK ———

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CONTROL STAND



NOTE:

ENGINE 2 THRUST LEVER IS SHOWN.
ENGINE 1 THRUST LEVER IS OPPOSITE.

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Thrust Reverser Control Switch Actuation Spring Installation
Figure 401/78-34-09-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-09-400-801-G00

3. Actuation Spring Installation

(Figure 401)

A. References

Reference	Title
78-34-04-400-801-G00	Control Switch Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Actuation Spring Installation

SUBTASK 78-34-09-420-001-G00

(1) To install the actuation spring [1], do these steps:

(a) Install the actuation spring [1] and spacer [4] on the plunger.

1) Make sure that the spacer [4] is inside the actuation spring [1].

2) Make sure that the actuation spring [1] is flat against the bottom surface of the housing.

(b) Install the adjustment nut [3] on the plunger.

(c) Install the locking nut [2] on the plunger.

SUBTASK 78-34-09-420-002-G00

(2) Install the thrust reverser control switch.

(a) Do this task: Control Switch Installation, TASK 78-34-04-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER CONTROL SWITCH ACTUATION SPRING - INSPECTION/CHECK

1. General

A. This procedure checks the free angle of the thrust reverser control switch actuation spring. The free angle occurs when the spring is not installed in the thrust lever and no load is applied to the spring.

B. This procedure can be used as part of a thrust reverser deploy fault isolation task. Intermittent thrust reverser control switch actuation may occur if the actuation spring free angle is below the minimum tolerance.

C. The thrust reverser control switch actuation spring is referred to as the actuation spring.

TASK 78-34-09-200-801-G00

2. Actuation Spring Free Angle Check

(Figure 601)

A. References

Reference	Title
78-34-09-000-801-G00	Actuation Spring Removal (P/B 401)
78-34-09-400-801-G00	Actuation Spring Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Check

SUBTASK 78-34-09-020-003-G00

(1) Remove the actuation spring from the applicable thrust lever.

(a) Do this task: Actuation Spring Removal, TASK 78-34-09-000-801-G00.

D. Actuation Spring Free Angle Check

SUBTASK 78-34-09-980-002-G00

(1) Make sure that the actuation spring free angle (Angle A) is within the tolerance (Table 601).

Table 601/78-34-09-993-801-G00 Actuation Spring Free Angle

Part Number	Angle A	Dimension B*[1]	Dimension C*[1]
69-78782-3	46 ±2°arc (0.80 ±0.03 rad)	0.86 in. (21.84 mm)	0.45 in. (11.43 mm)

*[1] Use dimension only to determine the part number.

E. Put the Airplane Back to Its Usual Condition

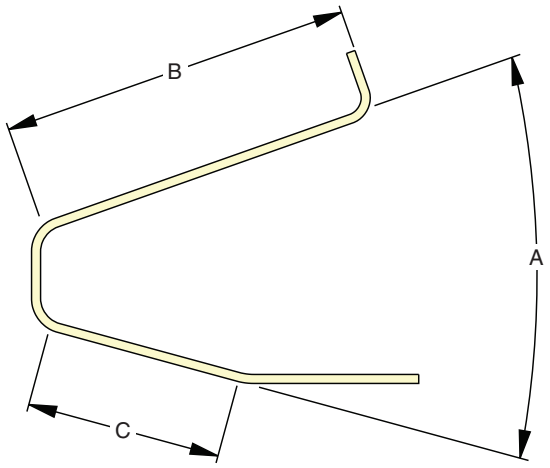
SUBTASK 78-34-09-420-003-G00

(1) Install the actuation spring.

(a) Do this task: Actuation Spring Installation, TASK 78-34-09-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Actuation Spring Free Angle Check
Figure 601/78-34-09-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SYNC LOCK - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the thrust reverser sync lock.

(2) The installation of the thrust reverser sync lock.

TASK 78-34-10-000-801-G00

2. Thrust Reverser Sync Lock Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the sync lock from the left or right thrust reverser on the applicable engine.

(2) There is a sync lock on the left and right thrust reversers on each engine.

(a) The sync locks are installed on the lower actuators.

(3) The sync locks are interchangeable between the left and right thrust reversers.

(4) The thrust reverser sync lock is referred to as the sync lock and the sync lock bracket is referred to as the bracket.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

D. Prepare for the Removal

SUBTASK 78-34-10-860-001-G00

(1) For engine 1, open this circuit breaker and install a safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-002-G00

(2) For engine 2, open this circuit breaker and install a safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-10-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

SUBTASK 78-34-10-040-002-G00

(4) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-34-10-040-003-G00

(5) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-34-10-040-004-G00

(6) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-34-10-010-001-G00

(7) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

E. Thrust Reverser Sync Lock Removal

SUBTASK 78-34-10-020-001-G00

(1) Disconnect the electrical connector from the sync lock [1]:

(a) For the left thrust reverser, disconnect electrical connector D1008 from the sync lock receptacle.

(b) For the right thrust reverser, disconnect electrical connector D1016 from the sync lock receptacle.

SUBTASK 78-34-10-020-002-G00

(2) Do these steps to remove the sync lock [1] and the bracket [5]:

(a) Put a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, below the sync lock [1] to catch the hydraulic fluid.

WARNING MAKE SURE THAT YOU WEAR THE PERSONAL PROTECTIVE EQUIPMENT WHEN YOU DO THIS TASK. PERSONAL PROTECTIVE EQUIPMENT WILL PREVENT INJURIES TO PERSONNEL.

CAUTION USE TWO WRENCHES TO DISCONNECT OR CONNECT TUBES. IF YOU DO NOT USE A WRENCH ON EACH SIDE OF THE CONNECTION, DAMAGE TO THE TUBES CAN OCCUR.

(b) Loosen the coupling nut on the sync lock [1].

1) Wrap a cloth rag around the sync lock [1] and the wrench to catch the hydraulic spray.

2) Let the hydraulic fluid drain into the container.

(c) Remove the nut [2] and washer [3] that attach the bracket [5] to the adapter plug [4].

(d) Remove the sync lock [1] and the bracket [5].

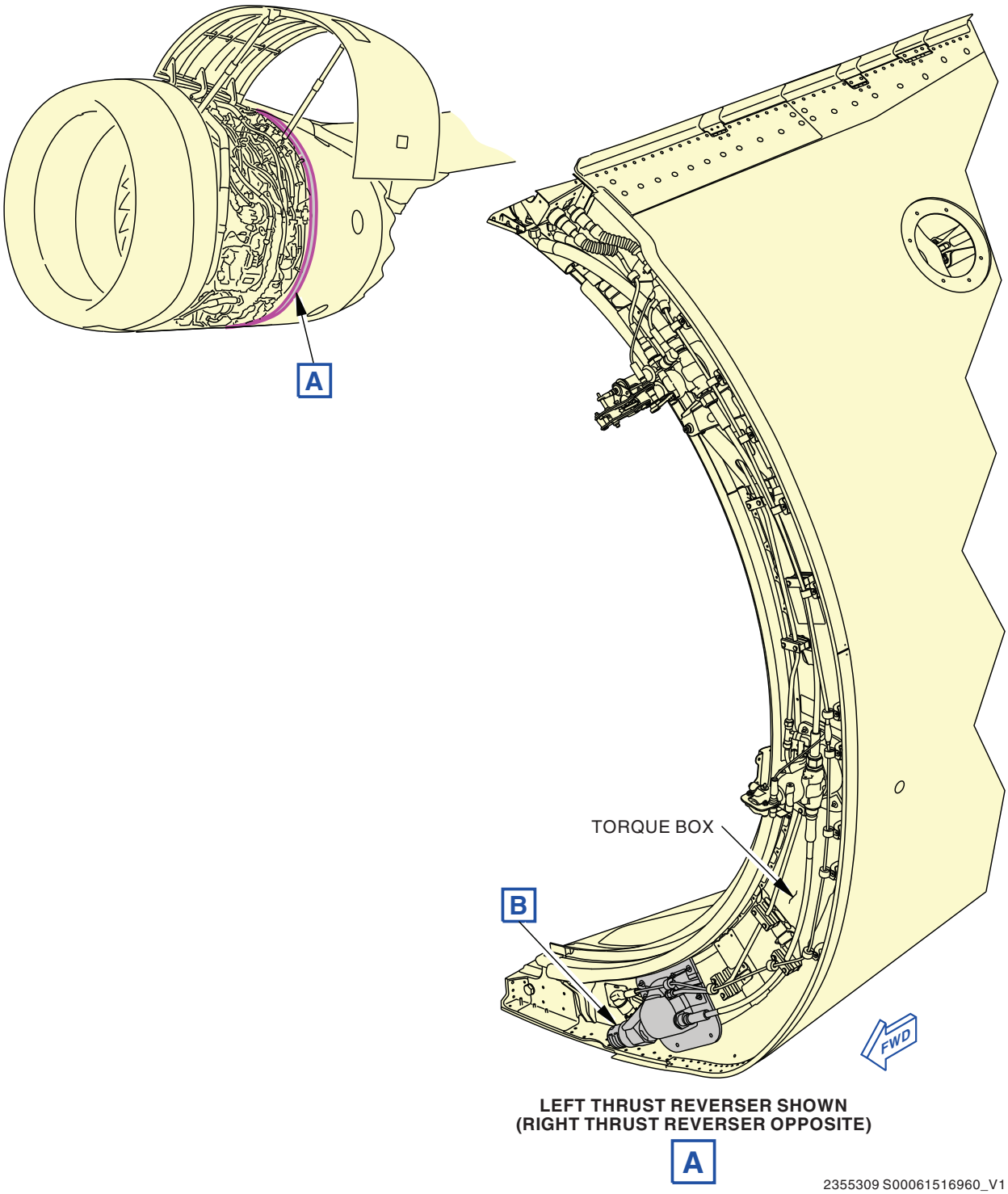
(e) Remove the two bolts [6] and two washers [7] that attach the bracket [5] to the sync lock [1].

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(f) Install protective covers on the ports of the sync lock [1] and the hydraulic actuator.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

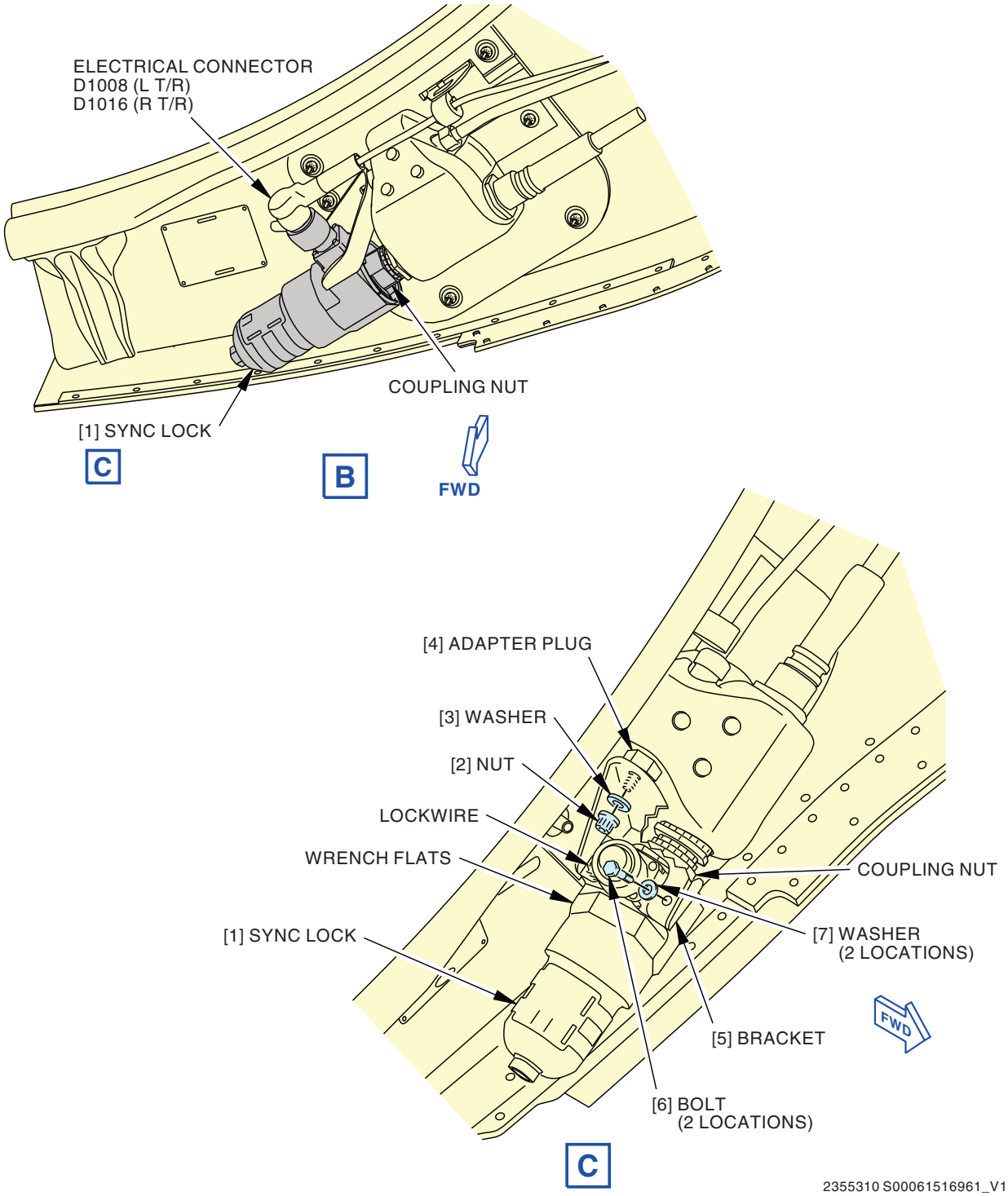


LEFT THRUST REVERSER SHOWN
(RIGHT THRUST REVERSER OPPOSITE)

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Thrust Reverser Sync Lock Installation
Figure 401/78-34-10-990-801-G00 (Sheet 1 of 2)

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Thrust Reverser Sync Lock Installation
Figure 401/78-34-10-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-10-400-801-G00

3. Thrust Reverser Sync Lock Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the sync lock to the left or right thrust reverser on the applicable engine.

(2) After you install the sync lock, you must do these tasks: Thrust Reverser Operational Test (Selection), TASK 78-31-00-700-801-G00, and Thrust Reverser Operational Test (Standby Hydraulic System), TASK 78-31-00-700-803-G00.

(3) The left and the right sync locks are identical and interchangeable.

B. References

Reference	Title
29-00-00-790-801	Hydraulic System External Leakage Check (P/B 601)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-21-00-000-801	Standby Hydraulic System Pressurization (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-801-G00	Thrust Reverser Operational Test (Selection) (P/B 501)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)
78-31-00-700-803-G00	Thrust Reverser Operational Test (Standby Hydraulic System) (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Sync lock	78-34-10-01-055	PTF ALL
		78-34-10-02-055	PTF ALL

E. Prepare for the Installation

SUBTASK 78-34-10-420-001-G00

(1) Do these steps to install the bracket [5] on the sync lock [1]:

(a) Align the fastener holes.

(b) Install the two washers [7] and two bolts [6].

1) Tighten the two bolts [6] to 32.5 ±2.5 in-lb (3.7 ±0.3 N·m).

2) Install the MS20995NC32 lockwire, G01912.

F. Thrust Reverser Sync Lock Installation

SUBTASK 78-34-10-420-002-G00

(1) Do these steps to install the sync lock [1]:

(a) Remove the protective covers from the sync lock [1] and the hydraulic actuator.

(b) Put the sync lock [1] in the correct position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure the fastener hole in the bracket [5] aligns with the stud on the adapter plug [4].

(c) Hand tighten the coupling nut on the sync lock [1].

(d) Install the washer [3] and nut [2] that connect the bracket [5] to the adapter plug [4] stud.

1) Tighten the nut [2] to 82.5 ±17.5 in-lb (9.3 ±2.0 N·m).

CAUTION

USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(e) Do these steps to tighten the coupling nut:

1) Tighten the coupling nut to 900 ±45 in-lb (102 ±5 N·m).

2) Loosen the coupling nut.

3) Tighten the coupling nut again to 900 ±45 in-lb (102 ±5 N·m).

SUBTASK 78-34-10-420-003-G00

(2) Connect the electrical connectors to the sync lock [1] receptacle:

(a) For the left thrust reverser, connect the electrical connector, D1008, to the sync lock [1] receptacle.

(b) For the right thrust reverser, connect the electrical connector, D1016, to the sync lock [1] receptacle.

G. Thrust Reverser Sync Lock Installation Test

SUBTASK 78-34-10-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-34-10-860-003-G00

(2) For engine 1, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

For engine 2, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-016-G00

(3) Make sure that the hydraulic reservoirs are pressurized. If not, do this task: Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 78-34-10-860-004-G00

(4) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801

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SUBTASK 78-34-10-860-006-G00

WARNING OBEY ALL OF THE WARNINGS AND CAUTIONS IN THE REFERENCED PROCEDURE. IF YOU DO NOT OBEY THE WARNINGS AND CAUTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do this task: Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

(a) Operate the thrust reverser through the extend (deploy) and retract (stow) cycle until the sleeves move smoothly.

(b) Examine the thrust reverser area for hydraulic fluid leaks.

1) If you find leaks, do this task: Hydraulic System External Leakage Check, TASK 29-00-00-790-801.

NOTE: Refer to the Tube Connections, Static Seals, and Other Dynamic Seals sections of the procedure for the leakage limits.

SUBTASK 78-34-10-860-005-G00

(6) Do this task: Standby Hydraulic System Pressurization, TASK 29-21-00-000-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-10-860-008-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SYNC LOCK - ADJUSTMENT/TEST

1. General

A. This procedure has two task:

(1) Sync Lock Manual Integrity Test When Hydraulic Power Is Not Available

(2) Sync Lock Manual Integrity Test When Hydraulic Power Is Available.

TASK 78-34-10-700-801-G00

2. Sync Lock Manual Integrity Test When Hydraulic Power Is Not Available

(Figure 501 and Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do a manual test of the locking ability of the sync lock.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-980-804-G00	Thrust Reverser Operation - Retract (Manual Procedure) (P/B 201)
78-34-10-000-801-G00	Thrust Reverser Sync Lock Removal (P/B 401)
78-34-10-400-801-G00	Thrust Reverser Sync Lock Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-6839	Wrench - Torque, Inspection, 0-150 in-lbs, +/-5% accuracy, Analog or Digital Readout (non-clicker)

D. Location Zones

Zone	Area
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Test

SUBTASK 78-34-10-010-002-G00

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 78-34-10-910-001-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

G. Sync Lock Manual Integrity Test When Hydraulic Power Is Not Available

SUBTASK 78-34-10-860-010-G00

(1) Unlock the upper and middle actuators as follows:

(a) On the upper thrust reverser actuator, move the manual unlock handle forward to the unlock position.

NOTE: The manual unlock handle moves forward approximately 30 degree.

(b) Hold the manual unlock handle in the forward position as you push the detent pin in.

(c) Release the manual unlock handle.

(d) Do the above 3 steps again for the middle thrust reverser actuator.

SUBTASK 78-34-10-720-002-G00

(2) Do a test of the sync lock locking feature as follows:

(a) Partially depress the manual drive pin on the end of the sync lock until the top of the pin is even with the counterbore in the manual drive.

NOTE: You partially depress the manual drive pin to engage the manual drive feature without releasing the sync lock locking feature.

NOTE: Do not fully depress the manual drive pin. If you fully depress the manual drive pin, you will release the sync lock locking mechanism.

1) Make sure that the manual drive teeth do not interfere with the clutch teeth.

NOTE: You may need to rotate the hex nut slightly (quarter turn).

(b) Use a inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension to apply 50 in-lb (6 N·m) to the hex nut on the end of the sync lock as follows:

1) For the left side thrust reverser sleeve (aft looking forward), apply the torque on the hex nut in the counter-clockwise direction.

2) For the right side thrust reverser sleeve (aft looking forward), apply the torque on the hex nut in the clockwise direction.

3) If you are unable to turn the sync shaft, then the sync lock unit is serviceable.

4) If the sync shaft turns when you apply 50 in-lb (5.6 N·m) or less, then the sync lock unit is not serviceable.

a) Replace the sync lock unit, do these tasks:

- Thrust Reverser Sync Lock Removal, TASK 78-34-10-000-801-G00,

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Thrust Reverser Sync Lock Installation, TASK 78-34-10-400-801-G00.

SUBTASK 78-34-10-720-005-G00

(3) Do these steps to check for the connection between the sync lock and the actuator shaft:

CAUTION

DO NOT HIT THE END OF THE SYNC LOCK OR THE 3/8 INCH DRIVE WRENCH TO PUSH THE LOCK RELEASE PIN. DAMAGE TO THE SYNC LOCK CAN OCCUR.

(a) Unlock the sync lock and engage the manual drive:

1) Make sure that the manual drive teeth do not interfere with the clutch teeth.

NOTE: You may need to rotate the hex nut slightly (quarter turn).

2) Put an open end wrench on the hex nut at end of the sync lock.

3) Put a 3/8-inch square drive ratchet into the 3/8-inch square opening.

4) Apply continuous inward pressure on the lock release pin with the ratchet.

5) Slowly move the open end wrench a small amount at a time in one or opposite direction until the lock release pin goes in.

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Use a inspection torque wrench, STD-6839, with a 6 in. (15 cm) extension to apply 50 in-lb (6 N·m) to the hex nut on the end of the sync lock as follows:

1) For the left side thrust reverser sleeve (aft looking forward), apply the torque on the hex nut in the counter-clockwise direction.

2) For the right side thrust reverser sleeve (aft looking forward), apply the torque on the hex nut in the clockwise direction.

3) If you are able to manually extend the thrust reverser sleeve, then the sync lock unit is serviceable.

4) If the sync shaft turns freely when you try to manually extend the thrust reverser with 50 in-lb (5.6 N·m) or less, the sync lock unit is not serviceable.

a) Replace the sync lock unit, do these tasks:

- Thrust Reverser Sync Lock Removal, TASK 78-34-10-000-801-G00,
- Thrust Reverser Sync Lock Installation, TASK 78-34-10-400-801-G00.

SUBTASK 78-34-10-980-002-G00

(4) Lock the upper and middle actuators as follows:

NOTE: If the thrust reverser sleeve is in the extended position, the manual unlock handle will not move to the locked position until the thrust reverser sleeve is fully stowed.

(a) On the upper thrust reverser actuator, move the manual unlock handle forward.

(b) Make sure that the detent pin returns to the extended position.

(c) Release the manual unlock handle.

(d) Repeat the previous 3 steps on the middle thrust reverser actuator.

SUBTASK 78-34-10-980-001-G00

(5) If it is necessary, manually retract the thrust reverser sleeve (TASK 78-31-00-980-804-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the manual unlock handle on the upper actuator and middle actuator move back to the normal, locked position.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-10-860-011-G00

(1) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

SUBTASK 78-34-10-410-001-G00

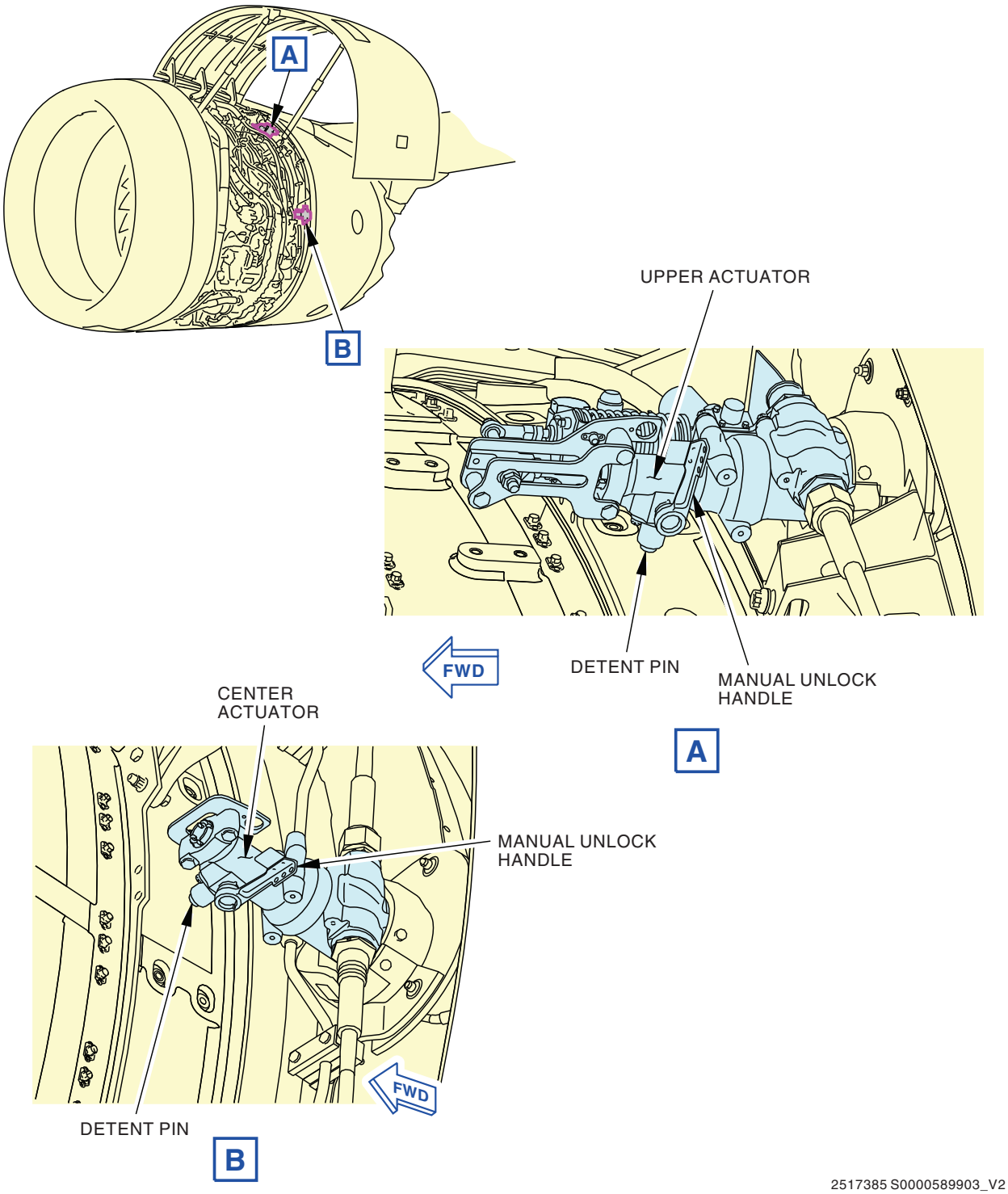
(2) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

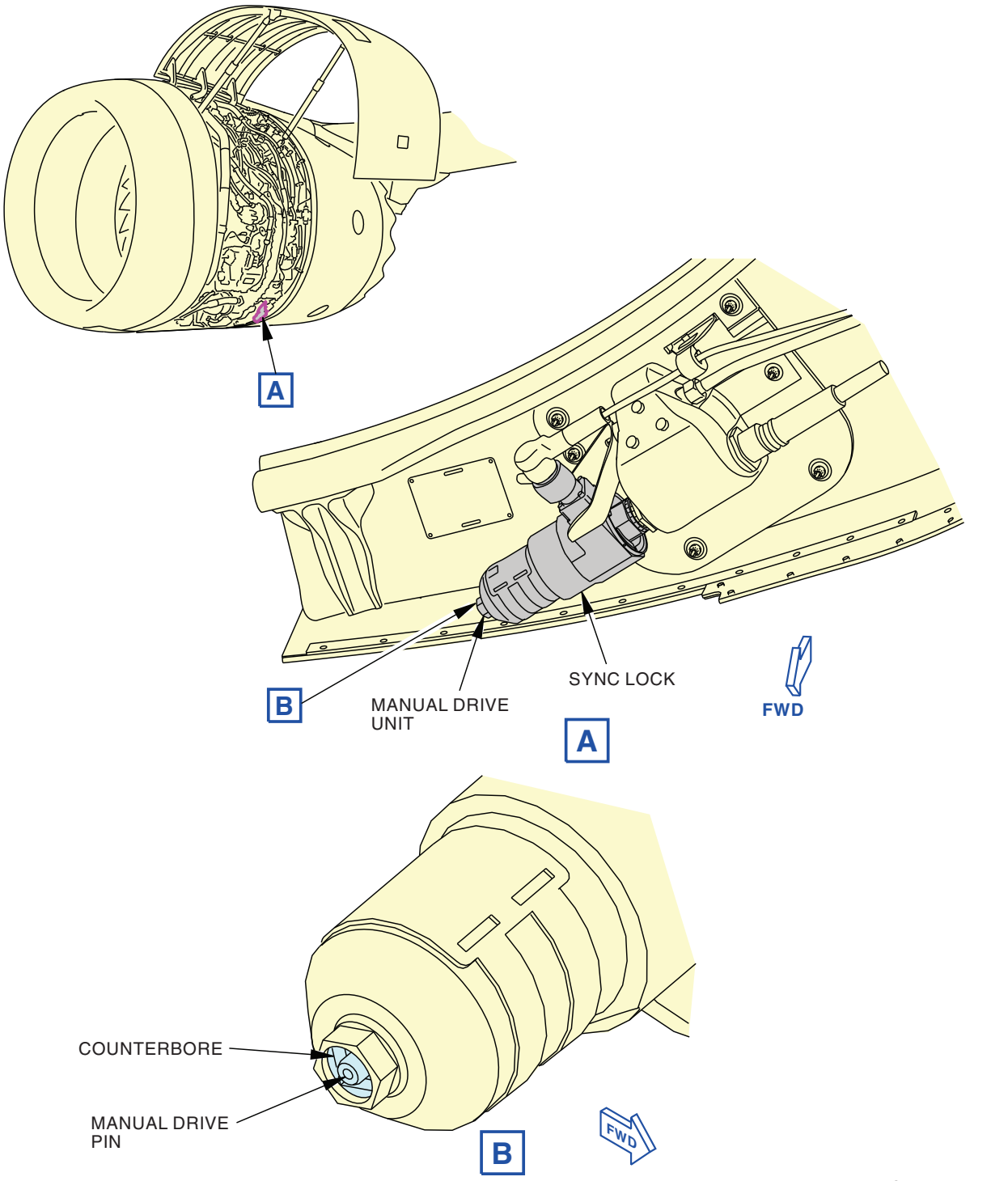
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Hydraulic Actuator Lock
Figure 501/78-34-10-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser Manual Drive Unit
Figure 502/78-34-10-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-10-700-802-G00

3. Sync Lock Manual Integrity Test When Hydraulic Power Is Available

A. General

(1) This task gives the instructions to test the locking ability of the sync lock with hydraulic power available.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-02-860-802	Special Functions (P/B 201)
71-00-00-700-816-G00	Test No. 38 - Thrust Reverser Cycle Special Functions (P/B 501)

C. Sync Lock Manual Integrity Test When Hydraulic Power Is Available

SUBTASK 78-34-10-860-017-G00

(1) Do this task: Test No. 38 - Thrust Reverser Cycle Special Functions, TASK 71-00-00-700-816-G00.

NOTE: This task will permit ground operation of the thrust reverser.

SUBTASK 78-34-10-860-018-G00

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Move the applicable reverse thrust lever up and aft to fully deploy the thrust reverser.

SUBTASK 78-34-10-860-019-G00

(3) For the left engine, open this circuit breaker and install a safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(4) For the right engine, open this circuit breaker and install a safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-020-G00

(5) Move the applicable reverse thrust lever down and forward to the stow position.

(a) Make sure that the thrust reverser sleeves do not move to the stowed position.

NOTE: This is the indication that the sync locks are in the locked position.

SUBTASK 78-34-10-860-021-G00

CAUTION

MAKE SURE THAT YOU PUT THE THRUST REVERSER LEVER TO THE DEPLOY POSITION BEFORE YOU CLOSE THE CIRCUIT BREAKERS. IF YOU DO NOT PUT THE THRUST REVERSER LEVER TO THE DEPLOY POSITION, YOU CAN CAUSE DAMAGE THE SYNC LOCK UNIT.

(6) Move the applicable reverse thrust lever up and aft to the deploy position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-10-860-022-G00

(7) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(8) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-023-G00

(9) Move the applicable reverse thrust lever down and forward to move the thrust reverser to the fully stowed position.

SUBTASK 78-34-10-860-024-G00

(10) After you stow the thrust reverser, select the STOP FUNCTION to complete the special function (TASK 46-13-02-860-802).

SUBTASK 78-34-10-860-025-G00

(11) Check the Onboard Maintenance Function (OMF) for any new fault messages, and, if present, clear the faults (TASK 31-65-00-750-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

THRUST REVERSER SYNC LOCK - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) A check of the sync lock control relay hot short protection

(2) A check of the thrust reverser sequence relay electrical resistance.

TASK 78-34-10-710-801-G00

2. Sync Lock Control Relay Hot Short Protection Check

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the instructions on how to ensure the functionality of the hot short protection for the sync lock control relay (R1158/R1159).

B. References

Reference	Title
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-603	Multimeter

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

F. Prepare for the Operational Check

SUBTASK 78-34-10-860-012-G00

(1) For engine 1 do these steps:

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

(b) Position the engine 1 thrust reverser lever in the stowed position.

1) Wait 10 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-10-860-013-G00

(2) For engine 2 do these steps:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

(b) Position the engine 2 thrust reverser lever in the stowed position.

1) Wait 10 seconds.

SUBTASK 78-34-10-010-003-G00

(3) For engine 1 open this access panel:

(Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 78-34-10-010-004-G00

(4) For engine 2 open this access panel:

(Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

G. Sync Lock Control Relay Operational Check

SUBTASK 78-34-10-710-001-G00

(1) Do these steps to do the operational check for engine 1:

(a) Remove the T/R Sync Lock Control Relay, R1158, located in the J22 Junction Box inside of the nose landing gear wheel well.

(b) Use a multimeter, STD-603, to measure the resistance between pin D16046–C1 and pin D16046–X2.

1) The measured resistance should be between 0 and 5 ohms.

(c) Install the T/R Sync Lock Control Relay, R1158, back into the J22 Junction Box.

SUBTASK 78-34-10-710-002-G00

(2) Do these steps to do the operational check for engine 2:

(a) Remove the T/R Sync Lock Control Relay, R1159, located in the J24 Junction Box inside of the nose landing gear wheel well.

(b) Use a multimeter, STD-603, to measure the resistance between pin D16048–C1 and pin D16048–X2.

1) The measured resistance should be between 0 and 5 ohms.

(c) Install the T/R Sync Lock Control Relay, R1159, back into the J24 Junction Box.

H. Put the Airplane Back to its Usual Condition

SUBTASK 78-34-10-410-002-G00

(1) For engine 1 close this access panel:

(Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-10-410-003-G00

(2) For engine 2 close this access panel:

(Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

SUBTASK 78-34-10-860-014-G00

(3) For engine 1 remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-015-G00

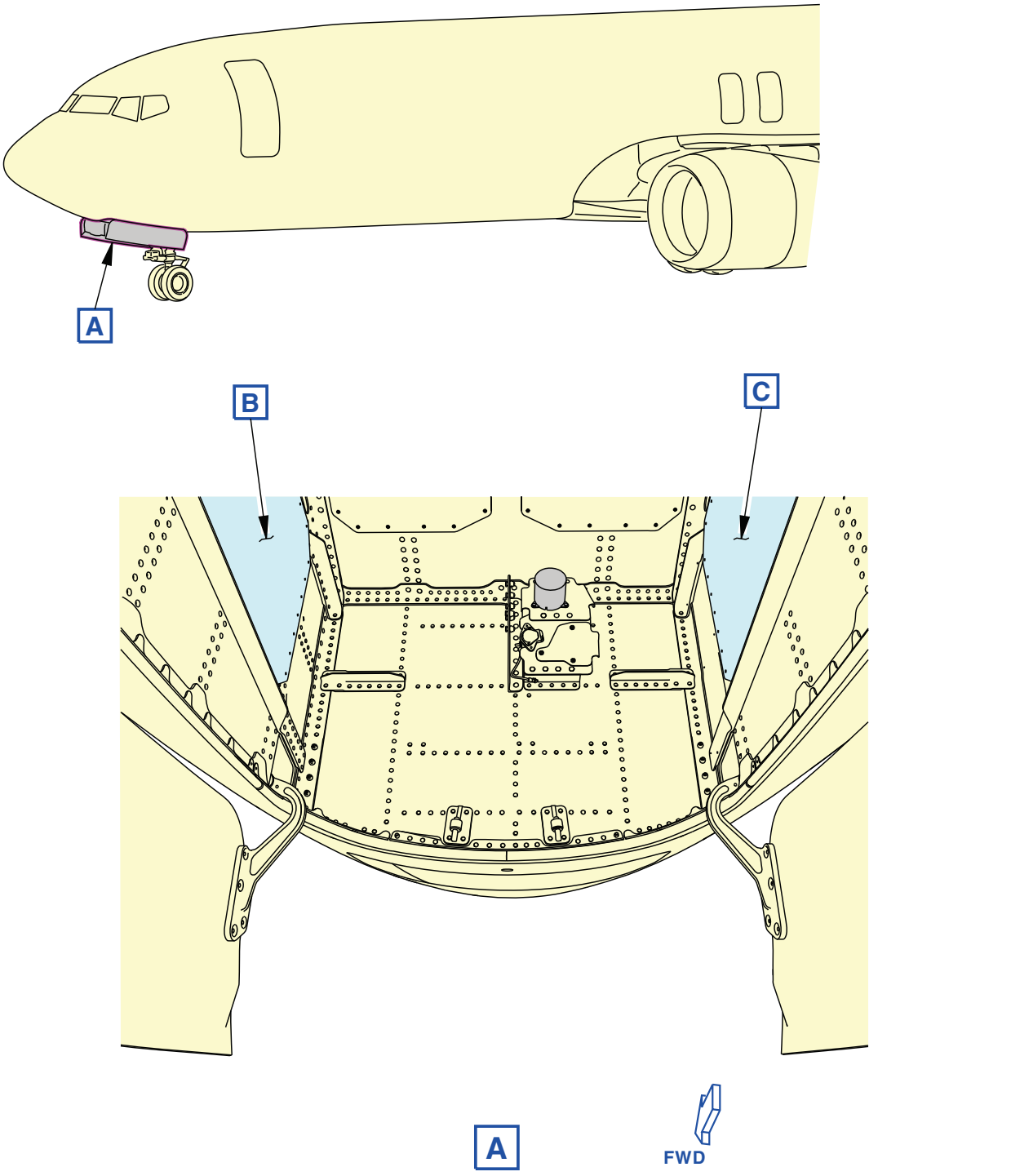
(4) For engine 2 remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

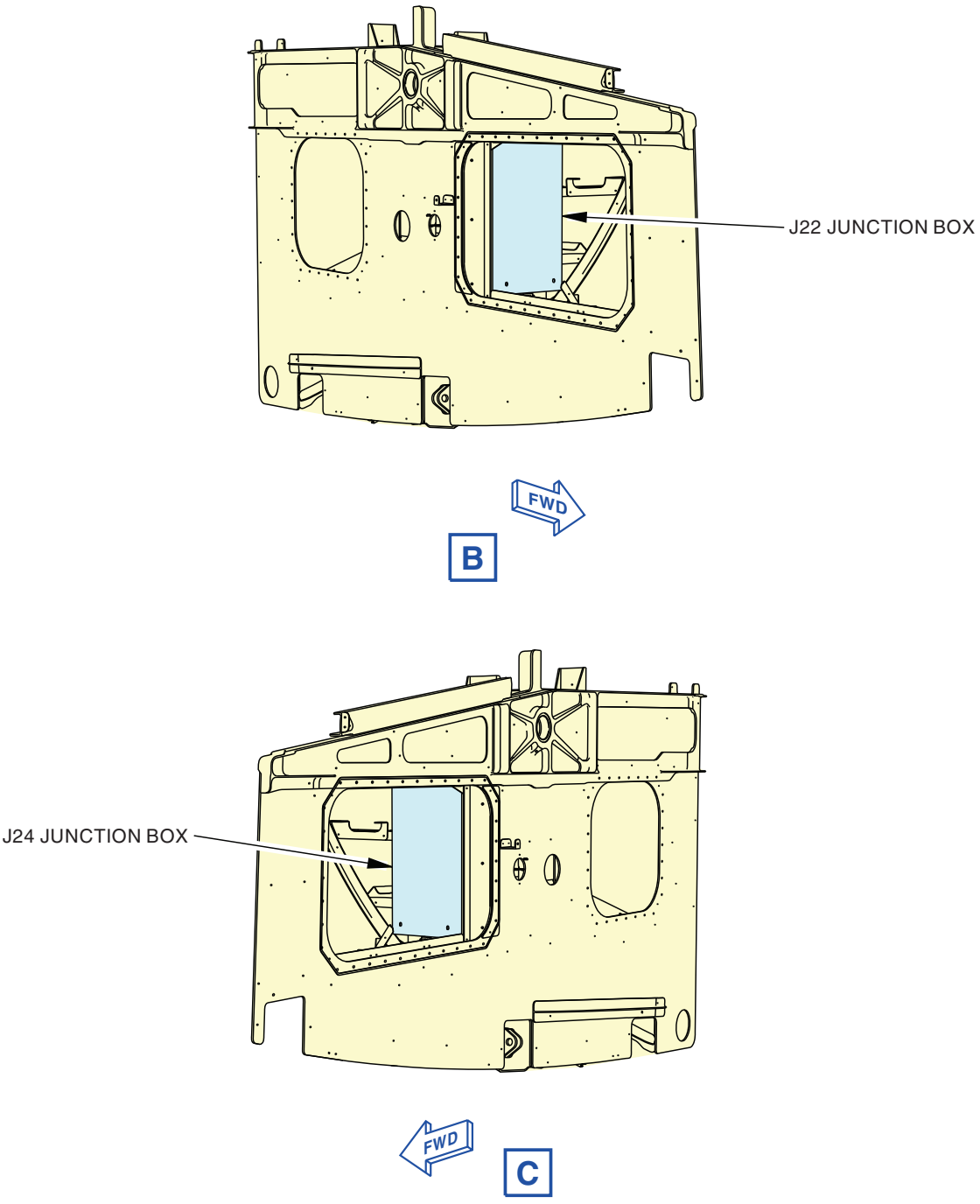
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Junction Box Location
Figure 601/78-34-10-990-804-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Junction Box Location
Figure 601/78-34-10-990-804-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-10-710-802-G00

3. Thrust Reverser Sequence Relay Electrical Resistance Check

(Figure 601)

A. General

(1) This task provides the instructions to operationally check the airplane in air mode while verifying that there is no ground at the T/R Sequence Relay (R495/R496).

B. References

Reference	Title
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-603	Multimeter

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

F. Prepare for the Check

SUBTASK 78-34-10-860-026-G00

(1) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 78-34-10-860-027-G00

(2) For engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-028-G00

(3) For engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-34-10-010-005-G00

(4) For engine 1, open this access panel:

(TASK 53-14-01-020-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 78-34-10-010-006-G00

(5) For engine 2, open this access panel:

(TASK 53-14-01-020-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

SUBTASK 78-34-10-020-003-G00

(6) For engine 1, disconnect the electrical connector from the relay R495.

SUBTASK 78-34-10-020-004-G00

(7) For engine 2, disconnect the electrical connector from the relay R496.

G. Thrust Reverser Sequence Relay Electrical Resistance Check

SUBTASK 78-34-10-710-003-G00

(1) To do the operational check for engine 1, do these steps:

(a) Use a multimeter, STD-603, to measure the resistance between pin D3527-A2 and airplane structure.

(b) Make sure that the resistance value between the relay R495 and airplane structure is 1 megaohm or more.

SUBTASK 78-34-10-710-004-G00

(2) To do the operational check for engine 2, do these steps:

(a) Use a multimeter, STD-603, to measure the resistance between pin D3529-A2 and airplane structure.

(b) Make sure that the resistance value between the relay R496 and airplane structure is 1 megaohm or more.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-10-420-004-G00

(1) For engine 1, connect the electrical connector to the relay R495.

SUBTASK 78-34-10-420-005-G00

(2) For engine 2, connect the electrical connector to the relay R496.

SUBTASK 78-34-10-410-004-G00

(3) For engine 1, close this access panel:

(TASK 53-14-01-420-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-34-10-410-005-G00

(4) For engine 2, close this access panel:

(TASK 53-14-01-420-801)

Number	Name/Location
114AW	Forward Nose Wheel Well Panel

SUBTASK 78-34-10-860-029-G00

(5) For engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	5	C00276	ENGINE 1 THRUST REVERSER CONT
B	7	C01266	ENGINE 1 THRUST REVERSER SYNC LOCK

SUBTASK 78-34-10-860-030-G00

(6) For engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	5	C01267	ENGINE 2 THRUST REVERSER SYNC LOCK
C	7	C00277	ENGINE 2 THRUST REVERSER CONT

SUBTASK 78-34-10-840-001-G00

(7) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STOW HYDRAULIC PRESSURE SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) The removal of the stow hydraulic pressure switch.

(2) The installation of the stow hydraulic pressure switch.

TASK 78-34-11-000-801-G00

2. Stow Hydraulic Pressure Switch Removal

(Figure 401)

A. General

(1) This task is the removal of the stow hydraulic pressure switch.

(2) The stow hydraulic pressure switch is located in the forward area of the engine fan cowl support beam.

(3) There is one stow hydraulic pressure switch for each engine.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-21-00-000-802	Standby Hydraulic System Power Removal (P/B 201)
54-52-01-010-801	Forward Fairing Removal (P/B 401)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-7394	Cap - Dust, Electrical Connector
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Prepare for the Removal

SUBTASK 78-34-11-040-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 78-34-11-040-002-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(2) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 78-34-11-860-001-G00

(3) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 78-34-11-860-002-G00

(4) Do this task: Standby Hydraulic System Power Removal, TASK 29-21-00-000-802.

SUBTASK 78-34-11-860-003-G00

(5) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 78-34-11-860-004-G00

(6) Do this task: Forward Fairing Removal, TASK 54-52-01-010-801.

E. Stow Hydraulic Pressure Switch Removal

SUBTASK 78-34-11-020-001-G00

(1) Remove the stow hydraulic pressure switch [1]:

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTIONS ARE NOT DIRTY BEFORE YOU CONNECT OR DISCONNECT THEM. SOLID OR LIQUID CONTAMINATION OF THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Disconnect the electrical connector [11] from the stow hydraulic pressure switch [1].

(b) Put dust caps, STD-7394, on the electrical connector [11] and the electrical receptacle on the stow hydraulic pressure switch [1].

WARNING MAKE SURE THAT YOU WEAR THE PERSONAL PROTECTIVE EQUIPMENT WHEN YOU DO THIS TASK. PERSONAL PROTECTIVE EQUIPMENT WILL PREVENT INJURIES TO PERSONNEL.

CAUTION USE TWO WRENCHES TO DISCONNECT OR CONNECT TUBES. IF YOU DO NOT USE A WRENCH ON EACH SIDE OF THE CONNECTION, DAMAGE TO THE TUBES CAN OCCUR.

(c) Loosen the B-nut that attaches the stow hydraulic pressure switch [1] to the thrust reverser stow hydraulic pressure line.

1) Wrap a cloth rag around the B-nut and the wrench to catch any hydraulic fluid.

(d) Remove the screw [3], nut [5], two washers [4], and bonding clamp [6] from the bonding adapter [2] on the stow hydraulic pressure switch [1].

(e) Loosen the four screws [9] that attach the two clamps [8] to the fan cowl support beam bracket.

(f) Remove the stow hydraulic pressure switch [1].

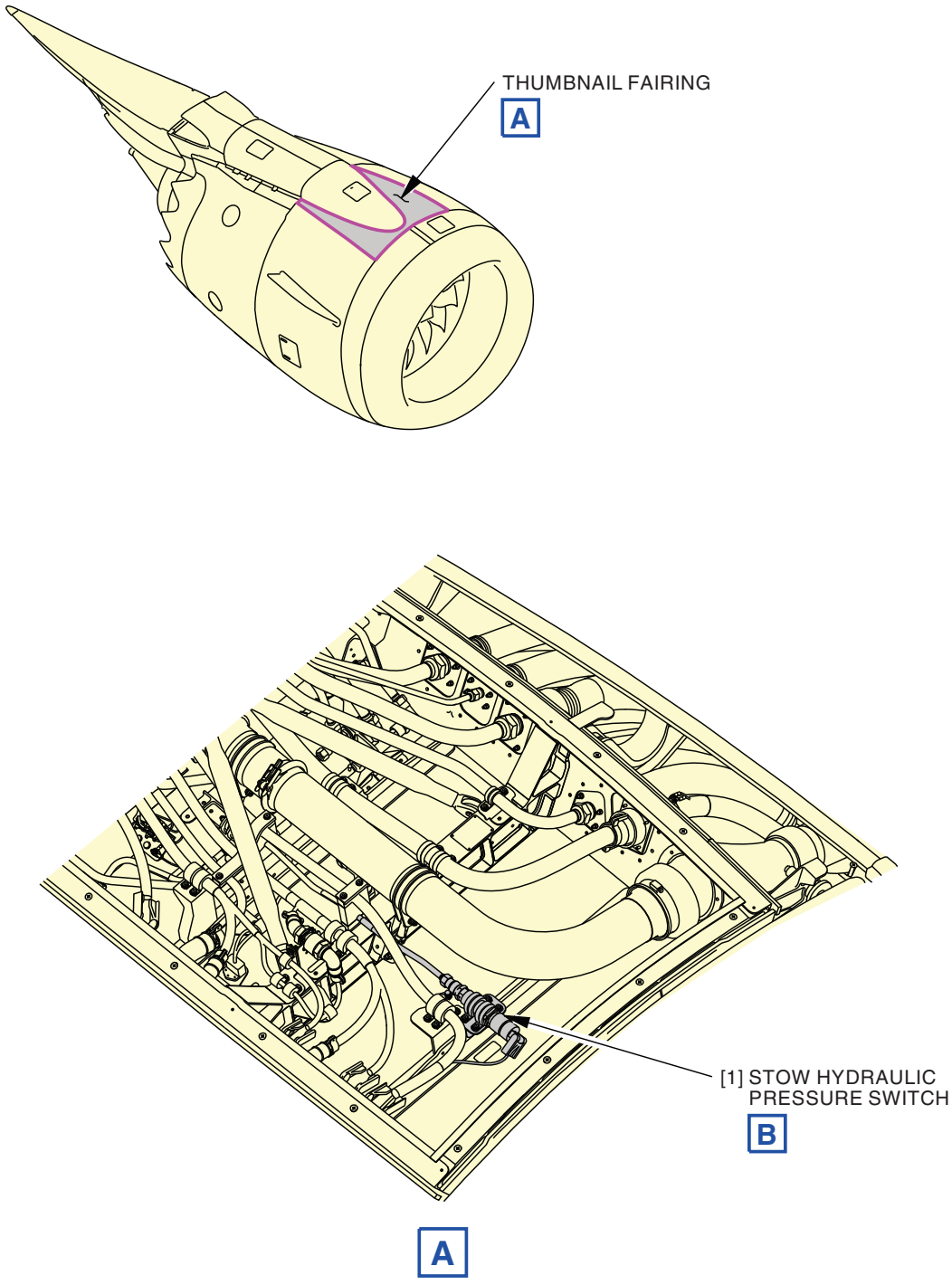
(g) If it is necessary, remove the bonding adapter [2] and packing [7] from the stow hydraulic pressure switch [1].

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(h) Put caps/plugs, STD-14686, on the open hydraulic line and the open ports on the stow hydraulic pressure switch [1] and bonding adapter [2].

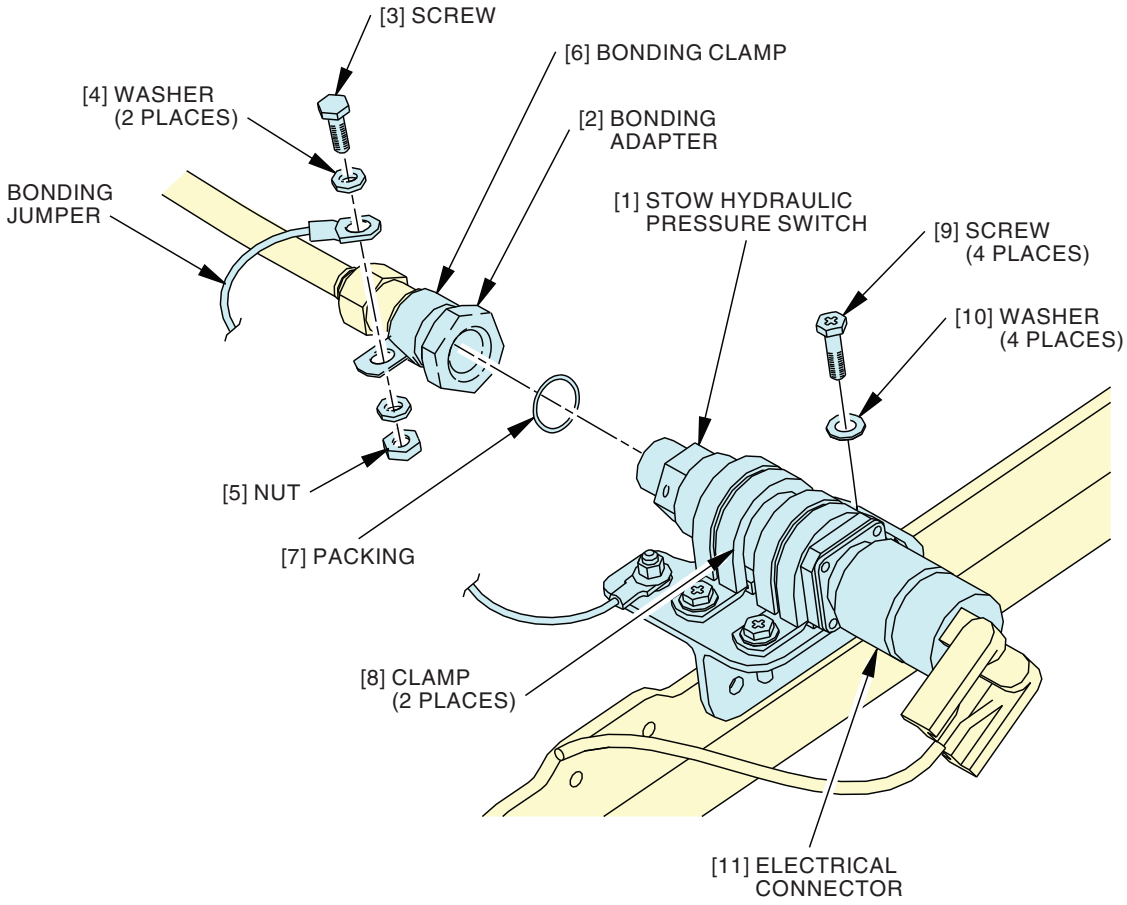
——— END OF TASK ———

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Stow Hydraulic Pressure Switch Installation
Figure 401/78-34-11-990-801-G00 (Sheet 1 of 2)

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Stow Hydraulic Pressure Switch Installation

B

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Stow Hydraulic Pressure Switch Installation
Figure 401/78-34-11-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-34-11-400-801-G00

3. Stow Hydraulic Pressure Switch Installation

(Figure 401)

A. General

(1) This task is the installation of the stow hydraulic pressure switch.

(2) The stow hydraulic pressure switch is located in the forward area of the engine fan cowl support beam.

(3) There is one stow hydraulic pressure switch for each engine.

B. References

Reference	Title
29-00-00-800-802	Hydraulic Power System - Activation (P/B 201)
54-52-01-410-801	Forward Fairing Installation (P/B 401)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
78-31-00-700-802-G00	Thrust Reverser System Normal Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50096	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63 Type II
B50073	Alcohol - Isopropyl	ASTM D 770

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Stow hydraulic pressure switch	54-51-51-11-315	PTF ALL
7	Packing	54-51-51-11-325	PTF ALL

F. Stow Hydraulic Pressure Switch Installation

SUBTASK 78-34-11-420-001-G00

(1) Do these steps to install the stow hydraulic pressure switch [1]:

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CAUTION USE TWO WRENCHES TO TIGHTEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

(a) If it is necessary, install the bonding adapter [2] and packing [7] to the stow hydraulic pressure switch [1]:

1) Make sure that the interfacing surfaces of the bonding adapter [2] and the stow hydraulic pressure switch [1] are free of oil, grease or other contaminants.

WARNING DO NOT GET ISOPROPYL ALCOHOL (IPA) IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM IPA. PUT ON GOGGLES, AND GLOVES WHEN YOU USE IPA. KEEP IPA AWAY FROM SPARKS, FLAME, AND HEAT. IPA IS POISONOUS AND FLAMMABLE. IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

2) If it is necessary, clean the interfacing surfaces with alcohol, B50073.

3) Install the packing [7] onto the stow hydraulic pressure switch [1].

4) Apply sealant, A50096, to the threads and interfacing surfaces of the stow hydraulic pressure switch [1] and the bonding adapter [2].

5) Install the bonding adapter [2] onto the stow hydraulic pressure switch [1] and tighten to 170 ±17 in-lb (19 ±2 N·m).

6) Use an intrinsically safe approved bonding meter, COM-1550, to do a check of the bonding resistance between the stow hydraulic pressure switch [1] and the bonding adapter [2] (SWPM 20-20-00).

7) Make sure that the bonding resistance between the stow hydraulic pressure switch [1] and the bonding adapter [2] is 0.0025 ohm (2.5 milliohms) or less.

8) Loosely attach the stow hydraulic pressure switch [1] to the bracket with the two clamps [8], four washers [10] and four screws [9].

9) Orient the so that the master key clocking of the electrical connector [11] is at the 12 o'clock position.

10) Make sure that the interfacing surfaces of the bonding adapter [2] and the bonding clamp [6] and the bonding jumper are free of oil, grease or other contaminants.

WARNING DO NOT GET ISOPROPYL ALCOHOL (IPA) IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM IPA. PUT ON GOGGLES, AND GLOVES WHEN YOU USE IPA. KEEP IPA AWAY FROM SPARKS, FLAME, AND HEAT. IPA IS POISONOUS AND FLAMMABLE. IT CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

11) If it is necessary, clean the interfacing surfaces with alcohol, B50073.

12) Put the bonding clamp [6], screw [3], two washers [4], bonding jumper and nut [5] into their position on the bonding adapter, but do not tighten the screw [3] at this time.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION USE TWO WRENCHES TO TIGHTEN THE TUBE COUPLING NUTS. USE ONE TO HOLD THE MATING PART, AND THE OTHER TO TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE TUBES AND MATING PART CAN OCCUR.

13) Install the hydraulic tube onto the stow hydraulic pressure switch [1].

14) Make sure to maintain the orientation of the stow hydraulic pressure switch [1], and tighten the hydraulic tube B-nut to 130 ±6 in-lb (14.7 ±0.7 N·m).

15) Loosen the B-nut so that it can be turned by hand, then retighten the B-nut to 130 ±6 in-lb (14.7 ±0.7 N·m) and apply a torque stripe.

16) Tighten the screws [9] to 35.0 ±3.5 in-lb (4.0 ±0.4 N·m).

17) Make sure that the bonding clamp [6] is in the correct position, and tighten the screw [3] to 35.0 ±3.5 in-lb (4.0 ±0.4 N·m).

18) Use an intrinsically safe approved bonding meter, COM-1550, to do a check of the bonding resistance between the bonding jumper terminal and the bonding adapter [2] (SWPM 20-20-00).

19) Make sure that the bonding resistance between the bonding jumper terminal and the bonding adapter [2] is 0.0017 ohm (1.70 milliohms) or less.

20) Connect the electrical connector [11] to the stow hydraulic pressure switch [1].

G. Stow Hydraulic Pressure Switch Installation Test

SUBTASK 78-34-11-860-005-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-34-11-210-001-G00

(2) Do this task: Hydraulic Power System - Activation, TASK 29-00-00-800-802.

(a) Do a check of the stow hydraulic pressure switch [1] for hydraulic leaks.

SUBTASK 78-34-11-710-001-G00

(3) Do this task: Thrust Reverser System Normal Operational Test, TASK 78-31-00-700-802-G00.

(a) Operate the thrust reverser a minimum of three cycles.

(b) Do a check of the stow hydraulic pressure switch [1] for hydraulic leaks.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 78-34-11-860-006-G00

(1) Do this task: Forward Fairing Installation, TASK 54-52-01-410-801.

SUBTASK 78-34-11-440-001-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the LVDT

(2) An installation of the LVDT.

TASK 78-36-02-000-801-G00

2. LVDT Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Linear Variable Differential Transformer (LVDT).

(2) There is one LVDT on each of the upper locking actuators on the torque box of the left and right thrust reversers on an engine.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Removal

SUBTASK 78-36-02-010-001-G00

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

SUBTASK 78-36-02-040-001-G00

WARNING DO THE DEACTIVATION PROCEDURE TO PREVENT THE OPERATION OF THE THRUST REVERSER. THE ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 78-36-02-860-001-G00

(3) For Engine 1, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK

SUBTASK 78-36-02-860-002-G00

(4) For Engine 2, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F. LVDT Removal

SUBTASK 78-36-02-020-001-G00

(1) Disconnect the two electrical connectors from the LVDT [5] on the applicable engine.

(a) For the left thrust reverser, disconnect the electrical connectors D30072 and D30076.

(b) For the right thrust reverser, disconnect the electrical connectors D30074 and D30078.

(c) Install the protective caps on the electrical connectors.

SUBTASK 78-36-02-020-002-G00

(2) Remove the LVDT [5].

(a) Remove the nut [2] and washer [1] that attaches the LVDT rod end to the actuator feedback spindle.

(b) Remove the bolts [8], washers [4], and nuts [3], or collars [15] that attach the LVDT [5] to the bracket [13].

1) Discard the bolts [8] and nuts [3] or collars [15].

(c) Remove and discard the safety cable or lockwire from the three screws [6].

(d) Remove the three screws [6] and the washers [7] that attach the LVDT [5] to the actuator.

(e) Remove the LVDT [5] from the actuator.

(f) Install the protective covers on the LVDT [5] receptacles.

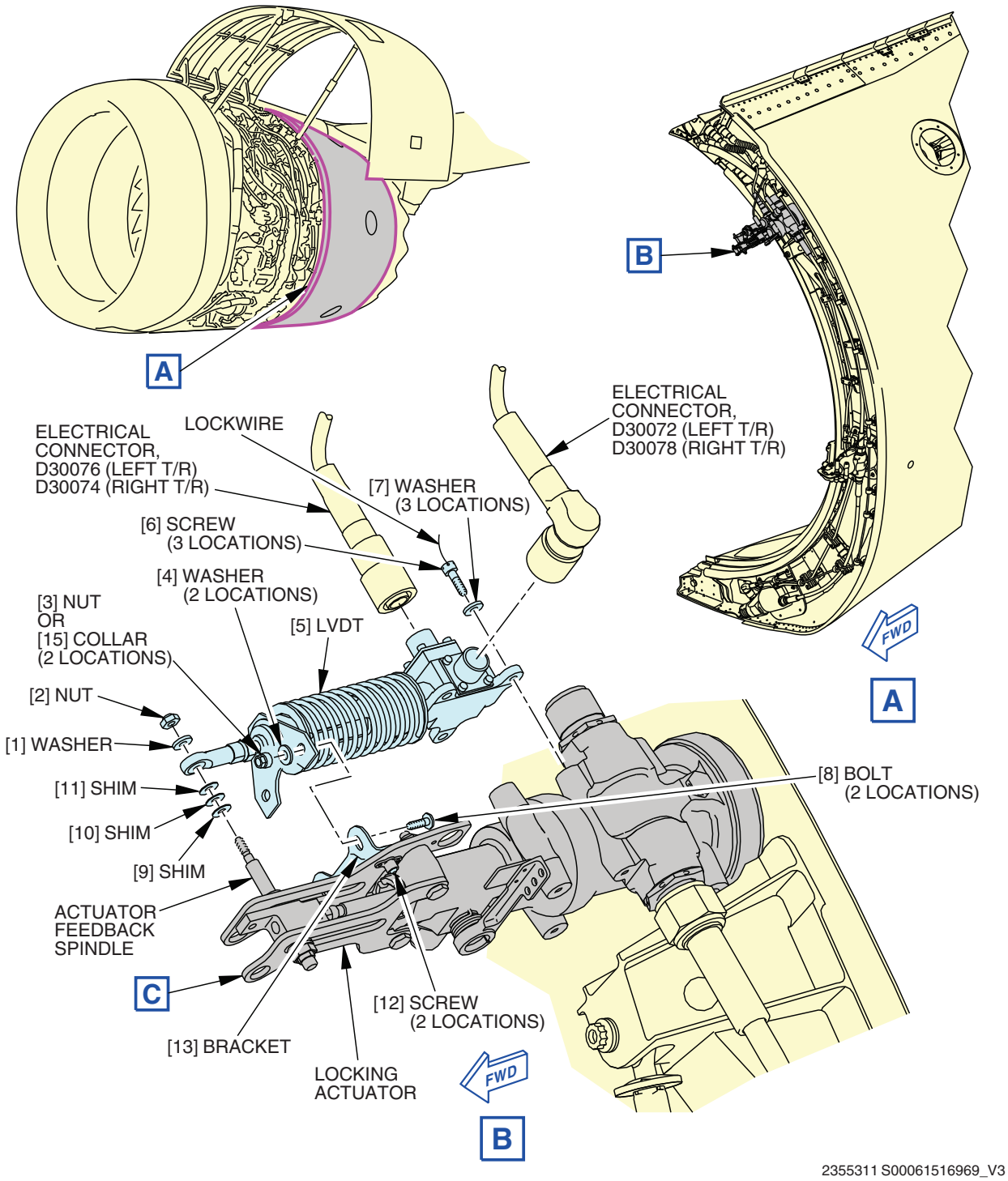
(g) Remove the shim [9], shim [10], or shim [11] from the actuator feedback spindle.

NOTE: It is important to note the quantity and the configuration of the shims for the subsequent installation.

(h) Clean all the remaining sealant off the bracket [13] with solvent, B00062.

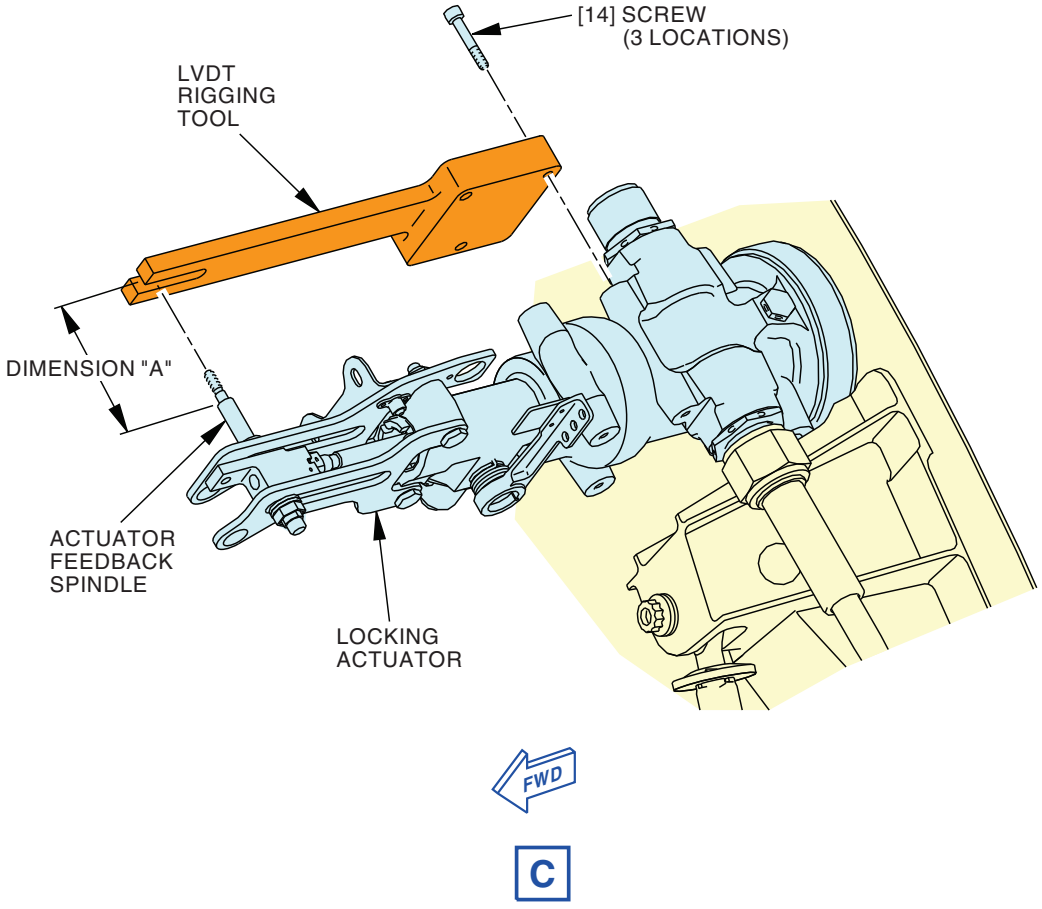
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Thrust Reverser LVDT Installation
Figure 401/78-36-02-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Thrust Reverser LVDT Installation
Figure 401/78-36-02-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 78-36-02-400-801-G00

3. LVDT Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Linear Variable Differential Transformer (LVDT).

B. References

Reference	Title
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-710-806-G00	Test No. 17 - Electronic Engine Control (EEC) System Test (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-7480	Rigging Bar - Thrust Reverser, Position Feedback LVDT Part #: B78006-7 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Nut	78-36-02-01-045	PTF ALL
5	LVDT	78-36-02-01-065	PTF ALL
8	Bolt	78-36-02-01-020	PTF ALL
15	Collar	78-36-02-01-047	PTF ALL

F. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left

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(Continued)

Zone	Area
426	Engine 2 - Thrust Reverser, Right

G. Prepare for the Installation

SUBTASK 78-36-02-420-001-G00

(1) Do these steps to find the correct thickness of the shims to be installed on the actuator feedback spindle:

(a) Install the LVDT rigging bar, SPL-7480, on the actuator using three 10-32 X 1 inch screws [14].

(b) Examine the actuator feedback spindle for axial free-play.

1) If free-play exists, set the actuator feedback spindle at the approximate middle position of the free-play.

(c) Measure the distance between the shoulder of the actuator feedback spindle and the rigging tool with a feeler gauge to get dimension A.

(d) Record dimension A that you measured.

NOTE: This dimension will be used to calculate the correct shim thickness.

(e) Remove the three screws [14] and the rigging bar, SPL-7480.

H. LVDT Installation

SUBTASK 78-36-02-390-001-G00

(1) Install the LVDT [5].

(a) Remove the protective caps from the electrical connectors.

(b) Remove the protective covers from the LVDT [5] receptacles.

(c) To prepare the area for the sealant, clean the bracket [13] and LVDT flange areas, that will touch when you install the LVDT, with solvent, B00062.

(d) Apply sealant, A00247, to the area that was cleaned.

(e) Identify the quantity and configuration of the shim [9], shim [10], and shim [11] that will equal dimension A ±0.002 in. (0.051 mm) that you recorded above.

NOTE: The shim [9] is 0.010 in. (0.254 mm) thick, the shim [10] is 0.005 in. (0.127 mm) thick and the shim [11] is 0.003 in. (0.076 mm) thick.

1) If the thickness of the shim [9], shim [10], or shim [11] that you removed during the removal task equals dimension A ±0.002 in. (0.051 mm), use the shims.

2) If the removed shims do not give the correct thickness, use the minimum quantity and configuration of shims that is necessary to give a thickness that is equal to dimension A ±0.002 in. (0.051 mm).

(f) Install the shim [9], shim [10], or shim [11] on the actuator feedback spindle.

CAUTION

MAKE SURE THAT YOU DO NOT BEND THE ROD END ON THE LVDT OR SHAFT WHEN YOU INSTALL THE LVDT. IF YOU BEND THE ROD END OR SHAFT, AN INCORRECT INDICATION OR DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Install the LVDT [5] on the locking actuator with the rod end on the actuator feedback spindle.

(h) Install the three screws [6] and the washers [7] that attach the LVDT [5] to the actuator.

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1) Tighten the screws to 27 in-lb (3.1 N·m) - 30 in-lb (3.4 N·m).

2) Install MS20995NC32 lockwire, G50347, or MS20995C32 lockwire, G01048, with double twist method, or the safety cable kit, G50375, on the three screws [6] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

(i) Push the bracket to make sure that the flange on the LVDT [5] is firmly against the bracket [13].

1) If the flange is not firmly against the bracket [13], loosen the two screws [12] and move the bracket.

2) Re-tighten the two screws [12] to 27 in-lb (3.1 N·m) - 33 in-lb (3.7 N·m).

(j) Install the new bolts [8], washers [4], and new nuts [3] or collars [15] that attach the LVDT [5] to the bracket [13].

NOTE: One extra washer can be used under each nut if it is necessary to adjust the grip length of the bolts.

1) Tighten the nuts [3] to 34 in-lb (3.8 N·m) - 36 in-lb (4 N·m).

2) Remove the unwanted sealant from the bracket [13].

(k) Install the nut [2] and washer [1] that attaches the LVDT rod end to the actuator feedback spindle.

1) Tighten the nut to 20 in-lb (2.3 N·m) - 30 in-lb (3.4 N·m).

SUBTASK 78-36-02-420-002-G00

(2) Connect the two electrical connectors to the LVDT [5] receptacles.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 78-36-02-440-001-G00

(1) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 78-36-02-860-003-G00

(2) For Engine 1, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	6	C01412	ENGINE 1 THRUST REVERSER INTLK

SUBTASK 78-36-02-860-004-G00

(3) For Engine 2, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	6	C01413	ENGINE 2 THRUST REVERSER INTLK
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

SUBTASK 78-36-02-710-001-G00

(4) Do this task: Test No. 17 - Electronic Engine Control (EEC) System Test, TASK 71-00-00-710-806-G00.

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SUBTASK 78-36-02-410-001-G00

(5) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT) - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Linear Variable Differential Transformer (LVDT) - Adjustment.

TASK 78-36-02-820-801-G00

2. Linear Variable Differential Transformer (LVDT) - Adjustment

(Figure 502)

A. General

(1) This task gives the instructions to adjust the thrust reverser linear variable differential transformer (LVDT).

(2) There is one LVDT on each of the upper locking actuators on the torque box of the left and right thrust reversers on an engine.

(3) Use the display values on the Multi-Function Display (MFD) screen in the flight compartment to adjust the LVDT's one at a time.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-980-801-G00	Thrust Reverser Operation - Extend (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Prepare for the Adjustment

SUBTASK 78-36-02-860-005-G00

CAUTION

DO NOT OPERATE THE THRUST REVERSER WHEN ELECTRICAL-POWER INTERRUPTIONS CAN OCCUR (FOR MORE THAN A NORMAL BUS-TRANSFER). IF YOU INTERRUPT THE ELECTRICAL POWER WHILE THE THRUST REVERSER IS IN TRANSIT, DAMAGE TO THE SYNC LOCKS CAN OCCUR. IF THAT OCCURS, YOU MUST COMPLETE THE SYNC-LOCK OPERATIONAL TEST.

(1) Do not operate the thrust reverser if there will be electrical power interruptions (for more than a normal bus transfer) while the thrust reverser is in transit.

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SUBTASK 78-36-02-860-019-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 78-36-02-860-020-G00

(3) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-36-02-860-021-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-36-02-860-022-G00

(5) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-36-02-860-023-G00

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-36-02-860-007-G00

(7) Make sure that the applicable thrust reverser is closed.

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(a) If it is necessary, do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 78-36-02-860-029-G00

(8) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the ENABLE position.

SUBTASK 78-36-02-860-030-G00

(9) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 78-36-02-860-008-G00

(10) Select the left or right Multi-Function Display (MFD).

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 78-36-02-860-011-G00

(11) Make sure that the applicable thrust lever is in the idle position.

(a) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 78-36-02-860-012-G00

(12) Make sure that the applicable reverse thrust lever is in the retract (stow) position.

(a) Make sure that the thrust reverser is in the retracted (stowed) position.

SUBTASK 78-36-02-860-024-G00

(13) Make sure that the EEC Data Load and Trim Balance Functions are inactive.

SUBTASK 78-36-02-860-006-G00

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(14) Put the applicable HYD PUMPS A ENG 1 or HYD PUMPS B ENG 2 switch to the ON position.

F. Linear Variable Differential Transformer (LVDT) - Adjustment

SUBTASK 78-36-02-710-002-G00

(1) Use the Multi-Function Display (MFD) screen to do a check of the LVDT position values with the thrust reverser in the retracted (stowed) position:

(a) Select the EPCS page 1 from the Maintenance Data Pages on the Multi-Function Display (MFD).

(b) Make sure that the position values that are shown for the applicable thrust reverser sleeve are in the limits that follow:

NOTE: The range limit is -5.0% to 112.0%. If the indication is out of range, less than -5.0% or greater than 112.0%, the field will be blank.

1) The position channel A limit, is 0.0 ±4%.

2) The position channel B limit, is 0.0 ±4%.

(c) If the position values are not in the limits, do the steps that follow to adjust the LVDT in the retracted (stowed) position.

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(d) If the position values are in the limits, continue at the steps below to do a check of the position values with the thrust reverser in the extended (deployed) position.

SUBTASK 78-36-02-820-001-G00

(2) Do these steps to adjust the LVDT on the applicable thrust reverser in the retracted (stowed) position:

(a) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(b) Loosen the jamnut [1] on the LVDT transducer.

(c) Turn the rod a small amount in one direction or the other to get the correct value.

NOTE: After the rod is turned, the new value will show on the MFD in approximately two seconds.

1) Wait for approximately two seconds.

2) Make sure that the position channel A limit, for the applicable thrust reverser sleeve, is 0.0 ±4%.

3) Make sure that the position channel B limit, for the applicable thrust reverser sleeve, is 0.0 ±4%.

(d) If the position values are not in the limits, continue to turn the rod until the values are correct.

(e) If the position values are in the limits, tighten the jamnut [1] to 20 in-lb (2.3 N·m)-30 in-lb (3.4 N·m).

1) Make sure that the position values are still in the limits.

SUBTASK 78-36-02-710-003-G00

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE AREA AFT OF THE APPLICABLE THRUST REVERSER. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

DO NOT EXTEND THE TRANSLATING SLEEVES IF A THRUST REVERSER HALF IS OPEN. DAMAGE TO THE THRUST REVERSER AND ADJACENT STRUCTURES CAN OCCUR.

(3) Do these steps to do a check of the LVDT position values with the thrust reverser in the extended (deployed) position:

(a) Do this task: Thrust Reverser Operation - Extend (Selection), TASK 78-31-00-980-801-G00.

(b) Make sure that the REV light turns amber when the thrust reverser sleeves are in transit and then turns green when the thrust reverser is fully extended.

(c) Make sure that the position values that are shown for the left and right thrust reversers are in the limits that follow:

NOTE: The range limit is -5.0% to 112.0%. If the indication is out of range, less than -5.0% or more than 112.0%, the field will be blank.

1) Make sure that the position channel A limit, is 100.0 ±5%.

2) Make sure that the position channel B limit, is 100.0 ±5%.

(d) If the position values are in the limits, continue at the steps below to do a check that the position values are still in the limits in the retracted (stowed) position.

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(e) If the position values are not in the limits, do the steps that follow to adjust the LVDT in the extended (deployed) position.

SUBTASK 78-36-02-710-004-G00

(4) Do these steps to adjust the LVDT on the applicable thrust reverser in the extended (deployed) position:

(a) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(b) Loosen the jamnut [1] on the LVDT transducer.

(c) Turn the rod a small amount in one direction or the other to get the correct value.

NOTE: After the rod is turned, the new value will show on the MFD in approximately two seconds.

1) Wait for approximately two seconds.

2) Make sure that the position channel A limit, is 100.0 ±5%.

3) Make sure that the position channel B limit, is 100.0 ±5%.

(d) If the position values are not in the limits, continue to turn the rod until the values are correct.

(e) If the thrust reverser sleeve is in the limits, tighten the jamnut [1] to 20 in-lb (2.3 N·m)-30 in-lb (3.4 N·m).

1) Make sure that the position values are still in the limits.

SUBTASK 78-36-02-820-002-G00

(5) Do these steps to make sure that the position values for the applicable thrust reverser sleeve are still in the limits in the retracted (stowed) position:

NOTE: If the TR Special Function has timed out because there was no movement of the throttle levers for greater than 5 minutes, the TR Special Function will have to be restarted.

(a) Move the applicable reverse thrust lever to the retract (stow) position.

(b) Make sure that the position channel A limit, is 0.0 ±4%.

(c) Make sure that the position channel B limit, is 0.0 ±4%.

(d) If the position values are not in the limits, do the adjustments at the retracted (stowed) and extended (deployed) positions again.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 78-36-02-410-002-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

SUBTASK 78-36-02-860-031-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

SUBTASK 78-36-02-860-032-G00

(3) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

SUBTASK 78-36-02-860-013-G00

(4) Put the applicable HYD PUMPS A ENG 1 or HYD PUMPS B ENG 2 switch to the OFF position.

SUBTASK 78-36-02-860-016-G00

(5) Remove the DO NOT OPERATE tag, STD-858, from the applicable thrust lever.

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SUBTASK 78-36-02-860-018-G00

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 78-36-02-860-025-G00

(7) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 78-36-02-860-026-G00

(8) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 78-36-02-860-027-G00

(9) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 78-36-02-860-028-G00

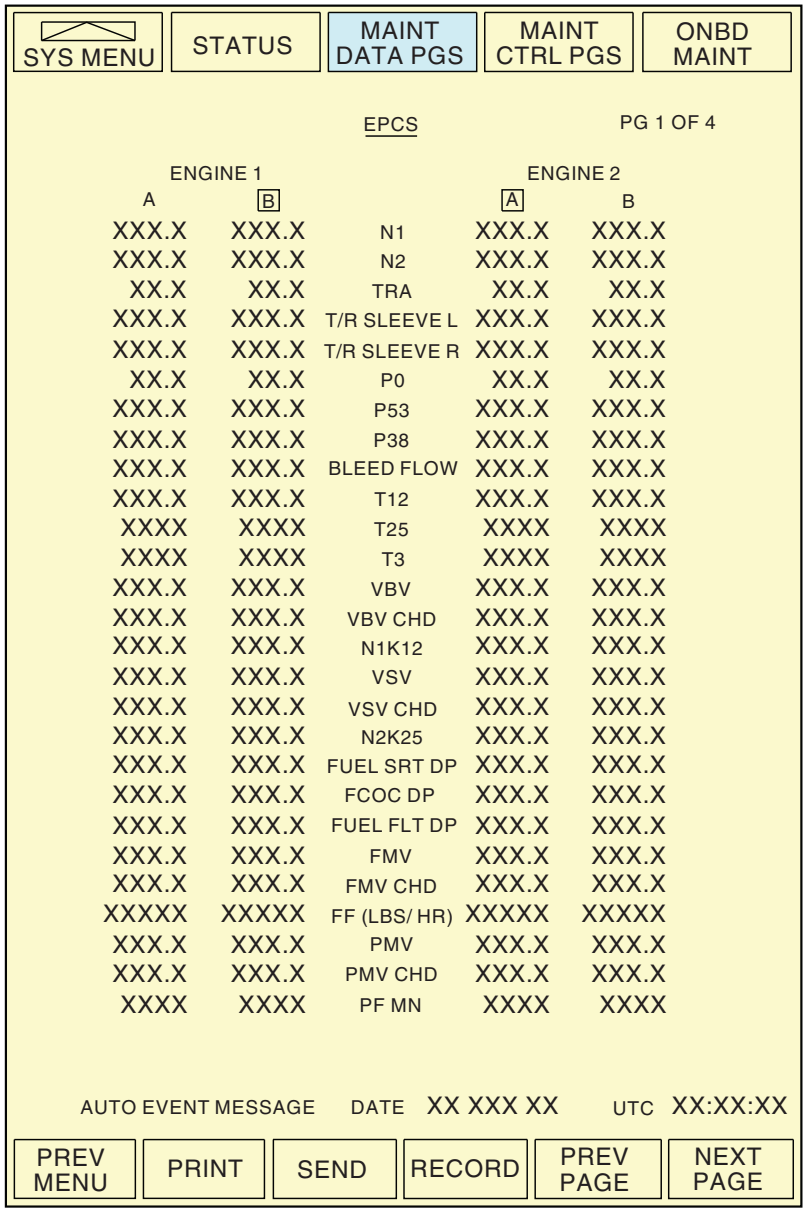
(10) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

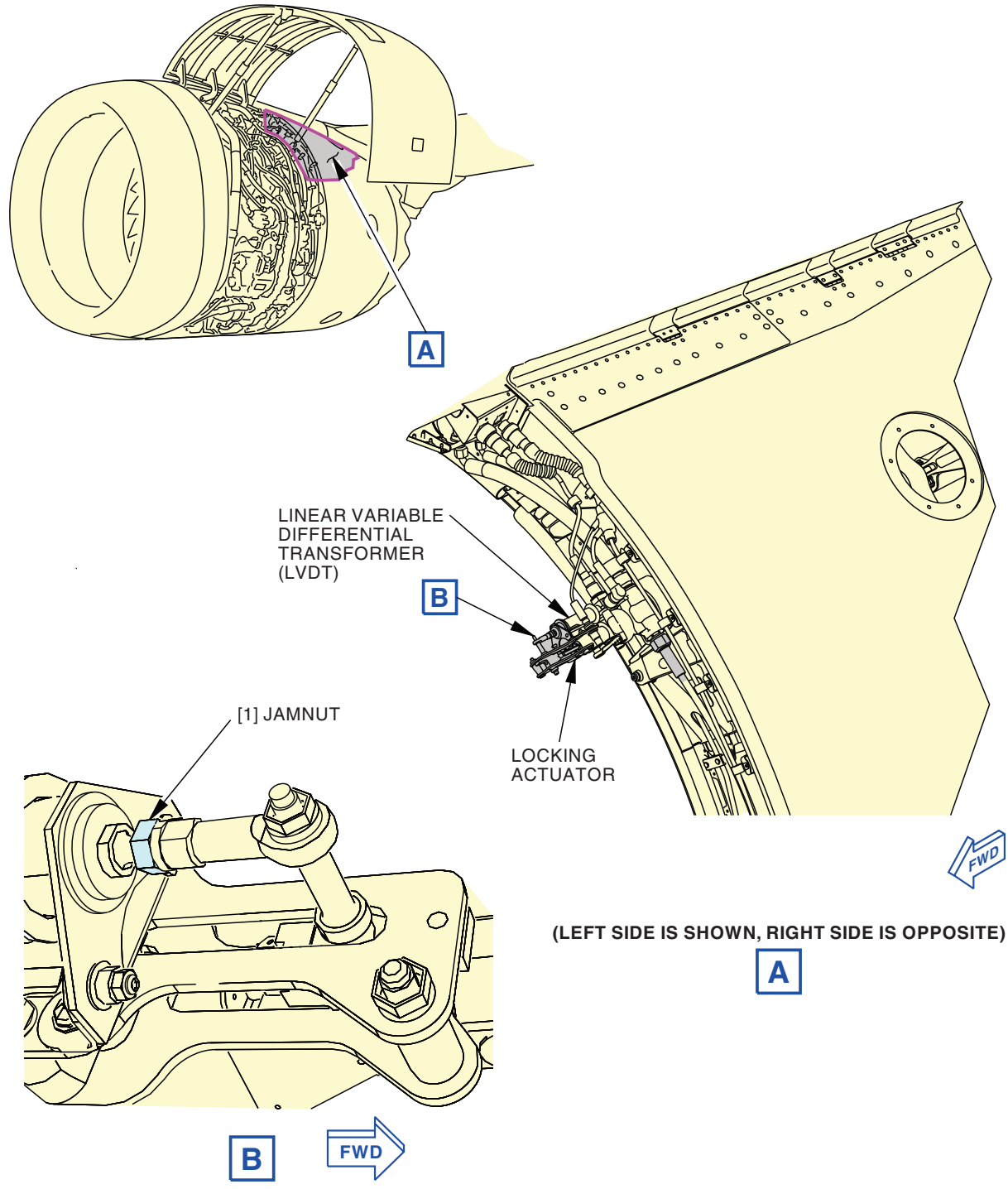
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EPCS Page 1 Status Display
Figure 501/78-36-02-990-804-G00

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Linear Variable Differential Transformer Adjustment
Figure 502/78-36-02-990-803-G00

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